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TRADING GLOBAL MACRO MARKETS

DIRK WILLER
ALEX SAUNDERS

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CHAPTER 1

Introduction

The earlier book on emerging market fixed income trading (Willer et al., 2020) was very well received. Many readers, both novices and, encouragingly, also seasoned practitioners in emerging markets (EM), called it the best book of its kind. The reason is, no doubt, that it is also the *only* book of its kind: truly written by practitioners for practitioners. The most common request from readers was for pointers to similar books for assets outside of emerging markets. As we were unable to find any similar books for most other asset classes, we resolved to write one ourselves. The effort was helped by two factors: First, Dirk's day job had shifted away from emerging markets and now focuses on global macro strategy and asset allocation. Second, Alex Saunders joined Citi's Global Macro team, heading Citi's Quant Global Macro group. This led to several years of cross-fertilization between our day jobs and this book. While the asset classes discussed in this book differ from the last one, the goals are still the same:

This book will make you a better portfolio manager and risk taker, and the key to success in markets is twofold. First, a solid methodology is crucial. And second, conviction to press the bets when the stars align is maybe even more important. The two issues are obviously linked. A solid methodology is, after all, what creates conviction. Such a methodology can come from participating in markets for a long time and internalizing what "works" along the way. The gut feeling of an experienced portfolio manager may be what gives conviction. But there is a shortcut. Data can substitute for hands-on experience and is likely even superior. After all, backtesting trading rules and analyzing what works across many countries, instruments, and time periods is more reliable than just using data subconsciously collected during any individual trading career, which is necessarily more limited in terms of instruments traded and timespans covered. This book is going to do the data analysis for our readers to generate a robust trading methodology for the important global macro markets. The result is a set of empirically validated rules of thumb that we found helpful to generate alpha. The rules are valuable for benchmarked, long-only investment funds, as well as for hedge funds. We establish both what works and what does not work, even though many think it does. Often, it is of equal importance to understand what does not work, as noise reduction is important to generate conviction.

This book will make you a better policymaker. While the book is written with the perspective of making money in markets, some chapters are also very relevant for policymakers. In particular, it is very important for policymakers to understand how markets will react to their various measures, be it foreign exchange (FX) intervention or interest rate policies. Actions of policymakers generate many of the opportunities for markets. But at the same time, understanding markets creates opportunities for policymakers.

This book will make you a better academic and quantitative researcher. This book is not at all written with an academic mindset. And while writing the book involved a significant amount of coding, we are also not doing the work of hardcore quants. We are willing to live with small sample sizes and simplifying assumptions in our backtests. In order to bring the ideas across, we typically assume zero transaction costs, and we don't worry about risk management overlays. But the book is about systematic ideas and about figuring out how markets work. We trust that academics and quantitative researchers will find many ideas interesting and worthy of further study. This book will make academics' output more relevant to the real world. For this subset of readers, we apologize in advance that some of our studies are based on data that only institutional clients of Citi have access to.

Combined, the authors have more than 55 years of experience in markets on the buy and sell side. At Citi, we have thousands of clients we constantly interact with and give investment advice to, including real money funds, hedge funds, pension funds, sovereign wealth funds, corporates, as well as policymakers. And over the years, we have learned a lot from our clients as well as from our colleagues, who cover most liquid assets under the sun, as well as many illiquid ones. We would note, though, that the views expressed in the book are the views of the authors and do not reflect the views of Citi Research or the Macro Strategy Group within Citi Research. The long-term performance of our investment recommendations has been outstanding, throughout bull and bear markets. We have highlighted the performance in EM in the EM book (for an update, see Willer et al., 2024d), but our newer global macro effort is also performing well, with our discretionary portfolio recommendations realizing an information ratio of 1.1 since January 2023 and until the time of writing in October of 2025 (see Willer et al., 2025c and Massy-Collier et al., 2025). As such, the methodology that we are proposing in this book has stood the test of time, though past performance never guarantees future returns. One reason for the resilience of our framework is that we care mostly about "deep" fundamentals of human psychology and behavioral patterns that are unlikely to change, rather than just economic fundamentals. We therefore expect many of our frameworks to keep doing well in the future, even though much will change, too. The book is not and should not be seen as an offer to purchase any financial products in general or Citi products in particular.

So why did we write this book? First, we see the book as our contribution to finding the "truth" of how markets work, and we hope that it leads to increased collaboration on this quest. Second, we enjoy teaching, and while we derive much satisfaction from teaching the junior analysts at Citi, we would like to equip a broader audience with a monetizable skill set. However, since our last book, AI has grown in leaps and bounds. No doubt, this book will also, in some shape or form, teach AI, not just humans and the advent of AI has made the idea of writing essentially a "how-to" book more questionable. However, over time, AI is likely to lead to superior trading principles irrespective of whether or not we write a book. Furthermore, markets always change, and a constant research effort is required to stay on top of them. Although we expect that the half-life of the principles covered in our book is reasonably high, the information content will decay over time. We are excited to combat this decay by continuing to research financial markets, getting closer to an ever-changing truth.

How the book is organized: In our experience, global macro markets are not particularly forward-looking or able to deal with complex multivariate interactions. Empirical evidence on common events such as elections or recessions shows that markets rarely price the event more than a few months before it happens and can rarely focus on more than a couple of narratives simultaneously. Geopolitical events tend to fade quickly, and the largest impact of monetary policy cycles is around the beginning or toward the end. Some of the most important and pervasive narratives are around the economic growth and inflation cycles and the knock-on effects these have on monetary policy. We therefore start the meat of the book with a discussion of the economic cycle in Chapter 2, where we go into some detail, also from a cross-asset perspective. We then do the same for geopolitics, which more recently has started to rival the economic cycle as an important big picture driver (though we still think economics does rule in the end). We then follow up with chapters that deal with the asset-class-specific effects, both on a strategic level, which may lend itself to longer-term trading decisions or asset allocation, as well as with more tactical, shorter-term analysis, that can be implemented for shorter-term trades. We go through all major asset classes: government bonds, equities, credit, commodities, and currencies. We first focus mostly on the United States, not so much because of home bias but because the US economy, the Fed, and the USD are still setting the stage for all the other countries. We then offer frameworks for non-US asset markets as well, which are a very important part of the opportunity set for macro hedge funds. We do not spend too much time on emerging markets, largely because the EM frameworks were already covered in Willer et al. (2020). Following the asset class discussions, we spend some time on asset allocation before ending the book with a brief discussion of two nontraditional asset classes: crypto and private assets. By chapter, we proceed as follows:

Chapter 2: Global Macro Rules. The chapter discusses the historical performance of macro funds and outlines why the future is bright: expensive equity markets, the end of the era of QE, and interest rates meaningfully above zero.

Chapter 3: It's the Business Cycle, Stupid. Macro investing is mostly business cycle investing. We discuss what is leading and what is lagging in the business cycle, and how to think about trading the business cycle in general and recessions in particular.

Chapter 4: Temporary Turmoil—Markets and (Geo)politics. With geopolitics having become more important, we discuss the link to oil and the global business cycle. We propose some trading rules linked to geopolitical events, before discussing how to trade elections.

Chapter 5: How to Trade Rates: A Framework. Performance of Treasuries in the long run. Discussion of how to trade government bonds and swaps in relation to the Fed cycle. How to trade curves and real rates. Does supply matter? Discussion of valuation and momentum models.

Chapter 6: How to Trade Rates: The Tactics. How to identify a peak in rates. Do data releases and FOMC meetings have follow-through? How to use economic surprises. Can risk aversion hurt Treasuries? Bonds and seasonals.

Chapter 7: How to Trade Rates: Globalizing. We study whether international markets can substantially diverge from the US cycle. We investigate trading patterns in Bunds, the European periphery, Gilts, JGBs, and CGBs more closely. We update some thoughts on EM rates.

Chapter 8: How to Trade Equities: A Framework. Performance of equities in the long run. Equities and the Fed cycle. What is needed for a bottom? Equities and earnings. How to use earnings revision indexes. Equities and liquidity. Equities and bubbles. Valuation is not working; momentum is.

Chapter 9: How to Trade Equities: The Tactics. How to identify a short-term top. How to buy the dip. Sentiment indicators and insider data, buy-backs, and positioning: What works? Seasonals in equity markets.

Chapter 10: How to Trade Equities: Globalizing. Can global markets decouple from US equities? Where are the most cyclical markets, where are the most FX-sensitive ones? We investigate trading patterns in European, UK, Japanese, Chinese, and EM equity indexes in more detail.

Chapter 11: How to Trade Credit. Performance of credit in the long run. Trading the default cycle. Credit and the Fed. Does valuation work, does carry work? Rising angels and fallen stars. Does credit lead equities? How do levered loans fit in? How to choose between US and EM credit. Ignore supply.

Chapter 12: How to Trade FX: A Framework. The USD smirk. The dollar and the business cycle. A composite model to trade EUR. JPY is special. Carry and value strategies both work. Economic surprises generate alpha.

Chapter 13: How to Trade FX: The Tactics. The Fed and the USD. Intervention at work. We investigate use cases for FX flows. We study intraday reversals.

Chapter 14: How to Trade Commodities. Commodities—not for the long run. Commodities and the business cycle. Commodities and carry. A fundamental discussion of energy, base metals, and gold markets.

Chapter 15: How to Allocate. The regime investing approach and how to protect against stagflation. Strategic asset allocation should be regime agnostic. The choice of equities versus bonds. The role of positioning and CTAs.

Chapter 16: Nontraditional Assets: Crypto and Private Assets. What type of asset is Bitcoin? How much to allocate to crypto? Trend following and Bitcoin. Making private asset returns comparable to public assets. Which factor exposures do private assets have, and in what regimes are they generating alpha?

Thanks for reading this book. Please feel free to get in touch to discuss the research agenda going forward.

1.1 ACKNOWLEDGMENTS

Writing acknowledgments for a second book is not as straightforward as it may seem. All the obvious people have been thanked, especially parents, family, and early mentors. Still, I have no hesitation in repeating myself: thank you very much, Mom and

Dad, for your unwavering support over all these decades. You remain a constant source of inspiration. This book was written after sufficient time had passed for me to forget what the first book had required. Thank you to Ulrike, Charlotte, Caroline, Sophie, and Anna for being a guiding light in all aspects of life in general, and for supporting the decision to embark on this book, despite remembering enough to know that it would be a multi-year endeavor once more.

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Richard Samson and Alice Hadaway at Wiley have expertly guided this book from raw draft to publication with remarkable patience, while Gemma Valler believed that we still have enough worthwhile ideas for a second book on trading. All remaining errors are, as ever, the sole responsibility of the authors. However, I want to thank the clients, friends, and colleagues who tried hard to minimize them. I am especially grateful to Michael Anderson, Aman Bansal, Scott Chronert, Susannah Gullette, Max Layton, Drew Pettit, Adam Pickett, Ernesto Revilla, Nathaniel Rupert, Jamie Searle, and Jason Williams. I would also like to thank Tobias Adrian, Al Breach, Paul Tudor Jones, Jim Leitner, Marcos Lopez de Prado, Ben Melkman, and Andy Merton for fighting their way through early drafts. Rob Dwyer, Bilal Hafeez, Tom Keene, Damian Sassower, and Patrick Reid deserve a shout out for being strong supporters of the EM book, and I trust this book will prove itself a worthy successor.

D. W.

Writing a book is a significant time commitment, and the first people to thank are my parents and family. Mum and Dad for the examples you set, and Athena, William, and Evelin for the joy and motivation you bring and the patience you've endured during the process.

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Lastly, thanks are due for the opportunities made available and chances taken by Janet, Bob, and Tom and for their valuable advice and mentorship early in my career. The final word again goes to Athena and my family for making the project worthwhile.

A. N. S.

CHAPTER 2

Global Macro Rules

Global macro-trading traces its roots to the 1970s, when the collapse of Bretton Woods and the rise of floating exchange rates created a fertile ground for cross-border speculation. More recently, macro-trading benefited from the rise in volatility on the back of the Asian financial crisis, the Great Financial Crisis (GFC), and the COVID pandemic. This chapter outlines the more recent performance of macro funds and makes the case for a profitable future.

2.1 LONG-TERM PERFORMANCE OF MACRO FUNDS IS STRONG

Macro funds have recently made a comeback, both in terms of returns but also in terms of mind share among asset allocators. Although this has been partly a function of the rise of the pod aggregators, which successfully dampen volatility for their investors, it was also helped by what happened in 2022. That year saw a rise in inflation, which decimated long only 60–40 portfolios. Although inflation has been in retreat since then, we think there is a more durable case to be made for the return of macro investing into the limelight.

In Figure 2.1, we analyze returns of macro hedge funds from 1997 to mid-2025. We use BarclayHedge as our source, which may underestimate returns, as some of the largest and most successful funds have not been included in the index due to data availability. For the whole period, annual geometric returns were 6.8%, less than the S&P returns of 9.4%. Returns have been strong before and during the financial crisis, but then were more meager for a period, before accelerating again in the aftermath of COVID. Adjusting for volatility, macro hedge funds have outperformed, though, clocking in a risk/reward of 1.23 (which is calculated as annualized returns over volatility), compared to the S&P at 0.65. Given that managers typically invest excess cash in Treasuries, we also calculate the excess performance over T-bills, analogous to a Sharpe ratio. The macro hedge fund index comes in at 0.83 versus S&P at 0.51 over the same period. Given that macro manager returns can generally be levered up, this Sharpe ratio is in many ways more meaningful for asset allocators than returns.

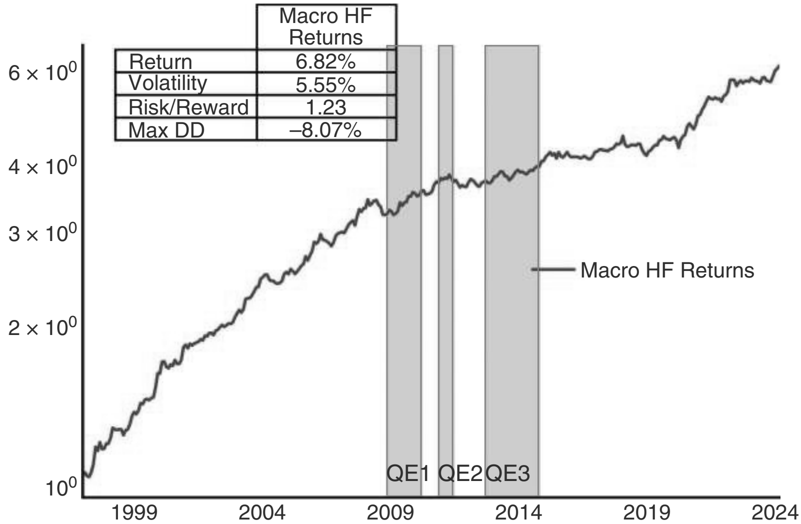


FIGURE 2.1 Macro hedge fund returns strong outside of QE.
Source: BarclayHedge.com and authors' calculations.

2.2 HOW DO MACRO FUNDS MAKE MONEY?

Before setting out to show various frameworks to macro traders, it is important to understand how macro hedge funds make their money. To this end, we regress macro fund returns on the various asset classes (equities, bonds, commodities, and DXY), as well as on Barclay's commodity trading advisor (CTA) index. CTAs are, in some sense, the original macro hedge funds and typically employ mostly trend-following strategies. This is illustrated in Figure 2.2. We find that the various asset classes and CTA variable explain between 5% and 95% of returns (the R^2 of a rolling 18m regression), with the average running at 70%. The only beta that is reasonably stable is that for the CTA index, which is, as expected, almost always positive and explains, on average, 20% of the macro hedge fund returns. The equity factor swings between positive and negative, suggesting that hedge funds really do hedge, in the sense that they do position for bear markets. The average explanatory power of equities for the overall returns is 36% in univariate regressions. Bonds, surprisingly, given that many of these funds were created by what were originally fixed income and FX traders, are less important for macro hedge funds in a directional sense, with an average R^2 of 8%. Given that we are using a monthly dataset, it may also be the case that traders are trading rates more tactically than the regressions can pick up. As with equities, the beta swings from positive to negative, as macro hedge funds can and do position for bear markets in bonds. FX exposure is even more tactical; the absolute exposure to FX (using the DXY index) is low over the full sample. In fact, it is the only asset class with an insignificant coefficient when estimated over the total 1997 to 2024 period, but it is highly time-varying. The average absolute t-statistic for rolling periods comes in above 2. Commodities add to returns in a similar fashion and tend to be traded from the long side.

Going forward, there are several reasons to remain optimistic on the asset class.

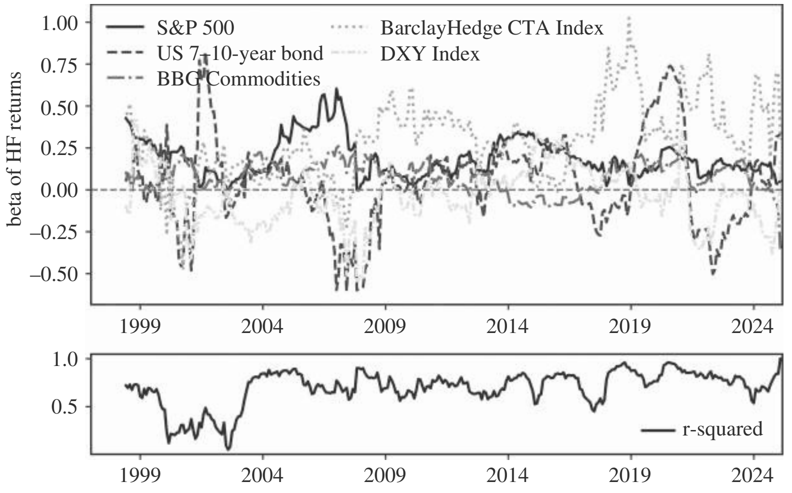


FIGURE 2.2 Hedge fund returns mostly explained by a blend of the big four asset classes and more than a dash of CTA returns.
Source: Barclayhedge.com, Bloomberg, and authors' calculations.

2.3 A BRIGHT FUTURE 1: EQUITIES ARE EXPENSIVE

During powerful equity bull markets, macro investing typically takes a back seat, as it is quite rare that macro managers outperform the SPX in a raging bull market. Even though the SPX is rarely an explicit benchmark for any macro fund, implicitly, investors only have limited patience for multi-year underperformance versus equity markets. Fortunately, for macro investors, at the time of writing, equities are relatively expensive, suggesting subpar returns in the long term. This can be seen by focusing on the cyclically adjusted price earnings (CAPE) ratio, popularized by Campbell and Shiller (1988), which is currently very elevated. To be precise, the current reading is not that different from the peak we have seen during the last days of the dot.com bubble. Expected future long-term returns are typically negative when the ratio is close to these extreme levels. This is illustrated in Figures 2.3a and b, which shows the CAPE ratio and subsequent 10-year real excess returns (calculated, as per Shiller's methodology, subtracting 10Y bond returns from the S&P and adjusting both for inflation and T-bill returns). Although it is difficult to use this ratio as a real-time market timing tool, partly because we may still be in the middle of a powerful AI bubble (more on this later), eventually markets are likely to reverse (see also Asness et al., 2015), suggesting a positive operating environment for macro traders over the next decade.

To go one step further, we also investigate the beta of the BarclayHedge index to the SPX. We find a positive, but relatively small beta of 0.17 in our monthly data sample. Furthermore, there is some evidence that global macro funds tend to have positive convexity. Macro hedge funds tend to perform much better, in relative terms, at times when equities are falling, as shown in Figure 2.4 (though they may also not be able to keep up with very strong equity returns on the right-hand side of the distribution). Using the Barclay index data is likely underestimating this effect, as anecdotally, many macro hedge funds have made their mark during bear markets.

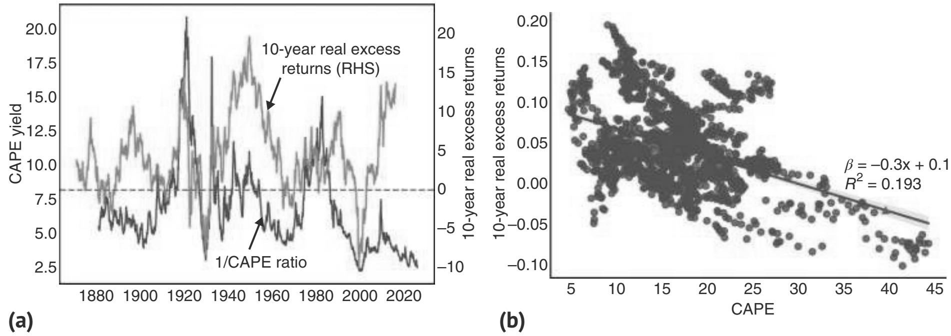


FIGURE 2.3 (a) High valuations today mean lower equity returns over the long term. (b) Current CAPE levels are associated with negative long-term equity returns.
Source: Bloomberg and authors' calculations.

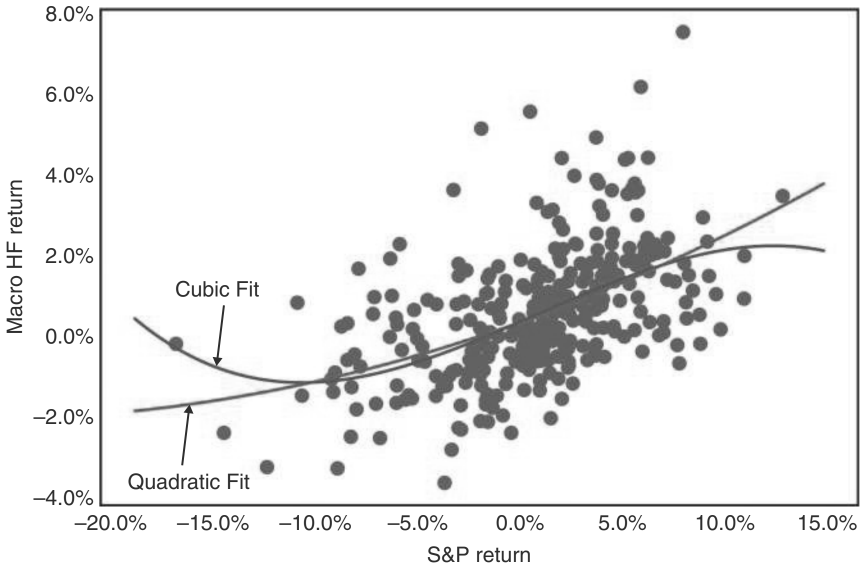


FIGURE 2.4 Hedge funds have a positive beta to equities but outperform on a relative basis when equities decline source Barclay.
Source: Bloomberg, BarclayHedge.com, and authors' calculations.

2.4 A BRIGHT FUTURE 2: VOLATILITY RISES DURING QT

The macro strategy tends to do well during periods of high volatility. The lack of dispersion in interest rates and suppression of volatility in the quantitative easing (QE) period led to a lack of opportunity, given that short-term fixed income and FX trading is a staple of macro investing. Figure 2.5 shows the impact of QE on the rolling annual return of macro hedge funds. The average in the post-GFC period, which brought QE to the forefront, was markedly lower than pre-financial crisis.

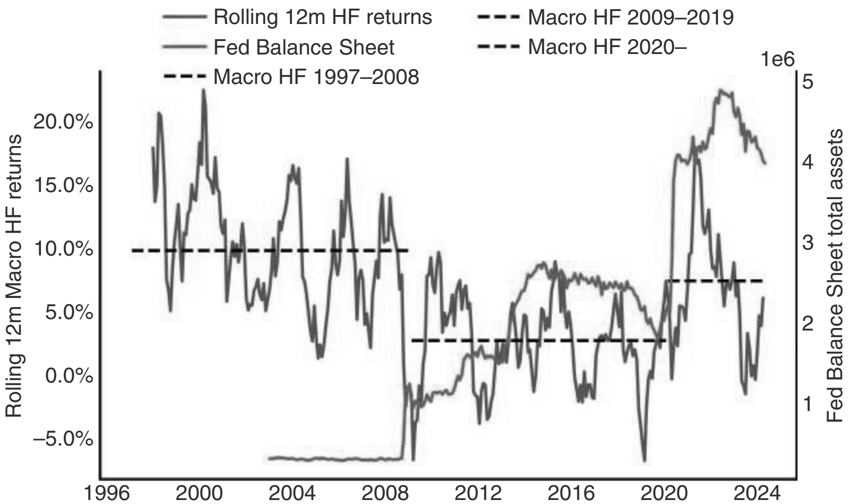


FIGURE 2.5 Hedge fund returns were lower in the post-GFC period of QE and anchored short-term rates.
Source: Bloomberg, BarclayHedge, and authors' calculations.

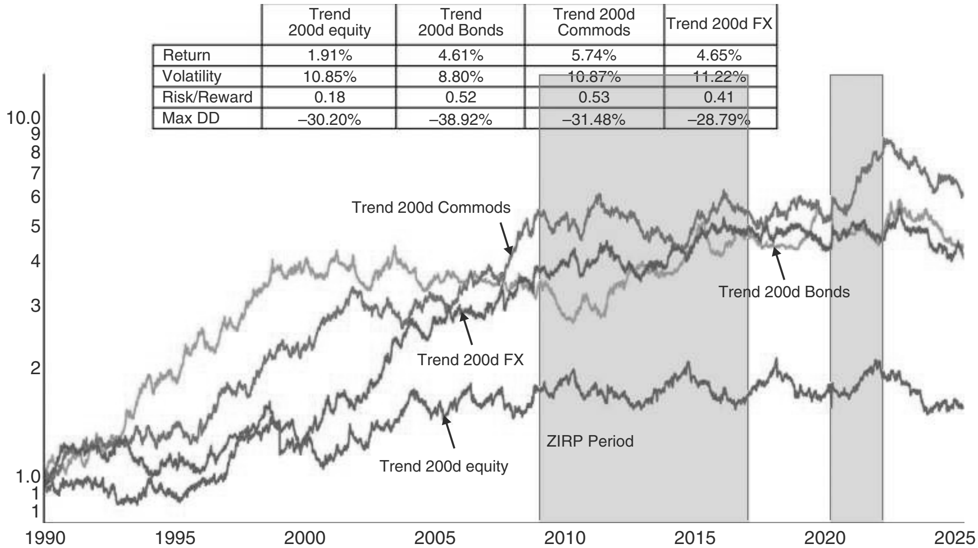
Quantitative tightening (QT) then coincided with the volatility from the COVID shock to lead to better returns again. We expect that the current episode of QT (or at least absence of QE) will again be helpful for the returns of macro hedge funds.

2.5 A BRIGHT FUTURE 3: RATES BETS WITH HIGHEST SHARPE RATIO AND MORE PROFITABLE WITH RATES SUBSTANTIALLY ABOVE ZERO

Rates are typically an area where macro fixed-income measures can gain alpha. When short rates were pinned to zero or below in the aftermath of the financial crisis, and back ends were depressed by QE, there was less opportunity for fixed-income trading. This is illustrated in Figure 2.6, where we use a simple trend-following strategy to proxy hedge fund returns by asset class. To be precise, the strategy goes long or short based on the 200d MA across asset classes, and positions are volatility adjusted. Each asset class strategy is adjusted to target 10% ex ante volatility. Fixed-income has delivered the strongest risk-adjusted returns over time but has not been as profitable when US rates were glued to zero.

2.6 A BRIGHT FUTURE 4: PERFORMANCE CUSHION FROM NON-ZERO SOFR

Global macro investors typically use leverage to gain exposure to investment themes and global markets. Futures and swaps are liquid and cheap to trade, and margin requirements are a fraction of the market exposure gained. This capital efficiency means that many managers will maintain relatively large allocations to cash-like



	Trend 200d equity	Trend 200d Bonds	Trend 200d Commods	Trend 200d FX
Return	1.91%	4.61%	5.74%	4.65%
Volatility	10.85%	8.80%	10.87%	11.22%
Risk/Reward	0.18	0.52	0.53	0.41
Max DD	–30.20%	–38.92%	–31.48%	–28.79%

FIGURE 2.6 Rates trend-following struggled in zero rates period.
Source: Bloomberg and authors' calculations.

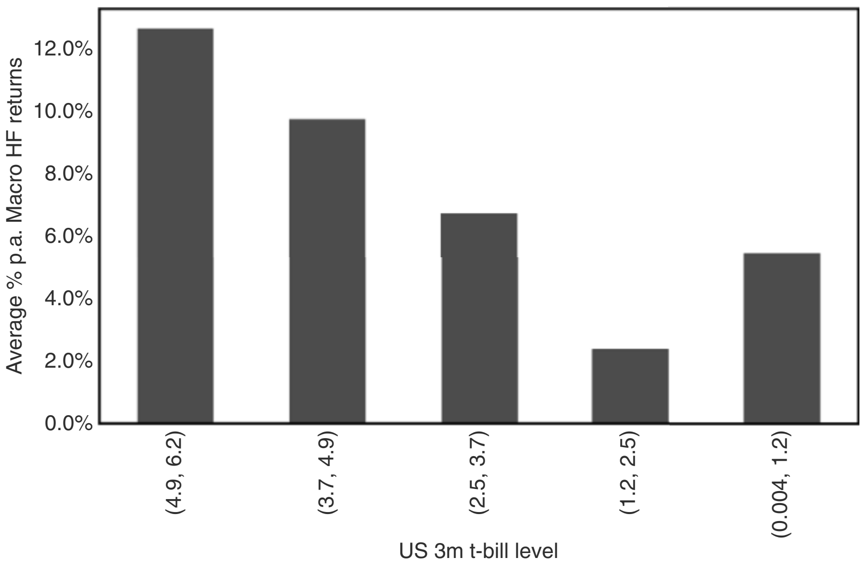


FIGURE 2.7 Hedge funds returns are higher at higher cash rates.
Source: Bloomberg, BarclayHedge.com, and authors' calculations.

instruments and will earn higher returns from these balances when short rates are high. Figure 2.7 illustrates this regularity by plotting the average returns (annualized) by quintile of short-term US T-bill yields. At the highest quintiles (close to levels seen today but more often observed in the 1990s), average annualized returns were over

12% compared to the post-GFC period of zero rates, low macroeconomic volatility, and barely 2% annualized macro fund returns. The effect may be compounded by two other aspects. First, higher rates typically do not stay high for long, and macro funds often are able to benefit from a period of high but falling rates. Second, there is a nexus between risk-taking abilities and T-bill yields. When T-bills are high, the income from T-bills can be used to leverage macro bets. After all, it can be very difficult to be an aggressive trader when in a drawdown, and to the extent that higher T-bill income can cushion drawdowns, it is helpful for keeping risk appetite high. Of course, not all hedge funds are judged against the zero line, and some may be judged against T-bill returns, which makes the last point moot for that subset of funds. But it is typically good to be a hedge fund when returns on cash are high.

2.7 A NOTE ON PROCESS AND DOES HISTORY REALLY RHYME?

The basic methodology of the book is to analyze economic history. This can be event studies that analyze past events and circumstances and apply them to the new events in question. Or it can be an investigation of which financial factors have historically worked in which macro regimes. The implicit assumption is that developments will unfold in a similar way to how they always have. The key criticism is, of course, that the future is the future, and lessons from the past may be somewhat limited, maybe especially in financial markets, where rewards for learning from past mistakes can be exorbitant. We acknowledge that there are limitations to the historical analysis, as history rarely repeats itself (despite allegedly rhyming). Still, in our minds, a thorough study of the past is crucial to building an adaptable framework to be ready for the challenges of the future.

This is true for several reasons. First, there is a tendency for analysts, especially specialists in one particular asset class, to adopt an "every time is different" view, based on the specific circumstances and their deep knowledge of the nuances of a given situation. This sometimes misses the forest for the trees. The second point, eloquently made by Kahneman (2011), is that we often neglect to get our priors right. We may mistakenly think that a person we know to be a Greenpeace activist is more likely to be a librarian than to work in finance, even though in the United States, 9 million people work in finance, compared to less than 150,000 in libraries. At the very least, the frameworks we lay out in this book will help to get the priors right, which must be the right starting point to explore what could be different this time around. Lastly, there is no alternative. As Yogi Berra observed, "It is tough to make predictions, especially about the future," which, by the way, was another case of history rhyming, as Nils Bohr had said something to this effect much earlier, possibly himself rehashing an old Danish proverb. Maybe there is nothing new under the sun, after all. While the future is the future, humans are humans, and many human biases, like overconfidence or a recency bias, for example, will likely remain in play in human history in general, as well as in financial markets in particular.

Furthermore, we often focus on the relationship of one variable with one or two other variables. Of course, in reality, processes are much more complex, and there may be many instances of omitted variables at play. Still, we find that even simple frameworks can be quite useful because the market is typically only trading two or three stories at a time, and for short periods, it is often the case that there is just one factor at play. While one of the authors can rightfully call himself a quant, this book is not overly quantitative. Our backtests do not necessarily have to be statistically significant, and we are willing to live with small samples. As long as results are in line with economic logic and make sense intuitively, we are willing to run with them.

In summary, while the inherent difficulty of forecasting will not stop people from making predictions, we should be modest with respect to our own forecasting abilities. Therefore, a lot of our work focuses on establishing the base-rate or typical path, or expected path under common circumstances. Once established, useful analysis can be done on why this time may be different or segmenting the previous history to a specific scenario or circumstance. We found in our own trading that having a strong framework increases the ability to withstand drawdowns. Trades based on predictions without a framework will lead to second-guessing of the predictions much faster, and often as quickly as prices start moving against the view. We hope that this book will help readers create strong frameworks for their macro-trade.

2.8 WHAT IS GOOD PERFORMANCE?

Throughout most of this book, we analyze returns as excess returns over cash (typically US T-bills). This enables comparison across different periods of time with different inflation and cash rates and, in certain periods, approximates real returns, as higher inflation episodes usually have higher T-Bill rates. It also has the benefit of replicating the return available to a leveraged macro-investor who, via swaps, futures, or other derivatives, implicitly or explicitly (through repos) finances positions at close to the risk-free short-end rate. In addition, we typically analyze investments through the lens of the information ratio, i.e., the ratio of the excess return on an investment (or strategy) to its volatility (which is very similar to the Sharpe ratio). There are many criticisms and alternative measures to assess performance, but none have been able to replace the Sharpe ratio. To our mind, the Sharpe ratio is a simple and efficient measure of the relative attractiveness of an investment using leverage, and we continue to use it as a first-order measure to compare performance. Of course, the path of returns matters, as do the timing, depth, and duration of drawdowns and (conditional) correlations with other asset classes. Lo (2002) describes useful adjustments to account for various non-normal statistical properties such as skew and kurtosis. In our experience, implementing these modifications makes little difference in practical applications. López de Prado et al. (2025) provide several extensions for conducting formal statistical inference and correcting for multiple tests. In any case, a simple timeseries plot of cumulative performance gives a much richer starting point to investigate the properties and relative attractiveness of a strategy. We rely heavily on intuition and economic priors and tend to use consistent parametrization (55d moving averages feature prominently for example) to avoid some of these pitfalls. This book will show many such simulated return series. For some strategies that have relatively short holding periods, we often report active information ratios. Active means that they are only calculated for the time period when the strategy has an active position. Lastly, our purpose is to build frameworks to generate insights on how to think about a given asset in a given circumstance. We do not aim to provide off-the-shelf trading strategies per se. We therefore typically do not adjust for transaction costs. We also do not formally distinguish in and out-of-sample periods. But we have been using many of the strategies for many years, which implies that the more recent decades are approximating an out-of-sample test. Still, we do expect that some of the strategies will break down going forward, and we hope to still spend many years researching, fine-tuning, and, yes, discarding the various frameworks in this book. History rules, but flexibility dominates.

2.9 SUMMARY

This chapter makes the case for macro-trading. From the late 1990s, macro hedge funds have had a strong information ratio of 1.23. After a period of worse-than-usual performance during the QE and low-interest rate years, returns have recently started to improve again. We expect that to continue, as volatility should remain high with QE firmly in the rearview mirror. Furthermore, rates well above zero also favor the macro strategy, with short-term interest rate trading being an important contributor to PNL. In relative terms, the fact that US equities are expensive positions macro-trading well for the medium term. The chapter closes with some remarks on the methodologies applied.

CHAPTER 3

It's the Business Cycle, Stupid

On a very basic level, macro investing is about extracting signals from fundamental data as well as from price action and, in particular, from the interaction between the two. In terms of fundamental data, it is all about getting the business cycle right. The business cycle is what drives central banks, allowing direct bets on the direction of the business cycle in interest rate derivatives and bonds. Growth-sensitive commodities also allow for fairly direct bets on the business cycle. The link of the business cycle to risky assets (equities, credit) is not as direct as it is for rates, but it is also very strong, as many of the large drawdowns in risky asset returns happen going into recessions. However, risky assets have a larger valuation component, which can create bull markets or drawdowns that are not as directly linked to the business cycle, even though valuation also has a cyclical component. FX is more complicated because it is about the differential between business cycles in two different countries, along with a whole host of other factors. Still, broadly speaking, none of the main macro asset classes can be traded from a top-down perspective without understanding the business cycle. This chapter will go through some of our frameworks for translating the business cycle into investments or trades. Given the importance of recessions for all asset classes, a fair amount of time will be spent discussing how to spot them and how they impact markets. A more granular discussion of how to trade the business cycle will follow in the asset class-specific chapters. The focus will initially be mainly on the United States. Although the share of global GDP of the United States has been declining, peaking around 40% in the 1960s before falling to around 25% currently (at current prices and exchange rates), the US economy still dominates, partly because the Fed and the USD dominate the global macro scene. We will discuss the impact of other regions on the global macro cycle in the following chapters.

3.1 FROM BOOM TO BUST AND BACK

The two main variables that policymakers and, therefore, markets care about are growth and inflation. The resulting four quadrants are the standard way to segment the business cycle: Goldilocks (high growth, low inflation), overheating (high growth, high inflation), stagflation (low growth, high inflation), and recession (low growth, low inflation). We found in our statistical work that to explain asset price performance, recessions need to be split into early and late parts, as risky assets are forward-looking and can perform quite well in late recessions. Returns of risky assets

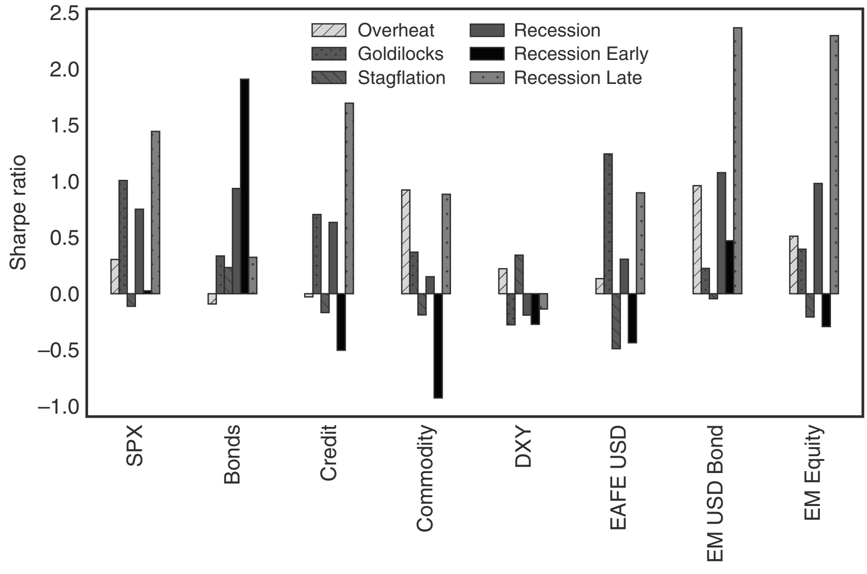


FIGURE 3.1 Risky assets do well late in recessions, after poor performance early in recessions.
Source: Bloomberg and authors' calculations.

are worst in early recessions and in stagflation, which is why it is key to identify periods of very weak growth in general and recessions in particular. This is illustrated in Figure 3.1, which shows the Sharpe ratios of the various macro asset classes by regime.

The National Bureau of Economic Research (NBER) is responsible for officially declaring recessions in the United States. Specifically, the NBER's Business Cycle Dating Committee, composed of eight economists, has the call. According to the NBER, a recession is defined as "a significant decline in economic activity that is spread across the economy and lasts more than a few months." It is important to realize that there is no fixed formula for the NBER's assessment, as the committee takes a holistic look at the economy during potential recession periods. However, it seems clear that the labor market plays an important role. All US recessions had negative nonfarm payroll (NFP) prints, either just before the recession started or in the early months of the recession. Because the NBER awaits both the original publication of all relevant data and also the subsequent revisions, the determination of a recession is quite lagged, frequently coming 6–12 months after the fact or even later. This means that investors cannot rely on the recession label as an input. In any case, given that markets are forward-looking, the key is to anticipate a recession by a few months. It is therefore crucial to find the leading indicators for a recession.

Table 3.1 shows some candidates for leading indicators among fundamental variables. One quick look at the various indicators at the onset of the US recession suggests that purchasing manufacturing indexes (PMIs) are typically already below 50, while the median NFP is still running at over 100k (positive!), industrial production (IP) is still expanding, and the EPS for the S&P 500 is still rising. Initial unemployment claims have moved up from the lows, but only by around 50k, which is more noise than signal. Housing starts are typically already reliably negative.

TABLE 3.1 PMIs and Housing Tend to Lead

Recession Start	PMI	NFP	Claims from min	IP YoY	Housing starts YoY	EPS YoY
Jan-80	46.2	305	70.5	0.81%	–17.73%	27.35%
Jul-81	46.7	383	30.5	6.53%	–17.97%	–4.94%
Jul-90	46.6	–219	36.0	2.54%	–17.94%	–20.75%
Mar-01	41.9	–86	121.8	–0.87%	2.68%	4.78%
Dec-07	47.7	18	47.2	2.16%	–38.73%	9.89%
Feb-20	50.9	273	8.2	–1.45%	34.85%	0.72%
Average	46.7	112	52.4	1.62%	–9.14%	2.84%
Median	46.7	146	41.6	1.49%	–17.83%	2.75%

Source: Bloomberg and authors' calculations

In spite of the NFP still often running strong at the outset of a recession, whether a recession eventually gets called or not does largely depend on the labor market. The characteristic feature of recessions is that they have negative NFP prints either just before or in the early months of a recession. Although NFPs are not necessarily negative at the beginning of a recession, they can move fast. The fastest NFP moves from very strong (300k plus) to negative were March 98 when we saw a drop from 310k to –36k, which recovered back to 200k+ by April, and 2010, when May's 431k report gave way to –125k the following month (starting a four-month negative streak). This is illustrated in Table 3.2. The volatility and speed of adjustment potentially explain the difficulty of forecasting a recession for both economists and equity markets, even when it is already close. The other difficulty is that the first negative NFP print can be the start of either a more extended weakness or a short-lived affair that bounces back very fast. The longest period of negative NFPs was the GFC (790 days), but in 6 out of 13 events, a return to positive numbers was observed in the next print. This makes the use of NFP numbers for recession forecasting purposes quite difficult.

3.2 WHAT IS LEADING—FUNDAMENTALLY?

Purchasing managers indexes (PMIs) are surveys of purchasing managers across various industries and are typically split into various subsections, asking, for example, whether new orders, inventory levels, or employment have risen, fallen, or stayed flat over the previous month. Aggregating the results leads to a diffusion measure, which often has leading properties compared to indices that add up the actual values for orders, etc. In the United States, the manufacturing ISM is the PMI with the longest history, starting in the late 1940s. The rule of thumb is that a manufacturing PMI below 45 is a recession, while a manufacturing PMI below 50, but above 45, is typically a less serious inventory correction. Of course, this rule is not overly helpful in forecasting recessions, given that inevitably the manufacturing ISM will be in the 45–50 range

TABLE 3.2 The First Negative NFP Print Is a Noisy Signal

Start	End	No. of days	Actual	Prior	Next	Days from last -ve
04-03-1998	05-08-1998	30	–36	310	262	
10-08-1999	11-05-1999	31	–8	124	310	518
08-04-2000	10-06-2000	61	–108	11	–105	274
04-06-2001	03-08-2002	334	–86	135	–223	182
10-04-2002	02-07-2003	123	–43	39	–5	214
03-07-2003	10-03-2003	214	–308	143	–108	28
10-07-2005	11-04-2005	31	–35	169	56	731
09-07-2007	10-05-2007	30	–4	92	110	669
02-01-2008	04-02-2010	790	–17	18	–63	123
07-02-2010	11-05-2010	123	–125	431	–131	91
10-06-2017	11-03-2017	31	–33	156	261	2,526
04-03-2020	06-05-2020	61	–701	273	–20,537	882
01-08-2021	02-05-2021	31	–140	245	49	214

Source: Bloomberg and authors' calculations

before hitting levels below 45. Every recession initially could be a soft landing, which makes it so difficult to distinguish between the two early on. An indicator leading the headline ISM is the delta between new orders and inventories. But this leading property comes at the cost of higher volatility, which can lead to incorrect signals more frequently. The service-based cousin of the manufacturing ISM was only introduced in 1997. Although manufacturing, especially in the United States and other developed market economies, has become a much smaller part of the economy in recent years, falling from a high of 28% in the early 1950s to just above 10% of GDP more recently, we still think it is important to focus on the manufacturing ISM, as it punches above its weight. First, volatility is higher for the manufacturing sector than for the service sector, which is why even a smaller sector matters for the changes in overall GDP. Second, the two PMIs are highly correlated. In particular, whenever the manufacturing ISM fell below 45, the service PMI also decreased below 45, and the US economy experienced a recession. One reason is that many of the more cyclical services depend on the manufacturing cycle too, such as transportation or ad spending, for example. Even post-pandemic, when supply chain disruption and constraints on economic activity (no tourism or restaurants) led to an unheard of rotations from services to goods and back, the two measures were still positively correlated, as per Figure 3.2. The other reason to continue to focus on the manufacturing ISM is that the earnings and price action of the S&P 500 correlate relatively well with it, again in part because of

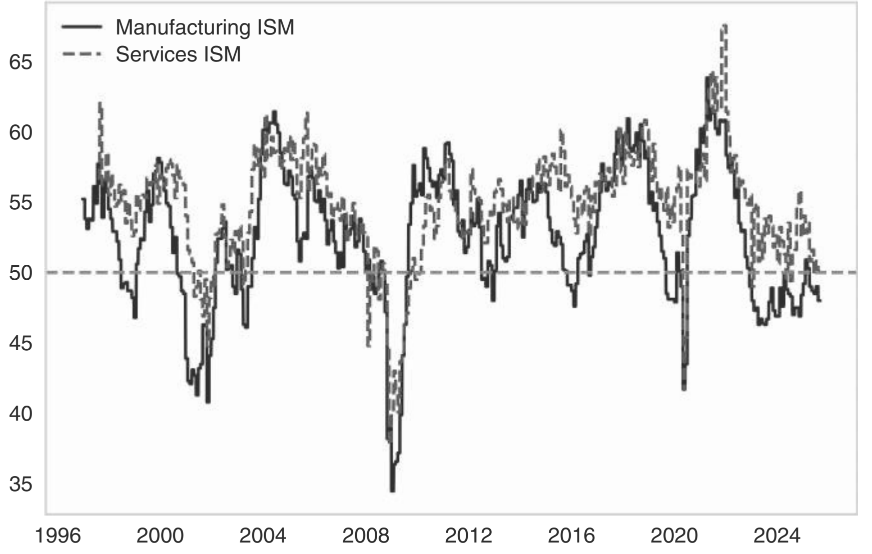


FIGURE 3.2 Manufacturing and services remain highly correlated.
Source: Bloomberg.

the cyclicality of some services. This was true, for example, in 2015–2016, when the SPX traded poorly, in line with the weak manufacturing ISM, even though the services ISM remained strong. In another example, in 2022, the manufacturing ISM decelerated quickly and fell below 50, which was seen as giving too bearish a read on the economy, but the S&P 500 traded very poorly that year, to which a softening ISM surely contributed. Still, the services ISM is also relevant. Lower manufacturing PMIs that do not also lead to the service index falling below 50 are usually not overly severe for the economy, which happened in 2015. PMIs will therefore feature prominently in our business cycle discussions.

Housing is the other important leading indicator. The reason is that many downturns are driven by the Fed hiking rates, and housing is the most interest rate-sensitive part of consumer spending, given that the vast majority of structures are mortgage-financed. New housing starts falling below 1.1 million in SAAR terms have fairly reliably coincided with a recession, with the 2000 recession, which was driven by the deflating Nasdaq bubble, being the main exception. The leading properties of housing were illustrated by Leamer (2007). Housing plays an important role in the business cycle.

3.3 BUT WHO IS REALLY IN FRONT?

Obviously, the previous discussion only relates to the leading indicators with respect to the economy. Fundamentals are extremely unlikely to ever lead markets. This does not make fundamentals useless, as we will show later, but it raises the question of what market indicators are leading. Various market prices are indeed the only thing clearly leading PMIs. This makes the exercise somewhat circular, as we then want to use the PMIs to make assessments about markets. Having said that, there is still a use case for this approach, as, at the very least, it captures the self-reinforcing nature of both upcycles and downcycles. For example, strong equity markets create a wealth effect and rising animal spirits, which improve the economic data, which impacts earnings, which in turn justify/magnify the asset market bullishness. One indicator that has worked relatively well for us is a leading indicator for US consumption includes gasoline prices, mortgage rates, and S&P 500 Index (SPX) returns. We add the five-year rolling z-scores for all three prices, which all impact an important part of the US consumers' cash flows. Gasoline prices can become a major drag, especially for low-income consumers. Rising mortgage rates can impact middle- and high-income consumers significantly (though sometimes with a lag), and SPX returns impact mostly high-income consumers. Our composite indicator seems to have an approximate 10m lead over the weighted ISM (manufacturing and services[1]). Although we do not propose a direct trading rule for this chart, it generally informs our view on the next big move in the ISM and where we are in the business cycle, as is illustrated by Figure 3.3.

Another market factor that has consistently been one of the best predictors of recessions is the yield curve. The logic is that when yield curves invert, the market is indicating that rates are too high, i.e., well above neutral levels. Rates above neutral

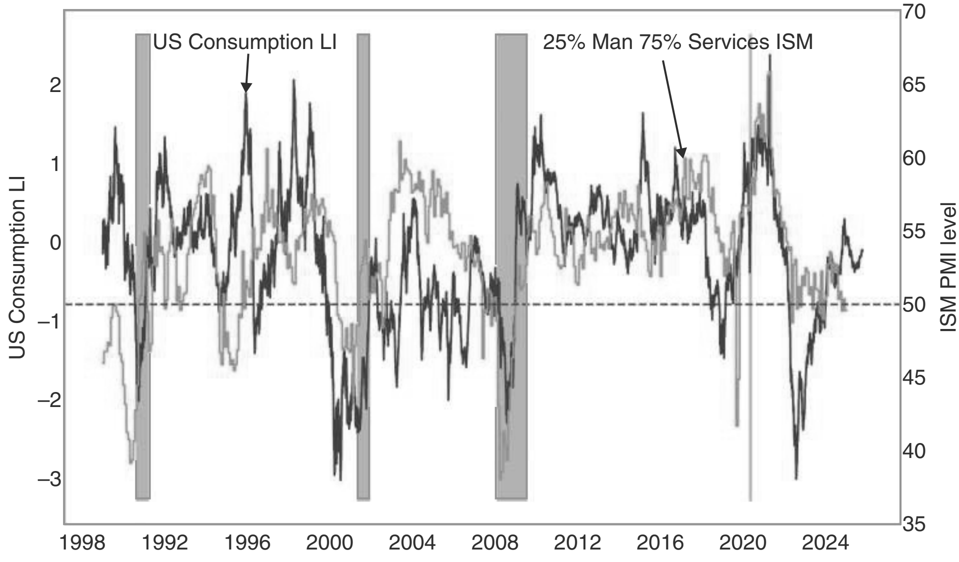


FIGURE 3.3 A leading indicator for the ISM.
Source: Bloomberg and authors' calculations.

[1]starting in 1997, before that only manufacturing ISM.

typically slow down the economy, eventually requiring cuts, which is reflected in an inverted curve. Of course, the same logic in theory also applies to soft landings, where insurance cuts may be necessary. The yield curve is really just forecasting coming Fed cuts rather than a recession per se. Because soft landings are very rare, we view inverted yield curves as an indicator of an easing cycle, which typically, but not always, involves a recession. But maybe that distinction does not matter much, at least from a rates trading perspective and in the early part of the cycle. Although easing cycles are much shallower for soft landings than for recessions, investors can still benefit from receiving (i.e., entering a swap agreement to receive a fixed rate in exchange for a variable one) into the first rate cut of a cycle. For equity investors, the distinction between soft and hard landings is, of course, much more important. But even for fixed-income investors, the main problem is that lags are long and variable. A leading indicator that is leading by too much can be as useless as a lagging one. The analysis is not made any easier by the fact that there are plenty of different curve segments to analyze. Table 3.3 shows the lags for the 3m/10yr, which is the segment the Fed has highlighted as the best predictor of a recession (see Table 3.3; Estrella and Mishkin, 1996). Other than in the 2022 soft landing cases, the predicted easing cycle coincided with a recession. But long and variable lags indeed.

TABLE 3.3 Long and Variable: 3m10yr Yield Curve Inversion Predicts Cuts with a 1m to 23m Lag

1st inversion	Recession start	Inversion months to recession	FF 1st cut	Inversion to cut	CPI start
Dec-1968	Jan-1970	13	Nov-1970	23	4.7%
Jun-1973	Dec-1973	6	Dec-1974	18	6.0%
Nov-1978	Feb-1980	15	May-1980	18	8.9%
Mar-1989	Aug-1990	16	Jun-1989	2	4.9%
Sep-1998	Apr-2001	31	Sep-1998	1	1.4%
Apr-2000	Apr-2001	12	Jan-2001	9	3.0%
Jan-2006	Jan-2008	24	Sep-2007	20	4.0%
Mar-2019	Mar-2020	12	Aug-2019	4	1.9%
Oct-2022			Sep-2024	23	7.8%
Median		14		18	4.7%

Source: Bloomberg and authors' calculations

The Inversion That Cried Wolf: The Case of 2022

One important question is whether the yield curve has become less useful over time. While there is no clear pattern in the data of increasing lags between inversion and Fed cuts over time, the 2022 inversion episode has certainly raised eyebrows, as the inversion was used by many observers to predict a recession that never happened. It also occurred while the hiking cycle was still going strong. Although receiving upon the inversion signal would initially have been profitable, all those gains (and more) would have been quickly given back as the hiking cycle continued, as is illustrated in Figure 3.4.

FIGURE 3.4 The 2022 inversion had a longer lag.
Source: Bloomberg.

Fundamentally, there are two explanations for the earlier-than-normal inversion. In theory, it could have been that the Fed overhiked more than usual. If the Fed hikes move more above neutral than is typical, it would be natural to see a deeper and earlier inversion. However, there is no evidence for this. While the inversion was in line with a move lower in housing starts as well as a move lower in the manufacturing PMI, neither of these fundamental indicators reached levels that typically indicate a recession, making this episode a case of a soft landing. If the Fed had overhiked more than usual, a recession would presumably have been the more likely outcome. In the event, the soft landing outcome created less urgency for the Fed to cut, making the inversion premature. But even in a soft landing scenario, past inversions would have worked much better to capture the subsequent insurance cuts. The more important consideration was likely that inflation was extremely high at the time the curve inverted. There seems to be a correlation between the time lag between inversion and the first cut, and the level of inflation at the time of inversion (see Figure 3.5), which the market did not take into account, maybe because high inflation had not happened in the working life of most market participants.

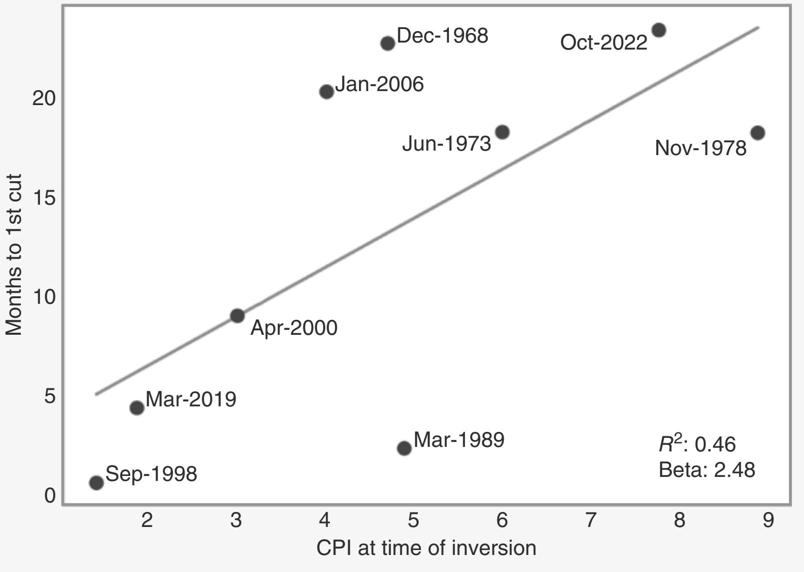


FIGURE 3.5 Cuts come quicker when inflation is lower.
Source: Bloomberg and authors' calculations.

There is also a more technical explanation for the poor performance of the yield curve in the 2022 cycle. It is plausible that too much money went into hedge funds with strategies that try to trade the interest rate cycle, which may have led to more front-running and larger positions. It is indeed noteworthy that the rate cuts priced for the subsequent 12 months were much larger than in any previous cycle at day zero, which is the day of the first cut. Of course, part of the problem was that the market expected earlier cuts, but there is also a component of deeper cuts than usual being priced, even allowing for uncertainty of the precise onset of the easing cycle. We will discuss that issue in more depth in the bond chapter. But in the bigger picture, we would expect that cycles that have less extreme inflation spikes than the post-COVID one will trade again more in line with the typical patterns of the past.

3.4 CRYSTAL BALLS OR FALSE PROPHETS: HOW USEFUL ARE THE LEADING INDICATORS?

The interesting question is how to link the various leading indicators with investment positions, especially since market prices usually lead even the leading indicators. We will go into more detail in the relevant asset class chapters, but at this stage, we want to shed some light on the leading indicators that we have already discussed. We start with the PMIs. For equities, as already alluded to earlier, in general, the manufacturing ISM is more relevant than the service ISM. It is also published earlier. Therefore, we focus mainly on the manufacturing ISM. While survey data has come under some scrutiny post-COVID, we think that the post-COVID period underlined the importance of the manufacturing ISM. In our view, the severe 2022 bear market was driven not just by the beginning of the Fed's hiking cycle, but the sharp move lower in the ISM manufacturing probably played an important role too. Although recessionary levels below 45 were never reached in 2022 or subsequently, the manufacturing sector has been quite weak in the United States, which was reflected/front run by US equities in 2022.

Figure 3.6 shows a detrended version of the SPX next to the manufacturing PMI. We use the deviation from the 200dma to detrend the SPX. It is actually one of our pet peeves that analysts often use year-over-year (yoy) numbers to detrend a series. The SPX in yoy terms is not tradable, though, given base effects. The deviation from the 200dma is tradable in the short term, as the moving average is sufficiently slow-moving. At first glance, the chart suggests a strong correlation between the two series. However, that does not make it easy, as the lags are once again highly variable, and often markets are leading the data, as should be expected.

We first study ISM bottoms. The first observation is that, when analyzing all manufacturing ISM bottoms, the median case is for the SPX to bottom around two months prior. Sadly, the pattern is not overly consistent, though. In the last few troughs, the

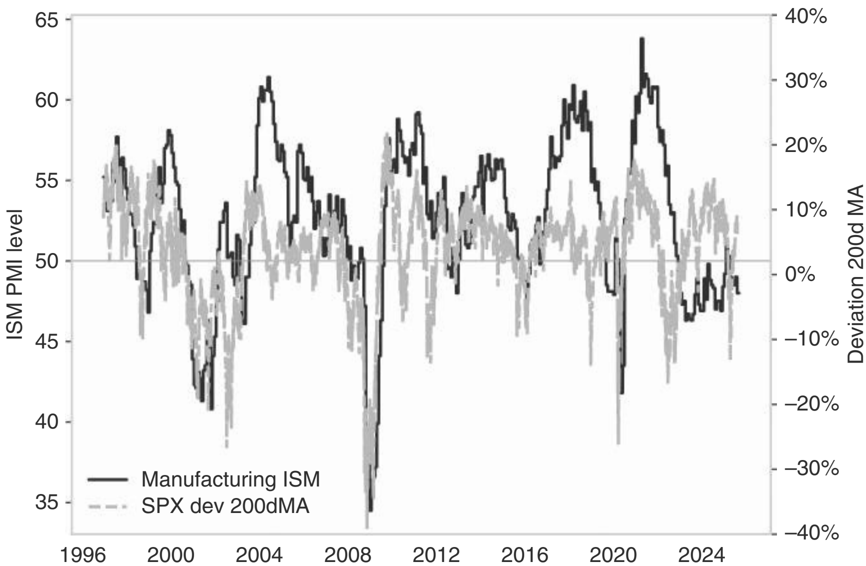


FIGURE 3.6 SPX and the manufacturing ISM are highly correlated.
Source: Bloomberg and authors' calculations.

ISM bottoms were either much too early (2001), slightly early (2008), or slightly late (by one month in 2020 and by four months in 1990) relative to the trough of the SPX, as indicated by Figure 3.7. However, it is a relatively consistent pattern that the SPX moves strongly higher in the months after the bottom. For peaks, the patterns are not as consistent. Although the median SPX performance is negative, starting around two

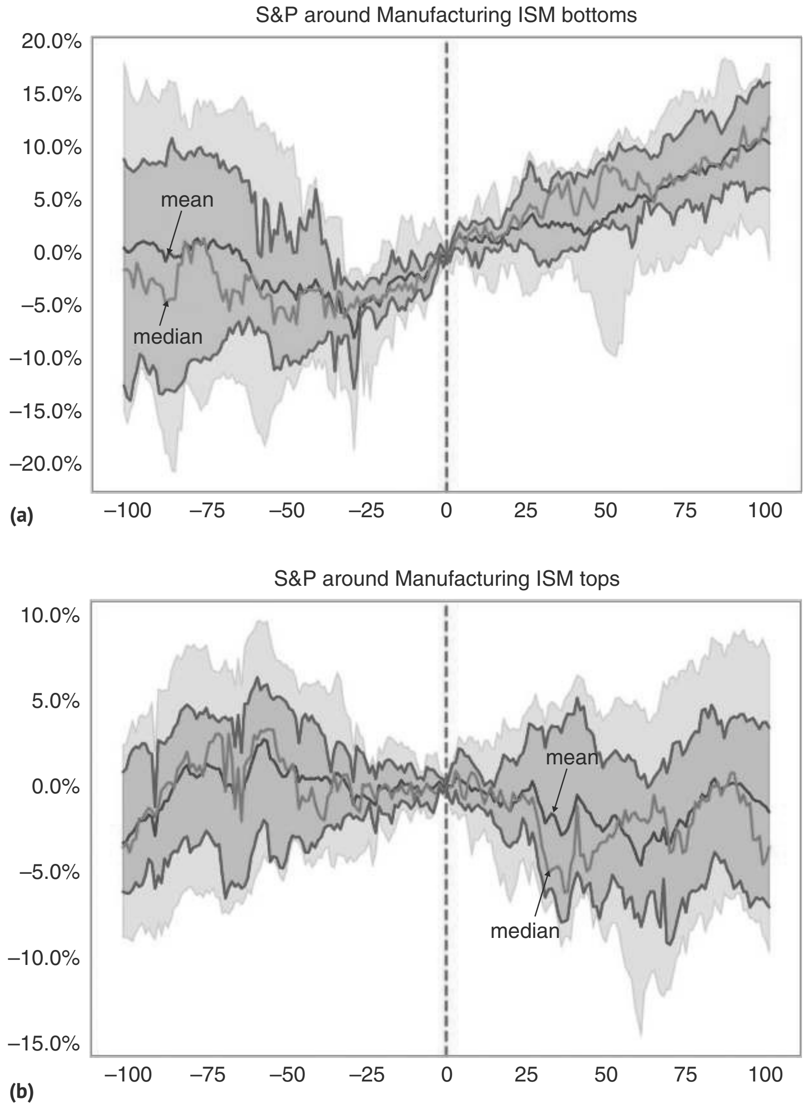


FIGURE 3.7 S&P bottoms two months ahead of ISM … (b) … and the S&P falls ahead of the ISM peaks.
Source: Bloomberg and authors' calculations.

months before the peak and until at least one month after, the dispersion of returns is much broader, making it more difficult to bet on.

The bigger problem is that we only know ex post when the bottoms and tops for the ISM are in. We therefore also study various rules that could be used in real time. Table 3.4 shows that the active IR for the one-month return after a release of the last manufacturing ISM print, which comes in below the 50 level, is slightly higher than is the case above the 50 level. This is all driven by prints below the 40 level, though, which have been very bullish. Combining ISM levels with momentum, we find that the single highest 1m forward information ratio comes when the print is below 50 and the three-month moving average points higher. However, the sample is quite small. Prints below 45, where only the one-month print is up, also do very well, with the ones below 45 and upward momentum also being stronger than normal. The only negative active information ratio is observed when the manufacturing ISM is above 60. Again, the sample is quite small, though. Overall, the ISM is best used as a counter-indicator at extremes, especially when momentum is changing.

With respect to whether inversion is a useful indicator, we focus more on the fixed income market than the equity market. As already mentioned, inversion is consistent with a soft landing that requires insurance cuts, which is equity bullish, as well as

TABLE 3.4 ISM More of a Counter Indicator

	Return	Volatility	Sharpe ratio	Active sharpe	Percent on
Unconditional	7.2%	17.2%	0.42	0.42	100.0%
PMI < 50	2.9%	10.7%	0.27	0.50	29.6%
PMI 45 to 50	1.3%	8.0%	0.16	0.37	20.3%
PMI < 45	1.8%	6.9%	0.26	0.86	9.1%
PMI 40 to 45	0.6%	3.9%	0.14	0.71	4.2%
PMI < 40	1.2%	5.7%	0.21	0.95	4.8%
PMI Mom +ve	3.3%	11.3%	0.29	0.43	46.3%
PMI Mom +ve from <40	1.4%	3.7%	0.38	1.92	3.9%
PMI Mom +ve from 40 to 45	0.2%	2.6%	0.06	0.40	2.5%
PMI Mom +ve from 45 to 50	0.8%	6.5%	0.12	0.34	12.0%
PMI > 50	4.1%	13.5%	0.31	0.37	68.1%
PMI 55 to 60	2.1%	7.1%	0.29	0.57	25.3%
PMI > 60	–0.1%	4.3%	–0.01	–0.05	8.7%
PMI Mom -ve	3.8%	12.8%	0.30	0.42	49.7%
PMI Mom -ve from >60	–0.1%	3.7%	–0.02	–0.07	5.9%

Source: Bloomberg and authors' calculations

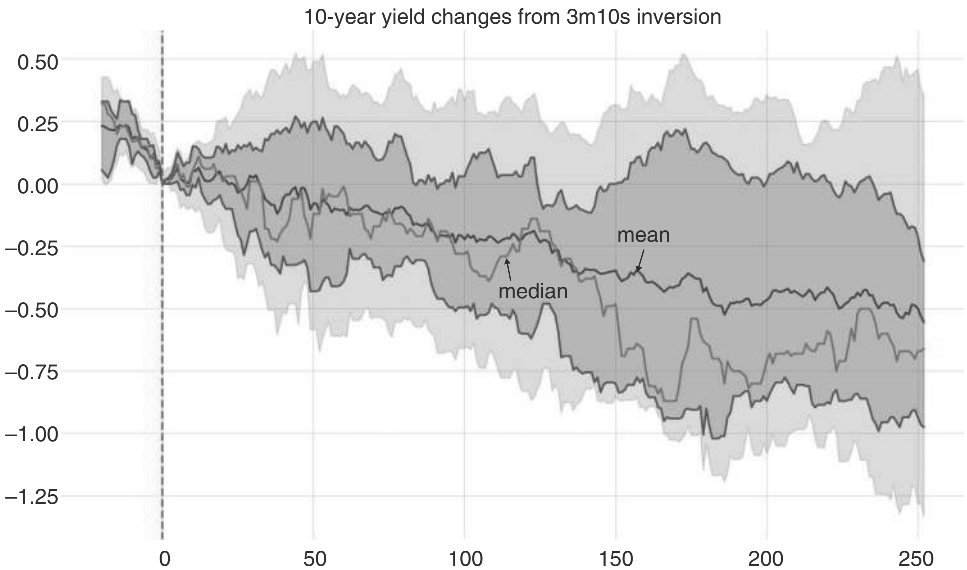


FIGURE 3.8 Inversions see yield declines.
Source: Bloomberg and authors' calculations.

with a recession, which is equity bearish. For fixed income, the difference is not as stark, though the extent of rate cuts is different in the two cases. This makes inversion more useful for fixed-income markets. Figure 3.8 shows the performance of UST after the 3m/10yr inverts, which is one of the more reliable curve segments. Clearly, in the median case, 10-year rates fall significantly post-inversion. We will investigate inversion-based rules in more detail in our bond chapter.

For equity markets, it has been argued that the problem is not so much the inversion, but rather the subsequent disinversion, which leads to sell-offs. We agree that given the long and variable lags between yield curve inversions and recessions, yield curve inversions are not necessarily a good timing tool for equities. However, the question is whether disinversions work any better. We focus here on 2s/10s disinversions, as the most classic expression of the yield curve. It turns out that after such disinversions, US equities are mostly trading in line with their unconditional performance but turn lower on a three-month and six-month horizon, as can been seen in Figure 3.9; 3m/10y disinversions (not shown) lead to negative one-month SPX returns but have positive 3m returns.

3.5 TRADING IN THE REARVIEW MIRROR

There are also some lagging indicators that generally get a lot of market attention in a downturn. We quickly describe why investors would be ill-advised to pay too much attention. As described previously, whether a recession is called by the NBER typically depends on the labor market. It is, therefore, somewhat ironic that labor

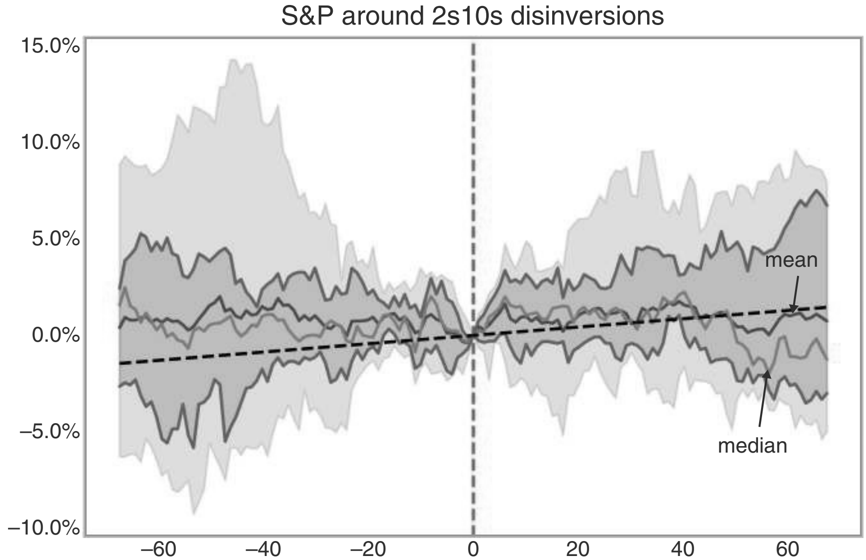


FIGURE 3.9 Disinversion of 2y/10y are a slow-burning warning signal for equities.
Source: Bloomberg and authors' calculations.

market indicators are quite lagging and are not overly helpful for trading. The most timely labor market indicator in the United States is initial claims. This has led to a strong market focus on this measure. The rule of thumb for claims is that by the onset of a recession, they have risen by at least 50k from the lows and typically are above 350k. They tend to peak out above 500k. The problem is that these weekly numbers can be quite volatile, as they can result from temporary issues, like extreme weather patterns or strikes. Any trigger based on initial claims would, therefore, require claims to remain above a certain trigger level for three to four weeks in a backtesting context. Even with smoothing, it is hard to mechanically correct for the impact of Hurricane Katrina in 2005, for example. If we use as a trigger that claims have to be at least 50k above the low of the previous year, and have to be above 350k, the SPX has already fallen through its 200dma in every instance where there is a subsequent recession. This suggests that the 200dma may be a better risk management tool than claims.

Another labor market indicator that is typically lagging is the so-called Sahm rule. According to the rule, a recession is likely already in its early stages when the three-month moving average of the national unemployment rate rises by 0.5 percentage points or more relative to its low during the previous 12 months. The rule was originally designed by its author as an indicator to signal that the United States was already in recession, pushing policymakers to increase stimulus (see Sahm, 2019). The simplicity and reliability of the rule (admittedly calibrated for a high hit ratio, like most rules) have led financial markets to follow the exact specification carefully. The broader point the rule tries to capture is that labor markets deteriorate in a nonlinear fashion. Once they reach a critical point, the weakness often becomes self-reinforcing. The Sahm rule captures the important regularity that in the early days of a recession, unemployment tends to move up gradually but eventually accelerates, possibly as self-reinforcing effects start to kick in once unemployment impacts overall consumption more significantly. The rule uses original (unrevised) data and would have identified each recession since 1953 relatively early on, compared to the NBER. However, the Sahm rule is also lagging. On average, US equities peak 126 trading days (i.e., six months) before the Sahm rule, and typically already fell through their 200dma by the time the rule triggers, as is illustrated in Figure 3.10. The median returns for the SPX post-Sahm rule trigger are flattish but slightly below unconditional on a 3m+ horizon. Where these morph into really big recessions, such as 74 and 2008, or bear markets like 2001, we see a further leg down, but most of the damage has already happened when the rule triggers, as is illustrated by Figure 3.10. The other fun fact about the Sahm rule is, of course, that just as markets really tried to pay attention in 2023, it stopped working. The rule was triggered in July 2024, without a recession in sight. Eventually, it was actually untriggered in October. Although an expanding labor force may have contributed to the demise of the Sahm rule, it was likely never all that useful for trading purposes in the first place - which was of course never Ms. Sahm's intent to begin with. Moving on.

The same applies when using negative NFP prints as an indicator of a recession. In the following study, we selected events where NFP printed less than –10k when the previous month was positive. Except for one print in 2003, these periods were separated by at least three months. As illustrated by Figure 3.11, the SPX performed poorly leading up to these prints and lost almost 10% in the 75 trading days prior. However, it typically bottomed when the first negative NFP print was in. The results are even worse when using a –50k threshold for this study (not shown).

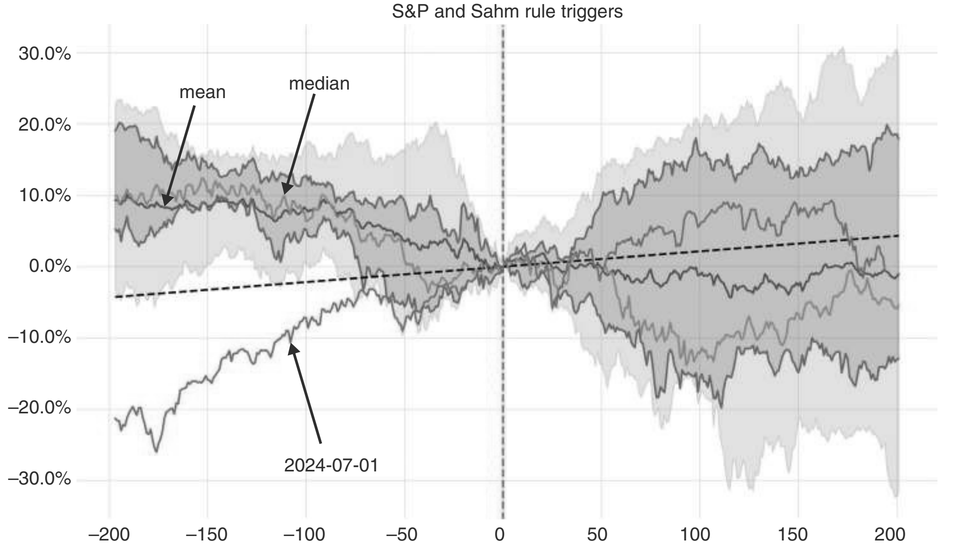


FIGURE 3.10 When the Sahm rule triggers, equities are already meaningfully lower.
Source: Bloomberg and authors' calculations.

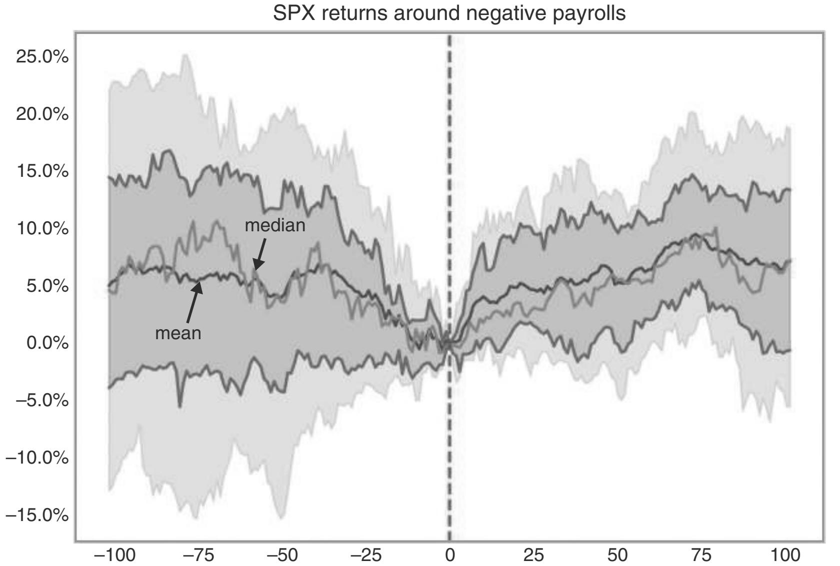


FIGURE 3.11 A negative payroll number is not a great signal for equity downside.
Source: Bloomberg and authors' calculations.

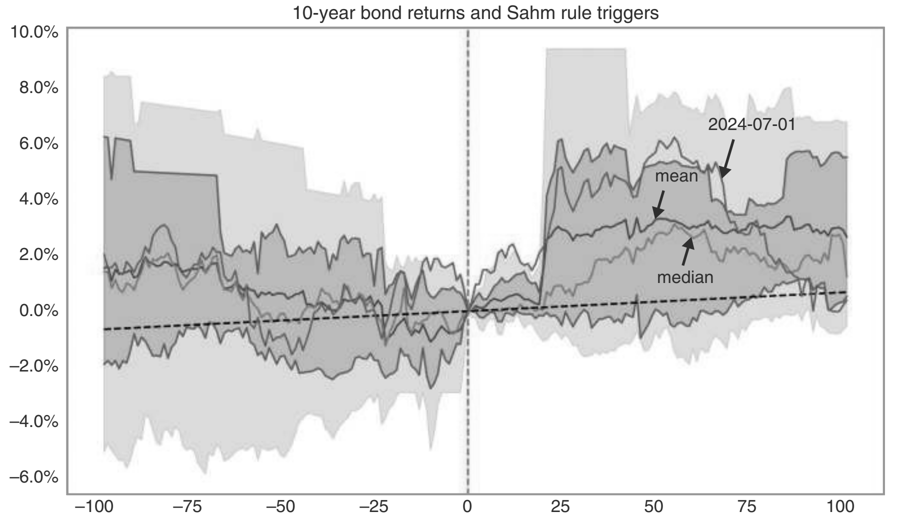


FIGURE 3.12 Bonds do well after the Sahm rule triggers.
Source: Bloomberg and authors' calculations.

Bonds (as usual, in excess return space) are better behaved in the aftermath of the Sahm rule triggering and, in the median case, also trade well into the trigger, as is illustrated by Figure 3.12. The 2024 episode was atypical in that the Sahm rule untriggered in October, shortly after it triggered in July, without unemployment accelerating higher. A triggering and untriggering without a recession has not happened before, though in the 1950s and 1960s, the rule triggered twice somewhat prematurely. If you have to use the Sahm rule, use it for bonds.

Overall, we believe that the Sahm rule is too lagging to be overly useful, even though we think that the false signal of the 2024 episode is not that likely to repeat itself. It may have been because the post-COVID labor market remained extremely distorted. Sahm herself had cautioned that this time things may be different, as the rise in the unemployment rate was not driven by people losing their jobs, but by a rise in new entrants or re-entrants. Both a high share of re-entrants and some labor hoarding may be specific to the post-COVID economy and may not materialize in the same way in future cycles.

3.6 HOW TO TRADE A US RECESSION

Recessions are special. While equities typically go up, at least in the United States, recessions are one of the few regular events that usually cause equities to fall by 20% plus. They therefore deserve their own thorough investigation. Interestingly, equities are not overly forward-looking. In the median case, the SPX range trades six months before the start of the recession, and only starts to sell off more sharply two to three months before the recession starts, as illustrated in Figure 3.13. The main exception is the bursting of the equity bubble in 2000, where the market peaked almost a year before, though the SPX came very close to retesting the highs six months before the recession. In any case, we think 2000 may have been special, as the downdraft in equities arguably caused or at least significantly contributed to the recession.

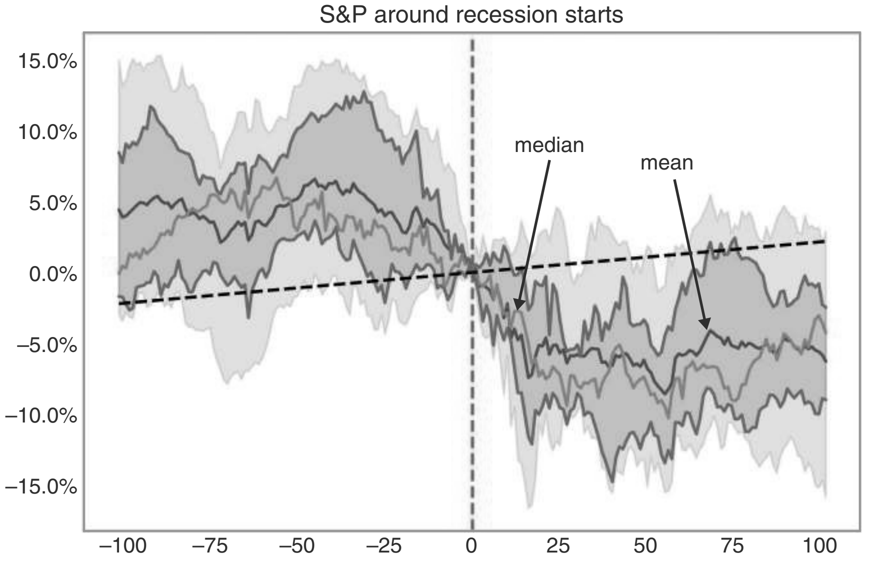


FIGURE 3.13 SPX only looks forward by around three months.
Source: Bloomberg and authors' calculations.

The main reason for the decline in equities during recessions is, of course, the negative impact on earnings. As per Figure 3.14a, 12-month forward EPS typically falls by at least 10%, with the GFC being a much more substantial 35%. Perhaps more interestingly, equity valuation also tends to contract in and during the early parts of the recession, as in 3.14b. It is *not* the case that multiples rise as EPS contracts because the equity market does *not* look through the valley. It does *not* see current EPS as overly depressed and, therefore, does not attach a higher multiple to the earnings. Instead, the typical pattern is that valuations compress first, then forward EPS estimates start

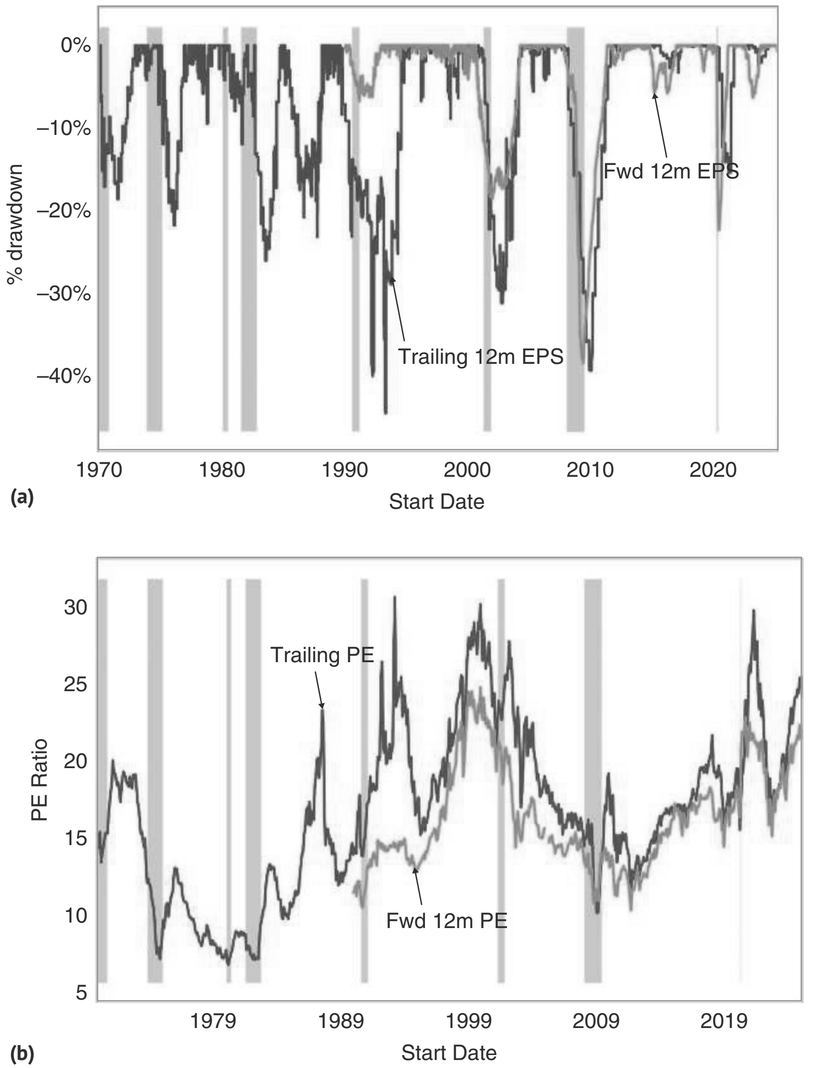


FIGURE 3.14 (a) EPS comes off during recessions. (b) Valuations also contract.
Source: Bloomberg and authors' calculations.

to compress, initially with valuation still compressing, and only late in the recession, when EPS still tends to fall, valuation starts to recover. Overall, the analysis suggests some caution to not buy back the market too early once a recession has begun.

US bond yields peak much earlier relative to a recession than US equities, often more than a year earlier, but only if it is a low-inflation recession. This is illustrated in Figure 3.15a, which plots the median path for US rates relative to the start of the recession for the low and high inflation cases (inflation above 7% at the start of the recession). In the 1970s and 1980s, the recessions were driven by the reaction of

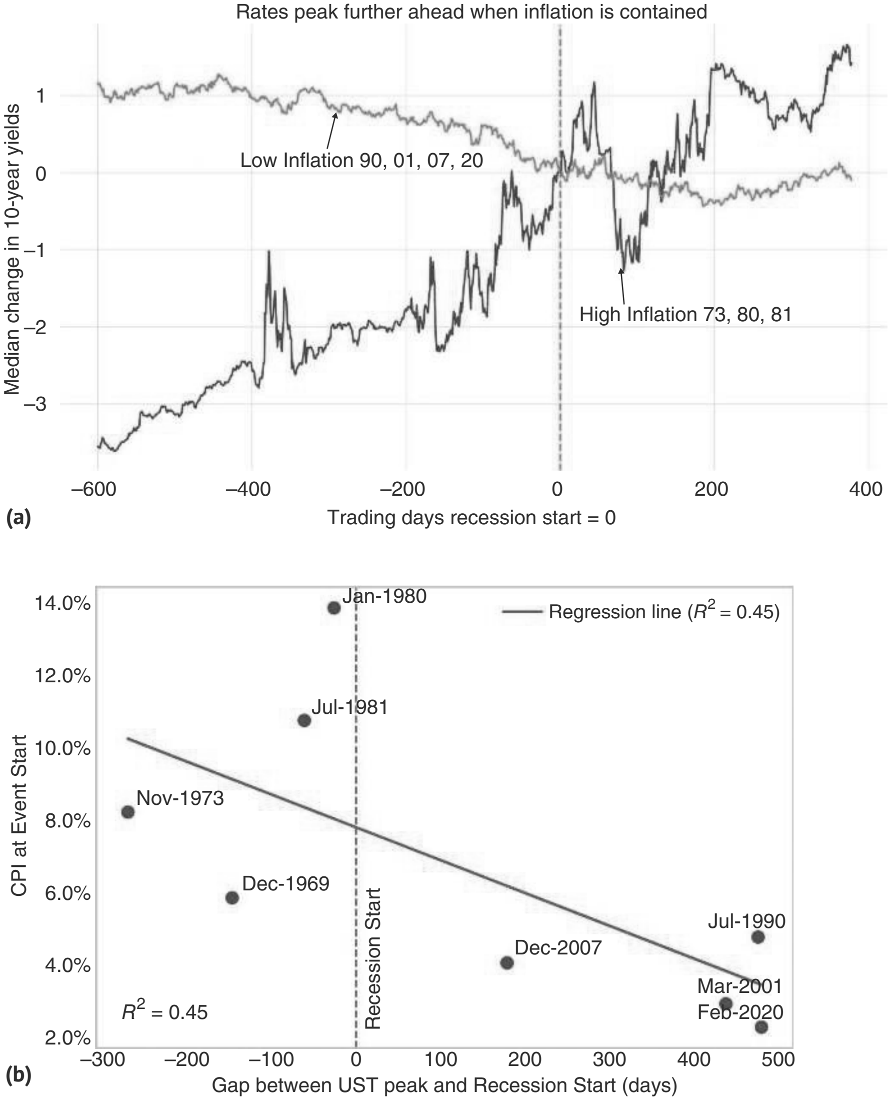


FIGURE 3.15 (a) US rates peak well before the start of the recession, if inflation is low. (b) The lower inflation at the onset, the earlier markets trade the recession, pushing rates lower.
Source: Bloomberg and authors' calculations.

the Fed to high inflation and, therefore, clearly did not result in an early peak in rates. In those cases, US rates peaked well after the recession had started. This is illustrated in Figure 3.15b, which shows a relatively strong correlation between how forward-looking rates are with respect to a recession and the level of inflation at the time of the start of the recession. This makes sense, as in the high inflation scenarios, the Fed has to hike more aggressively and cannot easily stop even as the recession likelihood rises. The other interesting aspect is that for every single recession, inflation was still above 2% at the start of the recession, i.e., clearly above target. This was true even in the "low inflation" cases. But those overshoots were not enough to prevent the Fed from cutting.

If we do enter a recession, it is worth noting that the SPX typically only bottoms well into it. The record fast turnaround was 23 days after entering, in the case of the COVID recession. The longest period was 557 days in the aftermath of the 2001 recession, largely due to the bursting equity bubble. Both the maximum and the minimum times are likely driven by very specific circumstances. The median of 226 days, i.e., around three quarters of a year, is not that useful, as lags are either quite short or quite long. We think investors should start to have an open mind about bottom fishing starting two months in (which requires, of course, correctly identifying when we enter a recession, given that the NBER is unlikely to be overly fast with this designation). Size-wise, the median SPX drawdown around a recession is 34%, with the highest clocking in at 57% in the 2007 GFC episode and the lowest at 17% around the 1980 recession. These are large numbers, and it would be a good idea to avoid equity risk around recessions if possible!

Credit shows a similar pattern. For US IG, the spread widens by a median of 124bp, while the high yield spread widens by around 785bp (but with fewer observations). The bottom in credit also occurs after the start of the recession. In the most recent recessions, credit often bottomed at the same time as the equity market, though in the GFC it bottomed earlier. Prior to the 2000s, credit tended to trough later. HY bottoms roughly at the same time as IG, though the bottom was slightly delayed in the GFC.

To get out of a recession, which typically causes an equity bear market, Fed cuts (or at least the expectation of imminent Fed cuts) are often a necessary condition. In recessions since the mid-1980s, when Fed funds became the main policy instrument, equity markets usually bottomed only after Fed cuts, as per Table 3.6. The exception was the recession of 1990, when the market bottomed on 10/11, while Fed cuts started on 10/29. But as two-year rates fell into that meeting, the market was likely strongly anticipating the cut. In the other recessions, the market bottomed only after the Fed cuts, at times shortly thereafter (20 days in 2020) and at times only much later (539 days in 2001). The additional downside for the SPX between the first cut and the bottom was substantial even in 2020, when the lag was not that long. Overall, we are biased to not buy the market before Fed cuts are very close if there is a recession. Having said that, we cannot derive much from this study in terms of how much remaining downside there is, even when the Fed is starting to move. All we can say is that historically, the first cut was usually not enough to put a bottom in for the SPX as per Table 3.5.

As an aside, we would also note that bear markets that are not coinciding with a recession do not necessarily require a Fed cut for a bottom (Table 3.6). The 2022 bear market is a case in point, as the market troughed on October 13, 2022, when the

TABLE 3.5 Both Equities and Credit Bottom Well After the Start of a Recession

	SPX drawdown	IG spreads max widening	HY spreads max widening	SPX trough days	IG spread peak days	HY spread peak days
Dec-1969	–26.9%	71		146	365	
Nov-1973	–46.0%	142		307	455	
Jan-1980	–17.4%	99		56	121	
Jul-1981	–32.9%	132		377	426	
Jul-1990	–20.8%	67		72	184	
Mar-2001	–44.9%	117	431	557	558	559
Dec-2007	–56.6%	486	1,602	434	338	351
Feb-2020	–34.0%	280	785	23	23	23
Average	–34.9%	174	939	246	309	311
Median	–33.5%	124	785	226	352	351

Source: Bloomberg and authors' calculations

TABLE 3.6 Fed Cuts a Necessary Condition for an SPX Bottom During Recessions

	First cut	SPX bottom date	Draw-down to first cut	SPX draw-down	SPX trough days	Start to first cut (days)	First cut to SPX low (days)
Jul-1981	Nov-1981	Aug-1982	–19.8%	–32.9%	377	94	283
Jul-1990	Oct-1990	Oct-1990	–20.8%	–20.8%	72	90	–18
Mar-2001	Apr-2001	Oct-2002	–20.3%	–44.9%	557	18	539
Dec-2007	Jan-2008	Mar-2009	–16.9%	–56.6%	434	22	412
Feb-2020	Mar-2020	Mar-2020	–12.8%	–34.0%	23	3	20
Average			–20.0%	–34.9%	246	109	137
Median			–20.0%	–33.5%	226	76	57

Source: Bloomberg and authors' calculations

equity market sold off in the morning, but then ended the day violently higher despite a higher-than-expected CPI print. Other than fully relying on the price action in equities, it would have been very difficult to call the bottom, especially as the bear market had a significant inflation component and as US rates did not show the same counterintuitive reaction on that day, with bond yields ending higher (though maybe not as much as feared). Ex post, it became clear that the stock market had bottomed exactly on the day when the highest CPI print for the cycle was published. The bottom did not require Fed cuts but, on the contrary, was put in with substantial remaining Fed hikes.

Back to the recession story, the frustrating thing is that it is very hard to distinguish soft from hard landings early on. Past hard landings had the manufacturing ISM fall below 45, while soft landings had the ISM bottom out somewhere in the 45 to 50 zone. However, going into a recession, both economic data and markets look quite similar. Conditions are basically weakening, but observers cannot be sure whether the weakening will stop in time to avoid a recession. The other frustrating issue is that there are not that many datapoints to study, as recessions are relatively rare, with soft landings even more so, given that it is tricky to analyze the more distant past, either due to data availability or because the likelihood of a structural break in the data increases the further we go back. Nevertheless, there are some interesting patterns to discuss.

First, it turns out that usually soft landing optimism runs high going into a recession (but also going into a soft landing). This is illustrated by Figure 3.16, where we count stories on Bloomberg for soft versus hard landings. Stories for soft landings spiked going into every single one of the last four hard landings, clearly suggesting that the majority view and sentiment on soft versus hard is not overly predictive.

Second, we found that the delta between soft data (surveys, like the ISM surveys), as compared to hard data (actual economic activity, like IP or housing starts), may have some salient information, though this finding comes with a severe small sample warning. In our small sample, whenever the soft data momentum falls below the hard data momentum (measured by the Citi data change indexes, see Connell [2025] for other applications), it pays to reduce risky asset positions. Investors should then reenter

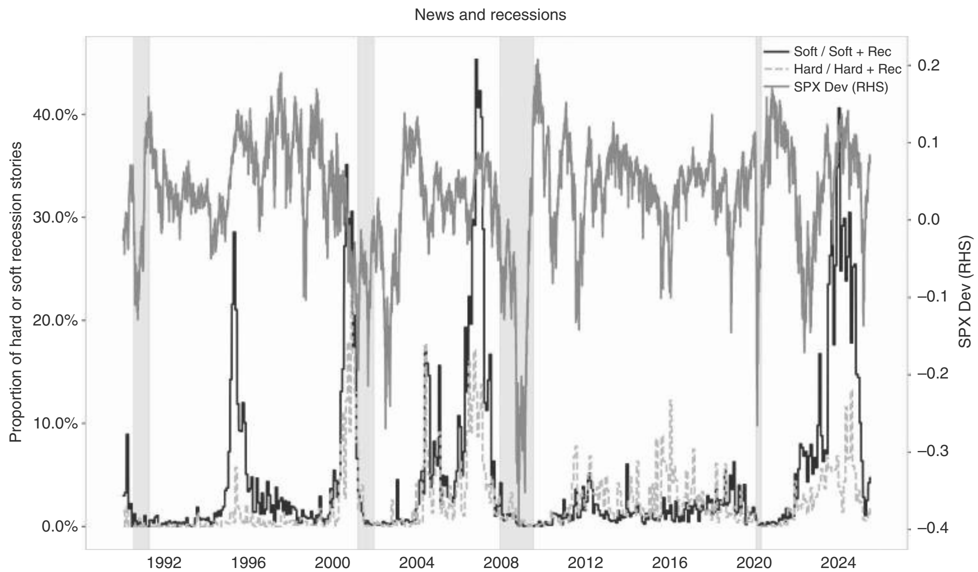


FIGURE 3.16 Hard landings at first look like soft landings.
Source: Bloomberg and authors' calculations.

risky assets when soft data is back above hard data momentum again. While this is a general rule, it also works well going into recessions. Recessions have always been preceded by periods where soft data was doing worse than hard data, though we would note that the data only starts in 2004. Like with any rule, there are a considerable number of false alarms. The rule overall improves the information ratio for the SPX from 0.55 to 1.04, as per Figure 3.17a, and outperforms in both of the recessions in

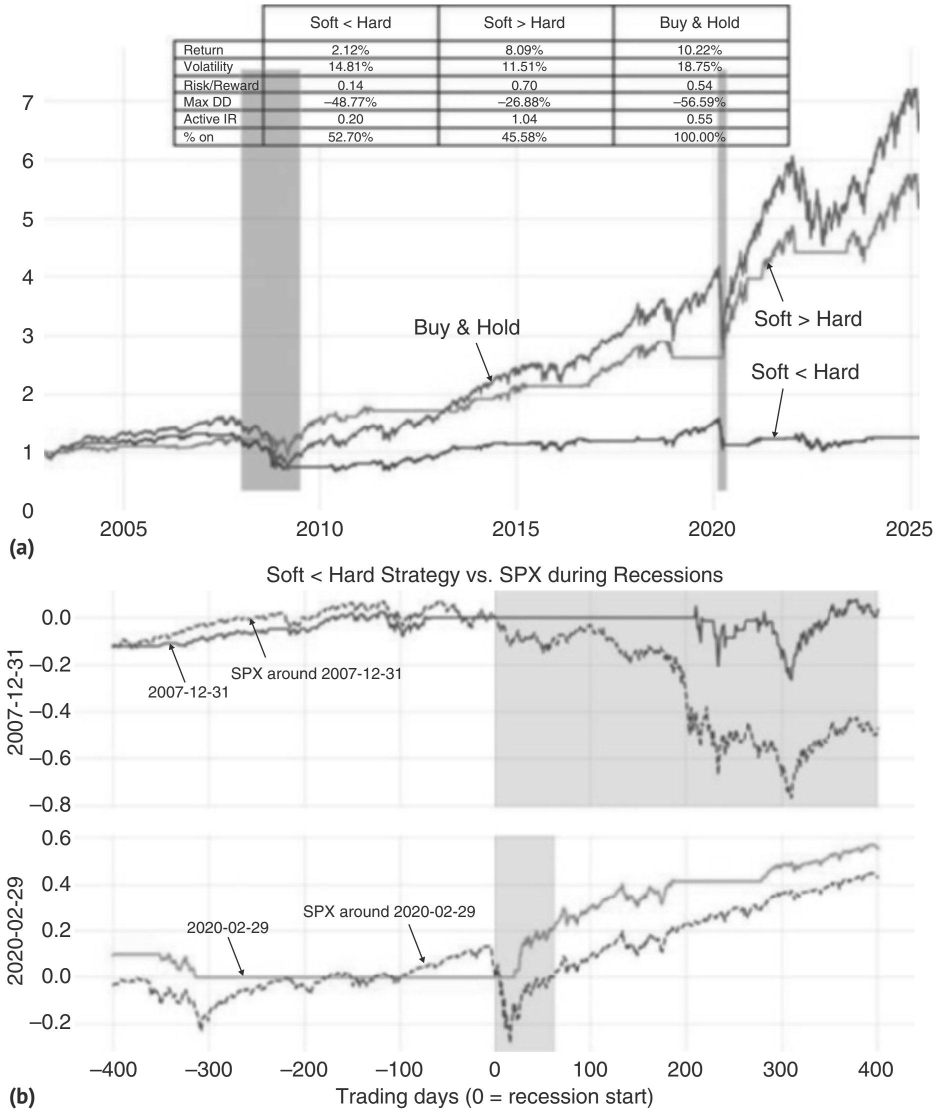


FIGURE 3.17 (a) Soft data momentum worse than hard data mom is a warning signal. (b) The signal works well going into recessions.
Source: Bloomberg and authors' calculations.

the sample, as per Figure 3.17b, but does not maximize total returns. We would take the rule more seriously when we expect a recession for other reasons. We would take it less seriously when the SPX is trading above its 200dma.

3.7 PRICING PRICE CHANGES

So far, we have woven inflation into the discussion of growth and recessions rather than letting it take center stage. This is the case because most central banks often have focused more on growth than inflation, even though most central banks have an inflation target, with the Fed being one of the very few central banks with a dual (at least) mandate, i.e., targeting both growth and inflation. The main reason is that, for most of the time, central banks in general and the Fed in particular have believed that inflation is lagging growth. Therefore, when growth weakens, most central banks have looked through higher inflation readings, expecting inflation to eventually follow growth lower. Even starker, inflation shocks, like oil shocks, were largely seen through the lens of reducing consumer incomes and, therefore, as a negative growth shock, which in the end would more than offset the initial one-off price hikes. Central bank orthodoxy only required banks to act in response to second-round effects, i.e., if prices outside the goods that were subject to the shock were getting impacted, or when inflation expectations were destabilized by the initial shock. This meant that most of the time, central banks were implicitly more focused on growth than inflation shocks, especially in the aftermath of the GFC.

However, in the aftermath of the COVID shock, inflation was high enough that it moved center stage, similar to what had occurred in the 1970s. Back then, it was largely the outcome of two oil shocks (the 1973 OPEC embargo in response to the Yom Kippur war and the output loss in response to the Iranian revolution in 1979), combined with the depeg of the USD from gold (in 1971). These shocks were large enough to lead to very major second-round effects, forcing the Fed to become extremely aggressive in squashing inflation. In 2022, it was a case where supply chains were mangled by COVID, made worse by major fiscal stimulus, which more than offset the growth hit from COVID. Again, it was an outsized shock, which had major second-round effects, forcing the Fed to eventually address it, after having initially remained in the post-GFC mindset of fading "temporary" inflation spikes.

Going forward, it is unclear how frequent large inflation shocks will be. The Fed's stance of seeing inflation as a temporary shock is, on average, justified. Inflation does contain the seeds of its own demise, as high inflation usually undermines real incomes, leading to weak growth, and eventually to lower inflation. The main exceptions to this rule are central bank mistakes (i.e., money printing) or mistakes by finance ministries (adding fiscal stimulus), where both mistakes happen in reaction to an attempt to blunt the pain from inflation, thereby making this adjustment channel inoperable.

We therefore keep the inflation discussion relatively brief but want to share two observations. First, we note that oil is a very important ingredient of most inflation shocks. As seen in Figure 3.18, many of the large inflation spikes have an important oil component. Although this is very well understood for the headline CPI, it is also true for the core CPI. The typical pattern is that oil leads the PPI, which leads headline CPI, which leads core inflation. This is likely the case, as core inflation also includes

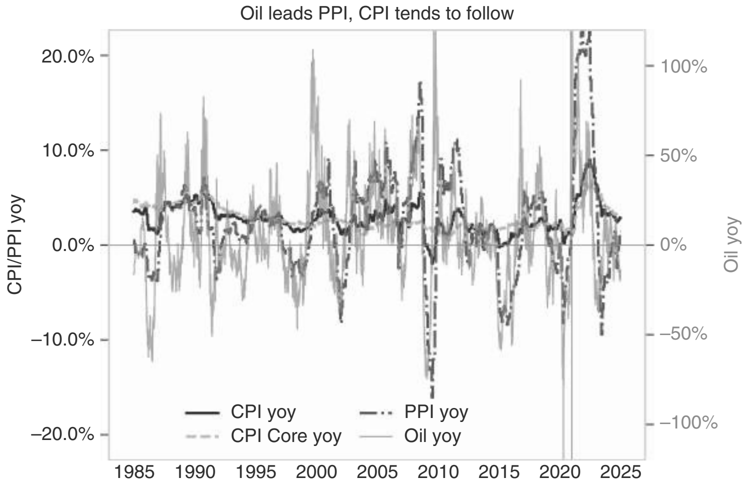


FIGURE 3.18 Oil leads PPI leads CPI leads core CPI.
Source: Bloomberg and authors' calculations.

oil, given how important an input energy costs are for many products or services like travel.

Having stressed the importance of commodity prices, we also show some evidence that labor markets are relatively less important for inflation, although labor is, for many goods, a larger share of costs than raw materials. The reason is that labor costs are much less volatile than commodity prices. Figure 3.19 illustrates that nominal wages are highly correlated with the CPI. Moreover, the CPI is at times leading. Although the wage jump in 2020 appears to be leading, it was just a compositional effect, as lower-paid workers lost their jobs, while higher-paid workers, who were able to work from home, did not. More broadly, there is virtually no period of high inflation where nominal wages were not rising as well. However, there have been numerous periods of low unemployment rates that did *not* trigger higher inflation. Academic studies found that the impact of unemployment on the CPI (Phillips curve) has weakened since the 2000s, even though the impact on wages is still measurable (see, for example, International Monetary Fund, 2022). While we would not neglect the labor market, a larger focus on commodity prices seems justified when assessing the inflation outlook.

3.8 THE THIRD MANDATE

Outside of growth and inflation, the Fed also targets financial stability. This is not overly controversial in a narrow sense, and it is widely expected that the Fed comes to the rescue when money markets or US Treasury trading freeze up, for example. More controversial is the question of whether the Fed should also stabilize the back end of the curve (which is explicitly listed as a mandate of the Fed in the Federal Reserve Act). Actions by the Fed are even more controversial when targeting financial stability more broadly, which can be interpreted as a put for equities, as the Fed typically springs

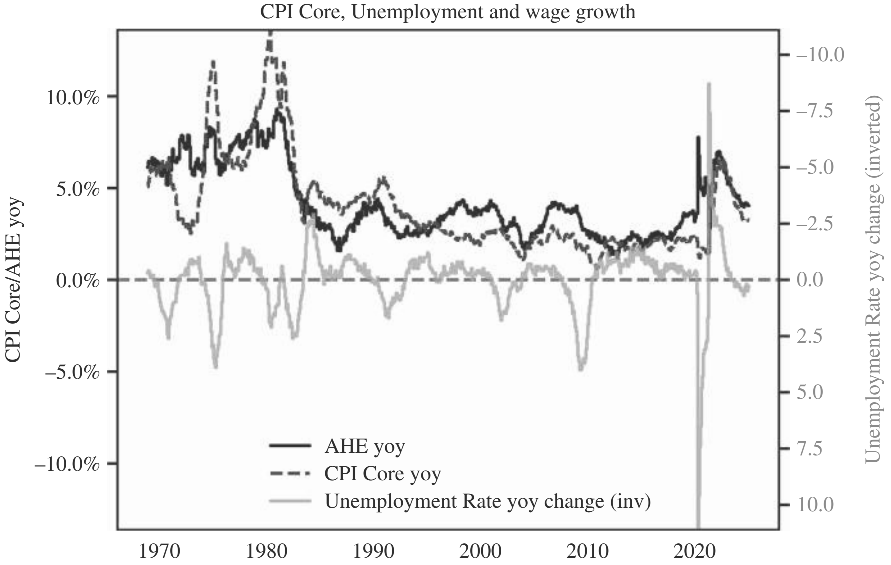


FIGURE 3.19 CPI drives wages more than unemployment does.
Source: Bloomberg and authors' calculations.

into action during financial market accidents. We briefly discuss how to trade such accidents.

Financial sector accidents have a long history of significantly impacting asset prices. This is true when the accident in question is quite severe and pushes the real economy into recession (think GFC), or when the accident is benign and quickly gets addressed by the Fed (as in the LTCM crisis in 1998 or, to a lesser extent, Silicon Valley Bank, SVB hereafter, in 2023). We define a financial sector accident by a three-standard-deviation underperformance of banks versus the SPX, over three months, in the context of a SPX downmarket. Some interesting takeaways are illustrated in Figure 3.20. First, banks continue to underperform after these events, even if equities themselves bounce higher relatively quickly, which is typically the case: The SPX usually recovers sooner rather than later and, in the median case, makes back all lost ground within three months at the very latest. The GFC is obviously a major exception. Credit spreads take much longer to make it back to pre-event levels. Two-year rates continue to drop after the event. This is shown in Figure 3.21. The reason is that the Fed plays it quite cautiously after the event, even if it turns out to be a fairly benign one. This was also evident in the 2023 SVB episode. The equity market bounced back quickly, but two-year rates initially barely moved up, eventually following equities higher, but with a substantial lag. The behavior of the USD (not shown) is unclear, though. As a rule of thumb, the larger the SPX downside, the stronger the USD upside. On average, the best trade if our financial accident trigger is hit is therefore to receive rates and, with a slight delay, buy the dip in equities, possibly when the volatility risk premium model tells us to (see the equity chapter).

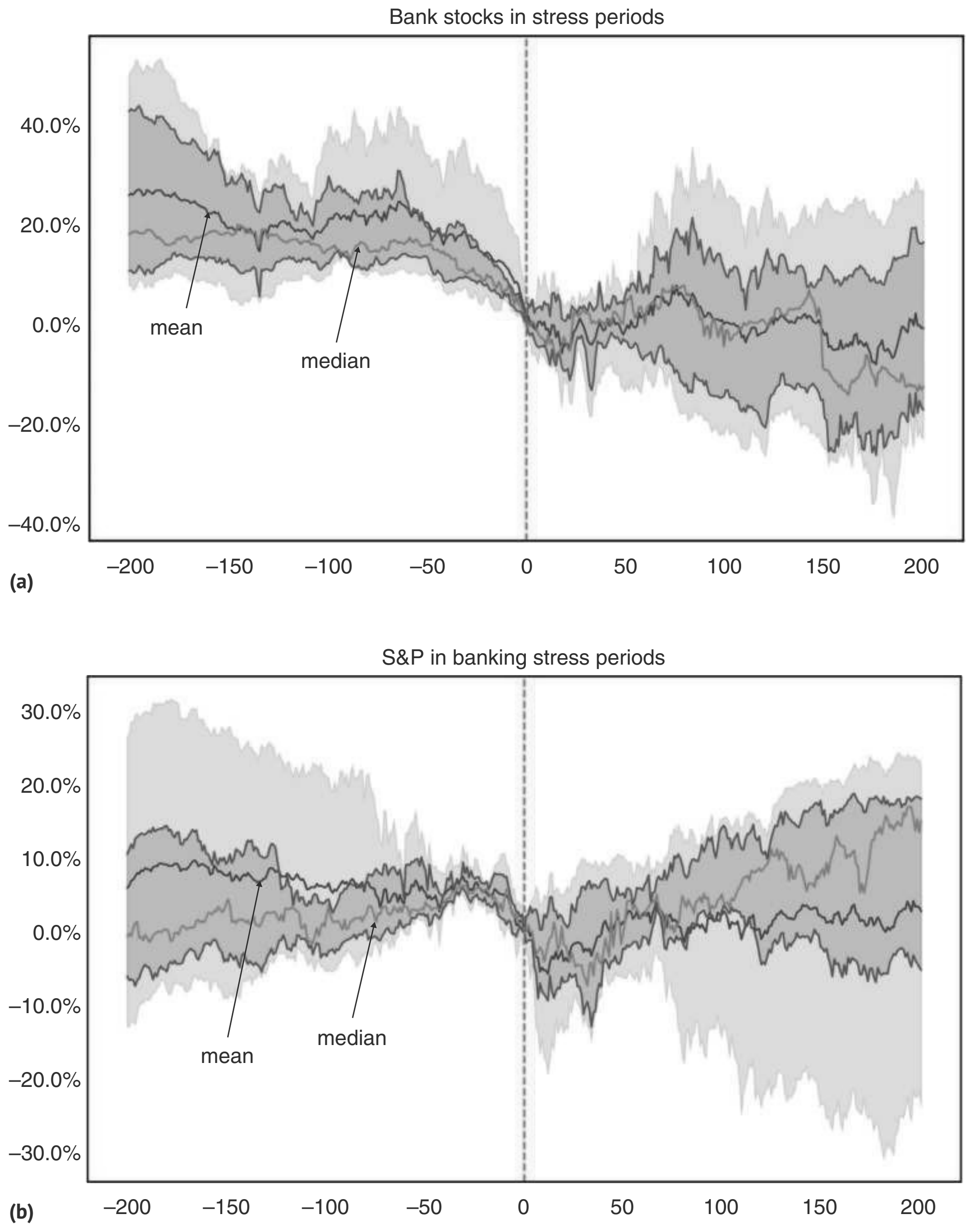


FIGURE 3.20 (a) Banks keep underperforming after the accident. (b) Even as the SPX recovers fairly quickly.
Source: Bloomberg and authors' calculations.

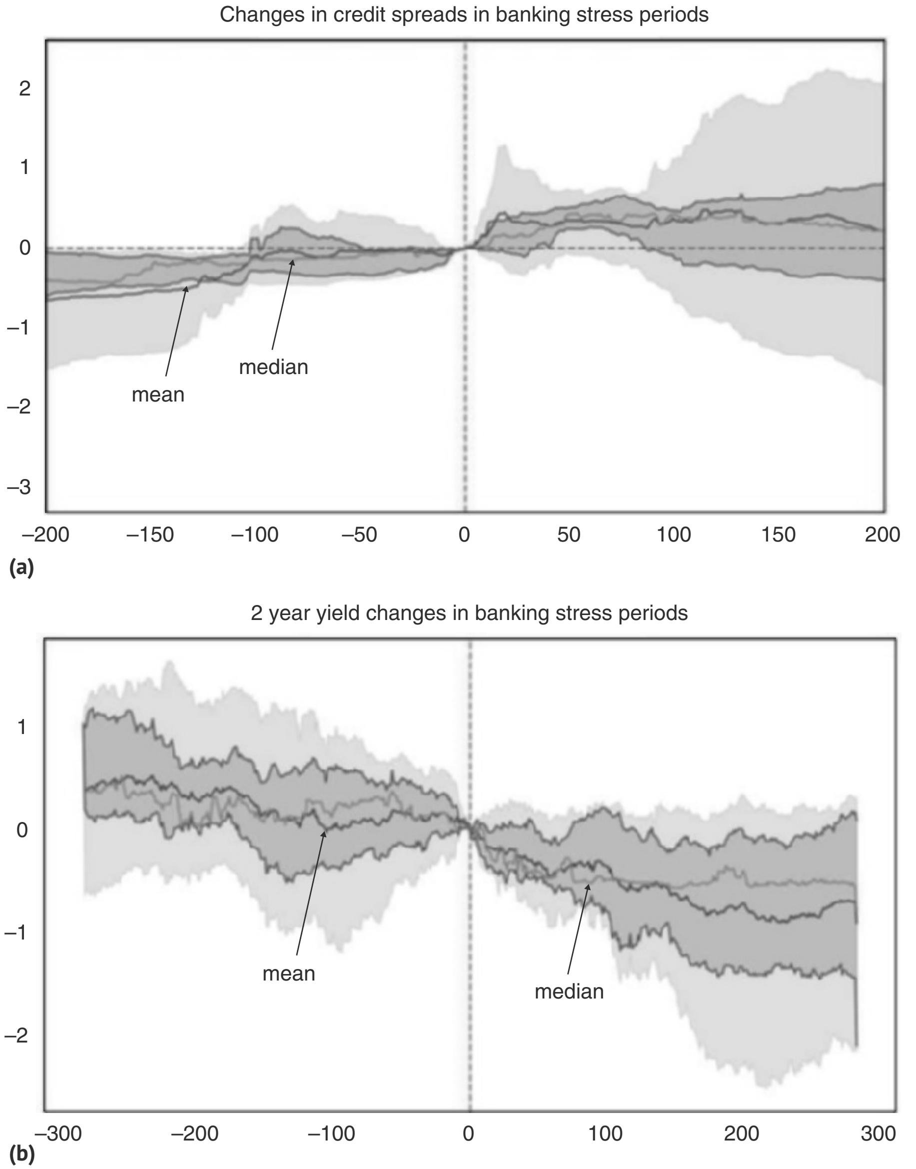


FIGURE 3.21 (a) Credit spreads recover more slowly. (b) 2Y yields decline significantly.
Source: Bloomberg and authors' calculations.

In the end, much depends on the extent to which financial conditions tighten, and how protracted that tightening is. This determines the impact on the real economy and therefore on earnings and on the equity market in the more medium term. In 2008, the tightening was off the charts. When our financial accident indicator hit in July 2008, the financial conditions index was already quite extended on the downside, but then took another leg lower. When financial conditions are in their tightest quintile, the

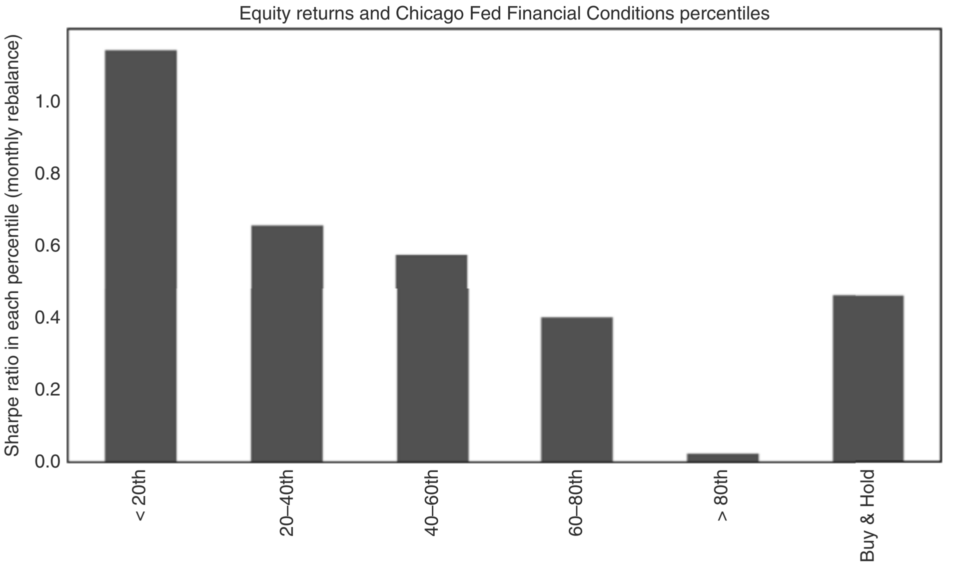


FIGURE 3.22 Tighter financial conditions mean worse forward returns.
Source: Bloomberg and authors' calculations.

active information ratio of one-month forward returns is the weakest, being very close to zero. The opposite is true for the quintile with the loosest financial conditions. This is illustrated in Figure 3.22, which shows results for the Chicago Financial Conditions Index. Although this index gives only a very small weight to equity indexes, it also includes equity volatility as well, it does include equity volatility as well as various credit spreads, which can make results somewhat tautological. By focusing on forward Sharpe ratios, we mitigate this problem (and maybe benefit from the trending behavior of equities). We would suggest following the financial conditions index in real time, especially during financial accident episodes.

3.9 RHYMING IT DOES, BUT NOT REPEATING

Although the methodology in the book is largely based on establishing the priors for how to position for the archetypal cycle, it is, of course, true that no cycle is the same. Although above we distinguished between soft and hard landings, it is also extremely important to understand the driver of the cycle at hand. Knowing where the weakness in a particular cycle comes from not only allows investors to more surgically choose the right assets to position for stress but also gives a better sense of what measures policymakers have at their disposal to address the issue at hand. In the twentieth century, many US recessions were blamed on tight monetary policies and/or oil price shocks. Post-war demobilizations and asset bubbles (1929 crash) also make a (less frequent) showing. In the twenty-first century, it was again mostly asset bubbles (dot.com in the 2000s, housing in 2007), followed by the more atypical COVID recession, as per

TABLE 3.7 A Brief History of Recessions

Recession	Cause	Fed hikes	Oil spike yoy	Equity Mcap % GDP	Fiscal tightening %GDP
01-02-1970	Fed hikes	4 to 9%	9%	N/A	–3%
11-30-1973	Oil and rates shock	5.5 to 11%	180%	N/A	–1.4%
01-31-1980	Oil and rates shock	5 to 15.5%	140%	N/A	–1%
07-31-1981	Rates shock	9.5 to 20%	N/A	N/A	N/A
07-31-1990	Oil shock	N/A	79%	N/A	N/A
03-31-2001	Equity bust	4.5 to 6.5%	144%	70 to 150%	N/A
12-31-2007	Housing bust	1 to 5.25%	98%	100 to 140%	N/A
02-29-2020	COVID	N/A	18%	130 to 250%	N/A

Source: Bloomberg and authors' calculations

Table 3.7. Overall, this suggests that investors should be on the lookout for overtightening by the Fed, large oil price moves, and asset bubbles. It could be argued that over time, asset bubbles are becoming more important, in line with the increasing financialization of the US economy, while oil price shocks may become less important, given a lower energy intensity of US GDP, coupled with the fact that the United States is now a large energy exporter, which should cushion the negative impact on the US consumer.

To identify which asset markets can cause stress, a useful starting point is to investigate which assets boomed the most during the preceding good times. Table 3.7 shows how much market cap was added, as a percentage of GDP, in the last three asset bubbles to give a sense for how big a bubble needs to get before it becomes a macro event. The size of the bubble alone is, of course, not enough to know where the stress comes from. It is also crucial to understand just how much leverage is involved in financing the bubble and where it sits. The more leverage, the more likely it is that there is meaningful forced selling, which will make the crisis more acute.

There were also global crises that did not lead to a US recession, though, at times, they triggered Fed rate cuts. The most famous case is the Asian crisis, which generated Fed cuts in 1998. However, it is important to note that the Fed really only acted once the Asian crisis, through the Russian debacle, impacted US markets by bringing down LTCM. It is not obvious whether the Fed would have been forced to react if the global stress had not made itself felt onshore, interestingly, through a channel that was different from the obvious economic trade linkages. Although globalization over the last decades has made the United States more vulnerable to global stresses, it is still correct that the hurdle for the Fed to react to global stress with rate cuts is high, and the hurdle for the United States to go into a recession (generating a big Fed easing cycle) in response to global stresses is even higher.

3.10 BORN IN THE USA

Macro cycles are often born in the USA. The reason for the dominance of the United States in global macro-trading is not just the central role of the USD and the Fed, but it is also the US economy. Although its share of the global economy has fallen sharply, to around 25%, the US economy punches above its weight, as its growth rates are generally higher and more volatile. Since 2001, the United States has contributed around 35% to global growth, the single most important locomotive. However, it has not been the sole engine. China comes a close second, contributing close to 30%. The EU is still a relevant 15%, but then it falls quickly to just above 5% for India, and less than that for the United Kingdom, Japan, and other large emerging markets, as is shown in Figure 3.23. To get the global business cycle right, investors also need to understand China and, to a lesser extent, Europe.

The other important issue for macro-trading is whether growth among the large contributors is positively correlated. Clearly, uncorrelated growth can make a country more important for global macro-trading than correlated growth. This is illustrated in Table 3.8, which suggests that European and US growth are positively correlated, given the contributions have the same sign, with the exception being the European crisis in 2012 and 2013—a major macro event. With respect to China, it is noteworthy that the correlation is generally not very high. In particular, during the crisis in 2008 and 2009, China famously bounced back much earlier and more strongly than the United States or Europe, given a major stimulus. In 2020, China was also still contributing positively to global growth. We therefore first briefly discuss China before discussing Europe. We also add India, which has aspirations to become the next China. Rather

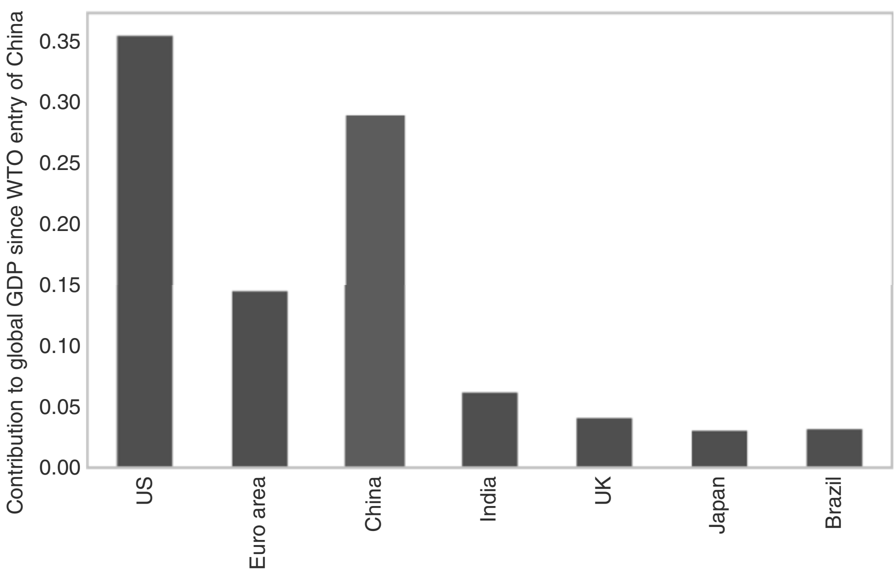


FIGURE 3.23 China has been a large driver of global GDP since WTO entry.
Source: Bloomberg, World Bank, and authors' calculations.

TABLE 3.8 Contributions to GDP: China Offers Less Correlated Growth

Date	US	Euro area	China	India	UK	Japan	Brazil
2006	25.0%	20.9%	34.0%	5.2%	3.6%	3.3%	3.1%
2007	18.1%	19.2%	42.3%	5.2%	3.9%	3.6%	5.0%
2008	2.6%	6.7%	79.3%	5.5%	–0.9%	–7.3%	10.9%
2009	–61.7%	–75.7%	91.2%	15.9%	–18.3%	–35.3%	–0.3%
2010	21.6%	11.9%	38.0%	6.2%	2.9%	8.4%	6.2%
2011	17.4%	13.7%	51.0%	5.6%	2.1%	0.1%	4.7%
2012	30.9%	–8.5%	54.3%	7.3%	3.3%	4.7%	2.8%
2013	25.7%	–1.4%	51.0%	8.0%	3.5%	6.1%	3.9%
2014	26.4%	10.0%	44.2%	8.3%	5.4%	0.8%	0.6%
2015	29.4%	13.9%	41.7%	9.0%	3.6%	3.8%	–3.6%
2016	21.2%	13.7%	49.3%	11.4%	3.6%	2.1%	–3.7%
2017	22.7%	15.5%	41.6%	7.8%	3.9%	3.7%	1.1%
2018	28.5%	11.0%	43.8%	8.0%	2.2%	1.5%	1.6%
2019	29.7%	11.9%	47.7%	5.9%	2.9%	–1.1%	1.3%
2020	–24.7%	–42.9%	18.5%	–8.6%	–17.6%	–10.6%	–3.4%
2021	29.5%	18.9%	30.8%	6.2%	6.1%	2.7%	2.1%
2022	26.2%	22.7%	23.3%	9.9%	7.5%	2.1%	2.8%
2023	31.1%	2.6%	43.5%	12.5%	0.5%	3.8%	2.8%

Source: World Bank and authors' calculations

than going into a deep analysis of those jurisdictions outside of the United States, a good part of the discussion is designed to illustrate how important the US cycle really is for global macro and just how big a local shock is needed to impact the US cycle.

3.11 FROM CHINA WITH LOVE

China has been the second most important economic force for the global macro cycle, starting when it entered the WTO in late 2001. Rapid urbanization, combined with substantial gains in market share in global manufacturing, meant that China dominated the global commodity cycle and, as a result, the global cycle with respect to emerging markets. As China has already been discussed in detail in the EM book, we keep the discussion here brief.

Our main point is that China matters for the global cycle, and not just for EM. This was most clearly seen in the aftermath of the GFC. Going into the GFC, global GDP was 61trn USD, of which the United States was roughly 15trn, the Eurozone was 13trn, and China was 3.5trn. Thanks to aggressive fiscal and credit stimulus in early 2009, China's growth accelerated from 6.4% in Q1 of 2009 to 12.1% in Q1 of 2010. For 2009 overall, growth was 9.4%, adding 320bn to global GDP (in real terms), at a time when the United States detracted 390bn and compared to a US "run rate" at the time of +300bn or so (at 2% growth). China had essentially single-handedly filled the hole left by the United States. At this stage, China's economy is around 17.5trn, and growth is just below 5%, still leading to an addition of 875bn to global GDP, whereas the run rate for the United States is now around 540bn. Clearly, China is still extremely important for the global cycle, even at reduced growth rates.

The second point to make is that the risk of a significant impact of China on the global economy is at this stage more symmetric than in the past. The reason is that China is facing several structural problems, which create both upside and downside risks. While those problems are not new and are fairly well understood, it is nevertheless important not to take our eyes off China with respect to the global macro cycle. The first risk is the balance sheet recession as a result of the real estate bust. The deleveraging of household balance sheets has been ongoing for several years. The late stages of deleveraging can either create downside stress or, alternatively, an end to the deleveraging process may eventually lead to a positive growth shock. The second structural risk is the decoupling from the United States, and maybe others, which can undermine the formidable export engine that China has relied upon during its deleveraging process. This creates more downside than upside risks at this stage, as we are early in this process. The impact of China on global inflation is also important. The weak demand for energy by China during the Russia/Ukraine war has arguably had an inflation-dampening impact. As such, China will remain key for macro investors, as also discussed in more detail in our emerging markets book.

3.12 TRAILING THE TIDE: EUROPE USUALLY A FOLLOWER

In the past, it has been relatively rare for Europe to be the main driver of the global business cycle. This is not so much an issue of size because the European economy is not that much smaller than the US economy. It is more because Europe's economy is generally growing less, partly because of lower population growth, but partly also because it is less innovative. Interestingly, on the downside, it is not true that Europe is less volatile. European crises are often as deep as their brethren in the United States, and sometimes even deeper, even when the crisis originated in the United States (like in the dot.com area or the housing bust). On the inflation side, there is also more volatility in the United States than in Europe, especially on the upside. This may be one of the reasons why policymakers in the Eurozone often hike later and less forcefully than in the United States, though it does not explain a slower easing cycle. The more compelling reason is that joint European decision-making across many countries is much harder and more time-consuming, partly because many decisions, especially during a crisis, have major distributional consequences across nations. It is therefore not overly surprising that the European Central Bank (ECB) tends to be a follower, rather than a leader, in the global interest rate cycle. Since the start of the ECB as a monetary policy institution, on most interest rate turning points, the Fed moved meaningfully before the ECB, both in upcycles and downcycles. Most famously, ECB President Trichet decided to hike interest rates in the middle of 2008, at a time when the global economy was already close to the abyss, making the ECB very late to the 2008 cutting cycle. But it is not just a timing issue. The ECB also typically raises rates by less and cuts rates by less, partly because the long-term neutral rate, R*, is likely meaningfully lower than in the United States. The current cycle is the exception rather than the rule.

In spite of a more activist Fed, it is interesting that the chart of Bunds and UST have in the last decades been extremely similar, both in terms of direction and volatility. US rates were meaningfully higher during the late 1970s and the Volcker years, but Bunds then caught up during German reunification (see following box). It took the European crisis (more on that too later) for that gap to reappear, but even then, the directionality stayed largely the same, just the level downshifted for Bunds, as is illustrated in Figure 3.24.

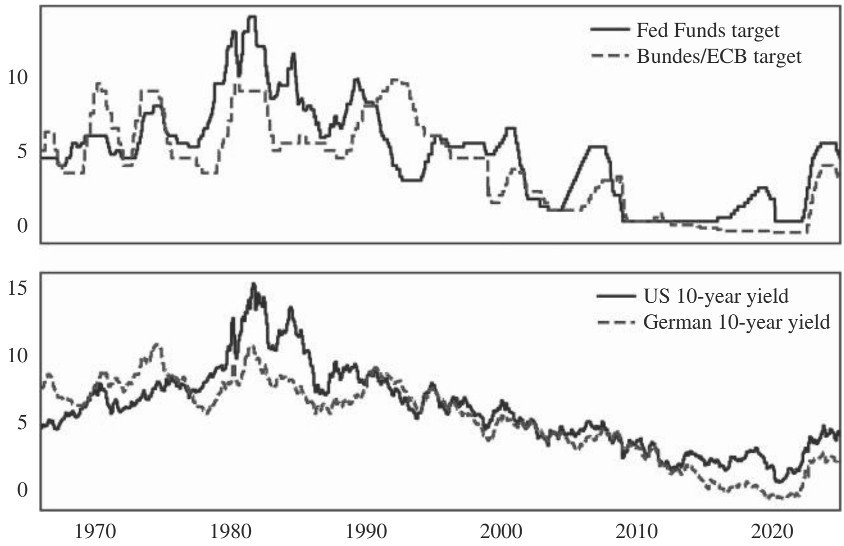


FIGURE 3.24 US policy rates and bond yields tend to lead Europe.
Source: Bloomberg.

When Europe Mattered, Take I: German Reunification

There are exceptions that prove the rule. One is the German reunification in 1989–1990. This event has been discussed at length elsewhere (see, for example, International Monetary Fund, 1991), and, therefore, we just narrowly focus on how it impacted the global business cycle. The German reunification was basically a large fiscal shock, where the West German government transferred around 180bn DM per year to Eastern Germany, or around 6.5% of Western German GDP, starting in 1990. Unsurprisingly, this fiscal shock led to inflation in Germany shooting higher, to 3.3% in late 1990, triggering a hawkish Bundesbank, which raised interest rates in 1990 and 1991 at a time when the Fed was already cutting rates in reaction to the US recession. This is the only time there has been a long-lasting divergence in monetary policy actions across the Atlantic in recent economic history. Needless to say, the DEM surged, from close to 2.00 just before the first Fed cut, to around 1.48 when the last Bundesbank hike came into the markets' field of vision. After much volatility, the DEM peaked at around 1.40, when the last Bundesbank hike occurred. Eventually, the initial boom faded as the staggering costs of reunification and the extent of the desolation of the Eastern German economy became apparent, bringing the reunification boom to an end. While the German reunification is at this stage ancient economic history, it illustrates just how big a European-centric shock is required for Europe to meaningfully decouple from the United States and to become an independent driver of global macro. The other lesson is that the United States itself was barely impacted. In spite of German growth accelerating from 3% to 7% in the reunification boom, the US economy went into recession, as is illustrated in Figure 3.25.

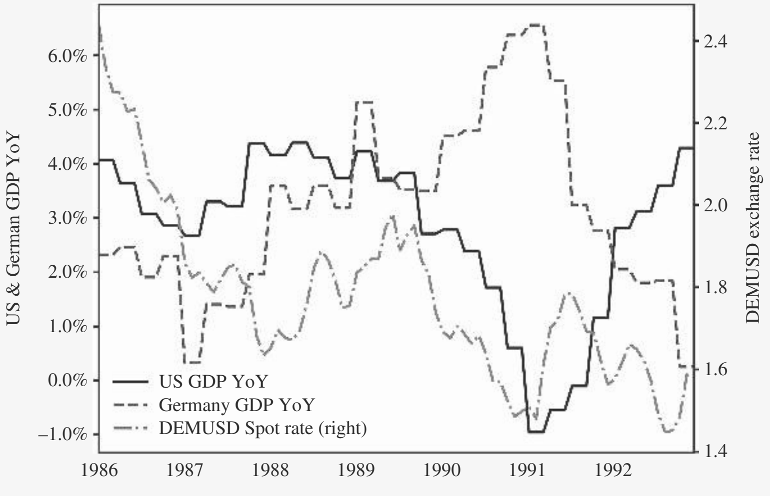


FIGURE 3.25 Germany reunification—big impact on DEMUSD, no impact on US GDP.
Source: Bloomberg.

When Europe Mattered, Take II: The European Crisis

The second major episode in which the European economy followed its own path was the European crisis in the aftermath of the GFC. The introduction of the EUR in 1999 had depressed borrowing costs for the European periphery, leading to a significant economic boom, fueled largely by borrowing and unsustainable spending. The banking crisis during the GFC led to increased government spending, in part on bank bailouts, and lowered government revenues, resulting in ballooning budget deficits and rapidly rising sovereign debt levels. As a result, countries like Greece, Ireland, and Portugal faced soaring borrowing costs, as investors demanded a higher risk premium. Higher interest rates, in turn, exacerbated fiscal deficits. This was a classic vicious cycle that happens in emerging markets relatively frequently, but that had been quite rare in developed markets. At the same time, with European banks holding substantial amounts of sovereign debt, the banking crisis also became worse, as banks faced massive losses due to falling government bond prices, requiring even more fiscal spending in the form of bank bailouts. The linkage to the banking system made the cycle even more vicious. The remedy proposed by the owner of the largest balance sheet, Germany, stressed austerity measures to bring public deficits under control. Such cuts likely worsened the recessions in the countries of the periphery, putting even more pressure on revenues, and offsetting the positive impact of expenditure cuts at least partially. This created fears about the future of the Eurozone as a single currency block, further adding to the widening spreads. Then, ECB president Mario Draghi delivered his famous "whatever it takes" speech on July 26, 2012, emphasizing the ECB's commitment to preserving the EUR and signaling the ECB's intention to stabilize the Eurozone economy. The speech reduced risk premia in sovereign bonds and was followed up by the OMT (Outright Monetary Transactions) instrument, which aimed to purchase sovereign bonds of crisis countries. This was the turning point in the Eurozone crisis. Once more, the size of the European shock had to be enormous for it to truly matter on the global macro scene. The ECB started to cut rates again in late 2011, while the Fed, unable to cut interest rates further from its prevailing level of 0.25%, launched QE2 in November 2010 and QE3 in September 2012. Interestingly, Bunds clearly outperformed US rates, but only in 2014–2015, after the European crisis had well peaked, while the ECB was on its way to cutting to zero. And even then, directionally, both Bunds and UST kept moving in exactly the same way. The EUR remained in a very volatile range, largely following credit spreads in the periphery, and only collapsed in late 2014, when the ECB had cut rates to below the 0.25 Fed funds. Furthermore, US GDP chugged along relatively nicely during the years of the European crisis from 2010 to 2012, as is shown in Figure 3.26. The lesson is once more that a European shock has to be enormous to matter for the global cycle, and that US GDP is typically quite robust with respect to foreign affairs. US rates do set the tone for global rates even in the face of large local shocks.

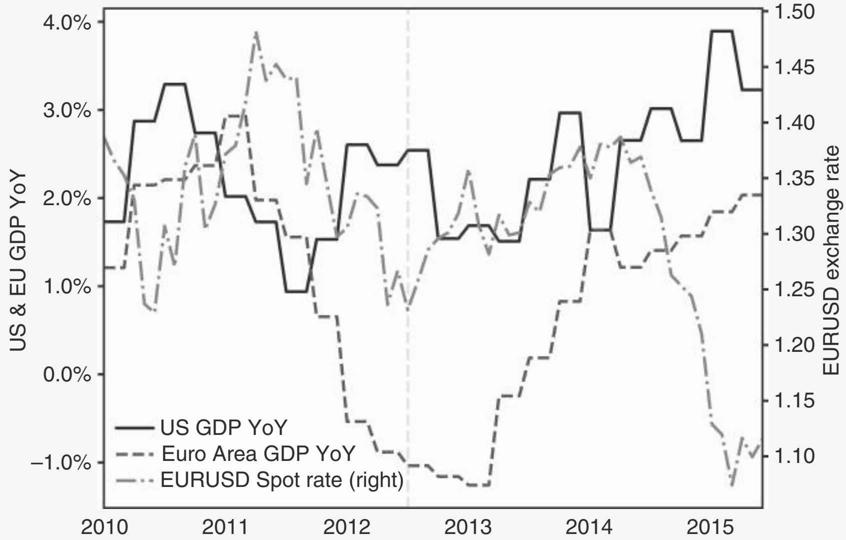


FIGURE 3.26 Euro crisis without much impact on US growth, and EUR negative with a long delay.
Source: Bloomberg.

The other interesting difference between the Eurozone and the United States is that Europe is much more bank-based. This means that monetary aggregates, an old staple of the Bundesbank, are still somewhat relevant, even though they lost any relationship with the business cycle in the United States on the back of financial innovation. For Germany, for example, the year-over-year change in real M1 still works as a decent leading indicator for the business cycle (see Figure 3.27), which is not the case for the United States, as has been frequently illustrated by Christian Schulz (Schulz et al., 2025).

3.13 INDIA AS THE NEXT CHINA?

Given the Chinese malaise due to its real estate bust, compounded by the demographic bust, the question is, which country could be the new China? India is a candidate. This is largely based on the idea that India is one of the few large countries that will still benefit from population growth, if only in the poorer regions. Furthermore, and perhaps more importantly, it is also on the back of Prime Minister Modi, aiming to improve the competitiveness of the Indian economy. However, at this stage, India's GDP is only around 3.6trn USD. Growth of around 8% of GDP adds around 290bn to real global GDP, much less than what even a challenged China adds. From a global GDP dynamics perspective, at this stage, macro investors can therefore safely ignore India (and any other emerging market outside of China). The more interesting question is whether India could, over the longer term, grow into China's shoes, and if so, whether macro investors can benefit. One way to think about it is that China, when it entered

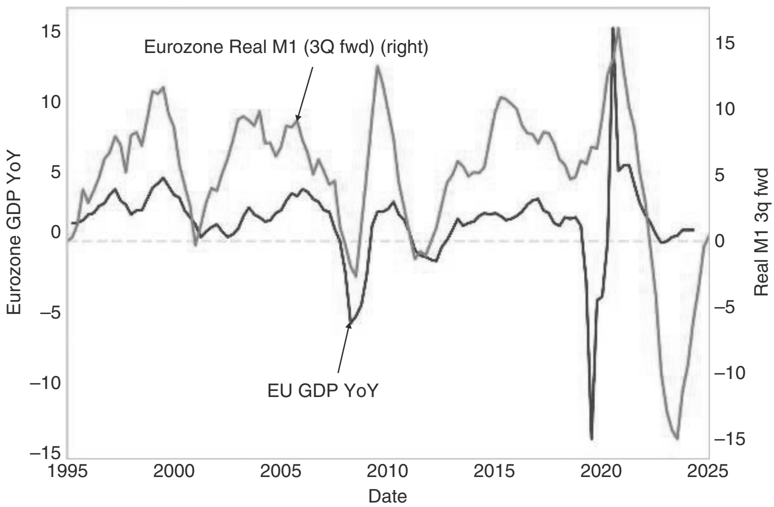


FIGURE 3.27 Monetary aggregates still work in bank-based systems.
Source: Bloomberg and authors' calculations.

the World Trade Organization (WTO) and its rise began, was around 2.5% of global GDP. It then proceeded to grow by around 6.6% on average for the next 20 years. At this stage, India's GDP is around 3.4% of global GDP. While we do not have long-term growth estimates for India, growth rates around 6% may well be plausible in the next few years. The question is how investors would best benefit from this development.

The experience of China is somewhat disillusioning in this regard. To come to this assessment, we focus on the Shanghai Composite, which includes more than 2,000 shares listed in Shanghai, rather than the much narrower HSCEI, which includes 60 shares listed in Hong Kong. The Shanghai Composite Index only returned about 5% per year between China's entry into the WTO through December 2024, although GDP grew at 12% in nominal USD terms (real GDP 8%). This is much less than the SPX, which returned slightly above 9% p.a., even though US GDP growth only averaged around 4% per year in nominal terms. Clearly, China's equity investors did not participate in the Chinese success story, though we note that the HSCEI index performed more similarly to the SPX. The currency performance did not add much to the return of foreign investors either. The CNY (in spot terms) strengthened by 0.6 percent on average over the time China was in the WTO. One plausible reason is that profits went to a larger extent to company insiders, as well as to the government, and much less to external shareholders. SHCOMP also suffered from the fact that more than 50% of the market capitalization is state-owned enterprises, which could have been expected to benefit less from fast GDP growth, which was largely driven by the private sector. This is one reason why the HSCEI performed better. The disappointing result for Chinese investors in equities is not unique to China. Dimson et al. (2002) document a negative correlation between per capita economic growth and stock returns over a 100-year period. Studies by Ritter (2005) and Ritter (2012) have confirmed the absence of a reliable link between GDP growth and stock returns. Countries with higher long-run real GDP growth do not necessarily have higher long-run real stock market returns. We find the same pattern in the more recent data, as is illustrated in Figure 3.28. The correlation is weak, and the chart highlights just how disappointing the Chinese experience has been. Still, there are likely countries where the institutional set-up means that more of the profits generated from GDP growth end up with shareholders.

As an interesting aside, the big picture correlation between currency returns and GDP growth is also very low. This suggests that investors may not necessarily benefit from the FX leg of their investment either. Again, China stands out as a disappointing case, as per Figure 3.29. Here, the performance is even worse, as it accounts for the negative carry.

So far, the Indian experience has been much more positive. The Nifty has returned around 13% (in USD terms). Although the INR has weakened in spot terms, the total returns were also positive. One reason for the better performance of Indian equities versus China may be that the weight of SOEs (mostly banks) is smaller than in China. Although India may not be overly relevant yet to get the global macro cycle right, if India continues in the footsteps of China in terms of gaining market share in global GDP, it is plausible that foreign investors can participate in it more fully than was the case in China.

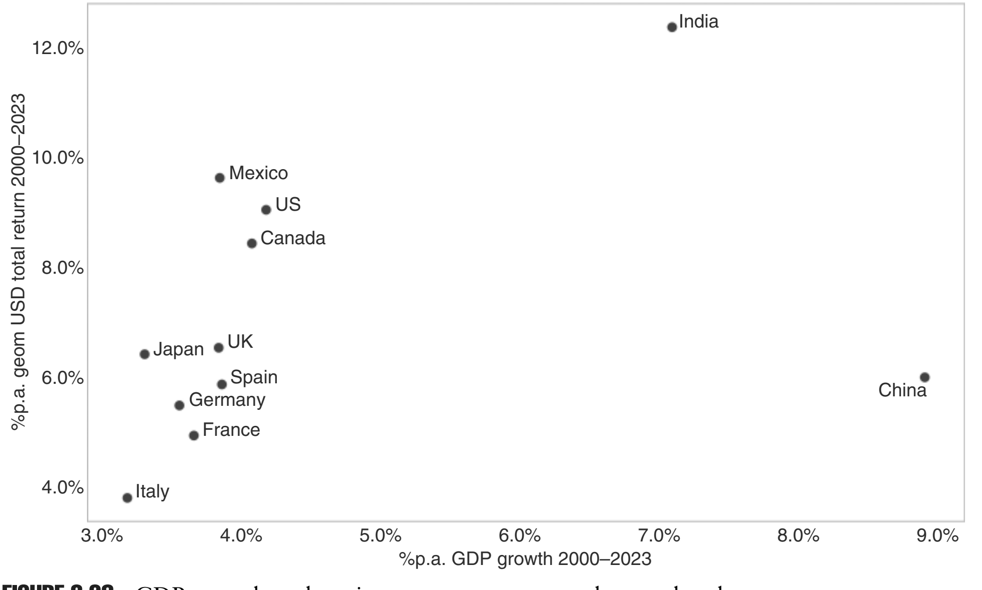


FIGURE 3.28 GDP growth and equity returns are not overly correlated.
Source: Bloomberg and authors' calculations.

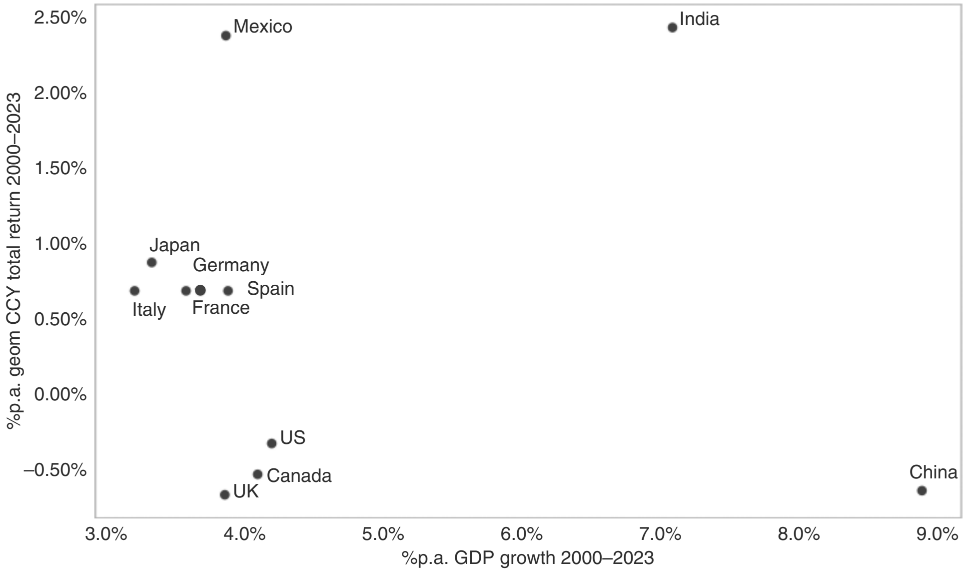


FIGURE 3.29 GDP growth and FX returns are uncorrelated.
Source: Bloomberg and authors' calculations.

3.14 SUMMARY

The chapter investigates the leading indicators for the business cycle. Of the official data releases, the manufacturing ISM is one of the most useful. One-month forward SPX returns are strongest when the ISM is below 50 and pointing upward. The sample size is small, though. With respect to market indicators, the inversion of the yield curve has been the most prominent leading indicator. Duration performs particularly well after the 3m/10yr inverts, while equity markets are overall displaying slight subpar returns on disinversion, rather than inversion. The rules of thumb for trading a US recession are to avoid equities two to three months before it starts and to remain bearish for the first half of the recession. EPS typically falls by more than 10%, and valuation compresses. Fed cuts are a requirement to put in a bottom during recessionary times. Bond yields react much earlier than equities, often more than a year ahead of a recession. The higher the inflation, the shorter the lead time. Although it is very difficult to distinguish between soft and hard landings, periods when the soft data weakens aggressively versus the hard data have increased recession risk. We also study financial market accidents, proxied by a significant underperformance of banks. We find that in the median case, the SPX bounces back quickly, but US rates continue to fall, and banks remain underperformers for longer. We briefly discuss regions outside of the United States and discuss their importance for the global macro cycle.

CHAPTER 4

Temporary Turmoil: Markets and (Geo) Politics

For most of the last 30 years, it was a tough job to be a geopolitical consultant, given that market-relevant conflicts in the aftermath of the fall of communism were extremely rare and short-lived. However, this has changed after the 2022 attack by Russia on Ukraine. Geopolitics has become a major buzzword in global macro, with geopolitical consultants mushrooming, proclaiming a brand new world where investors are unable to survive without their services. Although it is probably correct that geopolitics will be less harmonious than it was in the direct aftermath of the fall of communism, it is still important to remember that it is very rare for geopolitics to impact global macro-markets in a significant and long-lasting way. Most of the time, it is a case of fading the news, with the main analytical effort best spent on just how quickly this can be safely done. In this chapter, we establish frameworks for trading three different types of (geo)political events. We start with old-style military confrontations and follow with the newer style of geopolitical conflict, which uses tariffs as its main tool. We end the chapter by analyzing how to trade elections.

4.1 GEOPOLITICAL RISK THE BAD OLD DAYS

Geopolitical risk has become a catch-all term for a variety of events in recent years. How we use the term and how it is commonly understood is that it describes risks arising from (or related to) competition and posturing by sovereign states. The most obvious of these are military conflicts, but they can also include acts of terrorism (especially if undertaken by state-sponsored proxies), cyberattacks, and even trade policy actions such as sanctions and tariffs (which we will discuss later). From an investment perspective, we focus on the impacts of such events on the global economy and financial markets.

The economic transmission to the real economy tends to occur via trade through the current account, but often more importantly through the capital account of the balance of payments. Citi's economists (Sheets et al., 2024) view geopolitical pressures as a supply shock, being associated with slower growth and, at least initially, higher inflation and rising commodity prices. The scope of a geopolitical event is then determined by the nature, linkages, geographic scope, and novelty. Our work on geopolitical risk is derived from a measure developed in Caldara and Iacoviello (2022). The geopolitical risk index (GPR) the researchers calculate is based on the share of (Western-centric) news articles discussing geopolitical risk. They use the frequency of broad terms and combinations of words, such as war, threat, peace, crisis, or invasion, to measure risk, normalized by the total number of articles in the sample. Figure 4.1 shows the 30-day moving average of the index with some of the large spikes annotated.

Interestingly, spikes in the GPR index quickly mean-revert. To investigate the impact of GPR spikes on the SPX, Figure 4.2 plots the 12-month change in the log of the GPR index next to the detrended SPX. There clearly is a correlation between the SPX and the major geopolitical shocks (Gulf War, 9/11, Russian invasion), but there is not much of a reaction when it comes to the smaller shocks. Furthermore, the SPX recovers quickly and, at times, slightly leads the GPR index.

Zooming in on the more significant shocks, Figure 4.3 shows the behavior of equities into and out of the largest stress events, where day zero is a spike in the GPR index, defined as either a 100% jump in a month or to the 99th percentile level. The peak of the index typically occurs within a few days, although for September 11 and in the 1990 Gulf War, the peak was 14 and 20 days after the initial trigger, respectively. The more detailed analysis confirms our earlier impression. At times, markets may retreat going into large geopolitical episodes, but typically, the impact of even the large events is short-lived. Even after 9/11, the largest spike in the GPR index by far, the market (after having been closed for a week), sold off for only 5 days (and 10%), bottomed, and then raced higher. It made it back to pre-9/11 levels by 10/11. Following the invasion of Ukraine, markets recovered initially quickly too, then retreated again, but this subsequent drop likely had less to do with geopolitics and more to do with recession risks (we will cover the link between oil price spikes and recessions later in this chapter).

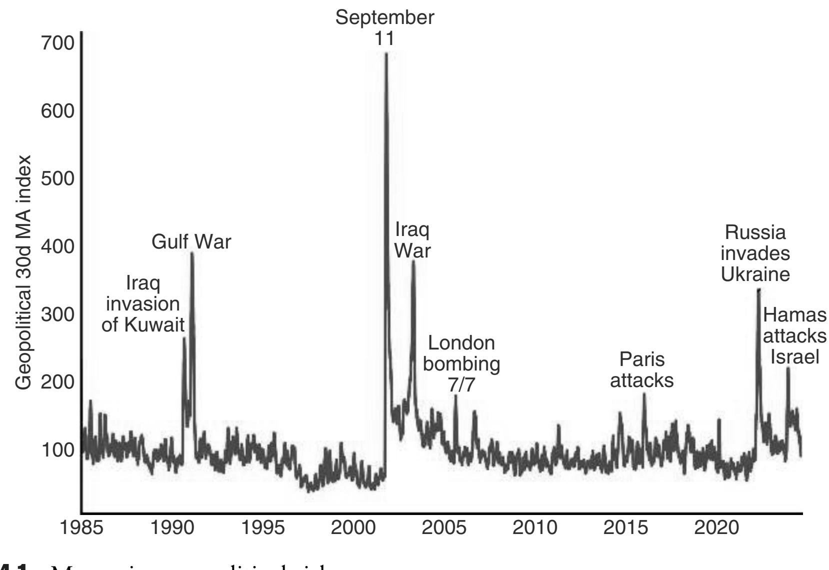


FIGURE 4.1 Measuring geopolitical risk.
Source: Matteo Iacoviello website and authors' calculations.

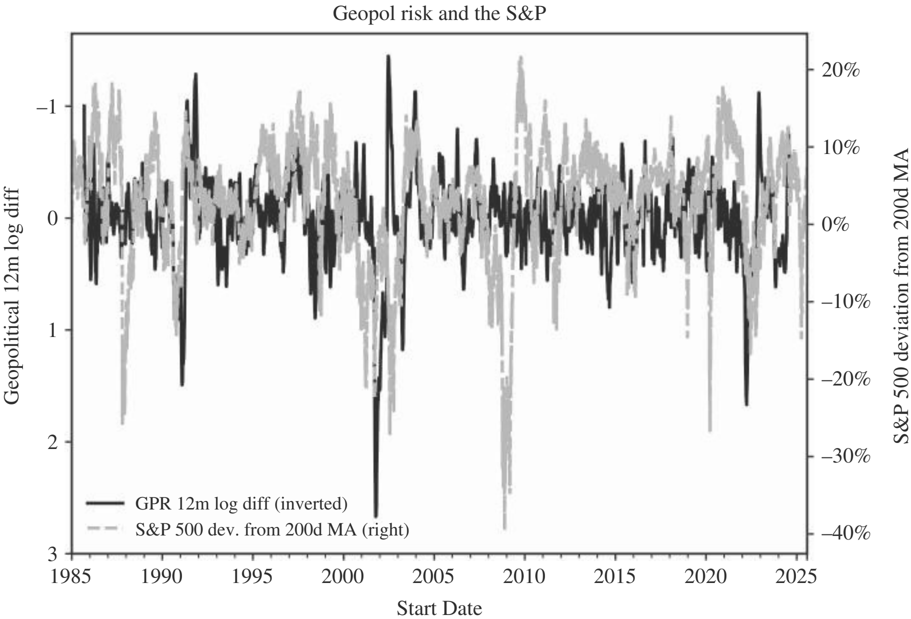


FIGURE 4.2 S&P falls as geopolitical risk rises, but not for long.
Source: Matteo Iacoviello website and authors' calculations.

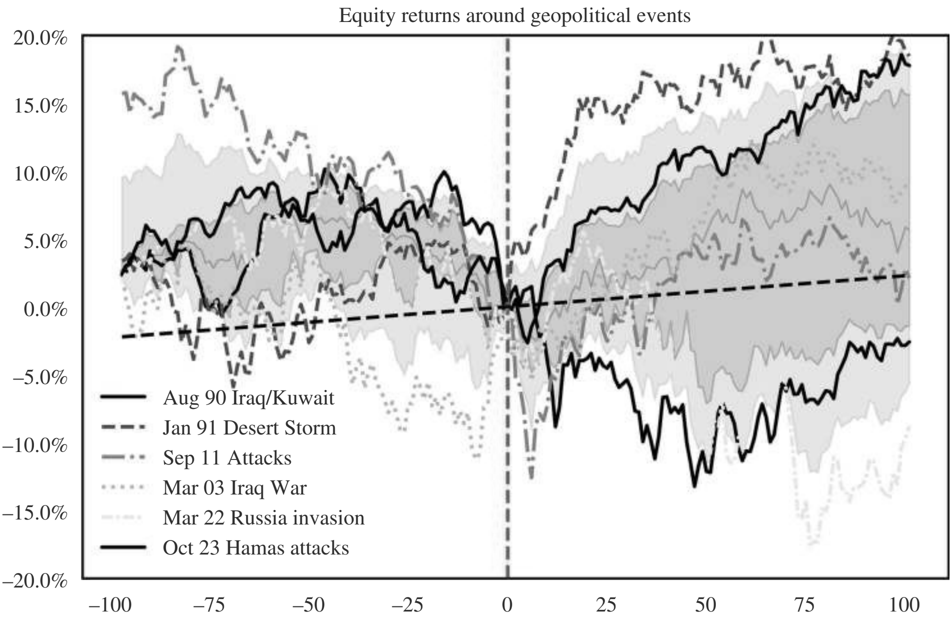


FIGURE 4.3 Markets usually shrug off even large geopolitical shocks relatively quickly.
Source: Matteo Iacoviello website and authors' calculations.

4.2 FROM CONFLICT TO BARREL

Interestingly, most of the large spikes in the GPR index have an oil angle. 9/11 was the only exception. Although 9/11 eventually led to increased conflicts in the Middle East and higher oil prices, the first reaction was lower oil prices, given the substantial hit to US growth and oil demand. Still, oil is typically the key mechanism by which geopolitical tensions can impact economic variables in the United States and the global economy. Without an impact on oil, global markets typically do not pay attention for long. Why would they, given that the earnings of the vast majority of companies are unlikely to be impacted? We therefore fade geopolitical events quickly if there is no oil impact.

However, even conflicts that lead to a spike in oil are quite quickly pushed to the side by markets. One reason for this is that oil spikes also often do not last particularly long. Table 4.1 shows the relevant geopolitical events and how long oil spikes lasted. The median duration of the oil rally after the start of the event was 38 days and was sometimes extremely short. For example, Russia attacked Ukraine on February 24, 2022. Spot oil (we use WTI for consistency across events) shot up by a bit more than 40%, and then peaked on March 8, 2022, just 12 calendar days (and seven trading days) later. The SPX put in a short-term bottom at that time, in line with the

TABLE 4.1 Geopolitically Driven Oil Spikes Are Sharp and Short-lived

Event	Start date	Oil peak move	Oil peak days	Oil 3m change
Israel Arab War / Oil Embargo	10-16-1973	134.6%	77	134.6%
Iran Hostage Crisis	11-04-1979	27.6%	89	19.4%
United States bombing Libya	04-15-1986	67.1%	34	–0.3%
Iraq invasion of Kuwait	08-2-1990	149.1%	70	43.2%
Gulf War	01-16-1991	29.6%	0	–34.6%
Kosovo Bombing	03-24-1999	58.2%	41	20.3%
Iraq War	03-20-2003	11.4%	–22	7.0%
Libya Intervention	03-19-2011	34.9%	44	–7.7%
Crimea	02-20-2014	11.5%	11	1.8%
Russia/Ukraine	02-24-2022	42.8%	12	23.9%
Mean		56.7%	36	20.7%
Median		38.8%	38	13.2%

Source: Bloomberg, Matteo Iacoviello website, and authors' calculations.

oil peak, but subsequently retraced to a new low, as oil made a new run for the highs (but failed on June 8th of 2022—a bit more than 3.5 months after the initial attack). At the time, Russia was roughly on par with Saudi Arabia as the second-largest oil producer after the United States. Interestingly, fears of taking out such a significant oil supplier for an unpredictable length of time only pushed prices up for seven days, at least initially. If oil spikes are short-lived, then clearly the impact of geopolitics on global macro-markets will also be short-lived. Three months after the events, oil had typically retraced significantly from the highs set in the direct aftermath of the event. At times, it was also lower than on the day before the crisis broke, largely because the market already had a meaningful risk premium priced in going into the events.

The more general point is that oil spikes are not necessarily a negative for macro-markets. We have studied episodes where oil goes up by 30% or more within three months. We use WTI here, but the results hold for Brent. For US equities, we find that they go up into day 0 (the day when the 30% mark is reached), and then continue to move higher in the median case, as can be seen in Figure 4.4.

We note that bonds, on the other hand, typically underperform their unconditional returns when oil is rising fast. Figure 4.5 shows that to the left of day zero, when oil is rising 30% over three months, bonds go mostly sideways to down. After day zero, bonds perform again in line with their unconditional returns, which is in line with the fact that oil, on average, trades sideways after the spikes we study.

Of course, the just discussed study does not distinguish between supply and demand shocks that drive oil spikes. It could be argued that only supply-driven oil spikes should hit equities, since very strong demand leading to much higher oil prices

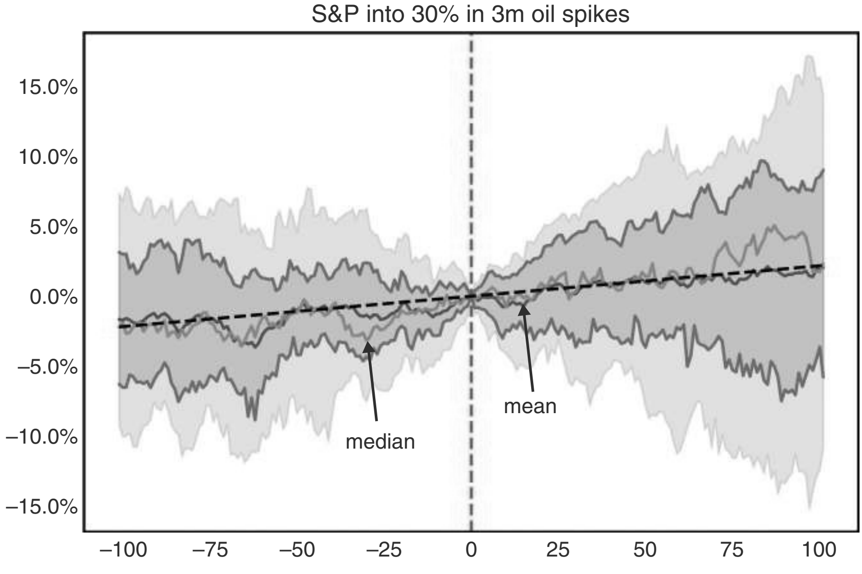


FIGURE 4.4 No major impact of oil spikes on SPX.
Source: Bloomberg and authors' calculations.

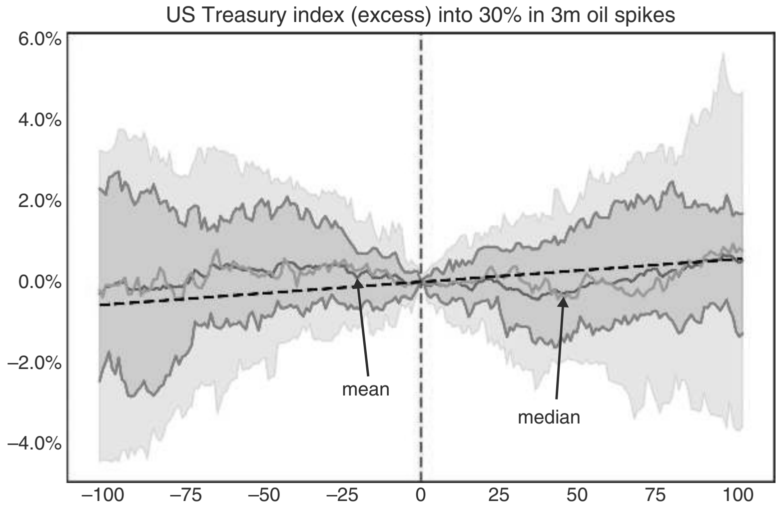


FIGURE 4.5 Bonds underperform during oil spikes.
Source: Bloomberg and authors' calculations.

is likely positive for growth-sensitive assets like equities. The main reason for not distinguishing between supply and demand shocks is that it is surprisingly hard to do. One model that has been proposed (see Groen et al., 2013) is to treat oil rallies on days when risky assets such as equities or cyclical sectors fall as supply-driven oil spikes, while oil rallies on days when these sectors rise as demand-driven. However, this segregation does not allow for a study of how asset markets behave during supply and demand shocks, as it ex ante assumes that risky assets do poorly during supply shocks.

Still, this methodology can be used to study the impact on other assets that are not directly related to risk aversion, such as FX. For example, Figure 4.6 shows the impact on FX during the 30% oil spikes, during which the SPX traded lower. We find 10 such episodes since 2000. Interestingly, even during oil spikes that are more supply-driven in this methodology, as indicated by lower equities, most oily and cyclical assets benefit. Most commodity currencies do well, especially in EM space, as does EUR and EUR proxies. JPY and CHF weaken, as they are famously countercyclical FX (and also large oil importers). Other oil-importing Asian nations are also trading somewhat poorly: TWD, KRW, and THB round out the currencies with negative returns during those periods. Currencies of other large oil importers, like India or China, are hovering around the zero line. Interestingly, they do not sell off much in spite of the negative terms of trade shock. Overall, we are not convinced that a methodology that uses equity prices to determine the nature of the oil shock is all that helpful.

One way out may be to combine big oil moves with big GPR moves. If the GPR spikes with oil, the oil spike is likely mostly supply-driven. We implement this study by analyzing oil moves of more than 15%, combined with GPR spikes to the 95th percentile. We find some SPX weakness going into these events, but Figure 4.7 shows that the SPX is not overly impacted by the event itself or its aftermath.

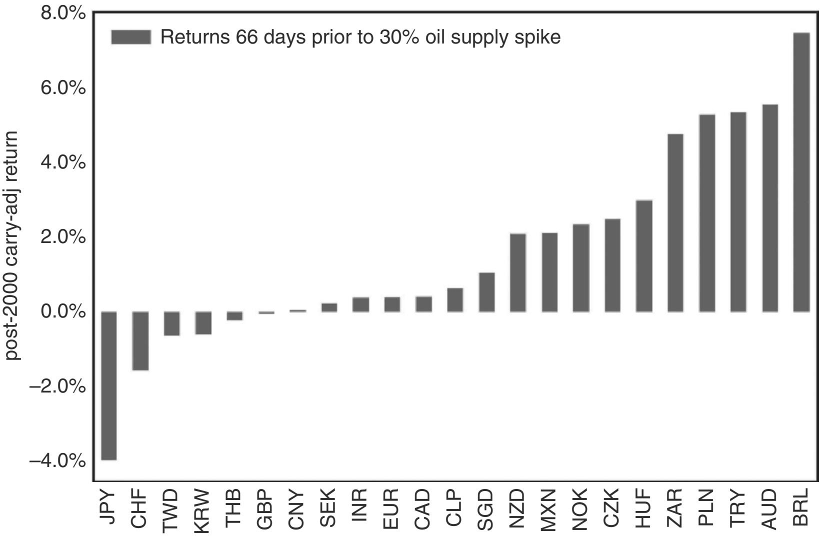


FIGURE 4.6 Cyclical FX strengthens even during "supply shock" driven oil price spikes.
Source: Bloomberg and authors' calculations.

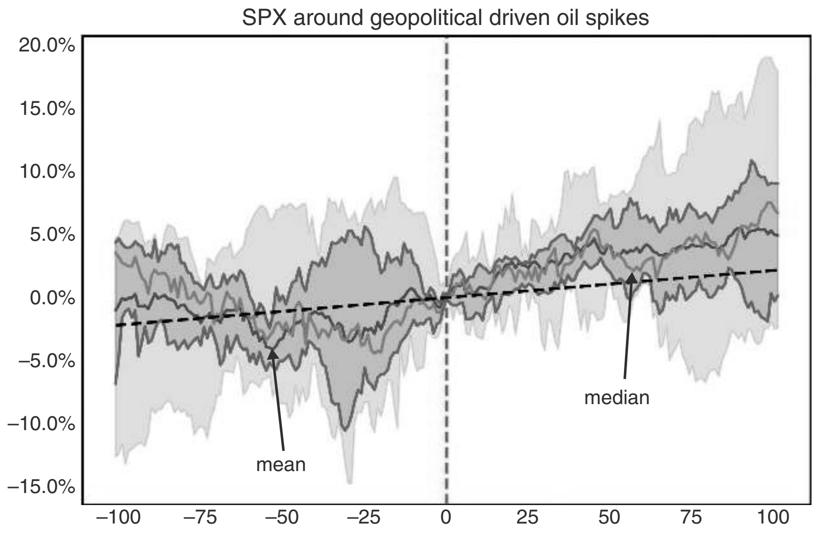


FIGURE 4.7 Geopolitically driven oil spikes not overly negative for equities, either.
Source: Bloomberg, Matteo Iacoviello website, and authors' calculations.

Interpreting geopolitical events through the oil lens will likely remain important going forward. However, semiconductors are the oil of the twenty-first century. Certainly, any potential conflict involving Taiwan, threatening the supply of chips, would also qualify as a situation where markets should be significantly impacted, more in line with the more distant oil crisis of the past, when oil still mattered more to the US economy, rather than the more benign recent examples.

Crude Consequences

A related question is whether crude spikes cause recessions. Although oil and rates are typically positively correlated, that is, higher oil coincides with higher rates, too large an oil spike can be recessionary, leading to lower yields. The question is how useful oil spikes are for forecasting recessions and what magnitude of spike makes a recession more likely. For our analysis, we define an oil spike as a 75% move in year-over-year terms, that holds for at least five days. It turns out that the empirical record is mixed. Even such large oil spikes do not necessarily cause recessions. This can be seen in Figure 4.8, which illustrates that oil spikes have preceded or coincided with five of the last six recessions (COVID excluded). But there were many false alarms. One common oil spike happens when the United States is *exiting* a recession, as oil can rise strongly from depressed levels. But even excluding oil spikes in the direct aftermath of a recession, the hit ratio of oil spikes is not that high. The lags are also not overly stable. In 1990 and 2007, the spike occurred when the recession had already started, while in 2001, the oil spike had occurred well ahead of the recession and may only have been a minor contributor. Overall, we conclude that oil spikes causing recessions is somewhat of an urban myth.

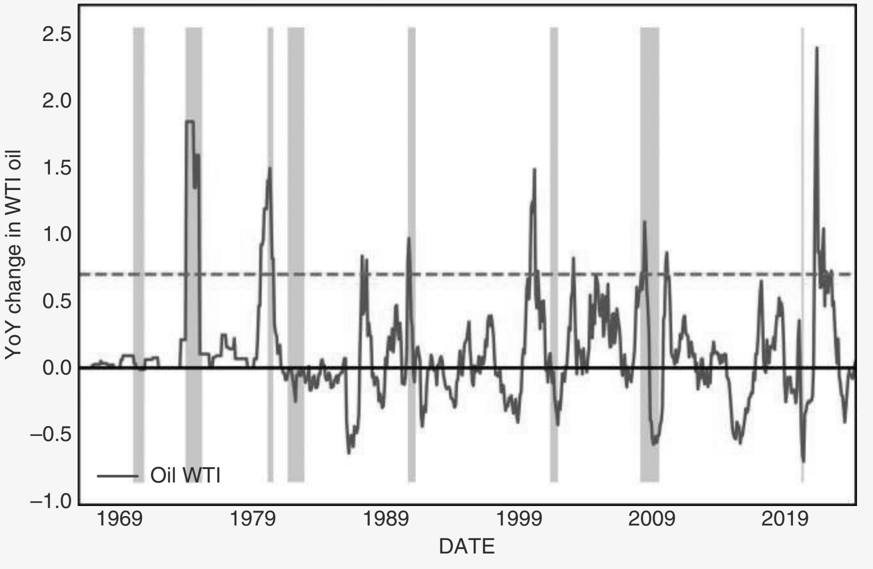
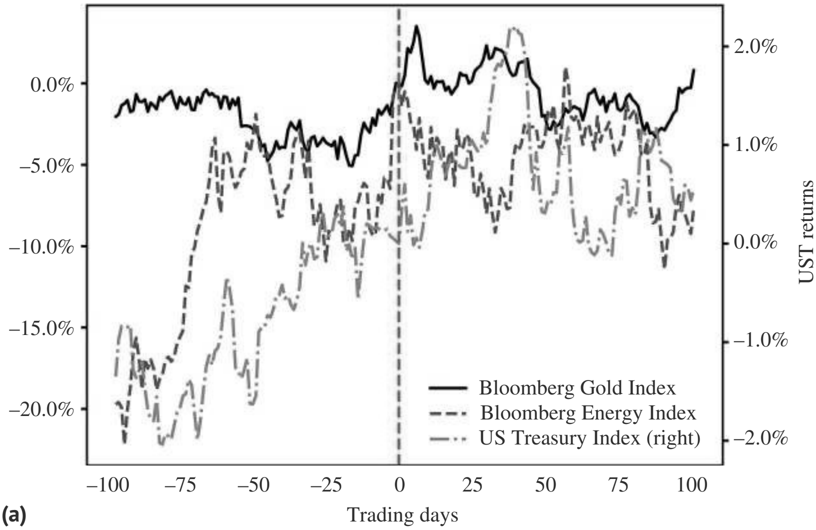


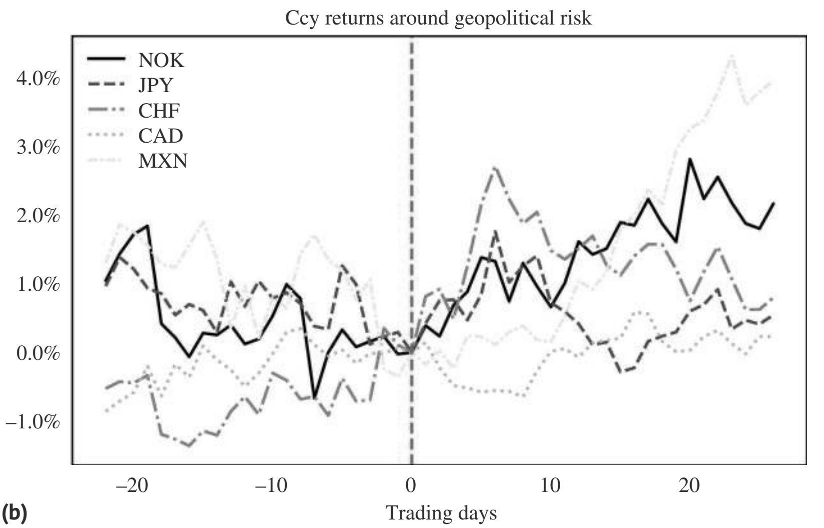
FIGURE 4.8 Oil spikes not great at forecasting recessions.
Source: Bloomberg and authors' calculations.

4.2.1 Hedging Conflict

Next, we study potential hedges for geopolitical risks. In Figure 4.9a, we show the median returns of various assets that could plausibly work as hedges going into and out of the large spikes of the GPR index. As markets anticipate newspaper headlines, we focus mostly on the left side of the zero line to investigate hedges. Given that the transmission mechanism to the real economy is often via oil prices, it is not surprising



(a)



(b)

FIGURE 4.9 (a) Best hedge for geopolitical risk is oil. (b) FX, even safe-haven currencies, are not a great hedge.
Source: Bloomberg and authors' calculations.

that energy works well as a hedge in the run-up and on the trigger day. Energy moves up by 10% over the 10 days prior to day zero. Energy then peaks out on the GPR spike and retraces. Gold similarly trades well in the 10 to 20 trading days into the GPR spike, but with less upside. Gold then continues to rally for a few days longer. US bonds mostly trade sideways in the few days ahead of the GPR spike, move up for a day or two, and then come back down. However, while bonds may not do overly well right around the GPR spike, in the bigger picture, US bonds do quite well going into day zero and for the next 50 trading days.

We also investigated the traditional FX safe-havens, as well as oily currencies, in Figure 4.9b. An upward line here means a weaker USD. We do not find a strong case for using JPY as a hedge, which is largely trading in a volatile range, probably because Japan is a major oil importer. JPY does appreciate, though, a few days after the GPR spike. CHF works better, trading well both into the event and in its aftermath, but the size of the move is relatively small. Oil exporters Norway and Mexico trade well after the event, but not necessarily into it. Overall, FX does not seem like the go-to hedge for geopolitical shocks.

4.3 WARFARE BY TARIFF

More recently, economic warfare has gained market share from military warfare. After having been mostly on a long downward trend since the 1930s, the average tariff of the United States has started to move up after the 2016 election of President Trump. During Trump 2.0, tariffs have made a strong comeback as tools for geopolitics. Sanctions have also gained share from military warfare. Tariffs are somewhat difficult to study, as the number of announcements is high and follow-through is not always guaranteed, especially during Trump 2.0. Therefore, we focus on the 11 tariff dates identified by Amiti et al. (2021) during 2018 and 2019, when Trump imposed tariffs and when reversals were less common. The broad finding is that equities did poorly on event day and in the 10 days afterward, with Chinese equities, which were targeted most frequently, performing the worst, while the United States was outperforming, albeit in a negative context. Interestingly, bonds did well, including in the United States, as markets did not focus on the potentially inflationary consequences, but were very much worried about a negative growth impulse instead. These growth fears also explain weaker base metals. Gold was not a great hedge, other than for the event day itself. The USD mostly strengthened against CNY, the main tariff target, as is illustrated in Table 4.2.

However, the tariffs during Trump 2.0 have not played out in the same way, as per Table 4.3. Given that there were countless announcements and much back and forth, we focus on the returns into and out of liberation day, April 2, 2025, when reciprocal tariffs were announced. This episode was also equity negative, but this time around, US equities were, in the big picture, trading in line with European equities, not clearly outperforming. US bonds were again trading well, but only for a few days, before selling off hard, which was much less the case for Bunds, leading to strong underperformance of UST. Furthermore, the USD weakened, as is illustrated in Table 4.2. This is almost the opposite reaction to the Trump 1.0 episode. The reason is likely that the

TABLE 4.2 During Trump 1.0, Tariffs Meant Weaker Equities, Lower Yields Higher USDCNY

	–10d	–5d	–1d	0d	1d	5d	10d	22d
S&P 500	–0.7%	–0.7%	–2.0%	–0.9%	–0.8%	–1.0%	–1.9%	–0.4%
Euro Stoxx 50 Future	–0.5%	–1.1%	–1.5%	–0.9%	–1.9%	–1.0%	–1.9%	–1.1%
Hang Seng (HK) Future	–1.7%	–3.7%	–1.6%	–1.0%	–1.7%	–3.2%	–5.0%	–4.0%
US 2Y Bond	0.0%	–0.0%	0.1%	0.0%	0.1%	0.0%	0.1%	0.0%
US 7–10Y bond	0.5%	0.1%	0.4%	0.3%	0.3%	0.5%	0.7%	0.7%
Bund (Ger) Futures	0.4%	0.6%	0.2%	0.2%	0.3%	0.5%	0.8%	1.2%
DXY	0.1%	0.2%	–0.1%	–0.1%	–0.1%	–0.1%	0.2%	0.2%
CNY FX Carry adj	–0.4%	–0.6%	–0.5%	–0.3%	–0.5%	–1.0%	–0.9%	–0.5%
EUR FX Carry adj	–0.7%	–0.2%	0.1%	0.1%	0.2%	0.1%	–0.4%	–0.3%
BCOM Energy	0.9%	–0.7%	–0.1%	–0.6%	–0.3%	0.6%	–0.1%	–1.8%
BCOM In. metals	–2.4%	–2.9%	–1.4%	–0.9%	–1.0%	–1.2%	–1.1%	–2.2%
BCOM Gold	–0.1%	–0.3%	0.3%	0.2%	0.7%	–0.1%	–0.4%	–0.2%

Source: Bloomberg and authors' calculations.

United States starting a trade war with just one country, China, under Trump 1.0 impacted China significantly, while the domestic US impact was relatively low. However, when the United States started a trade war with the rest of the world, it impacted its own economy more than the economies of its trading partners. At the time of writing, the jury is still out on how the trade war will play out. So far, the lesson is that investors should see such action initially as equity negative but not remain negative for too long, just as with the other geopolitical shocks. Furthermore, investors should not be overly dogmatic about which country is likely to outperform.

4.4 VOLATILITY AT THE BALLOT BOX

Moving from geopolitics to politics, we address elections next. Conceptually, there are two ways that elections may matter. The most obvious one is in case elections have two or more candidates with very different policy platforms, either concerning economic policies or concerning structural/industrial policies. The outcome must have an impact on macro assets, or at least on equity sectors. Of course, this is only the case if the election is close enough that the outcome is highly uncertain. The second way in which elections may matter is that, going into an election, the incumbent government may find ways to boost the economy, usually with fiscal policy but sometimes also with monetary policy. This can also impact macro assets, even if the election offers two or more candidates with a relatively similar policy outlook. These two cases are very

TABLE 4.3 Liberation Day Also an Equity Negative, But Short-lived, and with the Opposite Impact on the USD

	–10d	–5d	–1d	0d	1d	5d	10d	22d
S&P 500	0.9%	–1.9%	1.0%	0.7%	–4.2%	–3.2%	–6.5%	0.7%
Euro Stoxx 50 Future	–3.7%	–3.4%	0.9%	–0.5%	–4.1%	–13.4%	–7.0%	–0.9%
Hang Seng (HK) Future	–6.8%	–0.9%	0.2%	–0.2%	–2.0%	–13.3%	–8.9%	–3.5%
US 2Y Bond	0.2%	0.1%	–0.0%	–0.1%	0.3%	–0.0%	0.1%	0.0%
US 7–10Y bond	0.8%	0.9%	0.2%	–0.1%	0.9%	–0.9%	–0.6%	–0.8%
Bund (Ger) Futures	0.8%	0.7%	0.2%	–0.2%	0.4%	0.9%	1.4%	1.0%
DXY	0.5%	–0.4%	–0.4%	–0.4%	–2.1%	–1.3%	–4.7%	–4.1%
CNY FX Carry adj	–0.8%	–0.2%	–0.2%	0.0%	–0.2%	–1.1%	–0.6%	–0.4%
EUR FX Carry adj	–1.1%	0.4%	0.3%	0.5%	2.4%	1.3%	5.3%	4.1%
BCOM Energy	3.9%	3.8%	–0.1%	1.3%	–2.4%	–9.4%	–12.5%	–13.3%
BCOM In. metals	–3.2%	–4.0%	–0.9%	–0.6%	–3.3%	–12.5%	–6.2%	–6.5%
BCOM Gold	3.0%	3.6%	0.5%	0.6%	–0.8%	–2.2%	6.2%	2.7%

Source: Bloomberg and authors' calculations.

different in terms of trading strategy. In the former case, it can be expected that some risk is priced into the market going into the election. If a sufficiently large risk premium has been put into asset prices going into the election, assets typically start to recover shortly before the election day, which continues after the election. Implied volatility quickly returns to normal. In many cases, this is a tradable pattern, which we also highlighted in the EM book. In the second case, the election date itself is not so crucial, but the year before the election can benefit from higher growth and potentially stronger equity markets. This is harder to trade directly, but it is an important consideration for the macro view during election years.

Given the dominance of the US economic cycle, the US political cycle also deserves special attention. We therefore focus on the United States analyzing elections post-World War II, starting in 1952. As is the case for most developed markets, in the United States, the economic platforms of the candidates are generally not that different, at least when compared to emerging markets. In the economic area, there are typically disagreements on levels and distribution of both taxes and budget spending, but the overall economic system is rarely questioned. As such, on a macro level, there should be much less uncertainty than is the case in emerging markets, even though on a micro level, the choice can be highly consequential.

Given this set-up, we start by analyzing the policy impacts for the year going into the election. On the monetary policy side, there is little evidence that the Fed is trying to goose the economy on average. This is illustrated in Figure 4.10a, where we find that the median case for monetary policy does *not* include rate cuts going into the election. However, rates tend to move up after the election, suggesting a bias for policymakers

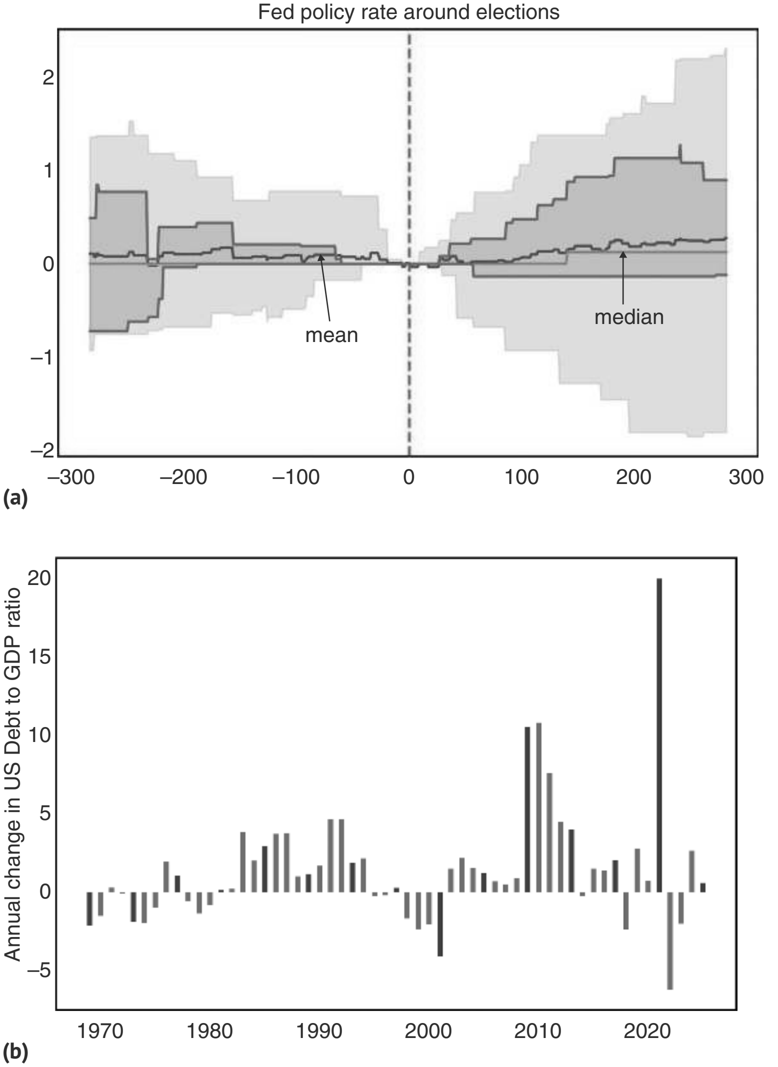


FIGURE 4.10 (a) The Fed typically does not goose the economy into elections. (b) Debt/GDP increased in two notable election years, but no broad pattern.
Source: Bloomberg and authors' calculations.

to delay rate hikes until after the election. On the fiscal policy side, Figure 4.10b shows the behavior of budget deficits into elections from 1968 onward. There are two notable recent election years, 2008 and 2016, where deficits ballooned, although both these cases were driven by an external shock: the GFC and the COVID pandemic, respectively. Both these elections also featured Republican incumbents losing to Democratic challengers. However, when considering the median case, the increase in the deficit to GDP ratio in election years (1.18%) is very similar to the long-term median of 1.16%. Moving to asset prices, there is a relatively consistent election cycle playing out in the US equity market. Election years are strong years for the SPX. The weak years are the years of midterm elections, not the years of presidential elections. This is illustrated in Figure 4.11. Mid-term election years only have a 50–50 chance of being positive, compared to 94% for pre-election years, as is shown in Table 4.4.

Going into the election, the worst two-month performance occurs when the market is sniffing out that a Democratic challenger may win. But the pullback is small, around 2.5% in two months, with a trough just a few days ahead of the election. In the first days after the election, the market has the strongest positive reaction to a Republican challenger winning, probably because incumbents winning is a lesser surprise, not creating much price action either way. However, a victory of Democratic challengers usually also leads to upside, partly as the market priced in a risk premium ahead of time. In the bigger picture, all of these effects are short-lived. One hundred days after the election, markets under Democratic challengers have done better than under Republican ones. Given a relatively small sample size (19 elections from 1952), subdividing this already limited sample means that a lot of the performance just depends

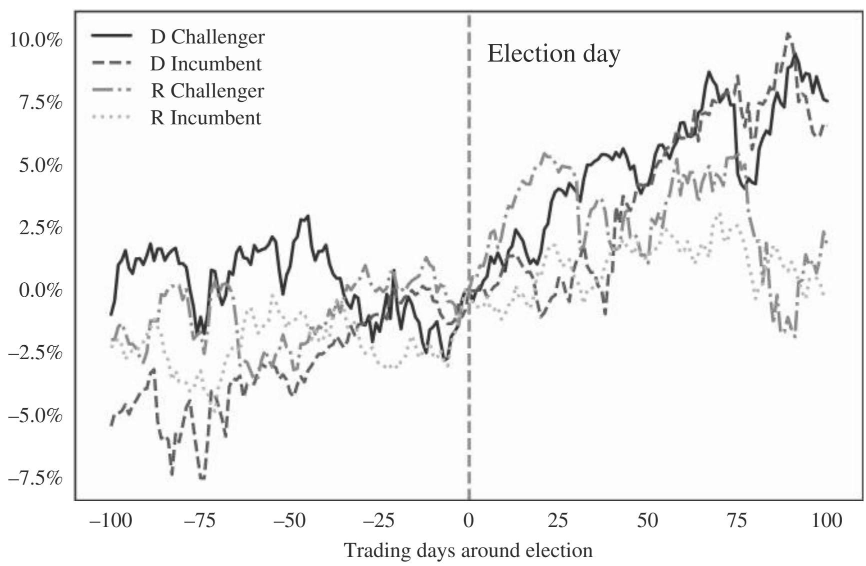


FIGURE 4.11 Short-term response to election best with Republican challenger winning.
Source: Bloomberg and authors' calculations.

TABLE 4.4 Beware the Mid-terms

	Median ret	1952 on hit ratio	Median ret	1972 on hit ratio
Election year	11.7%	78.9%	12.6%	78.6%
Post election year	9.1%	61.1%	18.9%	69.2%
Mid-term year	–1.5%	50.0%	0.8%	53.8%
Pre-election year	18.7%	94.4%	20.0%	92.3%

Source: Bloomberg and authors' calculations.

on how lucky or unlucky a given presidency is with respect to being hit by positive or negative macro shocks.

The other macro asset where the presidential election cycle matters is bonds. Here, it is less about who wins and more but about whether the election results in a unified government, where one party controls both chambers of Congress as well as the White House. In particular, US Treasuries underperform into elections that result in a unified government and continue to perform poorly in the aftermath of such elections (though with a smaller differential compared to divided governments). These differences become even more pronounced after the first 150 days in office, as budgets and reconciliations are made and spending and taxation preferences are revealed in more detail. This is illustrated in Figure 4.12a. The overall pattern is not overly surprising, given that unified governments have a much higher risk of larger fiscal expenditures, regardless of which party is in office.

The patterns in the USD are more muted, as per Figure 4.12b, but roughly in line with rates. The USD strengthens into elections, resulting in a unified government, which continues for a few days after the election. But then that pattern fades. Some of this may be due to the reversal of USD strength that occurred earlier due to elevated risk premia from recessions. Both 1992 and 2008 saw Democratic unified victories at a time of heightened uncertainty and recession, which subsequently unwound.

While all these patterns should be kept in mind for macro-trading, the stylized facts for the USD are by themselves not strong enough to justify a position, but are just one piece of the puzzle. The overall view will depend on the proposed policies in a given year. We now turn to shorter time frames, where the election can easily dominate the price action, leading to clearer trading opportunities.

On a shorter time frame, it is all about the release of uncertainty post-election. We find that elections in the United States are not all that different from emerging-market elections. There is evidence that equities pull back, all so slightly, around 40 trading days before the election. Although the down move on average is not overly sharp, the SPX underperforms its unconditional returns. VIX also moves higher into the elections. However, around six trading days before the election, the SPX starts to outperform,

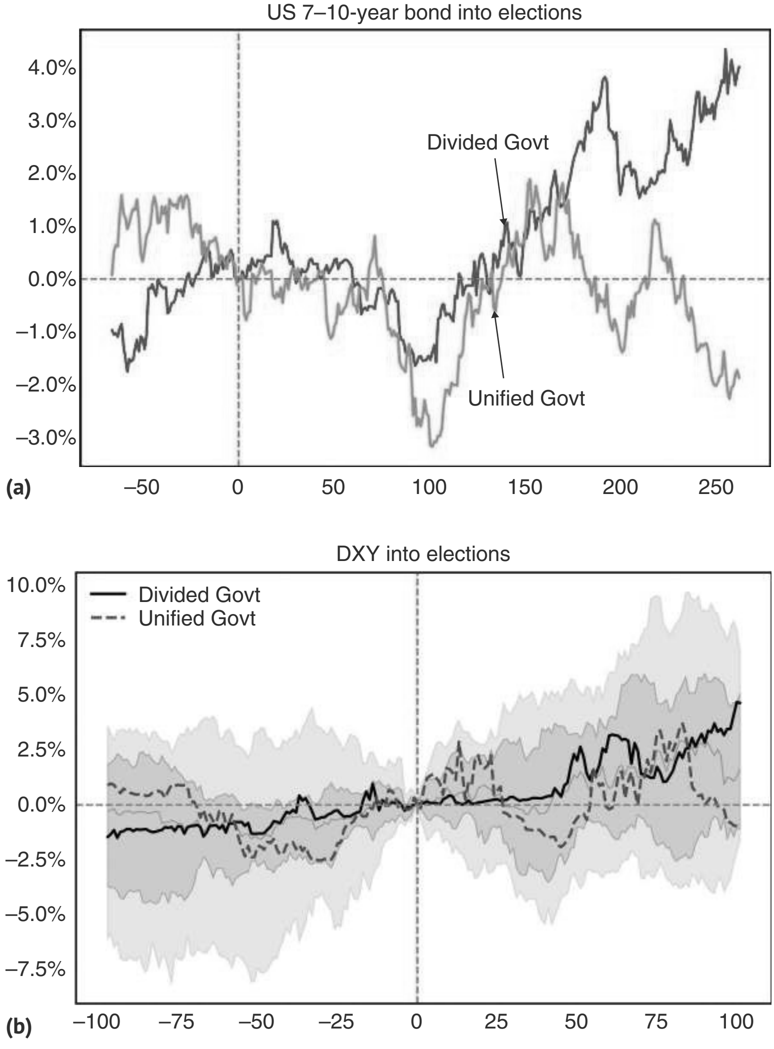


FIGURE 4.12 (a) Bonds sell off into unified govt victories. (b) USD strengthens into elections of unified governments.
Source: Bloomberg and authors' calculations.

which continues after the election, as the risk premium is unwound. On average, the SPX performs very well in the three months following the election, as is illustrated in Figure 4.13.

The pattern is even clearer in volatility, where VIX starts to move higher around two months before the election date, only to peak a few days before, and then fully unwind that move, reaching pre-election levels around two months later, as per Figure 4.14a. VIX has a constant 30-day maturity, and therefore, the pattern is more than just depending on whether the election day is in the option window or not. The pattern is

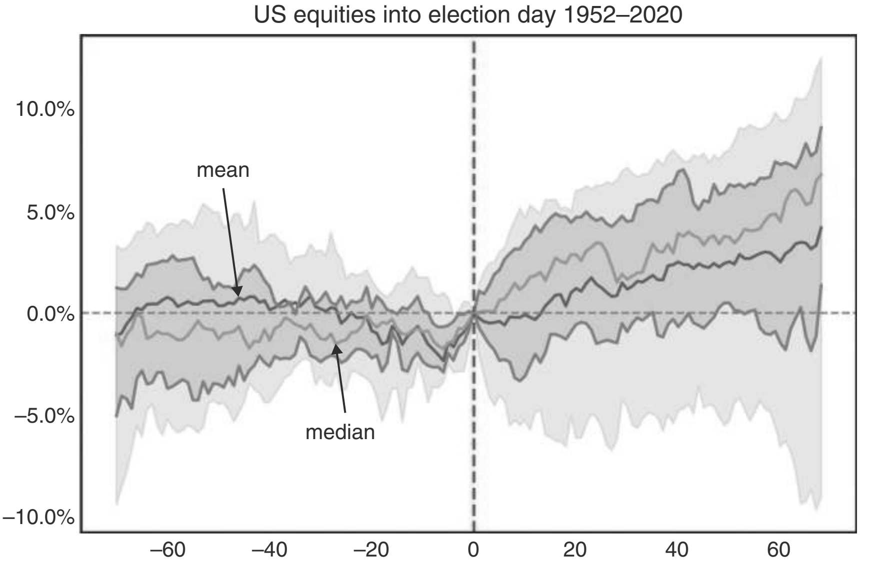


FIGURE 4.13 (Small) equity underperformance reversed the weak before election day.
Source: Bloomberg and authors' calculations.

similar to the one we found to generally hold for emerging markets (see Chapter 5.6 in Willer et al., 2020). Although the outcomes always depend on the candidate in a given year, our rule of thumb is to fade election-driven weakness in equities shortly before the election takes place. Typically, the market has priced in enough risk premium by the time the event rolls around. Furthermore, usually, even if the market-unfriendly candidate wins, policy outcomes are often much more centrist than feared, as newly elected administrations often move to the center, given the various constraints that they face.

There is also an interesting nuance. As VIX moves up into the election, if realized volatility does not go up, this can trigger our volatility risk premium (VRP) buying model. While this model is our favorite buy-the-dip model, it also works when it gets triggered by an event pushing expected volatility higher, even if there is no dip in sight. Our VRP model actually did trigger into the 2024 US election and worked well in this case. Volatility in FX and rates space behaved very similarly to equity volatility, as per Figure 4.14b.

The other, more tactical point to make is that after US elections, there is often follow-through with respect to the price action on election day. Although, importantly, there can be vicious intraday reversals, as was the case for the 2016 Trump election. It pays to wait for the end of the day before using this rule. The fundamental argument is that, typically, investors reduce risk into elections, at least if they are tight. Once there is a resolution of election uncertainty, investors will likely want to put risk back on, especially since November and December, i.e., the post-election months, are typically quite bullish seasonally for equities, as illustrated later in this book. Testing a strategy to go with momentum on election day results in high hit ratios in the most impacted

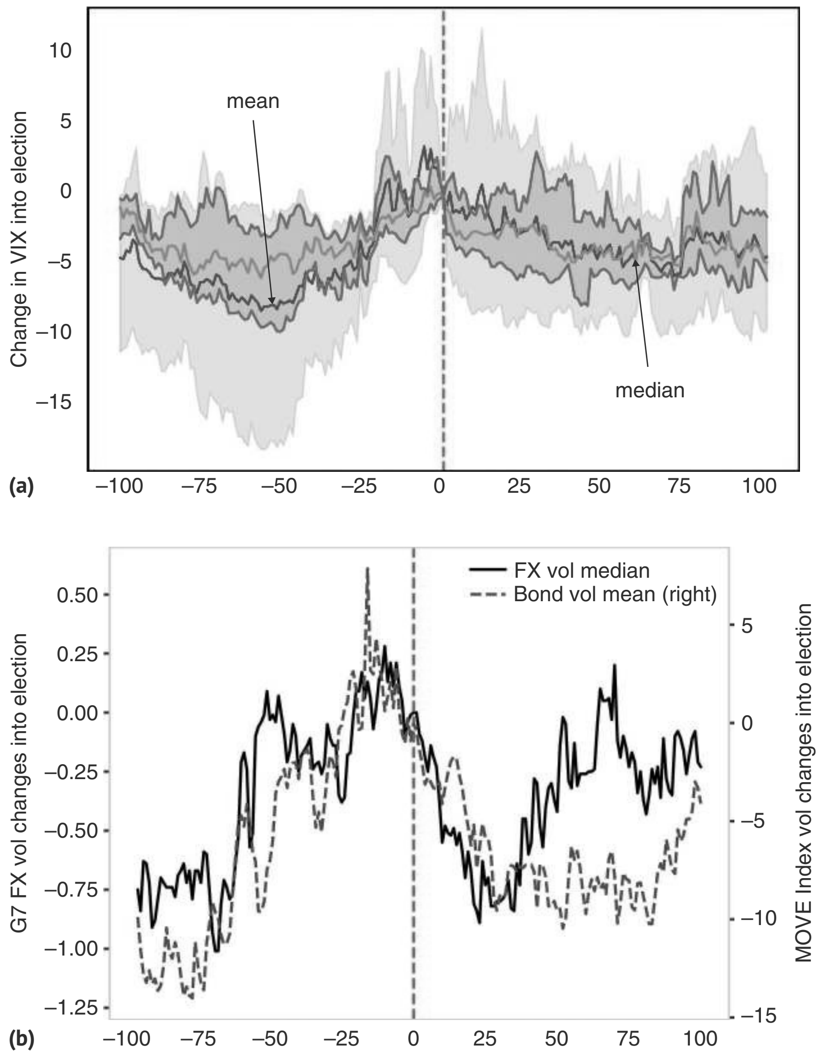


FIGURE 4.14 (a) VIX tends to start rising 50 days before election, then unwinds. (b) FX and bond vol rise 25–50d before election before unwinding.
Source: Bloomberg and authors' calculations.

assets, and especially in DXY (93% hit ratio after 10 days, with a focus on USDJPY), as well as SPX (86% after 10 days, with a focus on small caps). Interestingly, in rates space, we do not find that there is much follow-through in US rates, though there is in Mexican rates (which are closely tied to US rates, but have a risk component too). This is illustrated in Table 4.5.

TABLE 4.5 Chasing Works After US Elections

	1d	5d	10d	22d	Sample size
DXY	86%	93%	93%	64%	14
JPY FX Carry adj	89%	100%	67%	56%	9
S&P 500	93%	86%	86%	79%	14
RTY Index	83%	92%	92%	92%	12
Euro Stoxx 50 Future	86%	86%	86%	57%	7
Topix (Japan) Future	56%	56%	44%	44%	9
Hang Seng (HK) Future	44%	67%	56%	67%	9
10Y US Future	91%	64%	64%	64%	11
Bund (Ger) Futures	67%	67%	67%	67%	9
Mex. 10Y (yield changes)	100%	100%	100%	100%	4
BCOM Gold	77%	77%	77%	85%	13
BCOM Energy	73%	55%	73%	73%	11
BCOM In. metals	89%	78%	78%	89%	9

Source: Bloomberg and authors' calculations.

The 2016 US Election—Trump 1.0

The 2016 US election was one for the ages, where our rules mostly worked, in spite of the quite singular character of the election. Hillary Clinton was leading in the polls, and the clear consensus was that she would carry this election. However, polls were tight enough that the market did put some risk premium into prices, with the SPX pulling back around 4.5% in the two months before the election date, fearing the economic damage of a potential Trump victory. In the event, the SPX troughed only one day before the election, somewhat later than normal. When the surprise Trump win became clear, the SPX first dipped, before rallying strongly, ending the day higher, and then continuing its rally for the next weeks. Enough risk premium had been put in ahead of the event, and perhaps more importantly, the market concluded that Trump policies would be positive, rather than negative, for SPX earnings. Follow-through from the price action of the event day was strong. The VIX also peaked just before the election and then returned to its local lows over the next month. However, for some other assets, it is harder to make the case that a lot of the news was already priced, which would tend to put more of an emphasis on the explanation that the market revised its assessment of the likely impact of the Trump policies rather than on the idea that

(*Continued*)

(*Continued*)
everything was already priced. While US rates had also put some risk premium in, leading up to the election, rates exploded higher on election day and kept moving up for the next 1.5 months. The same was true for USDMXN, which also moved significantly higher, even with some election risk premium already in the price, only peaking two months later. Our rules on follow-through worked well, as per Figure 4.15, but if Trump had been fully priced, USDMXN would have moved lower on election day, not higher. What happened? The most obvious explanation is that most pundits, and arguably the market, got the election result wrong, and therefore, there had to be a strong reaction. Still, how come there was not enough risk premium in the market already, which is the more typical pattern? To our mind, the reason is that President Trump was a truly unknown quantity, coming from outside the political establishment. Therefore, it was arguably the first time in US history that such an unpredictable candidate won the Oval Office. Had US traders been more attuned to how the typical EM election plays out, they may have put a sufficient amount of risk premium into the market prior to the event. Interestingly, on the day Trump won the second time, USDMXN did not budge in spite of very tight polls. The second time around, the market had put enough risk premium into USDMXN. The other point to note is that, in volatility terms, the normal rules were holding more clearly in 2016. For USDMXN, 3m implied volatility came off right on cue, peaking four trading days before the election, though US fixed income implied volatility kept moving higher for another four days, before starting to come off again. This is shown in Figure 4.16. Even with truly unpredictable outcomes, volatility tends to peak shortly before election day.

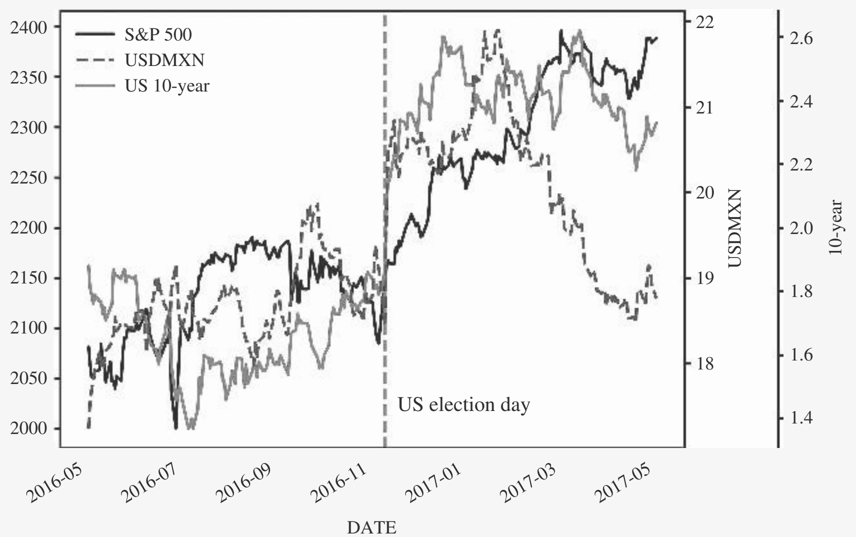


FIGURE 4.15 Strong follow-through on event day in all asset classes for Trump 1.0.
Source: Bloomberg and authors' calculations.

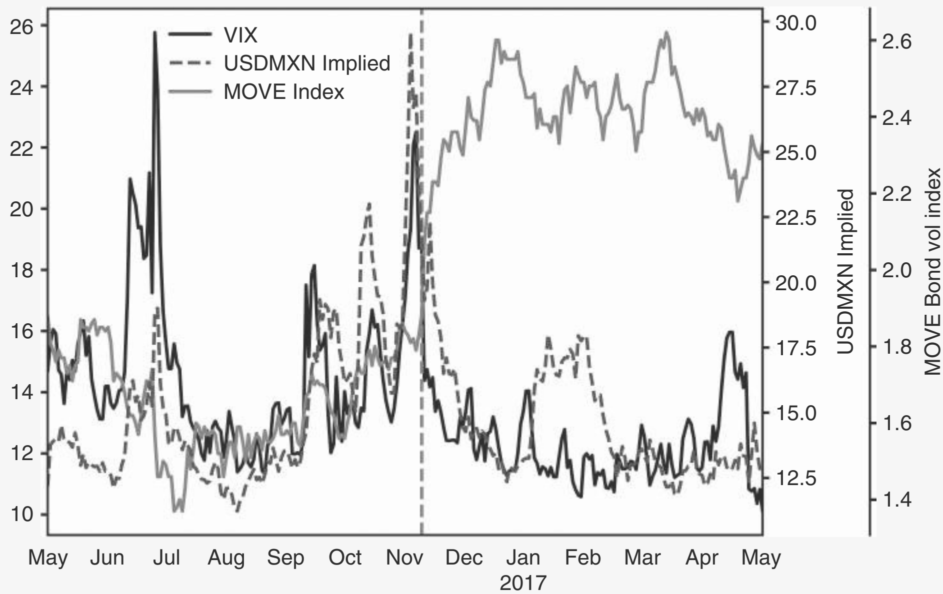


FIGURE 4.16 Volatility peaking around election day.
Source: Bloomberg and authors' calculations.

4.5 SUMMARY

Geopolitical risk, as measured by the GPR index, mean-reverts quickly. Oil is an important transmission mechanism of geopolitics to markets. Geopolitical events without an oil angle should be faded fast. In general, very large oil moves do not have a major impact on equities but do have a negative impact on bonds. However, just analyzing oil spikes where the GPR index also moves higher suggests that there is a slight negative impact on equities too. We find that oil spikes can contribute to recessions, but there were many false alarms too. Unsurprisingly, oil is a good hedge against spikes in the GPR index and is superior to gold or US Treasuries, which also work somewhat. Tariffs have recently become the key source of geopolitical risk, typically leading to equity downside and bond upside, though with respect to FX, a lot depends on how broad such tariff measures are. When trading US elections, in the big picture, election years are strong years for US equities. Bonds dislike unified governments. More tactically, the SPX slightly pulls back into elections and starts to rally around four trading days before the election. The rally usually continues post-election and into year-end. Price momentum on election day often has strong follow-through.

CHAPTER 5

How to Trade Rates: A Framework

Rates trading lies at the heart of macrotrading, and many macro funds were built around risk-taking in rates. There are many flavors of how macro funds engage in rates trading. Some are highly relative value driven, while others are positioning to catch the economic and central bank cycle. We feel that many, though not all, relative value trades still have a directional component and, therefore, focus here on how to get directionality right. This chapter starts with a discussion of long-term returns of US Treasuries, before going into some detail on how to trade the Fed cycle. We start, once more, with a US-centric approach and discuss international fixed income in a subsequent chapter.

5.1 A TREASURE TO BEHOLD

The US bond market has been a success story. Since 1973, when our data starts, the average annualized return of our US Treasury index has been 6.3% (6.1% using geometric returns). There were only eight drawdowns of more than 5% which leads to an attractive information ratio of 1.1. Most of the drawdowns, shown in excess of 3m T-bill, are associated with aggressive hiking cycles by the Fed, as per Table 5.1. Only two were meaningfully above 10%. Both the 1977 to 1981 and the 2020 to 2023 drawdowns coincided with very high inflation, which the Fed had to react to. Without a major inflationary problem, drawdowns have been on the order of 10% or less. In 1987, the Fed hiked to stem inflationary concerns resulting from the falling USD post Plaza accord. In 1994, the Fed surprisingly raised rates by 300bp to 6% leaning against strong growth and a quickly tightening labor market. In 1999, the Fed hiked rates by 175bp to 6.5% to cool an overheating economy, linked to the Nasdaq bubble. The only negative annual return without a Fed hiking cycle was in 2009, after the conclusion of the GFC-driven easing cycle. The sell-offs were typically short-lived, with the 2021–2022 cycle the only time US Treasuries sold off for two years in a row. We would note, though, that in Table 5.1, drawdowns are longer, as we show returns over 3m T-Bills.

The other reason why the US Treasury market has been such a treasure is its correlation structure with other markets. The correlation with risky assets is usually negative, which makes the bond market an interesting hedge, often with positive carry. This explains the popularity of the 60/40 strategy and more broadly of the risk parity concept. The only calendar years where both US Treasuries and the SPX were lower

TABLE 5.1 UST Drawdowns Driven by Fed Hiking Cycles

Peak	Trough	Recovery	Drawdown	Days to trough	Days to recover	FF Chg high from start	CPI peak (%)	CPI Chg start to peak	SPX Chg to trough (%)
1973-02-02	1974-09-27	1976-01-29	–7.4%	602	1092	3.0%	12.2%	8.3%	–46.8%
1977-01-03	1981-09-30	1986-02-27	–25.5%	1731	3343	8.8%	14.6%	9.4%	–11.4%
1986-09-01	1987-10-29	1989-07-28	–9.1%	423	1062	3.9%	5.3%	3.5%	–6.1%
1993-11-01	1994-11-21	1995-10-12	–9.0%	385	711	3.0%	3.1%	0.4%	–3.3%
1998-10-06	2000-01-12	2001-09-13	–10.2%	463	1074	1.2%	3.8%	2.3%	39.4%
2008-12-19	2009-06-10	2010-06-22	–7.3%	173	551	0.0%	2.8%	2.8%	7.1%
2016-07-11	2018-11-08	2019-08-02	–7.4%	850	1118	2.0%	2.9%	2.0%	34.1%
2020-08-05	2023-10-19	2025-03-28	–23.1%	1170	1697	5.2%	9.0%	7.7%	28.1%
Median			–9.0%	532	1083	3.0%	4.5%	3.2%	1.9%
Average			–12.4%	725	1331	3.4%	6.7%	4.6%	5.2%

Source: Bloomberg and authors' calculations

TABLE 5.2 S&P Usually Does Well During US Treasury Drawdown Episodes

Peak	Trough	Recovery	SPX change	SPX change (to trough)
02-02-1973	09-27-1974	01-29-1976	–19.4%	–46.8%
01-03-1977	09-30-1981	02-27-1986	36.1%	–11.4%
09-01-1986	09-30-1987	08-29-1991	30.3%	23.5%
11-01-1993	11-21-1994	10-12-1995	19.6%	–3.3%
10-06-1998	01-12-2000	09-13-2001	–1.1%	39.4%
12-19-2008	06-10-2009	06-22-2010	27.4%	7.1%
07-11-2016	11-08-2018	08-02-2019	39.6%	34.1%
08-05-2020	10-19-2023		55.3%	28.1%

Source: Bloomberg and authors' calculations

on the year in nominal terms are 1994 and 2022, i.e., two very aggressive US hiking cycles, one with and one without inflation. Table 5.2, which lists excess returns, has, on average, very strong SPX returns during UST drawdowns. The only major exceptions were in the highly inflationary 1970s, when equities sold off significantly during the UST drawdown periods. For the full post-COVID drawdown, lasting from August 2020 to October 2023, the SPX was up strongly, though during 2022 equities followed bonds lower given the aggressive hiking cycle. But overall, years of bond drawdowns are typically years when equities work well, cushioning the blow. We will discuss US Treasuries in a portfolio context in more detail in the asset allocation chapter.

We would add, though, that the long-term performance history is not as rosy in real terms. As a nominal asset with a fixed yield, US Treasuries are highly susceptible to unexpected inflation. In ex post real terms, returns were only 3% for an information ratio of 0.54. While this book mostly discusses returns in nominal space, many of our studies are based on excess returns over T-Bills. As T-Bills should react to inflation, or at least to the Fed dealing with inflation, this gets us somewhat closer to the concept of real returns. Excess returns over T-Bills since 1973 were 1.66% (1.46% using geometric returns), for an information ratio of 0.3. Still respectable, but not the same.

The list of drawdowns above has made it clear that the bulk of the analysis needs to go into hiking cycles, especially if those hiking cycles are driven by very high inflation. But before we go into the analysis of the cycle, there is a longer-term point to make on when it may be a good time to buy duration: given that yields are set at issuance, yields predict forward performance to maturity perfectly. Of course, when moving into real space, the picture changes as unexpected inflation spikes can undermine returns. Although unexpected inflation is by its nature unexpected, a cushion in terms of real rates is important to weather unexpected inflation. Figure 5.1 shows that the 10-year ex post real returns (using actual CPI outcomes) strongly depend on the real yields on offer at the time of the investment. This relationship is stronger at very high real returns, which are currently not on offer.

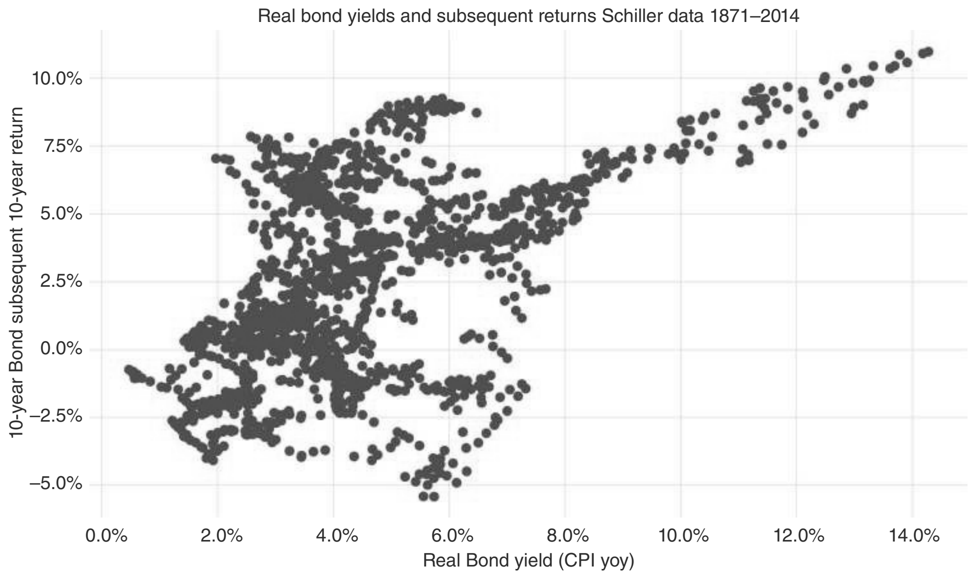


FIGURE 5.1 (Sky) high real yields equal higher real returns.
Source: Bloomberg and authors' calculations.

What constitutes an attractive real return may vary over time. How should we think of fair value for bonds? Before the GFC, there had been a widely held view that the equilibrium 10-year real yield in the United States should be roughly in line with the long-term potential US GDP growth rate. The reason is a very loose arbitrage argument. If rates were much lower, it would, in theory, be possible to lever up and buy the entire income stream of the US economy at a profit. It is, of course, very clear that this arbitrage condition is not exactly airtight, to say the least, but initially, the data fit the theory, and it was therefore widely accepted. After the GFC, rates fell significantly below potential growth, partly driven by QE. The post-COVID sell-off brought rates back in line with potential GDP growth, suggesting that the period of artificially low rates, which often led to subpar returns, is at this stage behind us. This is illustrated in Figure 5.2. With that longer-term perspective under the belt, we move on to analyzing the all-important central bank cycle.

5.2 IT'S THE FED, STUPID

In our business cycle chapter, we highlighted the importance of the business cycle for global macro. For rates trading, it is not so much the business cycle, but more the monetary policy cycle that is the key driver. This is obvious since a longer-term interest rate is just the average of many shorter-term interest rates over time, plus a term premium. Getting the average short-term interest rate right, therefore, means getting the long-term rate mostly right. Although the business cycle and the monetary policy cycle are highly correlated, as central banks aim to smooth out the business cycle, they are

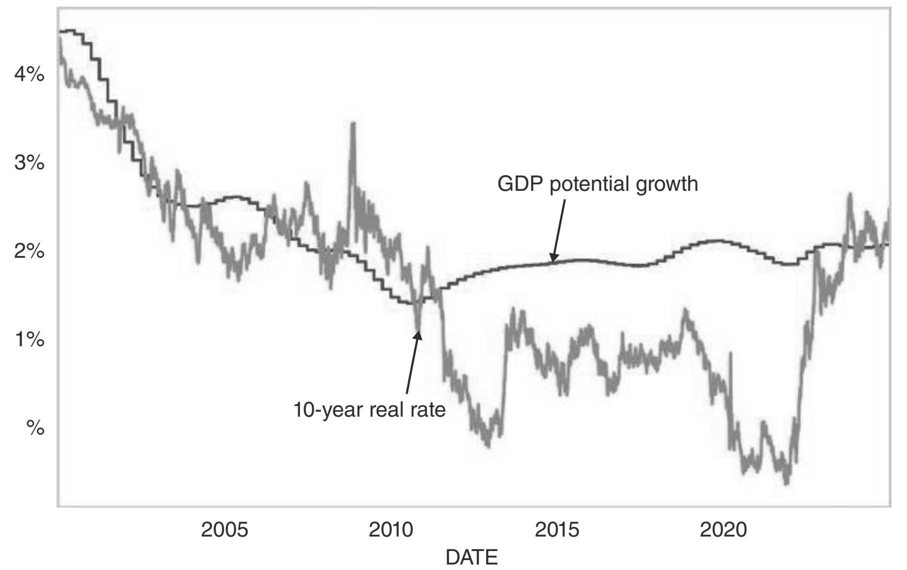


FIGURE 5.2 US rates back to fair.
Source: Bloomberg and authors' calculations.

not the same. For example, a central bank may ease in a soft landing scenario, where rate cuts can coexist with relatively strong growth. We therefore focus most of our energy on the monetary policy cycle. We start with the United States, as the Fed cycles are globally the most important. The reason is that US Treasuries set the benchmark for global rates. This is especially the case for back-ends, whereas front-ends are much more driven by local central bank policies. Of course, back-ends can trade at lower yields than US rates, even for quite risky credits, if the front-end rates in the country in question are low enough. However, *changes* in local back-end yields typically have a very high correlation to changes in US rates, even if central bank behaviors diverge. The Fed rules the global financial system.

We start with some stylized facts about monetary policy cycles and then discuss ways to trade both hiking and easing cycles. Fed cycles are very easy to label post-1986 but are less clear-cut in the prior Fed regime when monetary aggregates were the primary tool to set monetary policy. During that period, we define hiking and easing cycles by looking for months where the policy rate is either higher for tightening (lower for easing) than the previous two months and where the policy rate in the following quarter is higher (lower for easing). The latter condition is necessary to exclude fine-tuning moves. We do not impose any limits on the length of the cycle beyond the quarter look-ahead, in contrast to Forbes et al. (2024), as we want to focus on actual policy moves and their tradable implications. Given that the use of future data is problematic for trading rules, we use the large sample only for descriptive statistics and base our trading rules on the post-1986 data.

Tables 5.3 and 5.4 illustrate that easing cycles tend to be shorter and sharper than tightening cycles, especially in the modern era. Focusing just on the post-1986 cycles,

TABLE 5.3 Hiking Cycles Are Longer and Slower

Cycle	First hike	Last hike	No. hikes	Ave hikes	Total hikes	Hiking months	Pause months
1	12-16-1986	05-20-1987	3	0.29%	0.87%	5	5
2	03-29-1988	02-24-1989	8	0.41%	3.25%	11	3
3	02-04-1994	02-01-1995	7	0.43%	3.00%	12	5
4	03-25-1997	03-25-1997	1	0.25%	0.25%	0	18
5	06-30-1999	05-16-2000	6	0.29%	1.75%	11	8
6	06-30-2004	06-29-2006	17	0.25%	4.25%	24	15
7	12-16-2015	12-19-2018	9	0.25%	2.25%	37	7
8	03-16-2022	07-26-2023	11	0.48%	5.25%	17	14
Mean			8	0.33%	2.61%	15	10
Median			8	0.29%	2.62%	12	8

Source: Bloomberg and authors' calculations

TABLE 5.4 Easing Cycles Are Shorter and Sharper

Cycle	First cut	Last cut	No. cuts	Ave cuts	Total cuts	Easing months	Pause months
1	10-19-1987	10-19-1987	1	–0.37%	–0.37%	0	5
2	06-05-1989	09-04-1992	23	–0.34%	–6.75%	40	17
3	07-06-1995	01-31-1996	3	–0.25%	–0.75%	7	14
4	09-29-1998	11-17-1998	3	–0.25%	–0.75%	2	8
5	01-03-2001	06-25-2003	13	–0.46%	–5.50%	30	12
6	09-18-2007	12-16-2008	10	–0.62%	–5.00%	15	85
7	07-31-2019	03-16-2020	5	–0.56%	–2.25%	8	24
8	09-18-2024		0				
Mean			7	–0.41%	–3.05%	14	24
Median			4	–0.37%	–2.25%	8	14

Source: Bloomberg and authors' calculations

easing cycles had a median duration of eight months, and four cuts averaging 37bps for a total of 225bp, whereas the median tightening cycles lasted for 12 months, and involved eight hikes of just over 25bps each, for a total of around 262bp. Furthermore, the pause between the central bank ending its tightening cycle and the first cut is much shorter, coming in at eight months, than the pause between the easing cycle ending and the first hike, which comes in at 14 months in the median case. While some of the differences are driven by the fact that the zero bound became binding during the 2000s, cutting easing cycles short and delaying hiking cycles, there are also broader forces at play. First, emergencies that require quick and decisive action are most often associated with very weak growth or financial market accidents. Although there have been times when central banks panicked about inflation, that typically only happens when they previously dragged their feet for too long and therefore find themselves behind the curve and must worry about their reputation. Overall, financial markets are much faster-moving than inflation, suggesting that the likelihood of central banks having to react very quickly to a financial market accident is much higher than that they have to react inter-meeting to high inflation. Secondly, even though formally independent, the Fed is to some extent a political beast, and everyone likes a dove, especially politicians. Hiking cycles are much less popular, and therefore central banks often drag their feet and hike at a more measured pace.

Next, we show the returns of the UST index (excess over T-Bills) depending on the Fed regime. We delineate between active tightening and easing periods and periods of a hold, either after a tightening cycle or after an easing period. This is of importance for trading cycles. Figure 5.3 shows that the average monthly returns are negative for broad US fixed-income markets in periods of active hiking cycles and positive when

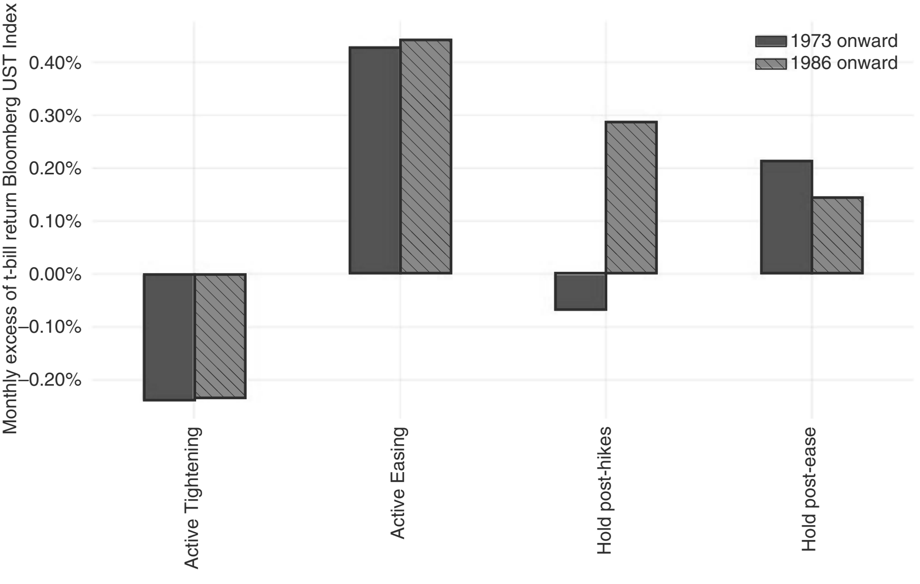


FIGURE 5.3 Negative excess returns only during active hiking cycles.
Source: Bloomberg and authors' calculations.

monetary policy is eased. More interestingly, for periods when the Fed is on hold after a cut, i.e., going into hiking cycles, excess returns are on average positive. This is even more so the case for holds after a hiking cycle, i.e., going into an easing cycle, but only for the cycles post-1986. In the high inflation cycles of the 1970s, this was not the case. The end of a hiking cycle is a dangerous time for receivers if inflation is high, but not otherwise.

Plotting Fed Funds next to both 2-year and 10-year rates shows some interesting patterns. Figure 5.4 illustrates the cycles with transitions from tightening to easing annotated on the chart. First, the two-year rate is typically leading Fed funds (having led to some discussion whether the Fed is truly needed, which does not fully take into account that the two year is partially just guessing what the Fed will do). However, Fed Funds and two-year rates peak roughly at the same time. More interestingly, even 10-year rates tend to peak around the time of the last hike, although at times those highs are retested (as in 2007) or even bested (as in 2024) during the pausing period between the last hike and the first cut. The other important observation is that the level of 10 year typically peaks at or above Fed funds, with the exception of the 2023 hiking cycle, where 10 year peaked well below Fed funds, which set us up for a very unusual easing cycle too. The same is true for the 30 year. This fact pattern is surprising, as it could be argued that the 10 year and especially the 30 year should not be overly influenced by where exactly the Fed is in the monetary policy cycle at any given point in time. But this is not how markets trade. In practice, the whole curve is largely trading in line with Fed funds, just to varying degrees.

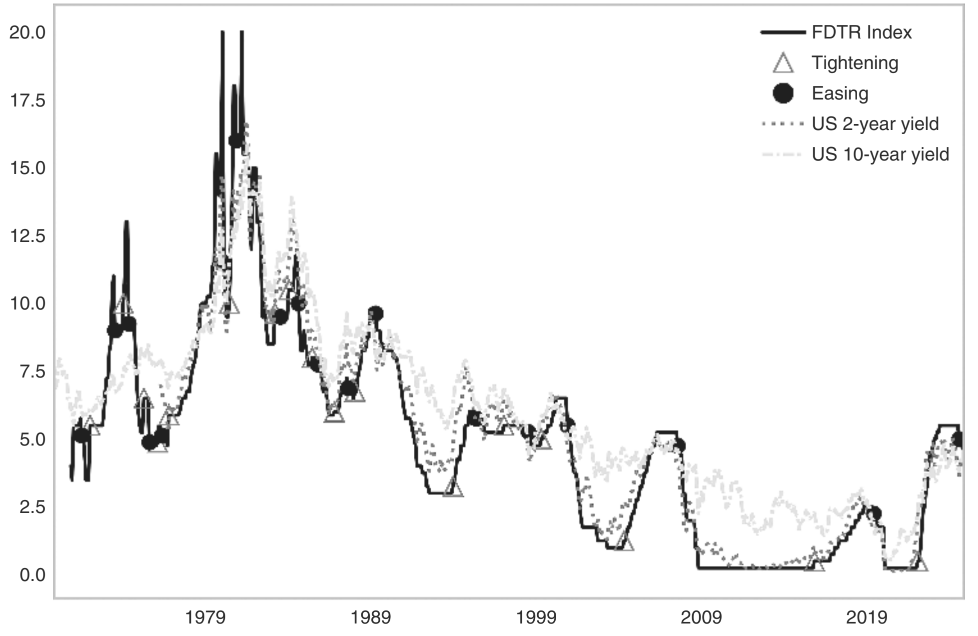


FIGURE 5.4 Fed cycles: 10s behave like 2s.
Source: Bloomberg and authors' calculations.

5.3 HOW TO TRADE A HIKING CYCLE

We start our more detailed analysis with Fed hiking cycles. The asset with the clearest link to Fed policy is the front end of the US rates curve, where the hikes have to be priced. For this analysis, we use two-year bonds as a benchmark. The typical hiking cycle is well telegraphed. The Fed does not wake up one day and hike in a panic. The reason is that the Fed is targeting core inflation. Although headline inflation can move rapidly, often in response to oil price spikes, core inflation tends to move more slowly. In addition, the Fed often ignores the initial price impact of exogenous shocks and instead focuses on slower-moving changes in wage-setting behavior and/or in the formation of inflation expectations. This is different from easing cycles, where panic cuts can and do happen, usually in response to severe equity market weakness. Figure 5.5 shows how total returns of the two year note behave into the first Fed hike of the cycle. We have used our algorithm to determine first hikes and focus on cycles post 1986. Our dates are similar to qualitative analysis of cycles, for example, by Blinder (2023) where the authors' reliance on drifts in the effective fund rate dates some cycles a few months earlier. As can be seen in Figure 5.5, returns over three months are poor going into the first hike, starting around 75 to 55 trading days (around 3 months) before the first hike. Returns remain subdued once the hiking cycle begins, especially relative to the unconditional positive average (rates have fallen, on average, since the late 1970s). But returns after the first hike are initially not disastrous, largely because they have been priced going into it. Markets eventually trade poorly again as the hiking cycles progress.

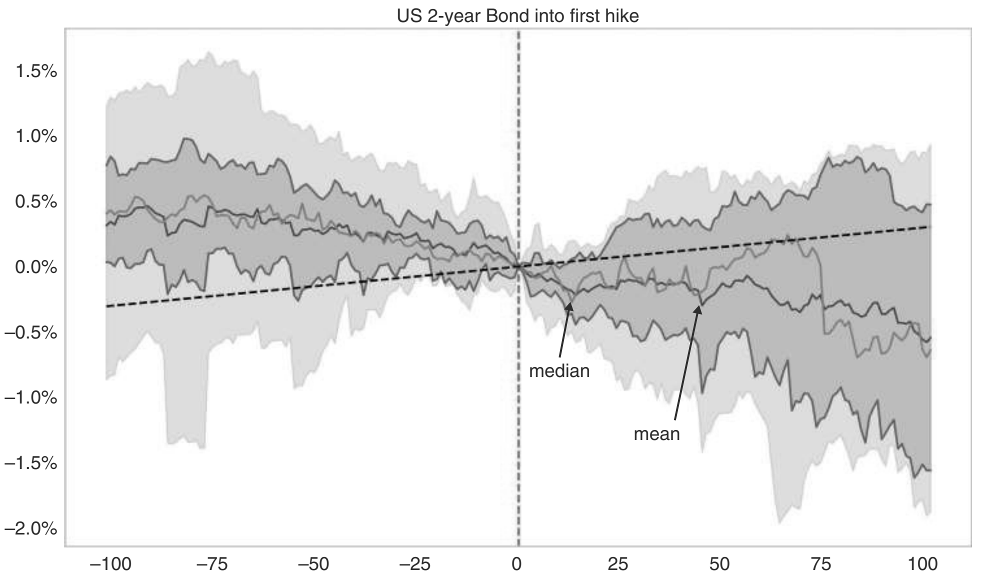


FIGURE 5.5 Two-year bond returns poor into and out of first hike.
Source: Bloomberg and authors' calculations.

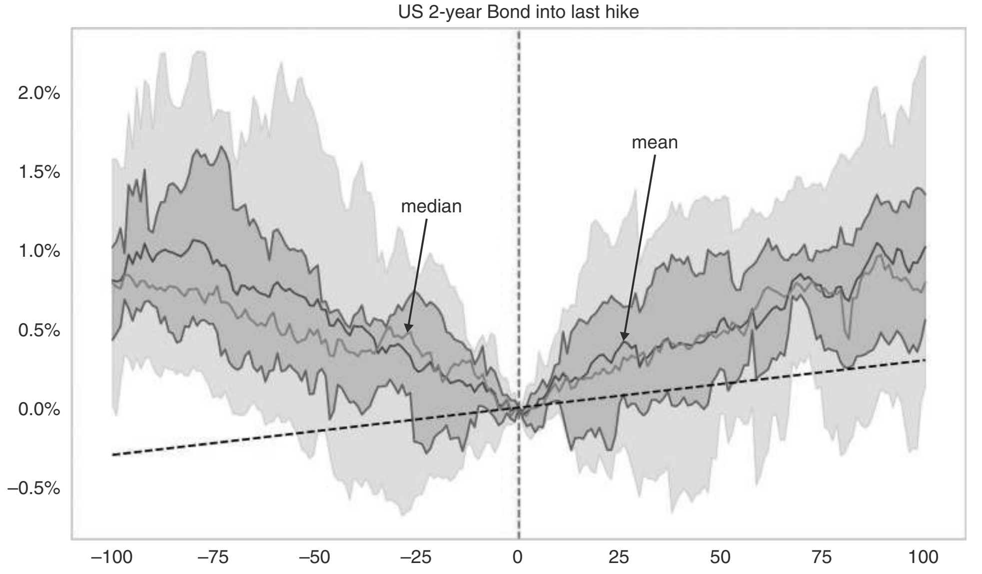


FIGURE 5.6 Two-year bond returns turn positive only after last hike.
Source: Bloomberg and authors' calculations.

Payers typically continue to work until the hiking cycle is quite mature. In fact, the trading patterns around the last hike of the cycle are even clearer than around the first hike. We show this in Figure 5.6. It turns out that the excess returns of the two year rates trough around the time of the last hike. For the 10th percentile (the lowest line shaded), two year bond returns bottom a bit more than a month (30 trading days) ahead of the last hike, which is roughly around the penultimate hike. 2s then rally strongly after the last hike.

What about the period between the first and last hike? The answer is that investors should just remain paid while hiking cycles continue. One question is whether pauses or a step down in the size of the hikes matter and can be traded. Figure 5.7a shows returns for two year US rates around events where the Fed is becoming more dovish in a hiking cycle, but where the cycle is not over yet. The events are Fed pauses that occur in a manner different from the prevailing rhythm of the hiking cycle under investigation. We find that pauses in an ongoing hiking cycle are not as helpful as we would have thought. Returns go mostly sideways into the pause and out of it. Figure 5.7b shows the two step-downs in the size of the hikes in 2022 and 2023, which came toward the end of the hiking cycle. The two step-downs in late 2022 and early 2023 (note they overlap on chart given their proximity) led to the market trading well into it, but then poorly out of it, with bonds trading poorly and two year yields making new highs after the two step-downs. We conclude that when step-downs or pauses in hiking cycles are expected, payers will probably not generate much positive PNL, but it is not clear that trading around such events can add much alpha either. Given the inherent uncertainties, it may be best to remain paid.

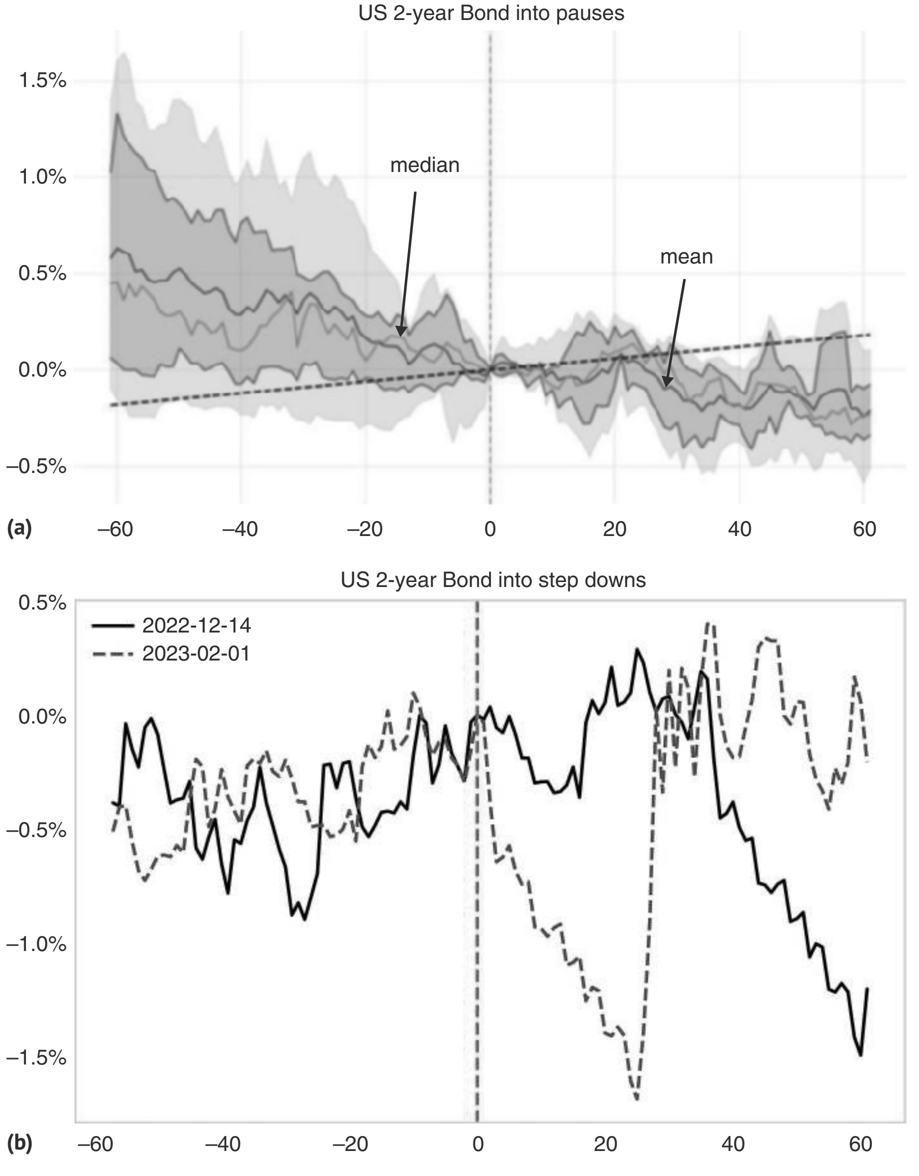


FIGURE 5.7 (a) Fed pauses not overly helpful. (b) And neither is a step-down.
Source: Bloomberg and authors' calculations.

Of course, positioning for these regularities is easier said than done. It would be good to know when the Fed is likely to start hiking rates, given that the market is pricing it a few months earlier. Furthermore, paying rates in expectation of a first hike is typically negative carry, because after the end of the preceding easing cycle, the market almost always prices at least some chance of a hike quite quickly. Clearly, a heads-up would be useful. The same is true for the last hike, even though the market appears less forward-looking on that side of the ledger. To get at least a rough idea of where the Fed stands, we can either employ Taylor rule models a la Taylor (1993), or we can listen to the Fed, or both.

The Taylor rule is based on the dual mandate of the Fed, as it is tasked by Congress to generate stable prices and maximum employment. The Fed has operationalized the inflation target by aiming for 2% over the longer run, measured by the core personal consumption expenditure price index (PCE). The core concept strips out volatile food and energy prices. The original rule used the GDP deflator, but Fed Chair Bernanke subsequently proposed a modification, using different coefficients and focusing on the PCE see Bernanke (2015). In an irony of historic proportions, the Fed actually tweaked its inflation targeting framework in August of 2020 to aim for average inflation targeting, meaning that following periods where inflation had been below 2% (as was the case going into August 2020), the Fed would allow inflation to run moderately above 2% for some time. As is often typical in wars and policy making, generals and policy makers fight the last war, not the war of the future. The decision by the Fed to react to persistent deflationary pressures in the decade before August 2020 by moving its goalpost in a dovish direction likely contributed to its very slow reaction to what was the largest inflationary shock since the 1970s. In addition to the two explicit goals of inflation and employment, the Fed also targets financial stability, both inside and outside the banking system, which justifies adding measures like equity performance or certain money market signals to the rule. However, in practice, the authors have not found a Taylor rule specification that lends itself to strategies with high out-of-sample Sharpe ratios. This is illustrated, for example, by a 2022 paper by former St. Louis Fed President Bullard, where his stable of Taylor rules would have given poor signals in 2018 and then again in 2021 and 2022 (see Bullard, 2022). In our day-to-day work, we are therefore more focused on Fed speeches or data releases rather than Taylor rules to get the start of the cycle roughly right.

Another way to get a sense of when to start positioning for hikes is to use market signals. In particular, we observe how quickly the market has priced hikes in previous cycles after the Fed appears to be done with an easing cycle. The exercise is complicated because the Fed can pause for a relatively long time after a cut, only to eventually continue with its easing cycle. The record for the longest such pause was held for a long time by the 2002 cycle (11 months) but was more recently bested as the Fed remained on hold for 12 months between 2024 and 2025. Table 5.5 shows what the market was pricing at the time when the Fed was either done, or paused for at least six months (using perfect foresight). At the time of that last cut, the market was already pricing hikes one year out, in 7 out of 11 cases, with the median coming in at 50bp. Six months later, the median hike had risen to 66bp, with only two episodes, where the market was still pricing a small cut one year out. Furthermore, the market is not overly good at distinguishing between final cuts and long pauses. All three episodes where the market was still pricing cuts on the day of the cut in question were episodes where the final cut was in, while three of the four episodes where more cuts were still to come were pricing meaningful hikes, with 2024 being the exception. For the market, being on hold for six months is usually just the same as being done. For our purposes, that means that if we expect the Fed to be on hold for at least six months, we should pay into the last cut.

TABLE 5.5 Market Quickly Prices Hikes After Cuts Are Done

	Final cut or pause	Fed funds	ED5-ED1 (or FF)	ED5-ED1 6m later	Change
1988-02-10 Final Cut	6.50%	99	66	–33	
1989-12-19	Pause	8.25%	17	42	25
1992-09-04	Final Cut	3.00%	79	99	20
1996-01-31	Final Cut	5.25%	–42	55	97
1998-11-17	Final Cut	4.75%	–20	72	91
2001-12-11	Pause	1.75%	188	187	–1
2002-11-06	Pause	1.25%	95	39	–55
2003-06-25	Final Cut	1.00%	59	130	71
2008-12-16	Final Cut	0.25%	50	94	44
2020-03-16	Final Cut	0.25%	–53	–6	47
2024-12-18	Pause	4.50%	–47	–84	–37
Mean		3.34%	39	63	24
Median		3.00%	50	66	25

Source: Bloomberg and authors' calculations

However, it is important to remember that the period between the last cut and the first hike is quite long, coming in at eight months in the median case (and 14 months in the average case). This means that rate hikes priced for one year after the last cut are, on average, exaggerated. The period between the last cut and the first hike is more of a traders' market, and more medium-term positioning for the hiking cycle should only commence when a hike is expected in the next three months.

5.4 HIKES ARE OILY

In addition to following the Fed, investors would be forgiven for also wanting to follow OPEC, or rather, the oil price. The reason is that in the past, rising oil prices have been contributing to a loss of economic momentum (while at the same time also increasing inflation). This is illustrated in Figure 5.8a, which shows a detrended version of Brent, next to the US manufacturing ISM. Of course, causality is not only going in one direction, as the correlation is also impacted by the fact that one omitted variable (a weakening economy) may cause both oil and the ISM to drop. But even if this is the dominant channel, it would still be important to watch oil, since it is a price that is updated daily, unlike the monthly ISM survey. We would therefore argue that significant oil spikes are an important reason to position for a weaker economy going

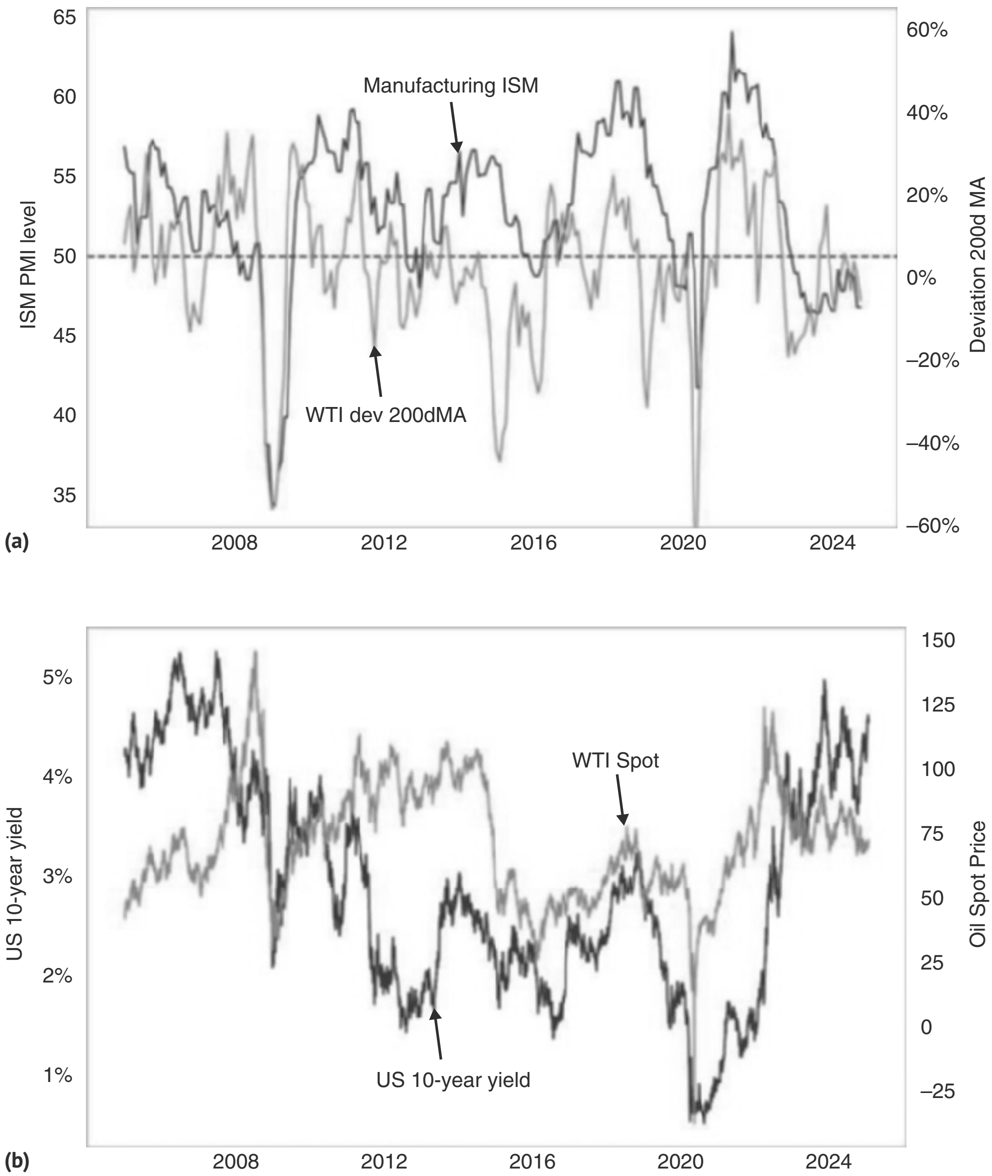


FIGURE 5.8 (a) Energy often contributes to weaker growth momentum. (b) But US rates unlikely to peak much before oil, unless there is a crisis.
Source: Bloomberg and authors' calculations.

forward. Unfortunately, leading the manufacturing ISM is not the same as leading bond prices. Relative to UST yields, the lead-lag relationship is less clear. We find it interesting that in spite of the important growth-negative impact of higher oil prices, it is generally *not* the case that yields can look through oil rallies. Although rates can, and often do, move slightly ahead of oil, the lags are not overly long. The initial response of rates is to sell off with oil and not to focus too much on the negative growth impact. For example, in the 2008 oil spike, US rates peaked well ahead of oil because the United States was in a recession and the outlook was very deflationary. But even in this episode, US rates moved up again for a while in the last inning of the oil spike in May 2008. Rates then reached their local peak almost in a coincident fashion with oil. And in the 2011 oil spike, which happened without a US crisis, US rates peaked only two months before the oil peak. Lastly, in the 2022 oil spike, US yields were also unable to look through the spike and went straight up with oil. The post-COVID inflation meant that yields kept going up even with oil falling, but overall the message is clear: rising oil prices are first and foremost leading to higher yields, even though a major oil spike in the end typically has a large negative growth impact, often reversing much of the oil and yield upmove.

5.5 THE WISDOM OF EQUITY MARKETS

We once had a colleague who got into trouble with the equity department for suggesting that fixed-income markets are smarter than equity markets when commenting on the pattern that credit markets seem to suss out trouble earlier than equities. But it turns out that if we look under the hood of the equity market, it is oftentimes leading the smartest of all markets, i.e., the Treasury market. In particular, cyclical versus defensive equity indexes are typically peaking ahead of the peak in yields. This is illustrated in Figure 5.9. By the time US 10-year rates peak, on average, cyclicals versus defensives are down almost 5% in the previous two months. The issue with this indicator is that equities are much more volatile than interest rates, which can lead to

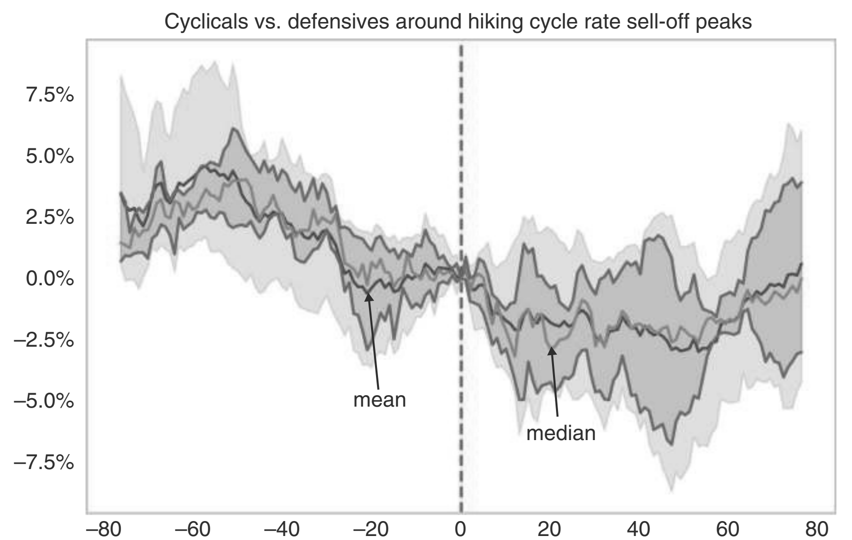


FIGURE 5.9 Cyc/def as warning sign in hiking cycles.
Source: Bloomberg and authors' calculations.

frequent false signals. Furthermore, our broader study of interest rate sell-offs, which not only analyzes cycle peaks but also includes other tradable interest rate sell-offs, finds that this ratio is more coincident. We would therefore only focus on this measure if we think that the Fed cycle is at a stage where a peak in interest rates is plausible. We will roll out the broader study in the next chapter, where we show that using the indicator can still deliver improvements when compared to buying and holding fixed-income indexes outside hiking cycles. However, it does not protect from inflation-led pullbacks such as 2022 (see Figure 6.6).

5.6 HOW TO TRADE AN EASING CYCLE

Unsurprisingly, the rules for trading easing cycles are very similar to trading hiking cycles. As mentioned previously, one difference is that the easing cycles can be very fast, especially in response to market accidents, while the hiking cycles are almost always much slower. Central bankers prefer cuts to hikes and, therefore, do so at the slightest provocation.

Given this asymmetric pattern, it is not surprising that rates need to be received before they need to be paid. We find that on average, two year US rates should be received just around the time when the last hike happens. We already illustrated this in Figure 5.6 in the section on how to trade hiking cycles. This is different from the case of hiking cycles, where payers should only be entered around 2.5 months before the first hike. However, we also note that countertrend sell-offs in rates going into the first cut can be significant if the first cut is still too far out. Once we are in the three-month window before the first cut, countertrend rate sell-offs are quite small, as can be seen in Figure 5.10. The largest one was 21bp in 2001, which was quickly reversed. In most other episodes, it was smaller. This makes easing cycles somewhat more symmetric to hiking cycles. Yes, investors should already start receiving around the last hike, but they should press the receiving bet around three months before the first cut. This part worked well even in the atypical 2024 easing cycle.

The receiving typically comes to an end with the last cut, as per Figure 5.11. Returns of bonds are strongly positive into the last cut, followed by a sideways pattern. It is around a month *after* the last cut that yields start to move higher. The reason is likely that at the time of the last cut, it is often not clear whether the last cut has indeed occurred or not. Either the market still assumes that cuts will continue in the next meeting, or it may be seen as just a pause in the cycle. The following Fed meeting often clarifies things, and on the first pause, rates move aggressively higher. The best point to pay is therefore not the last cut, but just before the meeting, where the Fed for the first time sits it out. This is just a tactical pay. Between the last cut and three months before the first hike, when positioning for the coming hiking cycle has to start in earnest, rates are often more of a range trade rather than a clear directional trade.

Given that it is quite difficult to forecast the first cut, we had proposed a rule for emerging markets that worked relatively well in getting investors into the trade, though not necessarily as early as we would like. The rule is to start receiving when the one year swap breaks through the floating leg of the swap, i.e., when the market is starting to price cuts one year out. The nice thing about this rule is that upon entry, the carry

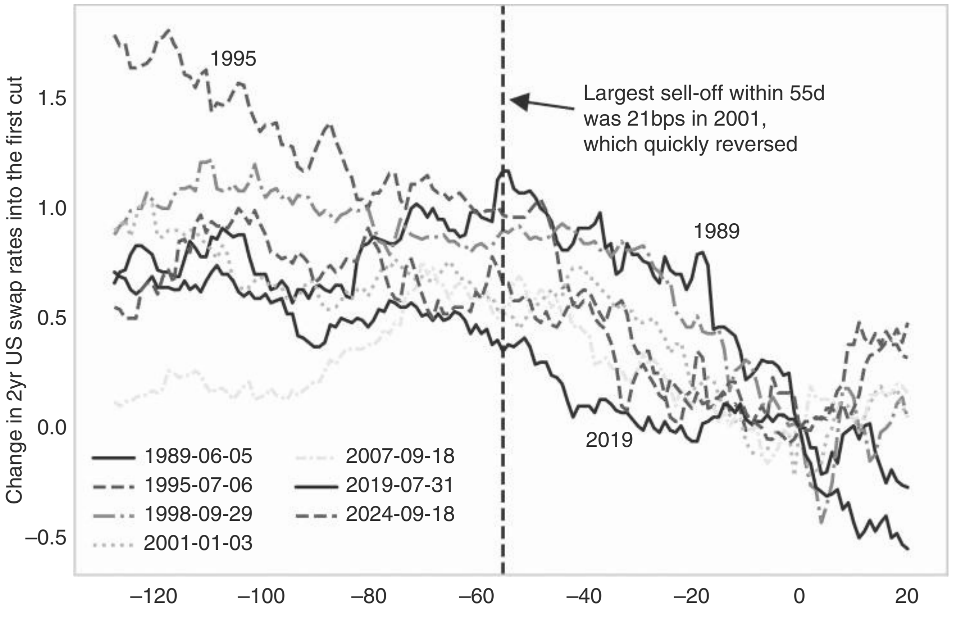


FIGURE 5.10 No large sell-offs in the last 2.5m into first cut.
Source: Bloomberg and authors' calculations.

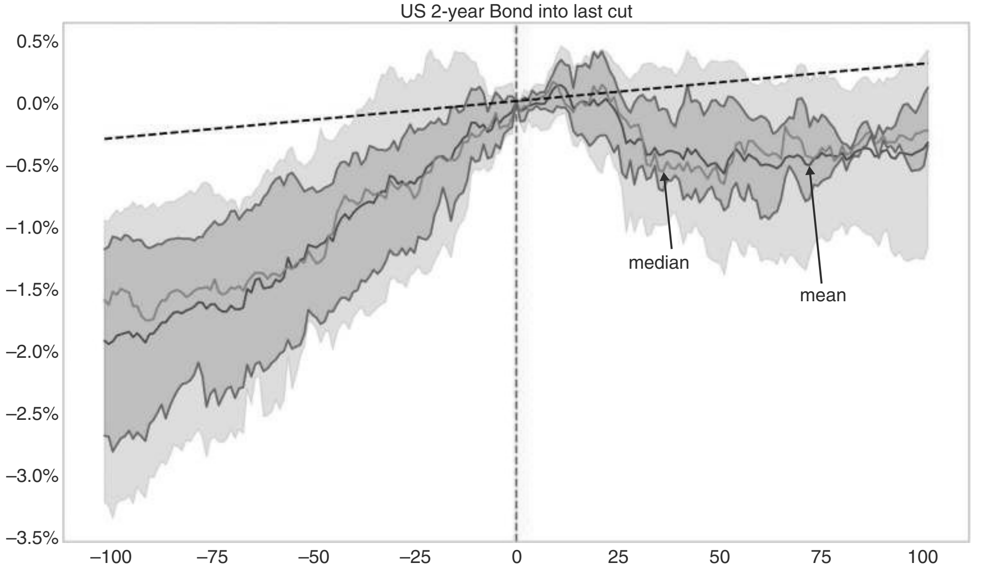


FIGURE 5.11 Exit upon last cut.
Source: Bloomberg and authors' calculations.

is in theory around zero. If there was no slippage on entry and exit, the rule would trade for zero profit or loss if there was a "false alarm," only to benefit tremendously when an easing cycle truly did start subsequently. In practice, there is slippage, there are transaction costs, and the floating rate may also jump up on a residual hike. Still, the rule is useful to monitor, and we illustrate in Figure 5.12, that the rule works for the United States too. The median rate keeps falling after the event, with the skew strongly to the downside. Results improve significantly when cutting out of the receiver when the one year trades back above the fix (here just shown for the mean). The US banking crisis of 2023 was a rare misfire for this rule, as the Fed bailout happened so quickly that rate cuts were not necessary, which was helpful to a Fed that was still held back by very high inflation readings.

Going cycle by cycle, it becomes clear that the 2022/2023 hiking cycle was a striking exception to past cycles for which we have data. In the past, when the market was pricing at least half a chance of a cut six months out (i.e., six month FF inverting versus one month by more than 12.5bp), cuts either materialized or, at the very least, no more hikes occurred over the next six months. However, as mentioned earlier, the banking panic in March 2023 did not halt the hiking cycle, as inflation remained above 5%. There was actually a second false alarm in May of 2023, ahead of the Fed pause in June of 2023, but then the Fed hiked one final time in July. This episode does not show up in Table 5.6, as we just show the first event over any three month period. The 2022/23 cycle clearly impairs the hit ratio of the various inversion rules. We think that high-inflation cycles may require a more hawkish Fed than low-inflation cycles. We therefore give our rules the benefit of the doubt if the next cycle should again be a low(er) inflation cycle, also because the pattern worked well again in 2024.

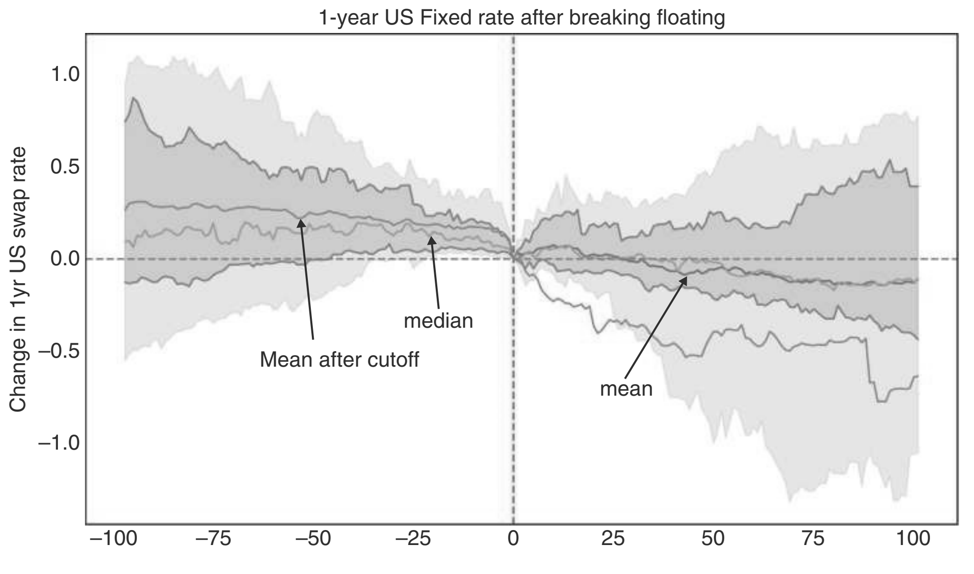


FIGURE 5.12 There is follow-through after breaking through the floating rate.
Source: Bloomberg and authors' calculations.

TABLE 5.6 Promises Kept: Fed Rarely Hikes Within Six Months When Pricing—12.5bps of Easing

	Level of FF6–FF1	Actual 3m FF change	Actual 6m FF change
05-24-1995	–19	–25	–25
02-13-1998	–13	0	0
08-20-1998	–13	–75	–75
09-19-2000	–15	0	–100
10-04-2006	–15	0	0
02-27-2007	–17	0	0
07-26-2007	–19	–50	–100
05-13-2019	–16	–25	–75
03-13-2023	–34	50	75
11-29-2023	–13	0	0
05-01-2024	–16	0	–50

Source: Bloomberg and authors' calculations

Although the one-year inversion rule works reasonably well, there are many parts of the curve that could be used as a signal, and the question is how often which part of the curve is right in predicting subsequent Fed cuts. We already briefly commented on a related question in our business cycle chapter, where we focused on returns for the US 10-year post-inversion of the 3m/10yr, which were generally strong. Here, we investigate both returns for receiving 10-year rates as well as the hit ratio with respect to predicting Fed cuts for a broader set of curve inversions. This is shown in Figure 5.12. On the return side, we find that the most profitable short-term signals come when the short leg of the inverted curve is quite short (three months, or even better, six months). Using six months has the advantage of working faster (i.e., on a one-month horizon). Inversions of 2s/5s also work well, and for longer, as returns keep rising after three months too. Hit ratios for the trade to be money-making are generally in the 60% range. If the short leg is longer (5 years or 10 years), the returns tend to be negative. The choice of the long leg of the inversion trade is less crucial. Six months works best with 2s, but it also works with 5s or 10s, as is illustrated in Table 5.7.

With respect to hit ratios in terms of whether a cut happens, we require an inversion of at least 12.5bp, suggesting at least a 50% chance for a full 25bp cut. We then count it as a "win" if a cut was delivered within the following six months, one year, or two years (where, admittedly, the two-year time frame is quite a loose condition for a win). With this definition, it turns out that on a six month horizon, hit ratios are overall low, as is shown in Table 5.8. The highest is for six months/2yr, coming in at 58%. Hit ratios obviously rise from there, and with a one-year window, they are often meaningfully above 50%. Again, 6m/2yr scores best at 74%. The two-year window has very high hit

TABLE 5.7 10-year Returns on Inversion Positive for Shorter End

	Returns			Hit ratios		
	1m	**3m**	**6m**	**1m**	**3m**	**6m**
3m10s	0.26%	1.11%	1.03%	65%	58%	62%
6m10s	–0.16%	0.34%	0.24%	47%	47%	60%
6m2s	0.78%	1.71%	0.48%	70%	70%	60%
6m5s	0.45%	0.76%	1.66%	50%	65%	72%
2s5s	0.23%	0.14%	0.55%	56%	56%	50%
2s10s	–0.79%	–1.74%	–2.15%	40%	33%	33%
2s30s	–0.53%	–1.63%	–1.09%	54%	38%	38%
5s10s	–1.16%	–1.21%	–0.01%	39%	43%	44%
5s30s	–1.43%	–1.27%	–1.49%	28%	39%	33%
10s30s	–1.19%	–1.58%	–0.38%	32%	32%	42%

Source: Bloomberg and authors' calculations

TABLE 5.8 Lags Between Inversion and the First Cut Long and Variable

	6m hit ratio	**12m hit ratio**	**24m hit ratio**	**10-year 6m yield chg**	**Avg months to cut**	**Max months to cut**
3m10s	39.1%	52.2%	100.0%	0.10	10.3	23.4
6m10s	39.1%	56.5%	95.7%	0.04	11.0	26.6
6m2s	57.9%	73.7%	84.2%	–0.17	12.0	54.1
6m5s	45.8%	66.7%	95.8%	–0.14	8.7	26.3
2s5s	53.3%	73.3%	93.3%	–0.14	7.7	26.9
2s10s	50.0%	57.1%	85.7%	0.28	11.4	30.1
2s30s	41.7%	66.7%	91.7%	0.16	9.4	26.6
5s10s	40.0%	63.3%	93.3%	0.16	10.8	32.4
5s30s	35.3%	58.8%	82.4%	0.31	11.7	30.1
10s30s	55.6%	72.2%	94.4%	0.12	8.4	24.3
Mean	45.8%	64.1%	91.7%	0.07	10.1	30.1
Median	43.8%	65.0%	93.3%	0.11	10.5	26.8

Source: Bloomberg and authors' calculations

ratios, with 3m/10yr reaching 100%. Interestingly, the most commonly quoted 2s/10s is not scoring as well. Furthermore, the time between inversion and the first cut is highly variable. 2s5s has the shortest average time to the first cut, while 3m/10yr has the shortest maximum lag. Overall, the time to cut is highly variable, with the longest wait on record coming in at 23 months for that particular measure. In terms of trading it, the strongest rally in the 10-year point comes after the inversion of 6m/2yr, with other yield curve slopes at the shorter end of the curve, also scoring respectably, while at the backend of the curve, 10yr typically do not rally after inversion.

Having investigated the start and end of the easing cycle, we now discuss what happens between the first cut and the last cut. In spite of the strong rally into the first cut, historically, the Fed has typically out-cut what was priced, with the 2024 easing cycle being an exception. This is illustrated in Figure 5.13, where we plot market pricing for the next 12 months next to the actual change of Fed Funds over 12 months. Going into the shaded recession periods, Fed funds clearly fall more than what is priced. That the Fed outcuts the forwards is unsurprising during hard landings. During the few soft landings on record in the 1990s, the Fed cut slightly more than was priced as well, supporting receivers. But for soft landings, the margin by which the Fed outcut market pricing was, of course, much narrower. And in 2024, the Fed undercut what was priced. We will investigate soft landings in more detail later.

As an aside, Figure 5.13 also shows that when the market is pricing hikes, it is more frequently the case that the Fed does not validate the market. The market often prices hikes that do not occur. Only if there actually is a hiking cycle, the same pattern holds: the hikes delivered over the next 12 months exceed what was priced, suggesting that the markets are not pricing hiking cycles aggressively enough either. This was most clearly the case in the 2022 hiking cycle, but also to a lesser extent in previous cycles.

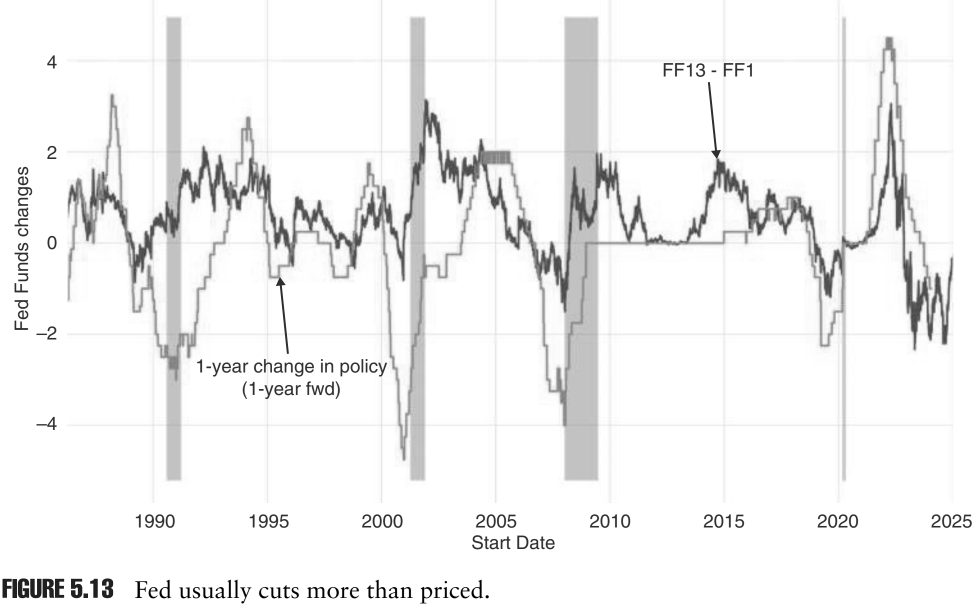


FIGURE 5.13 Fed usually cuts more than priced.
Source: Bloomberg and authors' calculations.

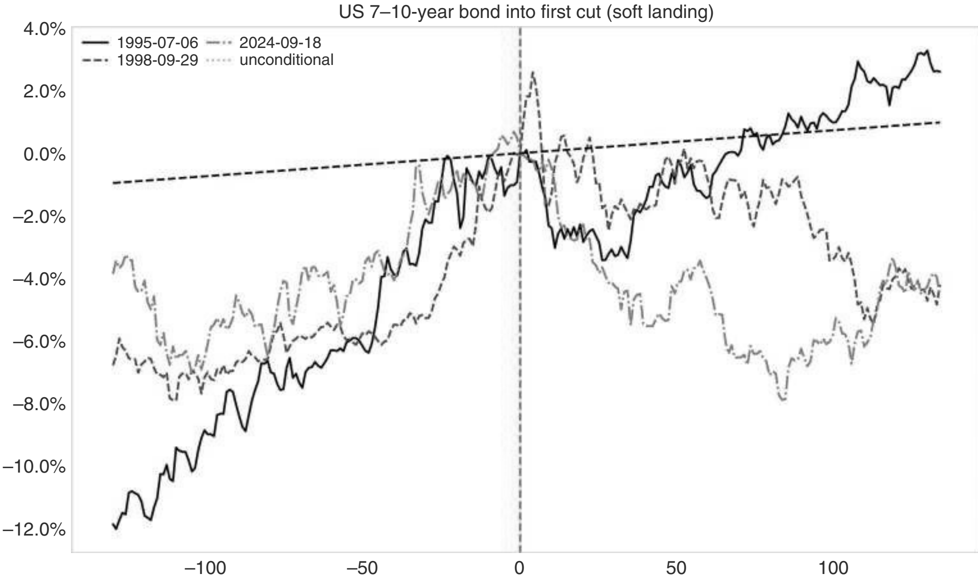


FIGURE 5.14 Receiving works into first cut, even with a soft landing.
Source: Bloomberg and authors' calculations.

Figure 5.14 illustrates the soft landing price action in more detail, going into the first cut. We exclude 1987 here since it was the equity crash that changed the economic "outlook" for the Fed overnight. In the classic soft landing of 1995 (of three cuts), 10-year rates only troughed around the last cut. In the 1998 easing cycle (also totaling three cuts), the market bottomed with the first cut, slightly ahead of our rule of thumb, which expects a trough when there is only one cut left. In this episode, the two-year rate only troughed when just one cut was left, though. The 2024 easing cycle also had three cuts, before a long pause, and led to a trough in the one year US rate shortly after the first cut. This was also slightly ahead of time. The lesson is that in all three cases, rates traded well into the first cut, just as in the case of hard landings. Coming out of the first cut, performance quickly deteriorated, although in 1995, only after a short burst of strong performance in the direct aftermath of the first cut. We conclude that in a soft landing scenario, investors should consider taking profits already shortly after the start of the easing cycle rather than sticking around for too long. This assumes, of course, that investors are able to figure out that a soft landing is coming.

We have already pointed out in the business cycle chapter how difficult it is to distinguish early on between soft landings (with small easing cycles) and hard landings (requiring large easing cycles). After all, every hard landing initially looks like a soft landing. One of the better predictors we found of how many Fed cuts will be required in an upcoming cycle is the SPX. Although this is a somewhat circular argument, when it comes to trading the SPX, it is helpful for trading rates. In particular, we found that a strong SPX, observed on the day before the first cut and measured as a deviation from its 200dma, is a good predictor of how many Fed cuts are going to take place over the next 12 months, especially when we ignore the 1987 episode, which is arguably somewhat special as the Fed cut in reaction to the equity crash. This is illustrated in

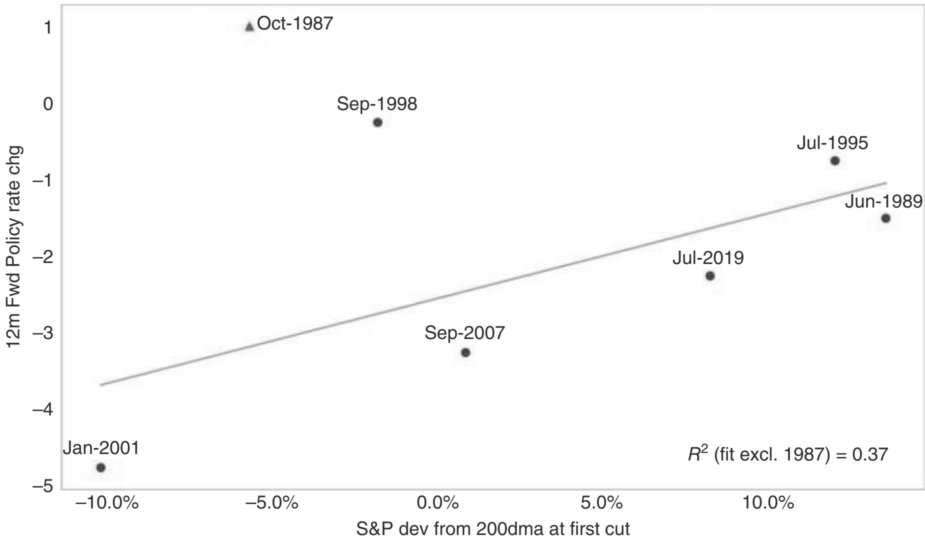


FIGURE 5.15 S&P strength and aggressive cuts don't mix.
Source: Bloomberg and authors' calculations.

Figure 5.15. For example, in the case of the 1995 cycle, the Fed only cut three times and paused after the first cut. A strong SPX going into that cycle was a warning sign for receivers. On the other hand, going into the 2001 easing cycle, the SPX was already very weak, and the Fed had to cut very aggressively over the next 12 months. Although the sample size is extremely small, the relationship passes the smell test, given that the SPX is one of the better leading indicators for whether we should expect a soft or hard landing. Whenever the market is pricing a hard landing type of easing cycle, but the SPX is strong, investors should be cautious.

What Went Wrong and What Went Right in the 2024 Easing Cycle?

Concerning the 2024 cycle, some of our trading rules worked well, while others played out slightly differently. It is important to understand what happened, to draw conclusions on whether the traditional rules laid out here are still going to work going forward. We start the analysis with the 2022 hiking cycle as it holds important implications for the subsequent easing cycle.

The 2022 hiking cycle became tradable around three months prior to the first hike, which is roughly in line with the typical rules. Although there was a big UST sell-off in late 2020, as the worst-case outcome from COVID was avoided, payers driven by the coming hiking cycle started to work well in December 2021, around three months before the March 2022 start of the hiking cycle. Then, in late 2022, US 10-year rates started to range trade, in line with the Fed slowing

(*Continued*)

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down its hiking cycle, first to a 50bp hike and then to steps of 25bp. However, a subsequent sell-off pushed rates to the cycle highs in October of 2023, three months after the last hike. That was much later than usual, as typically 10 year rates peak close to the last hike. The other unusual factor was that rates peaked well below Fed funds. Usually, 10-year rates, and certainly two year rates, are close to, or sometimes well above, Fed funds, at the end of the cycle. In the 2022 hiking cycle, the market seems to have never thought that high rates would last too long, pricing rate cuts much earlier than is typical. This is illustrated in Figure 5.16. US rates then rallied strongly into the first cut, and especially hard in the last three months into the first cut, which occurred in September 2024. This timing is in line with our typical rules. The size of the rally was also not too different from the typical rally at this part of the cycle. However, because the market had never sold off as much as it typically does during the earlier hiking cycle, pricing in cuts much earlier, the rally into the first cut made the priced-in cuts quite extreme. This is illustrated by Figure 5.17, which shows the extent of cuts priced one year forward into the day of the first cut (labeled zero). This extreme pricing made the market very vulnerable to a soft landing scenario, where much fewer cuts would be likely. As a result, with soft landing vibes getting stronger throughout 2024, US rates sold off aggressively in the direct aftermath of the initial 50bp cut by the Fed. While at the back end, fiscal fears may have contributed, the front-end sell-off was largely due to the fact that the soft landing narrative collided with the aggressive pricing of cuts.

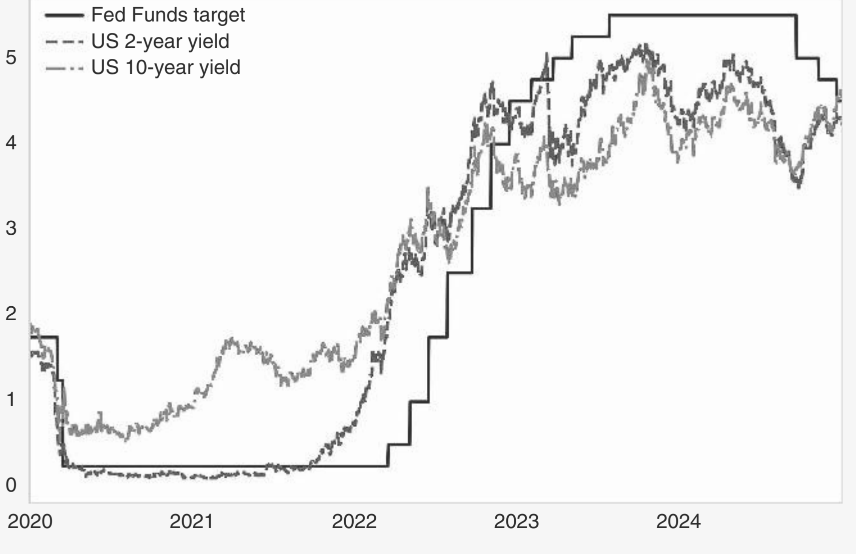


FIGURE 5.16 Sell-off in 2022 hiking cycle smaller than typical.
Source: Bloomberg and authors' calculations.

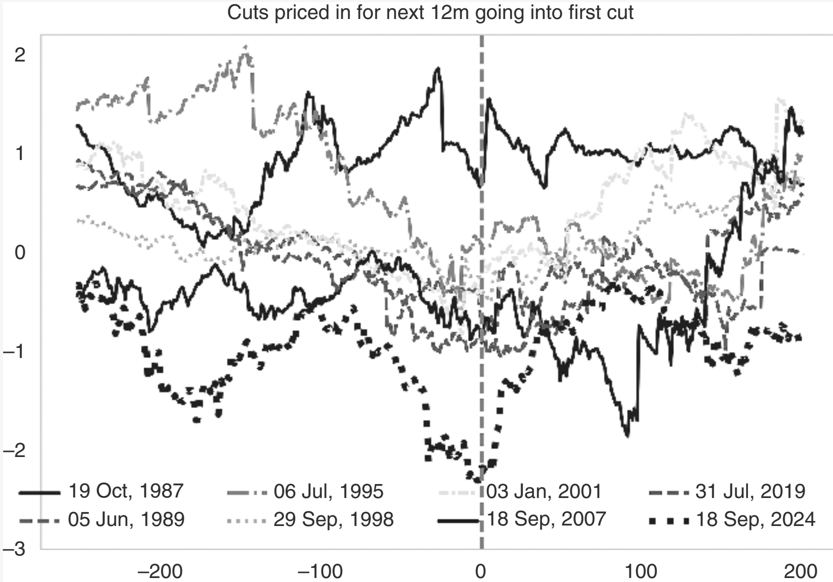


FIGURE 5.17 Pricing of cuts into the 2024 easing cycle the most aggressive on record.
Source: Bloomberg and authors' calculations.

What does this mean for our rules? It seems clear that paying three to six months ahead of the first hike still seems to work. Furthermore, receiving the three months before the first cut also still plays out quite nicely. However, to start receiving around the last hike was not the right call for 2023, even though at the front end of the curve, the initial damage was somewhat contained. Another lesson is that during the easing cycle, investors have to be more tactical on how much is priced in, especially if a soft landing is plausible. In the past, we could more or less rely on the regularity that the market would never price enough cuts early on in the cycle, likely because some risk premium is required in case the central bank does not reach the terminal rate. However, this may have changed, maybe because there are more macro funds with more capital to position for these type of trades early on. With more capital chasing these front-end trades, less risk premium is required. It seems to us that whenever the market is pricing close to a reasonable estimate of the terminal rate already early in the easing cycle, investors should be more tactical and not stay in the receiving trade until the last cut is close, especially if they have reason to suspect a soft landing.

The other rule of thumb that was dented by the 2022 episode is that the Fed did not wind down its hiking cycle when inflation peaked. Inflation peaked in June 2022, but the Fed kept hiking until July of 2023 (and rates only peaked in October of 2023). The Fed hiking for a full year after inflation peaked, is more hawkish than normal. However, Figure 5.18 illustrates that there were two reasons for that. First, the lag between inflation taking off and the hiking cycle

(*Continued*)

(*Continued*)
starting was extremely long. Starting later than usual presumably also meant it would continue for longer. Secondly, when inflation is extremely high, as was also the case in the 1980s, Fed hikes can coincide with falling inflation. Investors should take both these factors into account when starting to position for an end to a Fed hiking cycle.

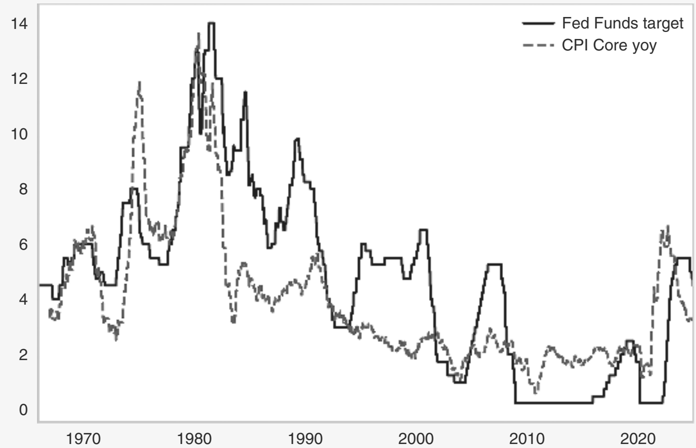


FIGURE 5.18 High inflation and a slow start mean a later peak in rates.
Source: Bloomberg.

5.7 TRADING THE CURVE

So far, we have mostly analyzed the short end of the curve, as this is the part of the curve that is most closely linked to Fed decisions, which in the end determine the cycle. We now discuss the curve. There are many segments of the curve that investors trade. We focus most of our discussion on 2s/10s. The reason is that many other curve slopes behave very similarly to 2s/10s. Table 5.9 shows the correlation of various parts of the curve (using daily data from 1965 to 2024). As can be seen, correlations are generally high, especially for 2s/10s with 2s/5s, 2s/30s, 5s/10s, and 5s/30s. The main exception is 1s/2s, as the very front end does have a life of its own. The very backend is also less strongly correlated, as indicated by 10s/30s, but to a lesser extent than the very front end.

The first point to make is that the back end typically trades with the front end, though with a beta lower than 1. Figure 5.19 illustrates this point by showing the strong correlation between the 10 year and SFR5 (the 5th SOFR contract) for the period since the front end moved away from zero to now (March 2022 to March 2025). As the front end falls/rises, so do 10-year rates. The R2 is just below 91%. The beta on daily data is 0.9, i.e., relatively high, but below 1.

This suggests that the monetary policy cycle is largely one where the front end does most of the work, which therefore either leads to bear flattening in a hiking cycle, or to

TABLE 5.9 High Correlation Across Curve Segments

	2s5s	2s10s	2s30s	5s10s	5s30s	1s2s	10s30s
2s5s	1.00	0.82	0.68	0.07	0.04	0.01	–0.01
2s10s	0.82	1.00	0.88	0.63	0.47	–0.01	0.09
2s30s	0.68	0.88	1.00	0.61	0.76	–0.09	0.55
5s10s	0.07	0.63	0.61	1.00	0.76	–0.03	0.16
5s30s	0.04	0.47	0.76	0.76	1.00	–0.13	0.76
1s2s	0.01	–0.01	–0.09	–0.03	–0.13	1.00	–0.17
10s30s	–0.01	0.09	0.55	0.16	0.76	–0.17	1.00

Source: Bloomberg and authors' calculations

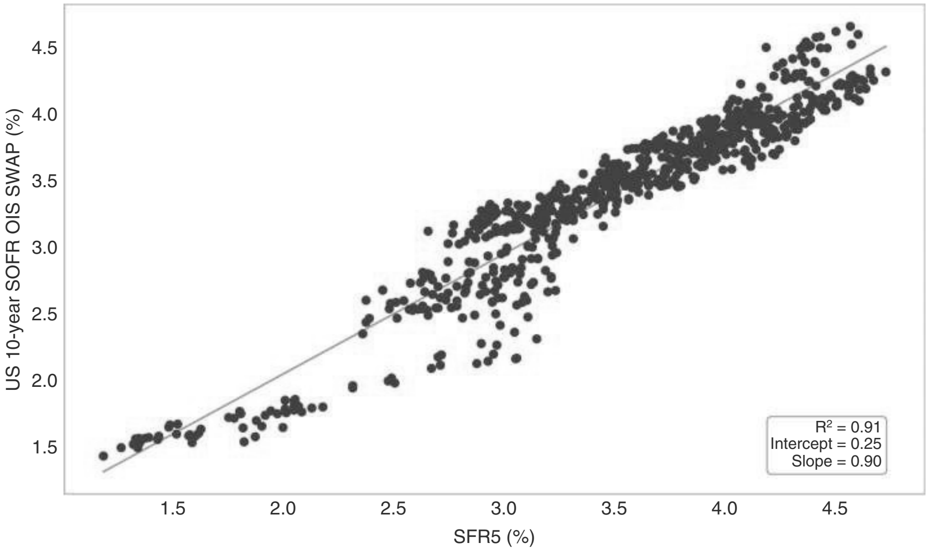


FIGURE 5.19 Back end trades with front end.
Source: Bloomberg and authors' calculations.

bull steepening in an easing cycle. This is exactly as it should be, given that the 10 year point needs to price the average Fed funds for 10 years, plus some risk premium, which clearly is less impacted by any given hiking cycle than the two year point is. This can also be illustrated by the trading patterns into the first cuts and hikes of the cycle. We first analyze 2s/10s into the first hike, with return data starting in 2000. Figure 5.20a illustrates that DV01 neutral flatteners are profitable, starting around 50 trading days before the first hike. Trading steepeners into the first cuts has been somewhat messier. Yes, steepeners make money, but in the median case, the trended move higher only starts well after the start of the easing cycle, as per Figure 5.20b.

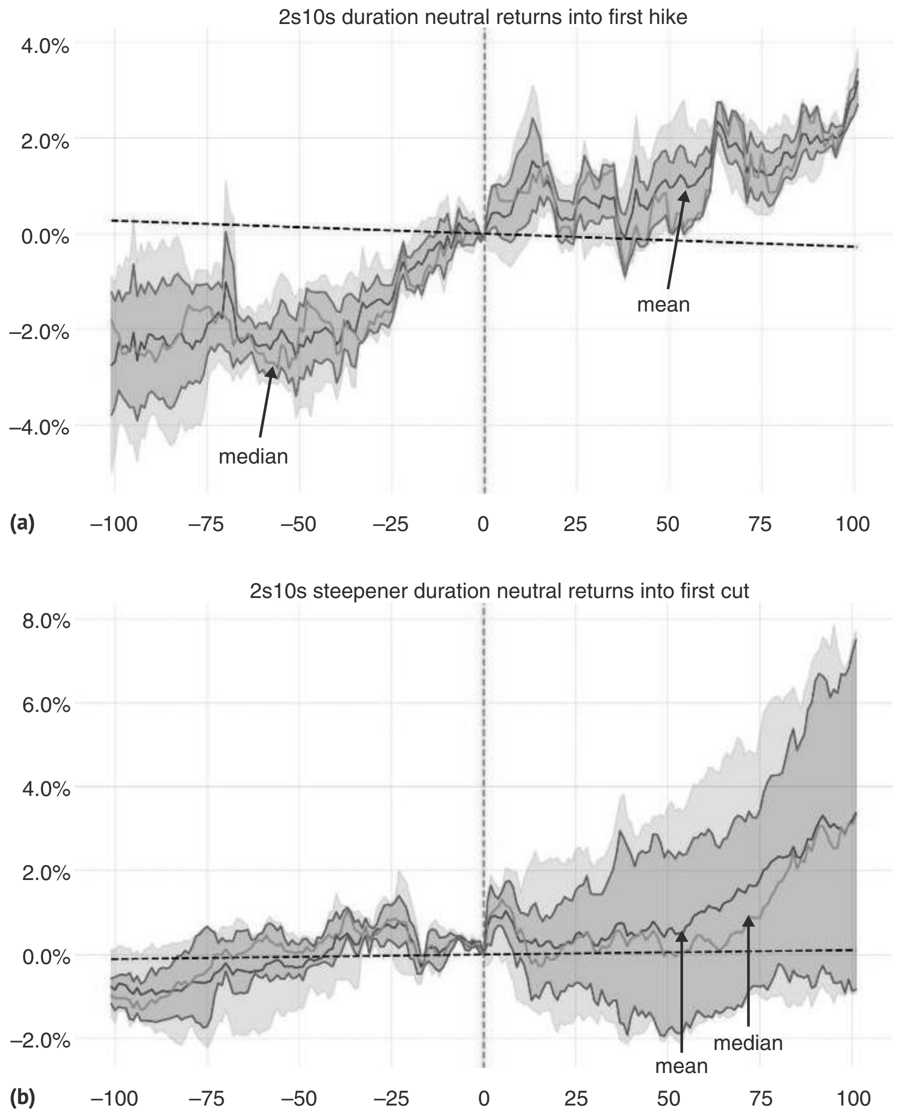


FIGURE 5.20 (a) Flatteners are profitable into the first hike. (b) 2s10s steepeners work after the first cut.
Source: Bloomberg and authors' calculations.

Before the first cut comes close, the curve is not overly directional. This is true for most segments of the curve, though the very front end (1s/2s) tends to flatten into the first cut, but then also steepens as the first cut occurs. Moving into forward space, the steepening happens earlier, and such trades can therefore be implemented earlier.

As an aside, for real money investors, there is always the question of whether it is possible to benefit from the bull steepening of the yield curve if no leverage is available. The question is whether the higher duration of the back end more than compensates for the bull steepening that the back end misses out on. Figure 5.21 shows how each

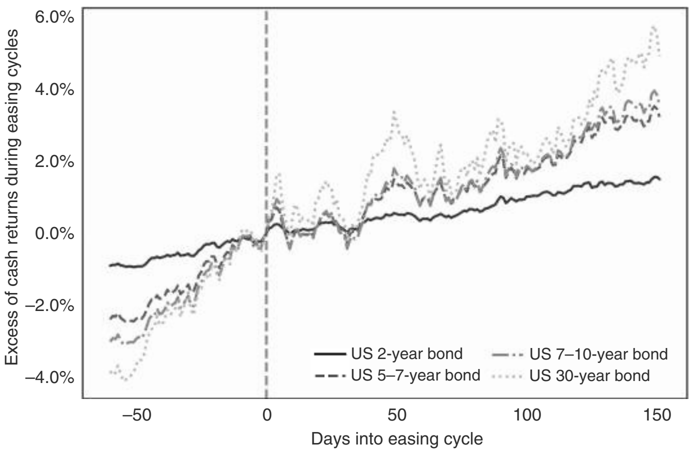


FIGURE 5.21 Without leverage, backends do better during easing cycles.
Source: Bloomberg and authors' calculations.

segment of the curve would have done during past easing cycles for the same cash investment. It turns out that the higher DV01 of the backend more than compensates for the bull steepening. Thirty year bonds are the best instrument to trade in easing cycles if investors are cash-constrained. Of course, this pattern will only continue if fiscal issues do not start to overwhelm the easing cycles. During hiking cycles, the same is true. It is generally better to hide in low-duration products, even though the curve bear flattens. Cash rules during hiking cycles.

However, life is somewhat more complicated than the stylized facts. It turns out that bear steepening and bull flattening are surprisingly common, and not just during the QE periods, when the front end of the curve remained pinned to zero. If we define bull and bear steepening/flattening by the move in the two year only, and only "count" it if the two year moves by more than 2bp either way, to cut out the noise, and only focus on periods when the two year is above 1% to cut out the QE periods, then we *still* find that bull steepening and bear flattening only account for 64% of moves. Thirty-six percent of the time, we get either bull flattener or bear steepener. This is illustrated in Figure 5.22. One reason may be supply, which we cover next.

5.8 THE WEIGHT OF SUPPLY

The reasons for bull flatteners, or bear steepeners, can be manifold. Bear steepeners are usually linked to a situation where the Fed is seen behind the curve, as was the case in the post-COVID inflation shock in 2021. Another prominent reason for bear steepening episodes is fiscal fears. We are especially interested in bear steepening during the late parts of the hiking cycle and while the Fed is on hold, as bear steepening while the Fed is on hold can easily stop-out investors, who follow our typical rule that receivers should be introduced around the last hike (even though we would always recommend to focus on the front end at those times). Late-cycle bear steepening is not as infrequent as widely perceived, but usually the moves are not overly large. The three most extreme late-cycle bear steepening events happened in 2006 (36bp), 2007 (28bp),

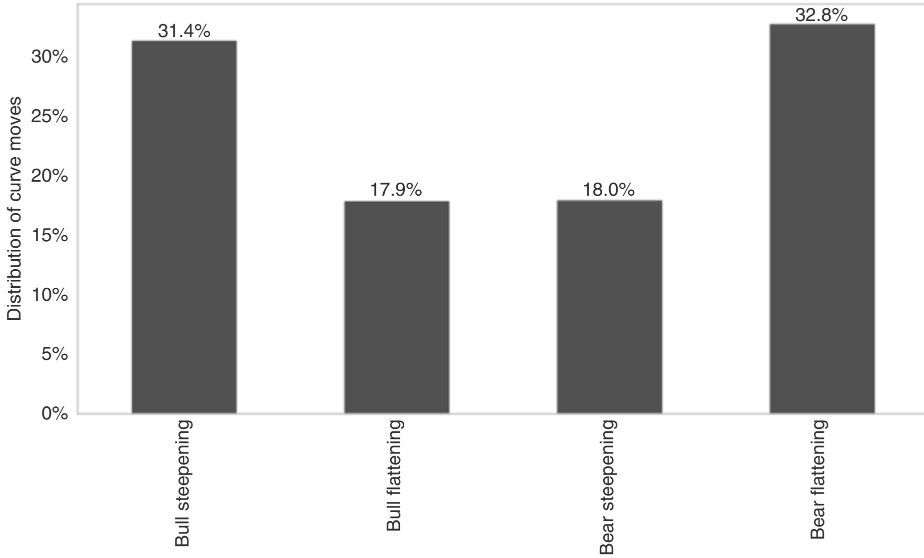


FIGURE 5.22 Bull flattening and bear steepening surprisingly common.
Source: Bloomberg and authors' calculations.

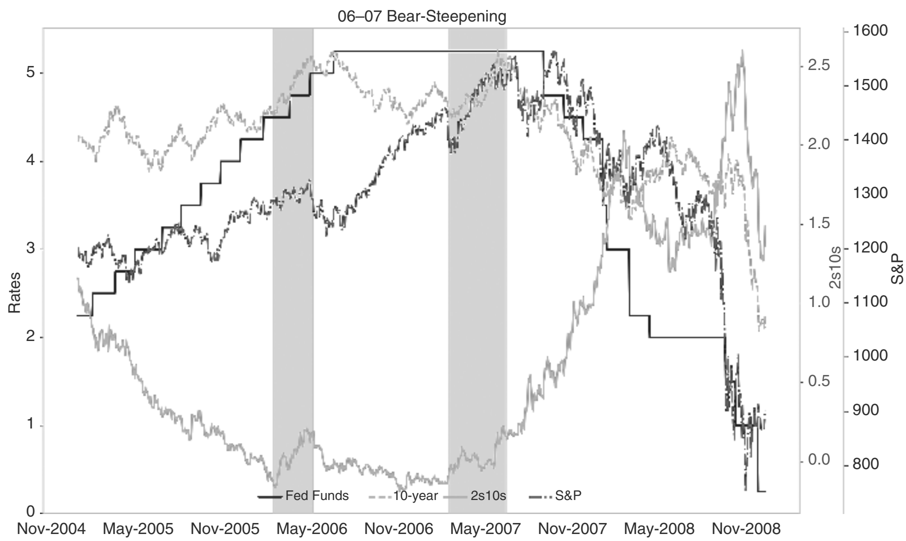


FIGURE 5.23 Bear steepening late cycle is typically small.
Source: Bloomberg and authors' calculations.

and then in 2023 (93bp). The moves in 2006 and 2023 largely stopped when the Fed turned dovish. In 2007, it was weaker equities that put an end to the bear steepening. The 2006 and 2007 case is illustrated in Figure 5.23.

The 2023 episode was larger than in the past because it had a strong supply flavor to it and will be discussed in more detail later. However, we note that in the past, the supply impact has been very inconsistent. Analyzing past data, using changes in yearly supply to explain the steepness of the US curve, does not suggest a strong relationship between supply and the shape of the curve, as per Figure 5.24a. The relationship is much stronger between supply and swap spreads, as per Figure 5.24b, as has been demonstrated by Mathai et al. (2025). The underlying logic is that supply often rises during recessions, when investors want to own bonds, and when the Fed cuts, which weakens the correlation. Swap spreads abstract from such monetary policy or growth

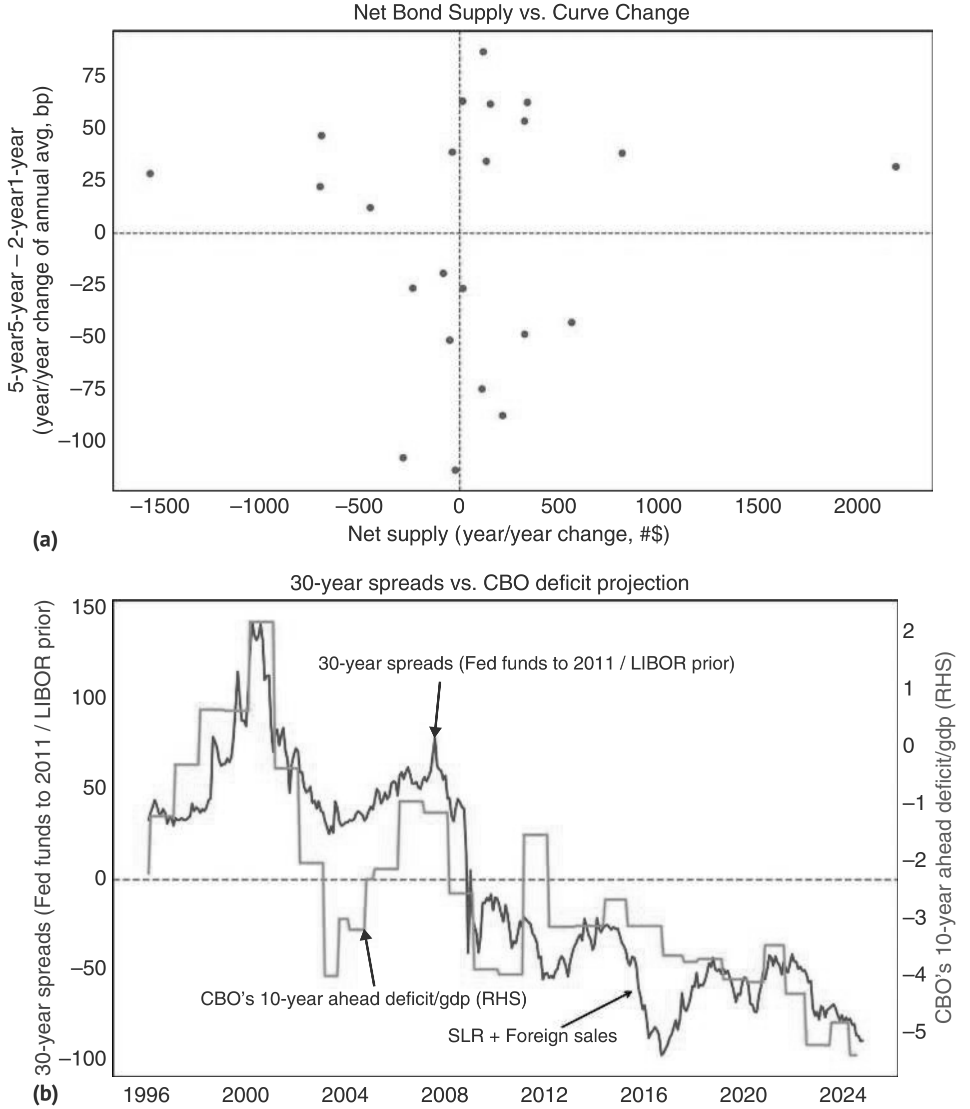


FIGURE 5.24 (a) No clear relationship between annual changes in net supply and term premium. (b) Swap spreads more clearly move with deficits.
Source: Citi research.

issues and capture demand and supply more directly through the cycle. As such, the tighter correlation of supply with swap spreads is not too surprising. Having said that, it appears quite likely that going forward, supply will become more of a market driver, if only episodically.

The Bear Steepening of 2023

The 2023 episode, when the market started to worry about supply, deserves further attention. The basic problem was that the fiscal largesse during COVID was not reversed in its aftermath, leading to significant increases in debt levels and annual supply in many countries. Going forward, it is feared that the fiscal response during COVID could be repeated when the next crisis hits, given that the repercussions from large fiscal deficits have been somewhat limited. Therefore, there may be important trading lessons from the 2023 episode for the future.

In Q3 of 2023, supply fears were the dominant narrative for the bear steepening, which was the largest on record during a time when the Fed was almost or fully done with its hiking cycle. To be precise, 2s/10s bear steepened by 93bp in Q2/Q3 of 2023, while the previous record bear steepening at a similar point in the monetary policy cycle was only 36bp. The move started on July 31, 2023, when the Treasury increased its marketable borrowing estimates (MBE) for the July-September quarter by 280bn USD. Subsequently, supply fears and a buyer strike of UST buyers remained the main discussion point to explain the price action. Although our analysis above suggests little evidence that supply mattered very much in the past, the question is whether there could be a threshold effect with respect to the size of the US debt load, or the extent of annual UST supply, when additional supply will impact pricing and bear steepen the curve more significantly. And it is indeed plausible that we have reached debt levels where supply will matter going forward. However, we note that in 2023, the supply concerns disappeared almost as quickly as they came. All it took was for the US Treasury to suggest in its November 1, 2023, TRA (Treasury refunding announcement) that they would raise coupons for only one more quarter and that such coupons would then stay on hold for the rest of the year, for the market to price out supply fears, as can be seen in Figure 5.25. Maybe even more importantly, the announcement of the borrowing plan by the UST came at the same time when Fed Chair Powell turned more dovish, which made the strategy to hide out in T-Bills less tenable for investors. Supply fears seem to matter mostly when the curve is inverted and the Fed is not close to cutting interest rates. This suggests that we will only find out whether we have indeed crossed some threshold where supply fears will become more prevalent when the Fed turns hawkish again.

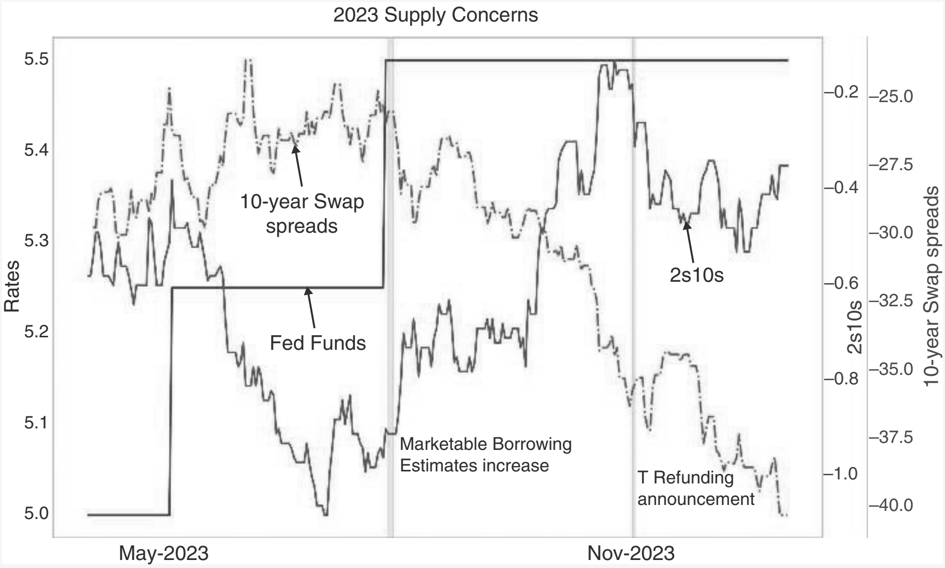
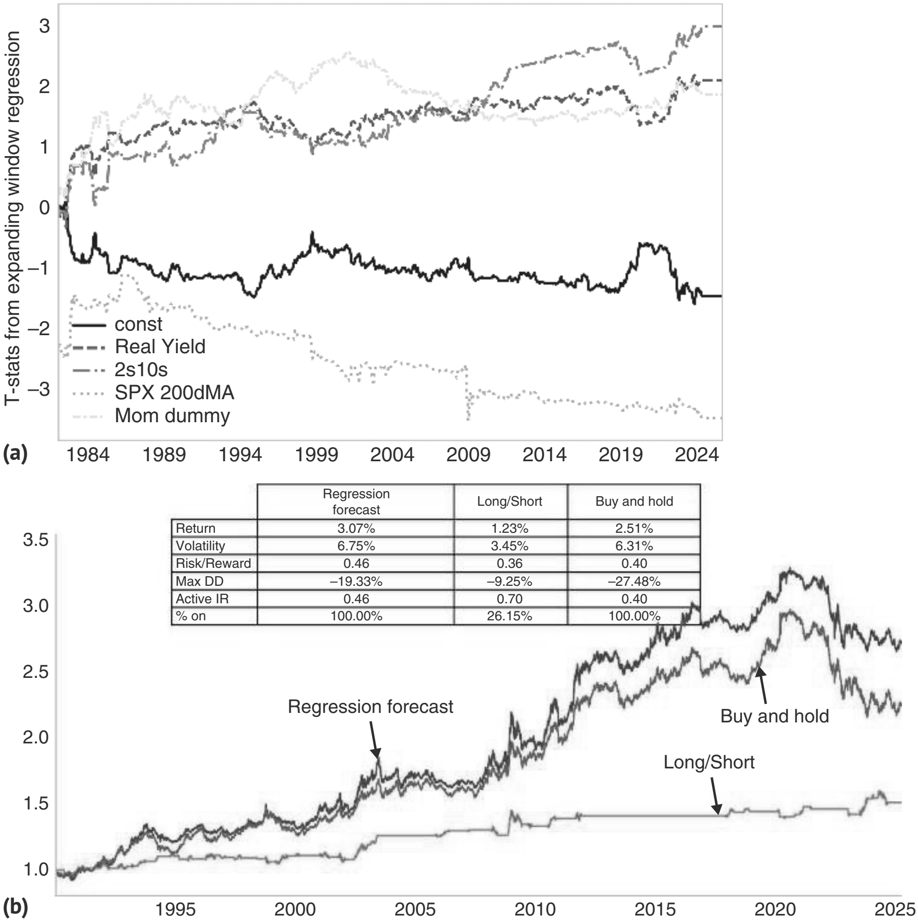


FIGURE 5.25 The bear steepening of 2023.
Source: Bloomberg.

5.9 VALUATION—USE SPARINGLY

There are different ways to think about valuation in the rates space. One way is to consider real rates relative to the neutral real rate. However, given the uncertainty in estimating neutral, often the underlying assumption is that the neutral real rates are mostly constant or, at the very least, slow-moving. This allows practitioners to use outright real rates, possibly relative to recent history, as a measure of value. Another measure is the slope of the curve. While there is a lot of literature trying to split the slope into expected monetary policy and term premium, in practice, changes in most term premium estimates are very similar to changes in the slope. This allows practitioners to use the slope directly. Both factors have played a role in traditional econometric models of bond returns. A practitioner model that uses those concepts is Ilmanen (1997). This model uses several simple variables to predict returns for the following month. One is the slope of the yield curve, with steeper slopes symptomatic of elevated risk premia, higher potential carry on bond investments, and therefore higher expected returns. We show the univariate forecasting relationship between the slope and returns and the t-statistics from an expanding window regression in Figure 5.26a. We are regressing monthly returns on the observed variables at the previous month's end. The t-statistic is consistently positive. The same is true for the level of real rates (yields minus lagged CPI). The model also uses stock market performance relative to



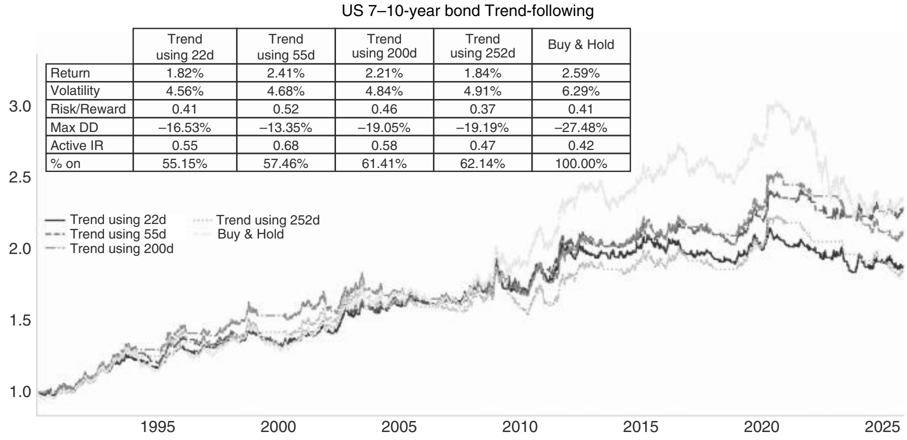
	Regression forecast	Long/Short	Buy and hold
Return	3.07%	1.23%	2.51%
Volatility	6.75%	3.45%	6.31%
Risk/Reward	0.46	0.36	0.40
Max DD	–19.33%	–9.25%	–27.48%
Active IR	0.46	0.70	0.40
% on	100.00%	26.15%	100.00%

FIGURE 5.26 (a) Valuation variables like real rates and steep curves significant for forecasting bond returns. (b) But value added is relatively small.
Source: Bloomberg and authors' calculations.

trend by taking the deviation from the 200d MA, which either proxies the strength of the economy or works through a wealth effect and harms future expected bond returns. However, we would not place too much emphasis on valuation-heavy strategies to trade bonds. According to Figure 5.26b, the information ratio using these forecasts is only slightly above the buy-and-hold strategy, mainly because volatility increases in line with returns.

5.10 TREND FOLLOWING—ONLY IN THE SHORT TERM

With valuation only modestly improving buy-and-hold strategies in bonds, the next question is whether trend following/momentum works any better. To answer this question, we look at very basic moving average models. While there are many smart ways



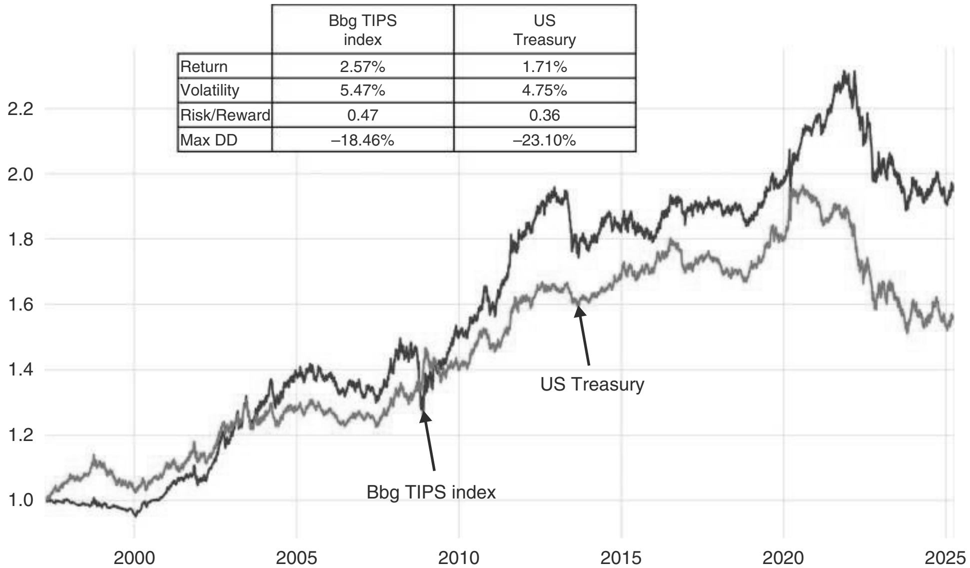
	Trend using 22d	Trend using 55d	Trend using 200d	Trend using 252d	Buy & Hold
Return	1.82%	2.41%	2.21%	1.84%	2.59%
Volatility	4.56%	4.68%	4.84%	4.91%	6.29%
Risk/Reward	0.41	0.52	0.46	0.37	0.41
Max DD	–16.53%	–13.35%	–19.05%	–19.19%	–27.48%
Active IR	0.55	0.68	0.58	0.47	0.42
% on	55.15%	57.46%	61.41%	62.14%	100.00%

FIGURE 5.27 Shorter-term moving averages work well.
Source: Bloomberg and authors' calculations.

to improve upon this basic setup, even a simple version is good enough to show that trend following works somewhat better than valuation. This is illustrated in Figure 5.27. Both the 22dma and the 55dma raise the active information ratio by a somewhat more meaningful 0.13 and 0.26, respectively, although the 252dma only breaks even with buy-and-hold. We are not overly surprised that the longer moving averages do not work as well as in stocks, for example. After all, higher stock prices are self-reinforcing, while higher rates are self-limiting. This is true because, ceteris paribus, higher rates lead to a slowdown, which would lead to lower rates. This likely makes trend-following strategies in bonds more successful using shorter moving averages.

5.11 TIPS—THE TRUE TREASURE

We started the chapter by calling Treasuries a treasure, due to their history of strong returns and volatility-dampening correlation properties. However, the true treasure is not US Treasuries, but TIPS, i.e., the inflation-protected bonds the Treasury offers. This is illustrated in Figure 5.28, which shows the information ratios of the return of nominal Treasuries and TIPS since the inception of the TIPS market. Clearly, TIPS are the way to go in terms of information ratio, but also absolute returns and max drawdowns. However, we note that the correlation properties with equities are not quite as strong as for nominals. The reason is two-fold. First, many SPX bear markets are deflationary in nature, where oil is falling, which means that breakevens are falling, and TIPS underperform US Treasury in a time of need (though the opposite holds true during the less common inflationary shocks). The second issue is illiquidity, which can lead to painfully large technically driven sell-offs when liquidity is shrinking, which also happens frequently during a crisis. Although such episodes were short-lived in both 2008 and 2020, it nonetheless somewhat undermines the long-term case for TIPS, at least for investors who care about monthly returns.



	Bbg TIPS index	US Treasury
Return	2.57%	1.71%
Volatility	5.47%	4.75%
Risk/Reward	0.47	0.36
Max DD	–18.46%	–23.10%

FIGURE 5.28 TIPS the better deal.
Source: Bloomberg and authors' calculations.

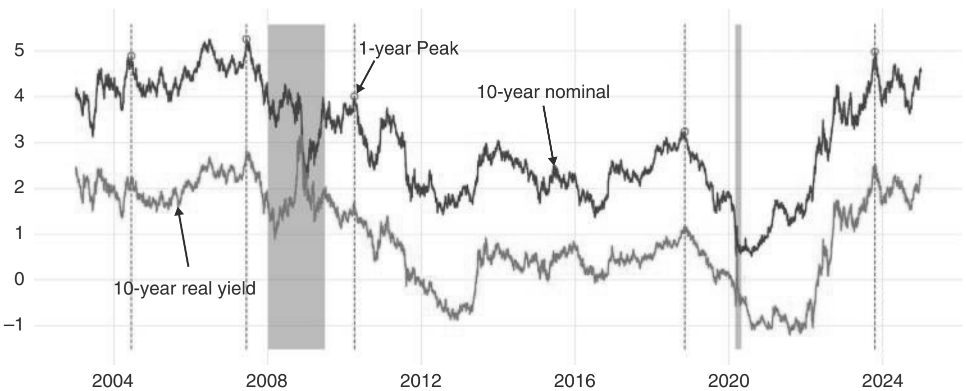


FIGURE 5.29 Real rates peak when nominal rates peak.
Source: Bloomberg and authors' calculations.

With those long-term asset allocation considerations out of the way, we quickly want to discuss how to trade TIPS. The first thing to realize about TIPS is that they are essentially duration products, and as such, they should be traded like nominal rates. Usually, real rates peak at exactly the same time as nominal rates, as can be seen in Figure 5.29. This is especially important to realize during inflation spikes. When high inflation is expected, investors often switch from nominal rates to real rates. However, this will not fully protect portfolios. Although it is true that losses will be smaller in real rates than in nominal rates when inflation increases, there will still be losses.

This was also illustrated in 2022. As soon as the Fed started to hike interest rates, pushing nominal rates higher, TIPS also started to lose money. High inflation was unable to overcome the headwind of higher nominal yields, other than at the very short end of the curve.

For true protection during inflation spikes, it is therefore better to trade breakevens than real rates, as breakevens can generate alpha independent of the general US rates view. As can be seen in Figure 5.30, the basic rule of thumb is that breakevens fall sharply into (deep) recessions, only to bounce back aggressively when markets smell the end of the downturn. The two recessions, for which we have TIPS data, were so deep (GFC and COVID) that the Fed cut rates to zero, pushing breakevens to close to zero or into negative territory. Outside of (deep) recessions, breakevens also start to rise well before the first hike and into the very early parts of the hiking cycle. When the Fed starts to hike in earnest, breakevens start to fall. During easing cycles going into a soft landing (2024) breakevens tend to widen.

The illiquidity of linkers is actually an advantage when trading breakevens. When real rates sell off during times of severe stress due to illiquidity, it happens at the same time when nominal rates rally. Therefore, breakevens become artificially depressed. The spikes in real rates are actually the reason why breakevens collapsed as much into the last Fed cut in 2008 and also in 2020. This feature is a great advantage when trading breakevens over trading linkers. In cutting cycles into hard landings, investors often position for breakevens to fall. If this gets amplified due to a rising illiquidity premium of linkers, that is helpful to the trade.

During more normal liquidity conditions, linkers have a big oil component given that they are linked to headline inflation, not to core. The oil impact is especially significant at the front end of the curve, but it is surprising that even the back end of the breakeven curve has a strong oil component. This is illustrated in Figure 5.31a. But it is not just oil that is relevant. It is the general business cycle that is important. This is why breakevens have a strong correlation with copper as well, as per Figure 5.31b.

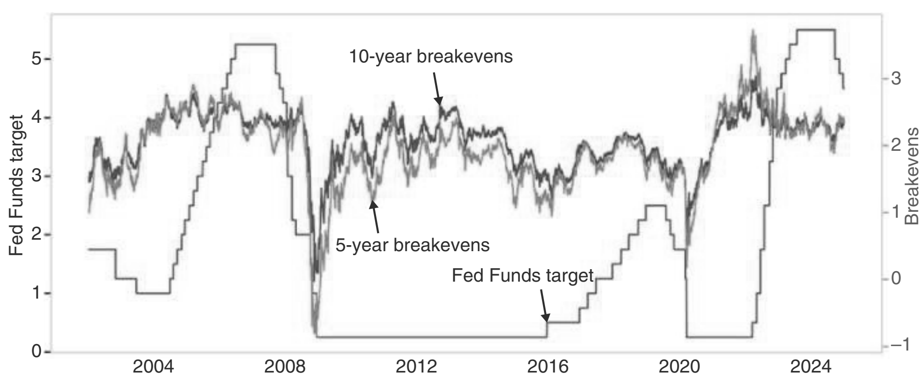


FIGURE 5.30 Breakevens collapse into recessions, then rebound quickly.
Source: Bloomberg.

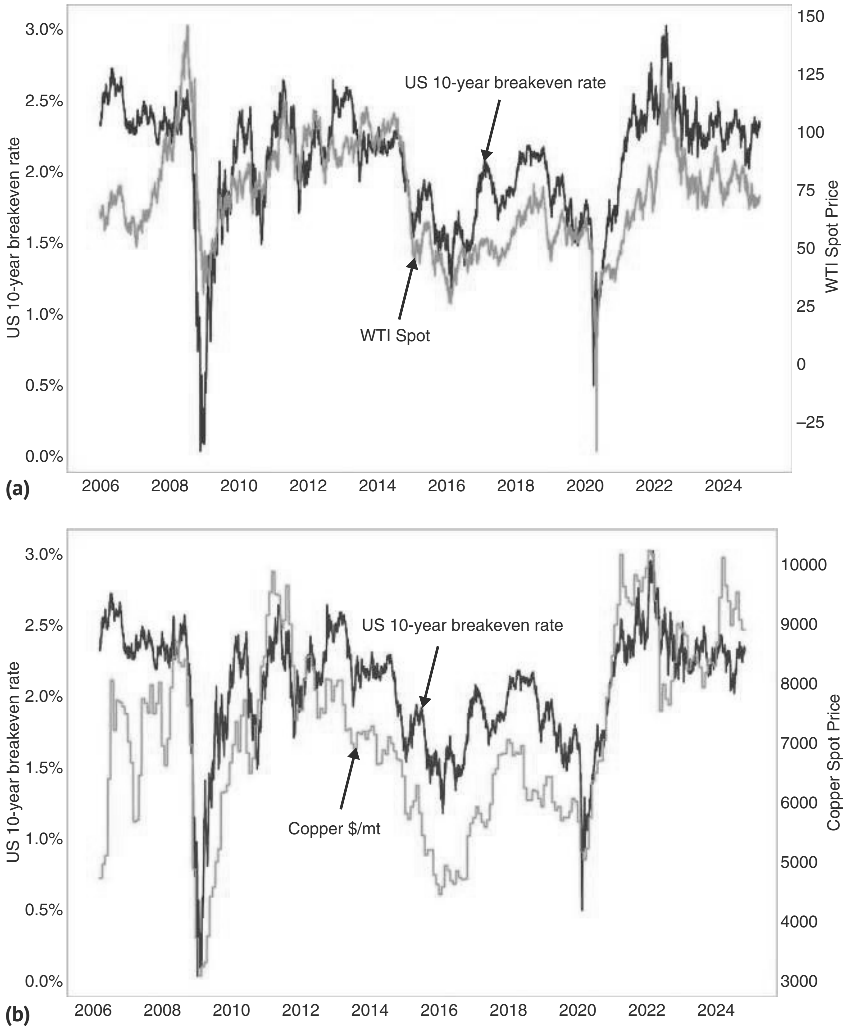


FIGURE 5.31 (a) Oil an important driver for breakevens. (b) Copper correlated to breakevens, too.
Source: Bloomberg.

Using these fundamental factors in a simple trading rule works quite well. Long the TIPS index versus the nominal US rates index has a positive IR of 0.23. The TIPS index is more volatile despite having a lower duration (averaging 4.95 since 1998 versus 5.7 for the Treasury index). Being overweight TIPS against nominals only when oil is above its 55dma (lagged one day), raises this to 0.63. Using the same strategy with copper rather than with oil improves the information ratio by even more, moving it to 0.81, as is shown in Figure 5.32. This is likely the case because our TIPS index has a duration of 4.5 years, and at that duration, the business cycle impact dominates over the direct price impact of oil, which is more relevant for the short end of the curve.

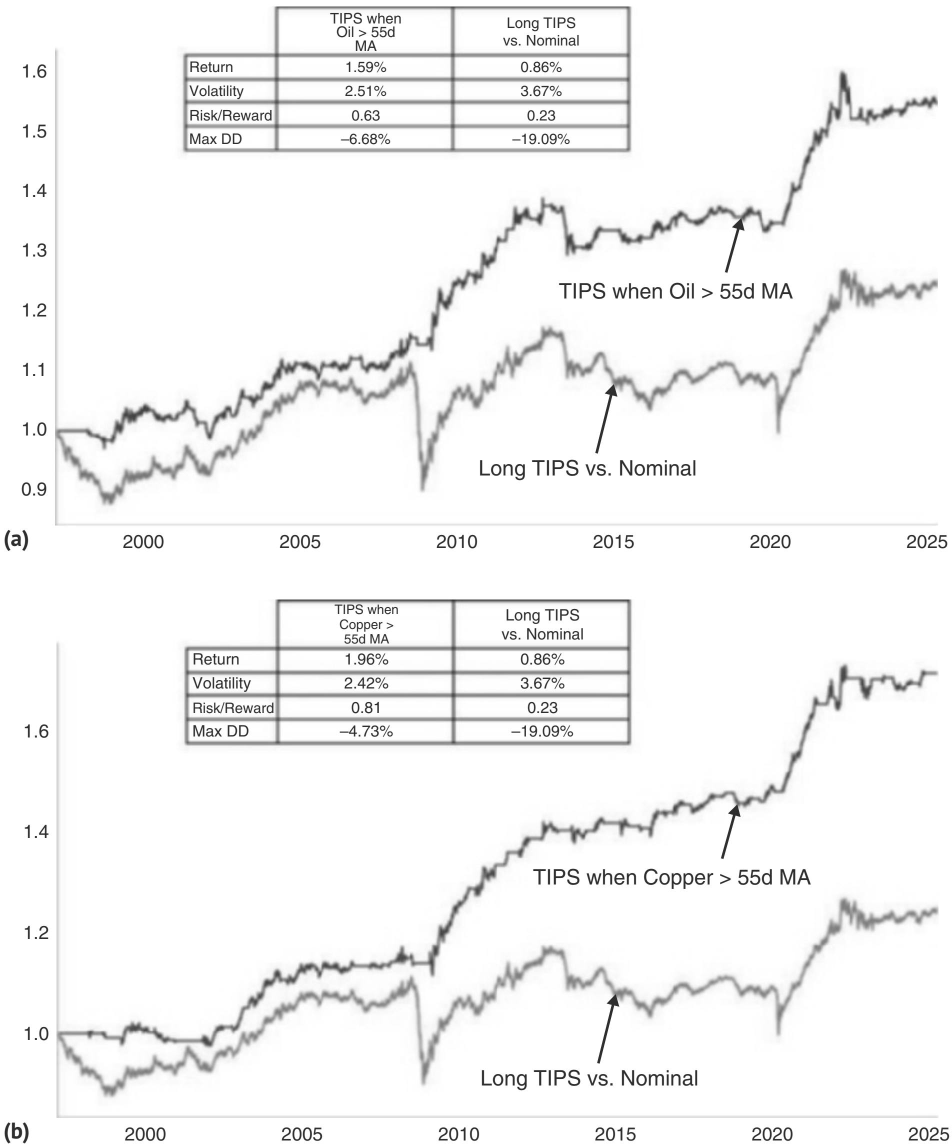


FIGURE 5.32 (a) Trading TIPS with oil almost triples IR. (b) Copper works even better.
Source: Bloomberg and authors' calculations.

5.12 WHEN RATES MARKETS ROAR

Before wrapping up this chapter, we want to share a quick observation on the interaction of the business cycle with interest rate volatility. Although bonds are often added to portfolios for their volatility-dampening characteristics, rate volatility can spike aggressively, which can serve both as a signal for other markets and can also be traded

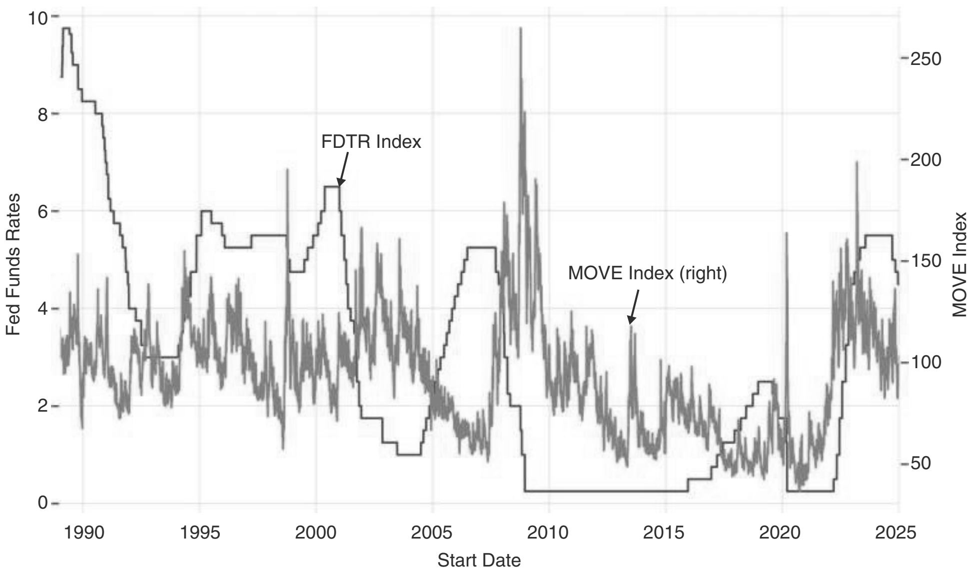


FIGURE 5.33 FI vol remains quiet in late part of hiking cycle.
Source: Bloomberg.

directly. With respect to the former issue, rate volatility often becomes a problem for risky assets when it spikes significantly. Interestingly, rate volatility is somewhat predictable, and more so than FX or equity volatility, which is more directly impacted by unforeseen shocks. Rate volatility is, to an important extent, driven by where we are in the Fed cycle. In particular, it tends to spike around the first cut, the last cut, or the first hike. Accelerations of the hiking or easing cycle by the Fed are also good candidates, as can be seen in Figure 5.33. Going into the final part of the hiking cycle and during the pause between the last hike and the first cut, volatility is typically relatively well-behaved. Of course, it is very difficult to know ex ante when the first cut is coming, but it is usually clear when the Fed is entering the final part of the hiking cycle. Knowing that fixed income volatility is likely well contained at that point is useful for many strategies, including the EM carry strategies.

5.13 SUMMARY

US Treasuries have strong long-term returns and information ratios. Studying Fed cycles, easing cycles are shorter and sharper than hiking cycles. Average excess returns over three months T-bills are negative during periods of tightening and positive during periods of easing, as well as during periods when the central bank is on hold post-easing cycles. We investigate how to trade Fed cycles. Returns of two year receivers are negative starting around three months ahead of the first hike. Payers typically continue to work until the hiking cycle is quite mature, only starting to lose money when the last hike is in. In easing cycles, receivers begin to work around the last hike. Three months ahead of the first cut, receivers work and countertrend selloffs are getting quite small.

The easy receiving trade ends with the last cut. For soft landings, rates trade well into the first cut, but post-cut, the risk-reward for receivers gets significantly worse. The strength of the SPX at the start of the easing cycle is a good predictor of how aggressive the easing cycle will be. We also discuss the shape of the curve with respect to the Fed cycle. Supply pressures are better traded in swap spreads than in the curve shape. Valuation in bonds works, but the value added is relatively small. Trend following works better than valuation. We also discuss TIPS.

CHAPTER 6

How to Trade Rates: The Tactics

Having laid out how to trade US rates strategically in line with the Fed cycle, we now move on to investigate a few more tactical frameworks. We start with a focus on how to call for a peak in rates, and then go through some of the major economic events that can generate good entry points into rates trades.

6.1 THE ANATOMY OF A RATES SELL-OFF

Trying to identify peaks in the rate cycle can be more rewarding than troughs, largely because rates typically retrace very quickly from peaks, often delivering "instant gratification," while troughs can be quite protracted affairs. As laid out in the previous chapter, we strategically look for an imminent end to Fed hiking cycles. Here, to fine-tune entry points, we study the average sell-offs to find patterns around peaks in yields. Table 6.1 uses dates from Plata-Vazquez and Mathai (2023), showing peaks after sell-offs of more than 70bp, as is visualized in Figure 6.1. Going back to 2000, we find 18 examples, ranging from 70bp to 214bp, with a median sell-off of 120bp over 135 days.

There are a few patterns that emerge in terms of leading indicators for a peak. First, in the median case, two-year breakevens start to fall a month before the peak in 10-year rates, as can be seen in Figure 6.2a. Furthermore, while 2s/10s typically peak at a very similar time to 10s, 5s/30s peak meaningfully earlier, as 30s often do not keep pace with the sell-off in shorter maturities. This flattening then reverses itself after the peak, as can be seen in Figure 6.2b. The 2023 episode was an exception, as the sell-off was more supply-driven than Fed cycle-driven.

DXY and WTI are mostly coincident, as per Figure 6.3a and b, respectively. Although neither will help investors identify a peak in US rates, this finding is nevertheless interesting. While DXY is driven mostly by rates and, therefore, is not overly informative, oil has been an important contributor to some of the US rates sell-offs under study. In the median case, oil rallied more than 20% in the three months going into the peak in yields. Therefore, a big break in oil would likely also result in a peak in rates.

It is also interesting to analyze what is *not* happening during those rates sell-offs. In particular, the SPX does *not* trade lower into it, but overall trades mostly sideways to moderately higher throughout the rates sell-off. However, under the hood, there is an equity impact. For example, real estate, relative to the S&P trades quite poorly

TABLE 6.1 10-year Rate Sell-offs

Start date	End date	Length	10-year sell-off	FF change
03-23-2001	05-15-2001	53.00	0.70	–1.00
11-07-2001	04-01-2002	145.00	1.22	–0.25
06-13-2003	09-02-2003	81.00	1.48	–0.25
03-16-2004	05-13-2004	58.00	1.15	0.00
06-01-2005	11-04-2005	156.00	0.75	1.00
01-17-2006	06-28-2006	162.00	0.91	0.75
03-02-2007	06-12-2007	102.00	0.74	0.00
03-17-2008	06-16-2008	91.00	0.91	–1.00
12-30-2008	06-10-2009	162.00	1.87	0.00
10-07-2010	02-08-2011	124.00	1.34	0.00
05-02-2013	09-05-2013	126.00	1.32	0.00
01-30-2015	06-26-2015	147.00	0.81	0.00
07-08-2016	03-13-2017	248.00	1.25	0.25
09-07-2017	10-05-2018	393.00	1.18	1.00
08-04-2020	03-31-2021	239.00	1.22	0.00
12-03-2021	06-14-2022	193.00	2.14	0.75
08-01-2022	10-24-2022	84.00	1.65	0.75
05-03-2023	10-04-2023	154.00	1.35	0.50
02-01-2024	04-25-2024	84.00	0.83	0.00
09-16-2024	01-13-2025	119.00	1.16	–1.00
		146.05	1.20	0.07
		135.50	1.20	0.00

Source: Bloomberg and authors' calculations

throughout the rates sell-off, only to bottom on the day of the rates peak. Furthermore, the cyclicals/defensive ratio peaks with US rates and pulls back in the aftermath of the US rates peak. This is shown in Figures 6.4a and b, respectively. While both indicators are mostly coincident (and cyclicals/defensives actually slightly lagging) and therefore less helpful in identifying peaks in rates, observing one or more of them rolling over during a sharp rates sell-off can act as a warning signal that the rates move is getting mature.

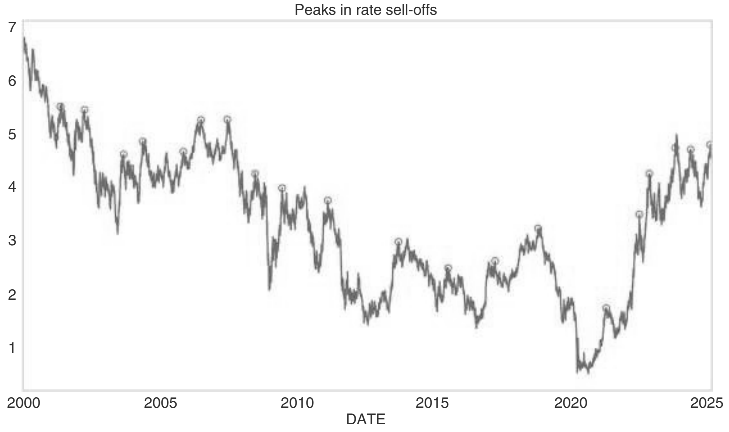


FIGURE 6.1 70bps sell-offs with end highlighted.
Source: Bloomberg and authors' calculations.

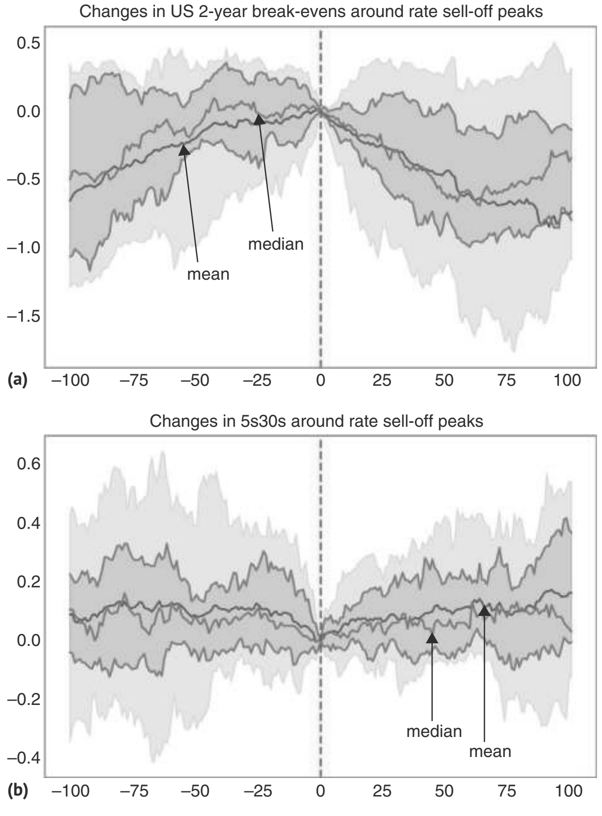


FIGURE 6.2 (a) Breaks peak slightly before USTs. (b) 5s30s tend to flatten before the peak in 10s.
Source: Bloomberg and authors' calculations.

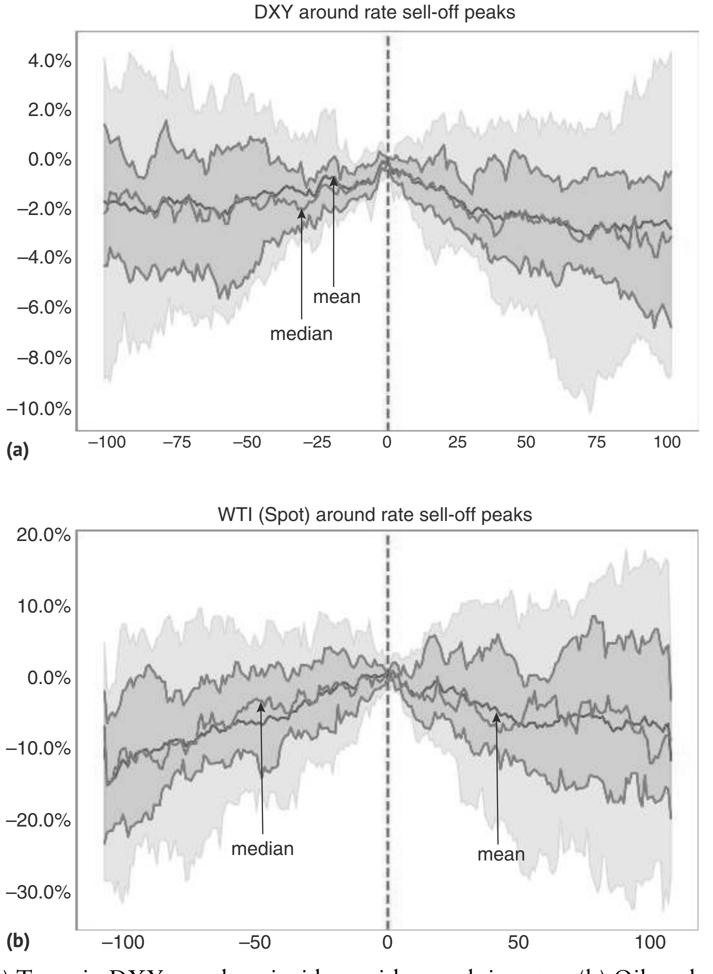


FIGURE 6.3 (a) Turns in DXY mostly coincident with a peak in rates. (b) Oil peaks coincident, too.
Source: Bloomberg and authors' calculations.

The cyclical versus defensive indicator deserves a longer discussion. Although our study above suggested that in the median case, there is a small lag for cyclicals/defensives, the lead/lag structure is extremely variable. In some of the major cycle peaks over the last decades, cyclicals/defensives were actually leading. Importantly, the indicator is mostly relevant when we are facing growth issues. If bond yields are rising due to a major inflation spike, the rule breaks down, as is illustrated in Figure 6.5. This has been the case in 2022. We would therefore disregard this ratio during high-inflation scenarios. As cycles with high inflation are rare, we do think that cyclicals/defensives should be part of the indicators consulted when positioning for a change of trend in US rates.

We operationalize this observation with our standard simple rule: if returns for cyclicals versus defensive are below their 55-day moving average, go long the 10-year

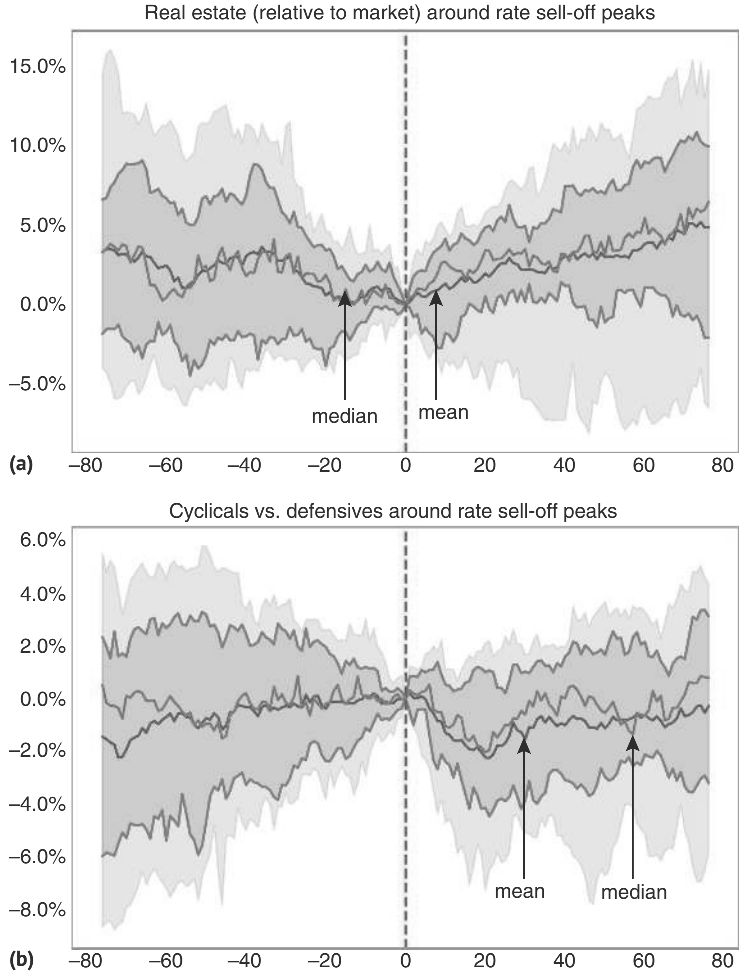


FIGURE 6.4 (a) Real estate troughs with the rates peak. (b) Cyclicals/defensives sell-off slightly after the rates peak.
Source: Bloomberg and authors' calculations.

future with a one-day lag. The results are shown in Figure 6.6. The risk-adjusted returns are higher with an active information ratio of 0.85, while the unconditional information ratio is coming in at 0.48. The improvement comes mostly from lower volatility when the signal does not take an active treasury future position. The rule protected investors only very partially against the inflation-driven pullback in 2022, though.

The next question to address is what news flow can lead to a change in trend in US rates. The main candidates are central bank meetings and economic data. Given the Fed's objective function, the key data points are nonfarm payrolls for the labor market and the CPI (published before the PCE), on the inflation side. We start with central bank meetings.

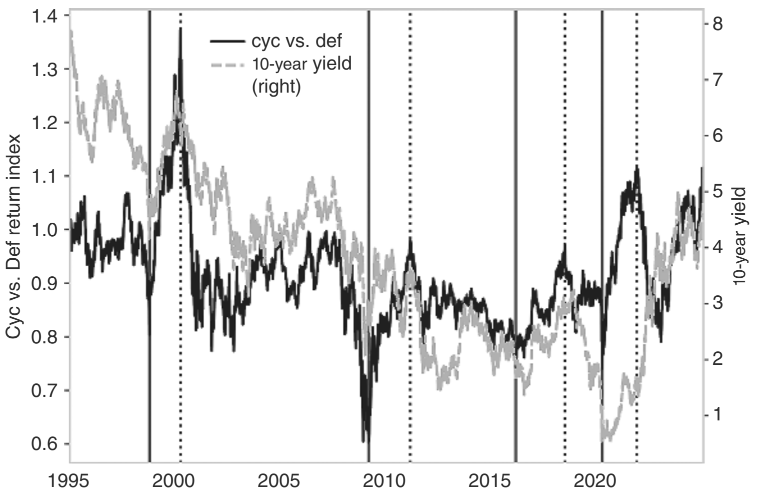
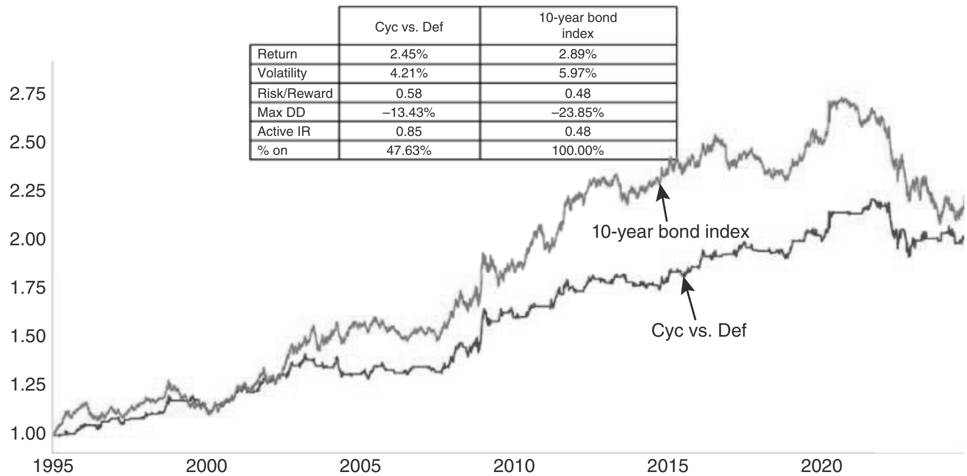


FIGURE 6.5 Cyc/def can lead rates—outside of inflation shocks.
Source: Bloomberg.



	Cyc vs. Def	10-year bond index
Return	2.45%	2.89%
Volatility	4.21%	5.97%
Risk/Reward	0.58	0.48
Max DD	–13.43%	–23.85%
Active IR	0.85	0.48
% on	47.63%	100.00%

FIGURE 6.6 Cyc/def a good trading tool for US Treasuries.
Source: Bloomberg and authors' calculations.

6.2 THE WISDOM OF THE FOMC

Oftentimes, it is not obvious how to score a central bank meeting, partly because there can be discrepancies between statements and press conferences, and partly because the press conferences themselves are long and can offer something for both hawks and doves. Furthermore, the question is always dovish or hawkish relative to what and where market participants may hold different priors going into the meeting. We think the best way to score a FOMC meeting is in line with the price action at the front end of the curve. Two year rates work well, as there is typically a reaction even at times when it is quite clear that no move on rates is imminent in the near term. Here we test we test rules where we receive/pay rates when there is a positive/negative market reaction on a FOMC day. We focus on a move on FOMC day that is at least 2 standard deviations using trailing one-month data.

Starting on the dovish side results in relatively large effects. Median returns are very positive and persistent after two-year rates fall by 2 standard deviations, as can be seen in Figure 6.7. Even in the worst 10th percentile, going with receivers after dovish FOMC decisions is not losing money 20 days later. Some of the dovish surprises happen late in the hiking cycle or as the easing cycle comes closer, and going with FOMC momentum may help to capture those trades. Fixed income returns are also attractive going into those meetings, though that part of the rally can be harder to capture. Interestingly, even during hiking cycles, there is some follow-through on receivers after FOMC day, maybe because those meetings come close to the end of the hiking cycle in question. In this case, the returns are less pronounced, though. Events that occur in periods when the Fed is on hold are mostly going sideways.

On the other side of the ledger, large sell-offs on FOMC days also tend to have follow-through, as per Figure 6.8. Interestingly, this is even clearer during easing cycles, probably because hawkish FOMC meetings come as a surprise and suggest either pauses or an imminent end to the easing cycle in question. The sample for hawkish surprises during easing cycles is small, though, with only two data points. One is the last cut in the 2003 cutting cycle, and one was the cut in March of 2008, probably because the Fed reduced the pace of the cuts in the subsequent meeting.

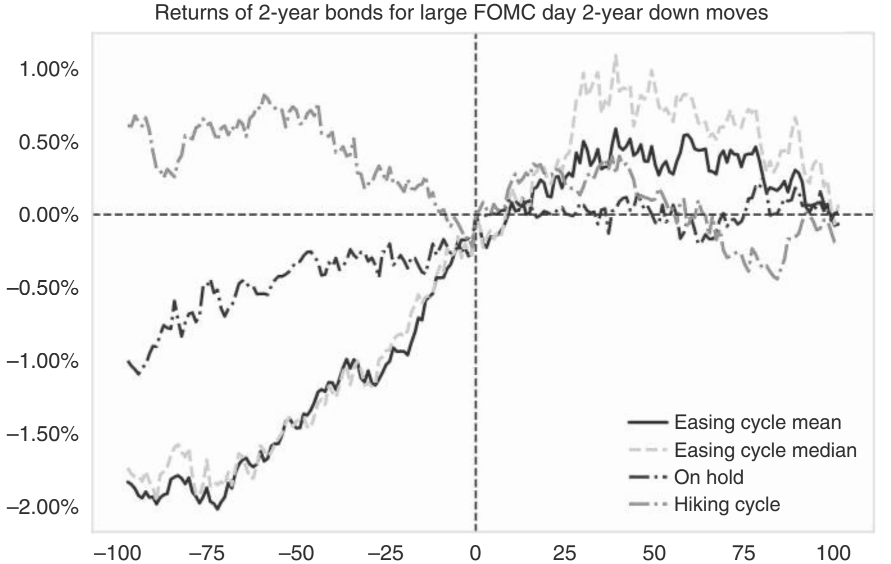


FIGURE 6.7 Large downward moves on FOMC day have follow-through.
Source: Bloomberg and authors' calculations.

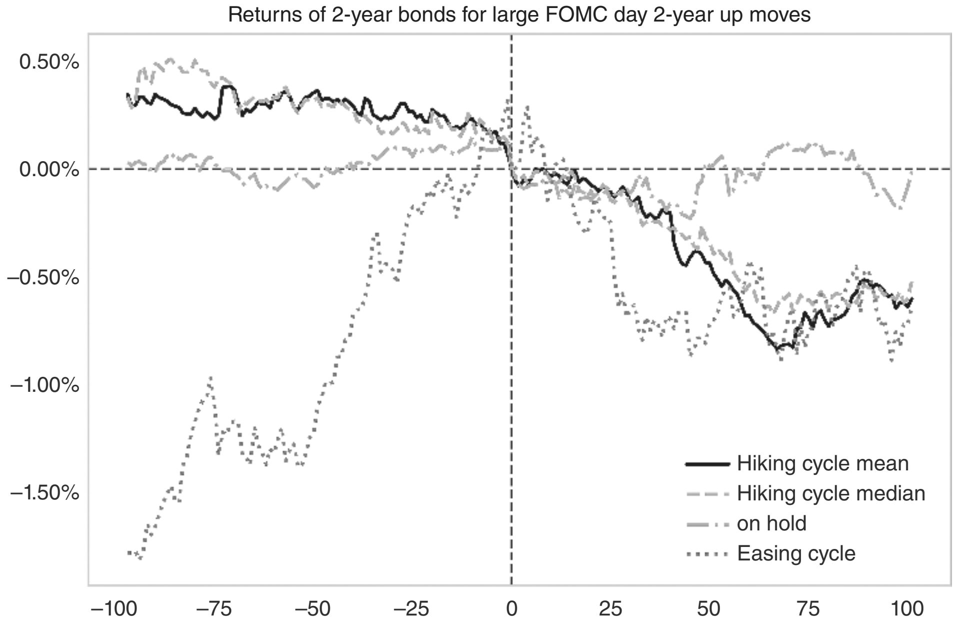


FIGURE 6.8 Large upward moves on FOMC day have follow-through, too.
Source: Bloomberg and authors' calculations.

6.3 TRADING THE NONFARM CATALYST (NFC)

We now apply the same idea to NFP prints. Generally, the impact is not as consistent as for the FOMC, which makes sense given that NFP is just one input for the FOMC and given that economic data can be quite noisy. Still, in the more distant macro past, momentum on NFP days used to have decent follow-through. At this stage, the pattern is more muted, but still noticeable. We find that bullish momentum on NFP day is more likely to be sustained than bearish.

On the bullish side, i.e. yields falling, we again use 2 standard deviation moves to account for time-varying volatility. We then invest in the 10-year US bonds (in excess of cash, as usual). We find 45 events (of 660 months in the study). Figure 6.9 shows that there tends to be a small amount of follow-through over the next few days. The more extreme the move, the greater the follow-through. Using 3 σ moves (13 events), we see median returns above 1% over the next 20 trading sessions (not shown). Rallies during Fed easing cycles are meaningfully larger than in the median case, suggesting that large rallies on NFP days during hiking cycles should be treated much more carefully, while large rallies on NFP days during easing cycles can be used as entry points.

On the bear-side, i.e., NFP days with significantly rising yields, we find more observations, 53 and 25, respectively. Follow-through is less pronounced, though, as shown in Figure 6.10. The pattern during easing cycles is not worse than in the median case, suggesting that even during hiking cycles, bearish NFP prints are not a good signal to pay.

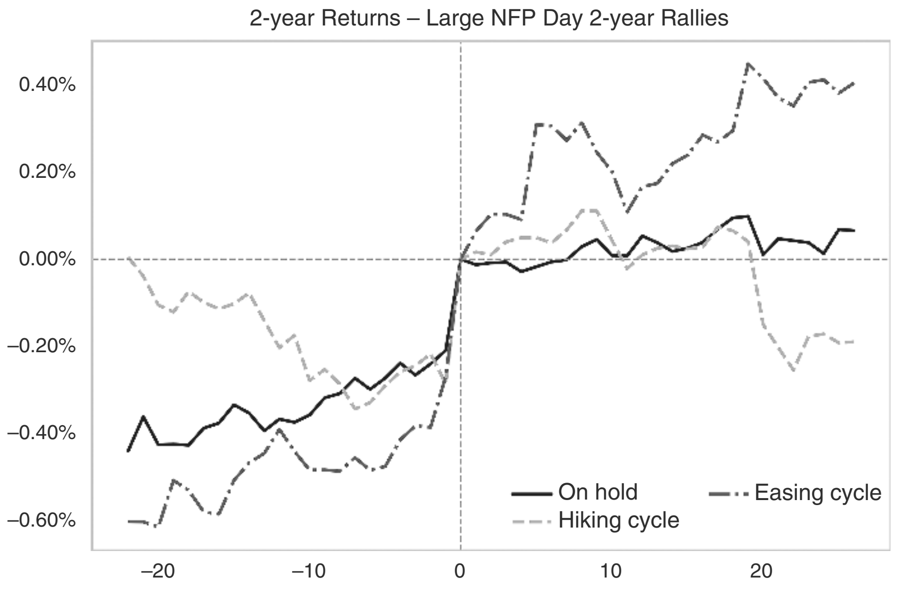


FIGURE 6.9 Large positive returns on NFP days with follow-through.
Source: Bloomberg and authors' calculations.

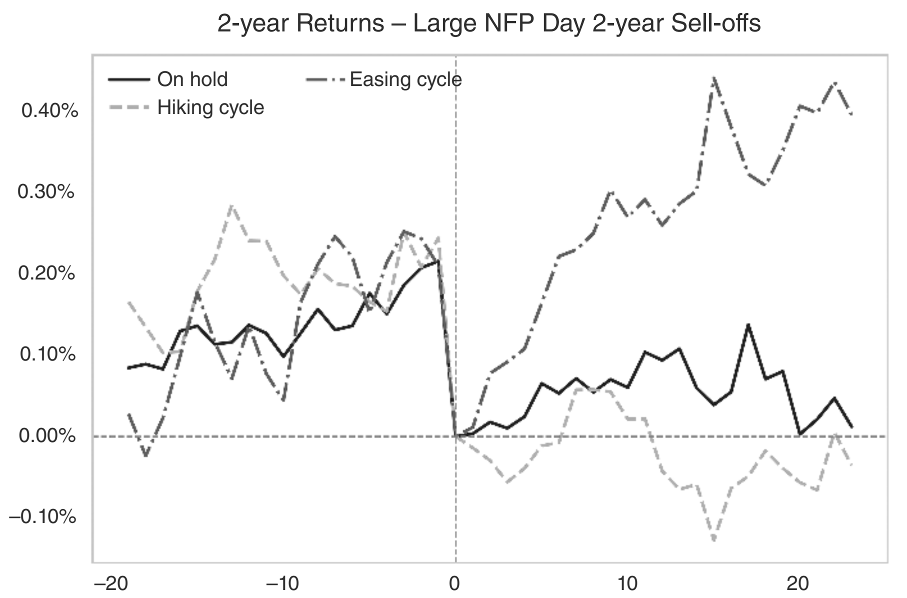


FIGURE 6.10 Rate sell-offs post-NFP moves have less follow-through.
Source: Bloomberg and authors' calculations.

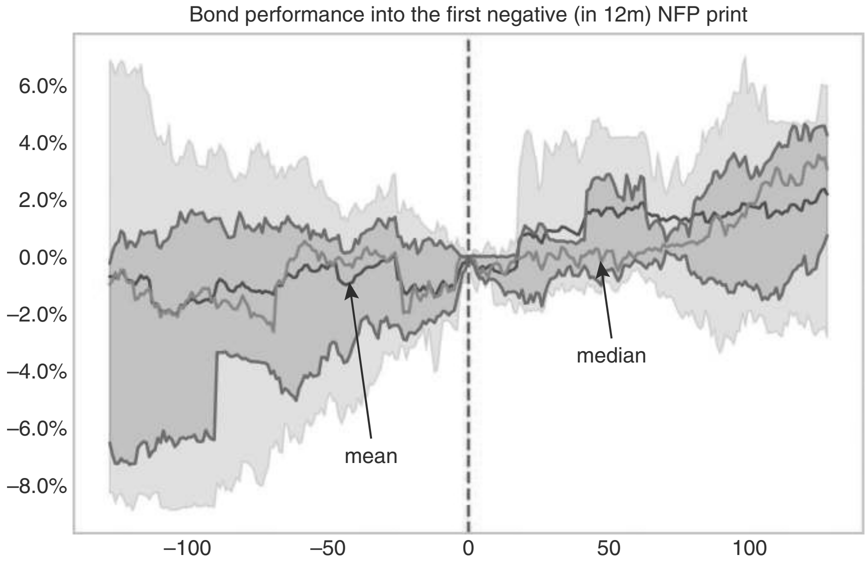


FIGURE 6.11 Negative NFP prints have follow-through.
Source: Bloomberg and authors' calculations.

Another bullish pattern emerges when we filter just for the first negative NFP print in a year, which is frequently associated with recessions. We only include the first event to occur after one year of positive prints. We have had 12 instances since 1974. Figure 6.11 shows the median returns for US Treasuries into and out of these events. Median returns are positive, but only after quite some choppiness. Performance is asymmetrically skewed to the upside, resulting in an even better (and faster) mean performance. While the bottom 25th percentile of returns is close to flat, the upper 75th percentile is at nearly 4%.

Payrolls are historically more volatile and can flip earlier than other labor market metrics. In fact, history shows that NFP at times turns negative for the first time in the cycle when the unemployment rate is still at the lows, as per Table 6.2. This makes NFP releases useful events to trade.

6.4 CESI—A BUSY MAN'S GUIDE TO THE ECONOMY

Having discussed NFP prints as an important economic indicator for trading rates, we now broaden the discussion by focusing on more generalized strength in economic data. We do this with the help of Citi's economic surprise index (CESI). The index is essentially a quick and dirty way to summarize the momentum of economic data versus expectations. Although originally designed for FX trading, with weights depending on how much FX tends to react to various releases, it also works reasonably well for rates. Figure 6.12 suggests a correlation between US economic surprises and rates, where CESI often leads.

TABLE 6.2 NFP Moves Earlier than Unemployment

Start Date	NFP actual	URate	Prior 12m low	Gap to low
March-1980	–140	6.3	5.6	0.7
June-1986	–89	7.2	6.7	0.5
July-1990	–219	5.5	5.2	0.3
April-1995	–9	5.8	5.4	0.4
March-1998	–36	4.7	4.6	0.1
September-1999	–8	4.2	4.2	0.0
September-2005	–35	5.0	4.9	0.1
August-2007	–4	4.6	4.4	0.2
September-2017	–33	4.3	4.3	0.0
March-2020	–701	4.4	3.5	0.9

Source: Bloomberg and authors' calculations

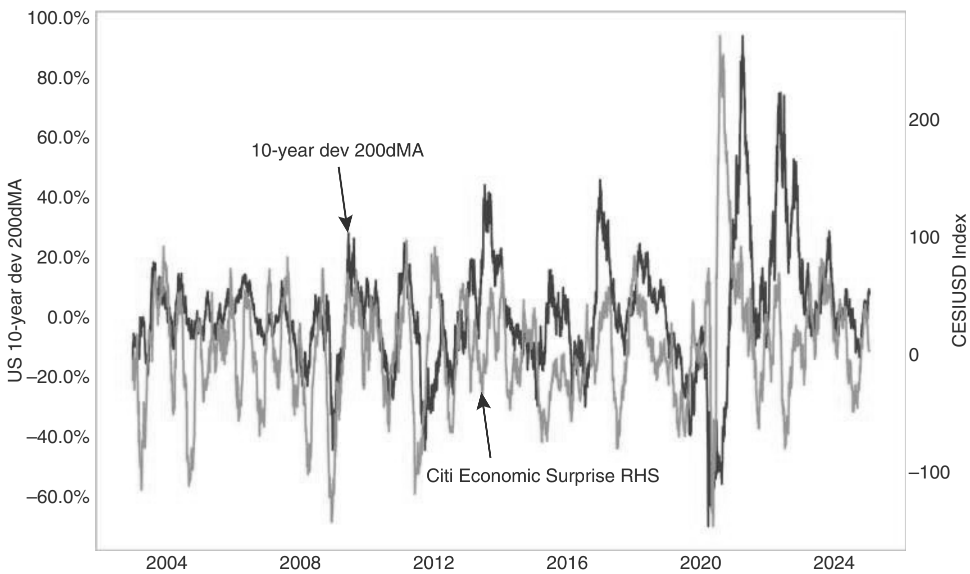
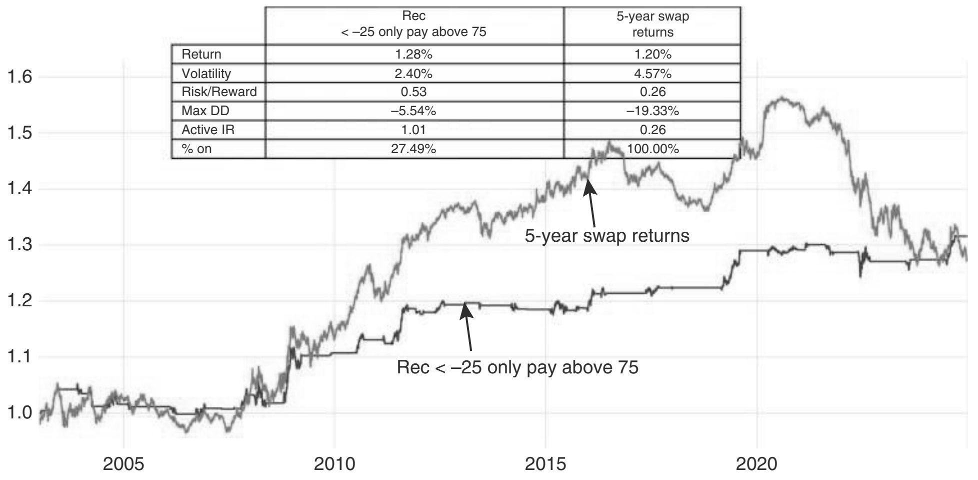


FIGURE 6.12 CESI and UST highly correlated.
Source: Bloomberg.



	Rec < –25 only pay above 75	5-year swap returns
Return	1.28%	1.20%
Volatility	2.40%	4.57%
Risk/Reward	0.53	0.26
Max DD	–5.54%	–19.33%
Active IR	1.01	0.26
% on	27.49%	100.00%

FIGURE 6.13 Using CESI to trade UST.
Source: Bloomberg and authors' calculations.

Somewhat frustratingly, the lead-lag structure between a turn in CESI and a turn in yields is not very stable. To see whether CESI adds any value, we analyze whether five-year US swaps can be traded using CESI extremes as a counter-indicator. This exercise (not shown here) did not result in much of a tradable strategy. Instead, it turns out that CESI works better as a trend indicator. In particular, receiving rates when CESI has already fallen to below –25, and paying when it has risen to above 75, raises the IR from 0.26 to 1.01, as per Figure 6.13. Symmetric signals of +/–75 gave equivalent results but were rarely activated, with investors missing out on capturing some of the bond risk premium. Given the recent bear market in fixed income, this surprise strategy has generated similar annualized returns despite only being active 25% of the time.

6.5 WHEN SAFETY BACKFIRES

An extremely interesting and concerning market regime is when US rates sell off at a time when risk aversion spikes. This phenomenon is quite frequent in emerging markets, where a spike in credit risk can lead to weaker bonds at a time of risk aversion. Although this regime is very rare in the United States, it can happen from time to time. Usually, it is related to liquidity problems, maybe because hedge funds are getting stopped out of relative value trades, or due to fiscal/credit worries. Those sell-offs are typically quite violent, as they are unexpected, catching investors off guard. Next, We study events where VIX was above 35, and US rates had daily sell-offs of more than 15bp. We do not find many episodes, but we did observe this pattern briefly in 1998 (LTCM), 2008 (Lehman crisis), and 2020 (COVID), as well as on a few other occasions. The main finding of our study is that US rates eventually often trade significantly lower, as these episodes are quite short-lived, as per Figure 6.14a. The reason for this is that the Fed acts when it sees market functioning under threat. As illustrated in Figure 6.14b, in 2020, the Fed had to implement QE to bring rates under control, and once the Fed acted, rates did move lower.

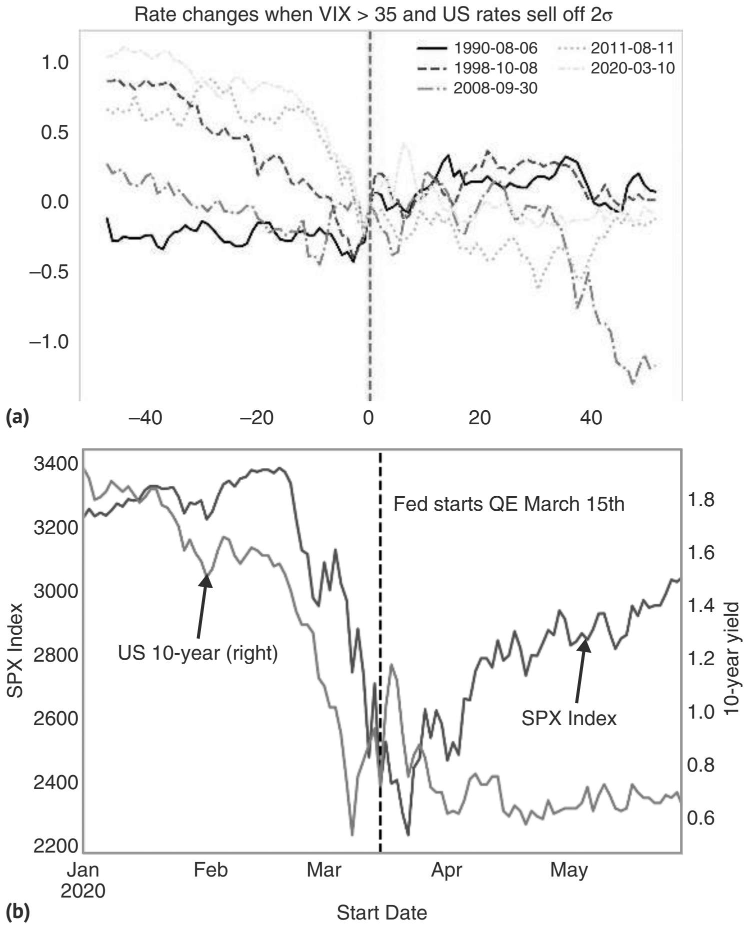
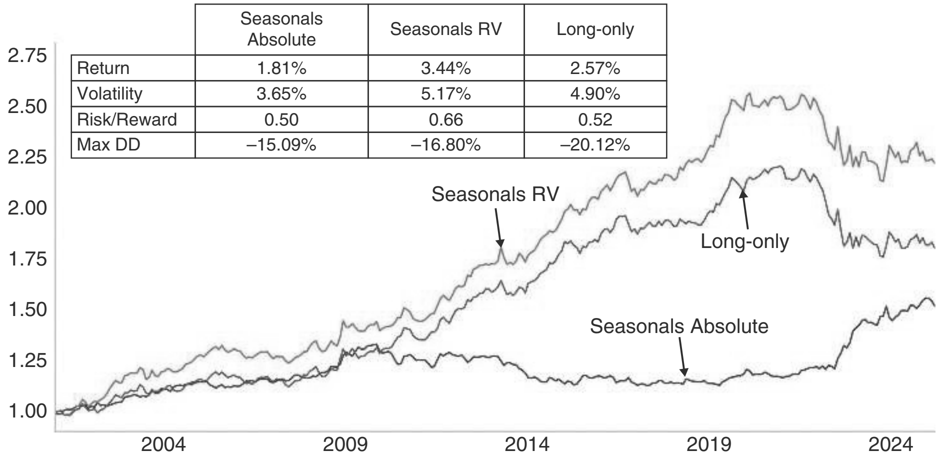


FIGURE 6.14 (a) Rates higher with VIX higher environments are short-lived. (b) Fed's QE in 2020 pushed rates lower.
Source: Bloomberg and authors' calculations.

6.6 A SEASON FOR BONDS?

Seasonals are one of the factors that tend to have a high mind share with practitioners, but that do not always backtest well. Trading seasonals in bonds is a case in point. Theoretically, bonds could work because de-seasonalization procedures employed by many governments do not fully remove seasonality from economic data. Bonds, which react the most directly to economic data, may therefore also display a stronger seasonality. We would also point out that it is famously difficult for de-seasonalization procedures to stay on course in the aftermath of very big shocks, like the GFC in late 2008, or the COVID shock in early 2020. Typically, a huge negative shock leads to the seasonal factors overcompensating in the subsequent years, resulting in artificial strength that only slowly unwinds over time.



	Seasonals Absolute	Seasonals RV	Long-only
Return	1.81%	3.44%	2.57%
Volatility	3.65%	5.17%	4.90%
Risk/Reward	0.50	0.66	0.52
Max DD	–15.09%	–16.80%	–20.12%

FIGURE 6.15 Seasonal RV works well for bonds.
Source: Bloomberg and authors' calculations.

However, results of the most basic seasonality strategies are not overly impressive. In Figure 6.15, we show the PNL of trading 10-year rolling seasonals for the large global bond markets (United States, Canada, United Kingdom, Japan, and the European markets), using bond futures. We show two different ways to translate seasonals into portfolios: absolute seasonals takes a long or a short position in the market depending on whether the median of the month is higher or lower than the median across all 12 months. Positions are scaled across markets to have a gross leverage of 100%. The relative value version takes long positions in markets where the seasonal median is higher and underweight positions where it is lower, where the total leverage sums to 100%, i.e., the portfolio is fully invested in fixed income. This enables us to compare the strategies to a long-only strategy that allocates equally to all bond markets. As can be seen in Figure 6.15, the RV seasonality strategies had added incremental value versus a long-only strategy in terms of information ratio, while the absolute strategy does not. Still, even the relative value strategy only results in modest improvements. Other specifications are not much better. Seasonality for bonds seems harder to use systematically.

6.7 SUMMARY

To identify peaks in 10-year yields, we find that two-year breakevens usually peak prior to nominal rates, and 5s/30s often peak prior to 10-year yields. Oil and DXY are mostly coincident with 10-year rates. The cyclicals over defensive ratio in equity space is at times leading and can be used for a trading rule for UST. Periods when US rates sell off in spite of rising risk aversion are usually very short-lived, and rates make new lows shortly after these events. The price action on FOMC day often has some follow-through and needs to be taken seriously. The same is true for NFP prints, but to a lesser extent. The CITI economic surprise indexes can be used for profitable trading of UST. Seasonals add only modest alpha in Treasuries.

CHAPTER 7

How to Trade Rates: Globalizing

So far, we have dealt exclusively with the US cycles and US rates. Even though the US rates picture sets the tone for the rest of the world, there are, of course, opportunities in other fixed-income markets. In the following, we provide a framework for how to think about some of the other markets, either outright or relative to the United States.

7.1 CAN GLOBAL FIXED-INCOME MARKETS DIVERGE FROM THE FED?

Global fixed-income markets are highly correlated. This is in part because central bank cycles are correlated. And central bank cycles are highly correlated because inflationary shocks are often global in nature, as are many growth shocks. The most recent examples are the Great Recession and the COVID shock. Inflation shocks are often globalized through the oil channel. However, there can also be local shocks. Most famously, the central bank of Mexico once hiked rates because of rising tomato prices in local markets. Another prime example is natural disasters, which can impact the growth outlook for a country in a very idiosyncratic way or, more frequently, fiscal policy choices. Although global monetary policy cycles are often synchronized, cycles of individual countries may not be. These local cycles manifest themselves in differential moves by the respective central banks, leading to divergences at the front end of the curve. When it comes to the backend of curves, divergences are less pronounced. The reason is that it is not only the expectation of future monetary policy that matters, but the spread to UST is also relevant, as that spread is important to drive investment flows and therefore impacts the term premium over and above the expected average future policy rate. As a result, the short end of the curve is typically less highly correlated with the short end in the United States than is the case for the backend of the curve. This is shown in Tables 7.1 and 7.2. Weekly changes of global two-year rates have a correlation with US two-year rates ranging from 0.06 in China to 0.59 in Canada, which, for obvious reasons, has a highly synchronized cycle with the United States. For weekly changes in 10-year rates, the correlation is higher, ranging from 0.10 (for Brazil) to 0.79 for Canada.

However, there are twists to the story. Sometimes it is difficult to distinguish between truly local shocks and local shocks that are global shocks in a trench coat, i.e., that become global shocks with a delay. This matters in particular when a shock

TABLE 7.1 Correlations of Changes in Yields with United States at Two-year Point Moderately High

	USD2Y	CAD2Y	DEM2Y	GBP2Y	JPY2Y	MXN2Y	BRL2Y	CNY2Y
USD2Y	1.00	0.59	0.51	0.43	0.18	0.42	0.08	0.06
CAD2Y	0.59	1.00	0.39	0.37	0.13	0.30	0.03	0.07
DEM2Y	0.51	0.39	1.00	0.55	0.22	0.29	0.03	0.06
GBP2Y	0.43	0.37	0.55	1.00	0.17	0.30	0.05	0.09
JPY2Y	0.18	0.13	0.22	0.17	1.00	0.14	0.08	0.05
MXN2Y	0.42	0.30	0.29	0.30	0.14	1.00	0.20	0.04
BRL2Y	0.08	0.03	0.03	0.05	0.08	0.20	1.00	0.04
CNY2Y	0.06	0.07	0.06	0.09	0.05	0.04	0.04	1.00

Source: Bloomberg and authors' calculations

TABLE 7.2 Correlations at 10yr Point More Significant

	USD10Y	CAD10Y	DEM10Y	GBP10Y	JPY10Y	MXN10Y	BRL10Y	CNY10Y
USD10Y	1.00	0.79	0.71	0.66	0.28	0.47	0.10	0.16
CAD10Y	0.79	1.00	0.65	0.63	0.25	0.40	0.07	0.18
DEM10Y	0.71	0.65	1.00	0.79	0.31	0.39	0.07	0.15
GBP10Y	0.66	0.63	0.79	1.00	0.25	0.40	0.09	0.16
JPY10Y	0.28	0.25	0.31	0.25	1.00	0.30	0.08	0.10
MXN10Y	0.47	0.40	0.39	0.40	0.30	1.00	0.32	0.09
BRL10Y	0.10	0.07	0.07	0.09	0.08	0.32	1.00	0.02
CNY10Y	0.16	0.18	0.15	0.16	0.10	0.09	0.02	1.00

Source: Bloomberg and authors' calculations

emanates from a large global economy, such as the United States, the Eurozone, or China. Other countries are quite unlikely to generate any shocks that become global. A good example to illustrate this point is the GFC, which started as a local shock to the US housing market. Growth in many countries was still reasonably high in late 2007, when the United States was already in a recession. Only after major US banks were significantly impacted by the snowballing housing crisis did the stress spread globally. This led to interesting opportunities to receive in countries outside of the United States, on the premise that the United States was hit by a big enough shock for it to become a global shock, while the market incorrectly assumed it would remain a local shock.

This example raises the question of whether it is a good strategy to receive the laggards when the Fed starts its easing cycles. We have data for a large number of global markets for the last three Fed cutting cycles, and we investigated this theory in a paper at Citi (see Willer et al., 2024g). There, we proxied market expectations for future central bank action by taking the difference of the one-year swap versus the floating rate, where positive numbers imply the market pricing hikes, and negative numbers imply cuts. Given that central banks, at the very latest, are typically guiding for the first cut around one month before it happens, we check what is priced into various global markets 1m prior to the first Fed cut. We then investigated the change in the one-year rate from that point of time to three months after the first Fed cut. This is illustrated in Figures 7.1a, b, 7.2a and 7.2b.

In 2001, one-year yields came off in all countries for which we have data, between the month ahead of the first Fed cut, and three months later. There is a slight correlation between how much was priced and how much one-year rates reacted, suggesting a slight preference to receive the leaders, rather than the laggards, but the bias was slight, and almost any receiver worked decently well. The case of 2007 pours cold water on the general idea of receiving everywhere when the Fed starts to cut. Six out of 18 markets priced hikes one month before the first Fed cut, and in five out of those six countries, one-year rates were even higher three months after the first Fed cut. Countries that already priced cuts one month before the first Fed cut also had mostly rising rates over that period, though they overall fared slightly better than was the case for the more hawkish countries. The reason that the Fed cuts were not overly helpful for global rates is that it took much longer than expected for the US housing trouble to spread to the US financial sector and from there to the rest of the world. Of course, eventually, the US troubles did spread globally, and every country cut rates aggressively. It did turn out to be just a matter of lags, but the start of the Fed cuts was not a good signal to receive in many countries.

In 2019, receiving into and out of the first Fed cut in global markets mostly worked. The relationship between pricing dovishness and subsequent moves in the one-year rates is fairly tight. Again, it was a better idea to receive the leaders than the laggards. Lastly, the case of 2024 was similarly good for receivers in most countries; 2024 also illustrates that it is better to receive the leaders, with Japan and Brazil the two laggards, where rates sold off meaningfully. Of course, the charts illustrate changes in one-year rates, not the PNL of a one-year receiver. As such, the results overstate the case, as receiving inverted curves accrues negative carry. However, eyeballing the magnitudes suggests that it is unlikely that the negative carry is sufficient to invalidate the results, given a three-month holding period.

Taking the experience of those four cycles together, the lesson is that a coming Fed cycle by itself is not always a sufficient case for receiving the rest of the world. Sometimes, it can take longer than expected for weakness in the United States to show up globally. Some patience may therefore be required when the Fed is starting to cut. Furthermore, leaders, i.e., countries that already price meaningful cuts by the time the Fed gets started, perform more consistently than the laggards, and investors should chase the leaders rather than the laggards.

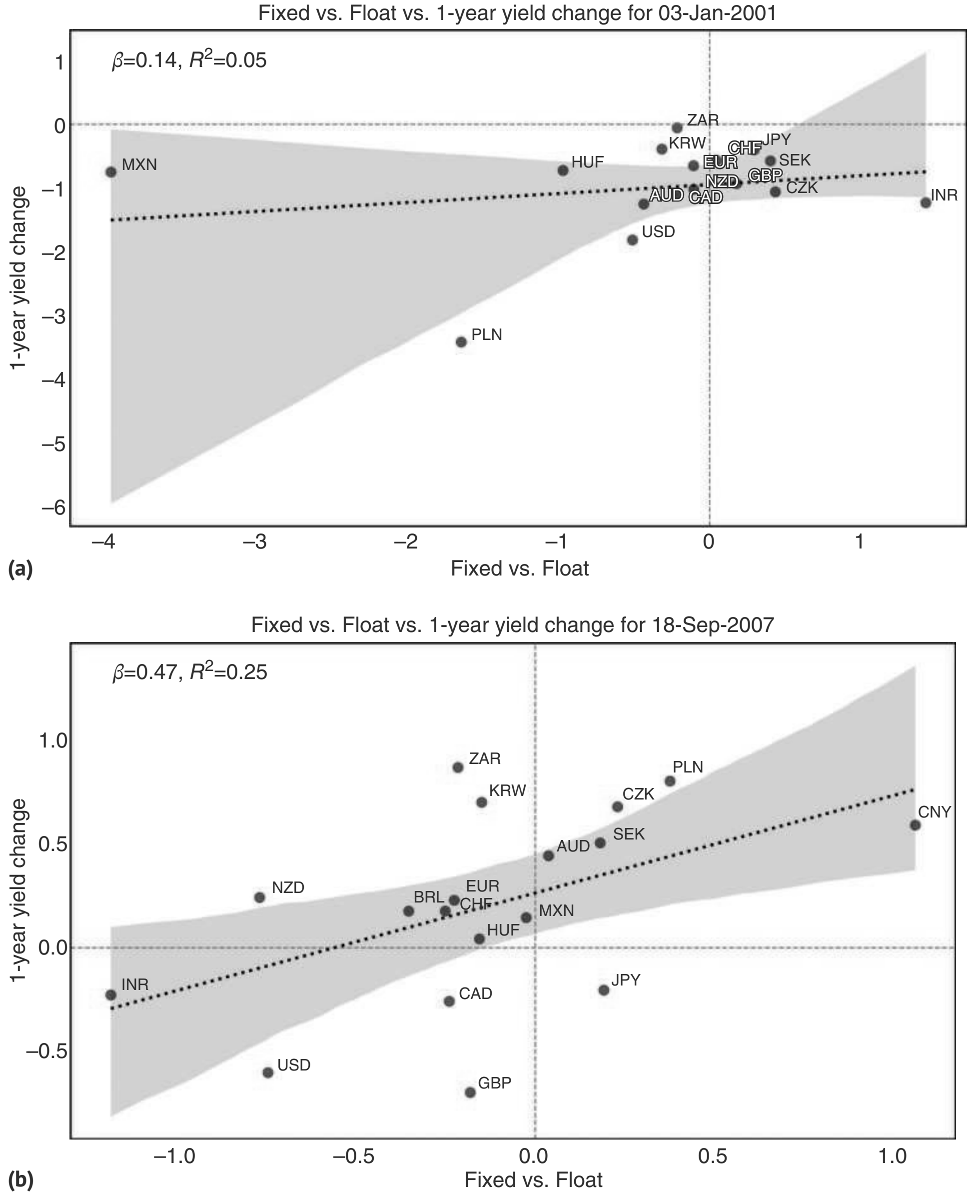


FIGURE 7.1 (a) Leaders perform very slightly better in 2001. (b) Leaders perform better in 2007. *Source*: Bloomberg and authors' calculations.

Having said that, focusing just on leaders and laggards among countries that are geographically very close and interconnected produces more interesting results. We investigated relative value opportunities between Sweden/Norway, Australia/New Zealand, and Canada/US. In particular, we focus on significant divergences,

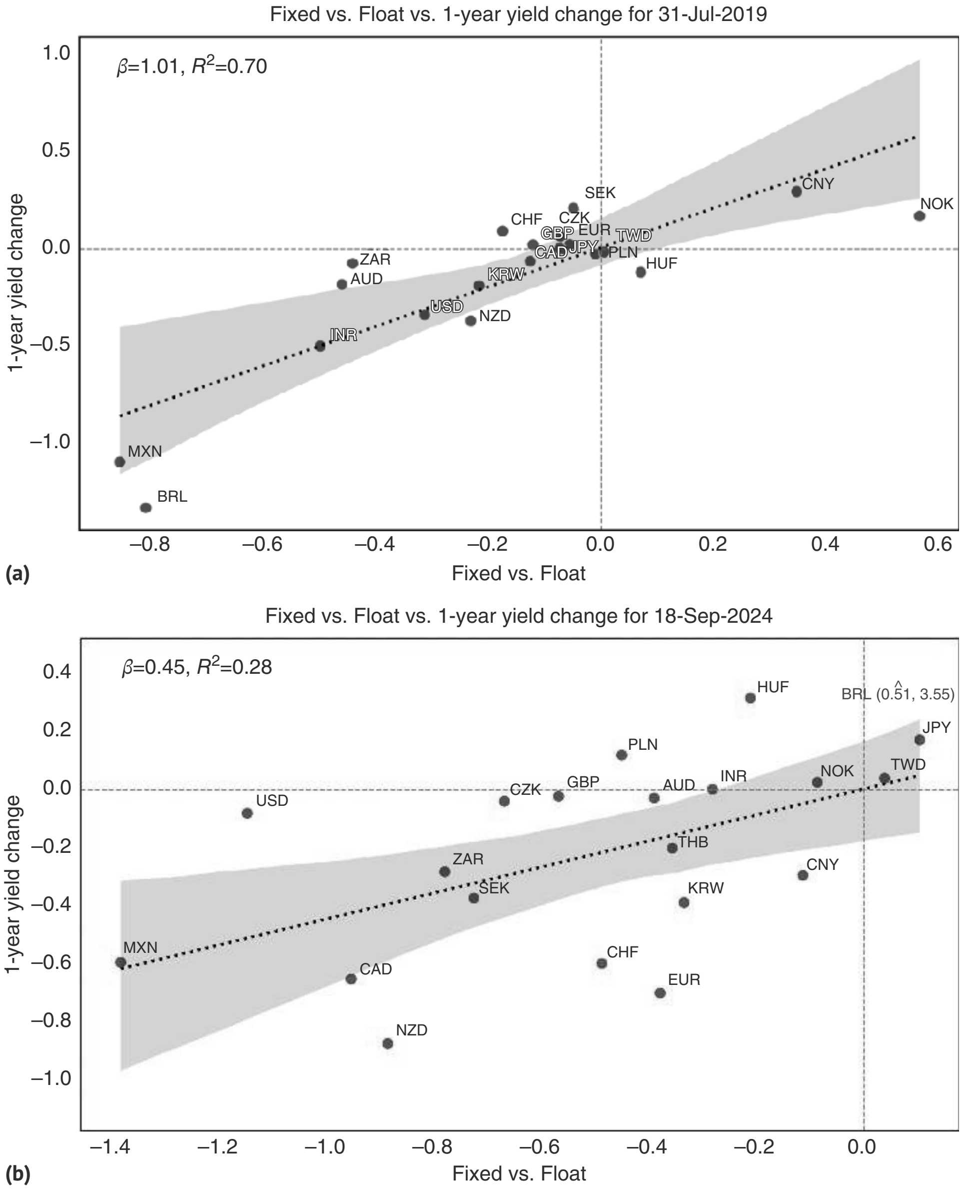


FIGURE 7.2 (a) Leaders perform much better in 2019. (b) Leaders perform better in 2024.
Source: Bloomberg and authors' calculations.

where five-year spreads are at their most extreme values for a 1,000-day lookback window. We find that reversal patterns exist but are not very pronounced. In Figure 7.3 we show the typical case for Australia/New Zealand. Eventually, rates do go in the opposite direction in the median case, and the skew is very favorable for a change in

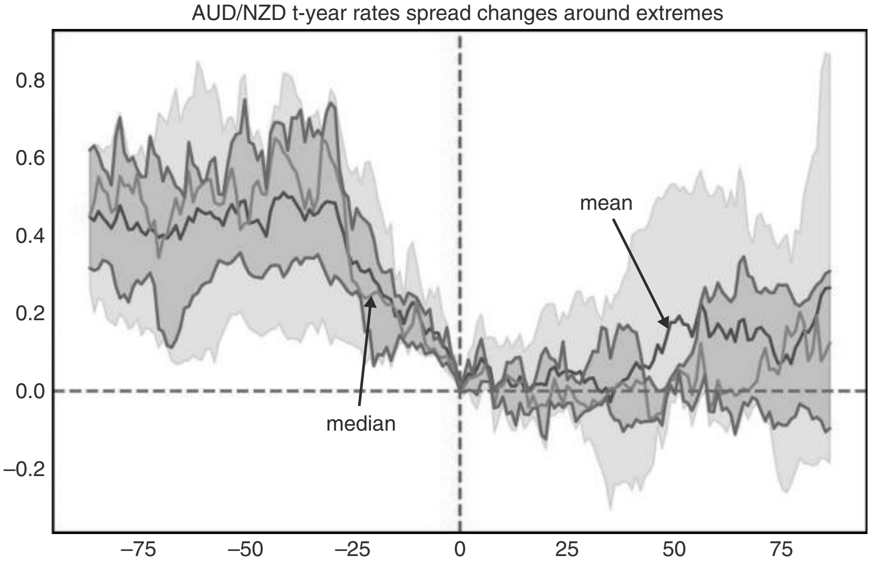


FIGURE 7.3 Aussie and Kiwi rates diverge only so far.
Source: Bloomberg and authors' calculations.

trend, too. But initially, when our condition is hit, it is more a matter of differentials stabilizing rather than going straight in the opposite direction. For the other two pairs (not shown), performance continues in the same direction for another two months or so, before also stabilizing. Overall, this suggests that investors should be on the look-out for any fundamental developments (data prints, central bank meetings) that may trigger a reversal, once the condition hits. But the results are not strong enough to invalidate the overarching thesis to go with the leaders.

7.2 BONDING WITH BUNDS

Outside of Canada, the correlation in the 10-year sector among major markets is highest with Bunds. This is partly the case due to business cycles being highly correlated, given relatively important trade links, as well as a somewhat similar industrial structure. Another reason is that Bunds are one of the more important competitors to US Treasuries as a high-quality store of value, given that Germany is one of the very few larger AAA-rated sovereigns left. Although the Bunds market is much too small to really compete with US Treasuries (at less than 1/10th of the market size), it is nevertheless one of the few avenues for diversification available for asset allocators.

Given this high correlation, many of the rules on how to trade US Treasuries also apply to Bunds. The more interesting question is how to switch between the two or trade the spread between the two. Spreads at the front end are clearly driven by leads,

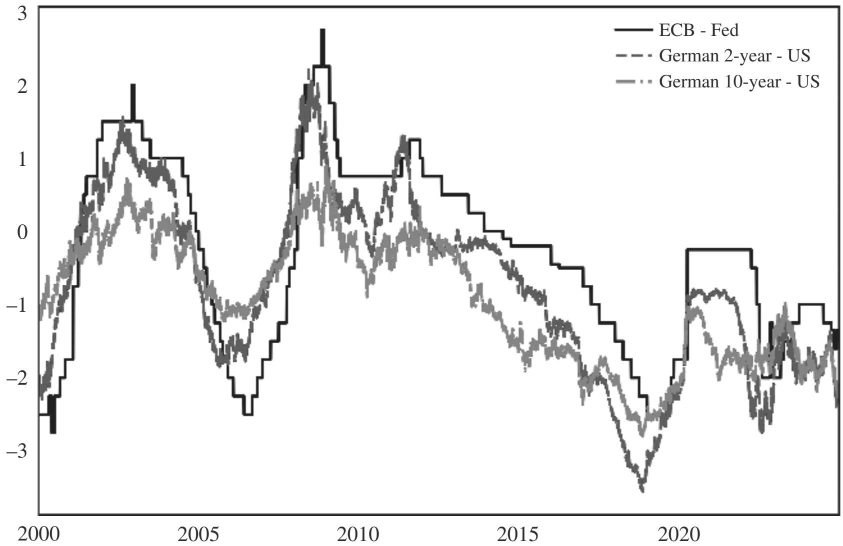
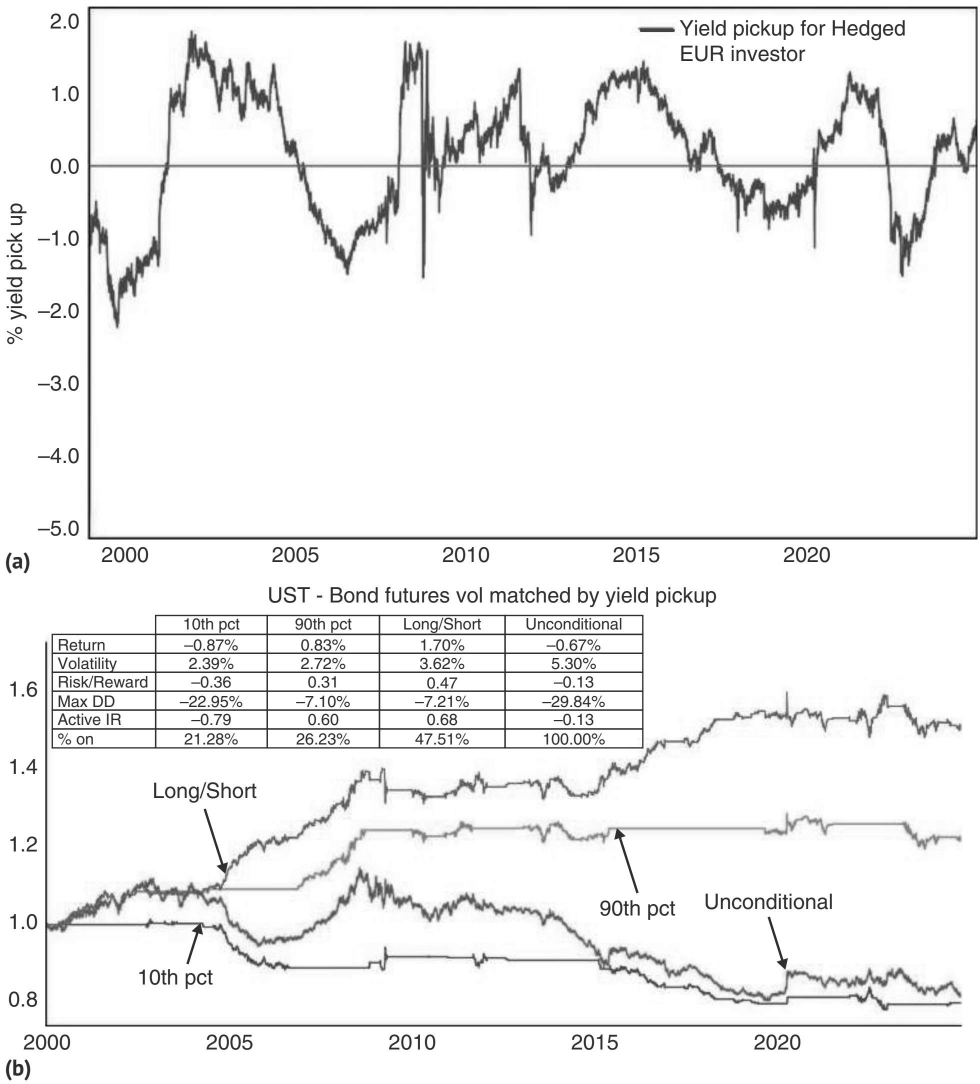


FIGURE 7.4 Ten-year spreads trade with differential monetary policy decisions.
Source: Bloomberg and authors' calculations.

lags or divergences of central bank policy action. Furthermore, the differential performance at the front end also determines the differential performance at the backend to a large extent. This is illustrated by Figure 7.4, which shows the spread in the monetary policy rate between the Fed and the ECB next to the 10-year spread of UST over Bunds. Markets are obviously forward-looking, and the turn in the 10-year spread tends to happen ahead of the peak in policy rate divergence, but at times markets are also slow and lags can be short.

So far, so logical. If investors have a strong sense of diverging central bank policies, then that will obviously be reflected in rates throughout the curve. However, at the back end, capital flows can also be driven by the level of rates, not just the changes. One rule that we backtest below is applying our more generalized fixed-income carry framework to German–US spreads. We first calculate the attractiveness of US Treasuries to Bunds, FX hedged. The FX hedging removes the carry at the front end of the curve. As a result, the FX hedged spread is relatively symmetric around the zero line, as per Figure 7.5a. In Figure 7.5b we backtest whether extreme values of the spread matter for forward performance. We find that UST outperform Bunds, duration-matched, and FX hedged, with an active information ratio of 0.60, when the spread is in the top 90th percentile. In the bottom 10th percentile, the information ratio is –0.79, i.e., Bunds outperform US Treasuries. This compares to an unconditional IR of –0.13. However, we note that the strategy has been going largely sideways since 2020. We think this is because it is essentially an application of a carry strategy in fixed income. During a bond bear market, in this case driven by the high inflation post-COVID, carry as a fixed-income factor is not performing overly well. Still, going sideways in bear markets and higher in bull markets makes it an attractive framework.



	10th pct	90th pct	Long/Short	Unconditional
Return	–0.87%	0.83%	1.70%	–0.67%
Volatility	2.39%	2.72%	3.62%	5.30%
Risk/Reward	–0.36	0.31	0.47	–0.13
Max DD	–22.95%	–7.10%	–7.21%	–29.84%
Active IR	–0.79	0.60	0.68	–0.13
% on	21.28%	26.23%	47.51%	100.00%

FIGURE 7.5 (a) Fx hedged carry of UST versus Bunds is symmetric around the zero line. (b) Trading the carry factor at extremes works for the UST-Bund spread.
Source: Bloomberg and authors' calculations.

7.3 GILDED GILTS?

The framework for the United Kingdom is very similar. It all starts with the front end, and the rules are reminiscent to the ones for trading UST outright. Figure 7.6 illustrates that monetary policy deltas drive changes in spreads of the backends for the UK versus the US. We next look at the recent history more carefully, which is illustrated in Figure 7.7. The 2015 to 2019 differential hiking cycle, where the Fed started early

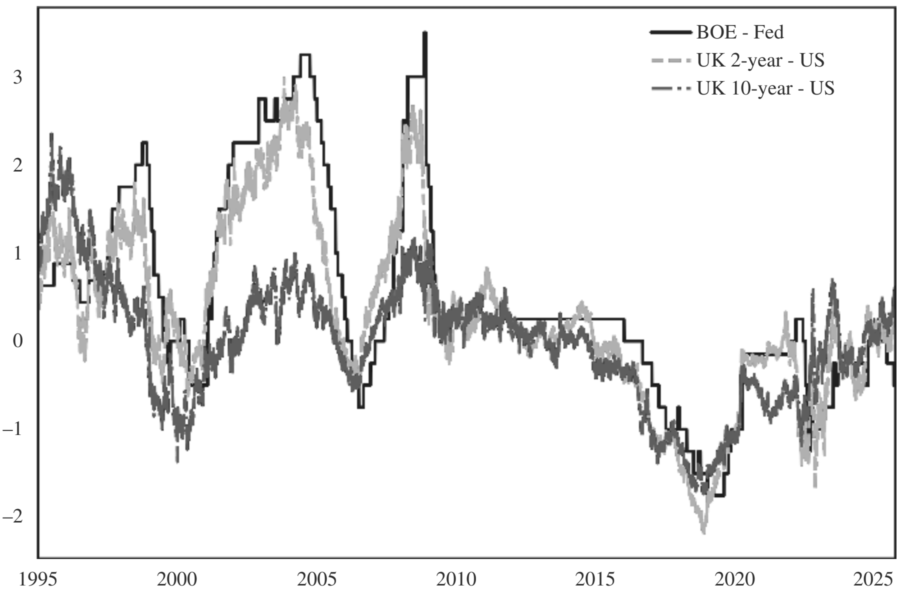


FIGURE 7.6 Back-end spreads follow front-end spreads, follow relative monetary policy.
Source: Bloomberg and authors' calculations.

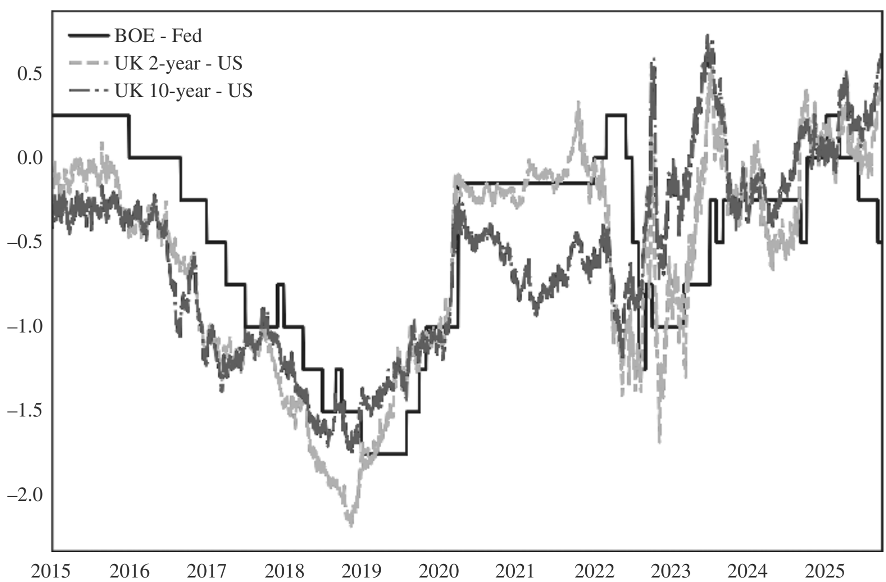
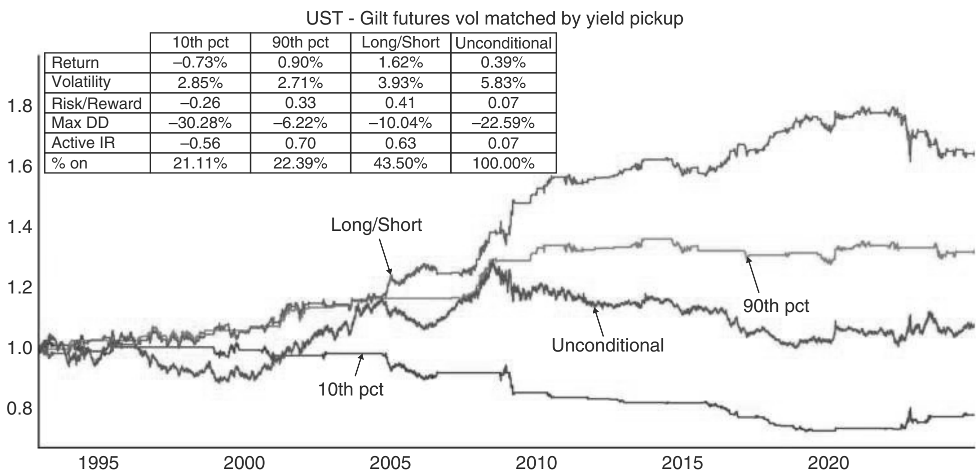


FIGURE 7.7 A closer look at the more recent cycles.
Source: Bloomberg and authors' calculations.

and hiked much more aggressively while the BoE first squeezed in a cut before moving to a more leisurely hiking cycle, largely behaved as expected. Investors should have paid the spread slightly before the start of the relative tightening by the Fed, and remain paid until shortly before the last differential hike was in, very similar to the rules we discussed for US rates. Spread differentials peaked in November 2018, while the last differential hike occurred in December 2018. In November, with the last hike in sight, investors should have done the opposite and received the spread. Of course, in this particular cycle, the BoE was so much less aggressive than the Fed that investors might as well not have bothered with the spread and instead just paid the United States. This is all easy to say in hindsight, but it also illustrates that the basic rules of how to trade fixed income outright can be used to trade spreads. While the charts do not take carry into account, the moves are large enough to generate profits after carry too, in most cases.

The 2022 cycle was more interesting, as both central banks were relatively aggressive in their hikes. The United Kingdom started early, but was then overtaken by the Fed's acceleration of hikes, and in the end stopped before the Fed. That cycle was relatively difficult to trade in spreads, partly because the Liz Truss moment in 2022 led to a large countertrend move for much higher UK rates, at a time when the Fed was still out-hiking the Bank of England (BoE). But in the end, the spread followed (or rather led) the relative monetary policy actions relatively closely. The two-year spread troughed very much in a coincident fashion with the maximum delta between the US and UK base rates. In emerging markets, it is often the case that FX is a secondary but also very relevant driver of rate spreads, especially at the back end, largely because FX is important for the inflation outlook, and also because it may capture general country risk, which also should be reflected in the curve. However, for G10, we do not find much evidence for such an effect, other than during very short-lived periods, such as the famous "LIz Truss moment." Outside of those brief periods, the directionality seems to be much stronger from rate spreads to FX rather than the other way around.

Using the same carry strategy that we proposed for the UST-Bunds spread also works for gilts. Figure 7.8 illustrates that trading in the top 10th percentile of FX hedged positive yield differentials has an active information ratio of 0.63 (versus unconditional of 0.07), while the bottom 10th percentile results in an information ratio of –0.56.



	10th pct	90th pct	Long/Short	Unconditional
Return	–0.73%	0.90%	1.62%	0.39%
Volatility	2.85%	2.71%	3.93%	5.83%
Risk/Reward	–0.26	0.33	0.41	0.07
Max DD	–30.28%	–6.22%	–10.04%	–22.59%
Active IR	–0.56	0.70	0.63	0.07
% on	21.11%	22.39%	43.50%	100.00%

FIGURE 7.8 Gilts outperform when pick up extreme enough.
Source: Bloomberg and authors' calculations.

A Loss of Truss

The 2022–2023 "Liz Truss moment" deserves a box because it is a rare occurrence, so far, where fiscal fears undermined rates in developed markets. Typically, it is mostly in emerging markets, where fiscal policy is a much more frequent headwind for rate markets. Analyzing this episode is important, as fiscal rectitude is still extremely unfashionable across the globe, and as commentators worry that the United States could remain a target of the re-awakening bond vigilantes for years to come. The Liz Truss moment refers to a sharp sell-off in gilts and GBP in reaction to the mini-budget announced on September 23, 2022, by then Chancellor Kwasi Kwarteng, under then brand new Prime Minister Liz Truss. The budget proposed unfunded tax cuts worth GBP 45bn, alongside substantial energy subsidies due to rising energy costs. Overall, the government suggested a need to borrow an extra 72bn GBP over the next six months alone, almost 3% of GDP. This happened at a time when inflation was already high, and with the BoE in the middle of a hiking cycle. As a result, gilts spiked dramatically by more than 100bp in just one week. The timeline is illustrated in Figure 7.9.

Interestingly, the curve initially flattened, suggesting that the market thought the BoE would hike more aggressively to offset the inflation impact of this fiscal stimulus. However, in the subsequent strong sell-off in the gilt market, the curve steepened aggressively, more than offsetting the early flattening, suggesting potential fiscal fears outweighing the monetary policy impact. Furthermore, GBP tanked during the volatile days after the mini-budget. This is also interesting, as higher rates are typically currency-positive in developed markets. This is true even during some episodes of rising fiscal risks, for example, during the debt ceiling worries in the United States. But it was not true this time around in the United Kingdom.

The market was calmed initially by the BoE intervening with a temporary gilt buying program, announced on September 28 for up to 65bn GBP, which was supposed to run from September 28 to October 14. The announcement led to a meaningful pull-back in yields, also helped by Kwarteng reversing one of the tax cuts on October 3. However, yields subsequently retested the highs only a few days later. As a direct consequence, Kwasi Kwarteng was replaced by Jeremy Hunt on October 14, who quickly reversed the remaining tax cuts that had been announced. Gilts rallied by 100bp post-Oct 14. Liz Truss resigned on October 20.

Although on the face of it, this seems like a successful story for the bond vigilantes, where market turmoil generates desired policy changes in a developed market, claiming a prime minister in the process, it is not clear that this is the correct reading. One key factor that caused the outsized sell-off in gilts was a large class of leveraged investors active in the gilt market, whose importance was not fully understood by market observers at the time. These were the so-called liability-driven investment (LDI) funds. LDIs are used by UK pension funds to

(*Continued*)

(*Continued*)

manage interest rate and inflation risks. The idea is to rely on leveraging gilts to hedge liabilities to allow pension funds to invest in other asset classes. These LDI funds were hit by margin calls as gilts started to have outsized moves post-mini-budget, leading to forced selling, which in the end forced the BoE into action. To us, this suggests that the problem was misunderstood pockets of leverage, rather than the fiscal announcement by itself. If it had not been for the forced selling, the budget may have led to higher rates, but not to a U-turn and the resignation of the chancellor and the prime minister. The other lesson is that the BoE was credible in stabilizing the bond market with purchases at the same time when it was continuing to hike interest rates, due to the high inflation environment. Most likely, this was the case because the gilt-buying program was limited in terms of both time and size. Overall, we see the episode as a partial validation of the bond vigilantes at best, and as somewhat reassuring in that the BoE was able to stabilize markets relatively quickly. Governments and central banks in developed markets do have tools to fight back against the bond vigilantes.

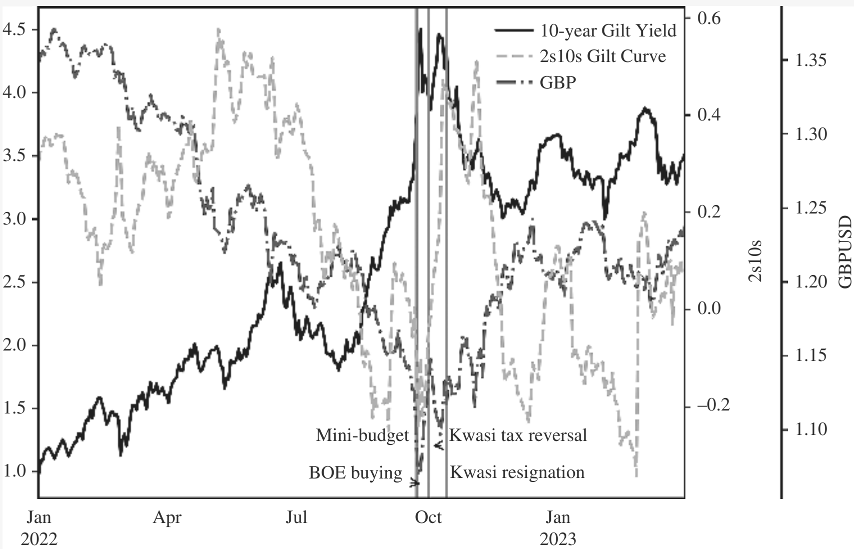


FIGURE 7.9 Bond vigilantes or LDI?
Source: Bloomberg.

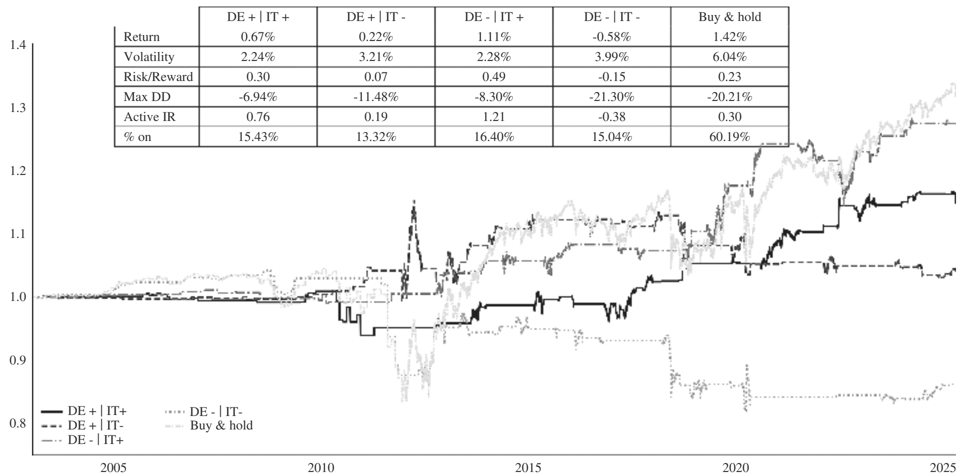
7.4 AT THE PERIPHERY

Bonds in the European periphery have been a major source of alpha in global macro-trading for decades. First, it was all about spread compression in the aftermath of the introduction of the euro, and then, in the aftermath of the GFC, it was all about the unwind of the spread compression trade, when the somewhat unflattering PIIGS acronym was coined as shorthand for Portugal, Italy, Ireland, Greece, and Spain. In the latter event, markets severely underestimated the primacy of politics over economics among European policymakers, which ultimately led to a severe re-tightening of peripheral spreads after Draghi's "whatever it takes" speech, given on July 26, 2012. Although we do not expect that markets will again test European policymakers' appetite for an exit of the structurally weaker economies from the EUR, we cannot rule it out either, given the rise of more populist parties in Europe.

The key issue for bonds in the European periphery is that they have both a monetary policy and a credit component in their rates. This means that when the ECB mostly cuts because of a deep recession or a financial market accident, the rates component may be supportive for receivers, while the credit component may not be. We have written about this in the rates chapter of the emerging markets book, but it is also, to a much lesser extent, a feature in the European periphery, where the spread over Bunds visualizes the credit component. This credit component is larger in the European periphery than in some other markets because the countries in the Eurozone have given up control of the printing presses to the ECB, which, at the very least, limits the ease with which politicians can bail themselves out.

Although the European crisis was all about credit, we note that even for explaining spread returns, monetary policy is a crucial input. This is because a hawkish ECB will raise the costs of funding, which is especially detrimental for the fiscally weaker countries in the periphery, undermining their credit quality and leading to wider spreads. It therefore stands to reason that the best environment for peripheral spreads (and peripheral bonds outright) is one where growth in the periphery is strong, leading to an improving credit quality, while growth in the core (with Germany and France being the focus due to their size, though going forward it is less clear whether France can remain core) is weak, causing the ECB to be relatively dovish, or at least not hawkish. We backtested this framework by focusing on Citi's data surprise indexes. Focusing on spread returns, the active information ratio moved from 0.30 to 1.21 when German surprises are negative, while Italian surprises are positive. Italy enjoys a weak Germany, and not just on the soccer pitch. The reverse also holds. During times when Italian data changes are negative and Germany is doing well, spread returns for Italian bonds have a much worse information ratio, barely above zero. The worst combination is negative data surprises for both. This result is not just driven by the European crisis, but holds more broadly, as can be seen in Figure 7.10. This suggests that the credit component is more important than the rates component, i.e., positive surprises for Italian growth are what really count for spread returns.

Given the importance of the credit component of the periphery, we also investigate whether there is value in the periphery relative to corporate credit. Using BTPs as an example, rates strategists at Citi have found that the asset swap spread (BTPs against European swaps) generally trades in line with European credit (see Bansal (2025)). In particular, post European crisis, a spread above the IBOXX EUR non-sov BBB Z-spread to Libor has suggested cheapness, while levels on par with the IBOXX EUR non-sov A Z-spreads to Libor have suggested that bonds are expensive, though recently credit quality improvements in Italy have shifted the lower bound to the AA spread. Sharp spikes in the BTP ASW can generate an ECB response. For example, in 2020,



	DE + | IT +	DE + | IT -	DE - | IT +	DE - | IT -	Buy & hold
Return	0.67%	0.22%	1.11%	-0.58%	1.42%
Volatility	2.24%	3.21%	2.28%	3.99%	6.04%
Risk/Reward	0.30	0.07	0.49	-0.15	0.23
Max DD	-6.94%	-11.48%	-8.30%	-21.30%	-20.21%
Active IR	0.76	0.19	1.21	-0.38	0.30
% on	15.43%	13.32%	16.40%	15.04%	60.19%

FIGURE 7.10 Periphery likes weakness in core, strength in periphery.
Source: Bloomberg and authors' calculations.

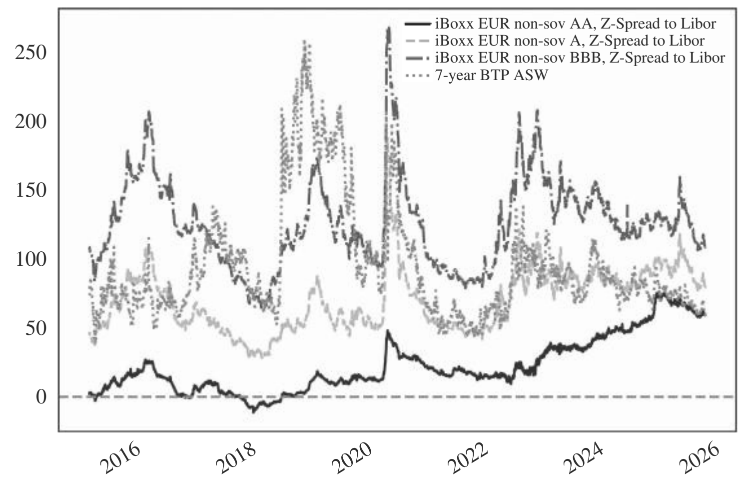


FIGURE 7.11 Italy trades in line with European credit spreads.
Source: Citi research.

the sharp widening triggered the ECB to announce the PEPP facility (pandemic emergency purchase program) during the COVID crisis, as illustrated in Figure 7.11. The PEPP was another reminder, if needed, that the countries of the Eurozone have not really given up control of the printing presses. A big enough crisis will likely get the ECB to act time and again, creating opportunities to fade those sell-offs.

7.5 IS JAPAN DIFFERENT?

Japan is holding a special place in the heart of many macro traders, partly because it is highly levered to the global cycle, partly because the JPY is a key funder for various carry trades, and partly because Japan has been the poster child of what happens after a large equity market bubble bursts and is therefore seen as a leading indicator of what could well happen to other regions in that eventuality.

The Japanese equity market bubble peaked in December 1989. Because of the relatively low inflation that accompanied the bubble (with inflation excluding food and consumption tax rising to around 3% only), rate hikes by the Bank of Japan (BoJ) were somewhat late, starting only in May 1989. Given the size of the bubble, they were also relatively limited, with the official discount rate peaking at 6% in August of 1990. For the following decade, the interest rate set by the BoJ was on a one-way street down, dealing with the bursting equity bubble, sitting out several Fed hiking cycles, and finally, the BoJ implemented a zero interest rate policy (ZIRP) in February 1999. However, after the internet bubble had sufficiently inflated Japanese stock prices again, the BoJ did tighten monetary policy, hiking the overnight call rate by 25bp in August 2000, following the earlier Fed tightening cycle at a time when the Fed was already done hiking. This hike almost perfectly coincided with the peak in the Japanese (and US) business cycle. The hike was quickly unwound, and quantitative easing was introduced, with the monetary policy focusing on the balance of current accounts at the BoJ, which was set 1trn JPY above required reserves, making sure the interbank rate would remain at zero. Purchases of long bonds were also ramped up. In September 2002, the BoJ also started to buy stocks. Only in June 2006 did it hike again, this time by 50bp, again at a time when the Fed was already done hiking. These hikes were also quickly unwound again, and this time the BOJ brought interest rates into negative territory (NIRP), to –0.1 percent for excess bank reserves. Having been burned twice by hiking, only to be forced to unwind those hikes in short order, the BoJ then sat out the 2015 to 2018 Fed hiking cycle. The 2024 hiking cycle is also one in which the BoJ started only after the Fed was already done. The first hike from negative territory back to zero happened in March of 2024 and was followed so far by two more hikes. Once more, the BoJ hopes that Japan has reached escape velocity with respect to deflation.

There are a few clear features when it comes to Japanese bond trading. First, the fact that the BoJ always starts late in the hiking cycle is no coincidence, but structural. Public servants who have spent most of their working lives fighting deflation will not be early in any global hiking cycle. This also means that by definition, the BoJ will start hiking when most other central banks, and importantly the Fed, are already done hiking, and often close to starting an easing cycle. Second, historically, the BoJ has not gotten very far in its hiking cycles once the Fed turned. This is also no coincidence. After all, the Fed is reacting to a weak US economy, which is often correlated with a weak global and Japanese economy. However, the case of 2024 could be slightly different, as the United States may achieve a rare soft landing. Furthermore, when the Fed starts to cut interest rates, USDJPY tends to weaken. USDJPY is a crucial driver of Japanese inflation, as can be seen in Figure 7.12. As a result, inflation pressures in Japan can come off the boil during Fed easing cycles. This results in the Japanese interest rate cycle lagging the US cycle, often with very substantial lags.

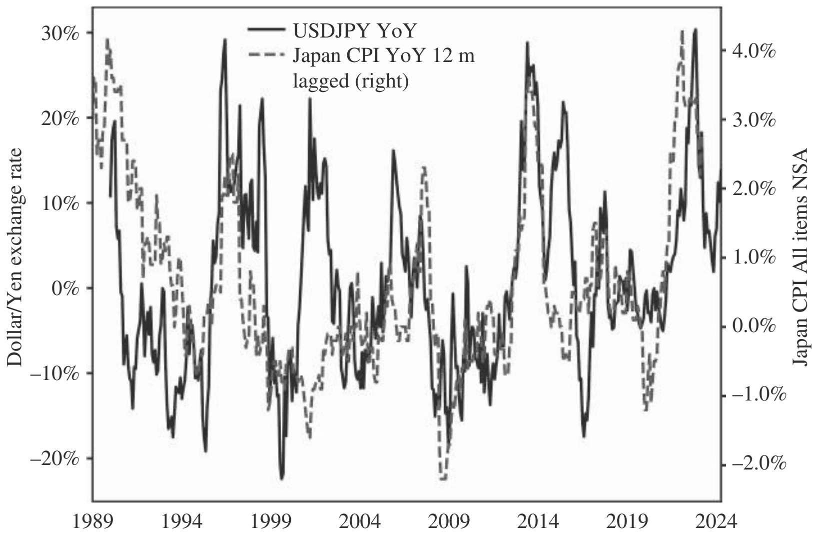
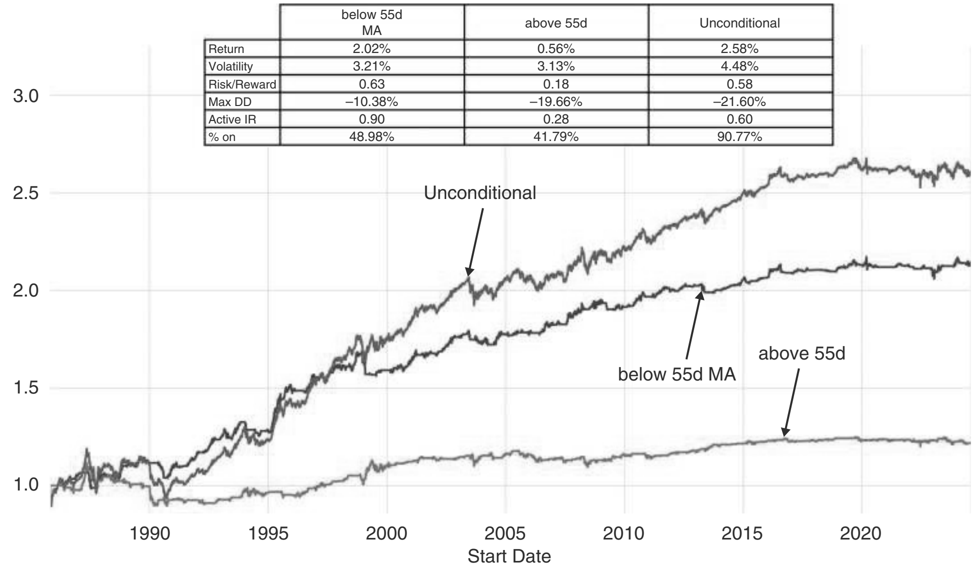


FIGURE 7.12 Japanese inflation is JPY driven.
Source: Bloomberg and authors' calculations.

Correlated monetary policy cycles, even with a lag, mean that back-ends in Japan are highly correlated with US rates, even though JGBs trade with lower volatility. This is illustrated by Figure 7.13, which shows that JGBs performance when US rates are below their 55dma is much higher than in the opposite case, with active information ratios clocking in at 0.9 and 0.28, respectively. While recently the performance has flattened out, this is still a good outcome for the model, given that a structurally tight labor market has led to longer lags than usual versus the Fed cycle.

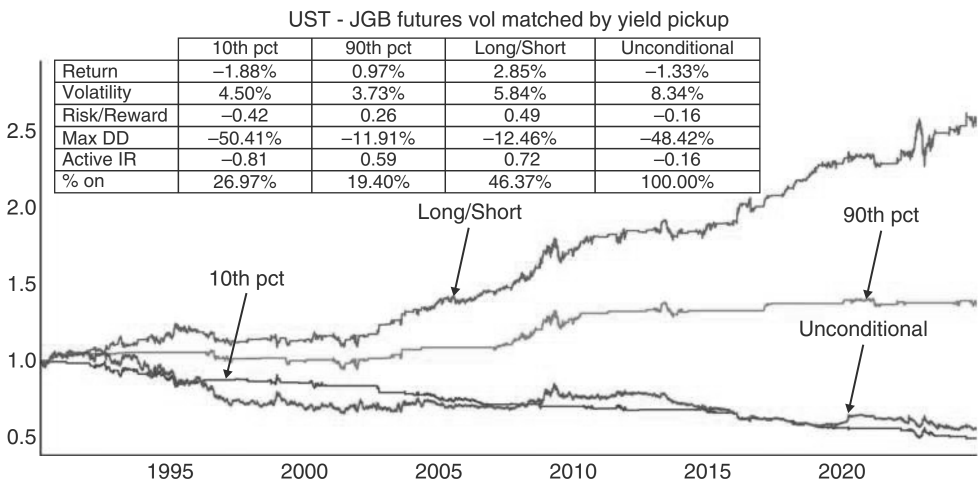
Relative to US Treasuries, the same carry strategy that we wrote about for Bunds also works for JGBs. In particular, when the yield pick up of FX-hedged JGBs versus UST is in the top percentile (over the last 12 months), the information ratio of long JGBs versus short UST, FX hedged, and volatility adjusted is 0.72. In the bottom percentile, it is –0.81. This compares with an unconditional information ratio of –0.16. It is always reassuring when the same rules work across multiple assets without any tweaks and is shown in Figure 7.14.

Positioning for a BoJ cycle has to happen at the front end of the curve, which is less subject to US rates. During a hawkish BoJ cycle, flatteners also often work and may even benefit from lower US rates, though only if fiscal worries do not take center stage. However, after Fed cuts have started, payers in Japan have to be implemented with more caution, as the risk rises that the BoJ will eventually lose its hawkishness and follow the Fed down a dovish path, though mostly if the Fed is cutting in a hard landing scenario. If a US soft landing appears more likely, payers in Japan can work well for longer. On the front end, the basic principles established for the United States hold too. Rates start to sell off less than a year before the first BoJ hike, and the peak



	below 55d MA	above 55d	Unconditional
Return	2.02%	0.56%	2.58%
Volatility	3.21%	3.13%	4.48%
Risk/Reward	0.63	0.18	0.58
Max DD	–10.38%	–19.66%	–21.60%
Active IR	0.90	0.28	0.60
% on	48.98%	41.79%	90.77%

FIGURE 7.13 UST are key for JGBs.
Source: Bloomberg and authors' calculations.



UST - JGB futures vol matched by yield pickup

	10th pct	90th pct	Long/Short	Unconditional
Return	–1.88%	0.97%	2.85%	–1.33%
Volatility	4.50%	3.73%	5.84%	8.34%
Risk/Reward	–0.42	0.26	0.49	–0.16
Max DD	–50.41%	–11.91%	–12.46%	–48.42%
Active IR	–0.81	0.59	0.72	–0.16
% on	26.97%	19.40%	46.37%	100.00%

FIGURE 7.14 JGBs outperform when pick up extreme enough.
Source: Bloomberg and authors' calculations.

is set somewhere around the last hike, even though it at times seems to take somewhat longer than for US rates to price a policy change by the central bank: the peak in the last cycle in 2005 came well after the last hike, while the low in yields in the last cutting period in 2015 came well after the cut. This is shown in Figure 7.15.

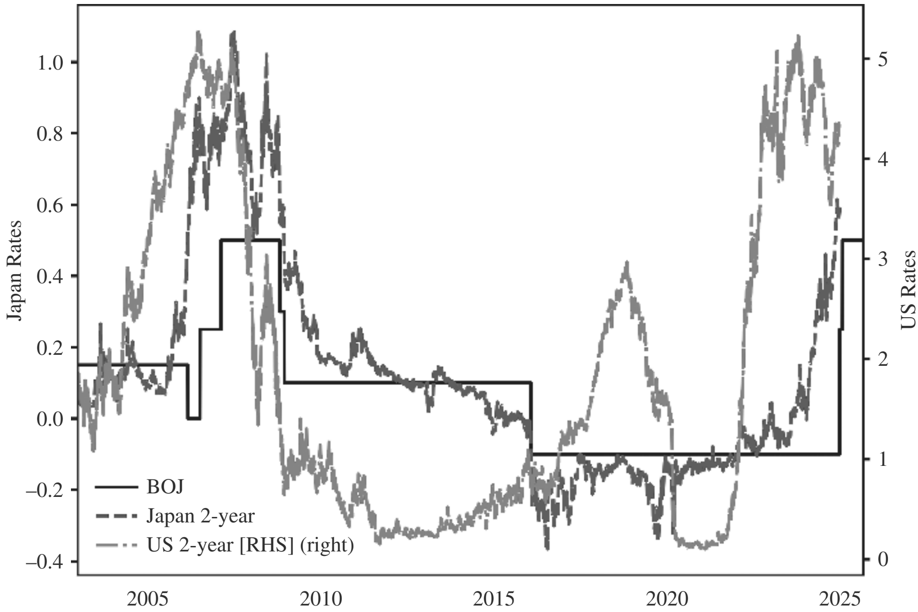


FIGURE 7.15 Front-end trades with BOJ, sometimes with long lags.
Source: Bloomberg.

7.6 IS CHINA DIFFERENT?

China is also a key global player in bond markets. We already discussed CGBs in the EM book and, therefore, keep the discussion brief. Unlike other countries of global importance, China still has a relatively closed capital account. This gives China more degrees of freedom to manage its economy. Although Chinese authorities carefully monitor the impact of any change of policy rates on USDCNY, Chinese bonds have a lower correlation with global bonds than other markets, partly due to these capital controls. Although US rates are an important factor, Chinese rates follow the local business cycle more closely. Just like in the case of the United States, rates bottom when the first hike is relatively close and then rally throughout the easing cycle. The main difference is that rate cuts in China are a much rarer occurrence than in the United States, and often quite spaced out. This is the case because the People's Bank of China (PBoC) uses a combination of quantitative and price-based measures. On the rates side, the medium-term lending facility (MLF) is the tool to provide medium-term funding to financial institutions. The PBoC also increasingly uses the 7-day reverse repo rate. On the quantitative side, it uses changes to the required reserve ratio (RRR) to adjust the amount of reserves banks must hold, tightening or loosening monetary policy. It also uses open market operations (OMOs), conducting repos and reverse repos to influence market liquidity. Given this somewhat more complex set of monetary policy tools, we show below both: the behavior of CGBs relative to RRR cuts (see Figure 7.16a) and, starting with the increasing usage of the MLF rate in 2016, relative to interest rate cuts (see Figure 7.16b). Both charts reinforce the message that rates are unlikely to trough before the last cut and/or RRR cut has occurred. Pauses in the interest rate-cutting cycle can also lead to large pullbacks in rates. For more details, please consult the EM book.

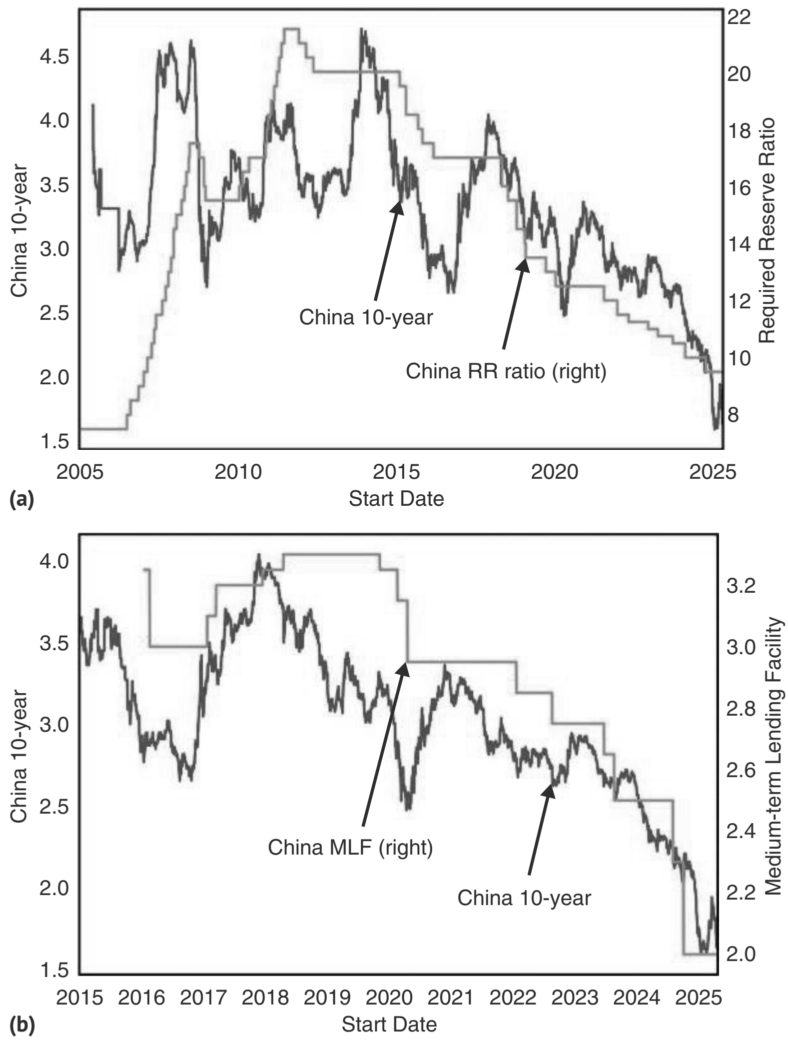


FIGURE 7.16 (a) CGBs peak after the last hike, and trough after the last RRR cut. (b) CGBs slightly more forward looking versus the MLF rate.
Source: Bloomberg.

7.7 EM RATES—WAGGED BY THE GLOBAL MACRO DOG

EM rates outside of China, we have also discussed at length in our EM book. Without going into too much detail, we briefly update our models on EM local on an asset class level. Broadly speaking, the information ratios have been quite poor on a FX unhedged basis, as FX generates most of the volatility for EM local, without adding a consummate return. We will discuss more about the EMFX angle in the FX chapter. FX hedging EM rates leads to much better information ratios. The global macro-environment is the key for when to own EM local hedged, as EM is often the tail being wagged by the global macro dog. In particular, both US rates and the USD have an enormous impact on EM rates, even once an FX hedge is in place. This is because the USD is an important input for EM inflation, which is the key target for most EM central banks. The significance of the USD stems from the high weight of traded goods in general, and commodities in particular (food, energy), in the CPI basket of many of the poorer countries, as services are a smaller part of the consumption basket. With traded goods usually having a USD component to their price setting, it is no surprise that FX matters more for inflation in poorer countries. US rates also obviously matter, setting the tone for global fixed income. The significance of those two factors is illustrated in 7.17a, where we show performance for four different regimes. DXY high or low (proxied as usual by the DXY above or below its 55dma), and US rates high and low (also proxied by whether US rates are above or below the 55dma). If we only own FX hedged EM local during times when US rates and DXY are below the 55dma, lagged by a day, the active information ratio is 2.47, while the active information ratio in the opposite case is –1.5. Of the two variables, US rates are more important, as low yields and a strong USD generate a very high information ratio, higher than with high yields and a lower USD. But the USD is also important. If higher rates do not generate/coincide with USD strength, the environment for EM local is not terrible, as EM local, FX hedged, still performs positively, albeit with a very low information ratio. Although it is the least likely scenario that higher US rates do not generate a higher USD, this scenario still occurs 21% of the time.

The picture is very similar if we do not FX hedge, but use spot FX to generate USD returns. Overall, information ratios are lower, given the FX volatility. The only strong return is for cases where both DXY and rates are well behaved, coming in at 1.4. FX is more relevant now, but rates keep the upper hand, as the case for a weak USD and higher rates does worse than lower rates and a higher USD. This is shown in Figure 7.17b.

7.8 ONE-WAY OR TWO-WAY TRAFFIC?

Most of our analysis treats the Fed as the agenda setter and investigates whether other central banks can decouple. In this model, US rates drive global rates, but not necessarily the other way around. Yes, the Fed will take global economic conditions into account, but only to the extent that they impact US fundamentals. But there is also the question to what extent global central banks or bond yields can impact US rates. This comes up most often in the context of Japan, as Japan is the single largest holder of US Treasuries, at just below 15% of outstanding, now larger than China, which has been slowly divesting. This discussion is distinct from our earlier discussion of whether JGBs outperform UST or not, as it is more concerned with a potential negative flow and price impact on USTs. One way to frame the discussion is that Japanese investors should really only buy US Treasuries when the yield, hedged into JPY, is higher than the yield on JGBs. If this were true, actions by the BoJ, which impact the cost of the FX hedge and Japanese yield levels, as well as the level of JGBs, should impact US rates through the demand channel. The reason to focus on hedged returns is that a large part of the bond holdings are typically FX hedged, given that most Japanese investors have

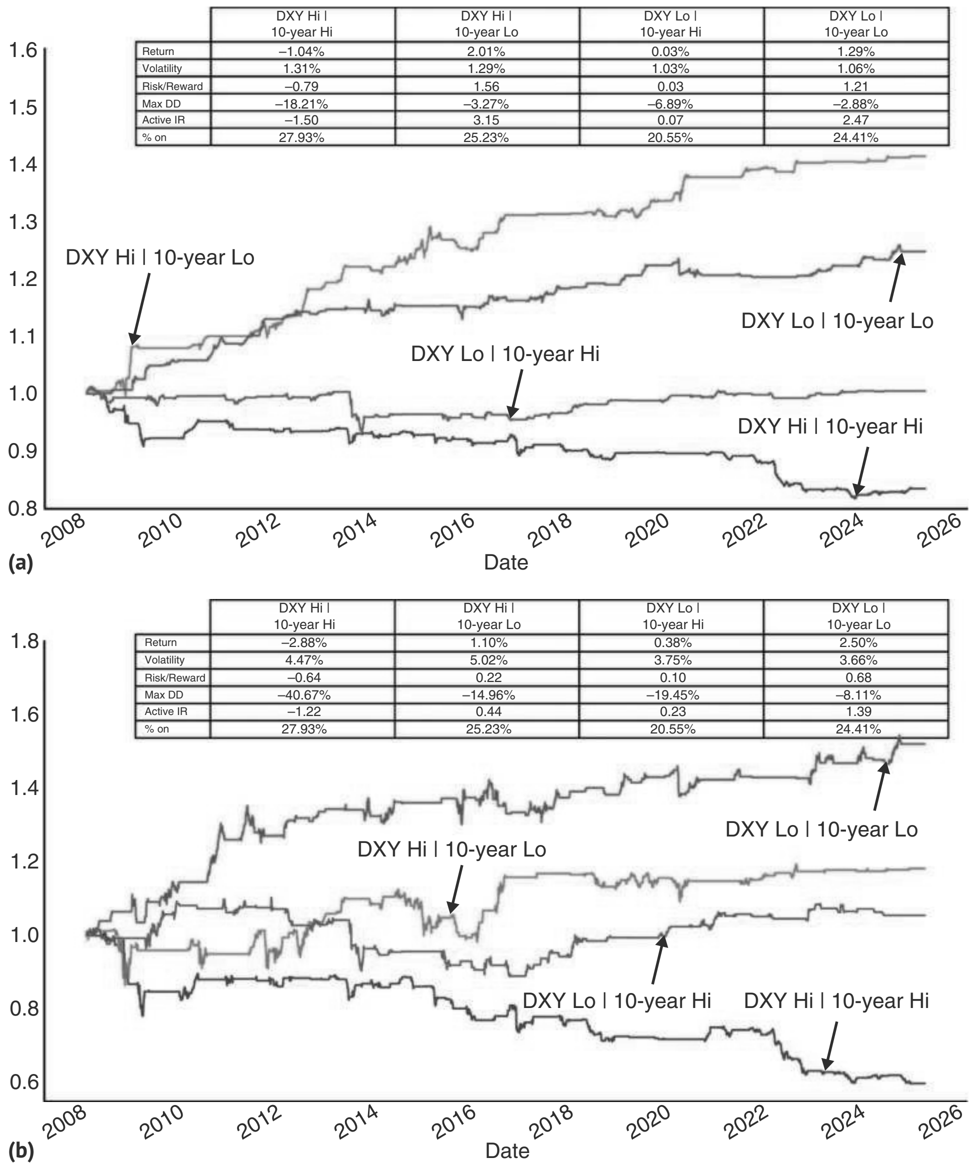


FIGURE 7.17 (a) EM bonds (FX hedged) require low US rates. (b) EM bonds with FX risk require low US rates and a weak USD.
Source: Bloomberg and authors' calculations.

JPY liabilities, and because the volatility of JPY is so much higher than the volatility of US rates, easily swamping any yield pick-up.

Figure 7.18 suggests that this is not fully correct, though. Japanese investors have been buying US Treasuries even with a deeply negative pick up both in 2019 and then again in 2023. The better way to explain Japanese flows into US Treasuries seems to be

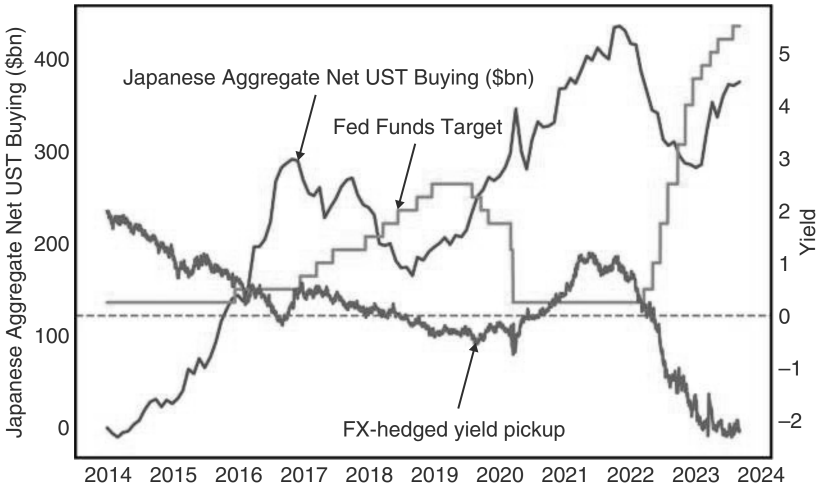


FIGURE 7.18 Japan trades the Fed cycle.
Source: Citi research.

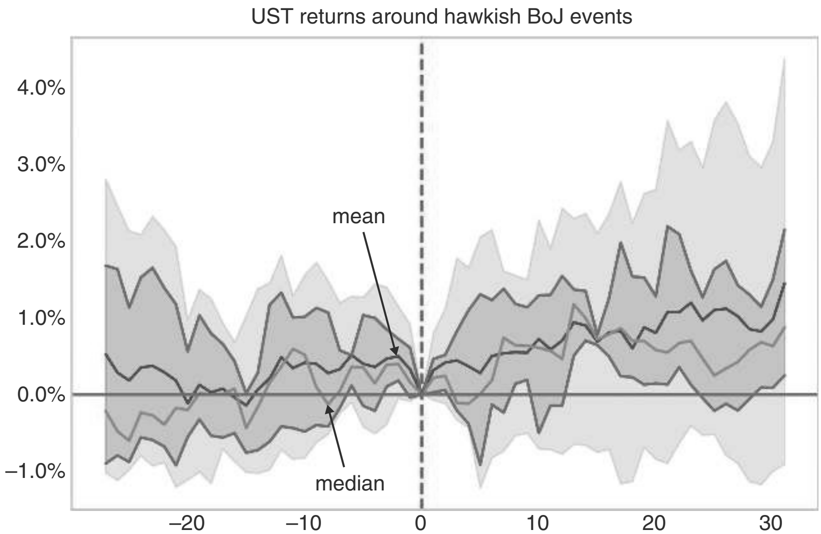


FIGURE 7.19 UST not overly impacted by BoJ.
Source: Bloomberg and authors' calculations.

the stage of the Fed cycle. Japanese investors start buying US Treasuries late in hiking cycles and keep buying throughout easing cycles, as well as when the Fed is on hold. They start to sell in the early parts of hiking cycles. They are apparently more driven by their expectation of capital gains than by the carry pick-up, outside of levels of extreme carry advantages discussed earlier.

Furthermore, US Treasuries generally perform well after BoJ hikes, as per Figure 7.19. This would suggest that the impact of the BoJ actions on US rates is limited. This is, of course, not the same as saying that global rates have no impact on the

United States. Under Greenspan, there was a long-lasting debate whether a global savings glut was leading to lower interest rates at the back end of the US curve, in spite of a 200bp hiking cycle by the Fed at that time. But even if true, this is not the same as saying that global rates or central banks are driving US Treasuries. It is more a case of a certain phenomenon, like the savings glut, impacting all global rates. Correlation is not causation, though the debate as to whether global central banks contributed to the savings glut has not been fully resolved.

7.9 SUMMARY

We investigate to what extent global bond markets can decouple from US Treasuries and find only limited evidence. We find that, on average, it pays to stay with the leaders, rather than with the laggards in a synchronized global easing cycle. Only between countries that are geographically close and highly interconnected, is it on average profitable to receive the laggard rather than the leader in the easing cycle, once divergence has become extreme. We discuss Bunds and find that when FX-hedged UST yields fall below Bunds, they outperform. The European periphery should be traded from the long side when German economic surprises are negative, and Italian surprises are positive, and vice versa. JGBs mostly trade with UST in a coincident fashion, but at the front end of the Japanese curve, the lags to the US front end can be very substantial and can be exploited. CGBs mostly beat to their own drum and yields rarely bottom before interest rate and RRR cuts are done. EM rates trade best in a regime of a weak USD and low US rates and vice versa.

CHAPTER 8

How to Trade Equities: A Framework

Equities are an interesting macro-asset. On the one hand, they are clearly impacted by macro cycles, by central banks, and by the USD. At times, equities are the key driver of the business cycle, like during the dot.com bubble or during the AI boom. On the other hand, for any given equity, the bottom-up picture is likely much more important than the overall macro, at least for medium-term returns. Therefore, we mostly focus on overall equity indexes, though we would note that macro is extremely important for many industries, and subindustry baskets, like banks. Moving to an industry level is also helpful in order to get around the fact that the SPX is constantly changing its composition, complicating historical comparisons. But that is a story for another book. For us, this changing industry composition of the SPX mostly means that we take the more recent performance of our backtests more seriously than the more distant past.

8.1 EQUITIES IN THE BIG PICTURE: THE ONLY WAY IS UP

The US equity market has been one of the best-performing long-term asset classes of recent decades. Even when starting in 1929, just before the onset of the deepest bear market on record, average annualized returns still amount to 7.9% (using monthly Shiller data excess of T-bill yields, 6.9% geometric returns). However, 10-year returns are highly dependent on the starting point and on the valuation at that point in time, as shown in Chapter 1. The other point to make is that the US market is somewhat unique. Sharpe ratios are significantly lower for many other equity markets (but not all).

To illustrate US exceptionalism, we use the underlying data in Jorda et al. (2019), which gives a very long-term history for a number of equity (and housing) markets, going back as far as the 1870s. In Figure 8.1 we show long-term, real (i.e., adjusted for CPI) returns for equity markets and their volatility, calculated using annual returns (see Jorda et al., 2017 for more details on the dataset). We start the study in 1901 such that each country has the same history, including a number of stress periods that have meaningfully reduced returns (especially in the case of France and Portugal) or increased volatility, like for Germany during the hyper-inflationary 1920s. We find that the United States, Australia, and a couple of the smaller Nordic countries stand out for their higher returns relative to volatilities. In absolute returns, Germany and Finland

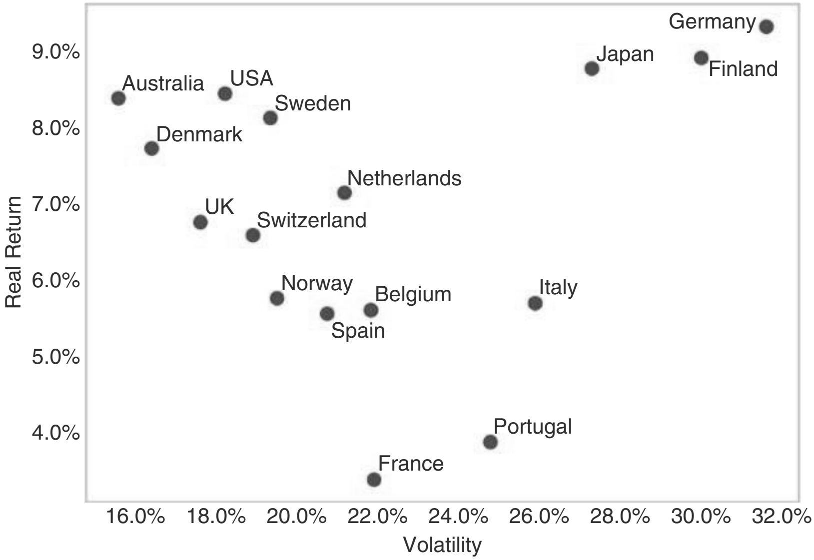


FIGURE 8.1 The US with strong, but not singular returns.
Source: www.macrohistory.net and authors' calculations.

score well over the very long term. Interestingly, in spite of its recent successes and the fact that World War II was much less devastating for the United States, it is not the single best performing large market.

In Figure 8.2 we show Sharpe ratios calculated using T-bill rates from the same database. On this basis, the United States is more exceptional, with only the much smaller Australia delivering higher risk-adjusted returns over the very long term. The United States has indeed been exceptional among large equity markets.

The question is whether the good fortune of the US equity market can persist. Until 2007, the US housing market had never turned down on a national level, even though it had happened frequently in other countries. Too narrow a focus on US historic performance could therefore create overconfidence. It is therefore important to understand why the US may be exceptional. To us, there are several factors at play. The single biggest one is that US EPS growth has also outpaced international markets. This could be a combination of higher GDP growth, maybe driven by more activist policymaking from both the US Treasury and the Fed, though the correlation between GDP growth and EPS growth is not as high as one would expect. Or it could be due to lighter regulation, potentially leading to more innovation, due to more oligopolistic industry structures, or due to more shareholder-friendly policies. The list of candidates is long. As some of these factors are structural, the good fortune of the United States may well persist. But we also note that a Sharpe ratio of 0.5 is not a lot of fun to trade, as drawdowns are relatively frequent and deep. We therefore investigate next how to actively trade equities from a top-down perspective.

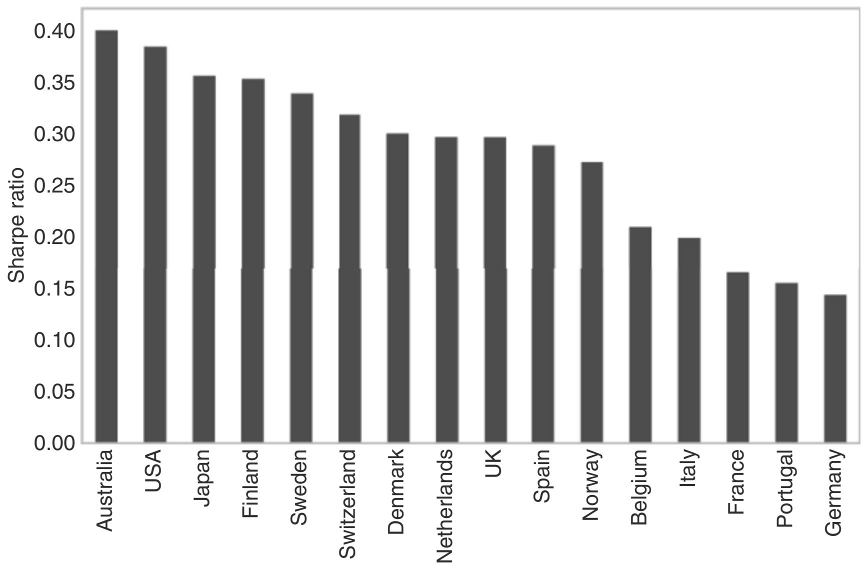


FIGURE 8.2 And a superior Sharpe ratio.
Source: www.macrohistory.net (use data since 1901) and authors' calculations.

8.2 EQUITIES AND THE BUSINESS CYCLE

Given the long-term track record, the default position on equities has to be long unless there is a compelling reason not to be. We have already shown in Chapter 3 that a coming recession is one reason not to be long equities, but here we broaden the approach by taking SPX drawdowns as a starting point for the analysis, rather than starting with recessions. Figure 8.1 shows all the episodes in which the SPX total return index had a drawdown of at least 20%, where a new drawdown can only start after a 25% rally from the lows (to get us back to where we came from). We check performance over T-bills. We start the analysis in the 1950s, as the 1930s and 1940s were a much more volatile time with a very different market micro-structure. We find that recessions are part of the problem. Of the 14 episodes where the SPX had a drawdown of more than 20%, seven were either coincident or directly preceding a recession, as is shown in Table 8.1. However, several recessions only led to downturns of the order of 15% and are therefore not listed here.

Avoiding recessions is only half the game, as is shown by Figure 8.3. The second most severe drawdown category was deflating bubbles (driven by the 2000–2002 event, as the 1961–1962 one was smaller). The 2000 event also had a (shallow) recession associated with it. Some drawdowns were financial market accidents/crises that did not lead to a recession: the 1987 crash, blamed partly on portfolio insurance exacerbating volatility, and the 1998 episode, driven by LTCM. More recently, the SPX experienced a sell-off of 25% in 2022. This sell-off probably coincided with a manufacturing recession, but more importantly, it coincided with an aggressive hiking cycle by the Fed, which was seen as likely to trigger a recession down the road, even though

TABLE 8.1 Bear Market Events

Peak	Trough	Max DD	Length (trough)	FF change	CPI change (trough)	10Y change	Reason
Jul-1957	Oct-1957	–21.2%	3.3	–1.2			Recession
Dec-1961	Jun-1962	–28.7%	6.4	0.0	0.6		Deflating bubble
Feb-1966	Oct-1966	–22.9%	7.9	0.0	1.2	0.3	Stagflation
Nov-1968	May-1970	–42.6%	17.8	0.3	1.6	2.4	Recession
Jan-1973	Oct-1974	–54.3%	20.7	2.1	8.2	1.6	Recession/ Stagflation
Nov-1980	Aug-1982	–35.6%	20.7	–2.0	–6.7	1.0	Recession
Jun-1983	Jul-1984	–21.5%	13.1	–0.6	1.8	2.5	Stagflation
Aug-1987	Dec-1987	–34.2%	3.3	2.4	0.0	0.2	Financial accident
Oct-1989	Oct-1990	–24.9%	12.1	–2.8	1.8	0.9	Recession
Jul-1998	Oct-1998	–22.8%	2.7	–0.5	–0.3	–1.0	Financial accident
Mar-2000	Oct-2002	–54.1%	30.5	–4.8	–1.7	–2.6	Deflating bubble
Jul-2007	Mar-2009	–55.8%	19.9	–5.0	–2.8	–2.2	Recession
Feb-2020	Mar-2020	–34.4%	1.1	–1.5	–0.8	–0.8	Recession
Nov-2021	Oct-2022	–26.0%	11.2	5.0	0.9	2.5	Stagflation
Mean		–34.2%	12.2	–0.6	0.3	0.4	
Median		–31.4%	11.6	–0.6	0.6	0.6	

Source: Bloomberg and authors' calculations

that broader recession never happened. We label it stagflation, as high inflation was an important characteristic of this bear market, which triggered the Fed hikes. 1966 was also based on stagflationary vibes, as those years saw the first inflation scare in the post-war period, even though it was on a smaller scale than 2022. 1984 saw an aggressive hiking cycle by Volcker, also due to rising inflation fears. We label the 1973–1974 episode recession, but it came with very high inflation too. Overall, the labels are not always clear-cut, but the lesson is that while it is crucial to avoid recessions, other factors can lead to major downturns as well. Environments of high inflation and weak growth can be very equity negative, even if a recession can be avoided. And the aftermath of bursting equity bubbles can see protracted bear markets. Financial market accidents are the other problematic regime, but those can rarely ever be anticipated.

As we had already discussed in Chapter 3, the good news is that the equity market is not overly forward-looking with respect to recessions. The bad news is that it is very difficult to forecast a recession even shortly before the recession starts—partly because the market is still close to the highs, and Wall Street feels like a happy place.

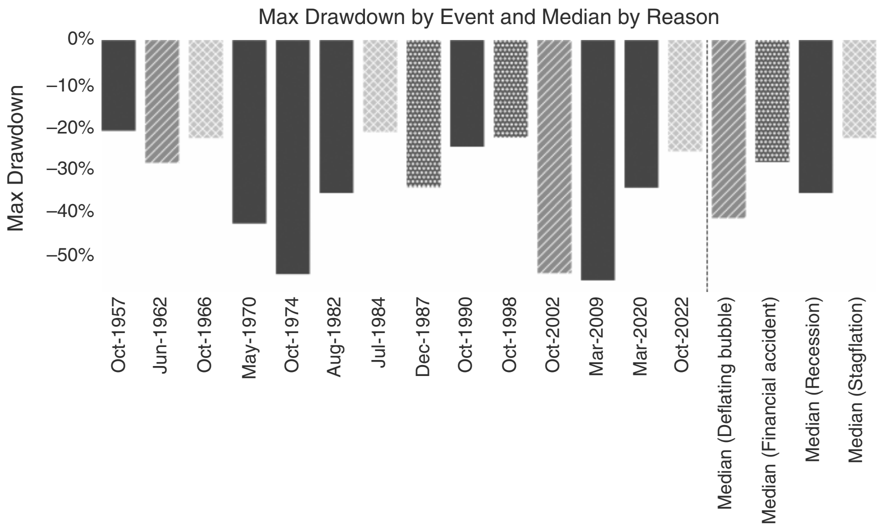


FIGURE 8.3 Recession and deflating bubbles the worst regime.
Source: Bloomberg and authors' calculations.

So what should investors do? As per Chapter 3, some promising indicators are the soft data versus hard data indicator, even though the sample size is quite small. We also commented on the yield curve and the long and varying lags between inversions and recessions, as well as on disinversions and recessions. In the end, we think that when any of these indicators suggests an increased probability of a recession, investors should trade more carefully, for example, by taking breaks of the 200dma more seriously as a risk management tool rather than necessarily reducing risk in a wholesale manner.

8.3 EQUITIES AND THE FED

The relationship between the Fed and equity markets is not obvious. A Fed cut itself is a positive factor as it lowers the discount rate for future profits. However, if the Fed cuts because of a coming recession, it often is not a sufficiently positive factor to detract from the overall recession dynamic, even though a recession *without* Fed cuts would likely be even worse. Given that it is not trivial to figure out whether a recession is indeed coming or whether it is a soft landing, at least early on, we analyze the empirical evidence. We focus here on the first and last cut, as well as the first and last hike, since a change in the overall strategy of the Fed should be the most meaningful.

First, we study the median performance of equities around first Fed cuts. As illustrated in Figure 8.4, the SPX pulls back three months before the first cut, probably as the first growth clouds appear, but then rallies into the cut the month before, likely expecting Fed relief. It then chops around the unconditional performance line, before eventually taking off again around two months later. The skew is to the downside, as the moderately higher median is the result of cases where the economy was still in relatively good shape at the time of the cut, and cases where the cuts were happening in

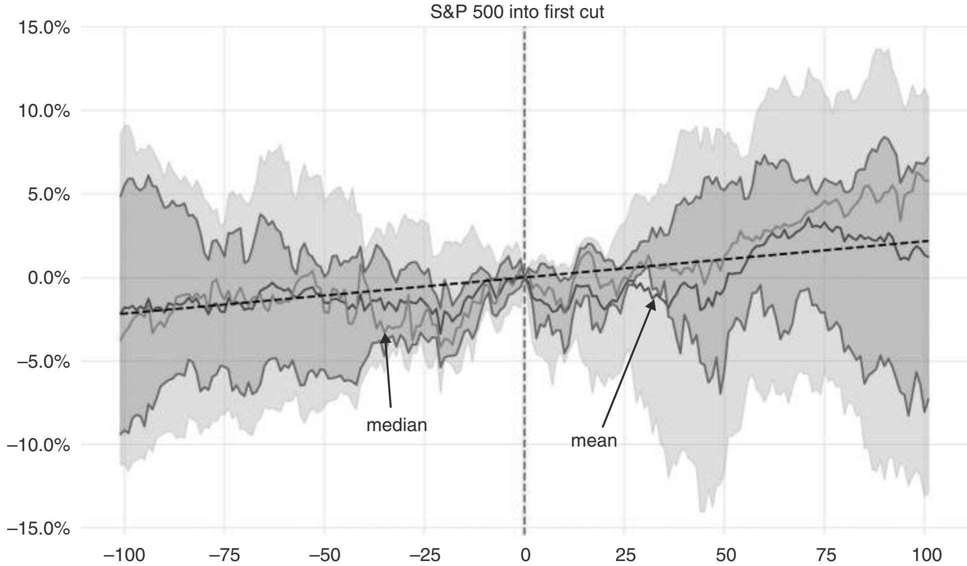


FIGURE 8.4 US equity returns strong just before first cut.
Source: Bloomberg and authors' calculations.

reaction to severe economic weakness, when market stress was starting to move substantially higher into very severe recessions.

When the Fed stops cutting, the opposite logic holds. Clearly, no more easing by itself is equity negative. But if it is done because of strong growth, these effects will likely outweigh the negative impact of the Fed's action. We studied how the SPX behaves around the last cut in Figure 8.5. We find that equities bottom around two months before the last cut, then rally into the last cut, and, after a month of sideways price action, do quite well again. The reason may be that it is often not fully clear whether the last cut occurred or not. A month later, when the next Fed meeting occurs, it becomes more obvious whether the last cut indeed has already occurred. The first meeting when the Fed pauses is a better entry point to buy the dip in US equities than the last cut, as by then less Fed stimulus is priced in.

If we focus on soft landings only, the number of sample points drastically decreases. We would consider easing cycles beginning in July 1995 (driven by rising US unemployment and falling inflation in the aftermath of an aggressive hiking cycle, with the Tequila crisis possibly contributing), September 1998 (fallout from Russian crisis and LTCM), and 2024 (inflation normalizing after an aggressive hiking cycle) as easing cycles coinciding with soft landings, as there was no imminent recession. 1987 could be seen as a candidate, but we exclude it as the equity crash triggering the cut was somewhat unique. Interestingly, all three soft landing cycles resulted in cumulative Fed cuts of 75bp, clearly less than the average Fed cutting cycle. The duration of those short cycles was not as uniform, though, ranging from two months in 1998 to seven months in 1995. In those three episodes, US equity returns were mixed into the first cut (driven by 1998, which was essentially financial market accidents). However, after

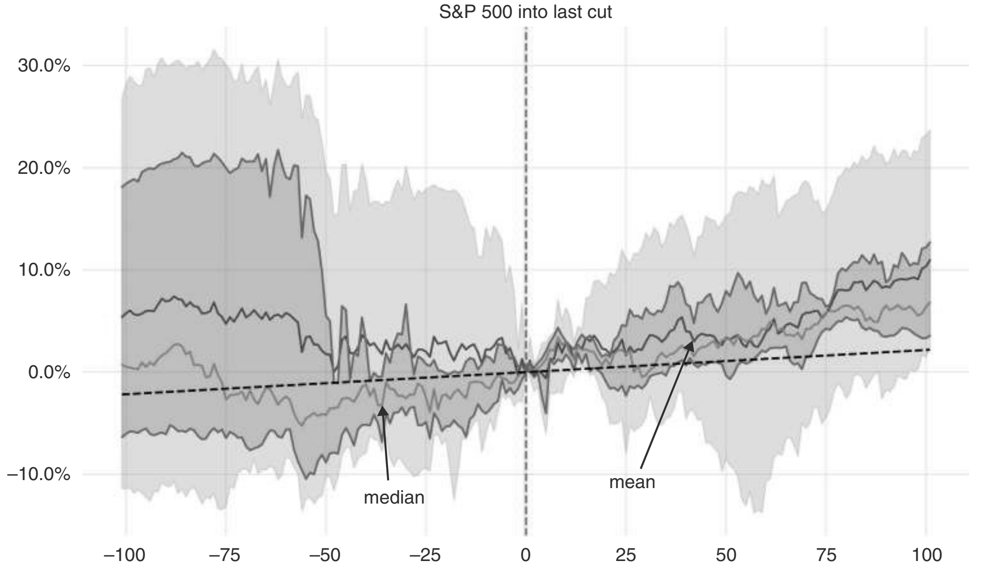


FIGURE 8.5 Best entry point is in Fed meeting after last cut.
Source: Bloomberg and authors' calculations.

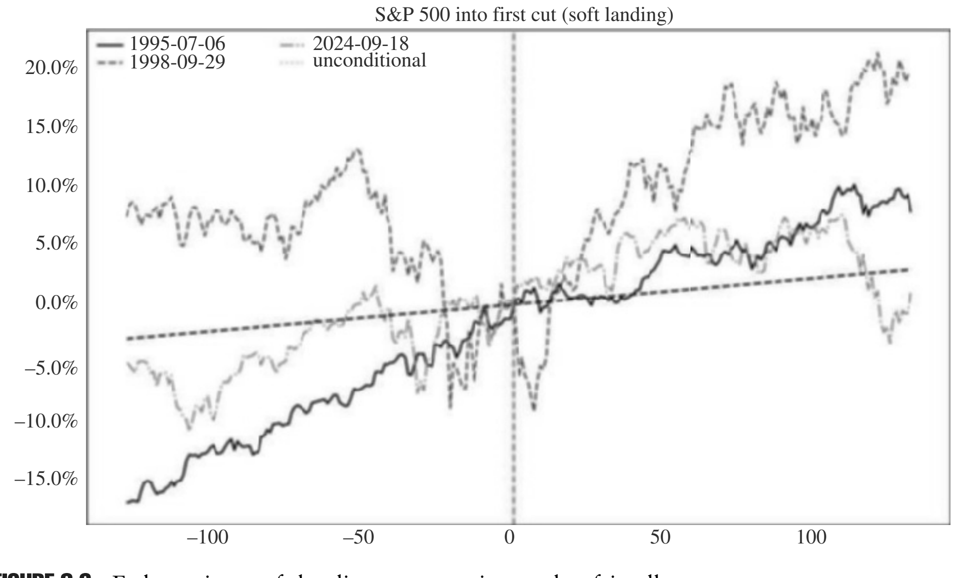


FIGURE 8.6 Fed cuts into soft landings very equity market friendly.

the first cuts, the returns were very strong, as can be seen in Figure 8.6. Cuts without a recession are the best-case outcome for equities.

What about the hiking side? Here, the same logic holds as during cuts. Although hikes per se are obviously not a positive, if they are driven by strong growth, they

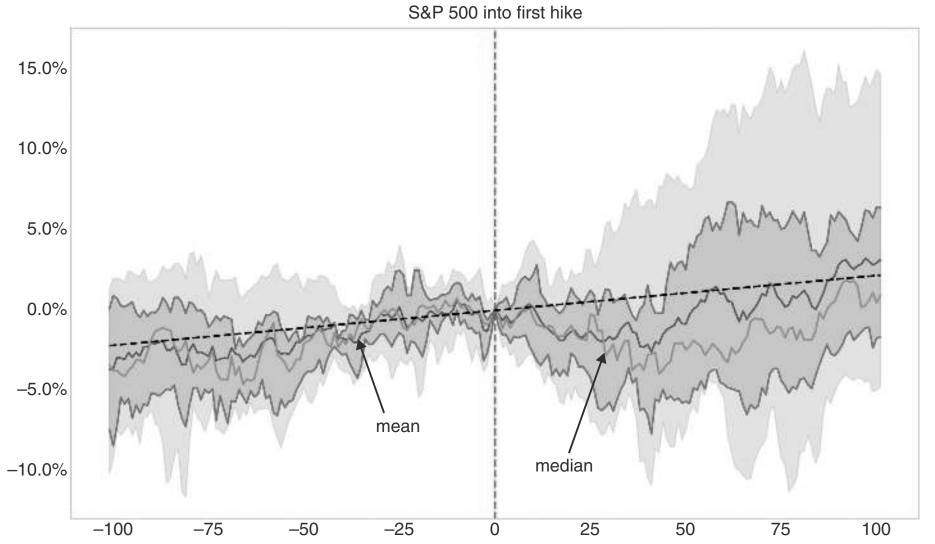


FIGURE 8.7 SPX pulls back after first hike.
Source: Bloomberg and authors' calculations.

are unlikely to end a bull market. Eventually, the Fed may well overtighten, hurting equities, but that is unlikely to be the case early in the hiking cycle. Still, as per Figure 8.7 there is a pullback after the first hike, in the median case, worth around 5%, which is quickly recovered. Interestingly, the skew is to the upside. Into the first hike, there is no clear pattern, and the median performance is close to the unconditional performance.

The same analysis for the last hike, and again starting in 1986, shows that equities have done well following the last hike, after a brief period of indigestion, as can be seen in Figure 8.8. In the 2023 episode, the equity market had rallied strongly into it but then had a longer-lasting and deeper pullback than usual after the cut, before restarting the bull market. The reason is likely that the Fed skipped a hike in June of 2023, before squeezing in one last hike in July. This is highly unusual, and a pause so late in the cycle is typically mistaken for the end of the cycle. Indeed, after the penultimate hike in May, the SPX rallied strongly. Last hikes are generally bullish after just a few days of indigestion.

Focusing on rates, rather than on Fed actions, we find that it is quite normal that, at various points during the hiking cycle, the equity market becomes jittery when rates are rising. Theoretically, rising rates due to higher inflation should be a bigger headwind than higher rates that are more driven by higher growth (though higher inflation will also be helpful to EPS growth, as EPS is a nominal number and, to some extent, price pressures are passed through to the consumer). We study this question below. First, we are interested in whether there is an absolute level of real rates that has historically been problematic for equity markets. The answer to this question is illustrated in Figure 8.9, where we split the level of 10-year real rates since the inception of the TIPS market into quintiles. The results are somewhat mixed. The first observation is that every

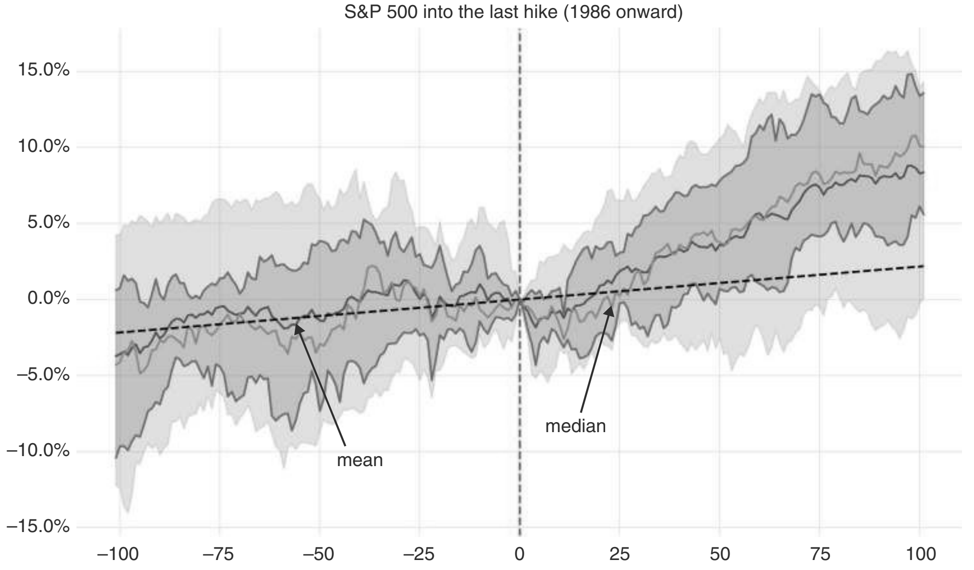
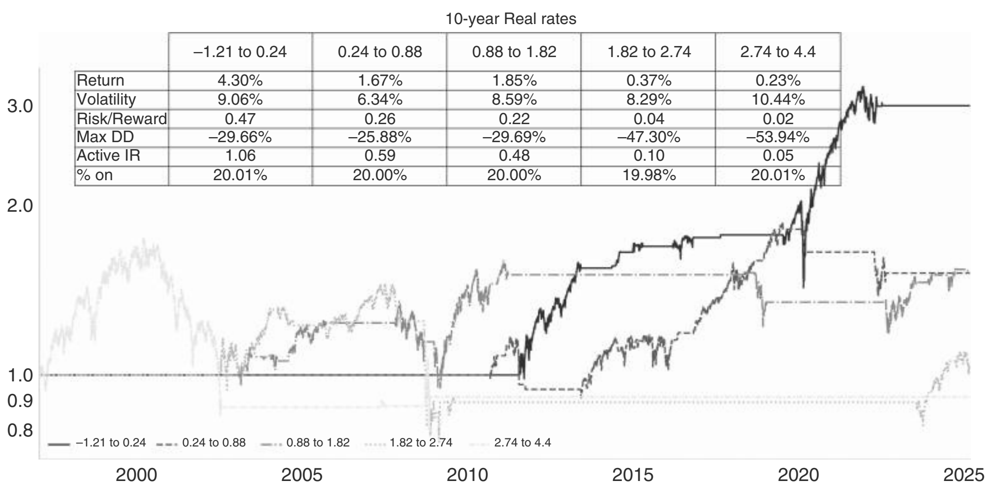


FIGURE 8.8 SPX likes last hikes.
Source: Bloomberg and authors' calculations.



	–1.21 to 0.24	0.24 to 0.88	0.88 to 1.82	1.82 to 2.74	2.74 to 4.4
Return	4.30%	1.67%	1.85%	0.37%	0.23%
Volatility	9.06%	6.34%	8.59%	8.29%	10.44%
Risk/Reward	0.47	0.26	0.22	0.04	0.02
Max DD	–29.66%	–25.88%	–29.69%	–47.30%	–53.94%
Active IR	1.06	0.59	0.48	0.10	0.05
% on	20.01%	20.00%	20.00%	19.98%	20.01%

FIGURE 8.9 Real rates above 1.8% a yellow light.
Source: Bloomberg and authors' calculations.

quintile has a positive active information ratio, though it is true that the ratio is lower, the higher the real rates are, and it deteriorates quite sharply when real rates exceed 1.8%. However, even the worst quintile included very meaningful bull markets, and the two worst quintiles are as negative as they are due to the aftermath of the Nasdaq

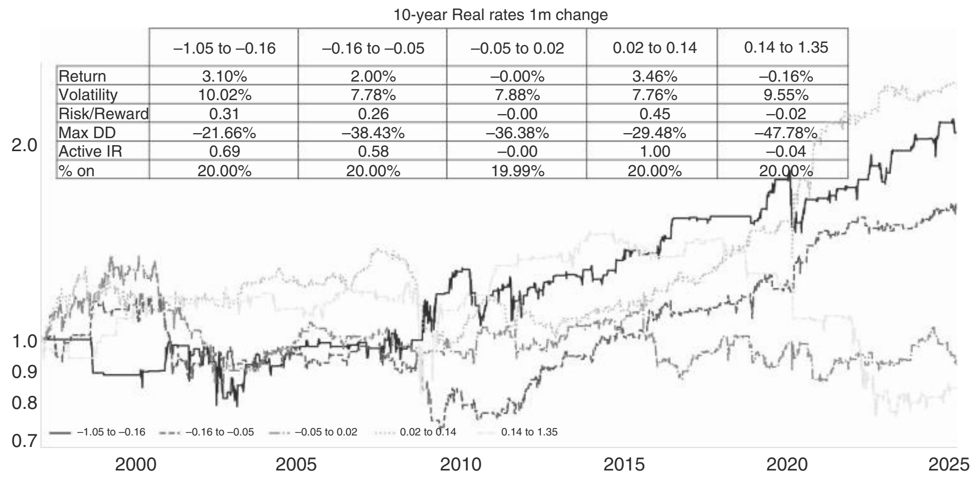
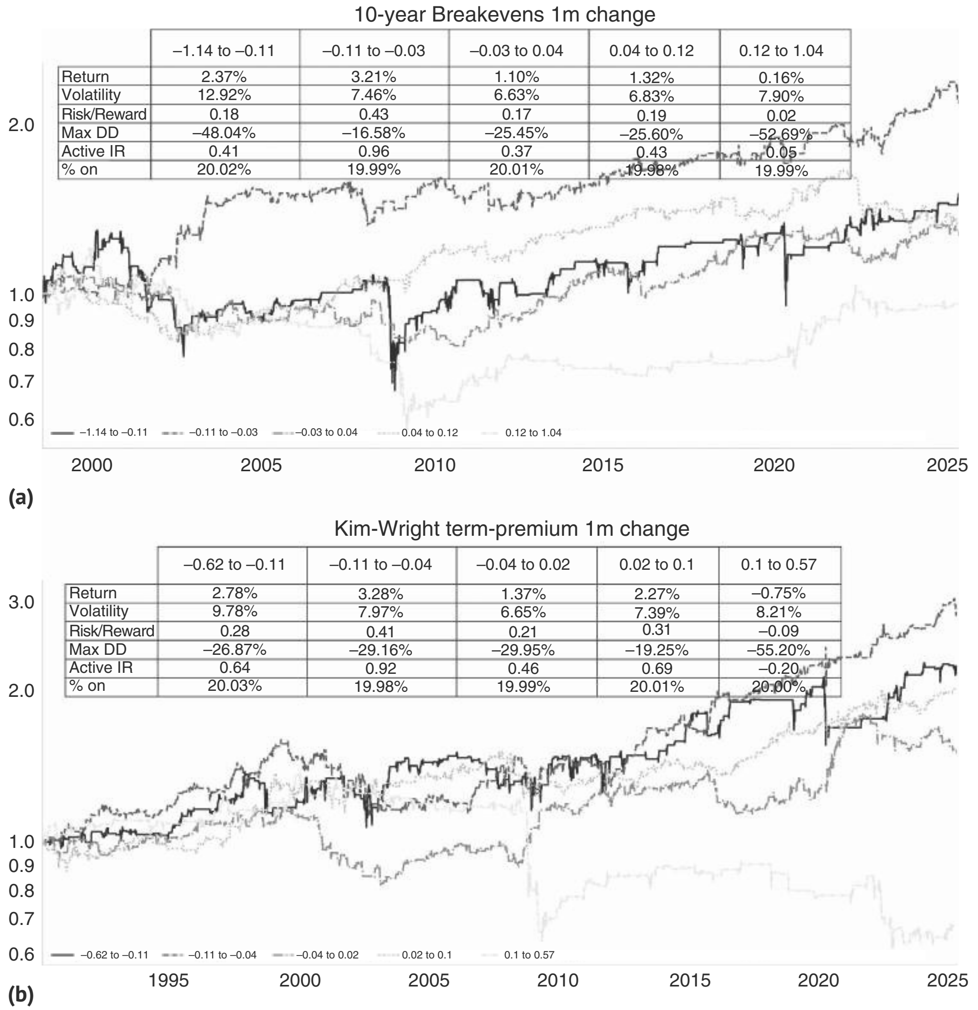


FIGURE 8.10 Large real rates move higher a clear equity negative.
Source: Bloomberg and authors' calculations.

bubble and the 2008 episode, respectively. We would therefore not make too much of the danger level of 1.8%, setting our warning light at best to amber.

Second, we investigate changes in real rates, focusing on the 10-year point. Changes are more highly correlated with equity performance but are, of course, not known ex ante. Using the same split into quintiles, the top quintile, when real rates increase by more than 14bp per month, is the worst performing and has a negative active information ratio, as is shown in Figure 8.10. The only big rally in that bracket is the sharp bounce back after the 2008 debacle. This move up in real rates back then started from negative territory, with real rates only becoming positive in Q2 of 2013, in the so-called "taper tantrum." A move toward positive rates was not harmful to the SPX (though, perhaps somewhat surprisingly, the SPX continued to ignore rising real rates until late 2014, i.e., even after rates had already moved to more reasonable levels). Since late 2014, sell-offs of more than 14bp per month have been relatively consistently negative for US equities. Although this rule of thumb may not be overly useful as those sell-offs are hard to anticipate, it is still helpful to know what type of sell-off is required before we have to worry about equity risk.

What about the split between growth and inflation as a driver? Re-running the study for one month changes in 10-year breakevens does not support the contention that rising inflation expectations are significantly worse than rising real rates, as is shown in Figure 8.11a. Here, all quintiles have positive active information ratios. Again, the highest quintile has the lowest ratio, but performance in this quintile has largely been positive since the end of the GFC in 2009 and is very slightly above the performance for real rates in that quintile. Although there are differences for the other quintiles, the overall conclusion from the study is that it is not *that* important whether a big sell-off in rates is driven by real rates or break evens. We also re-run the same exercise for the term premium, shown in Figure 8.11b. Here, we pick the Kim Wright



	–1.14 to –0.11	–0.11 to –0.03	–0.03 to 0.04	0.04 to 0.12	0.12 to 1.04
Return	2.37%	3.21%	1.10%	1.32%	0.16%
Volatility	12.92%	7.46%	6.63%	6.83%	7.90%
Risk/Reward	0.18	0.43	0.17	0.19	0.02
Max DD	–48.04%	–16.58%	–25.45%	–25.60%	–52.69%
Active IR	0.41	0.96	0.37	0.43	0.05
% on	20.02%	19.99%	20.01%	19.98%	19.99%

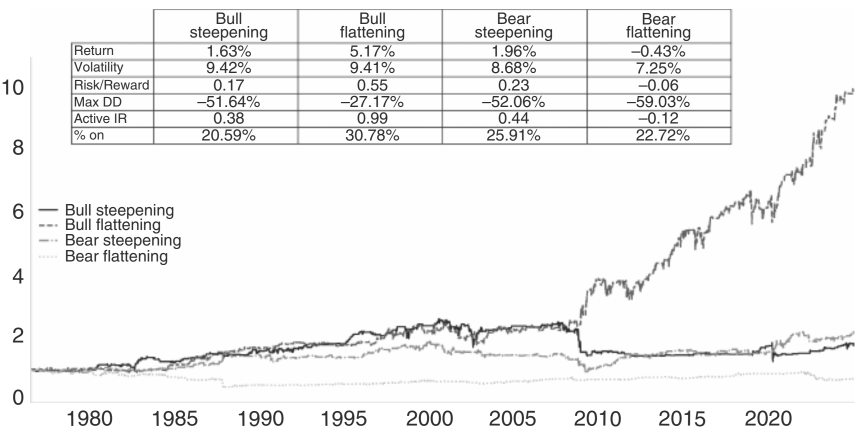
	–0.62 to –0.11	–0.11 to –0.04	–0.04 to 0.02	0.02 to 0.1	0.1 to 0.57
Return	2.78%	3.28%	1.37%	2.27%	–0.75%
Volatility	9.78%	7.97%	6.65%	7.39%	8.21%
Risk/Reward	0.28	0.41	0.21	0.31	–0.09
Max DD	–26.87%	–29.16%	–29.95%	–19.25%	–55.20%
Active IR	0.64	0.92	0.46	0.69	–0.20
% on	20.03%	19.98%	19.99%	20.01%	20.00%

FIGURE 8.11 (a) Performance with rising break evens not that different from rising real rates. (b) Large 1m term premia increases more equity negative.
Source: Bloomberg and authors' calculations.

model, which incorporates survey data to improve the measurement of the term premium. We find that the top quintile of changes has a more meaningful negative active information ratio than our previous two studies. A rise of more than 10bp in this measure of the term premium has resulted in a negative active information ratio of –0.20. While a significant part of this result is due to the GFC period, this measure has started to work well again since 2015.

Related to moves in term premia and real rates is what the curve is doing at the same time. In this section, we analyze equity performance in bull and bear steepening environments. There are many ways to measure the behavior of the curve. In this study, we use daily rolling 1m changes in both 2-year and 10-year US interest rates. If the long-end yield is rising and rising by more than the short-end, we classify it as bear steepening. If the short-end rises more, we label it bear flattening. We do the same for bull (10-year rate falling) steepening or flattening. As usual, we lag the SPX returns by one day to make it, in theory, tradable. As per Figure 8.12, bull flattening is the preferred environment, where the large majority of SPX gains accrue. The most negative regime is bear flattening, which typically occurs during Fed hiking cycles. However, the negative performance is really concentrated during only two episodes: the 1987 crash and the 2022 inflation-induced hiking cycle. Bear steepening is, somewhat surprisingly, not as negative as widely believed.

We now turn the question of how rates impact equities on its head and ask whether equity market sell-offs generate Fed cuts, i.e., whether the mythical "Fed put" exists and how to find out where it sits. We do this by showing what pullback from the one-year high in the SPX (reached on day zero in Figure 8.13a) is needed for the median Fed Funds to drop around day zero. It turns out that the answer to this exercise is 25%. After a 25% drawdown from the one-year high, the Fed is cutting rates in the median case. If we analyze episode by episode, Figure 8.13b illustrates that even a 20% drawdown usually is consistent with Fed cuts, though only if inflation is low. This 20% number coincides with the view that many of the corrections that are 20% or larger coincide with recessions, where the Fed cuts rates for other reasons than just equity market weakness. During high inflation episodes (1973, 2022), hikes continue for longer, which is why the median is a larger 25% drawdown. We would also note that other market factors also play a role, and maybe an even bigger one than equities per se. In particular, the Fed is very attuned to credit markets, liquidity conditions, and any market malfunction. Although these often coincide with equity weakness, these



	Bull steepening	Bull flattening	Bear steepening	Bear flattening
Return	1.63%	5.17%	1.96%	–0.43%
Volatility	9.42%	9.41%	8.68%	7.25%
Risk/Reward	0.17	0.55	0.23	–0.06
Max DD	–51.64%	–27.17%	–52.06%	–59.03%
Active IR	0.38	0.99	0.44	–0.12
% on	20.59%	30.78%	25.91%	22.72%

FIGURE 8.12 Bull flattening is the best environment.
Source: Bloomberg and authors' calculations.

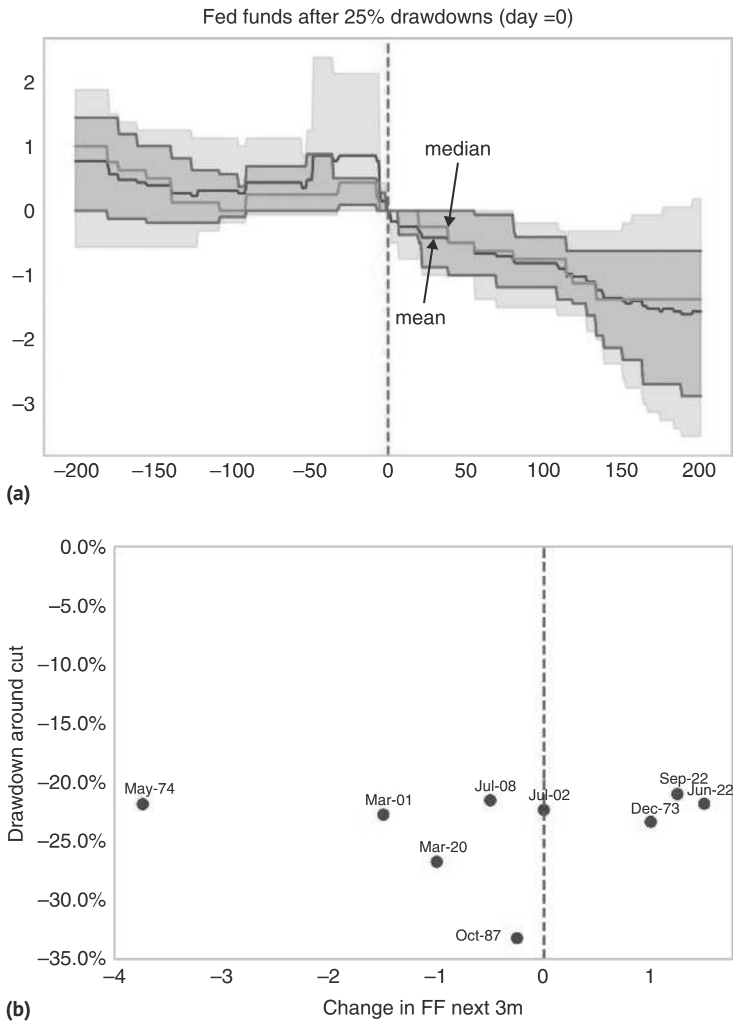


FIGURE 8.13 (a) 25% drawdown needed for immediate Fed Put activation. (b) In low-inflation environments 20% drawdown is often enough.
Source: Bloomberg and authors' calculations.

factors can certainly incentivize the Fed to move earlier or later than our equity-centric rule of thumb suggests.

8.4 EQUITIES AND EARNINGS

In our discussion about China and India, we highlighted that GDP is not necessarily a good predictor of equity performance in a cross-sectional setting. The exceptions are recessions, which are almost always equity negative, and extremely high growth of

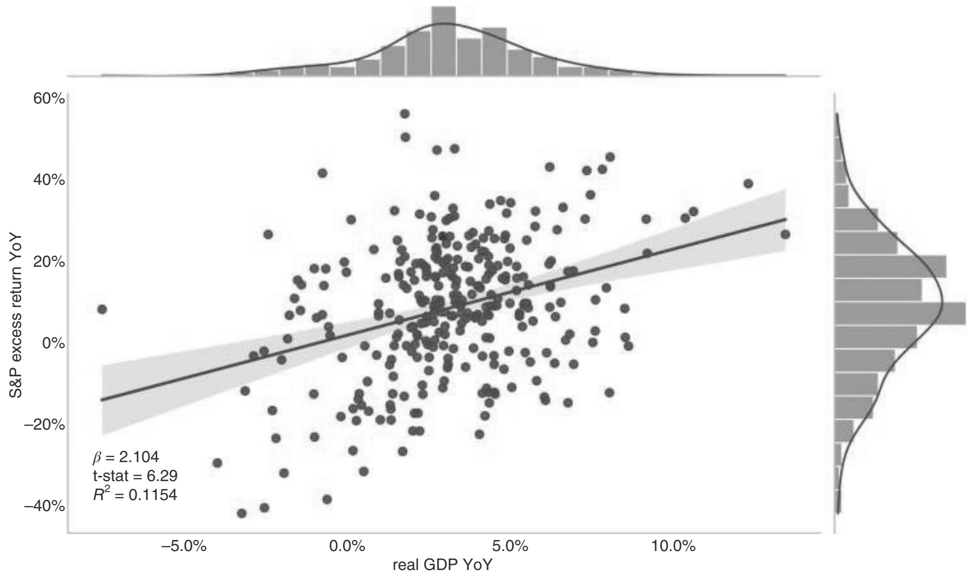


FIGURE 8.14 US equity returns and GDP only correlated at extremes.
Source: Bloomberg and authors' calculations.

more than 10%, which is often associated with a bounce back from a deep recession, and which is very equity positive. But outside of the extremes, real GDP growth is not overly correlated with equity performance. The equivalent correlation using nominal GDP is insignificant. This is illustrated in Figure 8.14.

The analysis is more interesting once we take leads and lags into account. Figure 8.15a shows the lead-lag relationship between nominal GDP changes and equities for up to eight quarters. We use the Shiller dataset and nominal GDP data from 1947 onward (starting later does not change the results). The bars are centered around the contemporaneous relationship (zero lag), which is weak (0.02) and insignificant. For negative lags, where a positive number would indicate GDP growth leads equity returns, we find a negative correlation, i.e., low quarterly GDP growth portends higher equity returns two quarters later. For the positive lags, i.e., equities leading GDP, we see a positive relationship; high equity returns tend to mean higher GDP growth one or two quarters in the future. With respect to earnings, we find a stronger contemporaneous relationship with GDP, with a significant correlation coefficient of 0.125, as per Figure 8.15b. And just like for equity returns, weak GDP leads to higher EPS growth with a three to five quarter lag.

Earnings are highly correlated with equity performance. As would be expected, equity prices are leading EPS. In Figure 8.16, we show the results of a regression of excess equity returns, with various monthly lags, on earnings. The coincident beta comes in at just over 0.1. The correlation is overall relatively high from zero to six months, with the highest correlation being the two-month lag. The publication lag would add to the lag, as earnings are only published a few weeks after the quarter end. Again, equity prices are forward-looking by around one to two quarters. Interestingly,

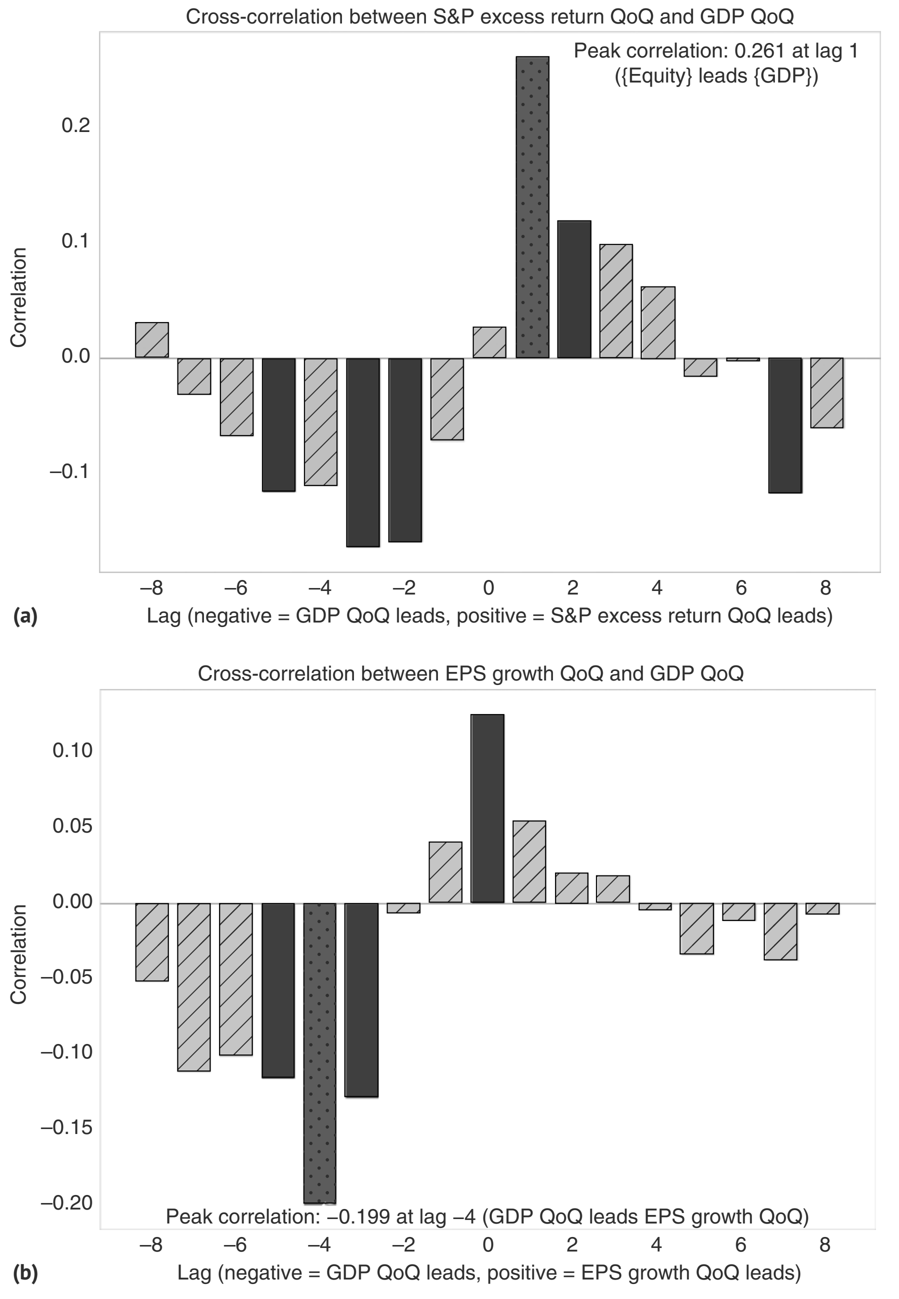


FIGURE 8.15 (a) High equity returns = higher future GDP growth. (b) Weak GDP means future higher earnings growth.
Source: Bloomberg, shillerdata.com, and authors' calculations.

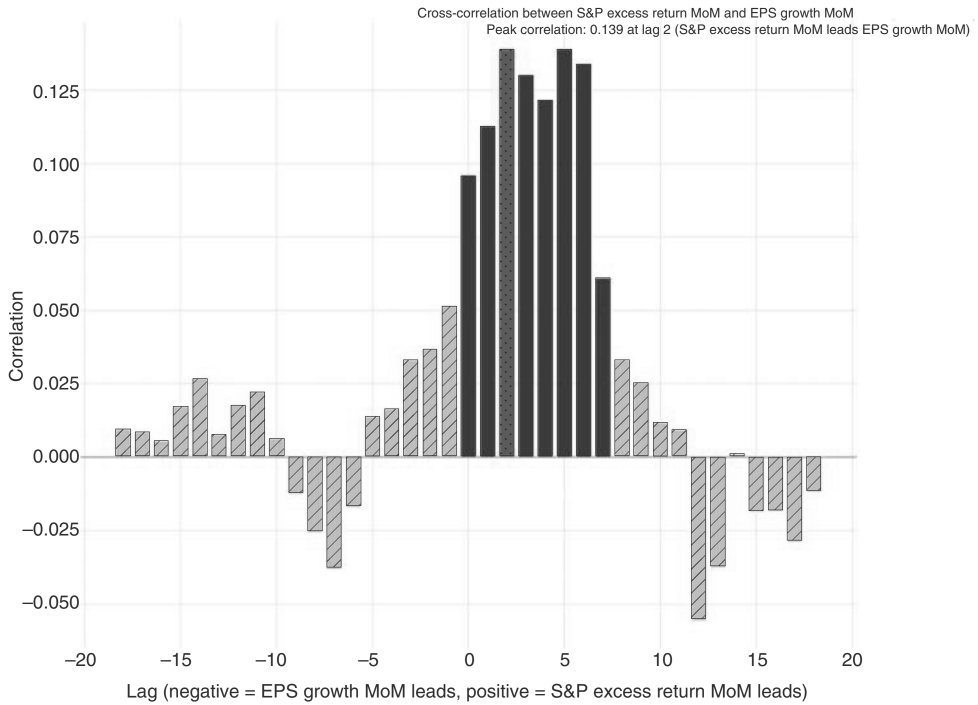


FIGURE 8.16 US equities lead earnings by one to two quarters.
Source: Bloomberg and authors' calculations.

at lags of more than one year, the relationship is negative, suggesting some of the price increases in earnings tend to come from overoptimism.

When observing how equities behave during earnings season, i.e., whether they go up or down on an earnings print, typically depends on whether the forward (full-year) earnings estimates are raised by analysts in the aftermath of the earnings call. Earnings estimates are, therefore, crucial to understanding the equity market.

There is a clear seasonal pattern when it comes to earnings estimates. We illustrate this in Figure 8.17, using Citi's earnings revision index (ERI). For the full year, earnings revisions are typically falling slightly. For more details, see Manthey et al. (2025). On a quarterly time frame, it is usually the case that earnings are revised lower during the pre-reporting season, only to rise during the reporting season itself. This is because corporates cut their estimates gradually going into the earnings call, lowering the bar. In particular, in the last month of the quarter, negative warnings come in from companies, where fundamentals deteriorated. On average, there are not enough positive warnings to offset those, as corporates are less incentivized to get the good news out ahead of the quarterly report. Therefore, earnings estimates typically move lower into the earnings release, only to go up after earnings are announced, as more corporates beat than miss.

The key question is whether the ERI is useful in generating alpha in equity markets. We find that earnings revisions are both valuable as a trend indicator *and* as a counter-indicator. In particular, buying equities when the earnings revisions break

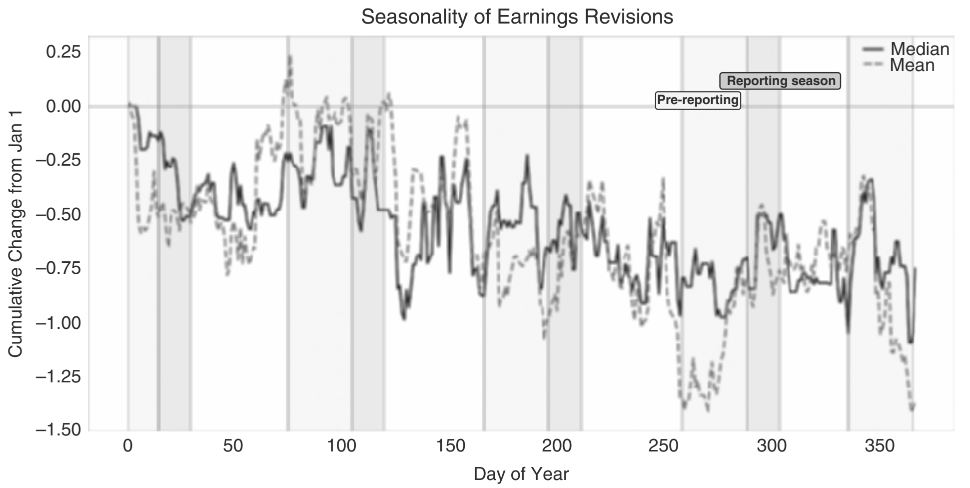
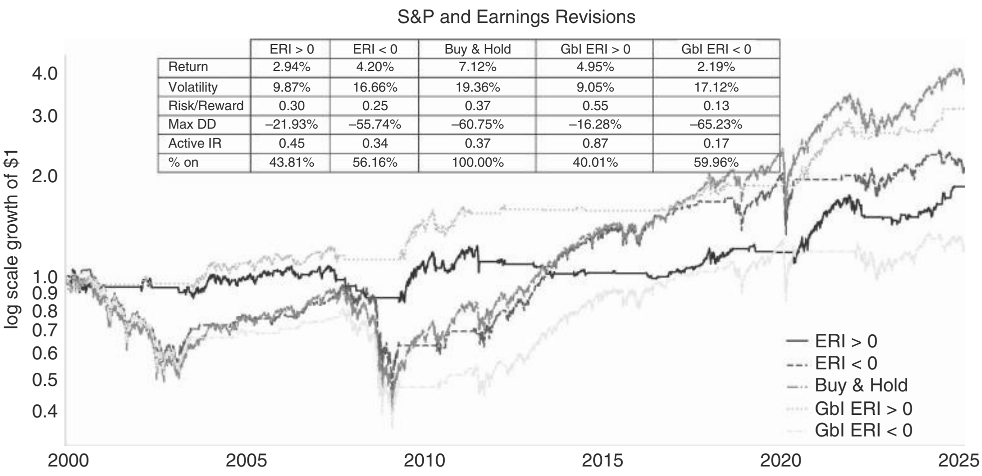


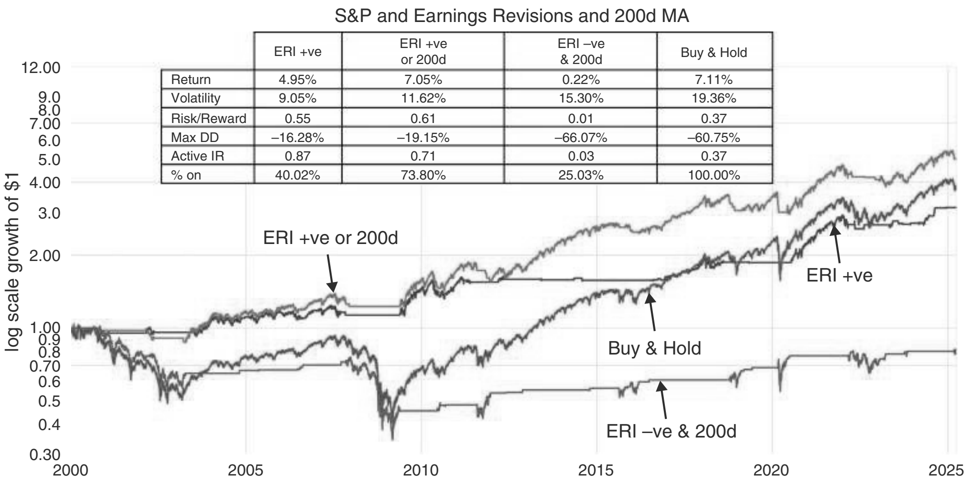
FIGURE 8.17 Earnings revisions lower in pre-reporting.
Source: Citi research and authors' calculations.



	ERI > 0	ERI < 0	Buy & Hold	Gbl ERI > 0	Gbl ERI < 0
Return	2.94%	4.20%	7.12%	4.95%	2.19%
Volatility	9.87%	16.66%	19.36%	9.05%	17.12%
Risk/Reward	0.30	0.25	0.37	0.55	0.13
Max DD	−21.93%	−55.74%	−60.75%	−16.28%	−65.23%
Active IR	0.45	0.34	0.37	0.87	0.17
% on	43.81%	56.16%	100.00%	40.01%	59.96%

FIGURE 8.18 Positive ERI helpful, mostly by avoiding bear markets.
Source: Bloomberg, Citi research and authors' calculations.

above zero is typically a good signal. The information ratio of equities with a positive earnings revision index is higher than with a negative earnings revision index. Interestingly, using global earnings revisions indices is superior to using US earnings revisions, even for US equities. The ERI is adding alpha mostly by avoiding major bear markets (2000, 2008). It also misses long bull markets, though, and would have kept investors largely out of the market from 2012 to 2016! The details of the study are illustrated in Figure 8.18.


S&P and Earnings Revisions and 200d MA

	ERI +ve	ERI +ve or 200d	ERI –ve & 200d	Buy & Hold
Return	4.95%	7.05%	0.22%	7.11%
Volatility	9.05%	11.62%	15.30%	19.36%
Risk/Reward	0.55	0.61	0.01	0.37
Max DD	–16.28%	–19.15%	–66.07%	–60.75%
Active IR	0.87	0.71	0.03	0.37
% on	40.02%	73.80%	25.03%	100.00%

FIGURE 8.19 Using ERI in combination with 200dma.
Source: Bloomberg, Citi research and authors' calculations.

We therefore also investigate how the ERI interacts with the 200dma in Figure 8.19. We find that the active information ratio is higher for positive global ERI only, rather than for a strategy that goes long if *either* the ERI is positive, *or* the market is above its 200dma. Having said that, the total return is better if we implement the either-or condition. Interestingly, insisting on both conditions on the downside (ERI negative and SPX below its 200dma) has a worse active information ratio than only a negative ERI and results in lower (but still slightly positive) returns. We would therefore use the ERI as a trend signal only if it is in line with the 200dma.

Interestingly, the ERI can also be a countersignal if the ERI is too oversold. As per Figure 8.20, the ERI is highly correlated with forward EPS estimates. Given that it is a breadth measure, it tends to move early but is also more volatile. We found that buying the market when the ERI is depressed is a superior countersignal to buying the market when the forward EPS is depressed.

In Figure 8.21, we show that buying the SPX when the global ERI is moving into the fifth lowest percentile is a good strategy. This is true for short holding periods of one month and for longer holding periods up to 12 months, where the active IR is still meaningfully higher than the unconditional one. This ERI indicator mostly buys after meaningful pull-backs in equities, given that equities react to lower earnings expectations. Sometimes the rule triggers too early, as in the deflating 2000 bubble or the 2008 crisis, and sometimes the trigger is close to perfect, occurring very close to the actual bottom of the market. It is interesting that such a simple rule, which is only in the market 32% of the time, comes relatively close in terms of total returns. This rule is likely just one example of a rule that buys a pull-back and holds it for one year. Our findings have a major caveat, though. Because our data only starts in 2000, i.e., right at the start of a major bear market, we cannot use an expanding window and instead use the full sample ERI for this exercise, something that we typically avoid like the plague. Here, it means that the model is unreliable as it essentially incorporates future

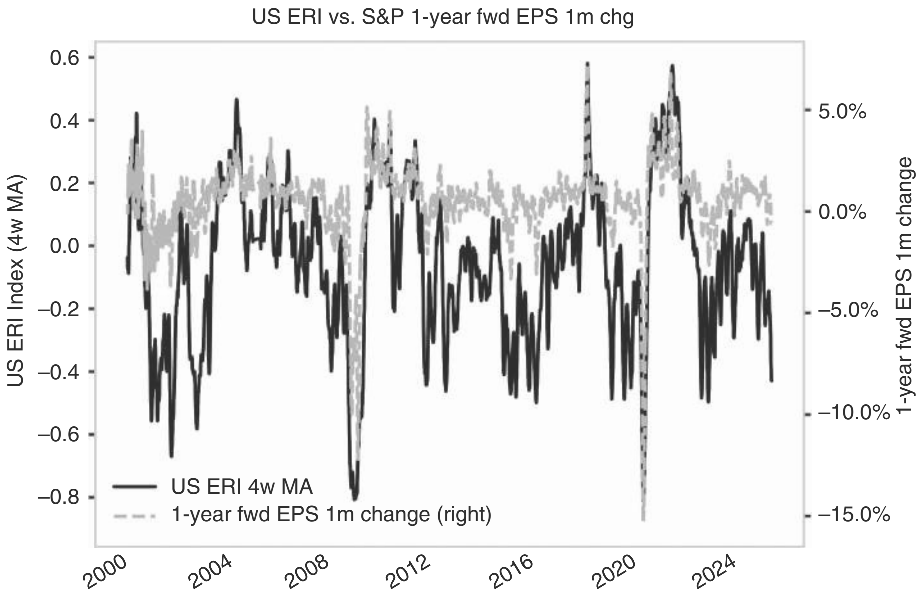
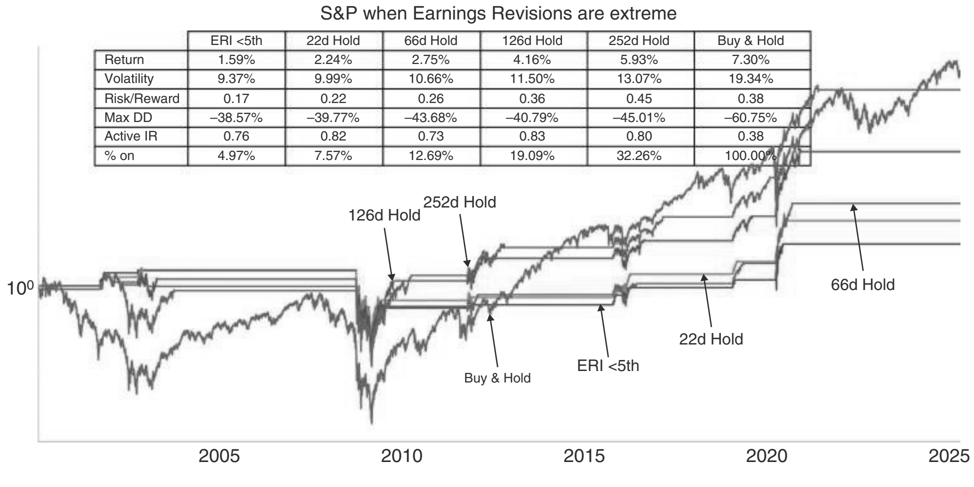


FIGURE 8.20 A depressed ERI is a buying signal.
Source: Bloomberg, Citi research and authors' calculations.



	ERI <5th	22d Hold	66d Hold	126d Hold	252d Hold	Buy & Hold
Return	1.59%	2.24%	2.75%	4.16%	5.93%	7.30%
Volatility	9.37%	9.99%	10.66%	11.50%	13.07%	19.34%
Risk/Reward	0.17	0.22	0.26	0.36	0.45	0.38
Max DD	–38.57%	–39.77%	–43.68%	–40.79%	–45.01%	–60.75%
Active IR	0.76	0.82	0.73	0.83	0.80	0.38
% on	4.97%	7.57%	12.69%	19.09%	32.26%	100.00%

FIGURE 8.21 ERI as a counter indicator.
Source: Bloomberg, Citi research and authors' calculations.

information. However, with both 2008 and 2020 now in the data, we imagine that going forward, the parameters should be more stable.

Using the ERI as a counter-indicator also works well internationally for most markets. This is in part true due to the high positive correlation of the various equity markets. The strategy works best for Asia x Japan and EM. For the United Kingdom

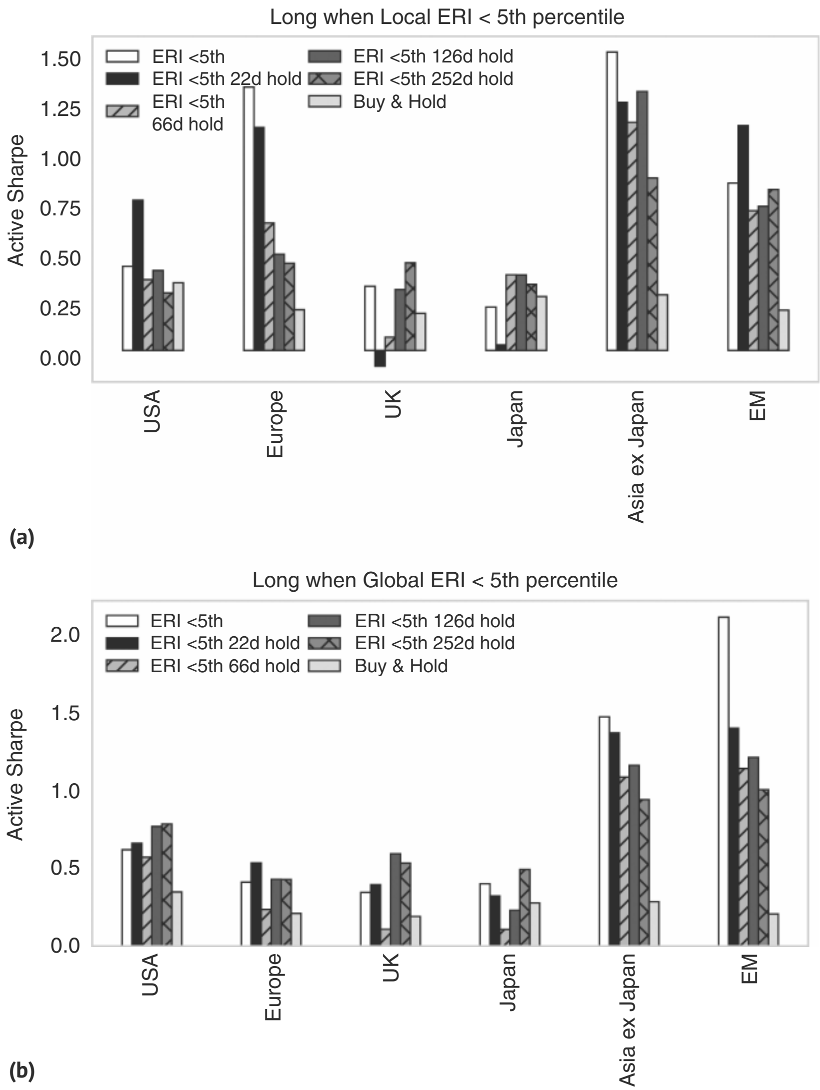


FIGURE 8.22 (a) ERI works as counter-indicator across regions. (b) Global revisions work better than local ERIs.
Source: Bloomberg, Citi research and authors' calculations.

and Japan, it largely depends on the holding period, but it works in most cases, as per Figure 8.22a. Once more, the global ERI works better than the local one, as can be seen in Figure 8.22b, probably because some regions do not have sufficient breadth of analyst coverage. Overall, this is evidence that the global business cycle may be more important for many equity markets than the local one. The same important caveat that we mentioned for the United States also applies here: we use the full sample ERI.

Another interesting question is how much information there is in the price action on earnings day. That stock prices have follow-through after the price action on the date of the earnings release was documented in the early 1980s by Rendleman et al. (1982), who found that stocks with the highest positive earnings surprises continue to outperform the market, while those with the highest negative earnings surprises continue to underperform. The most likely reason for this effect is behavioral biases, like conservatism, where investors just find it difficult to bid up a stock by too much on the basis of one strong quarter. Possibly, investors underestimate the autocorrelation of earnings, which is significant. Some may be hoping for a pull-back in a given stock after a large jump higher, but if this pull-back never comes, they may be forced to chase the stock higher regardless. The effect is larger for small caps, suggesting that liquidity and trading costs may also play a role. However, there is some evidence that the extent of the post-earnings drift has been falling over time. A related question is whether there is any information about the trading behavior of the aggregate index depending on how well the earnings season went, as measured by equities gapping higher after earnings results. We have, sadly, not been able to find a profitable rule that uses the aggregate outcome of the earnings season to trade SPX. While this remains an ongoing research project, for now, it appears that earnings drift is more easily expressed on a single stock basis.

8.5 EQUITIES AND LIQUIDITY

In the context of macro-trading, liquidity means the availability of cash and cash-like instruments in a sufficient quantity to facilitate economic activity and to meet short-term obligations. Liquidity is a very important concept for equity markets. It is no coincidence that before financial innovation made monetary aggregates less reliable, major central banks like the Bundesbank were targeting monetary aggregates. More recently, during the GFC in 2008, when central banks attempted to enhance monetary policy with quantitative easing (QE) and later by tightening (QT), liquidity made a comeback as a factor driving markets. Liquidity provided by the Fed is the key globally, even as other central banks have also been engaged in that game. We first loosely demonstrate the relevance of global liquidity for equities. This is achieved by Figure 8.23, which plots one-year changes of global USD reserves at the important central banks next to a detrended version of the SPX. The fit is surprisingly good for such a simple approach, as frequently discussed by Matt King (see, for example, King, 2023).

Next, we investigate the Fed balance sheet in more detail. The liability side of the Fed balance sheet basically consists of bank reserves, the checking account of the Treasury (Treasury General Account, TGA), the reverse repos, where money market funds keep balances at the Fed, and the currency in circulation. Interestingly, when investigating the various components of the Fed balance sheet, it turns out that the change of the overall balance sheet does not have a strong relationship with either risky assets or with US Treasuries. This is the case even on a coincident basis, not just in terms of using the balance sheet to forecast asset performance. Investigating the subcomponents, we find that bank reserves are the part of the Fed's balance sheet

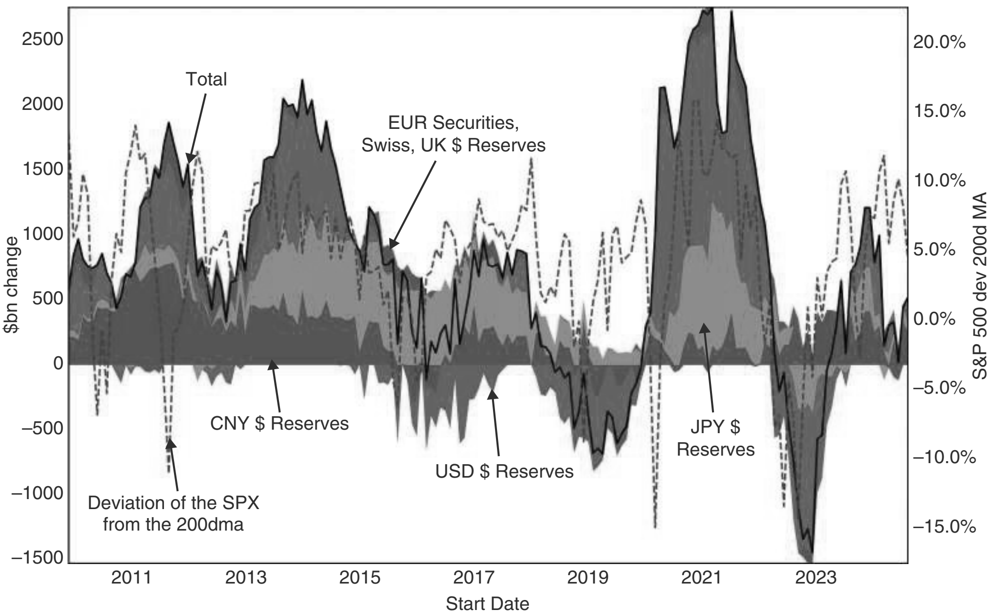


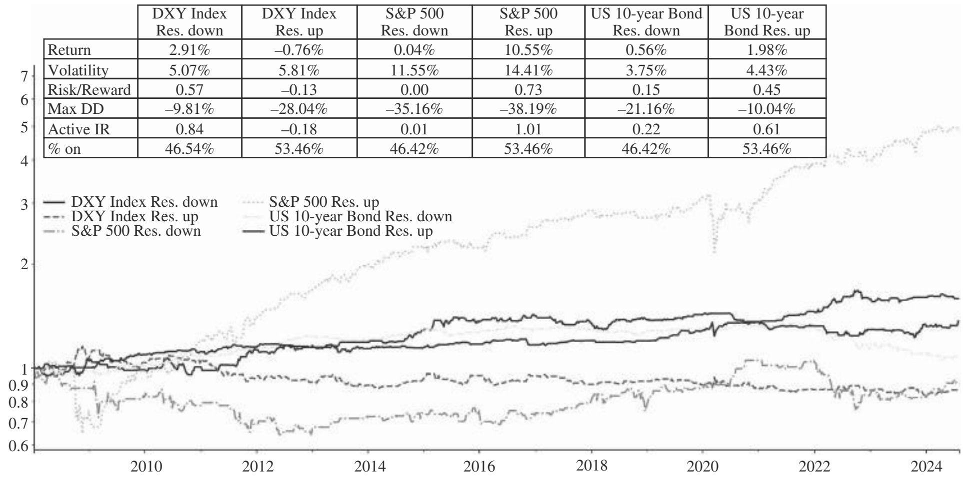
FIGURE 8.23 Global liquidity correlated with risky assets.
Source: Bloomberg and authors' calculations.

that is most reliably correlated with market performance. This is true not only on a coincident basis, which requires forecasting the behavior of bank reserves to generate trading rules, but also on a forward-looking basis. Both reverse repos and TGA have very little to do with equity returns.

This is illustrated in Figure 8.24, where we use the four-week moving average of weekly changes in bank reserves to forecast weekly returns for the SPX, bonds, and DXY. Long SPX after the four-week ma is positive, has an information ratio of just above 1, while it is roughly zero following a negative reading. US bonds also trade better following rising reserves, even though the improvement is not as stark, probably due to the bid that can result from weak equities. Following rising bank reserves, US bonds have an information ratio of 0.61, compared to 0.22 when falling. The divergence between the two return streams has mostly occurred since the bear market of 2022. For the USD, the difference is similarly stark as in the case for equities and quite consistent: When reserves are falling, the active IR to be long DXY is 0.84, compared to –0.18 in the opposite case. Falling reserves suggest a USD shortage, and rising risk aversion may also be supportive for the USD, in most cases.

8.6 EQUITIES AND BUBBLES

Bubbles can and do occur in any asset class. However, nothing captures the imagination of the investing public more than a bubble in equities. Clearly, bond yields close to zero or even negative could also qualify as bubbles, but an ever-rising equity market is



	DXY Index Res. down	DXY Index Res. up	S&P 500 Res. down	S&P 500 Res. up	US 10-year Bond Res. down	US 10-year Bond Res. up
Return	2.91%	–0.76%	0.04%	10.55%	0.56%	1.98%
Volatility	5.07%	5.81%	11.55%	14.41%	3.75%	4.43%
Risk/Reward	0.57	–0.13	0.00	0.73	0.15	0.45
Max DD	–9.81%	–28.04%	–35.16%	–38.19%	–21.16%	–10.04%
Active IR	0.84	–0.18	0.01	1.01	0.22	0.61
% on	46.54%	53.46%	46.42%	53.46%	46.42%	53.46%

FIGURE 8.24 Global liquidity helps with timing.
Source: Bloomberg and authors' calculations.

without doubt more fascinating. We therefore study bubbles in an equity market context. In terms of defining a bubble, we lean on the work of Jeremy Grantham at GMO (see Grantham, 2021), who suggested that a bubble is something where prices rise by more than two standard deviations above the long-term real trend, as illustrated in Figure 8.25. The alternative would be to define bubbles with valuations moving above the two standard deviation real trend, as not every large move in prices is necessarily a bubble. However, the price-based approach also picks up a bubble in fundamentals. What tends to happen is that bubbles usually have an underlying economic reason, which is often linked to some scarce resource. This was bandwidth during the internet revolution of the 1990s and is advanced computing power at the time of writing during the AI revolution. Scarcity and perceived scarcity lead to double and triple ordering and inflated prices for the goods in question, resulting in inflated earnings projections. This can lead to a bubble in earnings, and as a result, a purely valuation-based framework may not capture that. We therefore use the price-based GMO approach for bubble detection, but we certainly use valuation as an additional tool to understand how long the bubble may continue. In terms of exiting the bubble, we use a dual condition of either the market falling by 15% from the peak or exiting the two-standard-deviation condition, as we have previously discussed at Citi (see Willer et al., 2024e). The reason for the twofold condition is that after a 15% drop, we are getting close to what investors label a bear market. At times when the SPX is close to entering a bear market, we clearly have left bubble territory, even if the two-standard deviation rule should not have triggered yet.

Using these rules, we find nine bubbles in the SPX, starting with the first one in 1929, and with the last two linked to the current AI-driven bubble (interrupted by liberation day), as can be seen in Figure 8.26. The internet bubble, peaking in 2000, also shows up as two bubbles, interrupted by the EM and LTCM crisis of 1998. The first observation is that, after entering bubble territory (Figure 8.25), bubbles continued to

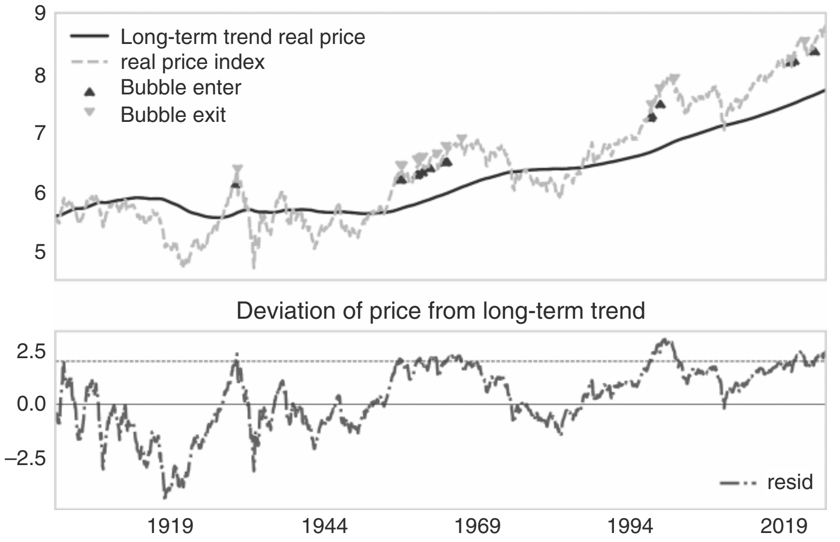


FIGURE 8.25 Bubble blown: at time of writing we were in bubble territory.
Source: Bloomberg and authors' calculations.

expand for quite some time. For example, the second leg of the internet bubble started in November 1998 and then lasted for more than 470 trading days, i.e., longer than a year, as per Figure 8.26. The only exception was September 1929, when the market turned down in very short order. This is likely the case, as 1929 was probably used as the episode to determine how to define a bubble. The trading rule is therefore not to sell equities when they enter bubble territory but to buy them. Some of the most explosive returns occur after the bubble warnings go off. Selling a bubble becomes safer after the exit of the bubble (shown as dotted red lines in the chart). Given how difficult it is to find any condition where the SPX reliably sells off, we find it noteworthy that in almost all the cases that our rule identified as bubbles, the large majority of the full rally that happened after entering bubble territory was eventually reversed, either while still in bubble territory or afterward. The only exception was 1958, when the market just went sideways for a prolonged time. This suggests to us that our bubble definition captures an important phenomenon.

As our bubble framework is not overly helpful in calling for a peak, the question is what to focus on to know that a peak is close. Fed hikes are typically part and parcel of a bubble, and eventually, those rate hikes contribute to the market's demise. It is very rare that the Fed eases into a bubble that is already extended. During the second leg of the internet bubble, the Fed hiked by 175bp, accelerating the pace of hikes at the end of the cycle. True, in 1998, the Fed cut rates in response to the Asian crisis that had spilled over into Russia and LTCM. But it took a crisis to get us there, and, according to our definition, we had already left bubble territory given the severe fall in equity markets at that time. Overall, we would treat an aggressive Fed hiking cycle as

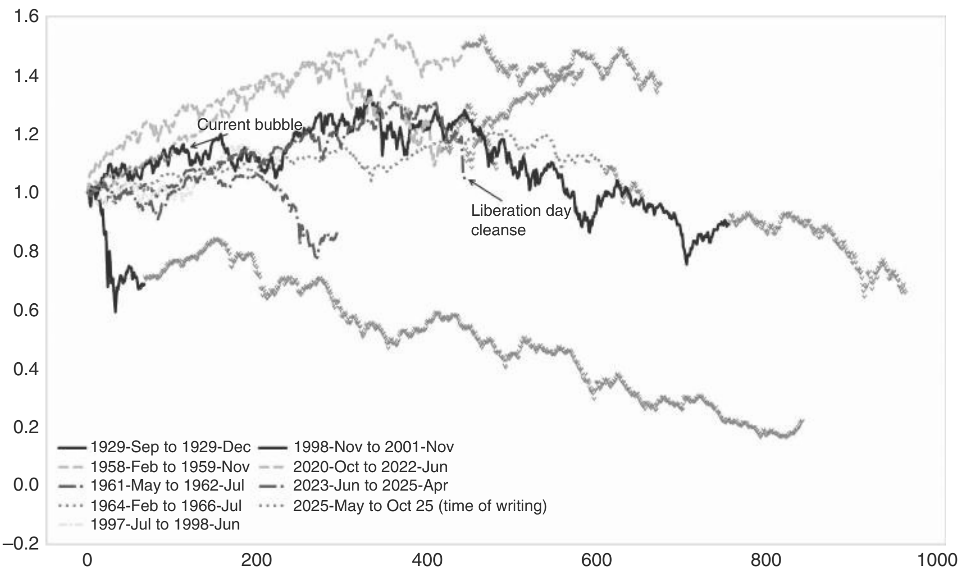


FIGURE 8.26 Bubbles reverse.
Source: Bloomberg and authors' calculations.

an early warning sign for an expanding bubble. Hikes have often been necessary but by no means sufficient to bring a bubble to an end. Other warning signs include falling liquidity (discussed later in the context of the 2000 episode), and extreme sentiment, for example, proxied by the individual investor bull-bear survey, which became very extreme in late 1999, as is shown in Figure 8.27.

Further signs of "peak bubble" have also been discussed by Ray Dalio (see Dalio, 2021) He has suggested screening for a high participation of retail investors (retail net purchases as percent of market capitalization), extended IPO activity (as percentage of GDP), and, very importantly, the use of leverage to finance equity positions (US household margin debt as percentage of GDP). As is often the case in markets, greed triggers the use of leverage. Without leverage, the risks of a bubble bursting are often not as extreme, as investors typically do not get forced out of positions unless they use leverage. Google trend searches with respect to the bubble narrative of the day can also help, though they suffer from the problem that there is no sufficiently long time series that gives a sense of just how extreme this indicator could become before peaking. In Figures 8.28a and b, we show margin leverage and IPOs as a percentage of market capitalization. It is not that obvious at what levels these measures of exuberance typically peak. But a quick move higher seems to be a condition that makes it more likely that we are in the late phase of a bubble.

One additional timing tool for the top of the bubble may be that the "generals" have to fail, where the generals are the large-cap stocks that drive most of the gains in a sharply narrowing market. This is somewhat speculative, as the analysis is grounded

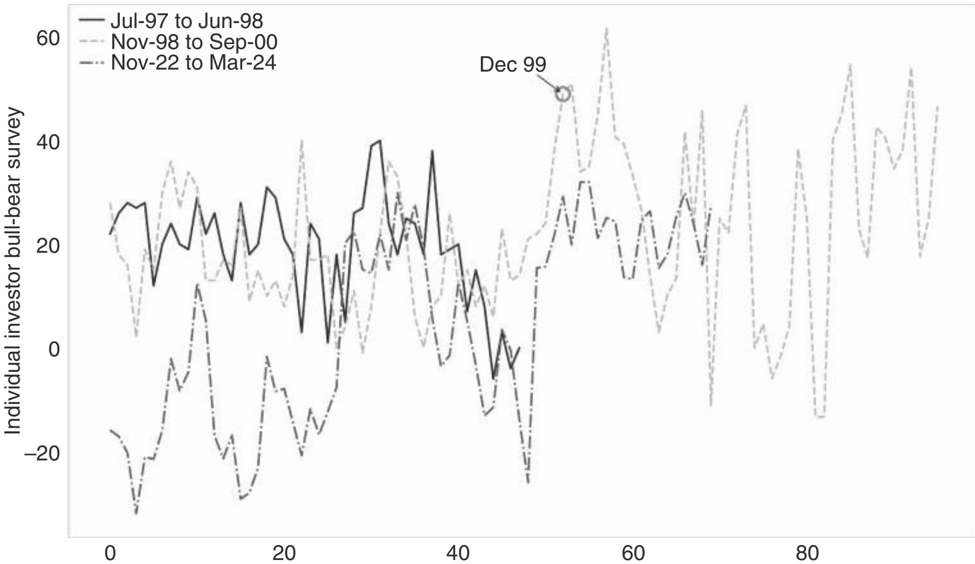


FIGURE 8.27 Peak bubble also means sentiment bubble.
Source: Bloomberg and authors' calculations.

very much in the specifics of the 2024 bull market, dominated by the Mag7. Although we can easily make comparisons to the 2000 bubble, a larger sample that can be formally backtested is harder to come by. Still, as the concept makes sense to us, we want to mention it here in the hope that history rhymes sufficiently going forward. The analysis is based on Willer et al. (2024f). Going into the 2000 peak, the seven largest cap stocks that drove the index higher were Microsoft, Cisco, GE, Walmart, Intel, Nokia, and Pfizer. Interestingly, Microsoft is the only company from the 2000 peak that made it into the Mag7. Furthermore, it is noteworthy that in 2000, the top stocks were less tech-heavy, which could make an eventual tech bust more problematic for SPX performance this time around than it was in 2000. For the sake of this study, we measure generals as "failing" when they break below their 200dma, and to reduce noise, we require them to stay there for at least five days. In 2000, it took three of the seven generals to put the bull market on notice, but four had to fall below the 200dma to call conclusively for the end of the bull market, as is shown in Figure 8.29.

Given that the "generals rule" is a tool for risk management during bubbles, we backtest this rule during our bubble periods only, but added one year after bubble exit, to make sure we capture deflating bubbles at least partially in the study, unless the SPX made it back above the 200dma faster than that. We find that the best information ratio is achieved if investors cut equity risk when the fourth general falls below its 200dma. This rule results in the information ratio moving from 0.67 (unconditional) to 1.04. For the case where four or more generals have fallen below their 200dmas, the information ratio is –0.6. This is superior to using the 200dma as a risk management tool on the SPX (which also leads to improvements). This is illustrated in Figure 8.30. Of course, there are also false positives. Still, if in a bubble, we think it pays to have

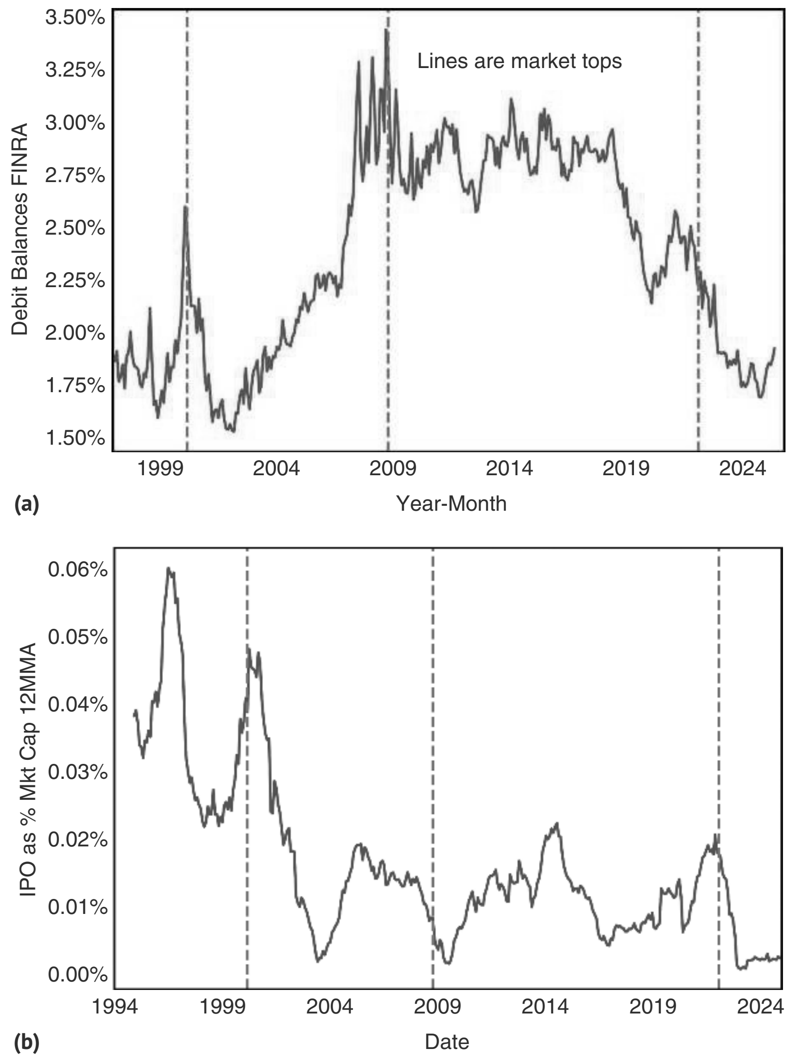


FIGURE 8.28 (a) Rising margin debt as a warning signal. (b) Rising IPOs a less reliable sign of exuberance.
Source: Bloomberg and authors' calculations.

an aggressive risk management, as downside risks are elevated. We therefore take the "generals" rule seriously during bubble times.

The "generals rule" is a highly specific version of market breadth. More broadly, market breadth indicators require that more equities are rising than falling. Typically, falling breadth is seen as a negative for markets, as market advances carried by only a few stocks are seen as vulnerable. However, we do not find breadth overly useful for timing reversals. In Figure 8.31a, we can see that most often low breadth is happening at market bottoms, while high breadth happens during various points of a

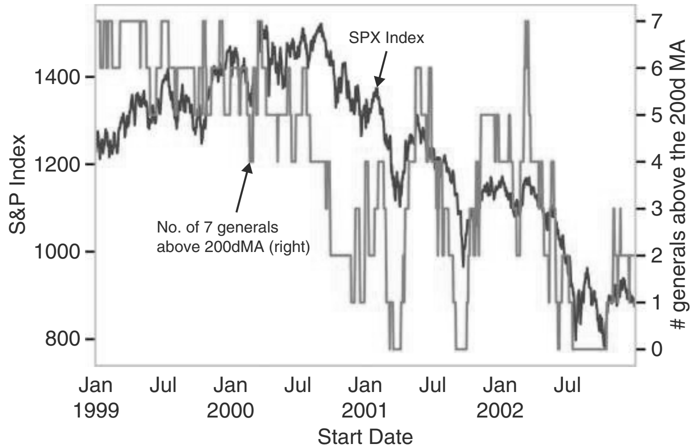
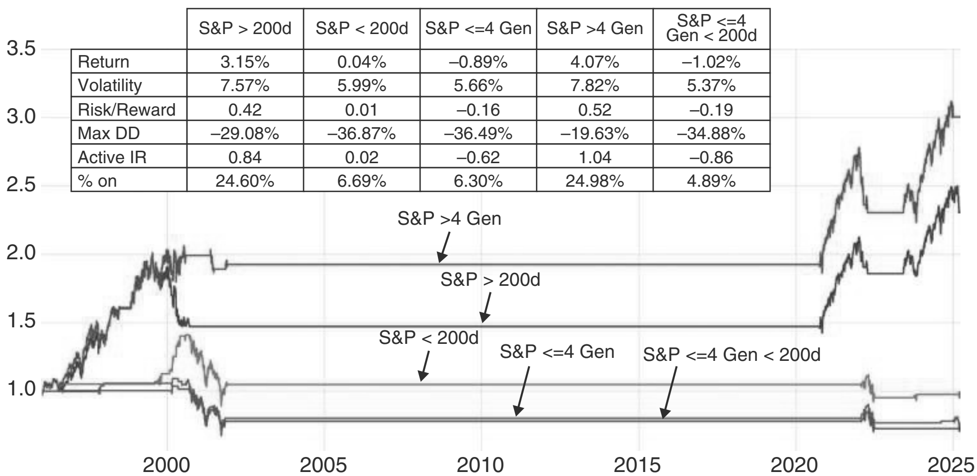


FIGURE 8.29 It takes four generals to fail.
Source: Bloomberg and authors' calculations.



	S&P > 200d	S&P < 200d	S&P <=4 Gen	S&P >4 Gen	S&P <=4 Gen < 200d
Return	3.15%	0.04%	–0.89%	4.07%	–1.02%
Volatility	7.57%	5.99%	5.66%	7.82%	5.37%
Risk/Reward	0.42	0.01	–0.16	0.52	–0.19
Max DD	–29.08%	–36.87%	–36.49%	–19.63%	–34.88%
Active IR	0.84	0.02	–0.62	1.04	–0.86
% on	24.60%	6.69%	6.30%	24.98%	4.89%

FIGURE 8.30 Generals rule performed well in 2000.
Source: Bloomberg and authors' calculations.

bull market. Just glancing at the chart suggests that there have been episodes where breadth deteriorated well ahead of a market top. We look at those tops in more detail in Figure 8.31b.

Of the major tops over the last 25 years, the only top with a fairly weak breadth was 2000. However, it is important to point out that breadth had been narrowing for a long time before that bubble eventually popped. Also, interestingly, breadth had started

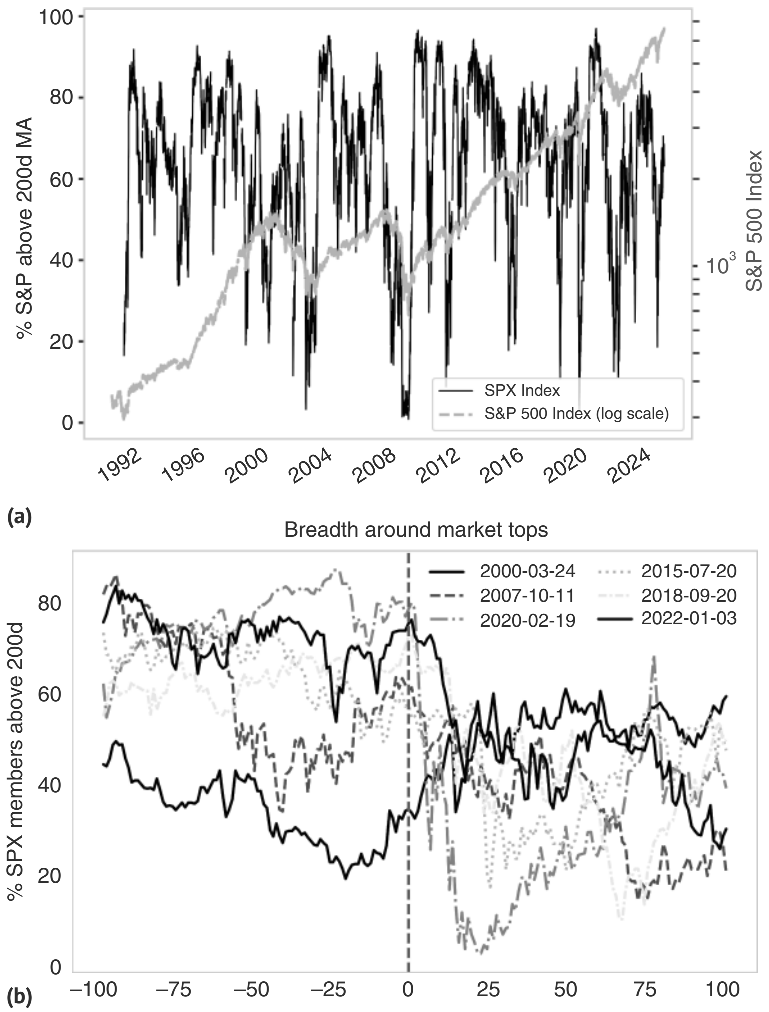


FIGURE 8.31 (a) Poor breadth at a market top is rare. (b) But not easy to use breadth to time market top.
Source: Bloomberg and authors' calculations.

to improve (from very low levels) about a month before the market peak. The second worst breadth was going into the 2007 top, but again, breadth had actually improved into the top, this time for two months. Overall, we do not find breadth overly helpful, even though breadth on par with the 2000 extremes may be worthwhile to take more seriously, understanding that a sample size of one will never make the concept overly compelling.

The Mother of All Bubbles

The internet bubble of the late 1990s was a time of intense speculation and investment in internet-based companies. Fueled by the widespread adoption of the internet and the emergence of online businesses, investors poured money into startups, many of which lacked viable business models and sustainable (or at times any) revenues. The Fed was broadly leaning against the expansion of the bubble by hiking aggressively. However, in the last quarter of 1999, the Fed did ramp up the supply of liquidity, mostly to counter fears of the Y2K bug. The underlying concern was that computer systems would fail to correctly process dates beyond December 31, 1999, as many programs were representing four-digit years with only the final two digits. Given fears that IT failures cause havoc in markets, the Fed increased its balance sheet by 19% in the last quarter of 1999, the largest increase on record at that time, which was only bested subsequently during the 2008 GFC and the 2020 COVID crisis. Adding so much oil to the fire unsurprisingly led to the Nasdaq having its best quarter ever, up 53% in Q4 of 1999, further helped in early 2000 by the fact that the Y2K bug came and went without major hiccups, as corporates had invested in patches ahead of time. When the Fed started to pull back that liquidity aggressively after the first of January, the NDX peaked a few months later at the end of March, at a trailing PE of around 200. This episode is illustrated in Figure 8.32. Interestingly, the SPX, which initially sold off with the NDX, recovered by September 2000 almost to its level in late March, partly as the Fed had concluded its hiking cycle in the aftermath of the April sell-off. But the NDX did not come even close to the March levels, and with such underperformance of the previous leaders, the writing was on the wall for the market more broadly. By August, it had become very clear that the bursting tech bubble would pull both tech companies and the overall economy lower, and the manufacturing ISM quickly fell below 50. In November, the CEO of HPQ, Carly Fiorina, suggested on an earnings call that “the lights went out” in answering an analyst question on how the quarter was going. By that time, fundamentals had clearly deteriorated, which was much less obvious at the time of the market peak. This is also interesting. Equities are not that forward-looking. The time between the equity peak and a peak in fundamentals, proxied by semiconductor sales, was only seven months (though we would note that there is a lot of reflexivity at work: the falling share prices contributed to semi sales peaking out). Subsequently, both NDX and SPX continued to fall as the bubble deflated until October 2002. We use the 2000 episode frequently to compare other bubbles to it. However, we note that it is not guaranteed that the next bubble will come close in size or sheer craziness, as the 2000 episode was very extreme, even by bubble standards. To our mind, the 2000 episode is therefore more a ceiling of where valuations can go, rather than a great analog to use for the average bubble.

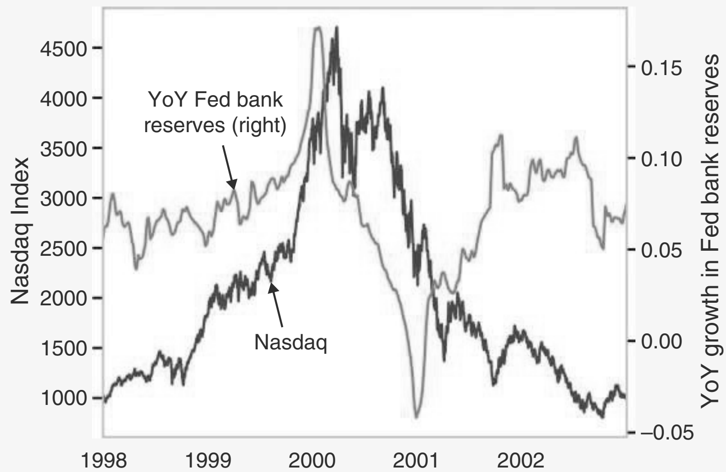


FIGURE 8.32 Last leg in NDX was liquidity driven.
Source: Bloomberg and authors' calculations.

8.7 VALUATION NOT VERY VALUABLE

Coming fresh from a discussion of bubbles, we now generalize the discussion of valuation and whether it is useful for equity trading. The short answer is that valuation is barely working: traditional market timing measures based on valuation have not generated alpha for the last 20 years. We start investigating valuation with a focus on the CAPE ratio. In Campbell and Shiller (1998), the authors use cyclically adjusted P/E ratios by taking a moving average of earnings over long time periods to adjust for the vagaries of the business cycle. They also adjust for inflation. Figure 8.33a shows the CAPE over time. Scatter plots of CAPE levels versus 10-year real forward returns show a clear correlation, as per Figure 8.33b, but are difficult to operationalize into an effective trading tool in real time, as we show next.

We use Professor Shiller's data to perform a backtest using the following rule: we calculate the CAPE yield (1/CAPE) and size our position based on the rolling 30-year percentile of that yield. In particular, we move the position size to 50% when the yield is in the lowest percentile and to 150% when it is in the highest. In theory, the average and median exposure should be fully invested, and we therefore compare the strategy to the excess returns (which are leverage-agnostic) of a buy-and-hold strategy. We show results from 1950 to avoid the war periods, though the precise starting point makes little difference to the results. The results are shown in Figure 8.34. It turns out that our version of the CAPE model underperforms not only in terms of actual returns but also with respect to the information ratio. If we modify the CAPE ratio to be seven years instead of the standard ten, we find a small benefit, in terms of the information

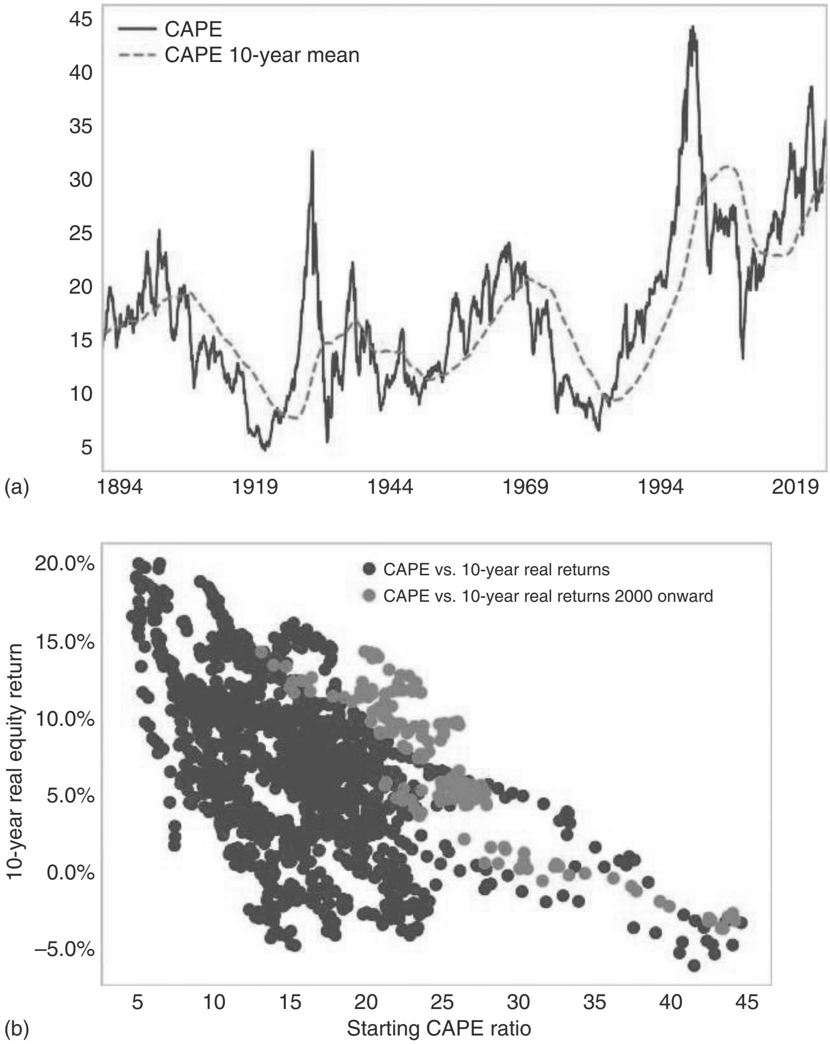


FIGURE 8.33 (a) CAPE levels may have moved structurally higher. (b) CAPE negatively correlated with future long run returns.
Source: Bloomberg, shillerdata.com, and authors' calculations.

ratio, but returns still lag the buy-and-hold strategy. This is because investors would tend to be under-invested as the CAPE ratio has moved upward over time. This can be clearly seen in Figure 8.33a. Ever since the 2008 financial crisis, markets have barely traded below the rolling 10-year average. For the strategy we test, this means that the rule is on average only 85% invested, instead of the expected 100%.

There have been extensions of the original work on CAPE, and some researchers have employed different earnings measures (see Siegel, 2016), though this has only

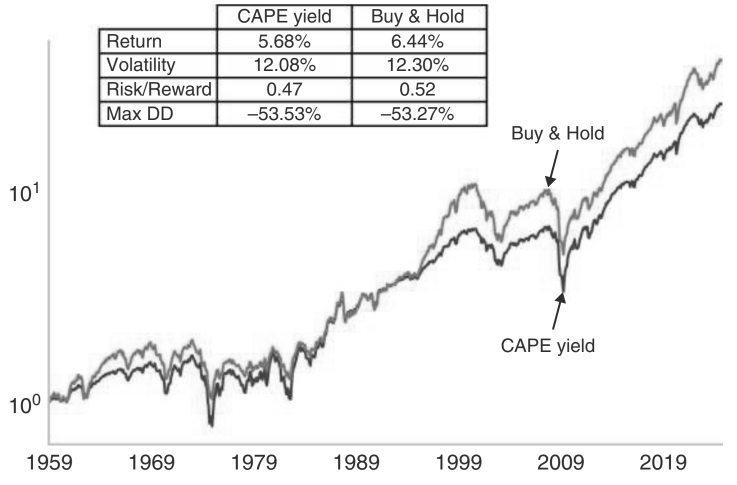


FIGURE 8.34 Using CAPE underperformed buy-and-hold.
Source: Bloomberg, shillerdata.com, and authors' calculations.

resulted in marginal improvements. Overall, conventional wisdom remains that structural changes over long horizons make valuation difficult to use (see also Asness et al., 2017).

8.8 WINNERS KEEP WINNING

Just like valuation, momentum is also an intuitively attractive strategy for equities. After all, a bull market can lead to a virtuous cycle. Higher stock prices generate a wealth effect and also higher CEO confidence, leading to better economic growth outcomes, and therefore to higher earnings, reinforcing the equity bullishness. In our experience, investors tend to self-select themselves into a more valuation-centric or more momentum-centric investment style. Although the two strategies are not necessarily the opposite of each other and most investors take both aspects into account, it typically makes a big difference whether investors put a bigger weight on changes (momentum) or levels (including valuation).

We are unashamedly favoring momentum, mostly because momentum is performing better in our backtests. For a fair comparison, we use the same methodology that we used for valuation. In particular, we use the x-month return as our momentum variable and invest 150% when it is above the 50th percentile and only 50% when it is below. Figure 8.35 shows that this methodology outperforms buy and hold in terms of both the information ratio and the absolute returns. This is the case for all shorter horizon momentum measures (3 months, 6 months, and 12 months), and only gets weaker for a very long one (84 months). The best version is the one using 3-month and 6-month returns, which is 5% overinvested over the backtested period.

	MoM 3	MoM 6	MoM 12	MoM 84	Buy & Hold
Return	8.08%	7.91%	7.30%	6.97%	6.44%
Volatility	11.69%	11.42%	11.59%	12.27%	12.30%
Risk/Reward	0.69	0.69	0.63	0.57	0.52
Max DD	–40.03%	–36.84%	–35.28%	–55.16%	–53.27%

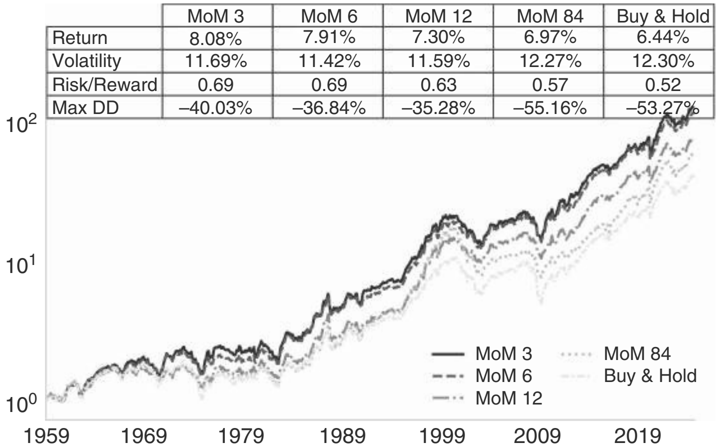


FIGURE 8.35 Momentum has outperformed buy-and-hold.
Source: Bloomberg and authors' calculations.

	MoM 6M	CAPE 7-year	Cheap Mom
Return	1.41%	0.39%	2.13%
Volatility	5.08%	11.06%	4.27%
Risk/Reward	0.28	0.04	0.50
Max DD	–15.95%	–72.78%	–19.14%
Active IR	0.80	0.06	1.51
% on	12.56%	35.48%	12.73%

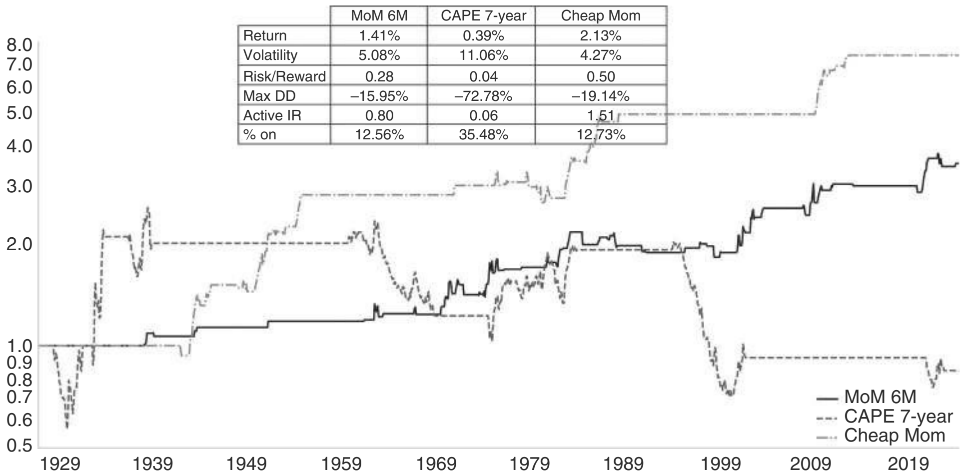
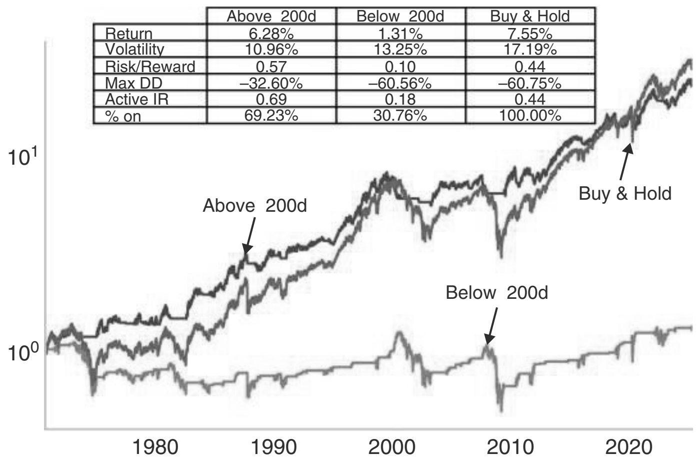


FIGURE 8.36 Extreme momo beats extreme valuation.
Source: Bloomberg, shillerdata.com, and authors' calculations.

For the tactical trader, momentum very frequently trumps valuation. In Figure 8.36, we use a very long monthly history to test trading momentum and valuation at the extremes. In particular, we only trade the top and bottom percentiles, positioning for mean reversion with respect to valuation, and for trend following with respect to very strong or weak momentum. As can be seen in Figure 8.36, the value strategy leads to shorting bubbles, which can inflate much longer than a macro manager can stay solvent. Buying strength, on the other hand, has been rewarded. Of course, having both value and momentum in your favor is the sort of A-trade that can



	Above 200d	Below 200d	Buy & Hold
Return	6.28%	1.31%	7.55%
Volatility	10.96%	13.25%	17.19%
Risk/Reward	0.57	0.10	0.44
Max DD	–32.60%	–60.56%	–60.75%
Active IR	0.69	0.18	0.44
% on	69.23%	30.76%	100.00%

FIGURE 8.37 Using a simple 200d MA avoids the worst drawdowns.
Source: Bloomberg and authors' calculations.

build generational wealth. Unfortunately, this seems to happen only once a generation. We see a strong active Sharpe of around 1.5 if both momentum and value signals are above the 50th percentile at the same time, but it happens only 13% of times and occurs only every 20 years or so. The last event happened in the aftermath of the GFC.

Another strategy, which is broadly in line with momentum, is using moving averages, which many CTA models are based on. We found much use over the years in the 200d MA. Although it can be frustrating whenever the market is gyrating around its 200d MA, giving many false alarms, respecting 200d MA makes sure that investors are on the right side of the market. Using the 200d MA for the SPX significantly improves the information ratio from 0.44 to 0.69, as per Figure 8.37. Although the total return is slightly less, the maximum drawdown is around half. The 200d moving average basically works by avoiding very nasty tail events. We always caution against the buy-and-hold strategy, even if it has higher returns. The reason is that very few investors are able to truly endure large drawdowns. Bear markets end when almost every market participant has given up on the market and sold, and investors who do not eventually throw in the towel are very rare. The best reason to stop out when the market breaks the 200dma is that it prevents investors from stopping out at even lower levels. If investors are comfortable with a 17% volatility that the SPX has to offer, they should just respect the 200d MA and lever up the SPX by 55% whenever the SPX is above the 200d MA.

8.9 SMALL, NOT OFTEN BEAUTIFUL

Although this book does not go into detail on sectors, we do want to investigate small caps, partly because there has been much written about the so-called small-cap effect. This effect was first documented in the early 1980s (for example, Banz, 1981, and later

Fama and French, 1992b) and posits that small-cap stocks generate higher returns, although often with higher volatility. The fundamental underpinnings for small-cap outperformance are that small caps have more room for growth, allowing them to expand more quickly. The limited analyst coverage and resulting lower market efficiency are also mentioned as a possible reason for the effect. Additionally, small caps may need to incorporate a higher risk premium due to the higher volatility, which is because small caps are more sensitive to the economic cycle, have less diversified revenue streams, and may have more difficulties accessing capital during market downturns, which can raise the default rate. Lower liquidity, resulting in higher transaction costs, may also have contributed.

However, time has not been kind to the small-cap effect, as can be seen in Figure 8.38. The all-time peak of the Russell 2000 over the SPX occurred in July of 1983, not too long after the original Banz study. Small caps then underperformed by a total of almost 50% into the recession of the early 1990s. While the post-recession recovery was sharp, the relative performance made new lows during the tech bubble of the second half of the 1990s, troughing around 60% off the peak in early 1999 (interestingly, almost one year ahead of the peak of the Nasdaq bubble, maybe because the Nasdaq bubble of the late 1990s also had small caps being swept up in it). Still, the bursting tech bubble was broadly positive for small-cap outperformance, largely because the recession it generated was very mild. These patterns have continued since then. The 2008 GFC led to small-cap underperformance, followed by a recovery, and the same was true for the COVID recession. Since March 2021, small caps have underperformed again, with a new tech bubble forming. This suggests that small-cap outperformance requires two ingredients: a positive economic cycle, preferably coming out of a US recession, and/or a deflating tech bubble.

To operationalize this trading rule, we check how small caps relative to the SPX perform when the Manufacturing ISM is below 50, but above its 3mm, to find episodes

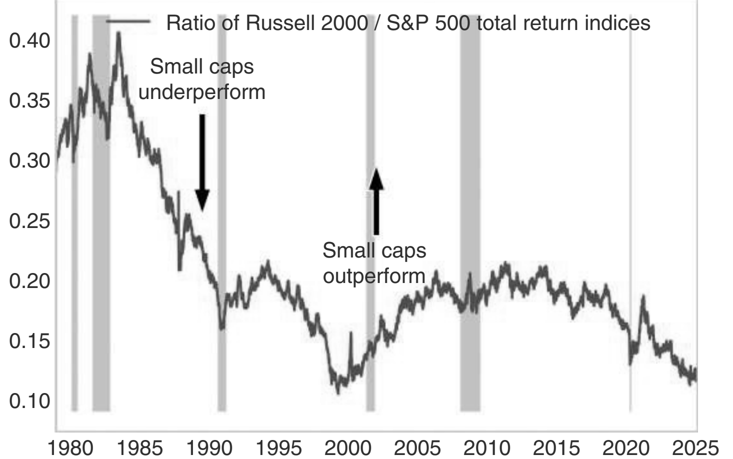


FIGURE 8.38 Small caps hate recessions and tech bubbles.
Source: Bloomberg and authors' calculations.

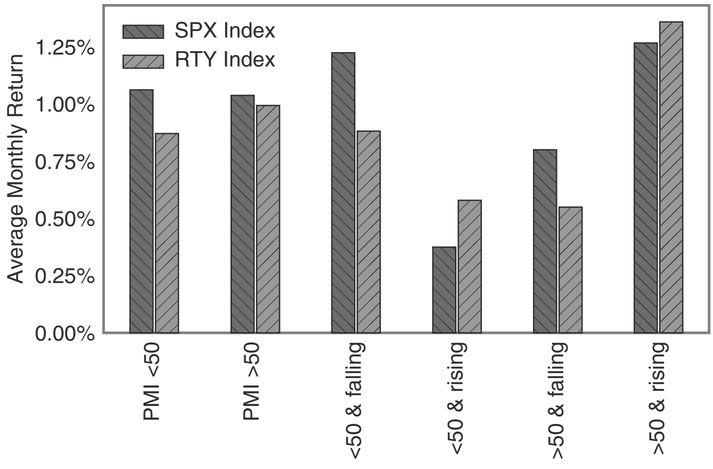


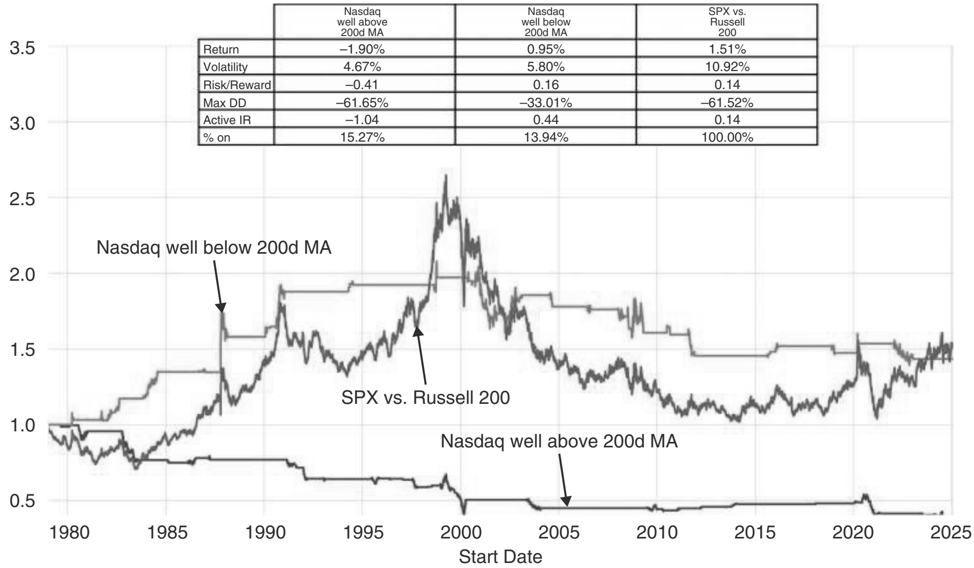
FIGURE 8.39 Small caps do well exiting recessions.
Source: Bloomberg and authors' calculations.

when the economy comes out of recession. Figure 8.39 shows that RTY mostly outperforms when the PMI is below 50 and rising (i.e., above its 3mma). It also outperforms somewhat when the PMI is above 50, and rising, but much less so. With the PMI below 50 it does much worse, driven by the regime of PMI below 50 and falling. This pattern is consistent with what we find in credit: coming out of the recession, it pays to buy "junk," i.e., the lower-rated credits. Information ratios of trading this pattern are quite weak, though, maybe because the framework does not take bubbles into account, which are often large-cap bubbles.

We therefore also investigate how RTY/SPX trades in relation to recent bubbles. Given that the most recent bubbles were tech-driven (which is probably more a feature than a bug), we investigate what happens when the Nasdaq is well above its 200dma. In particular, we use a deviation above the 85th percentile as a simple indicator of bubbly conditions in the tech-heavy index. Figure 8.40 shows the results. Small caps on average outperform when the NDX is in the bottom 15th percentile (which also captures some of the post-recession periods) and underperform when it is in the top 85th percentile, i.e., when the bubble is getting extreme. Having said that, more recently, small caps have not done well even when NDX was relatively weak.

8.10 SUMMARY

US equities have had a tremendous long-term performance, with a Sharpe ratio since 1926 using monthly data of 0.51. The default position, therefore, has to be long. However, drawdowns have been relatively frequent, with the ones above 20% often linked to recessions, but also to deflating bubbles, stagflationary episodes, and financial accidents. We discuss the interaction of the equity market with the Fed and find that equities rally shortly before the first cut. They also rally into and after the last cut. They



	Nasdaq well above 200d MA	Nasdaq well below 200d MA	SPX vs. Russell 200
Return	–1.90%	0.95%	1.51%
Volatility	4.67%	5.80%	10.92%
Risk/Reward	–0.41	0.16	0.14
Max DD	–61.65%	–33.01%	–61.52%
Active IR	–1.04	0.44	0.14
% on	15.27%	13.94%	100.00%

FIGURE 8.40 Small caps dislike bubbles.
Source: Bloomberg and authors' calculations.

underperform in the aftermath of the first hike and rally strongly in the aftermath of the last hike. The level of rates is not strongly related to equity markets, but large increases in the term premium are negative. The Fed put, on average, gets triggered around 25% off the highs of the equity market. On the growth side, equities are leading GDP by around two quarters. Using EPS, the coincident effect is larger, but there are also strong leading patterns for the next two quarters. We study earnings revisions (ERIs) and find that very negative revisions are a strong buy signal, but outside of excessive downgrades, US equities trade much better when the ERI is positive. On the liquidity side, we find a positive correlation with the change in bank reserves on the Fed balance sheet, one week forward. We also share some thoughts on equity bubbles: how to spot them and how to trade them. We demonstrate that momentum is a better strategy than value.

CHAPTER 9

How to Trade Equities: The Tactics

From the previous chapter, it is clear that investors need a very good reason not to be long the equity market on a strategic time frame. Expecting a recession is a good reason. However, even on a more tactical basis, it would be useful to also avoid shallower drawdowns. Although it is quite difficult to trade equities more tactically, there is some alpha to be had, mostly by fading euphoria and buying panic. The former seems harder than the latter. This chapter highlights some models that we found useful.

9.1 SPOTTING THE PEAKS

For short-term trading, Citi's POLLS index is useful. POLLS stands for Positioning, Optimism, Liquidity, Leverage, and Stress. The indicator was created by the equity trading strategy team with Citi markets (see Altmann and Raute, 2019). A high value indicates a higher risk of a pullback. As many of the underlying series are quite volatile, the indicator must stay above a certain threshold for more than a day. We found that levels above 18 for two days or more are a good warning signal. The median performance is for a 5% pullback after that trigger. While 5% is something many equity investors would happily sit through, the skew is sharply to the downside. Therefore, the main value of the index is that investors do not have to fight the temptation to reduce risk once the index is 5% lower. After all, it is much easier to resolve to sit through a 5% correction without reducing risk ahead of a pullback, when optimism is still riding high. When the market is actually down 5%, investors are typically more fearful and are more likely to belatedly reduce risk. It is therefore better to act early when the POLLS signal is triggered. Figure 9.1a suggests that one-month returns deteriorate quickly for POLLS above 18, with the likelihood of positive SPX returns dropping to well below 40%. Figure 9.1b illustrates the case for moving above 18. Although there may be some overfitting involved in the POLLS indicator, we note that the out-of-sample performance so far has been holding up quite well. This is also highlighted by the Macro Man write-up on Bloomberg in February 2023 (see Crise, 2023), which may have raised awareness of the indicator but did not undermine its performance. Since then, there were two triggers: July 2023, which was slightly early, but correct two months later, and in July 2024, which nailed the top. This was discussed in Willer et al. (2024b).

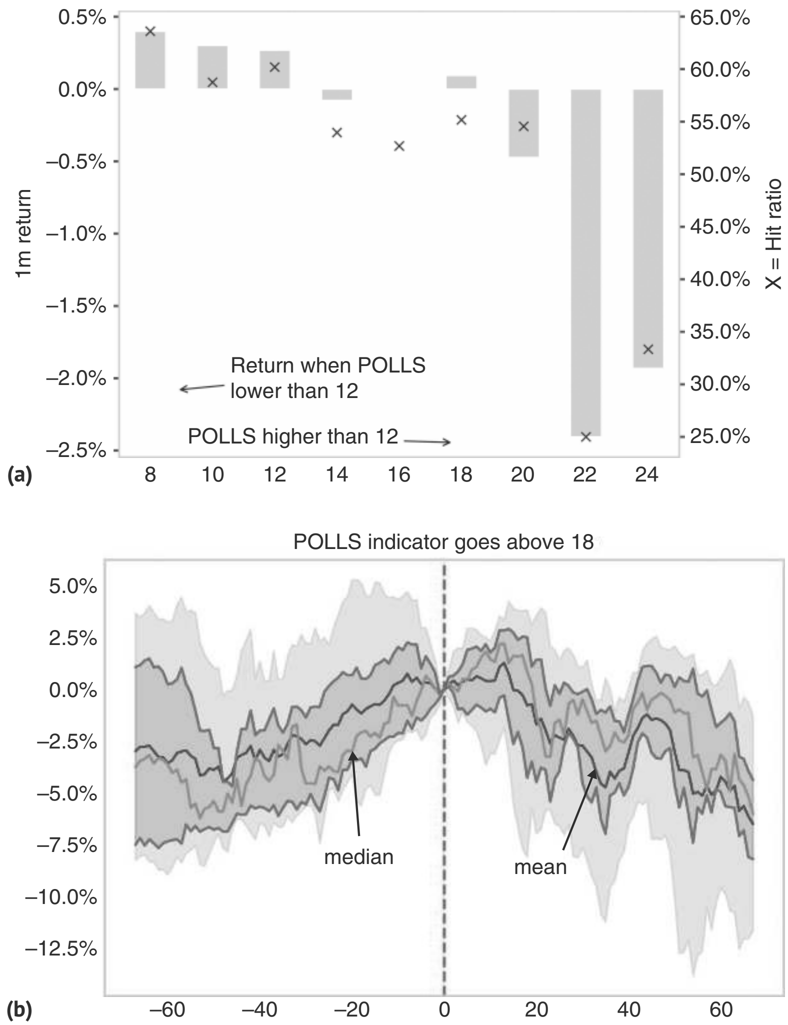


Figure 9.1 (a) POLLS a useful warning signal. (b) Polls > 18 with negative returns. *Source*: Bloomberg, Citi, and authors' calculations.

9.2 BOTTOM WATCHING

Given that equities typically go up, it is not surprising that there are more rules to successfully buy the dip than to sell a top. Bottoms are also in some ways easier to identify, but they are also often scarier to trade, as at the time investors are almost uniformly bearish, and the reputational risk from buying after a long bear market can be quite high. There are different techniques for identifying a tradable bottom. One is simply identifying the problem the market worries about and what a workable solution to said problem would be. Identifying the odds of whether the respective policymakers arrive at that solution in short order is then the key to becoming more positive on the market. Some issues are, of course, easier to fix for policymakers than others. If it is about a duration problem, like in March 2023, where some of the smaller US banks suffered large losses due to having invested in long-term Treasuries at a time of low interest rates, just to be caught out by aggressive Fed hiking cycles, the fix is simple. All that is required is for the Fed to become less hawkish. Although this is easier said than done in the context of high inflation, it is nevertheless still quite likely that policymakers can fix the problem, given that a banking crisis would likely be very deflationary, strongly mitigating the inflationary problem at hand. The March 2023 crisis was also helped by the fact that SVB was a relatively small bank that was easily bailed out by policymakers. If it is not about duration risk, but rising credit risk, the ask for policymakers is much larger. Building a political consensus on a bailout of investors who made poor credit decisions takes much longer than just getting the Fed to back off from its hawkishness. In either case, identifying the problem makes it much easier to understand what type of news is required to change the narrative and stop the sell-off.

We also study technical models to buy the dip. Our first point is that POLLS also works for bottoms (see Willer et al., 2025a). In particular, Figure 9.2 shows that POLLS below 8 is a good indicator to use, even though it typically triggers when the bottom is already well in. It is a good sign if an indicator works for both bottoms and tops.

For most of our other models, the key ingredient is the extent and volatility of the sell-off, measured either just by prices or by VIX. With respect to VIX, coincident returns are clearly negatively correlated with SPX prices, as per Figure 9.3a. However, with respect to monthly forward returns, the correlation becomes positive, though it is relatively weak. As per Figure 9.3b, VIX has to reach quite high levels (of close to 60 or above) before forward returns become higher than unconditional returns. At these higher VIX levels, the spread of returns is much wider, hence Sharpe ratios for buying equities are less compelling in high volatility environments. For the top 10th percentile, the median monthly return is 3.95%, though the Sharpe ratio is only 0.62 (annualized),

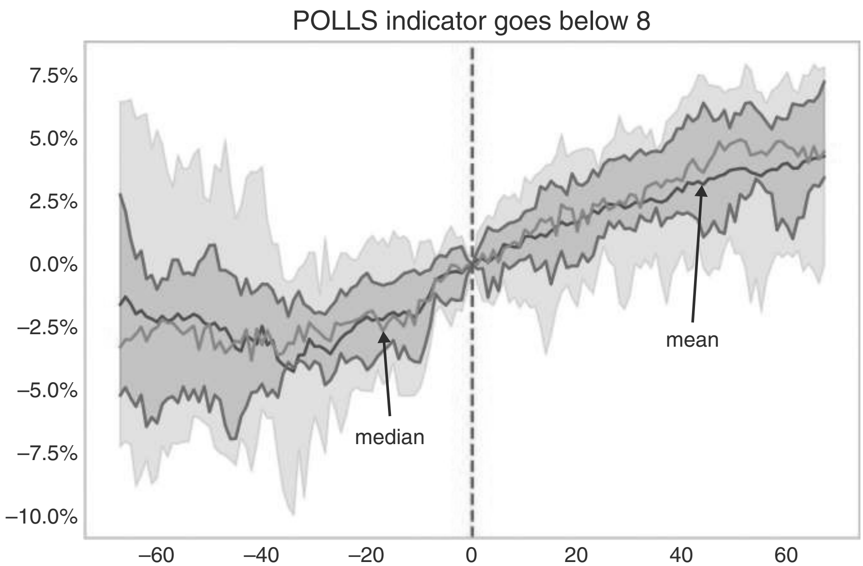


Figure 9.2 POLLS also works for bottoms.
Source: Bloomberg, Citi and authors' calculations.

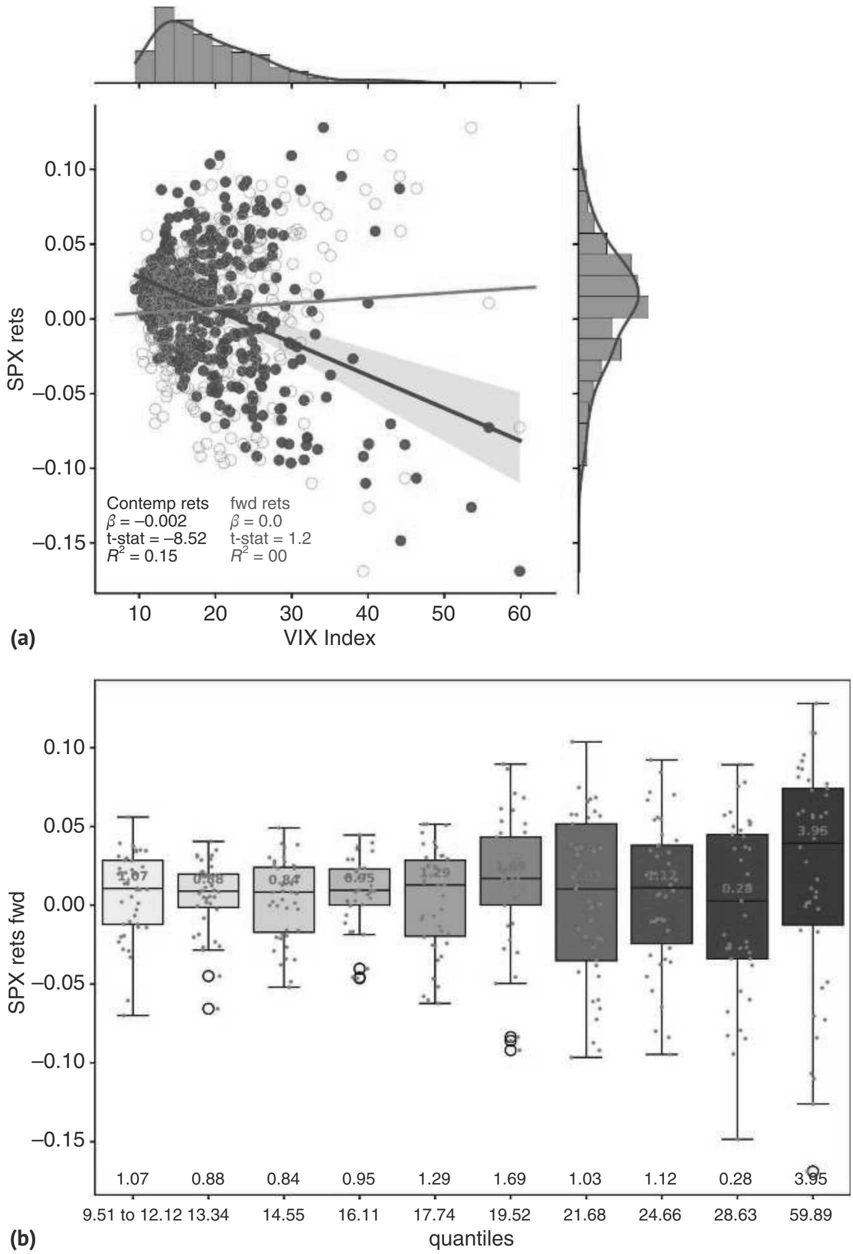


Figure 9.3 (a) Small positive correlation between VIX and forward SPX returns. (b) SPX returns higher than normal if VIX is very extended, but so is volatility.
Source: Bloomberg and authors' calculations.

compared to an unconditional of 0.49. However, the Sharpe ratio is north of 1 for the lowest quintile of VIX (below 12). The worst Sharpe ratios come in the marginally high VIX environments, with VIX between 17.5 and 21.5.

To improve on simple VIX strategies, which are quite volatile, we analyze volatility risk premium (VRP) strategies. These strategies are based on VIX versus one-month realized volatility, discussed, for example, in Bollerslev et al. (2011), as well as in Saunders et al. (2024b). Here, both coincident and one-month forward SPX returns are generally positively correlated with the VRP. The top decile has 2% median returns, while most other quartiles have returns closer to 1%. This is illustrated in Figure 9.4.

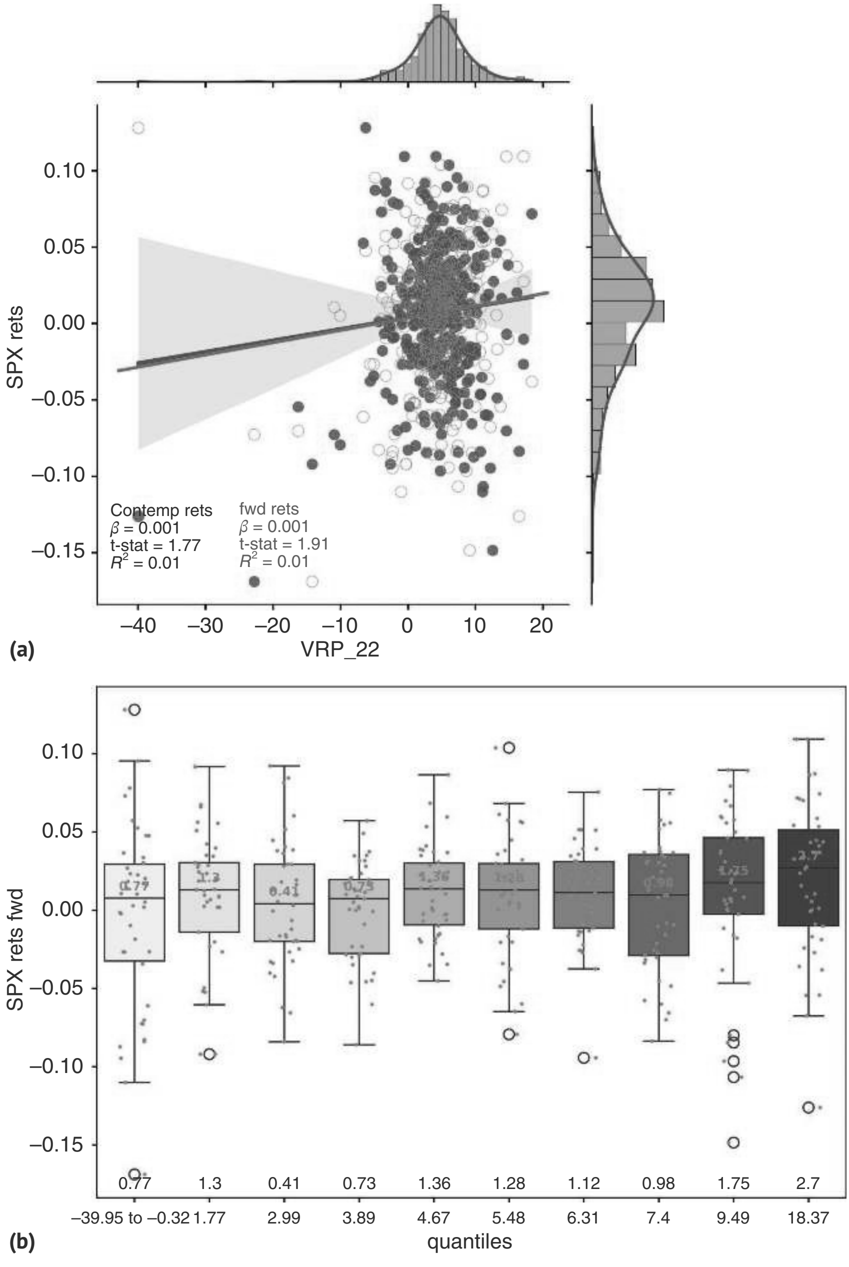
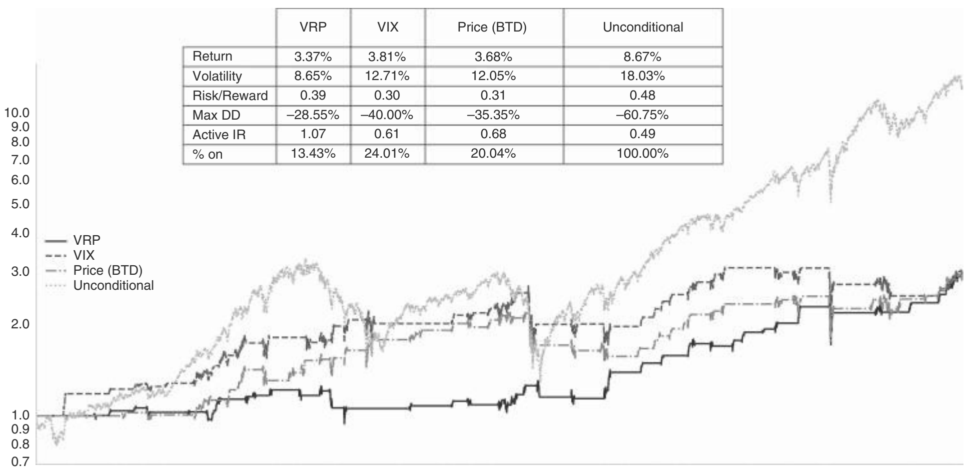


Figure 9.4 (a) VRP has a positive coincident and forward correlation with SPX. (b) High "fear" means higher returns.
Source: Bloomberg and authors' calculations.

To translate these observations into strategies, we backtest two strategies. The first one buys the SPX when VIX moves by more than 2 standard deviations (with a one-year lookback), and the second one when the VRP moves above the 2 standard deviation threshold. We hold a long SPX trade for one month (22 trading days). As is indicated by Figure 9.5, we find that the VRP-based rule has an active information ratio of almost 1.1, higher than the unconditional (0.49) and the active IR for the simple VIX rule (0.58). Its risk-adjusted return is also better than just buying two standard deviation price declines in the SPX (0.66), or than buying on VIX inversion (when the current VIX is above the three- or 12-month level—not shown in the chart). However, given the arbitrary one-month holding period, the overall returns are obviously much lower than buy-and-hold.

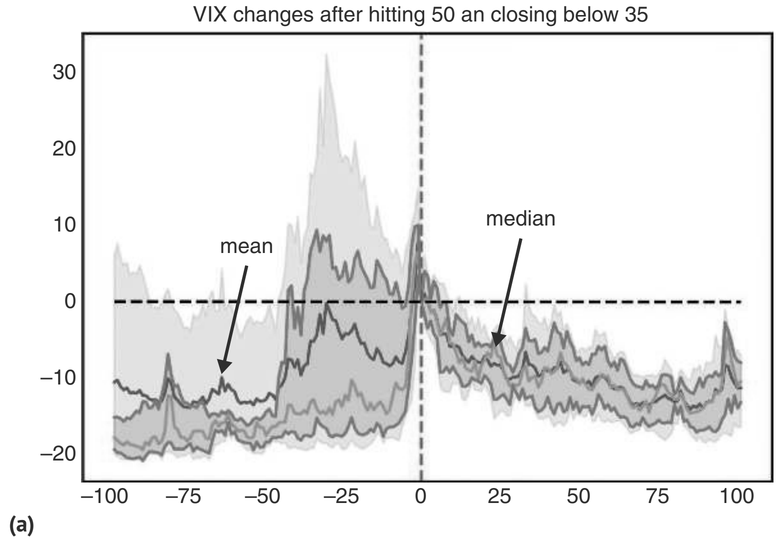
This does not mean, though, that we should disregard the outright level of VIX. One interesting regularity is that while VIX is very high, for example, above 50, many investors are unwilling to engage. At those volatility levels, essentially anything can happen, and investors are hesitant to commit more capital, as even being one day too early can lead to meaningful losses. This changes as VIX comes off. Although this is not helpful for identifying the bottom, as the bottom is usually already in when VIX is fading, a falling VIX leads to the "trade after the trade." This is illustrated in Figure 9.6a and 9.6b, where we show episodes in which VIX increases above 50, at least intraday, and then falls below 35. The finding is that VIX tends to come off further, and the SPX rallies afterward. At a high VIX level, a constant drip of bad news is needed to prevent a relief rally. On average, relief is forthcoming.

Another important way to measure sentiment is through surveys. One of the longer-running surveys is from the American Association of Individual



	VRP	VIX	Price (BTD)	Unconditional
Return	3.37%	3.81%	3.68%	8.67%
Volatility	8.65%	12.71%	12.05%	18.03%
Risk/Reward	0.39	0.30	0.31	0.48
Max DD	–28.55%	–40.00%	–35.35%	–60.75%
Active IR	1.07	0.61	0.68	0.49
% on	13.43%	24.01%	20.04%	100.00%

Figure 9.5 VRP beats VIX.
Source: Bloomberg and authors' calculations.



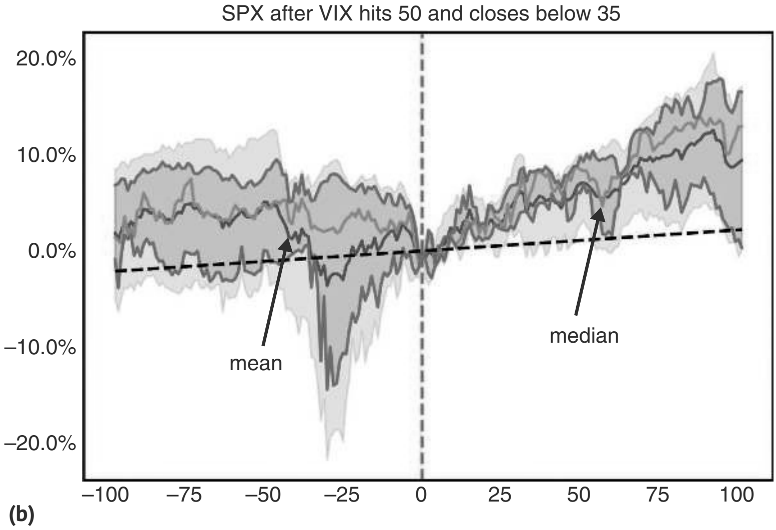


Figure 9.6 (a) VIX tends to keep falling once it hits 35 from a 50+ high (b) … and SPX keeps rallying.
Source: Bloomberg and authors' calculations.

Investors (AAII). Although we typically prefer market measures over survey measures, we note that the AAII bull-bear indicator has historically been quite useful. Analyzing the full distribution of the AAII indicator suggests that on a coincident basis, sentiment moves with prices, i.e., low returns are correlated with negative sentiment readings and vice versa. This carries through to the following week (the survey is published on a weekly basis). However, looking at forward returns, negative sentiment below −20 is typically associated with higher returns on a one-month basis. This is shown in Figure 9.7a with high hit ratios and high forward returns with the indicator between −30 and −10. Just focusing on extremes, we would note that the indicator is better at picking bottoms than at picking tops. The two single highest readings were in late December 1999, which was a good warning signal in the bigger picture, but then also in June 2003, which was a terrible signal. One reason for this is that at times investors can get very bullish early on in a recovery, which is typically a time when the bull market is not even close to topping out again. On the extreme negative side, readings at or below −40 work quite well, indicating that a bottom is close, but those readings are very rare.

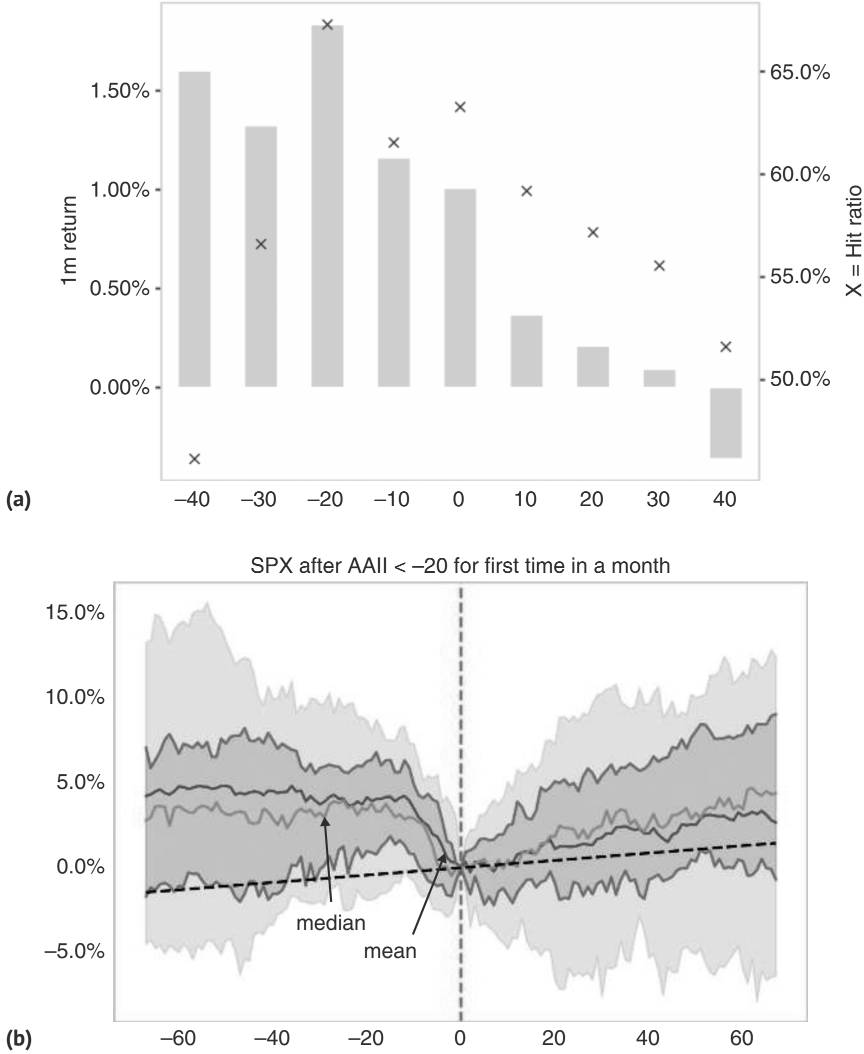


Figure 9.7 (a) AAII as good contrarian indicator. (b) AAII falling below 20 as buying signal. *Source*: Bloomberg and authors' calculations.

Given that the survey is published weekly and we look at one-month forward returns, some of the analysis involves overlapping periods. We therefore also run an event study where we implement a trade whenever the AAII falls below −20, excluding dates where the signal was already triggered in the previous month, to avoid overlapping episodes. With this methodology, we find 44 periods. As illustrated in Figure 9.7b, −20 is typically the bottom of the pullback, with median returns falling into it, but then starting to rise again. However, in the worst 10% cases, significant additional downside occurred.

9.3 POSITIONING PAIN TRADES

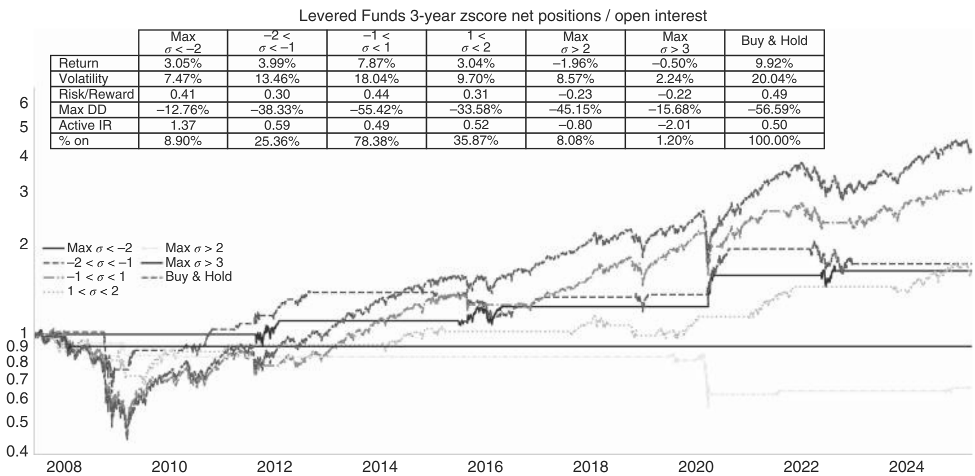
There is a whole cottage industry trying to figure out positioning. For equities, of the publicly available data, the CFTC data is one of the more popular series. Since 2008, this data has been disaggregated to distinguish long-only asset managers from levered funds. We focus here on the positioning of these two investor categories, but the results hold for the less well-defined noncommercial traders, which has a longer history (back to 2000). We impose a holding period of 22 days at each reading, with implementation lagged by five days as the data is released on Friday afternoons, covering up until Tuesday of that week. We find that on the negative side, moderately bearish sentiment results in active information ratios that are not too different from the unconditional one. However, z-scores of less than −2 for asset managers' positions show a meaningful performance improvement over the unconditional, with an active information ratio of 0.9 compared to the unconditional one at 0.5, as per Figure 9.8a. For leveraged funds that bracket results in an even better information ratio of more than 1.3, as per Figure 9.8b. On the long side, positioning has to be more extreme for asset managers before it works as a counter indicator: a z-score of 3, which only occurs 1% of the time, is required for negative returns. For leveraged funds, positioning above 2 standard deviations is already a meaningful negative, occurring 8% of the time. Overall, CFTC positioning is a surprisingly good counter-indicator, where leveraged investor positioning tends to be more useful than asset manager positioning.

9.4 INSIDERS HAVE INSIGHT

What about the sentiment of insiders? Insider sales and purchases must be reported in many jurisdictions, if only with a delay. Insider transactions have long been studied, given that they presumably contain some information about a company's prospects. Our investigation shows that the 12-month forward performance for the SPX is related to the number of insiders purchasing shares as a percentage of the total trades by corporate insiders. When more than 2.5% of insiders buy shares, the SPX is likely to print double-digit positive returns over the next 12 months. Sales are much more frequent than purchases, since most corporate insiders get allocated equity on a yearly basis as part of their compensation without having to buy in the market. Given that insiders could sell shares for many reasons, often to meet liquidity needs, the percentage of sales is much less directly linked to forward returns. Having said that, if less than

	Max $\sigma < -2$	$-2 < \sigma < -1$	$-1 < \sigma < 1$	$1 < \sigma < 2$	Max $\sigma > 2$	Max $\sigma > 3$	Buy & Hold
Return	1.02%	2.11%	8.75%	3.59%	1.05%	–0.10%	9.92%
Volatility	4.59%	11.30%	17.94%	13.31%	7.97%	2.18%	20.04%
Risk/Reward	0.22	0.19	0.49	0.27	0.13	–0.05	0.49
Max DD	–17.01%	–34.05%	–46.11%	–42.31%	–27.22%	–8.98%	–56.59%
Active IR	0.92	0.34	0.53	0.48	0.40	–0.44	0.50
% on	5.78%	29.40%	83.92%	31.14%	10.90%	1.12%	100.00%

(a)



	Max $\sigma < -2$	$-2 < \sigma < -1$	$-1 < \sigma < 1$	$1 < \sigma < 2$	Max $\sigma > 2$	Max $\sigma > 3$	Buy & Hold
Return	3.05%	3.99%	7.87%	3.04%	–1.96%	–0.50%	9.92%
Volatility	7.47%	13.46%	18.04%	9.70%	8.57%	2.24%	20.04%
Risk/Reward	0.41	0.30	0.44	0.31	–0.23	–0.22	0.49
Max DD	–12.76%	–38.33%	–55.42%	–33.58%	–45.15%	–15.68%	–56.59%
Active IR	1.37	0.59	0.49	0.52	–0.80	–2.01	0.50
% on	8.90%	25.36%	78.38%	35.87%	8.08%	1.20%	100.00%

(b)

Figure 9.8 (a) CFTC is an effective but rare contrarian indicator. (b) CFTC mean-reversion works better for levered investors.
Source: Bloomberg and authors' calculations.

17% of insiders are selling, returns tend to be very strong. This was demonstrated by Chronert et al. (2024), and is shown in Figure 9.9. Presumably, this is the case, as at these times the valuation of equities is so attractive that insiders will push out their liquidity needs to avoid having to sell. This is also a good sign for the equity market.

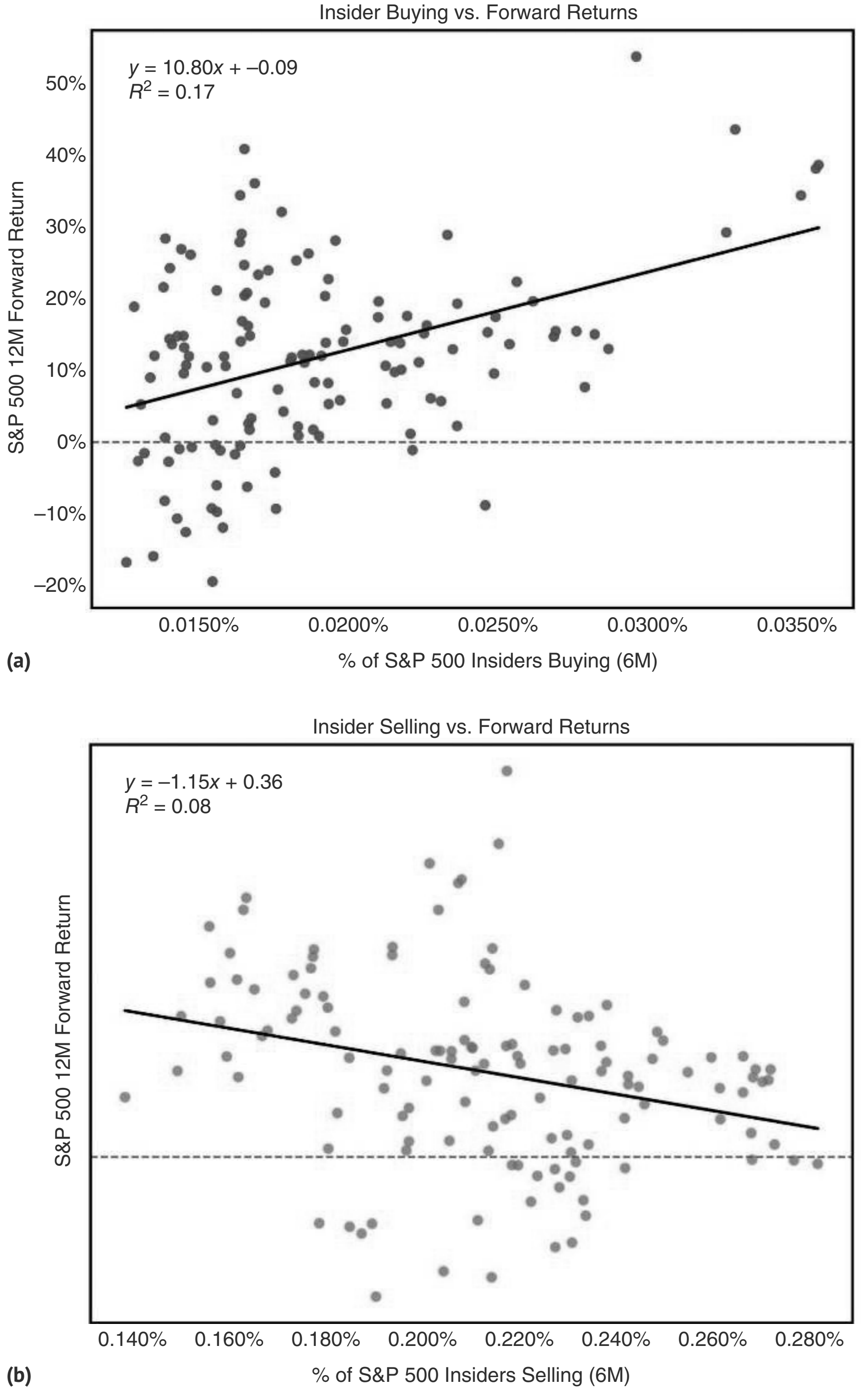


Figure 9.9 Insiders have information and trade.
Source: Bloomberg, Citi research, and authors' calculations.

9.5 BUYBACKS—DON'T HAVE YOUR BACK

Buybacks are a payout mechanism in which corporations repurchase their stock to return value to their equity holders. Buybacks are a more flexible payout than dividends, as there is no obligation by the company to repurchase shares by any given date. Many investors prefer buybacks to dividends, as buybacks will lead to a capital gain, which is typically taxed at a lower rate than dividends. Buybacks have historically been credited with being very supportive for equity prices. However, it turns out that on a contemporaneous basis, there is no relevant correlation between quarterly SPX returns and buyback activity (not shown). If anything, SPX returns are actually leading buybacks by a quarter, as can be seen in Figure 9.10a. Moving from buybacks

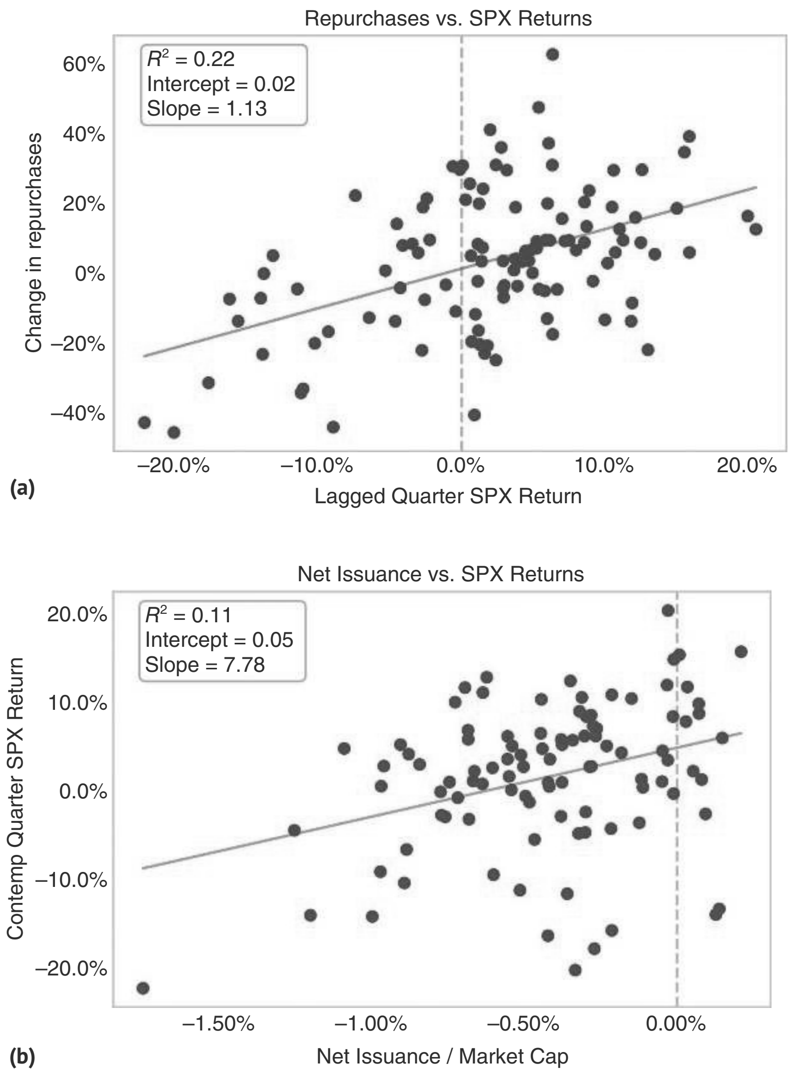


Figure 9.10 (a) SPX leads buy backs. (b) Net issuance more highly correlated, but coincident and not leading.
Source: Bloomberg and authors' calculations.

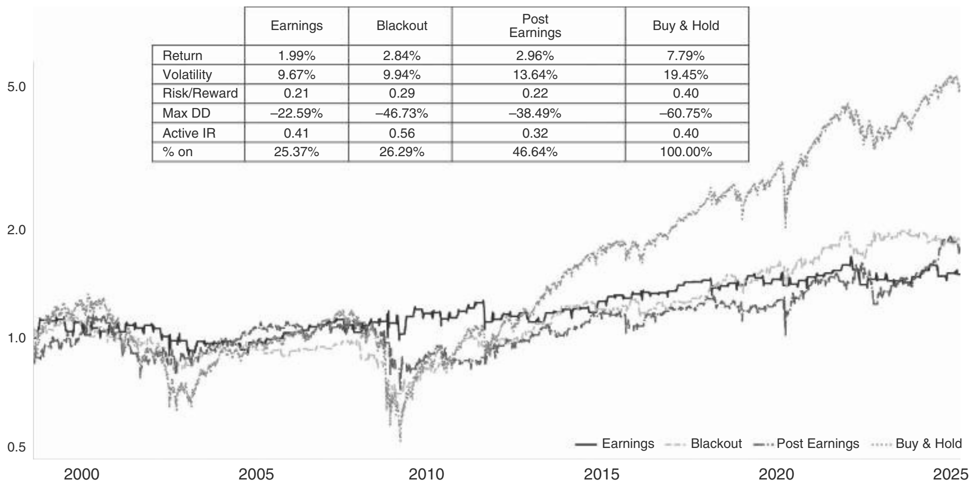


Figure 9.11 Blackout periods not overly harmful.
Source: Bloomberg and authors' calculations.

to net-issuance leads to a better contemporaneous fit, as per Figure 9.10b, but the R^2 remains well below the one when using SPX returns to explain future buyback activity. In any case, for net issuance, we also need a good sense of IPO activity, which can be challenging. This suggests that a focus on buybacks for trading the SPX may be misplaced.

Another question is whether the absence of buybacks can be a negative for the market. This usually comes up in relation to blackout periods. Companies typically suspend stock buybacks two weeks prior to reporting earnings (the blackout period), which has led to speculation that the S&P 500 realizes lower returns during these periods. To account for shifting reporting days, we split up the S&P 500 into three periods: Earnings weeks (when 20–90% of the S&P are reporting), Blackouts (two weeks prior to earnings weeks), and post-earnings (90–100% of the companies have reported already). As is illustrated in Figure 9.11, the blackout regime has a higher active information ratio than the unconditional case. Volatility is higher, though, than in the earnings regime, suggesting that buybacks do suppress volatility. Overall, it is not a good idea to avoid longs in the SPX during the blackout period. Interestingly, the post-earnings regime has the lowest information ratio, even though theoretically, post-earnings volatility should be lower.

9.6 THERE IS A SEASON FOR IT

Seasonals in equity markets have a long history. The most famous seasonal rule is "Sell in May and go away." This is usually followed by "but remember to come back in September." Similarly, the so-called Halloween indicator suggests that investors should invest from October 31 to April 30. Although the origins of these rules are not 100% clear, O'Higgins and Downes (1990) reported this rule in 1990, which gives us plenty of data for an out-of-sample test. Academic research, like Haggard and Witte (2010),

	Sell in May	Buy in May	Buy and Hold
Return	6.34%	2.30%	8.64%
Volatility	12.86%	12.63%	18.02%
Risk/Reward	0.49	0.18	0.48
Max DD	–37.85%	–52.51%	–60.75%
Active IR	0.71	0.26	0.48
% on	48.59%	49.45%	100.00%

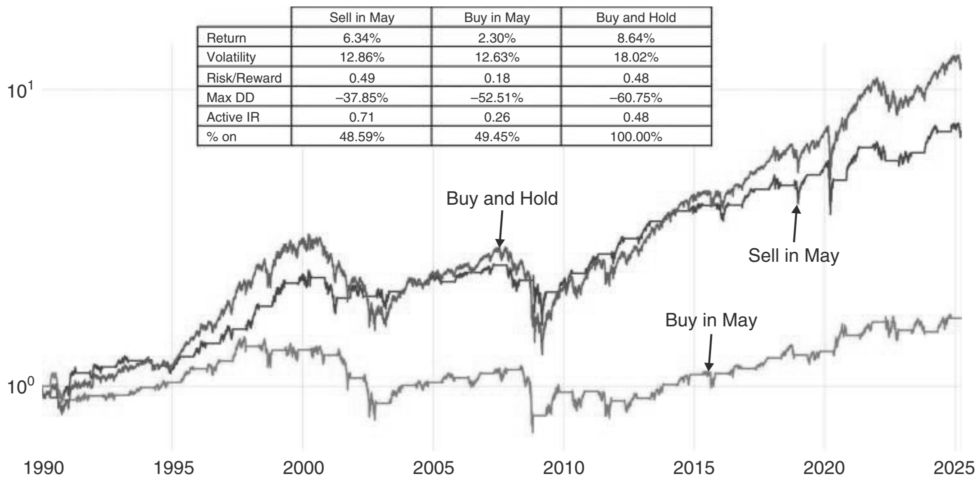


Figure 9.12 Returns tend to be lower May to Oct.
Source: Bloomberg and authors' calculations.

has confirmed the result and extended it to sectors; see also Jacobsen and Visaltanachoti (2009). For a cross-asset perspective, Citi found episodic performance (see Sheetz et al., 2022). But for the SPX, the results are relatively strong. Only starting in 1990, i.e., after publication of the rule, holding the SPX only from the end of October to the end of April results in an active information ratio of 0.71 compared to the buy-and-hold strategy of 0.48, as shown in Figure 9.12.

At Citi, we have publicized our own rule (Saunders and Willer, 2023a), with a better hit ratio: we position long in November and December, if the year-to-date returns, i.e., from January to the end of October, have been strong. In particular, if such returns are larger than 8%, it is difficult to lose money being long in November and December, as is illustrated in Figure 9.13a. We have used this strategy out of sample for more than five years now, with very good results.

We next examine plausible reasons for such outperformance, as we trust empirical patterns with an underlying rationale much more than without. One hypothesis is that institutional investors may be chasing returns into year-end. Investors may be keen to use the PNL under the belt to position more aggressively for additional upside. Furthermore, the subset of investors who have been underperforming the market could be aiming to make up for this underperformance before the year is over, adding to the strong performance into year-end. While intuitively plausible, we find little evidence for this line of thinking. Using investor surveys, there is no relationship between Jan-Oct returns and increased allocations. However, retail investors seem more prone to performance chase into the year-end. Although we only have 10 years of data, mutual fund and ETF inflows in November and December are stronger in up years. Another reason for this pattern of returns could be related to tax loss harvesting or window dressing into year-end. After all, we do find a strong seasonal pattern in momentum returns too, with December the strongest month, followed by a reversal in January. Momentum returns are from Ken French's website and constructed in the same way as Fama and French (1992a), based on the last 12-month performance,

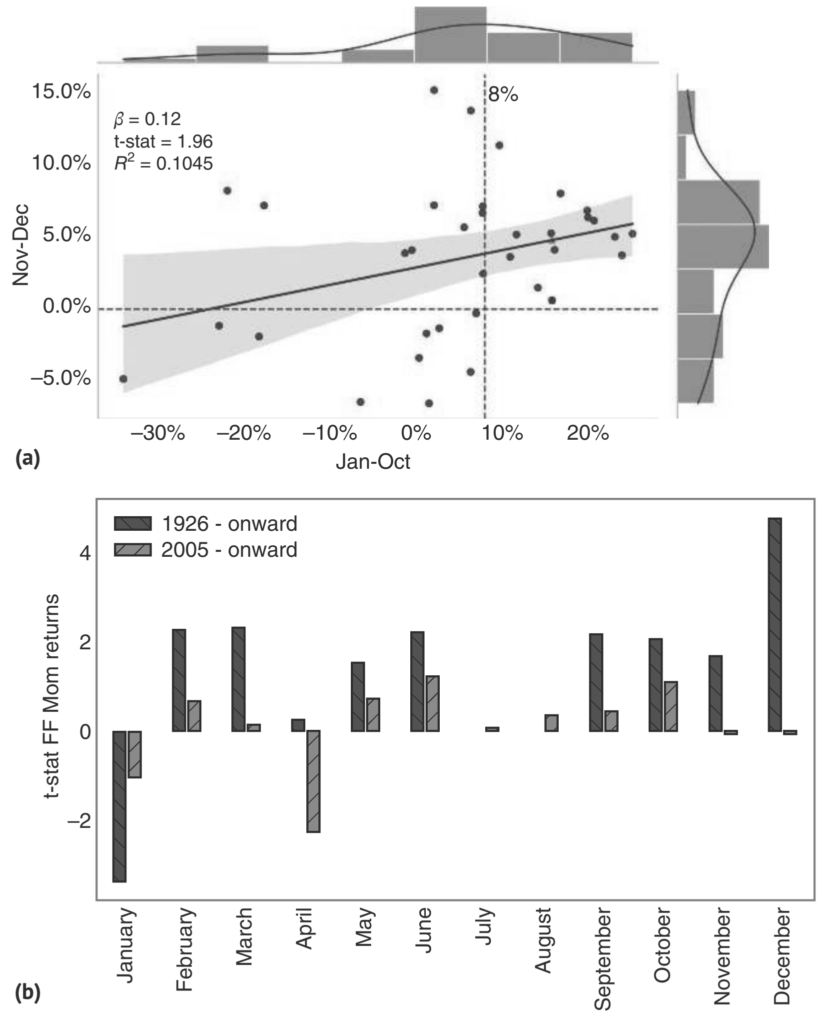


Figure 9.13 (a) Holiday rally is conditional. (b) Momo strategy highly seasonal.
Source: Bloomberg, Ken French, and authors' calculations.

skipping the last month (12–2). This pattern is consistent with the presence of tax-loss harvesting. YTD losers are more likely to be sold further to crystallize the losses and reduce tax payments for the year. This pattern is then unwound early in the new year, leading to a strong reversal in January, explaining the seasonal patterns of the momentum strategy, illustrated in Figure 9.13b. In strong years, there are fewer losers to harvest, reducing selling pressure and explaining strong December returns.

One interesting feature is that the December effect for the momentum strategy has weakened over time. In Figure 9.13b, we also show the t-stats from 2005 onward, which indicate that there is no major December effect for the momentum strategy anymore (even though weaker January returns remain in place). This may be because tax-loss harvesting has moved earlier in the year or because of hedge-fund de-grossing into year-end, as discussed by a hedge fund CIO (see Ahkong, 2024). In spite of weaker returns for the momentum strategy overall, our strategy positioning for year-end rallies conditional on the performance earlier in the year has continued to work well out of sample.

An interesting related question is whether the turn of the year makes a change in trend more likely, maybe because the PNL cushion overnight disappears on January 1. To investigate this question, we show the correlation of three month returns into and out of any given calendar day for the SPX. Going back 30 years, we find that the correlation is the most negative in late February/early March, as shown in Figure 9.14. It actually becomes quite negative, at more than –20%, which is interesting, given that over the last 30 years, the market mostly went up. Although this rule is not directional, it would be in line with selling in early March and going away, at least if the market went up into year-end as well as early in the year. It is also noteworthy that the correlation already starts turning negative in January, even though it is not as negative as in early March. This does provide some evidence that Q1 tends to be a tricky quarter when it comes to momentum trades. The other period where the correlation turns negative is in July and then again late September/early October, which lines up with the "buy back in September" narrative, tested earlier.

Another interesting seasonal effect occurs in April, when taxes are due. Typically, US equities do not perform well into April 11, which is in line with the April 15 tax deadline, when allowing for a few days for funds to settle. The underperformance starts around April 5. Unsurprisingly, the effect is much stronger when the previous year has been a 20%+ up year. Following down years, equities do quite well in early

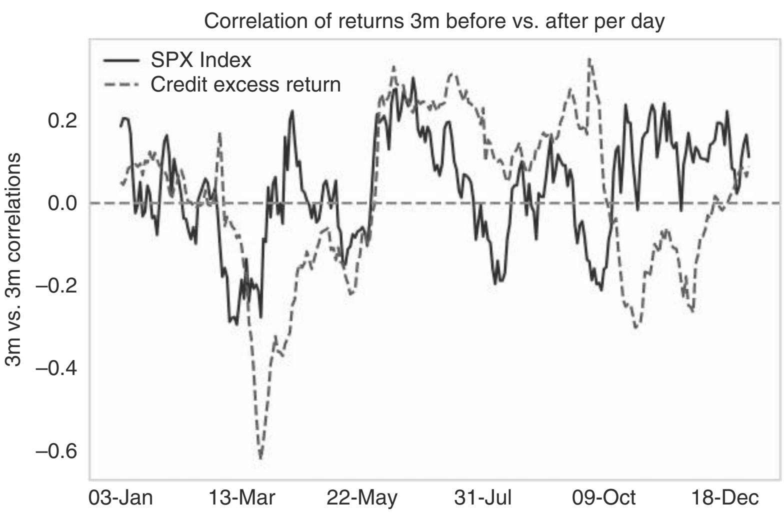


Figure 9.14 SPX trends change early in the year.
Source: Bloomberg and authors' calculations.

Figure 9.15 The Ides of April.
Source: Bloomberg and authors' calculations.

April. The negative effect after strong up years is quickly unwound after April 12. Although the effect is often too short-lived to trade, it can help timing bearish trades for investors who are already bearish for other reasons. The April effect is illustrated by Figure 9.15, which shows mean and median daily returns since 1970 and the medians after up and down years (see also Saunders and Glass, 2024). The April tax effect is still present in the recent data: focusing on the seven strong up years since 2004, the median return was negative (−65bps) through April 14 before recovering strongly to 1.25% by April 19.

9.7 OF GOOD NEWS BEARS—AND VICE VERSA

Equities do not always react in the same way to incoming economic data. A strong payroll number, i.e., good economic news, may lead to a big rally as markets are assured that economic growth is not as weak as is feared. However, on other occasions, the same data print may lead to an equity sell-off (the good economic news is bad market news) on concerns that the economy is overheating. Hence, tracking how prices react to incoming economic data gives an important window into how markets think about the interaction between growth, the economy, monetary policy, and the central bank reaction function. One way to operationalize this concept is by looking at the correlation of economic surprises with equity returns. We use the Citi economic surprise index, the original surprise tracker produced by the FX QIS team (see Kasikov, 2006). Figure 9.16 shows the surprise index next to the S&P index. The chart highlights several periods of divergence—either a string of positive surprises, coinciding with a sell-off (good news is bad news), or disappointing data coinciding with a market rally (bad news is good!).

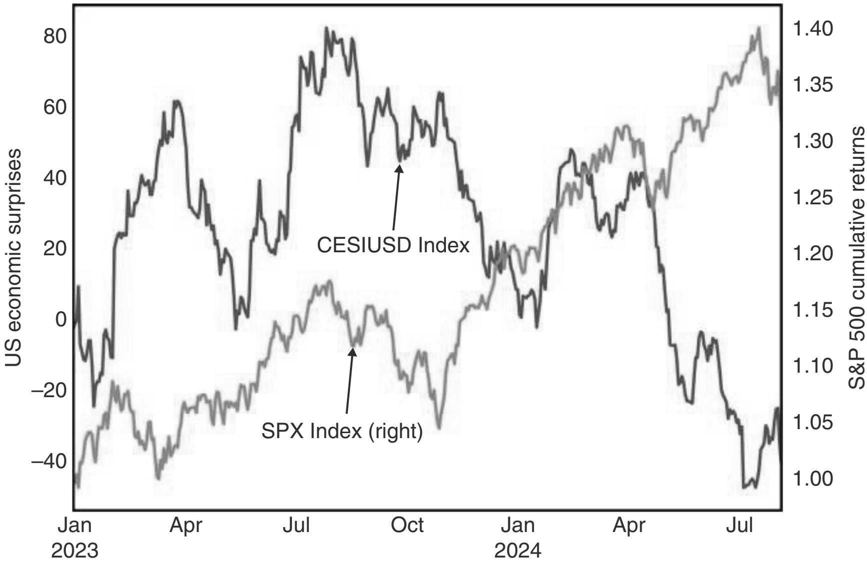


Figure 9.16 Equities can rally on good economic news, or sell-off.
Source: Bloomberg, Citi, and authors' calculations.

Figure 9.17 shows the correlation of the two series over time, with periods where the Fed is in a hiking cycle shaded. We developed this concept, for example, in Saunders and Willer (2023b). Periods of hiking cycles have, on average, lower correlations between changes in economic surprise and equity returns. This passes the intuition test. During hiking cycles, central banks are trying to slow down economic growth, and markets can therefore react positively to bad economic news. Recession risks are typically still quite low at that time, and the focus is on weak data that potentially leads to less aggressive rate hikes and a reduced risk that the central bank might "break something."

The surprise change/equity return correlation is therefore an important distinguishing feature for macro traders and informs how to position and react to changes in expected and incoming economic data. We create four distinctive regimes by examining the good-news–bad-news correlation indicator with how the economic data have been printing on balance. "Good news is good news," is the regime where the economic surprise index is positive, and the correlation of the SPX with the surprise index is positive too. If surprises are positive and the correlation is negative, we are in the good news is bad news regime. The opposite holds for the two regimes when the economic surprise index is negative. We find an interesting pattern when looking at the returns of equity markets in each of these regimes.

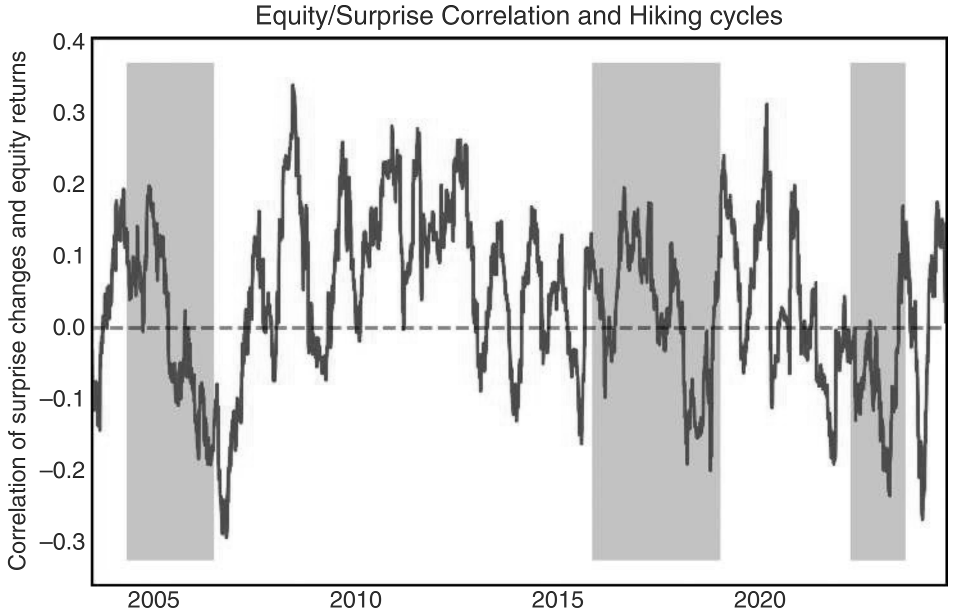


Figure 9.17 Correlation between equity returns and economic surprises is generally lower during hiking cycles.
Source: Bloomberg, Citi research, and authors' calculations.

As shown in Figure 9.18, the best environments are, not fully unexpected, when equities like good news and the market has received a string of above consensus data, i.e., the Citi surprise index is positive. Perhaps counterintuitively, nearly just as good is the environment when the equity/surprise correlation is negative and economic data is below consensus, i.e., bad news is good news and the news is bad. These patterns can be helpful for macro traders when they have a firm view on upcoming data.

This framework could also be used for trading equities more mechanically, as the relationships still hold when implementation is lagged by a day, which allows investors to observe the correlation and level of the surprise index. For many values, this lag is not necessary, as both numbers only change slowly, but given that there are times when the correlation and/or the surprise index is close to the zero line, adding a one-day lag is safer. The results are illustrated in Figure 9.19. When CESI is above zero, the active information ratio for the SPX is 0.61, compared to 0.47 when below zero. With the unconditional information ratio coming in at 0.55, the improvement is very small and potentially mostly noise. The results get much stronger when we add the correlation as well: When CESI is above zero and the correlation is above zero, the information ratio increases to 0.84. The other strong regime, CESI below zero with a correlation below zero, is also strong at 0.72. The worst regime is a positive CESI when the correlation is negative, though we would note that even in this regime, the information ratio is still positive, given that the SPX generally does not fall for long.

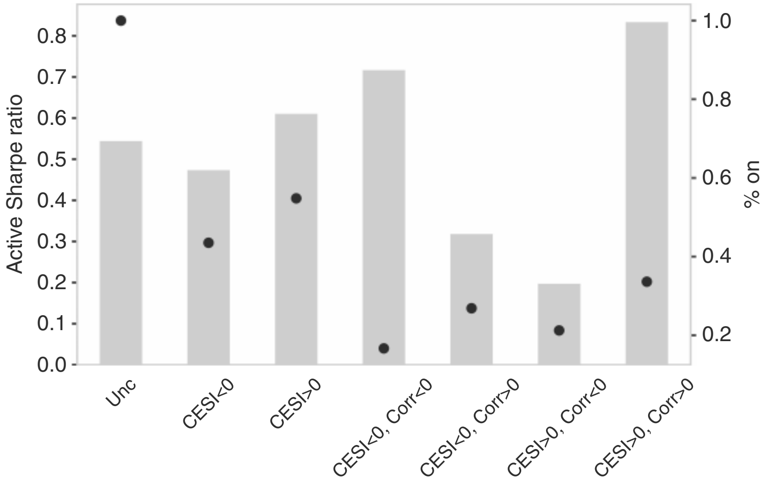
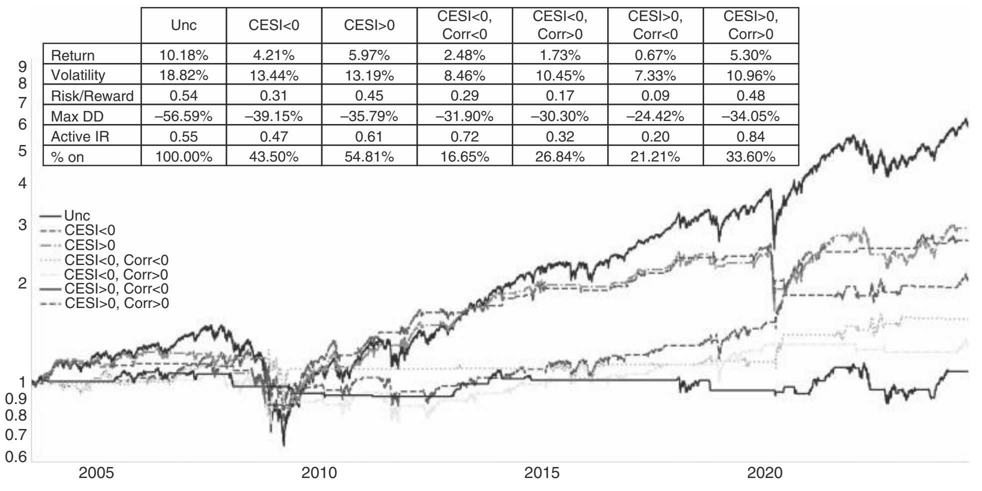


Figure 9.18 The best environment for Worst returns come in “good news is bad news” regime, with good news.
Source: Bloomberg, Citi, and authors’ calculations.



	Unc	CESI<0	CESI>0	CESI<0, Corr<0	CESI<0, Corr>0	CESI>0, Corr<0	CESI>0, Corr>0
Return	10.18%	4.21%	5.97%	2.48%	1.73%	0.67%	5.30%
Volatility	18.82%	13.44%	13.19%	8.46%	10.45%	7.33%	10.96%
Risk/Reward	0.54	0.31	0.45	0.29	0.17	0.09	0.48
Max DD	–56.59%	–39.15%	–35.79%	–31.90%	–30.30%	–24.42%	–34.05%
Active IR	0.55	0.47	0.61	0.72	0.32	0.20	0.84
% on	100.00%	43.50%	54.81%	16.65%	26.84%	21.21%	33.60%

Figure 9.19 The best environment for equities is good news is good news and the news is good!
Source: Bloomberg, Citi, and authors’ calculations.

9.8 THE FOMC EFFECT

We also examine trading around FOMC days. The first thing to mention is that equity returns are higher on FOMC days, as shown in Figure 9.20, coming in at 0.16%, compared to 0.03%, when focusing on mean returns. The hit ratio of positive returns is also meaningfully higher, coming in at 54.9%, compared to 51.1% on all days.

The second question is whether there is follow-through. We examine this in two ways. First, we evaluate this based on whether the monetary policy news is relevant, judged by the two-year yield moving up or down by 2σ. Second, we check whether equities move 2σ. We find that there is little read through from the monetary policy information, as shown in Figure 9.21a. We find that both cases show positive returns post-FOMC day. This is true for both large up-moves and large down-moves on FOMC days. This makes the study not overly compelling. What is interesting, though, is that in both cases when the SPX sells off (which can occur with or without a big rise in yields), we tend to see a small recovery in the next 10–12 trading sessions, as shown in 9.21b. Investors should fade downside momentum on FOMC day but, on balance, go with momentum after positive FOMC days.

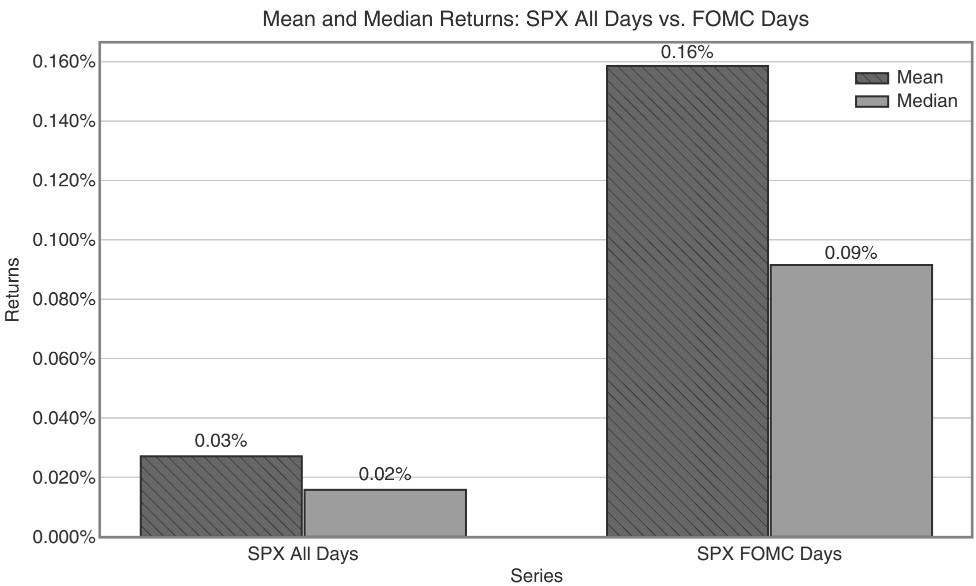


Figure 9.20 The SPX likes FOMC days.
Source: Bloomberg and authors' calculations.

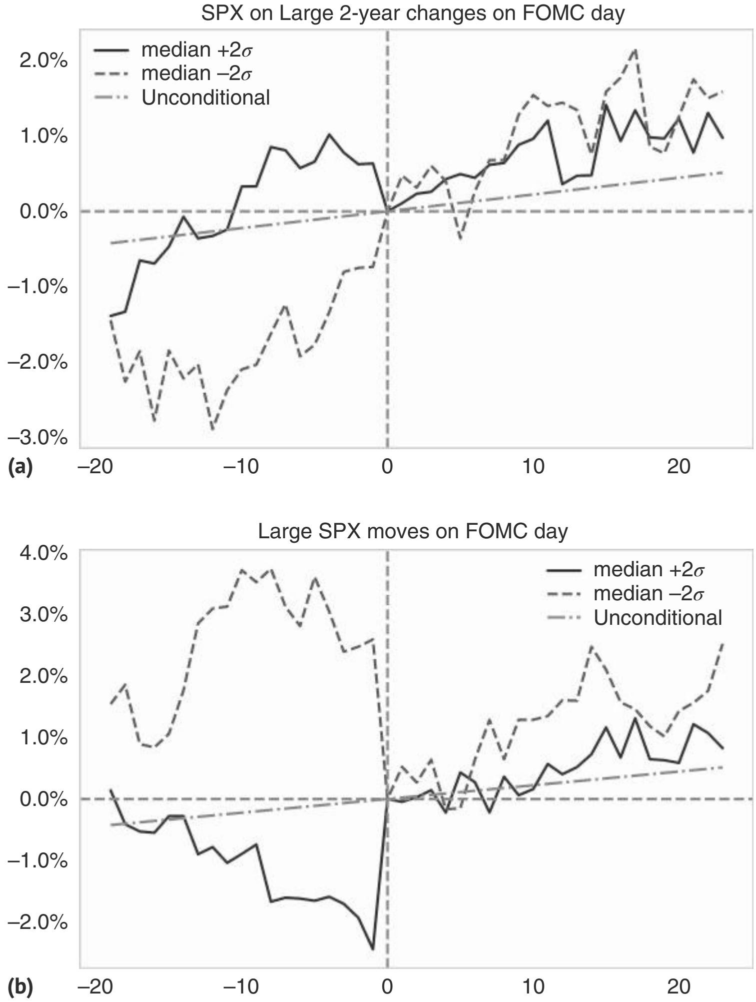


Figure 9.21 (a) Rising rates only with a short-term impact. (b) Fade downward momentum on FOMC days.
Source: Bloomberg and authors' calculations.

9.9 SUMMARY

Our favorite indicator to tactically sell equities is CITI's POLLS indicator. The best indicator to buy the dip is the volatility risk premium model (VRP). VIX spikes are generally there to be sold too, but it is much safer to do so only when VIX has fallen below 35. Sentiment-wise, the Individual Investor survey is a good counter-indicator, as is CFTC positioning. Insider equity purchases are a positive indicator, but buy-backs are not. Seasonal patterns work well in US equity markets. Returns tend to be higher in November to April than in May to October. We also found that combined returns for November and December are typically strong, but they are even stronger if the January-to-October return was strong already. The momentum strategy is also highly seasonal. We investigate regimes of good news and bad news and find that the SPX does best when we get good news (economic surprises (CESI) above zero), which the market treats as good news (correlation of CESI with SPX positive). FOMC days are typically bullish, as is their aftermath.

CHAPTER 10

How to Trade Equities: Globalizing

So far, we have dealt exclusively with US equities. At the time of writing, years of US exceptionalism have resulted in the US accounting for almost 75% of global market capitalization, measured at prevailing exchange rates, the only game in town. Some of it can be attributed to a higher valuation, but even in terms of earnings, the United States is outsized. US earnings are currently around 50% of global earnings. Despite this dominance, there are interesting macro opportunities in other markets, and following we discuss some useful frameworks for the other large global markets.

The first point to make when discussing international equity markets is that they are highly correlated with the United States, which limits any diversification benefits. The correlation of weekly returns ranges from 0.28 (Chinese A-shares, almost the only market that is still quite idiosyncratic) to 0.79 for Canada. The second largest market, Europe, also has a high correlation of 0.74, according to Table 10.1. Most correlations are well above 50%, with the Swiss market being the other important outlier in addition to China. Put differently, for any given week, the likelihood that local and US returns have the same sign ranges from the mid-50% (Asian equities) to the high 70% (Eurozone, Canada). This means that on any given week, losses in the SPX are unlikely to be offset by gains elsewhere in global equities. However, correlations in the 50 to 80 range still leave room for diversification and alpha generation. From a macro perspective, and before going into the local stories, country selection can be used to express views on the strength of the global cycle, on the view on interest rates, or to express views on FX.

10.1 THE CYCLE GOES GLOBAL

The easiest way to think about global equity markets is how sensitive they are to the global manufacturing cycle, here proxied by the US manufacturing ISM. The most trading-oriented way to do that is to calculate excess Sharpe ratios for the various markets for times when the global manufacturing PMI is above its 6mma. It is an open question whether we should correct for the unconditional Sharpe ratio of a given market or not. After all, we care about generating alpha, and a market that typically performs poorly, but less poorly when the manufacturing PMI is above its 6mm, is not necessarily helpful. Of course, this assumes that poorly performing markets have a structural reason to be poorly performing markets, which is a big assumption. We therefore offer both numbers. The analysis starts in 2000. Figure 10.1 shows that the

TABLE 10.1 Down Means Down

Asset	Corr	SPX up local up	SPX up local down	SPX down local up	SPX down local down	Same direction
S&P 500 Future	1.00	0.57	0.00	0.00	0.42	1.00
Nasdaq Future	0.86	0.51	0.07	0.06	0.35	0.86
TSX60 (Canada) Future	0.79	0.47	0.11	0.10	0.32	0.79
Bovespa (Brazil) Future	0.56	0.37	0.11	0.16	0.28	0.66
IPC (Mexico) Future	0.61	0.41	0.12	0.16	0.31	0.71
Euro Stoxx 50 Future	0.74	0.45	0.11	0.12	0.32	0.77
FTSE 100 Future	0.71	0.32	0.09	0.11	0.23	0.55
CSI 300 (China) Future	0.28	0.36	0.16	0.24	0.23	0.59
ASX 200 (Aus) Future	0.60	0.41	0.15	0.16	0.28	0.69
DAX (Germany) Future	0.74	0.37	0.08	0.09	0.26	0.63
CAC 40 (France) Future	0.73	0.34	0.08	0.09	0.24	0.58
IBEX 35 (Spain) Future	0.65	0.35	0.11	0.13	0.25	0.60
FTSE MIB (Italy) Future	0.66	0.44	0.13	0.15	0.28	0.72
AEX (Dutch) Future	0.73	0.35	0.08	0.09	0.24	0.59
SMI (Swiss) Future	0.34	0.35	0.10	0.12	0.24	0.59
Hang Seng (HK) Future	0.53	0.32	0.12	0.16	0.24	0.56
Topix (Japan) Future	0.51	0.33	0.11	0.13	0.22	0.55

Source: Bloomberg and authors' calculations

best absolute Sharpe ratios during good times for the global manufacturing sector are to be found in Japan and Hong Kong, followed by SPX and Italy. The cyclical underperformers are the United Kingdom and Brazil. If we compare the performance of the indexes relative to the unconditional, the winner is again Japan and Hong Kong, followed by Switzerland, Germany, Switzerland, and Italy, followed by the SPX, Germany and Switzerland. The UK clearly ranks the least cyclical, followed by Brazil and Mexico. While the UK market is famously defensive and Japan is famously cyclical, the low ranking of Brazil is a surprise, partly explained by the big run-up in 2004 to 2008, at a time when the manufacturing ISM was falling.

The case of Brazil raises the question whether local growth momentum matters more than global growth momentum. This is addressed in Figure 10.2, where we use

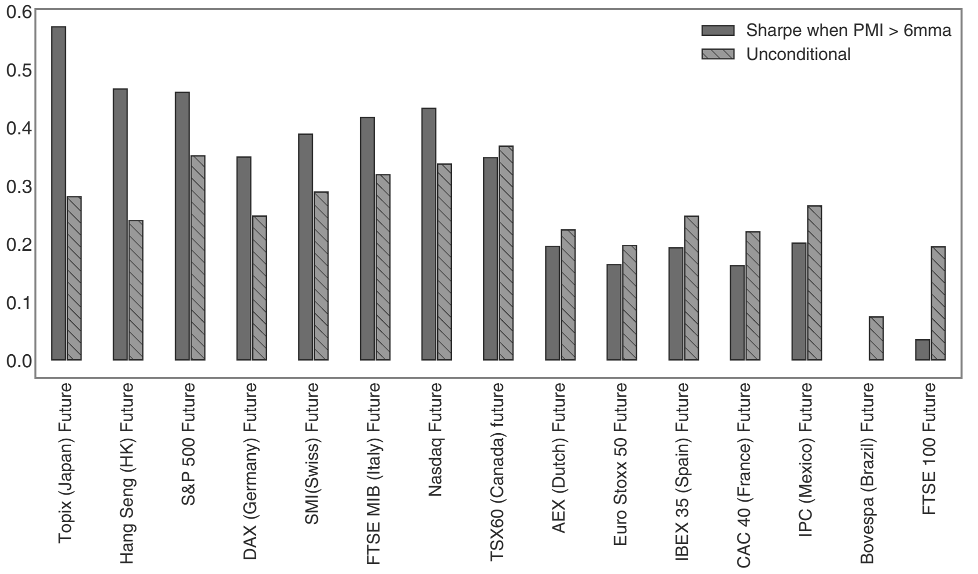


FIGURE 10.1 Japan is the most cyclical, the United Kingdom the least.
Source: Bloomberg and authors' calculations.

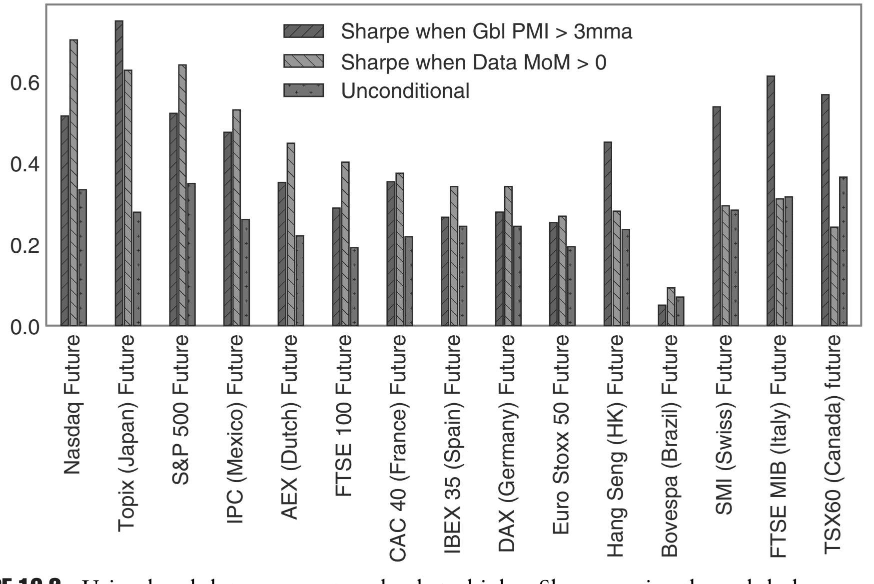


FIGURE 10.2 Using local data momentum leads to higher Sharpe ratios than global momentum.
Source: Bloomberg, Citi, and authors' calculations.

the local Citi data momentum indexes, rather than the global PMI, given data availability. It turns out that for the United States, Mexico, most of Europe (Netherlands, France, Spain, Germany, and, barely, Eurostoxx), as well as the United Kingdom, buying the market with positive local data momentum leads to higher Sharpe ratios than using the global manufacturing PMI. The opposite is true for Hong Kong, Switzerland, Italy, and Canada. Overall, we would think local data works slightly better. Ranking by local data momentum, similar countries show up as most cyclical in terms of most improved Sharpe compared to the unconditional: the United States and Japan. Brazil looks again not overly cyclical, and Switzerland, Italy, and Canada now also screen as less cyclical. In the end, investors should go with the highest Sharpe ratio on offer, and use both local and global momentum indicators, applying whatever works best.

Somewhat related to the question of cyclicality is the interest rate sensitivity of the various markets. More cyclical markets would be expected to cope better with higher interest rates. But even for cyclical markets, too quick and too large a rise in rates may detract from performance. Given likely nonlinearities, we bucket monthly US rate moves into five buckets. When rates fall by more than 20bp, all equity markets do poorly, with Hong Kong and Brazil being the least sensitive, with HK, as a China proxy, generally more idiosyncratic. The –20bp to –10bp bucket is the most mixed across markets and is generally still quite negative. The –10 to +10bp is one of the strongest, as equities prefer low volatility. Only the >20bp bucket is even more bullish. The growth factor clearly seems to dominate the rate factor when specified in this way. Certainly, markets do better when rates are rising fast than when they are falling fast. But the reason to show Figure 10.3 in our global equity chapter is Japan. Japan dislikes falling rates, in line with most other markets, and likes rising rates the most. This is true both in a time-series context and in a country comparison. This is in line with Japan being one of the most cyclical markets. Furthermore, higher rates usually mean higher USDJPY, which helps both Japanese exporters and the overall reflation story. When expecting higher rates, consider going long Topix.

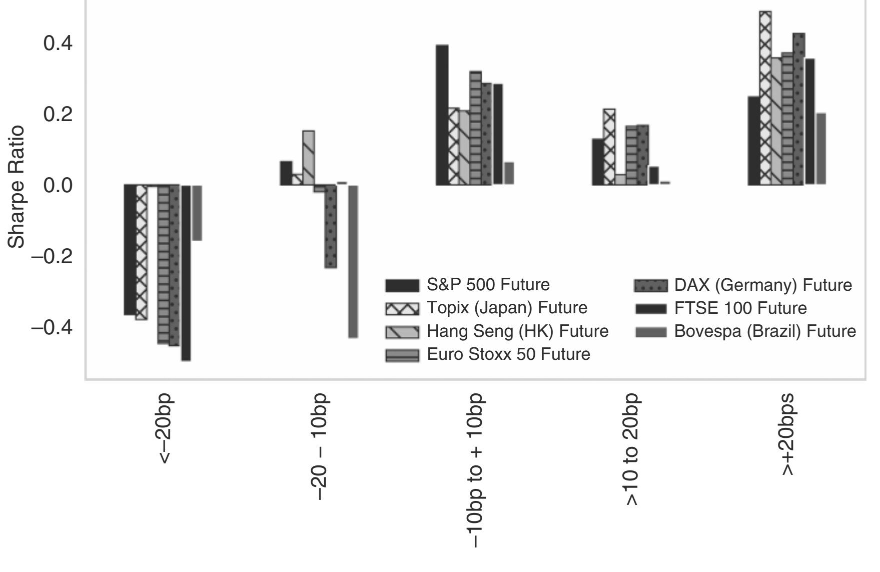


FIGURE 10.3 Japan likes higher rates the most.
Source: Bloomberg and authors' calculations.

10.2 GLOBAL EQUITIES MEET FX

The other macro factor to investigate is FX. We generally compare equity market performance on an FX-hedged basis, as we think of the FX trade and the equity trade as two separate investment decisions. But FX matters even for hedged indexes. Interestingly, different markets have a very different sensitivity to FX. Equity markets in Japan and the United Kingdom perform better, on an FX-hedged basis, when DXY is strengthening, and their currencies are weakening. This is due to the high exporter share in Japan and because the FTSE companies also have a very high share of international revenues. Europe benefits on average slightly from a weaker EUR, but the effect is not as strong. Emerging markets, even FX hedged, trade much better when their currency is appreciating. The reason is that emerging markets are often driven by an overriding macro risk factor, which will simultaneously strengthen the currency and create a bid for local assets, including equities, or vice versa. This effect is much stronger than the more marginal market share gain or loss by the export sector. Figure 10.4 shows that for Brazil as an EM proxy. See Manthey et al. (2025) for similar results. The performance of the United States is slightly negatively correlated with a stronger USD, likely also on the back of meaningful non-US income of US corporates. When bullish USD, buy Japan and the United Kingdom, when bearish USD, buy EM.

Having discussed the impact of the global macro factors on the various regional markets, we now go through some of the more important ones, laying out frameworks to think about the various idiosyncrasies.

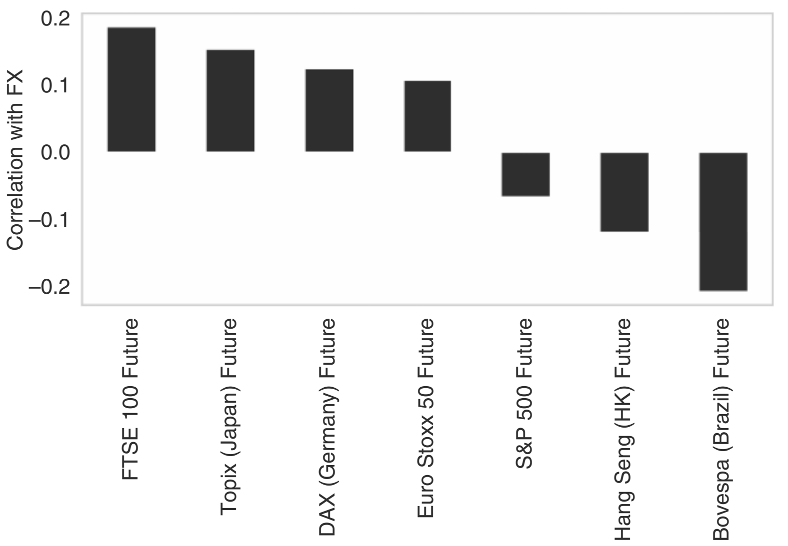
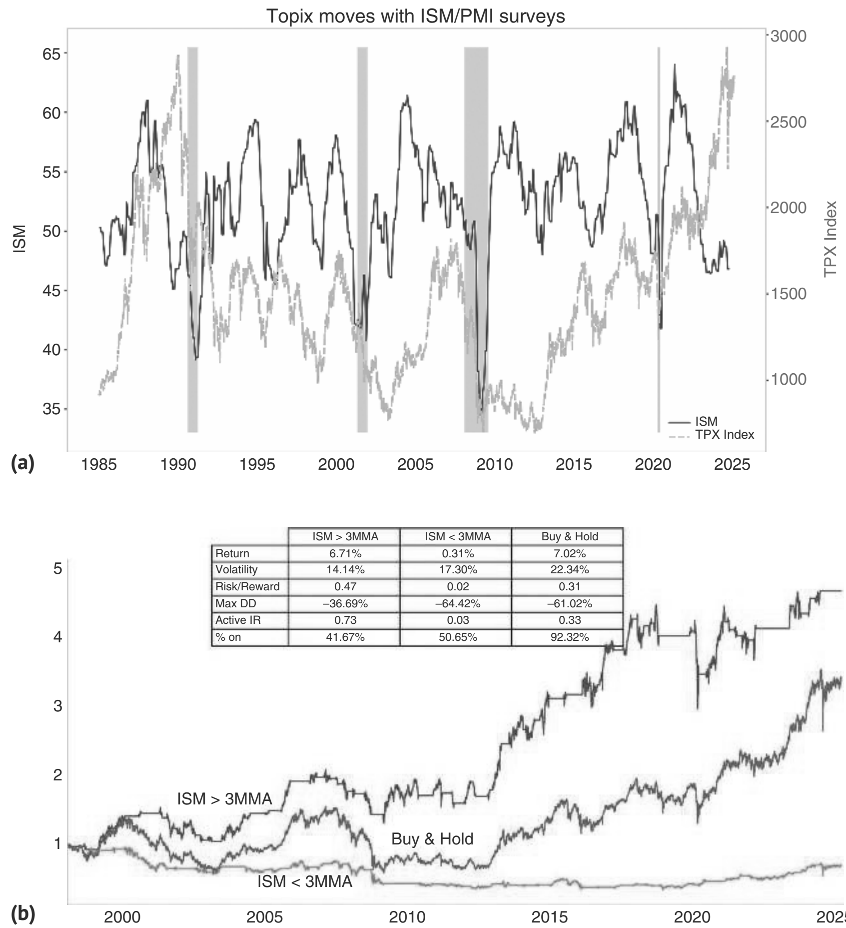


FIGURE 10.4 Japan and the United Kingdom like weaker FX, EM prefers it stronger.
Source: Bloomberg and authors' calculations.

10.3 TRADING TOKYO

We already discussed Japan in the fixed income section, where we briefly alluded to the history of the Japanese bubble in the late 1980s. Following the run-up during the bubble building and its subsequent demise, the market has become mostly correlated with the business cycle, as per Figure 10.5a, as multiple expansions and contractions had become much more muted. Building on the discussion of PMIs above, we test a very simple rule. Starting in 1999, we only buy the Topix when the US manufacturing ISM is above its 3mma and to be flat when the index has moved below (taking the publication lag into account). This rule results in an active information ratio of 0.73, as opposed to the unconditional one of 0.33, as is illustrated in Figure 10.5b.



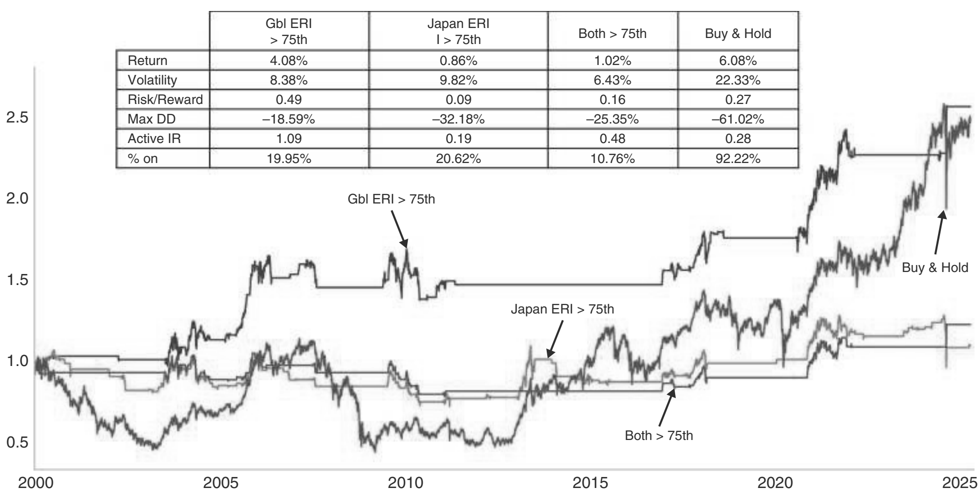
	ISM > 3MMA	ISM < 3MMA	Buy & Hold
Return	6.71%	0.31%	7.02%
Volatility	14.14%	17.30%	22.34%
Risk/Reward	0.47	0.02	0.31
Max DD	–36.69%	–64.42%	–61.02%
Active IR	0.73	0.03	0.33
% on	41.67%	50.65%	92.32%

FIGURE 10.5 (a) Topix moves with the global business cycle. (b) Only go long Topix if ISM is above its 3mma.
Source: Bloomberg and authors' calculations.

Similarly, given that earnings have become more important over time versus multiple expansion, Japanese equities trade much better when the global earnings revision index is positive. Interestingly, the global index works much better than the Japanese one, again highlighting the importance of global conditions for Japan. Starting in 2000, when the global earnings revisions are in the top 75th percentile, the active information ratio is 1.09, compared to the unconditional one of 0.28, as is shown in Figure 10.6.

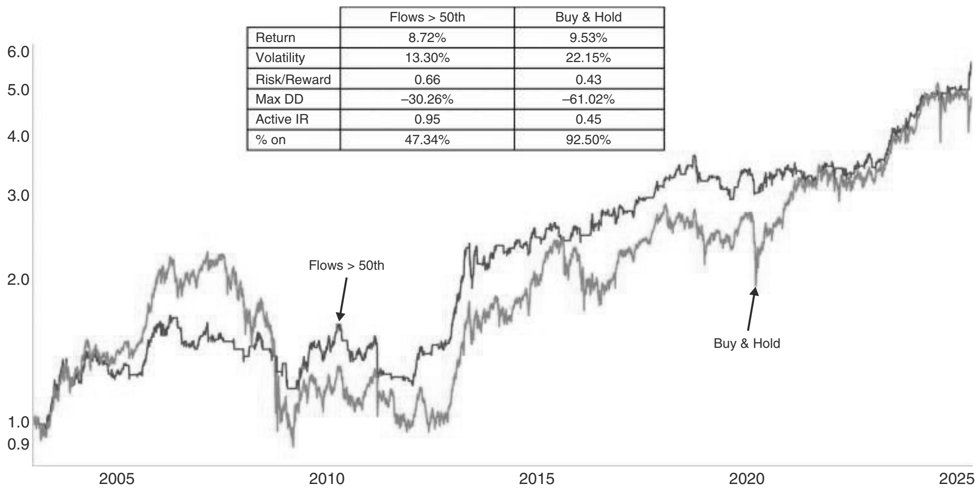
Flows are also helpful, especially foreign flows. Whenever weekly foreign inflows into Japanese equities are in the upper 50th percentile (with a two-year look back), buying the Topix the day after the data was published for the following week raises the active information ratio from 0.45 to 0.95, as per Figure 10.7.

Some of the previous models are not very Japan-specific, and the fact that some global indicators work better than purely Japanese ones suggests that at least the PMI and earnings revision ones should work for most of the cyclical markets. To explain outperformance against the global equity index, investors should mostly rely on the outlook for JPY. Given a relatively exporter-heavy mix in the Japanese indexes, FX-hedged Japanese equities outperform their global peers whenever the JPY weakens. It should be noted that our rule to prefer Japanese equities whenever USDJPY moves higher does not just rely on Japanese exporters doing well, although those are the firms that benefit most directly from a weaker JPY. It is also domestically focused firms, like banks, that do well. The reason is the link of USDJPY to reflation. A weaker JPY increases inflationary expectations, and Japanese banks are seen as a key beneficiary from reflation, both in terms of credit quality and in terms of rising expectations of higher interest rates down the line. We show in Figure 10.8 a study of the Topix when JPY is weak (below is 55dma), and the SPX is strong (above is 55dma), as usual lagged by one day. As can be seen, the active IR for Topix improves from 0.76 to 1.83, more than double. USDJPY is the clearest indicator for Topix outperformance.



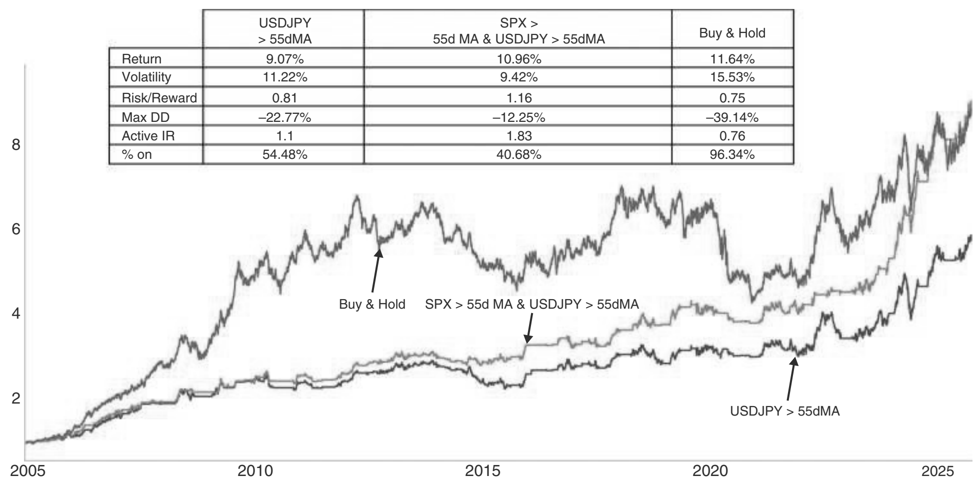
	Gbl ERI > 75th	Japan ERI I > 75th	Both > 75th	Buy & Hold
Return	4.08%	0.86%	1.02%	6.08%
Volatility	8.38%	9.82%	6.43%	22.33%
Risk/Reward	0.49	0.09	0.16	0.27
Max DD	–18.59%	–32.18%	–25.35%	–61.02%
Active IR	1.09	0.19	0.48	0.28
% on	19.95%	20.62%	10.76%	92.22%

FIGURE 10.6 Topix likes high global ERI.
Source: Bloomberg and authors' calculations.



	Flows > 50th	Buy & Hold
Return	8.72%	9.53%
Volatility	13.30%	22.15%
Risk/Reward	0.66	0.43
Max DD	–30.26%	–61.02%
Active IR	0.95	0.45
% on	47.34%	92.50%

FIGURE 10.7 Topix likes inflows.
Source: MoF and authors' calculations.

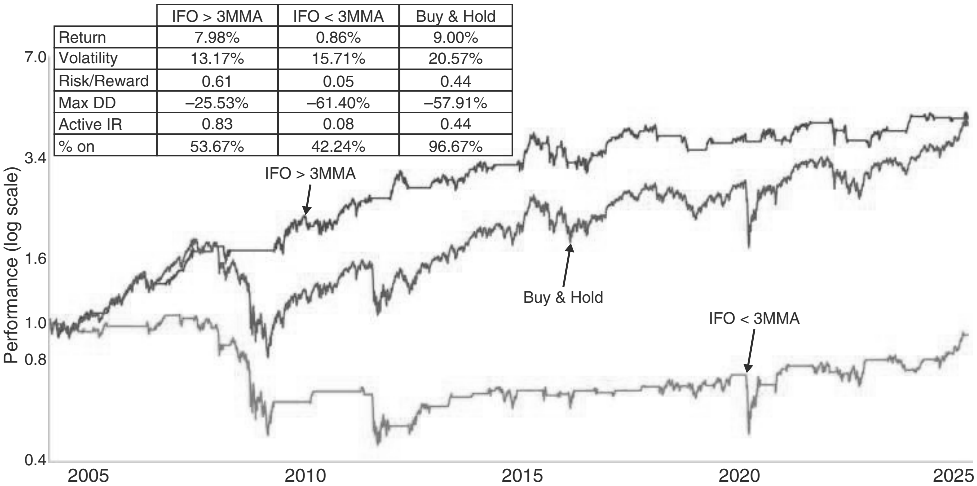


	USDJPY > 55dMA	SPX > 55d MA & USDJPY > 55dMA	Buy & Hold
Return	9.07%	10.96%	11.64%
Volatility	11.22%	9.42%	15.53%
Risk/Reward	0.81	1.16	0.75
Max DD	–22.77%	–12.25%	–39.14%
Active IR	1.1	1.83	0.76
% on	54.48%	40.68%	96.34%

FIGURE 10.8 Topix needs a weak JPY for outperformance.
Source: Bloomberg and authors' calculations.

10.4 TRADING FRANKFURT

In Europe, growth industries are hard to come by, limiting the extent of PE expansion. Cyclical industries tend to be more prevalent than in the United States. As a result, business cycle indicators also work relatively well, just as in the case of Japan. We therefore propose the same rule that we used in Japan, based on German manufacturing PMIs. Figure 10.9 shows that the information ratio of being long the DAX is about twice as high when the IFO survey for Germany is above its 3mma, than when it is below, something we have written about before (see Willer et al., 2025b). This is based



	IFO > 3MMA	IFO < 3MMA	Buy & Hold
Return	7.98%	0.86%	9.00%
Volatility	13.17%	15.71%	20.57%
Risk/Reward	0.61	0.05	0.44
Max DD	–25.53%	–61.40%	–57.91%
Active IR	0.83	0.08	0.44
% on	53.67%	42.24%	96.67%

FIGURE 10.9 European equities do well when IFO is moving higher.

on buying DAX the day after the release. Although on average the DAX also goes up when the IFO is below its 3mma, the returns are small and come with high volatility. It is therefore better to lever up a long DAX when the IFO signal is bullish, rather than to own it when the signal is in bearish mode.

In terms of relative performance versus the United States, we go back, once more, to Citi's economic surprise index. Plotting US economic surprises relative to European surprises does a relatively good job capturing some of the turning points in the relative equity performance, once more underlining that it is more about EPS growth, correlated with overall economic growth, than valuation changes, which can get and stay quite extreme for a long time. Of course, it is difficult to find a strategy to allocate toward European equities and away from the United States over the last 25 years, given the relentless outperformance of the United States. However, as shown in Figure 10.10a and 10.10b, both relative economic surprises (above the 55d MA) and momentum (relative outperformance of DAX over SPX) can pick times when the DAX has performed better than the United States. The information ratio remains quite poor (0.1 for CESI differentials only, and 0.21 if taking the 55dma of relative performance into account), but beats buy and hold, which is slightly negative (in terms of arithmetic mean, not the geometric mean). Post-COVID results are somewhat better. Only using relative momentum actually performs worse than buy and hold, underlining the importance of adding the relative CESI to the mix.

10.5 TRADING LONDON

The FTSE 100 is one of the most defensive large equity markets. It contains a large weight of defensive industries, like healthcare, gives a low weight to cyclicals, and has a very small weight of companies that are overly dependent on the UK business cycle, mostly because many companies are very global. This also means that the United

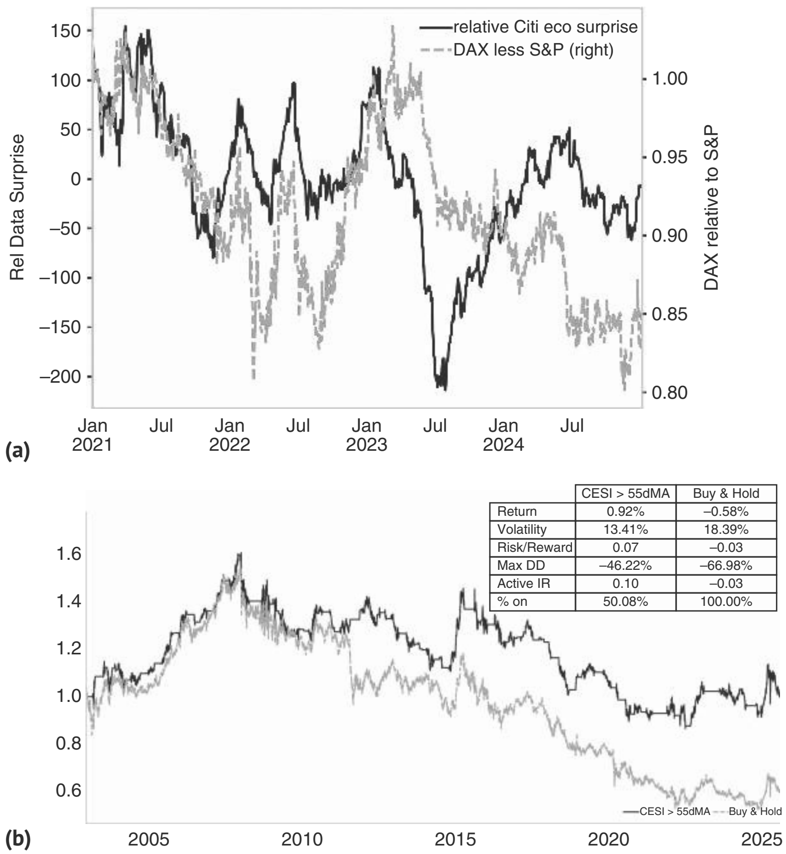
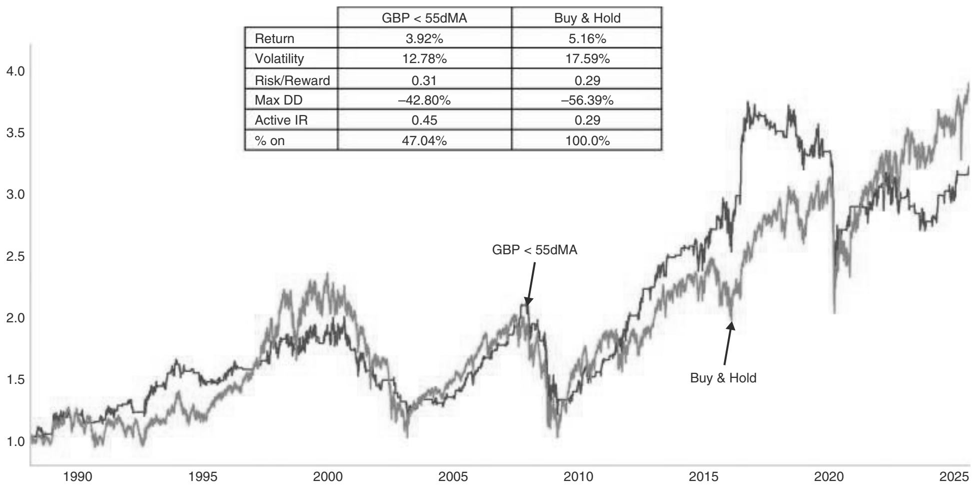


FIGURE 10.10 (a) Relative CESI is helpful for European equity call. (b) Europe has underperformed but when economic and price momentum are in favor DAX can outperform.
Source: Bloomberg, Citi, and authors' calculations.

Kingdom is typically a preferred underweight in a global equity bull market and a preferred overweight in a bear market.

One other important factor for the FTSE 100 is that it trades much better with a weak GBP, at least relative to the MSCI World, as already briefly discussed earlier. This is likely the case as the FTSE is highly export-oriented and does not contain many domestically focused companies. The USD is also a factor for the oil sector, which is important in the FTSE 100. Next, we show how the FTSE 100 trades depending on whether GBP is above or below the 55dma, as usual with a one-day lag. According to Figure 10.11, the information ratio moves from 0.29 for the case of buy and hold, to 0.45.

In spite of not being overly cyclical, there is one cyclical sector in the FTSE 100, which is oil. This sector has a larger weight than is typical in a developed equity market. The beta to oil is therefore unsurprisingly relatively high, once more tying the index much more to global conditions than to the United Kingdom. Interestingly, the beta is larger on the oil downside than the upside, as per Figure 10.12.



	GBP < 55dMA	Buy & Hold
Return	3.92%	5.16%
Volatility	12.78%	17.59%
Risk/Reward	0.31	0.29
Max DD	–42.80%	–56.39%
Active IR	0.45	0.29
% on	47.04%	100.0%

FIGURE 10.11 FTSE likes weak GBP.
Source: Bloomberg and authors' calculations.

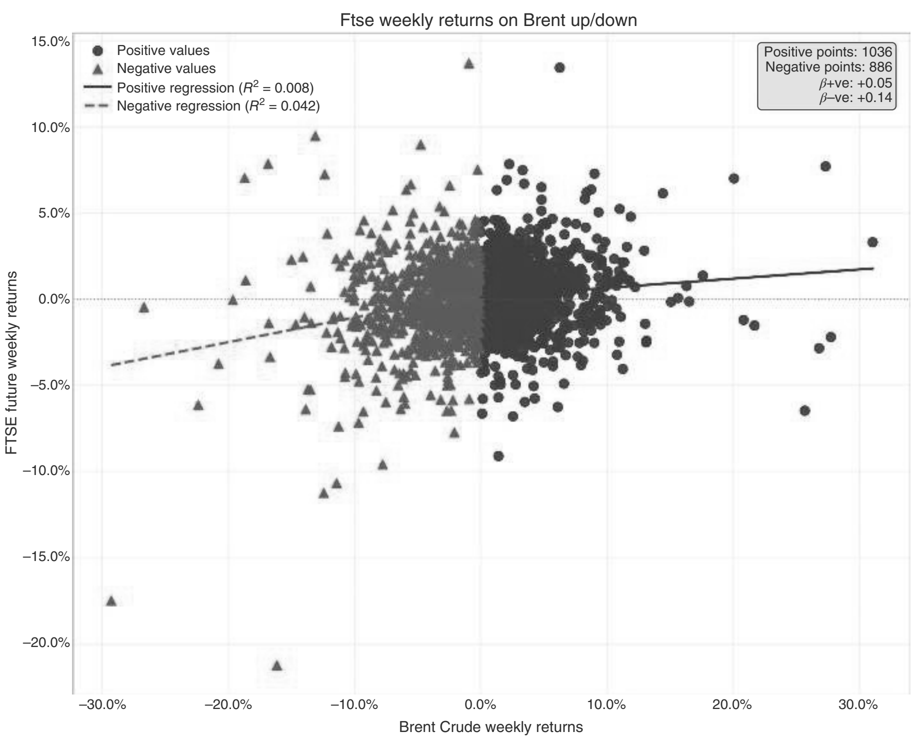


FIGURE 10.12 United Kingdom with a high beta to oil, especially on downside.
Source: Bloomberg and authors' calculations.

10.6 TRADING SHANGHAI

The Chinese market has become increasingly interesting, mostly because it has one of the lower correlations with US equities among the larger equity markets, as can be seen in Table 10.1. In downmarkets for the SPX, Chinese equities are up 24% of the time, ahead of runner-up Brazil (and a few others) at 16%. This makes it, in theory, a good market for large investors to add alpha.

Before studying how to trade Chinese equities, it is important to understand the different ways to gain access to China exposure. The most obvious ways are Chinese A-shares, traded in Shanghai and Shenzen (CSI 300), and Hong Kong-listed H-shares (HSCEI). Alternatively, there are indexes for the Chinese ADRs, which are at times listed in Hong Kong as well, and some investors prefer to take China exposure by trading baskets of US or EU stocks that have the highest profit contribution from China.

A-shares are typically dominated by Chinese retail investors, though foreigners can get access through the Qualified Foreign Institutional Investor (QFII) program or the Stock Connect scheme. Foreign ownership has been increasing over time. A-shares typically have a higher concentration of companies in traditional sectors such as manufacturing, finance, and consumer goods. A-shares have a high ratio of state-owned enterprises (SoEs), which are typically less efficient, have higher debt levels, and may struggle financially, but have less risk from regulatory agencies.

H-shares are more easily accessible to foreign investors and therefore have higher foreign ownership. They also tend to be more liquid and have fewer SoEs. The HSCEI includes a wider set of industries, including more multinational and Chinese tech companies. The HSCEI is therefore biased to growth over value.

ADRs trade in the United States, and some are listed in China too. They often trade at a premium, given the easier access for US investors. However, they face additional legal risks. The US Securities and Exchange Commission (SEC) has investigated Chinese issuers with respect to transparency and compliance with accounting standards. In 2020, the Holding Foreign Companies Accountable Act was legislated, which could result in delistings for companies that do not meet US auditing requirements. Furthermore, Chinese ADRs were typically structured as variable interest entities (VIEs) to overcome a Chinese restriction on foreign ownership in certain sectors, such as technology. VIEs give an offshore entity a claim on the earnings of the relevant companies but do not result in claims on the underlying assets. This creates additional legal risks, especially in the case of bankruptcies.

Given that China is less related to the US/global business cycle than other large markets, the focus needs to be on some China-specific variables to trade the Chinese market. An important indicator for the Chinese economy is the so-called Li Keqiang indicator. This indicator is an alternative measure of China's economic activity, named after former Chinese Premier Li Keqiang. The indicator is based on a 2007 conversation Li had with a US diplomat, where, according to WikiLeaks, he dismissed the official GDP numbers as too unreliable (and "man-made"), and instead suggested focusing on rail freight volume, electricity consumption, and bank loan growth. In addition to potentially increased reliability, the indicator is also more timely, as it is monthly, not quarterly. Interestingly, equities seem to do well after bottoms in this index, and it is a superior strategy to buy Chinese equities after a bottom in the Li-Keqiang index (defined as the 3mma index falling below 3.5%, and turning up from there) rather than buying after the bottom in the China credit impulse, which is the other important candidate for use in bottom fishing. The credit impulse tends to turn earlier than the Li-Keqiang index and, for equities, too early, as per Figure 10.13a.

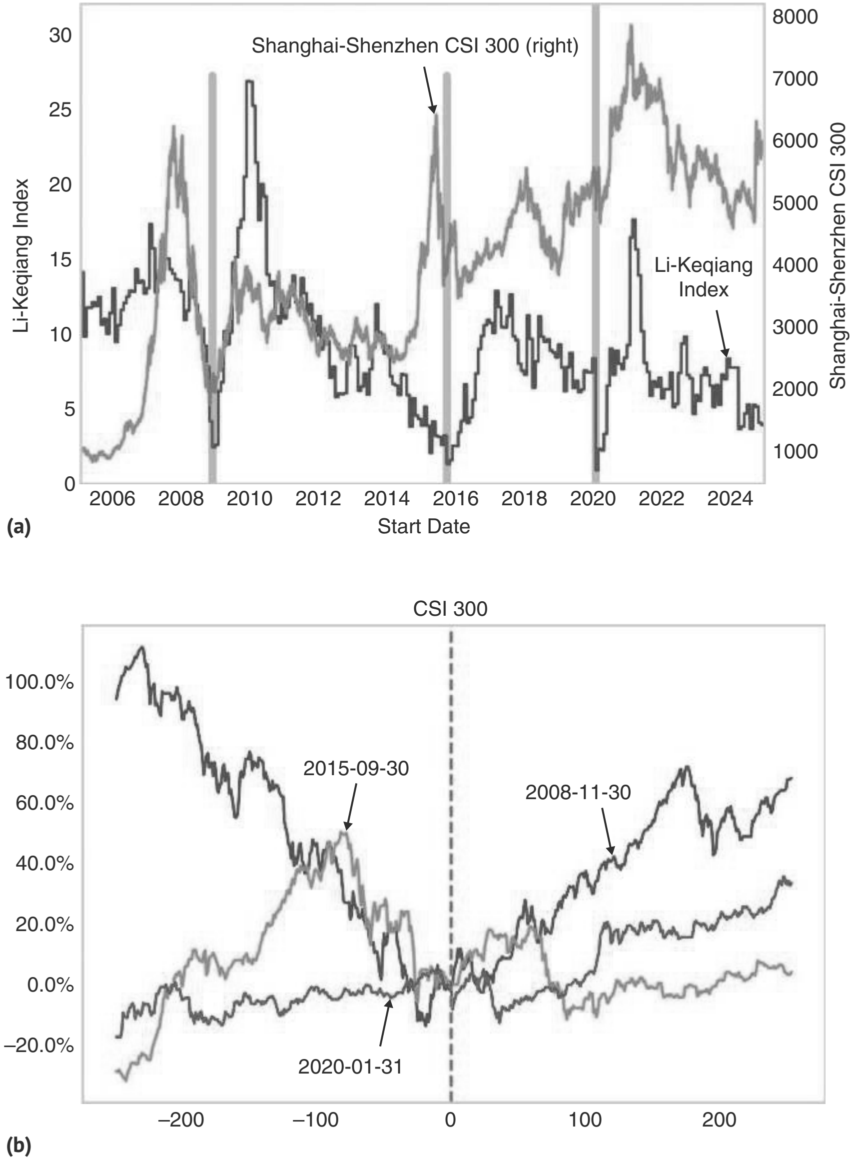
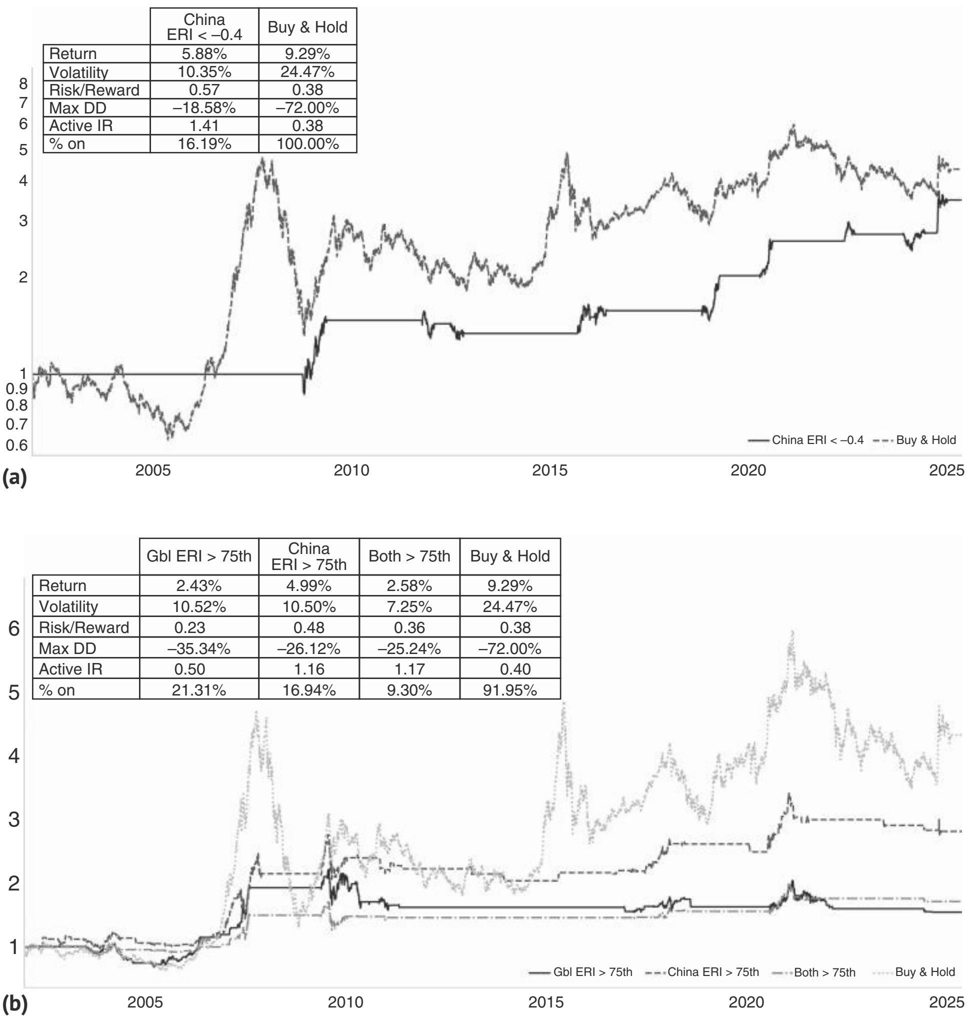


FIGURE 10.13 (a) Li-Keqiang bottoms are equity bullish. (b) Li-Keqiang bottom not too late for equity bottom.
Source: Bloomberg and authors' calculations.

Figure 10.13b shows double-digit returns 50 trading days after all three bottoms. The chart does not take publication lags into account, though, and is merely illustrating that a view on the bottom in the Chinese business cycle can be traded. But an ocular regression suggests that including a publication lag would not be too detrimental to performance.

Another signal, linked to economic strength, that works relatively well in China is to use earnings revisions. The strongest use case is actually as a countersignal. When earnings revisions are sufficiently depressed, it is typically a time to buy. We show this in Figure 10.14a, where we parameterized the rule by looking for earnings revisions



	China ERI < –0.4	Buy & Hold
Return	5.88%	9.29%
Volatility	10.35%	24.47%
Risk/Reward	0.57	0.38
Max DD	–18.58%	–72.00%
Active IR	1.41	0.38
% on	16.19%	100.00%

	Gbl ERI > 75th	China ERI > 75th	Both > 75th	Buy & Hold
Return	2.43%	4.99%	2.58%	9.29%
Volatility	10.52%	10.50%	7.25%	24.47%
Risk/Reward	0.23	0.48	0.36	0.38
Max DD	–35.34%	–26.12%	–25.24%	–72.00%
Active IR	0.50	1.16	1.17	0.40
% on	21.31%	16.94%	9.30%	91.95%

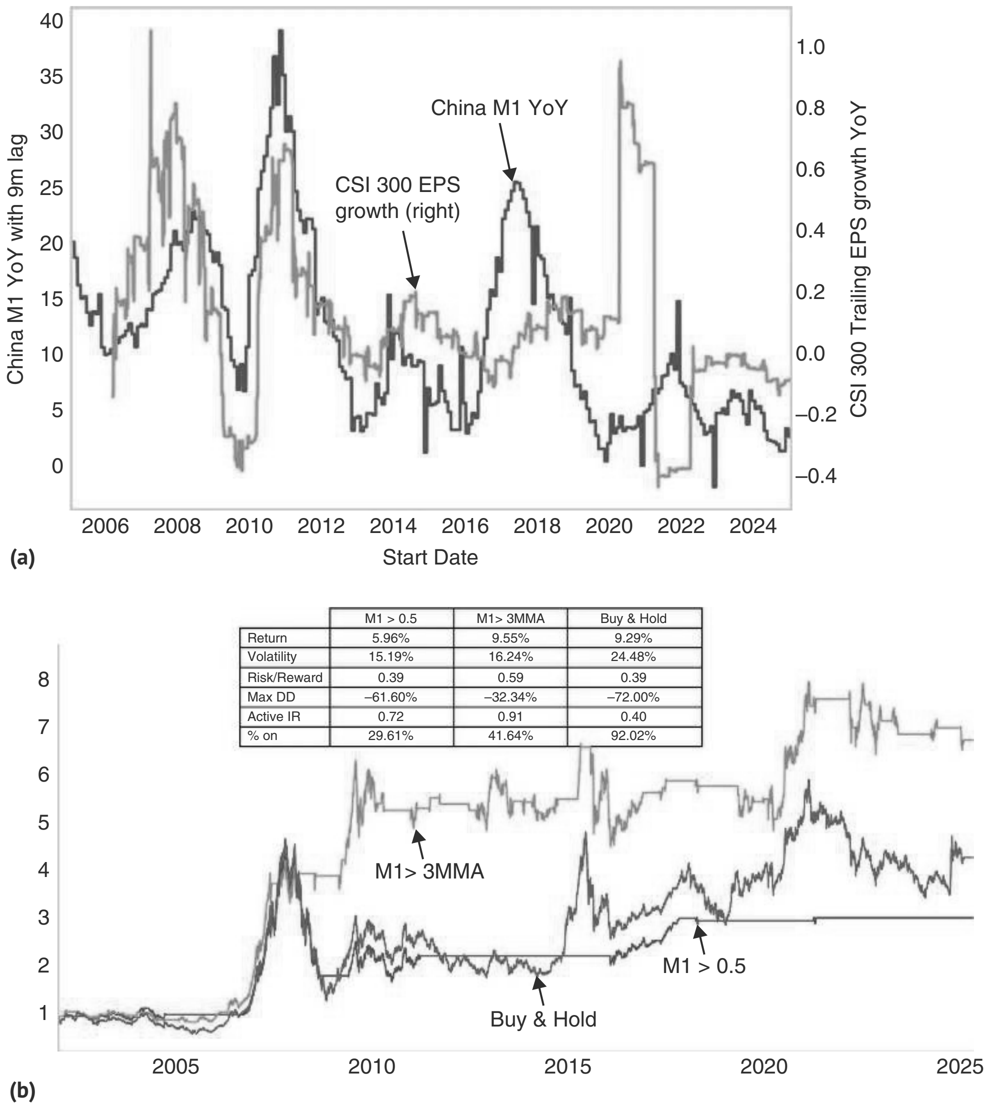
FIGURE 10.14 (a) Extreme weakness in ERI a counter signal. (b) But if not extreme, strength is better.
Source: Bloomberg and authors' calculations.

below the −0.4 level. The information ratio of buying at those levels and holding for three months is 1.4, meaningfully above the unconditional information ratio of 0.4. We use the total return index here rather than our usual excess returns to maximize the data availability. However, when it is not at an extreme, the index is better as a trend indicator. Chinese equities do well when the Chinese ERI is above its 75th percentile, as per Figure 10.14b. This increases the active IR to 1.16 (better than for the global ERI, which only comes in at 0.5). So the lesson from the Chinese ERI is that extreme weakness works, but that otherwise, strength is much better.

Monetary policy matters for Chinese equities, too. Monetary aggregates seem to do a better job than interest rates, which tend to be adjusted more infrequently. Monetary aggregates used to matter much more in the United States as well, but as the financial system developed and became quite complex, M1 in the United States has lost any correlation with asset pricing. Not so in China, maybe because it is still a more closed economy, where the financial system is more bank-based. M1 seems to relatively reliably lead earnings growth, as per Figure 10.15a, and earnings are an important driver of equities, if somewhat less so in the case of China than elsewhere.

We backtest two simple strategies based on the Chinese M1 money supply growth. The first strategy goes long whenever M1 is increasing, i.e., when the year-over-year (yoy) growth is above its three-month moving average. The second goes long when M1 growth is above average (strictly speaking, above its median, using an expanding window). We lag M1 by 40 business days to account for publication lags. The results are shown in Figure 10.15b. Both rules have been effective in trading the CSI 300. The yoy rule is more attractive to us, as it is switched on more often. It improves the information ratio to 0.91 from an unconditional one of 0.4. Recent years of lower money supply growth have kept the above median strategy on the sidelines, and the improvement is smaller, reaching 0.72.

The other interesting feature of the Chinese market is that it is highly momentum-driven, possibly because Chinese retail investors play an important role. This is especially true for market bottoms. There are presumably many momentum rules that work. We backtested days when the HSCEI index outperforms the SPX by 2.5 standard deviations during the post-COVID period. Such days typically have follow-through, as per Figure 10.16. These large moves often come with announcements of stimulative policy measures by the authorities. But attempting to judge whether a particular set of measures is sufficient for a market bottom is not always easy. The good news is that we don't have to make that call. Chinese equity markets seem to be a case of shooting first and asking questions later. This was, for example, true at the bottom of October 2024. At first glance, the measures announced on September 26, 2024, seemed subpar, in that authorities were judged to be adding liquidity into a liquidity trap. However, the market moved higher regardless. It then kept going up, partly as additional measures were announced, increasing the likelihood of a real U-turn in China. But the important point is that even in prior events when initial measures were not followed up by additional stimulus and therefore eventually failed, it paid off to jump on the train on the first signs of momentum having turned.



	M1 > 0.5	M1> 3MMA	Buy & Hold
Return	5.96%	9.55%	9.29%
Volatility	15.19%	16.24%	24.48%
Risk/Reward	0.39	0.59	0.39
Max DD	–61.60%	–32.34%	–72.00%
Active IR	0.72	0.91	0.40
% on	29.61%	41.64%	92.02%

FIGURE 10.15 (a) M1 growth leads China earnings. (b) M1 growth bullish for CSI300.
Source: Bloomberg and authors' calculations.

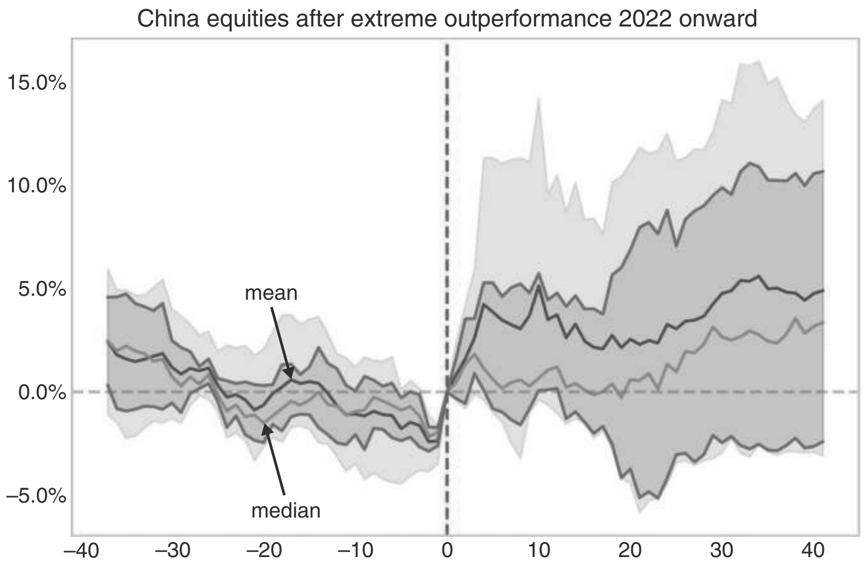


FIGURE 10.16 Significant HSCEI outperformance has follow-through.
Source: Bloomberg and authors' calculations.

The CSRC Indicator

One recent event that triggered our 2.5 standard deviation outperformance rule occurred on February 6, 2024, on the back of official intervention, as President Xi met with financial regulators and China's sovereign wealth fund announced it would increase its stock purchases. This was widely seen as the government meaning business when it comes to supporting equities. The most compelling signal came a day later, though, on February 7 when Yi Huiman, the chairman and party chief of the China Securities Regulatory Commission, was replaced. Equities rallied similarly strongly to the two times this had occurred in the recent past, as per Figure 10.17. When equities have fallen by enough to lead to the removal of a powerful bureaucrat, it means that we are due for at least a bounce. Of course, the sample size of this "CSRC indicator" is tiny, but the underlying message of using extreme official concerns with weak equity markets as a buy signal is likely valid.

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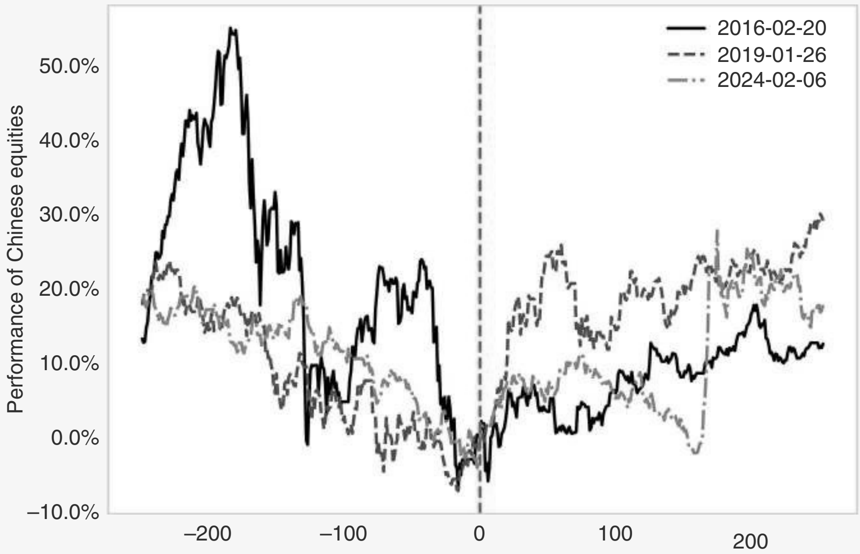
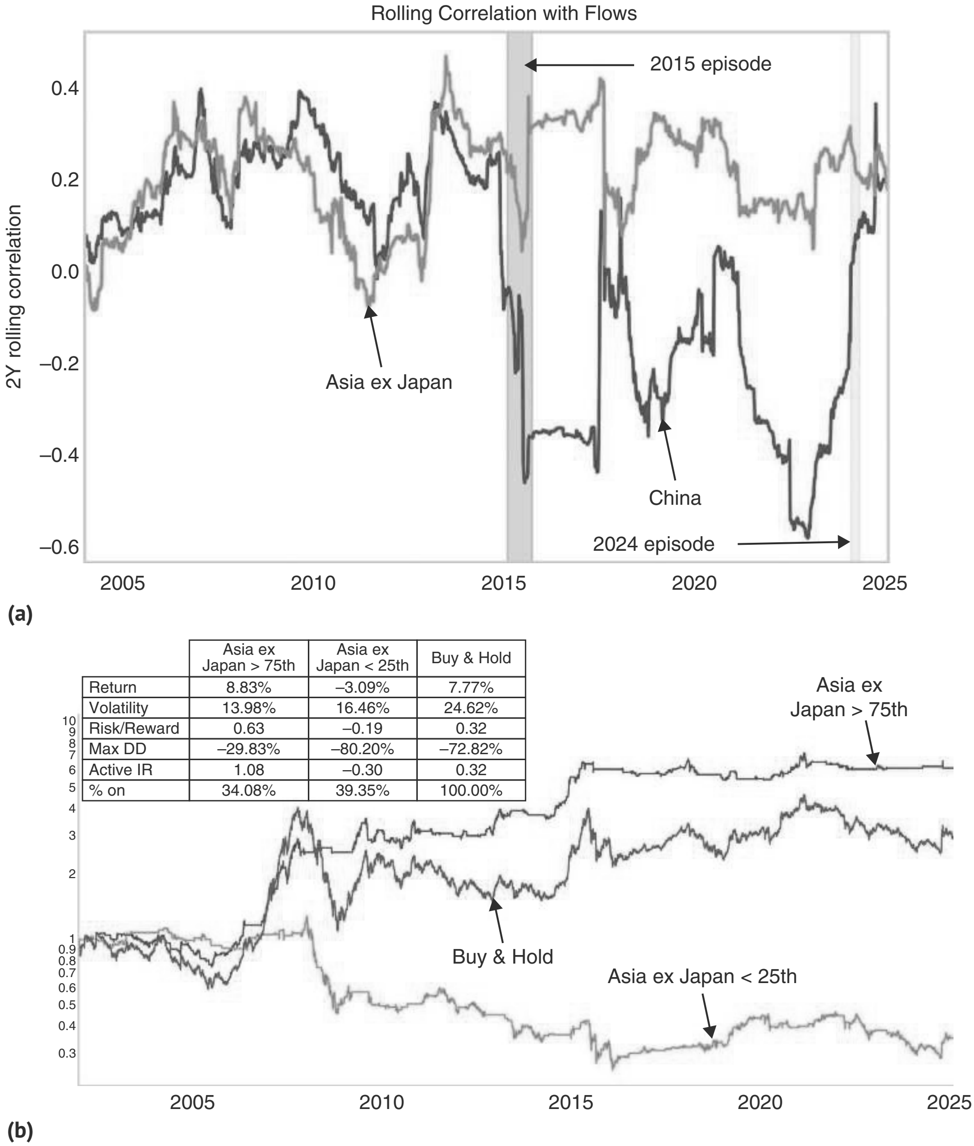


FIGURE 10.17 CSRC indicator works.
Source: Bloomberg and authors' calculations.

Flows also matter in the Chinese market, probably because it is such a momentum market. Interestingly, not all flows are created equal. In particular, it pays to distinguish between pure China flows and flows into Asia x-Japan funds. One reason is that pure China flows may just be driven by the national team, and therefore, more due to government orders rather than due to a sense of improving fundamentals. The problem with the national team is that the orders of the government can change overnight, without any fundamental change. It is riskier to bet on the continuation of flows from the national team than to bet on the continuation of more fundamentally driven flows. As a result, our backtests work better with Asia x Japan flows rather than with the China flows themselves, as illustrated by Figure 10.18a. As can be seen in Figure 10.18b, when those flows are in the 75th percentile, the active information ratio improves from the unconditional 0.32 to 1.08. When flows are below the 25th percentile, the active information ratio drops to –0.30.

Lastly, seasonals are quite important in China. Q4 has traditionally been very strong, while Q2 and Q3 often are quite weak, as can be seen in Figure 10.19. By itself, this seasonal pattern is not consistent enough to be an independent trading strategy. But on the margin, it can be helpful to take this into account when trading the other rules we discussed earlier.



	Asia ex Japan > 75th	Asia ex Japan < 25th	Buy & Hold
Return	8.83%	–3.09%	7.77%
Volatility	13.98%	16.46%	24.62%
Risk/Reward	0.63	–0.19	0.32
Max DD	–29.83%	–80.20%	–72.82%
Active IR	1.08	–0.30	0.32
% on	34.08%	39.35%	100.00%

FIGURE 10.18 (a) Flows into Asia x China as a better indicator for China equities. (b) Flows focus avoids major bear markets.
Source: Bloomberg, Citi, and authors' calculations.

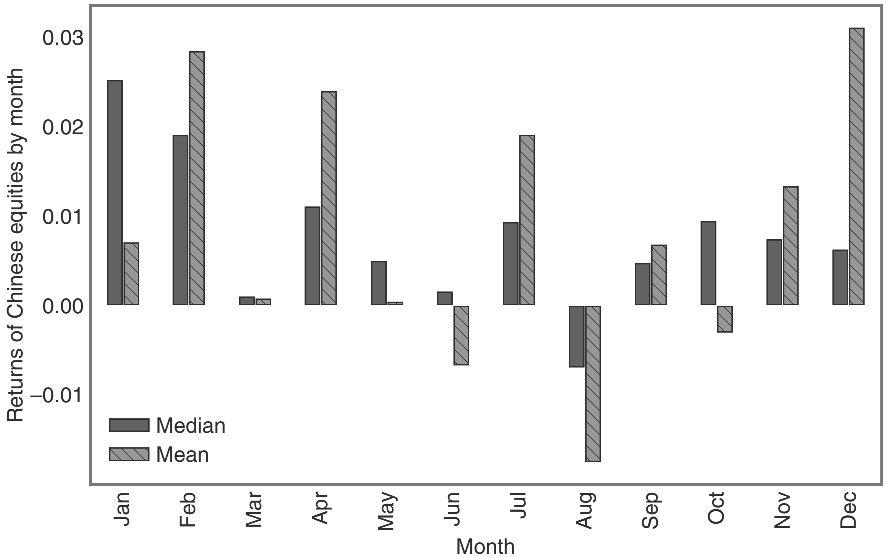


FIGURE 10.19 Q4 and Q1 tend to be stronger.
Source: Bloomberg and authors' calculations.

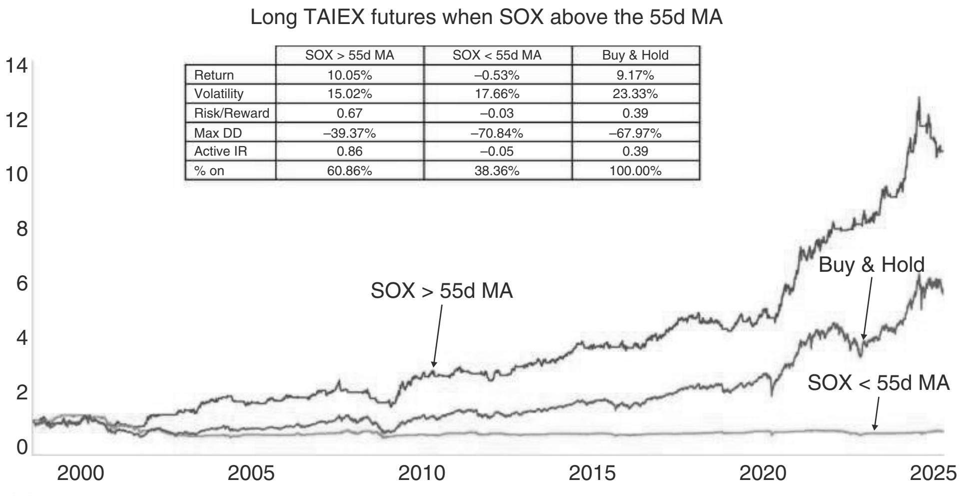
10.7 TRADING THE REST OF EM

Starting with emerging Asia, and having treated China separately, the index is dominated by South Korea, Taiwan, and India. The former two markets have a relatively high semiconductor weight, with TSMC dominating Taiwan with an index weight at the time of writing of more than 35%. Korea is less heavily biased toward semis, but between the Samsung and the Hynix chaebols, the weightings are also very substantial.

To translate the importance of the semi-cycle into a trading rule, we trade Taiwan and Korea in line with whether the SOX is above or below its 55dma, with a two-day lag, given time zone issues. We find that the information ratio for Korea improves from 0.24 unconditional to 0.41, and for the TWSE from 0.39 to 0.86, as illustrated in Figure 10.20. Clearly, it works better for the TWSE given the higher semiconductor dominance.

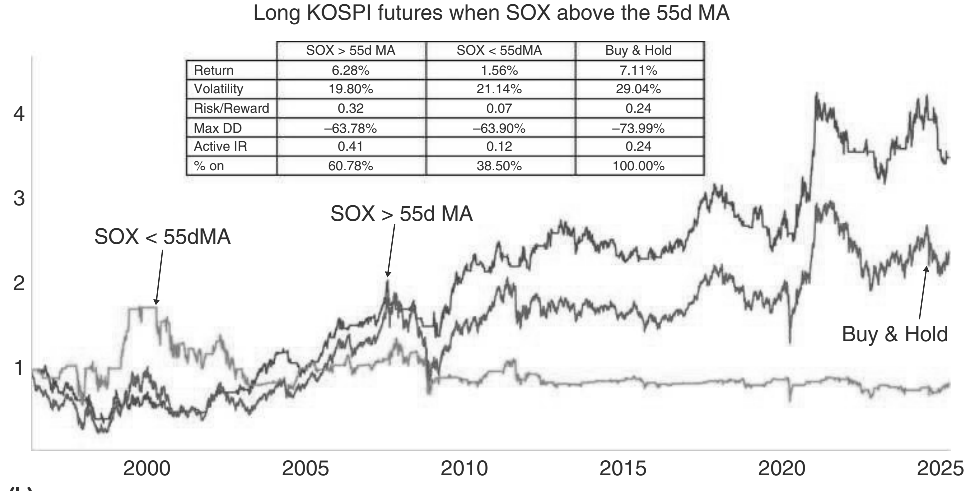
The SOX is correlated with the manufacturing cycle. It is therefore no surprise that emerging Asia is very sensitive to the manufacturing ISM. The bottom in Emerging Asia Equities typically only comes when the index is close to bottoming. In Figure 10.21, we show the returns from ISM bottoms, defined as the 12-month minimum, below 48 and higher next month. We exclude any dates that are within three months of the first bottom. We find that Kospi bottoms around two months before the ISM bottom, and then keeps going higher. This is just for illustrative purposes, given that our rule uses forward-looking information on defining the ISM bottom.

Moving on to Latam, the USD becomes a big focus. As discussed previously, the correlation with the USD is not the same across the various regional markets. Emerging markets generally like a strong local currency. The FX hedged outperformance of



	SOX > 55d MA	SOX < 55d MA	Buy & Hold
Return	10.05%	–0.53%	9.17%
Volatility	15.02%	17.66%	23.33%
Risk/Reward	0.67	–0.03	0.39
Max DD	–39.37%	–70.84%	–67.97%
Active IR	0.86	–0.05	0.39
% on	60.86%	38.36%	100.00%

(a)



	SOX > 55d MA	SOX < 55dMA	Buy & Hold
Return	6.28%	1.56%	7.11%
Volatility	19.80%	21.14%	29.04%
Risk/Reward	0.32	0.07	0.24
Max DD	–63.78%	–63.90%	–73.99%
Active IR	0.41	0.12	0.24
% on	60.78%	38.50%	100.00%

(b)

FIGURE 10.20 (a) Taiwan is mostly a SOX play. (b) as is Korea.
Source: Bloomberg and authors' calculations.

the local equity markets is positively correlated with the local FX. The reason may be that inflows into the equity market are typically mostly unhedged and can be a larger percentage of total capital flows than in developed markets. In that case, the inflows themselves are what lead to the positive correlation of FX with equity markets. Furthermore, other inflows are likely positively correlated with the equity market, as improving fortunes for a country, as reflected in the equity market, lead to broader bullishness. This seems to be especially the case in Latam, where credit quality is generally lower than in emerging Asia ever since the end of the Asian crisis.

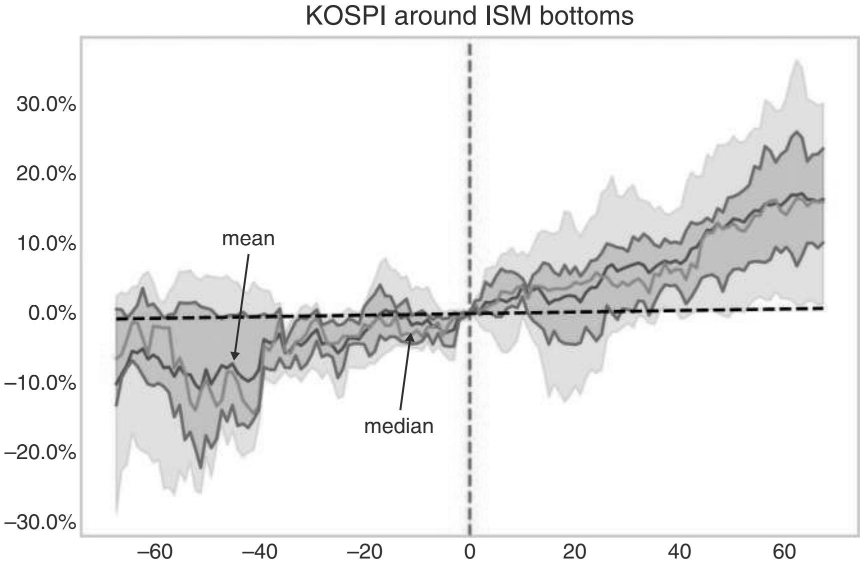
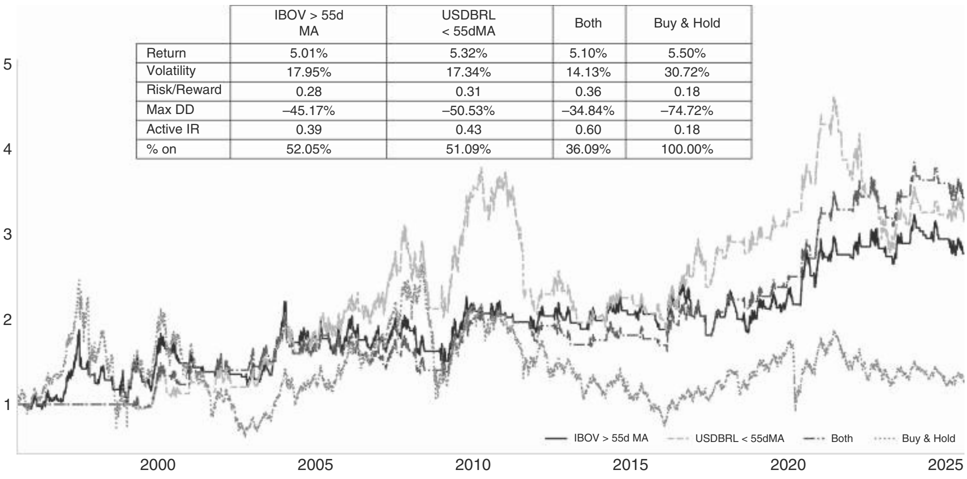


FIGURE 10.21 Kospi likes ISM bottoms.
Source: Bloomberg and authors' calculations.



	IBOV > 55d MA	USDBRL < 55dMA	Both	Buy & Hold
Return	5.01%	5.32%	5.10%	5.50%
Volatility	17.95%	17.34%	14.13%	30.72%
Risk/Reward	0.28	0.31	0.36	0.18
Max DD	–45.17%	–50.53%	–34.84%	–74.72%
Active IR	0.39	0.43	0.60	0.18
% on	52.05%	51.09%	36.09%	100.00%

FIGURE 10.22 Latam does well in strong FX environment.
Source: Bloomberg and authors' calculations.

The study in Figure 10.22 illustrates this pattern for Brazil. Just being long the Bovespa when the USDBRL is below its 55dma with a one-day lag raises the information ratio from an unconditional 0.18 to 0.43. Adding a condition that the Bovespa should also be above its 55dma moves the combined information ratio to 0.60.

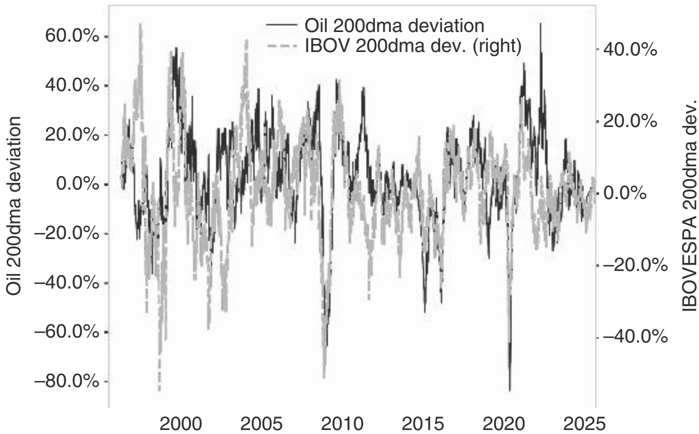


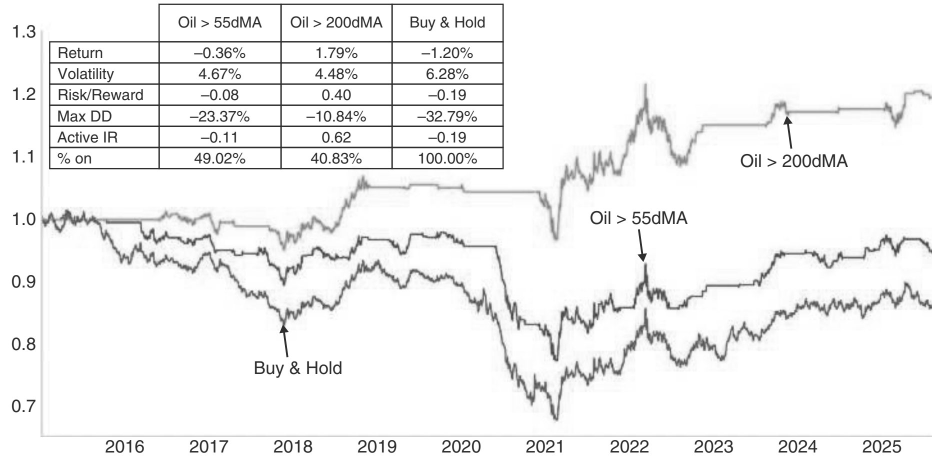
FIGURE 10.23 Brazil does well with strong oil.
Source: Bloomberg and authors' calculations.

Outside of FX, the other key to trading Latam equity markets is commodities. With the exception of Mexico, Latam is a large exporter of commodities. The positive correlation between commodities and Latam equities is illustrated in Figure 10.23. Here we use oil and Bovespa, but other indexes also work. Both equities and oil have been detrended by taking the deviation from their 200dma. As can be seen, the correlation is quite tight, though it is not clear which of the two is leading. The coincident nature makes a strong trading rule harder to come by, and investors therefore have to rely more on their commodity view than just observing oil price action.

Lastly, we briefly focus on CEEMEA. Historically, the bulk of CEEMEA equities has been oil-driven, initially given the large weight of Russia, and more recently, the large weight of Saudi Arabia. South Africa also has a commodity angle, which tends to be positively correlated with oil, even though it is not an oil exporter. This effect more than overpowers markets like Turkey, which is one of the largest oil importers. To make this relationship tradable, we show in Figure 10.24 the difference in the information ratio of trading CEEMEA equities with WTI above its 55dma and 200dma, and below, respectively (with one day lag for implementation). We take positions versus the MSCI CEEMEA index (currency hedged where we have daily data starting in 2015, restricting the backtest to the last 10 years). We find that when oil is trading above its 200dma, the active information ratio is 0.54, meaningfully better than the unconditional of −0.19. The improvement from using a shorter MA, like the 55dma, is less impressive.

10.8 TACTICAL CONSIDERATIONS

On a more tactical level, we note that many of the studies we illustrated for US equities work well globally. In Figure 10.25 we show this for the VRP model that we had previously introduced for the United States in Figure 9.5. Using the same methodology, we find that Sharpe ratios are generally high, especially when using exponentially weighted volatility, and are higher than buy-and-hold for every market. Outside of the



	Oil > 55dMA	Oil > 200dMA	Buy & Hold
Return	–0.36%	1.79%	–1.20%
Volatility	4.67%	4.48%	6.28%
Risk/Reward	–0.08	0.40	–0.19
Max DD	–23.37%	–10.84%	–32.79%
Active IR	–0.11	0.62	–0.19
% on	49.02%	40.83%	100.00%

FIGURE 10.24 Ceemea equities likes oil higher.
Source: Bloomberg and authors' calculations.

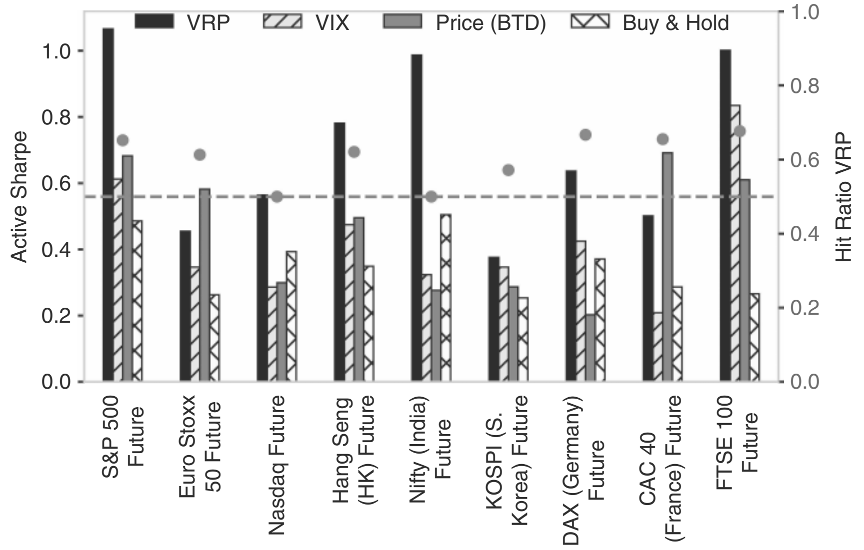


FIGURE 10.25 VRP also works outside of the United States.
Source: Bloomberg and authors' calculations.

United States it works best in India and the United Kingdom. In Eurostoxx and France, the simple price measure works better than VRP to buy pull-backs, though. Hit ratios are overall quite robust, with only India and Nasdaq close to 50%. However, we note that the trigger dates are highly correlated between countries. Often, it is a global crisis that makes the trades work (or not work) across regions at the same time. The diversification benefits from trading VRP across countries are, therefore, somewhat limited.

10.9 SUMMARY

Global equity markets are highly correlated with US markets, with China the least so. One top-down consideration to pick markets is to bet on cyclicality. The most cyclical markets are Japan and Hong Kong. In terms of rate sensitivity, Japan stands out as the market that outperforms when interest rates are rising. In terms of FX, Japan and the United Kingdom (FX hedged) do the best when FX is weak, while Brazil does well with a strong FX. Digging deeper into Japan, we find that buying the Topix only when the global PMI is above its 3mma improves the information ratio. Japan also does well when the global ERIs are strong. Foreign flows have momentum and can add to bullishness. Lastly, a weak JPY is also bullish. European equities are also cyclical and have a much better information ratio with the IFO above its 3mma. Chinese equities also benefit from a high ERI, but also bottom with a stressed ERI. The Chinese market is a major momentum market. Jumps of more than 2.5 standard deviations a day relative to the SPX, especially from oversold levels, have follow-through. Emerging Asia can be traded with the SOX, Latam markets with FX, and CEEMEA with oil.

CHAPTER 11

How to Trade Credit

Credit fits into the risky asset bucket together with equities. But it arguably may have a higher macro appeal than equities, as many of the business cycles are also credit cycles. The most famous business cycle that was clearly driven by credit was the GFC, but the European crisis was also largely a credit crisis. The nature of credit allows for placing very asymmetric bets, as the upside is typically limited, while the downside can be extremely significant relative to the upside. The asymmetric payoff in credit is what allowed for the famous bets against subprime that created a new class of macro hedge fund heroes.

11.1 CREDIT IN THE BIG PICTURE

When analyzing credit throughout the business cycle, the first question is whether to focus on total returns (i.e., the duration plus the credit component) or on spread returns (hedging the US Treasury risk). The drivers of credit are fundamentally different from those of duration. The duration component and the credit spread component are often negatively correlated. Some investors, especially asset allocators, therefore, think of corporate bond returns as a duration bet with some added yield in the shape of the credit spread. However, fixed-income managers can add value by either adding risk on the duration or the credit side. It is therefore instructive to analyze the return streams separately. Credit returns are related to the business cycle. When economic growth is strong, corporates are less likely to default, and earnings and balance sheets are improving. In bad times, government bonds typically do well as central banks are apt to cut rates, as discussed in Chapter 2. This pattern shows up in return correlations. We blend data from Asvanunt and Richardson (2016) with Bloomberg indices to generate a longer history. Figure 11.1 shows the rolling five-year correlation between monthly returns for government bonds, credit excess returns over the equivalent duration matched Treasury index, as well as equities. The excess return component of credit indices is nearly always negatively correlated with government bond returns and is strongly positively correlated with equities. Investors, therefore, should choose carefully which risk to take, rather than taking the joint duration and credit risk.

Given that we have already discussed duration at length, we focus on spread returns for the remainder of this chapter. Figure 11.2 shows that since 1971, excess returns for corporate bonds were much lower than both equity returns and US government bond returns. But, with volatility much lower too, the disadvantage

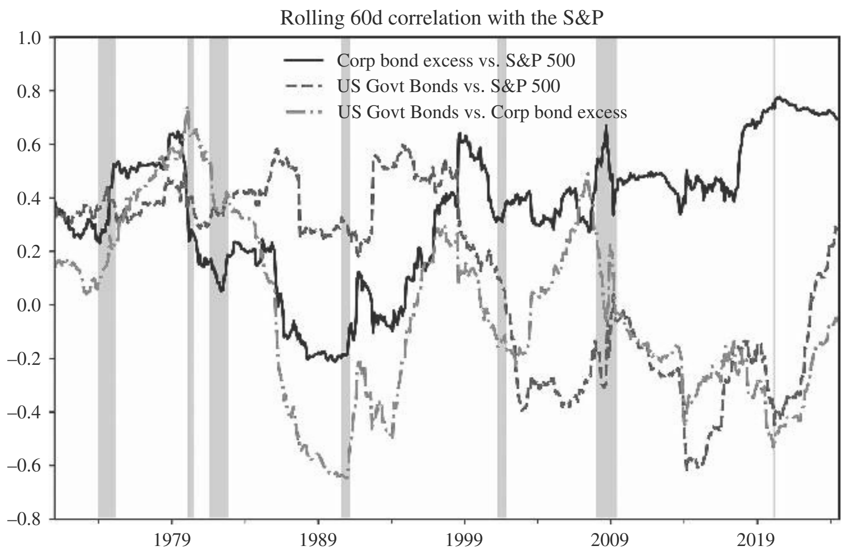
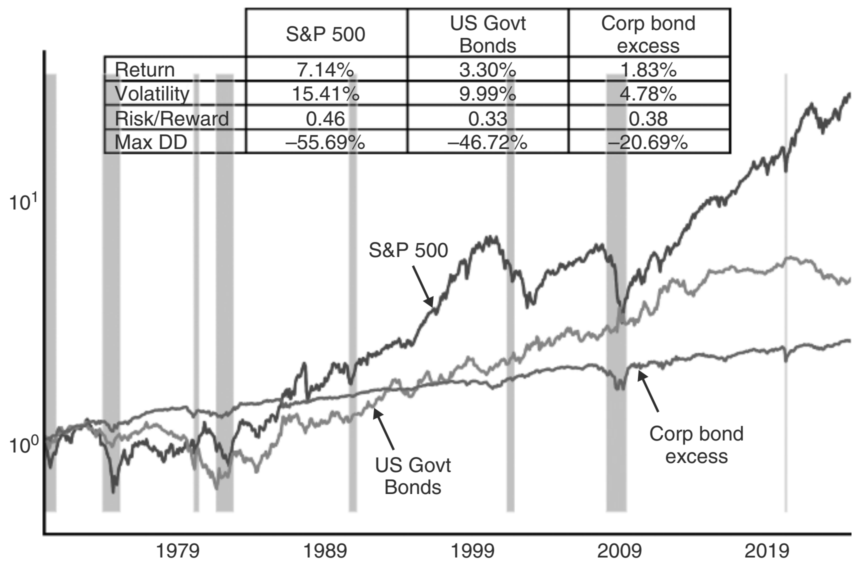


FIGURE 11.1 Credit excess returns negatively correlated with government bond returns.
Source: Bloomberg and authors' calculations.



	S&P 500	US Govt Bonds	Corp bond excess
Return	7.14%	3.30%	1.83%
Volatility	15.41%	9.99%	4.78%
Risk/Reward	0.46	0.33	0.38
Max DD	–55.69%	–46.72%	–20.69%

FIGURE 11.2 Information ratio for credit excess returns close, but lower than for equities.
Source: Bloomberg and authors' calculations.

against the other asset classes shrinks when focusing on information ratios. Here, the SPX comes in at 0.46 over that time period (as usual, measured in excess of 3m T-Bills), while excess IG returns come in at 0.38. Still not as strong as equities, but closer. The correlation with equities is high. Since 1999, when the duration hedged Bloomberg

US IG index starts, there were seven negative years for US IG spread returns. Five of those seven years also saw negative returns for the SPX, while two (2007 and 2014) were, with hindsight, warning signs that the SPX would not do overly well the following year. Later, we will investigate whether credit works more generally as a leading indicator for equities. On the other hand, there were also years when IG credit did well, even with equities doing poorly. This was the case in 2000 and 2001, when the bursting dot.com bubble only resulted in a shallow recession, which US IG was able to manage through, partly because there was little debt directly related to the main Nasdaq bubble names included in the US IG index. Credit was also able to print positive returns in the face of an equity downturn in 2022, although there was a reasonably large drawdown in US IG spread returns during that year.

The next question is in which macro environment credit can outperform equities. Figure 11.3a shows returns under different macro conditions. Corporate bond excess returns are typically best late in a recession when spreads are wide, but the recession

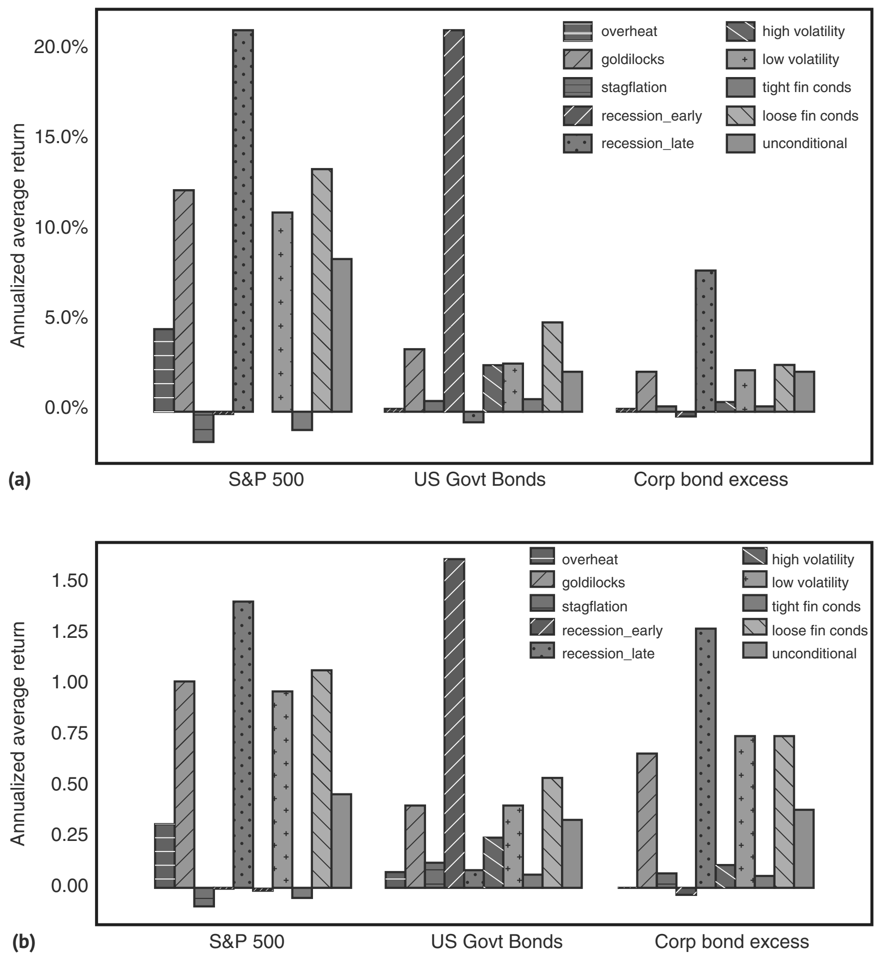


FIGURE 11.3 (a) Equity returns higher than IG excess returns in almost all regimes. (b) Equities also better in Sharpe ratio terms, outside of stagflation.
Source: Bloomberg and authors' calculations.

is already somewhat mature. Goldilocks returns are also decent, while early recession returns are the worst (though not that negative on average). Relative performance across regimes is similar to that of equities. However, returns are worse than for equities in almost all regimes other than stagflation, where IG credit performs slightly better. However, for a fair comparison, we need to move to information ratios, given the lower volatility of IG credit. The results are shown in Figure 11.3b. On an information ratio basis, credit is more comparable to equities but remains broadly inferior, with stagflation again the main exception. This limits the case for credit from a regime investing point of view.

11.2 DEFAULTS IN MOTION

The reason for the high positive correlation of excess credit returns with equities is that both markets react negatively to a downturn in the business cycle. In the case of equity markets, this is the case because earnings are expected to fall, while in the case of credit, it is the case because defaults and/or downgrades, a proxy for the likelihood of defaults, are likely to rise. The normal pattern is that firms overextend themselves during good times, borrowing aggressively to be able to invest in additional capacity to meet strong demand. When the business cycle turns, demand falls, and debt service becomes much more difficult. For credit, it is therefore key to figure out the default cycle.

Twelve month trailing defaults for IG tend to peak at a very low 0.4% to 0.6%, though largely because most IG gets downgraded before the default happens. More importantly, for high yield, the peak occurs at the same time as for IG, but default rates move to 10% to 15%, from their cross-cycle median of closer to 3.5% (and the low of 0.85% in December 2021, in the aftermath of the COVID stimulus). Defaults are obviously a lagging indicator, given that we already established that credit starts trading better late in the recession, whereas defaults only peak after the end of the recession (in the last three recessions we saw HY default peak in January 2002 versus recession end in November 2001, HY default peak in November 2009 versus recession end in June of 2009, and HY default peak in September 2020 versus recession end in April 2020). The Senior Loan Officer Opinion Survey (SLOOS) is slightly earlier and is typically a good predictor of defaults. When banks tighten lending standards, it tends to go hand in hand with decreased lending. This increases the rollover risk and, therefore, eventually, the default risk. However, SLOOS also lags market prices, as can be seen in Figure 11.4. Although lagging indicators should only be used with much trepidation, at times they can still be useful for understanding where we are in the cycle. For example, in the post-COVID cycle, banks started to tighten credit standards in late October 2021. The tightening continued until July 2023. Subsequently, the extent of tightening started to reverse, as became clear with the print for SLOOS for October 2023 (published in early November). Although spreads had already peaked out in October 2022 (retesting their peak during the banking crisis in March 2023), there was still a significant part of the rally in credit left, if investors concluded correctly that the credit tightening had peaked in November. Although spreads had never widened very aggressively to begin with, the argument was at the time that if credit did not react negatively to a tightening of lending standards, how could spreads possibly widen when the lending environment improves? And indeed, US IG spreads tightened by 40bp subsequently. When the first SLOOS is published, that suggests improving conditions after a spike higher, the credit market usually keeps rallying. An exception was 2002. In that episode, the equity market kept falling, even when economic conditions had already improved. This happened mostly on corporate governance (Worldcom, Enron) and valuation issues and dragged credit lower.

Given how lagging SLOOS is, it is surprisingly effective in a simple trading rule. Being long the credit market only if the previous SLOOS suggests a loosening of credit standards improves the information ratio from 0.46 to 1.10, as shown in Figure 11.5.

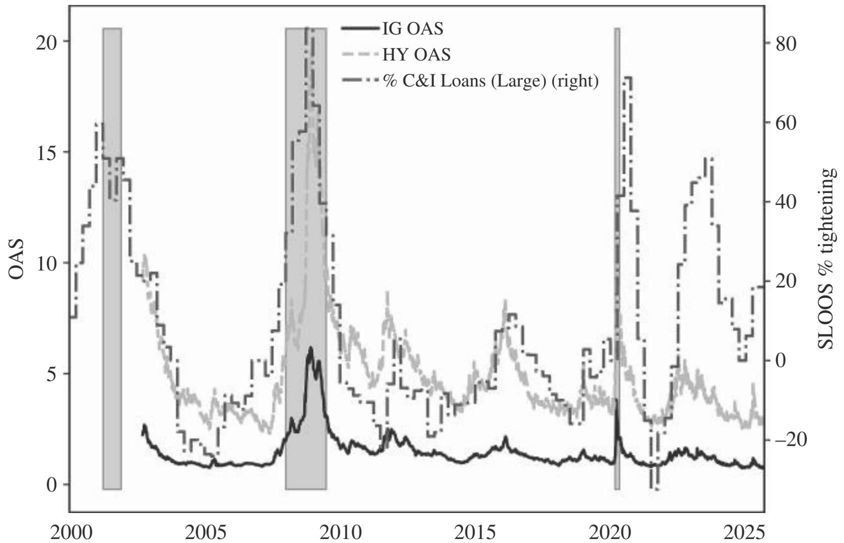
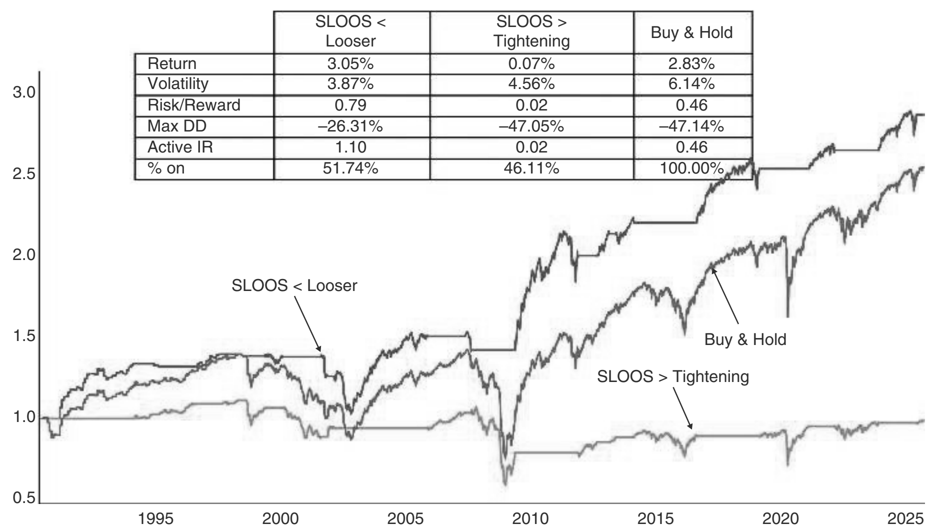
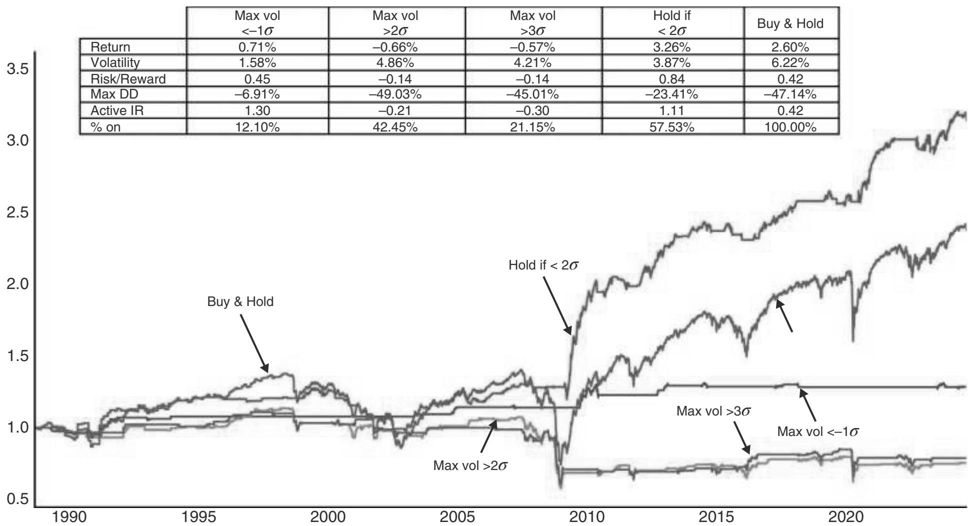


FIGURE 11.4 SLOOS is leading defaults, but lagging the market.
Source: Bloomberg and authors' calculations.



	SLOOS < Looser	SLOOS > Tightening	Buy & Hold
Return	3.05%	0.07%	2.83%
Volatility	3.87%	4.56%	6.14%
Risk/Reward	0.79	0.02	0.46
Max DD	–26.31%	–47.05%	–47.14%
Active IR	1.10	0.02	0.46
% on	51.74%	46.11%	100.00%

FIGURE 11.5 In spite of the lags, some money is typically left when SLOOS improves.
Source: Bloomberg and authors' calculations.



	Max vol <–1σ	Max vol >2σ	Max vol >3σ	Hold if < 2σ	Buy & Hold
Return	0.71%	–0.66%	–0.57%	3.26%	2.60%
Volatility	1.58%	4.86%	4.21%	3.87%	6.22%
Risk/Reward	0.45	–0.14	–0.14	0.84	0.42
Max DD	–6.91%	–49.03%	–45.01%	–23.41%	–47.14%
Active IR	1.30	–0.21	–0.30	1.11	0.42
% on	12.10%	42.45%	21.15%	57.53%	100.00%

FIGURE 11.6 Get out when trouble brews.
Source: Bloomberg and authors' calculations.

The rule works mostly by cutting out the very negative events, like part of the GFC as well as COVID.

But are there any more forward-looking indicators to warn us to reduce credit risk in the portfolio? One rule that has been useful is to cut credit risk when volatility starts to increase. We prefer to use a broad volatility indicator that we have already discussed in the EM book. We take the maximum value of the implied volatilities for equities (VIX), bonds (MOVE), oil (OVX), and FX (both G10 and EM), and high-yield credit spreads. We then calculate the one-year z-scores for each measure and take the maximum with the basic idea that we can never be certain in which corner of the market volatility strikes. We cut out of long credit exposure when the largest z-score of any volatility is 2 standard deviations above normal and buy it back when it falls below 2 again. Given the low liquidity of the credit asset class, we hold the position for at least a month and exit on the next day's close. As per Figure 11.6, holding credit risk only with that measure below 2 standard deviations increases the information ratio from 0.42 in the unconditional case to 1.11. Information ratios are negative with our indicator above 2, and even more so above 3. We also try other thresholds and find that the information ratio below –1 is even better, but the total returns are much worse, as it is switched on too rarely.

11.3 NO CREDIT TO THE FED

With respect to the Fed, the returns surrounding first Fed cuts should, in theory, be similar to equities. We have less data for credit, though, which starts only in 1989, initially monthly, before the series becomes daily from 1998 onward. The main takeaway from the more limited data for credit is that credit does not rally into the first cut and, if anything, underperforms the unconditional case, as can be seen in

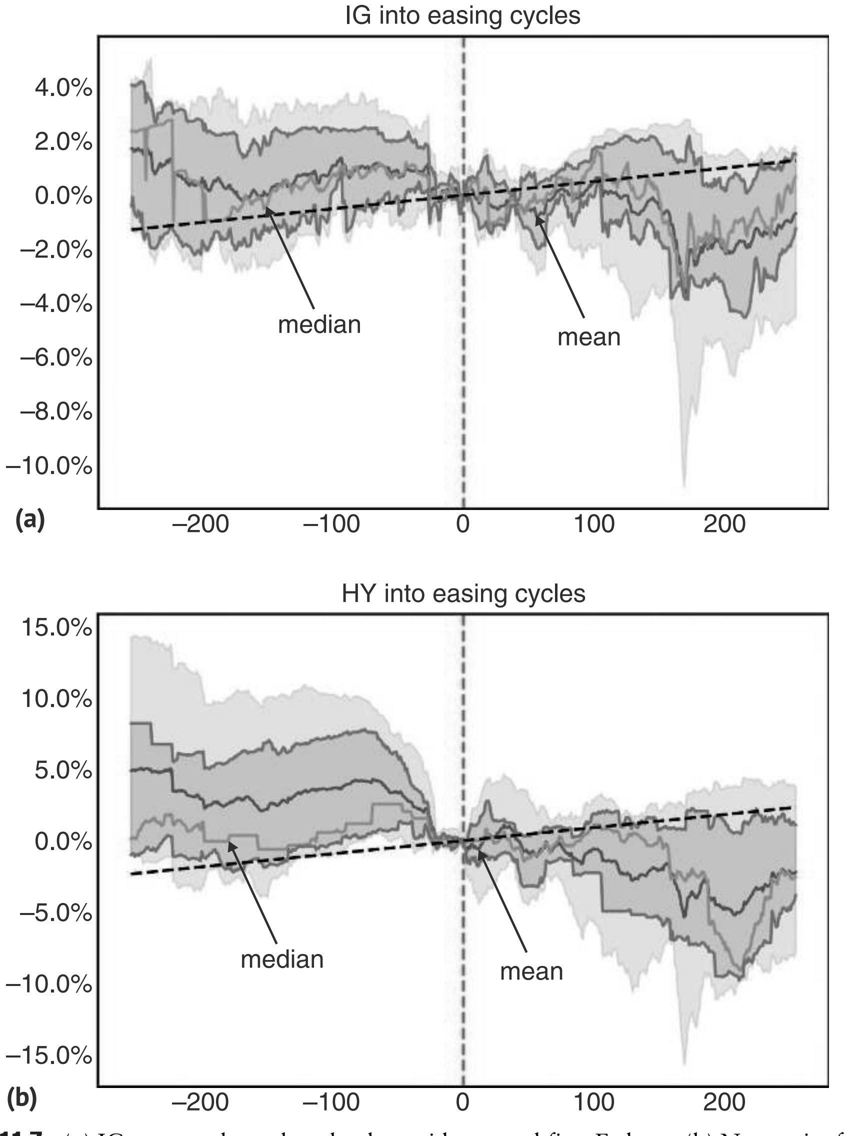


FIGURE 11.7 (a) IG returns skewed to the downside around first Fed cut. (b) No respite for HY when easing cycle starts.
Source: Bloomberg and authors' calculations.

Figure 11.7. The real underperformance only starts 100 days later, though, as recessionary fears rise. An important difference to equities is that the skew is much more negative. The average return a year after the first cut is still negative for spread returns in US IG, while the mean return is clearly positive for equities. Although this is partly driven by the GFC, which was at its heart mostly a credit crisis, it is also driven by the nature of credit, where upside is always capped, while downside is only capped at 100% of the investment.

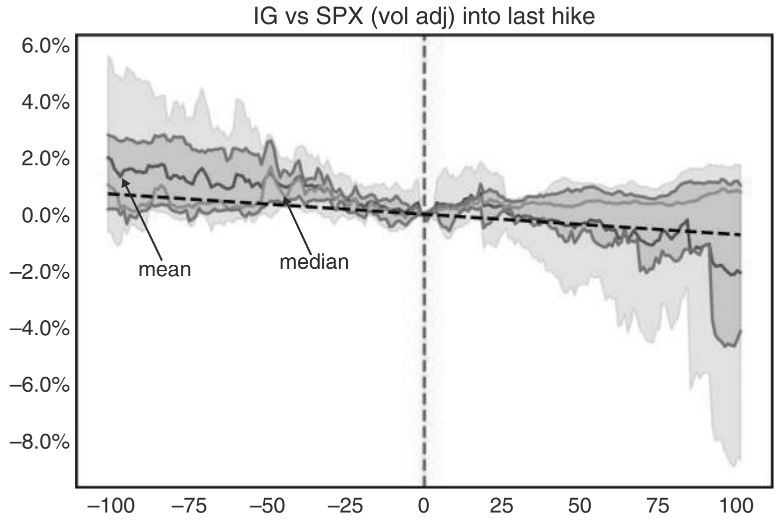
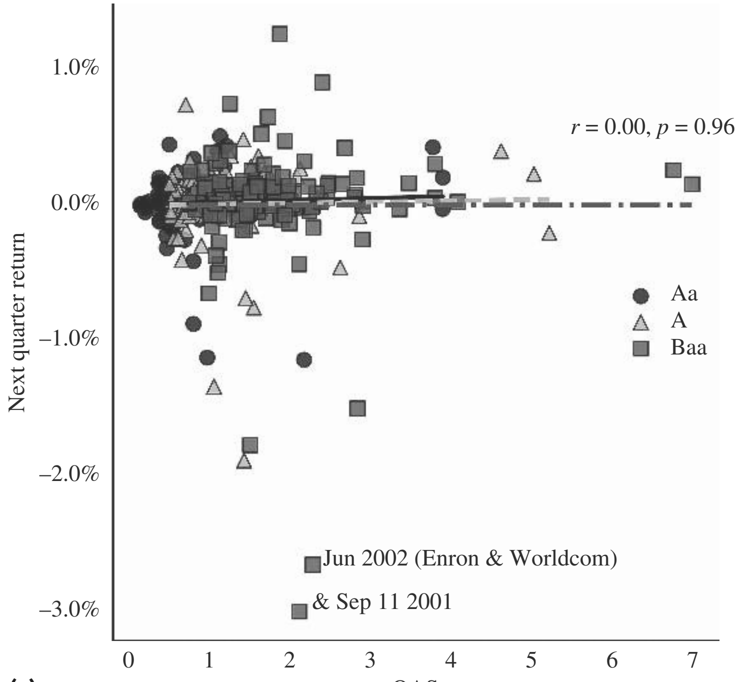


FIGURE 11.8 Credit versus equities with negative skew late cycle.
Source: Bloomberg and authors' calculations.

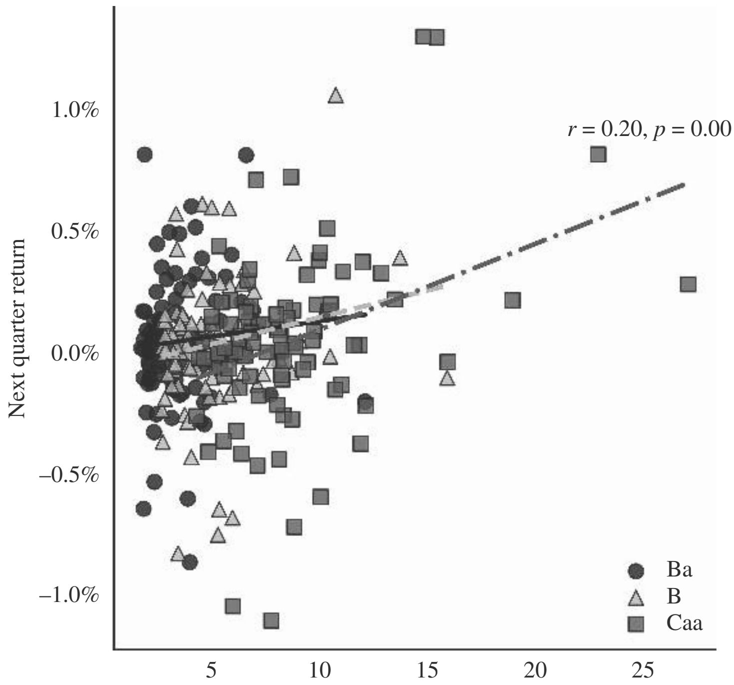
The situation around the last hike is similar. Equities tend to be better assets to own late in the cycle. We illustrate this in Figure 11.8, where we show spread returns in credit versus equities, volatility matched. Credit versus equities goes mostly sideways into the last Fed hike, and the flattish performance continues afterward. However, the skew is extremely negative. Credit is therefore not an attractive asset class for late-cycle investing.

11.4 (DIS)CREDITING VALUE

Valuation in credit can be thought of as the spread for a credit with the same credit rating and duration. While neither the credit rating nor the duration of the credit indices is constant, we use the OAS for the index as a proxy for valuation. One important difference between credit and equities is that the upside returns get very bounded when valuations are high/spreads are low, which is not the case for the equity market, where valuations can get much more stretched than appears possible. This means that on a one-year basis, there is little correlation between valuation and forward returns in equity space. What about credit? We find in IG credit, where bonds are a long way from default, there is very little relationship between the (option-adjusted) level of spreads and the subsequent quarter's returns. For high-yield bonds, the relationship is clearer, with times of higher spreads also being times of greater expected returns. This is shown in Figure 11.9. The difference in the relationships is somewhat exaggerated by the fact that the charts cover different time periods. IG OAS spreads are available from 1993, whereas the HY data starts only in 2002. However, even when comparing apples to apples, starting both analyses in 2002, the IG correlation of spreads with forward returns is still half the HY one.



(a)



(b)

FIGURE 11.9 (a) IG credit spreads with little relationship with next month's return. (b) High-yield bonds tend to outperform when spreads are wide.
Source: Bloomberg and authors' calculations.

11.5 CARRYING CREDIT

Carry is a concept that transfers well to credit. The question is whether riskier bonds, with higher spreads, outperform safer bonds with lower spreads, on a risk-adjusted basis. This is most easily done by trading HY against IG, or at least choosing one over the other index to take credit exposure. As per Figure 11.10, HY outperforms IG not only in terms of total returns, but also on an information ratio basis. In this simple sense, carry works.

When we volatility-adjust in real time to target the same ex ante risk by varying the exposure monthly using trailing three-month volatility, we find that the volatility of the IG index is higher, i.e., IG has more risk in such a strategy. This is because during bad times, the IG sell-off is usually larger given its low volatility in more tranquil times, just before going into a crisis. Although the information ratio for HY is much higher than that for IG, it is not impressive either. This is shown in Figure 11.11. Volatility

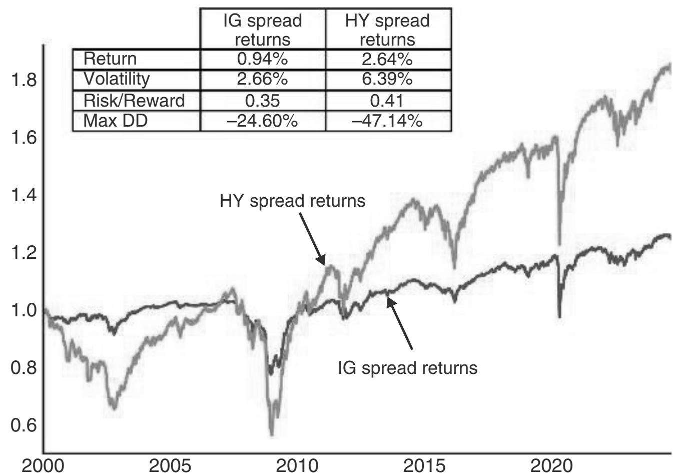


FIGURE 11.10 HY outperforms IG in terms of information ratios.
Source: Bloomberg and authors' calculations.

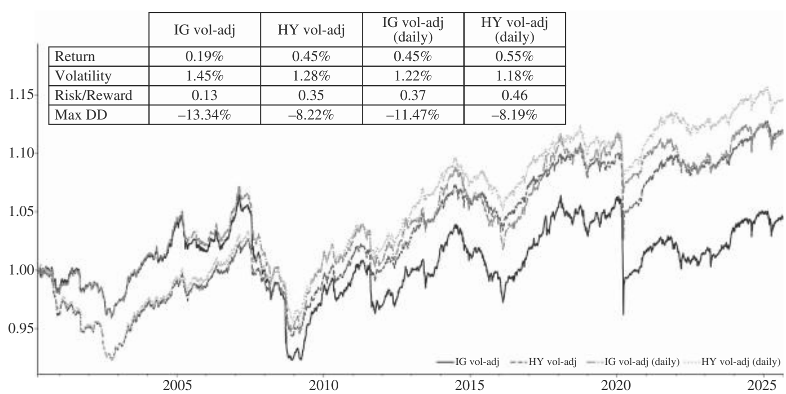


FIGURE 11.11 Volatility targeting does not work overly well.
Source: Bloomberg and authors' calculations.

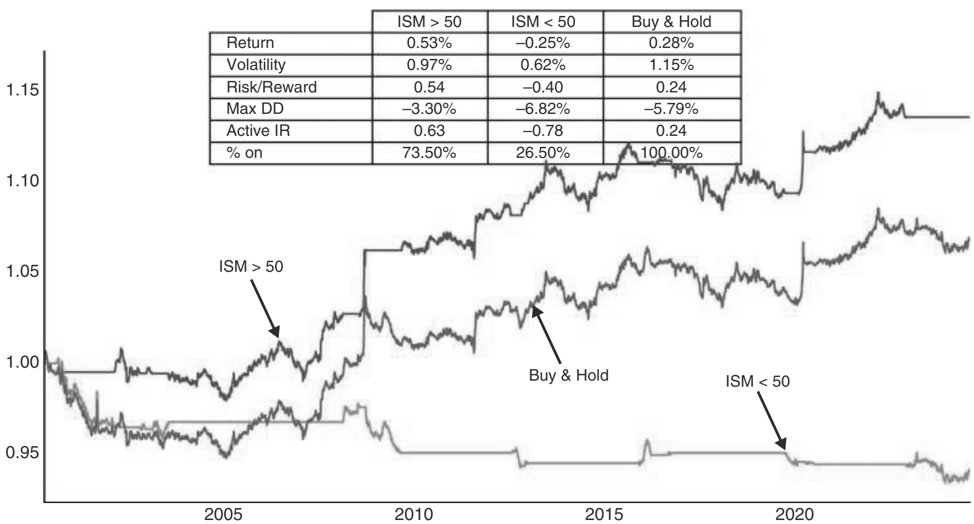
targeting does not work overly well in credit, even though carry does work. Adjusting positions to volatility daily, rather than monthly, improves information ratios, though none of these strategies can compete with using implied volatility as a cut-out mechanism, which we had shown already in Figure 11.6.

11.6 JUNK OR QUALITY?

We next address the choice between HY and IG US credit. In some sense, the question of how to choose between IG and HY has a very obvious answer: during good days spread returns in HY outperform IG, and during bad times, when bankruptcy risk rises more significantly for HY, the opposite is true. However, if we risk-adjust the returns, i.e., when we size positions in line with volatility (updated monthly), the answer is less obvious ex ante. Upon investigation, it turns out that even on a risk-adjusted basis, the same concept still holds. In particular, whenever the manufacturing ISM is above 50, US HY outperforms US IG, risk-adjusted, and vice versa, as can be seen in Figure 11.12.

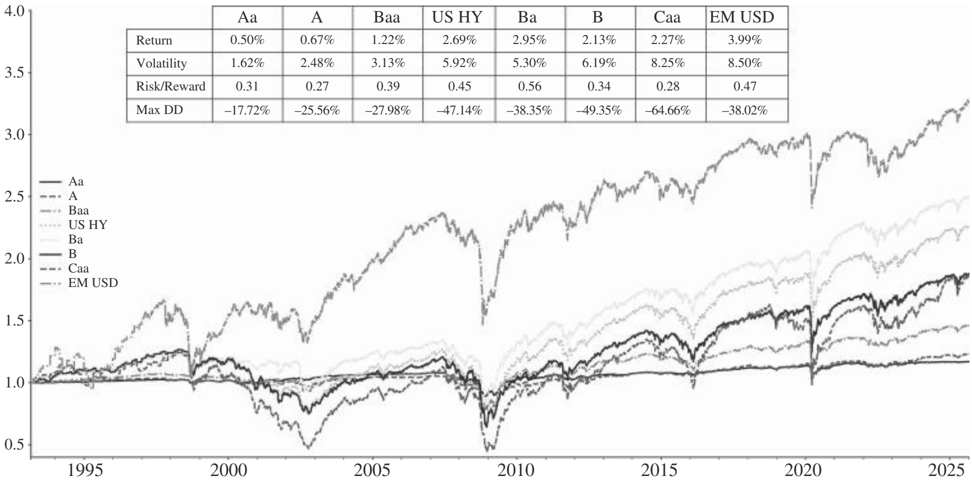
11.7 RISING ANGELS AND FALLING STARS

In this section, we investigate where investors get the highest returns across the credit spectrum. While outright returns have been highest in EM, information ratios for spread returns have been the best in Ba-rated credit, as is shown in Figure 11.13. The likely reason is that downgrades from IG to non-IG are typically events that lead to a meaningful negative price action going into it, but are often followed by positive price action after a short period of indigestion. This is the typical fallen angel price action. The initial price drop is primarily because many IG credit funds are prohibited from



	ISM > 50	ISM < 50	Buy & Hold
Return	0.53%	–0.25%	0.28%
Volatility	0.97%	0.62%	1.15%
Risk/Reward	0.54	–0.40	0.24
Max DD	–3.30%	–6.82%	–5.79%
Active IR	0.63	–0.78	0.24
% on	73.50%	26.50%	100.00%

FIGURE 11.12 HY outperforms IG , risk-adjusted, with ISM > 50.
Source: Bloomberg and authors' calculations.



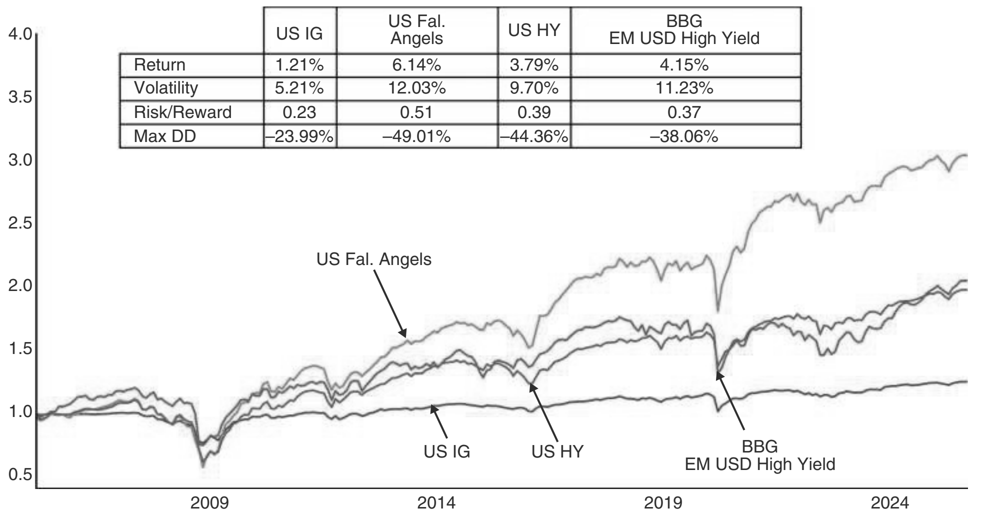
	Aa	A	Baa	US HY	Ba	B	Caa	EM USD
Return	0.50%	0.67%	1.22%	2.69%	2.95%	2.13%	2.27%	3.99%
Volatility	1.62%	2.48%	3.13%	5.92%	5.30%	6.19%	8.25%	8.50%
Risk/Reward	0.31	0.27	0.39	0.45	0.56	0.34	0.28	0.47
Max DD	–17.72%	–25.56%	–27.98%	–47.14%	–38.35%	–49.35%	–64.66%	–38.02%

FIGURE 11.13 Ba rated bonds with highest returns.
Source: Bloomberg and authors' calculations.

holding below investment grade debt and are therefore forced sellers. This selling pressure then hits the market over relatively short periods. Hedge funds try to front-run such flows, adding to the market pressures.

A paper by CAIA found that bonds entering a fallen angels index are priced about 150 basis points cheaper than their high-yield peers (see Benson and Hayes, 2019). After the initial price drop, the market tends to stabilize, and prices of these bonds often recover as the selling pressure subsides and new investors step in. The CAIA paper, using data from January 2005 to March 2019, found that over the 12 months after the downgrade, fallen angels outperform the BB Index by almost 10 percentage points. The biggest outperformance in this sample comes after down years, i.e., in 2009 and in 2016, though in the down year itself, there is also some outperformance. This may partially be because fallen angels typically have a longer duration (and bullet structures) than their newly found peers. They also likely outperform because they are large companies that operate in IG industries and need to return to IG in order to have similar financial flexibility as their competitors. Those fallen angels will therefore work hard to improve their balance sheet to make it back to IG. Larger scale also helps. However, an important aspect of the improved performance is likely just that the names in question are already quite oversold when the downgrade happens, suggesting some protection in a negative market environment. The fallen angel performance, which is superior to both IG and HY, as well as EM, is illustrated in Figure 11.14. For a deeper discussion, see Anderson et al. (2022).

What about rising stars? Rising stars are credits that are upgraded from HY to IG. *Insight Investments* notes that rising starts are almost twice as likely to come from the fallen angel segment as from the rest of the high yield index, which partially explains the outperformance of the fallen angels. We already mentioned that fallen angels are typically highly motivated to make it back to IG, explaining this phenomenon. The normal price behavior of rising stars is that they start to outperform their peers fairly consistently and for a longish time ahead of the final upgrade to investment grade.



	US IG	US Fal. Angels	US HY	BBG EM USD High Yield
Return	1.21%	6.14%	3.79%	4.15%
Volatility	5.21%	12.03%	9.70%	11.23%
Risk/Reward	0.23	0.51	0.39	0.37
Max DD	–23.99%	–49.01%	–44.36%	–38.06%

FIGURE 11.14 Fallen angels outperform.
Source: Bloomberg and authors' calculations.

In the final month into the actual upgrade, the outperformance accelerates, with a study by Columbia Threadneedle showing that the OAS compresses by 20bp in that period (see also Columbia Threadneedle Investments, 2022). After the upgrade, the remaining outperformance is very limited in size and duration, likely as the forced buying by index investors is offset by profit-taking from investors who positioned early for the upgrade event. This is similar to the pattern in EM. We found previously that in the case of EM credit, there was follow-through of narrowing spreads after the first upgrade to IG, but the trade was basically done by the time the second upgrade occurred.

Of course, it is much easier to react to falling angels (buy them) than to rising stars, given that there is not that much juice left after the second upgrade. Luckily, given that fallen angels are more likely to eventually be rising stars, buying fallen angels is one way to position for being long rising stars. Between 2000 and 2021 it took around 3.5 years to regain IG (for the ones that did). This is a long time. The good news is that this time has recently shortened, and not just as a result of COVID. In 2019, the average turnaround time from entering HY to exiting HY was less than one year, and the turnaround time has stayed well below average post-COVID.

11.8 THE SUPPLY FALLACY

Many credit investors (and analysts) spend much time figuring out supply. The aim is to come up with a supply/demand balance, with the idea that too much supply will require higher spreads to be taken down. However, this type of analysis is misleading. The reason is that both supply and demand are driven by the global environment for credit. If the environment is bullish, there is strong demand for credit, and supply is endogenous and rises to the occasion. If the environment is bearish, there are few

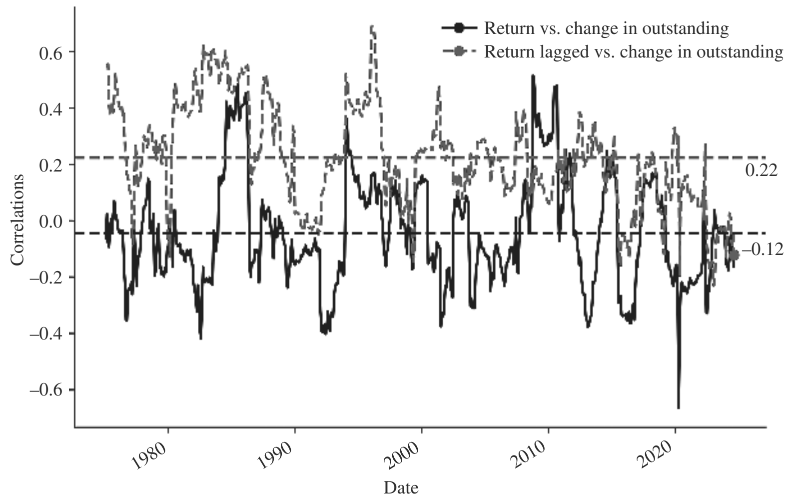


FIGURE 11.15 Supply a slight coincident negative but a more consistent positive the following month.
Source: Bloomberg and authors' calculations.

buyers of credit, and supply is often pulled. This suggests that forecasting supply to forecast returns is not overly useful. This is illustrated in Figure 11.15. Overall, the correlation of total returns versus supply is fluctuating around zero, averaging –0.12, suggesting that higher supply is a small negative, but that is not overly consistent. Furthermore, if returns are lagged by one month, the correlation is meaningfully positive, at 0.22, i.e., high supply leads to positive returns in the following month and vice versa. Overall, this suggests that supply is not a major problem, and if it is, it is quite short-lived and any negative effect is unwound the following month. In fact, the causation may be the other way around. In a throwback to Say's law (Say, 1821), times of abundant demand and positive credit conditions may lead to increased supply. More fundamentally, the primary market is an important source of price discovery for a relatively illiquid asset class, and supply helps companies to refinance and roll over debt, which is, of course, helpful to stay solvent. As such, we typically ignore supply forecasts with respect to evaluating credit.

11.9 THE CREDIT CANARY

One important focal point in cross-asset class investing and global macro is whether credit leads equities, especially in downturns. As already discussed in Chapter 2, credit starts to sell off shortly before the recession starts, with the largest part of the sell-off occurring only during the recession, as per Figure 11.16. This would suggest not covering credit shorts too early but to wait until well after the onset of a recession, though, as noted before, it is often unknown when a recession starts and ends.

Does the credit sell-off lead equities into recessions? The fact that credit seems to be riskier late cycle suggests that credit could be a leading indicator for equities. Theoretically, there are a few reasons to expect this to be true. First, fixed-income investors

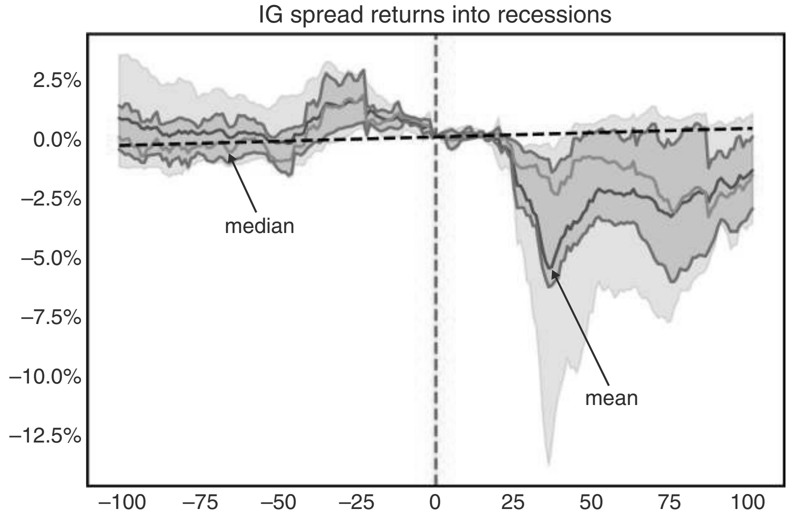


FIGURE 11.16 Credit sell-off stronger during recessions than into recessions.
Source: Bloomberg and authors' calculations.

tend to be more risk-averse, given that a rising default probability raises the risk of meaningful downside, compared to much less impressive upside from continuing to clip the coupons. Credit investors may therefore adjust portfolios more quickly when negative developments are anticipated, compared to equity investors, where upside and downside are more symmetric. Second, as many corporates rely on debt markets for (re)financing, higher borrowing costs or a more restricted supply of credit can put pressure on a company's credit very quickly. Third, wider spreads can be a symptom of declining market liquidity, which increases stress in the financial system, which eventually also catches up with equities. Lastly, equity investors may be more prone to look "through the valley," while for credit investors, there may not be another mountain to climb if bankruptcy hits. Although equities are wiped out by bankruptcies too, the constituents of credit and equity indexes are very different. The high market cap companies that dominate equity indexes are unlikely to face severe bankruptcy risk, while the index weights on the credit side depend on the size of the outstanding debt, which presumably is at least loosely correlated with the default likelihood. Of course, in the end, this is an empirical question and we investigate below.

In Figure 11.17, we show the performance of US IG (spread returns) over US equities, risk-adjusted, into US recessions. In the median case, credit clearly underperforms, both going into and during/after a recession. While this is a fairly uniform regularity into recessions, coming out of it/during it, has seen a few cases of credit outperformance, as the 75th percentile is moving aggressively higher. One such event was the 2001 recession, where credit stress did not play a major role, as the narrative was all about the wealth effect of a deflating equity bubble. Overall, we think that credit should work well as an equity hedge into most recessions, though care should be taken to take the precise nature of the recession into account. If the problem is a bursting equity bubble, credit is less well placed as a hedge for equities.

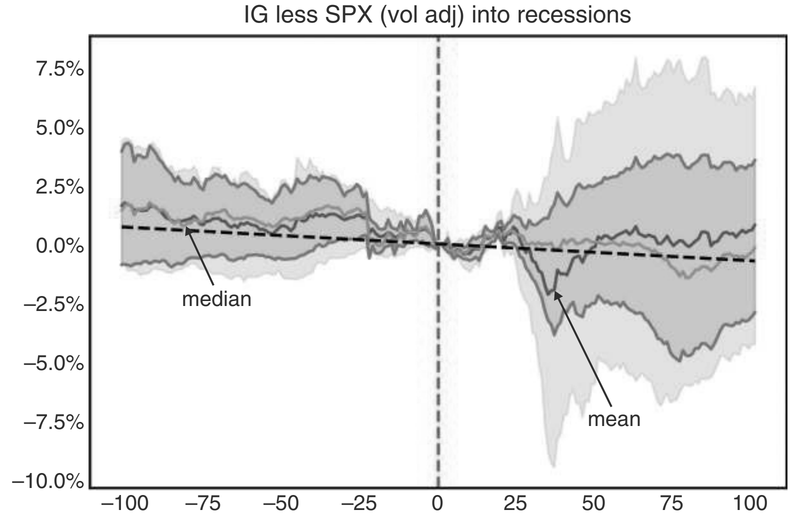


FIGURE 11.17 Credit underperforming equities around recessions.
Source: Bloomberg and authors' calculations.

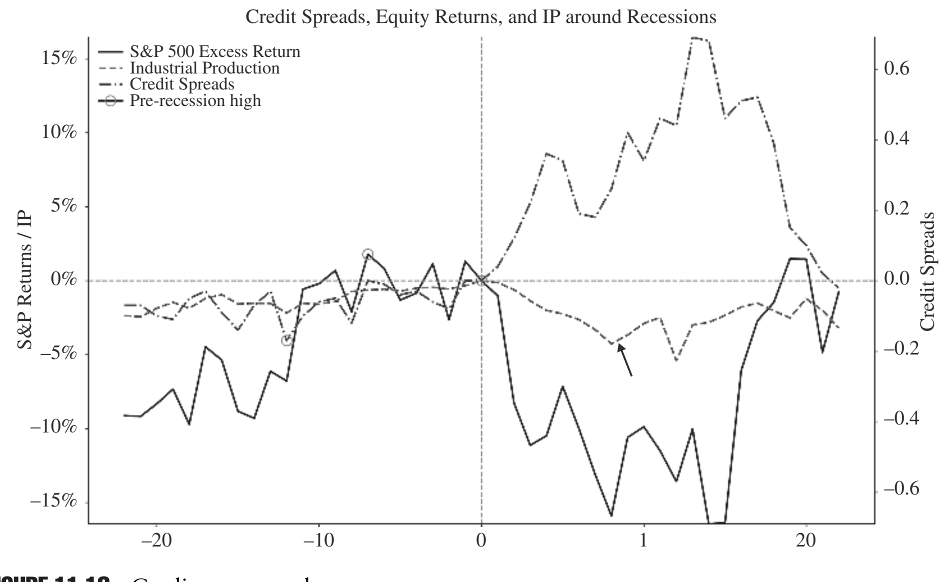


FIGURE 11.18 Credit warns early.
Source: Bloomberg and authors' calculations.

Investigating the timing of the sell-offs relative to recessions more carefully, we chart both equities and credit individually. As per Figure 11.18, credit spreads are at their tightest around 12 months before the onset of a recession, while the SPX peaks in the median case around six months before a recession, where the study is based on the seven recessions since 1973. On the other side of the coin, as we have already commented in Chapter 3, Table 3.5 shows that many recent bottoms during recessions were coincident with equities, while pre-2000, equities were typically leading credit coming out of recessions. The main exception was in 2001, when credit was leading. This analysis is all in terms of spread returns. Total returns bottom out before, given the fact that the rates component does well during recessions.

The other way to think about the relative attractiveness of equities versus credit is to see how well credit shorts protect against equity drawdowns in some of the major macro crises. This is shown in Figure 11.19, where we focus on major crisis-related equity drawdowns and show how various hedges would have performed in terms of realized Sharpe ratios, weighting the position sizes by the volatility of the relevant index in the previous six months. We do this for long Gold, long UST, short USDJPY (carry adjusted), short US IG, and short global HY. As can be seen, credit protects well against the equity downside in all episodes, and long equities, short credit adds meaningful alpha during the GFC and COVID. Treasuries work much less consistently and add to downside in the inflation-driven 2022 sell-off (as does USDJPY, which also does not work well in 2018). Gold is slightly worse than Treasuries and is also losing money in 2022. Whether IG or HY is the better hedge is somewhat unclear. On average, they offer very similar characteristics.

We have highlighted above that it is quite important to understand if a recession is substantially driven by a credit issue or not. A quick and dirty way to see if a cycle is likely driven by credit is to check how quickly credit has grown during the preceding bull market years. If there was no major credit growth either for households or corporates, the next downturn may not be credit-driven, in which case credit shorts are unlikely to protect fully against equity downside. For example, in the post-COVID cycle, the balance sheet of both households and corporates have so far remained relatively clean, largely because the government has socialized many of the liabilities during COVID. While this has raised risk for the government's balance sheet, there is a big difference between the balance sheet of the government, which can print money,

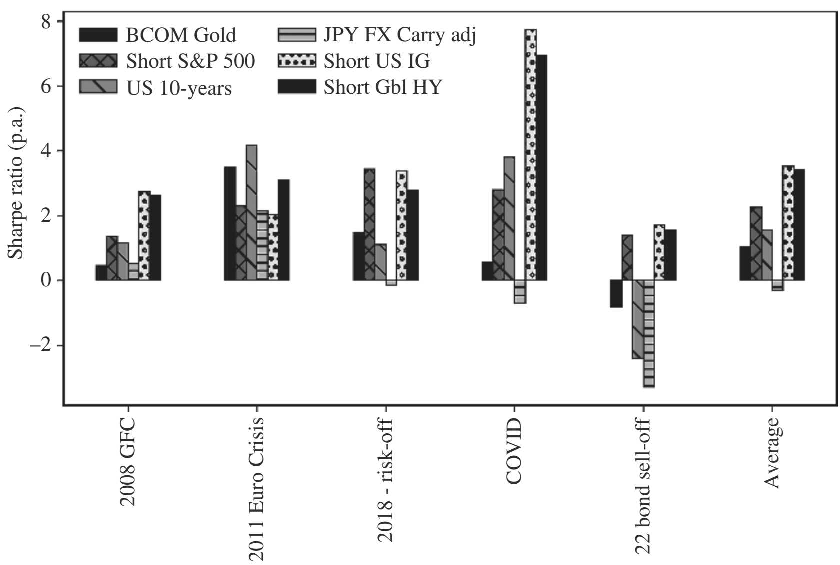
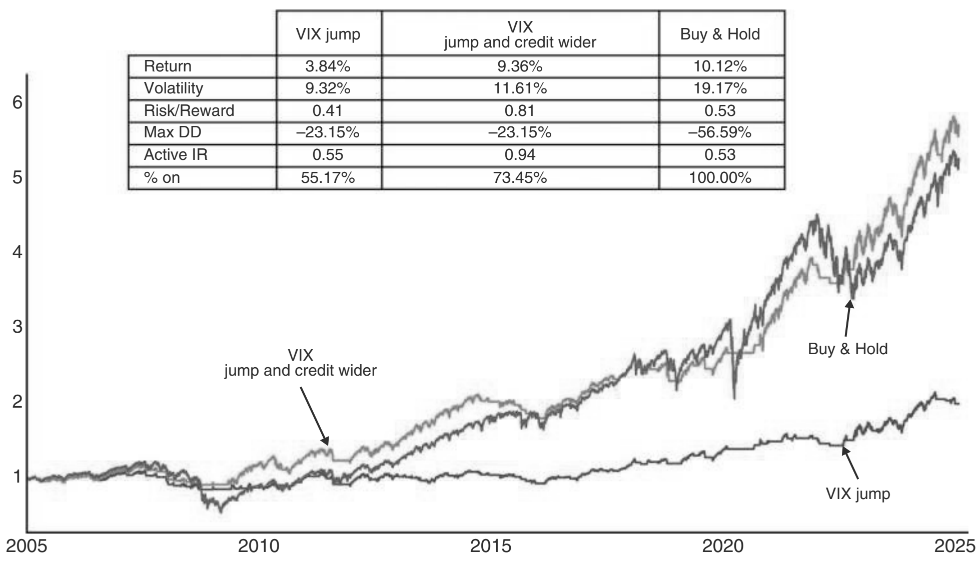


FIGURE 11.19 Credit underperforms in a crisis.
Source: Bloomberg and authors' calculations.

and the private sector, which cannot. The government may therefore hit binding constraints only much later than a private sector entity would (we have discussed some of these issues in the chapters on government bonds).

Abstracting from the government sector, the credit that has grown the fastest in private sector land has been private credit. Private credit refers to lending provided by nonbank institutions, typically negotiated directly between the lender and borrower. It has grown largely due to increased regulation of banks in the aftermath of the GFC. It typically consists of smaller loans to small and medium-sized enterprises, is floating rate, and has a maturity of three to seven years, though more recently the size of the loans has grown. The problem is that, because it is not traded, it is difficult to assess whether any of the typical problems that are often associated with the rise of a new credit class are accumulating. It is also not obvious how much leverage there is in the system, though the fact that the product is often held by insurance companies may suggest that leverage is not very high. We feel that the best we can do to monitor the risk from private credit is to observe the traded equities of private credit funds. When these equities trade below their 200dma, the risks could increase. But this is very speculative. We feel more strongly about the opposite: while the equities of the relevant private credit funds are above the 200dma, investors should probably not worry too much about private credit.

Having discussed credit during recessions and macro crisis, we next also include smaller pull-backs in risk in our analysis. Importantly, credit is not only working as a leading indicator for equities with respect to recessions. To come up with a tradable rule for this hypothesis, we tested one simple strategy. Figure 11.20 shows the PNL for a strategy that cuts out SPX risk if VIX jumps by 3 points or more (an almost 2 std deviation move), staying out for one month. We compare this strategy to one that only cuts equity risk on those same dates if, at the time of the local high in the SPX, HY credit spreads were wider over the last six months. Interestingly, just using



	VIX jump	VIX jump and credit wider	Buy & Hold
Return	3.84%	9.36%	10.12%
Volatility	9.32%	11.61%	19.17%
Risk/Reward	0.41	0.81	0.53
Max DD	–23.15%	–23.15%	–56.59%
Active IR	0.55	0.94	0.53
% on	55.17%	73.45%	100.00%

FIGURE 11.20 Credit warns of deeper SPX pull-backs.
Source: Bloomberg and authors' calculations.

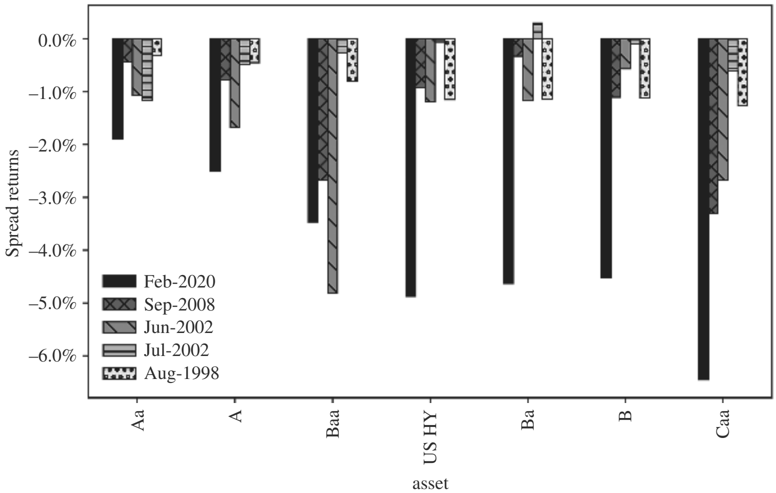


FIGURE 11.21 In stress moments, the lowest rates IG bonds can fall as much or more than higher quality HY, risk adjusted.
Source: Bloomberg and authors' calculations.

the VIX jump does not do much: the active IR is very similar to the buy-and-hold. Using VIX and credit in the way discussed above raises the information ratio further to 0.94 and leads to higher cumulative returns (and less volatility). This broadly confirms the important role that credit can play as a canary in the coal mine for equity risk.

11.9.1 Mind the IG (Tails)

When investigating which part of the credit universe is the best hedge during periods of stress, we find, somewhat surprisingly, that the lowest rating of IG underperforms the most. This is because in times of acute financial pressure, higher-rated tranches can feel the pain much more. The conservative nature of the typical investor base in IG and the fact that little default risk is priced into these issues mean that price adjustments can be abrupt and sometimes more severe than for the high-yield indices. Figure 11.21 illustrates the point. Even on a non-volatility-adjusted basis As and BAAs can perform as badly as Ba and Bs, or even worse.

11.10 THE LOANS LOGIC

Levered loans are a floating rate product, with credit risk that is not too dissimilar from US HY. Given that the interest rate is floating, credit risk can hit earlier than in other credit products. The reason is that while many Fed hiking cycles eventually lead to a downturn of the economy and credit stress, floating rate debt leads to a much faster increase in debt service, which can push firms into some level of discomfort, if not outright distress, much faster than their peers with fixed rate debt, whose interest rate bill may not move much for the first few years after the start of the Fed hiking cycle. Levered loans are therefore a product that does well early on in the interest

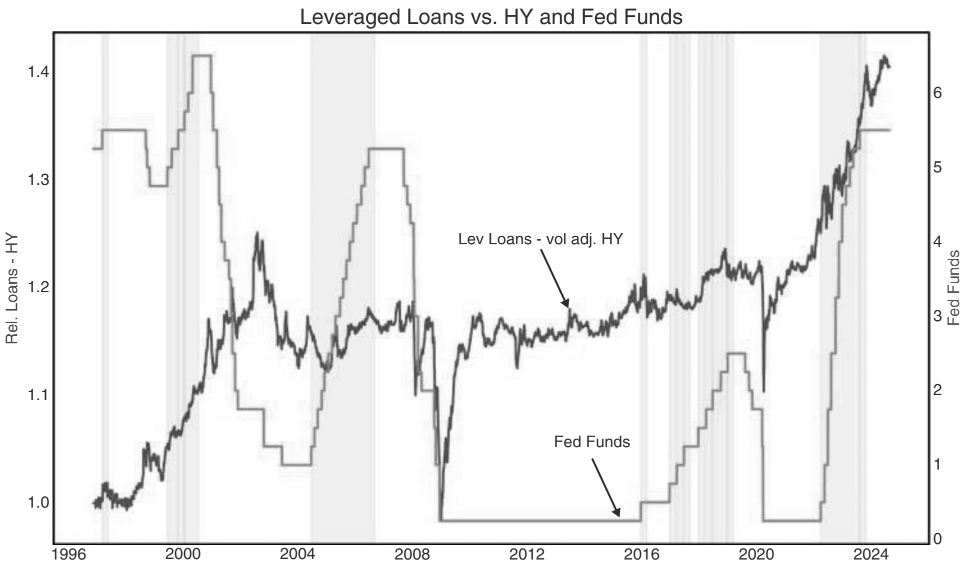


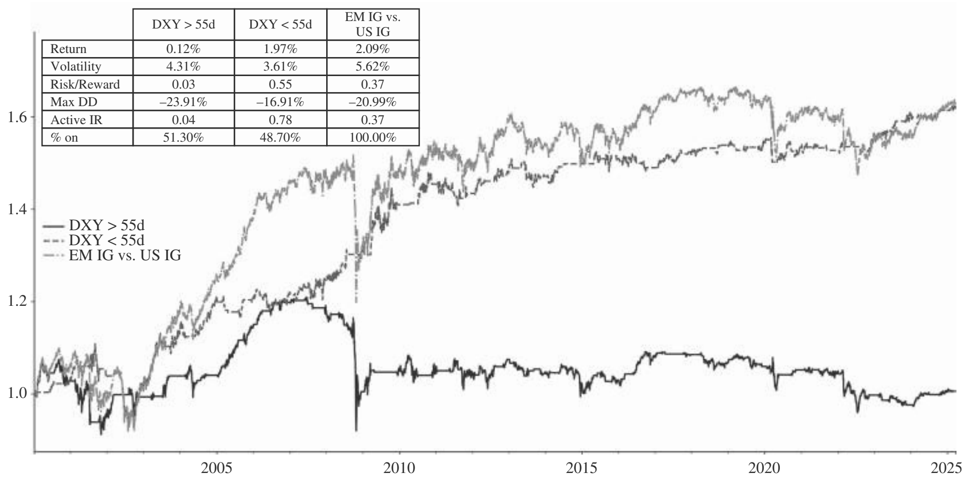
FIGURE 11.22 Leveraged loans like rising rates.
Source: Bloomberg and authors' calculations.

rate cycle, when investors want exposure to rising rates, but performance deteriorates when credit risk rises meaningfully enough to offset the advantages of being in floaters.

We illustrate this issue in Figure 11.22, where we show the performance of levered loans relative to US HY during Fed cycles. We volatility-adjust levered loans using the full sample volatility (which is roughly 60% of the high-yield index). Liquidity and accounting issues mean that loan prices move with more of a lag than HY bonds, which makes dynamically adjusting volatility (and trading to adjust position sizing) more difficult. We do find underperformance during easing cycles as high-yield bonds at least benefit from the lower interest rates, whereas leveraged loans just bear the credit concerns. There is a clear pattern of levered loans outperforming during rising rates, environments, which are the shaded areas in Figure 11.22, denoting times when Fed Funds are above their 55-day moving average. We also see a bounce-back post easing cycles as the economy recovers.

11.11 CHOOSING SIDES IN CREDIT: EM VERSUS US

Although we do not want to go into much detail for EM credit, as we covered it in the last book, we want to leave the reader with just one quick observation about how to choose between EM credit and US credit. The easiest rule of thumb to use is to prefer EM over US IG when the DXY is weak. We operationalize this rule of thumb in Figure 11.23, which goes long EM credit versus US IG, risk-adjusted, when the DXY is below its 55dma (lagged by one day). This improves the information ratio to 0.78 from an unconditional one of 0.37. This is in line with our rules of thumb in other asset classes, where EM also tends to outperform in a weak USD environment.



	DXY > 55d	DXY < 55d	EM IG vs. US IG
Return	0.12%	1.97%	2.09%
Volatility	4.31%	3.61%	5.62%
Risk/Reward	0.03	0.55	0.37
Max DD	–23.91%	–16.91%	–20.99%
Active IR	0.04	0.78	0.37
% on	51.30%	48.70%	100.00%

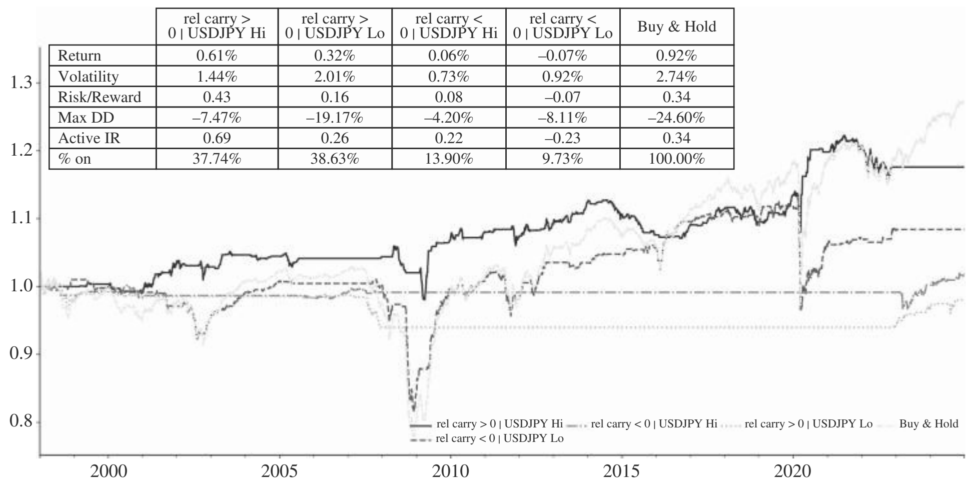
FIGURE 11.23 EM credit needs a weaker USD.
Source: Bloomberg and authors' calculations.

11.12 LIMITED FOREIGN SUPPLY, UNLIMITED DEMAND?

We do not write much about credit trading outside of the United States, as in many other jurisdictions, banks are still more dominant, making credit markets less liquid and also less important for macro traders. However, while supply is somewhat limited in many countries, international demand for US credit is important and is frequently discussed by credit analysts. In particular, Japanese institutional investors, such as life insurers and pension funds, are among the largest buyers of US IG credit. This is due to their need for higher-yielding, relatively safe assets to meet their long-term liabilities. With yields in Japan much lower than in the United States, with the corporate debt market in Japan quite small, and with large current account surpluses by Japan, substantial foreign investments are a logical consequence.

Much of the investments used to be fully FX hedged, given that USDJPY is very volatile and given the desire to keep assets as safe as possible. However, when USDJPY is strongly trending higher, or when the hedging costs are very significant, hedge ratios can be much lower than 100%. Post-COVID hedge ratios have therefore come down, though much depends on the type of investor. Banks and pension funds typically have higher hedge ratios, whereas life insurer ratios are lower.

Given a mixture of yield pick-up and FX as the main motivation for Japanese buying of IG credit, we study how US IG credit (in spread terms) trades when JGB yields are higher than the yield of US IG credit, hedged into JPY, *and* when USDJPY is moving higher (proxied by whether it is above or below its 55dma). In Figure 11.24 we find that the regime, when the IG yield is attractive relative to JGBs, and when USDJPY is moving higher, is the best regime. In this regime, IG is attractive either hedged or unhedged. The regime where the relative carry of IG is poor, and USDJPY is moving lower, is the worst of the four possible regimes, with a sharply negative information ratio. Some of it clearly has to do with the fact that US rates fall during



	rel carry > 0 | USDJPY Hi	rel carry > 0 | USDJPY Lo	rel carry < 0 | USDJPY Hi	rel carry < 0 | USDJPY Lo	Buy & Hold
Return	0.61%	0.32%	0.06%	–0.07%	0.92%
Volatility	1.44%	2.01%	0.73%	0.92%	2.74%
Risk/Reward	0.43	0.16	0.08	–0.07	0.34
Max DD	–7.47%	–19.17%	–4.20%	–8.11%	–24.60%
Active IR	0.69	0.26	0.22	–0.23	0.34
% on	37.74%	38.63%	13.90%	9.73%	100.00%

FIGURE 11.24 US IG performs better when yields are higher than JGB yields.
Source: Bloomberg and authors' calculations.

risk aversion, pushing IG yields lower. USDJPY also tends to trade lower during risk aversion episodes. But this does not detract from our Japan focus for trading IG, as the reason for USDJPY to move lower during risk aversion is intrinsically linked to Japanese investors not buying foreign assets, like US credit, as aggressively during such periods. Although the state of Japanese demand for US credit is unlikely to be the most decisive variable, it is nevertheless interesting how much the information ratio improves when IG is seen through the Japanese lens.

11.13 SUMMARY

This chapter focuses on how to trade credit spread returns, rather than total returns, and on this basis, many of the rules for trading equities also apply to credit, largely because the key for both asset classes is to avoid recessions. We find SLOOS is a useful indicator to trade credit risk, even though it is lagging. Our volatility indicator also improves credit trading. Valuation, measured by spread levels, is predictive for returns in HY, but not as much for IG. Carry works in the sense that HY tends to outperform IG in risk-adjusted terms, but barely so. When the manufacturing ISM is above 50, the outperformance of HY is much stronger. Within HY, Ba-rated credits have the highest information ratio, as they benefit from the fallen angel effect. We find that supply is only predictive for returns in the sense that periods of high supply lead to better returns afterward, even though supply does not depress current returns by too much. Negative credit returns often lead equity returns into recessions, and more generally, serve as a warning sign for equity markets. A jump in VIX combined with wider credit spreads is a good signal to cut equity risk. Volatility-adjusted, credit shorts have worked very well to protect against equity drawdowns. Levered loans are preferred over high yield in the early and middle part of hiking cycles. EM credit is preferred with a weak USD.

CHAPTER 12

How to Trade FX: The Framework

FX is a famously difficult asset class to trade as the link to the global business cycle can be very tenuous. One reason is that FX is explicitly a relative price, and therefore, it is not enough to trade it in line with a view on accelerating or decelerating global growth. Changes in relative growth differentials between countries are more important, and it is generally harder to have a strong view on relative performance than on the overall direction. Secondly, FX is traded by many different actors and for many different reasons. Pursuing profits in FX trading is only one of the objectives of market participants, as many other FX flows occur as a result of the normal business of exporting or importing goods, or due to corporate hedging needs. This means that flows can be less predictable, which makes FX a more frustrating asset class than most. It is no coincidence that there are very few FX dedicated funds with a good long-term record, which is different from bond, equity, or credit funds. It also means that there is a cottage industry of consultants that aim to suss out FX flows. “Technical” analysts are also most often found in FX space, given how poorly fundamentals are linked to FX price action. However, a less clearly defined link to fundamentals does not stop leveraged funds from trading FX aggressively in the pursuit of alpha. The attraction is a combination of cheap and plentiful leverage and low transaction costs, both of which mean that FX lends itself well to short-term trading strategies. And alpha opportunities do present themselves. In this chapter, we share some frameworks that we found helpful in the pursuit of FX alpha. For FX frameworks that are slightly more focused on shorter time horizons, also see the excellent book by Donnelly (2019).

12.1 USD SMILE, SMIRK, OR CLOUD?

A popular framework for the USD is the USD smile, originally developed by Stephen Jen. The framework explains the nonlinear relationship between the USD and the growth of the US economy. The idea is that the USD tends to strengthen in two extreme economic scenarios: when the US economy is performing exceptionally well or when it is going into recession. The USD strength of the first type is driven by rising interest rate differentials, while the USD strength of the second type is driven by risk aversion. A trading model for the USD, therefore, needs to include both business cycle measures to capture the USD strength that is correlated with strong US economic performance, but also needs to include a risk aversion measure to capture the other side of the USD smile. The “smile” is illustrated in Figure 12.1, which shows monthly DXY returns

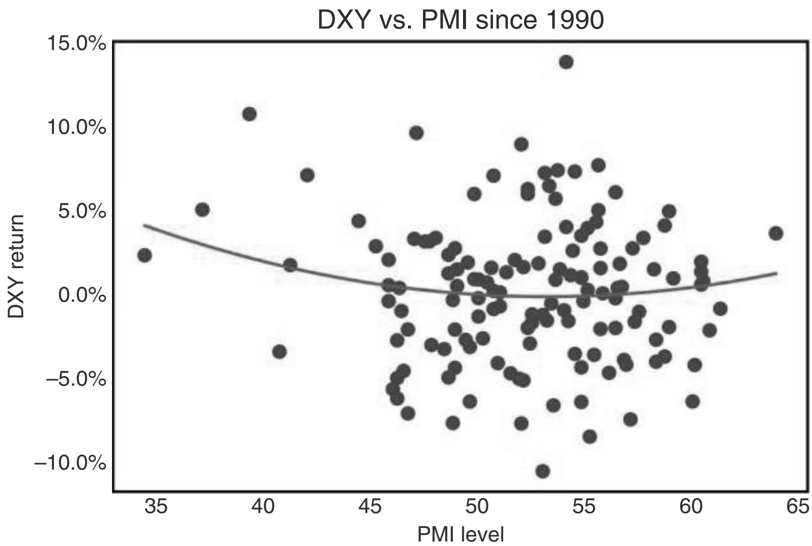


FIGURE 12.1 The USD smile is more of a smirk.
Source: Bloomberg and authors' calculations.

dependent on the manufacturing PMI. For PMIs below 45, a weak USD is very rare. The other side of the smile is not as pronounced, though. While the single largest PMI print was associated with USD strength, the hit ratio of long USDs when the PMI is above 60 is not good. It is therefore not fully surprising that most charts on the internet that try to illustrate the famous USD smile rely on a schematic illustration rather than on empirical data. Using GDP instead of PMIs, or US PMIs relative to the rest of the world, or using the change of growth, or using quarterly returns, does not improve results. We prefer the term USD smirk, but uncharitable commentators would probably call it the USD cloud. For almost all PMI outcomes, anything is possible in terms of USD returns.

The more visible left-hand side of the USD smirk across currencies is illustrated in Figure 12.2. The x-axis shows the average correlation of various currencies with the SPX and the y-axis is the average one-month implied yield. A positive correlation means that the currency in question appreciates with SPX strength. Data are from 2002 to 2024. The chart shows that there is usually a negative correlation between carry and the correlation to the SPX. High carry currencies are typically vulnerable to SPX pull-backs. The USD (here approximated by DXY) is the key currency not fitting this pattern, as it typically has positive carry but is negatively correlated with the SPX. This has made it an attractive hedge in the past, as other currencies that benefit during risk aversion are typically negative carry. This is especially the case for the JPY, but also for CHF. Of the remaining G10 currencies, the EUR came closest to having no significant beta to risk aversion.

However, we also note that this pattern is not ironclad, as can be seen in Figure 12.3. First of all, it was not really the case for long periods during the 1990s. Furthermore, in the aftermath of the bursting US equity market bubble in 2001 and 2002, there were plenty of months where a weaker SPX coincided with a weaker USD.

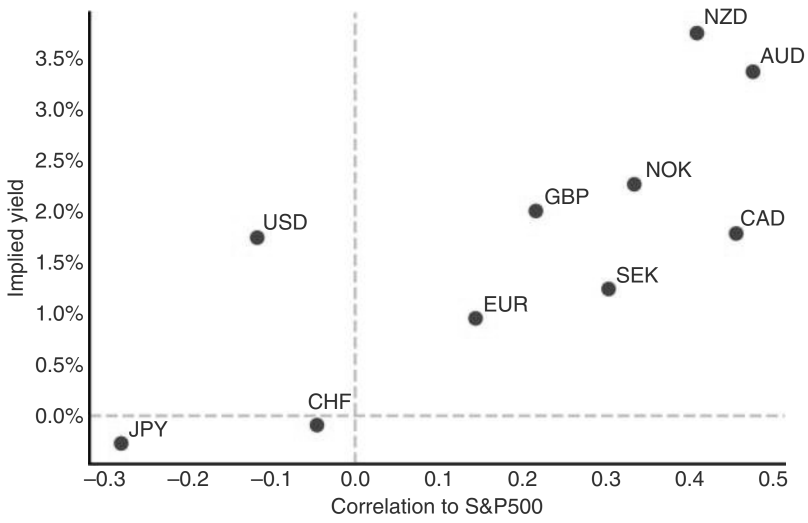


FIGURE 12.2 USD likes risk aversion.
Source: Bloomberg and authors' calculations.

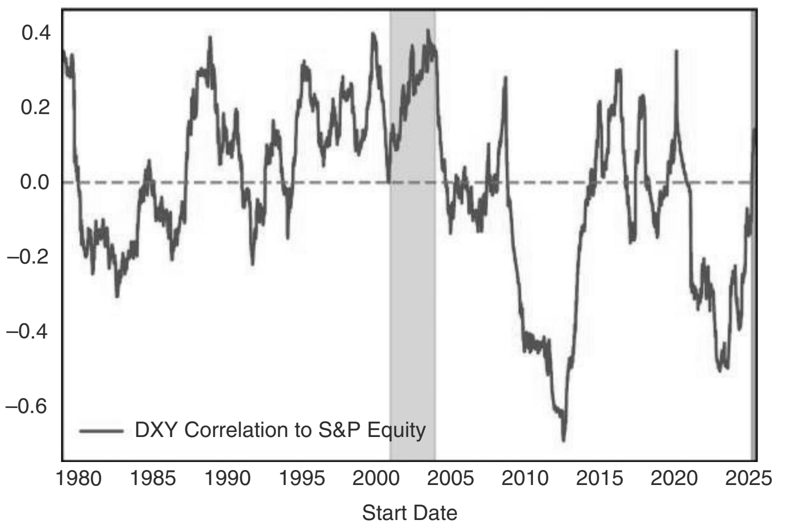


FIGURE 12.3 USD usually negatively correlated to equities—but not always.
Source: Bloomberg and authors' calculations.

The same has been playing out in early 2025 after "liberation day." Both were periods when the prevailing narrative of US exceptionalism was under pressure, which led to a broad selling of US assets, including the USD.

Next, we investigate how to make the left-hand side of the "USD smirk" tradable. When focusing on EUR, the risk aversion component is best measured by European measures of risk aversion. There are two candidates. We can either use the spread of

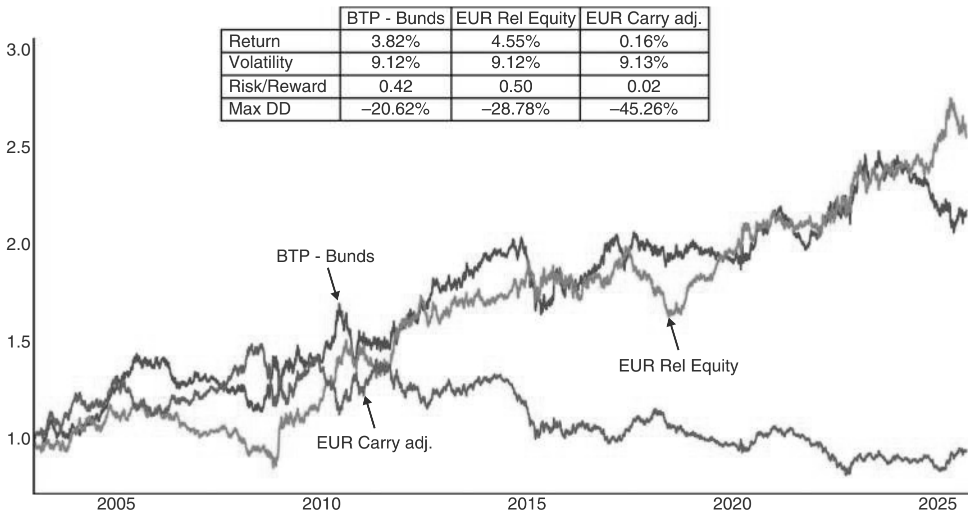


FIGURE 12.4 USD likes US equity outperformance.
Source: Bloomberg and authors' calculations.

peripheral bonds against bunds. As a shortcut, we can take the spreads of BTPs against Bunds, as Italy has been the country most consistently exposed to fiscal concerns and therefore has seen spreads most closely correlate with VIX and other measures of risk aversion. Trading the EUR in line with whether Italian spreads are below or above the 55dma creates significant alpha. Interestingly, this is not only true during the European crisis in 2011 and 2012, but also more generally, likely because the spread captures more general risk aversion, i.e., the right side of the USD smile.

An alternative version would be to use the relative equity performance of the United States over Europe. This will not so much pick up risk aversion, per se, but will pick up European-centric risk aversion, given that European equities will strongly underperform in such a scenario. Furthermore, it also picks up periods when strong US equity outperformance gives rise to large capital inflows into the United States, or the subsequent unwind of such flows. We implement this rule by using the Eurostoxx over the MSCI US, in local terms, and take our usual 55dma to generate a signal (with a two-day lag, to account for the timezone differences). We find that this signal works better than the risk aversion signal based on BTP-Bund spreads, as shown in Figure 12.4. But overall information ratios are relatively modest, at around 0.5.

12.2 THE USD AND THE BUSINESS CYCLE

As just laid out, the link of the USD to US growth on the right-hand side of the smirk, i.e., the strong US business cycle, is less clear-cut, or at least not as simple as just observing US growth momentum. Having said that, the business cycle is very important for the USD. Given that FX is about relative business cycles, we start the discussion with a focus on relative data momentum. We focus on EURUSD, which is the main component of the DXY index and has the advantage that we can more easily analyze relative data momentum than if we were trying to explain DXY. But we also discuss the trade-weighted USD. We start our discussion with Citi's data change indices (CECID) for the United States and the Eurozone. Those indices essentially score new US data releases against the levels of the previous 12 months. For a detailed description, see Noual (2014), and for other interesting use cases, see Kasikov (2018). The trading rule we are going to investigate is to go long EUR whenever the relative data change index crosses above the zero line, i.e., when data changes in the Eurozone are stronger than in the United States, and vice versa. Interestingly, rules based on the data change indexes generally work better than rules based on the data surprise indexes (CESI, which measure data releases against the Bloomberg consensus), even though the data surprise indexes are moving faster and are oftentimes leading CECID. The results are illustrated in Figure 12.5a for DXY and Figure 12.5b for EUR. The information ratio

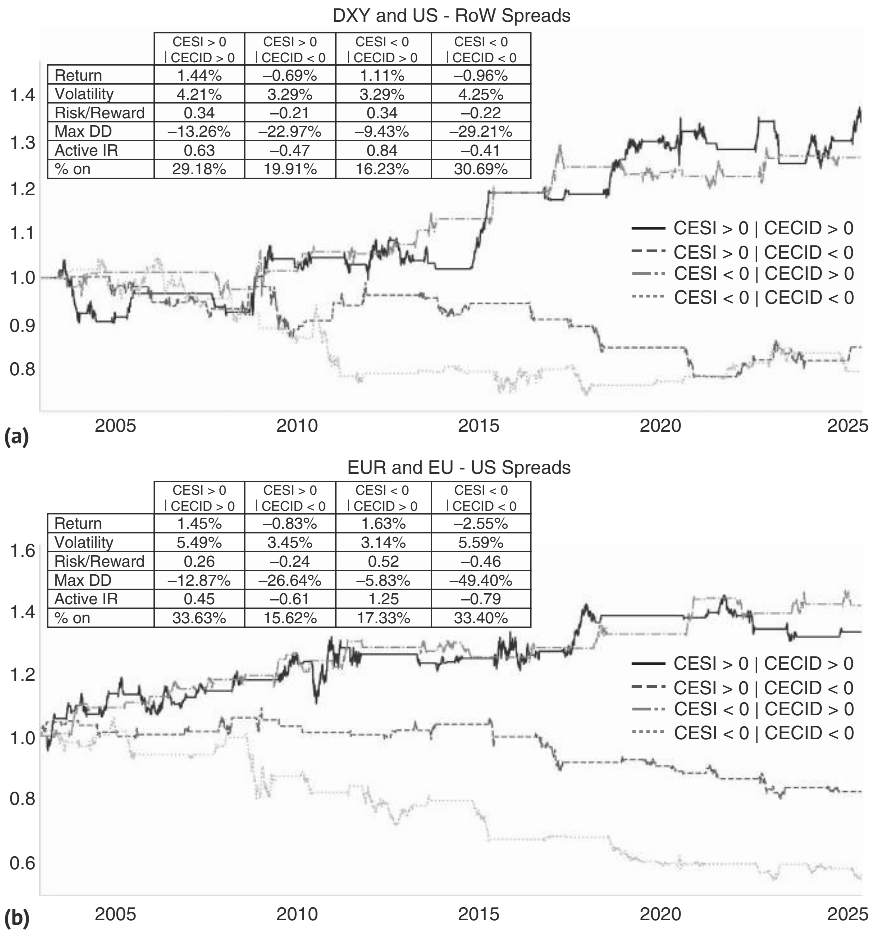


FIGURE 12.5 (a) Data change works better than surprises for DXY. (b) Model even better for EUR.
Source: Bloomberg, Citi and authors' calculations.

of EURUSD is negative if the CECID differential favors the United States and vice versa. Adding the surprise index as a secondary variable does not add any value. The results are similar for the broader USD. Trading the DXY only when the differential in the data change index is positive results in an information ratio of 0.84, while the information ratio is negative in the opposite case. It is often the case that models work better against a basket of assets than against a single asset, given idiosyncratic shocks that pack a smaller punch in baskets. Idiosyncratic shocks need to be adjusted for on an ad hoc basis rather than in a rules-based framework.

Of course, the question is why we do not just focus on rate differentials rather than on economic data. After all, typically stronger data should lead to a widening of rate spreads. And if US data are outperforming but do not lead to higher rate differentials, maybe it does not matter for FX. Interest rates could be the best summary statistics for economic data releases concerning FX trading. To test this hypothesis, an important question is which interest rate to use. There are several choices to make: real versus nominal, front-end versus back-end, and spreads versus other countries, or outright rates. To help us focus on the best interest rates expression, we start with an extremely stylized model of perfect foresight. In particular, we build a trading strategy based on realized changes in the rates in question and take a position each week based on the sign of the interest rate change. Of course, the information ratios are unrealistically high, relying on information not available in advance, but the results are instructive as to which interest rate to focus on. As per Figure 12.6, it is clear that rate differentials are more important than outright rates, but that US rates work better than European rates, if just focusing on one side of the spread. The information ratio is the highest for two-year nominal, though information ratios are also quite strong for the other parts of the curve, as well as for real rates.

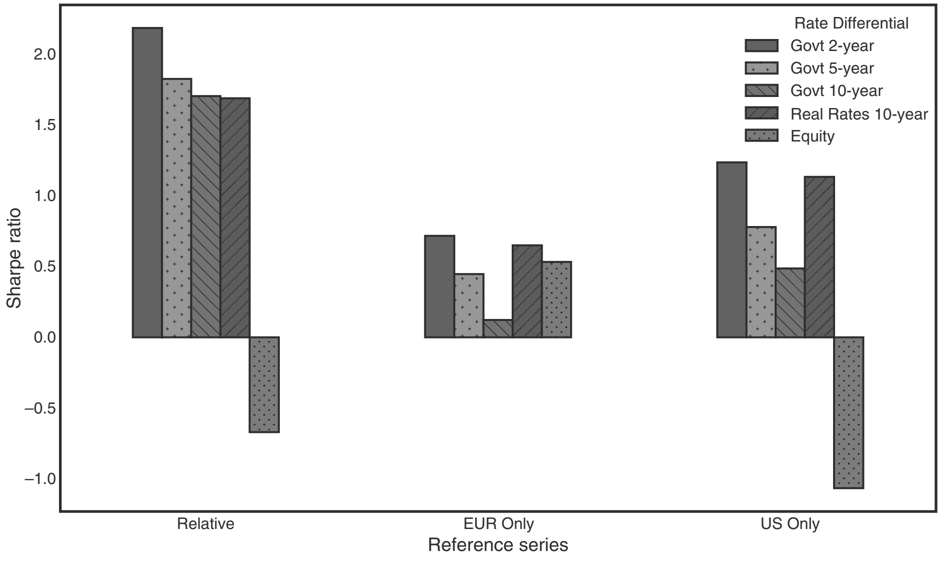


FIGURE 12.6 Front-end rates the driver.
Source: Bloomberg and authors' calculations.

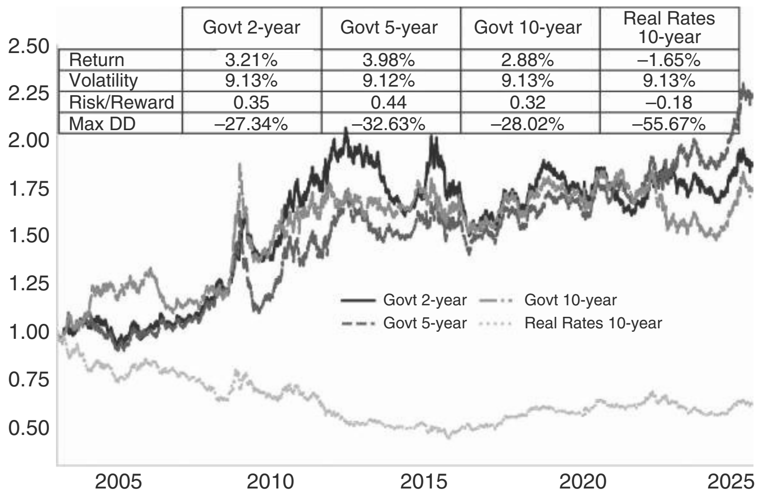
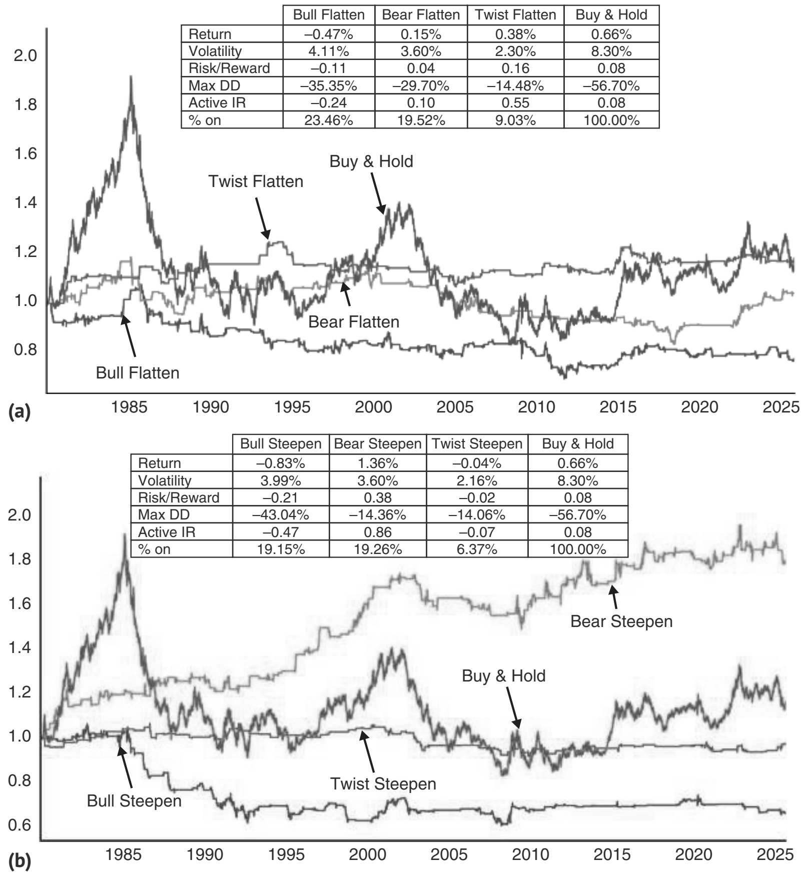


FIGURE 12.7 Rate differentials weak but positive signal.
Source: Bloomberg and authors' calculations.

Of course, what matters is a realistic trading rule rather than the perfect foresight analysis, which just informs what investors should focus on if they are willing to engage in a rates forecasting exercise. To get us there, we run our usual tests, where we show information ratios of strategies, where we trade FX in line with whether the relevant rates are above or below the 55dma, lagged by two days. We find that for this specification, the best rate driver is the five-year spread between the United States and Germany, followed by the two-year spread. The performance using real rate is not overly helpful. We would note, though, that even for the five-year spread, the information ratio is not overly impressive, coming in only at 0.40, as s shown in Figure 12.7.

For our discussion of rates as a driver for FX, it is also important to investigate the slope of the curve. Often, yield curve changes are driven by the front end. The slope is therefore somewhat similar to using two-year rate differentials. However, it is not quite the same, as we have discussed in the rates chapter. Next, we show a backtest for the 2s/10s yield curve. We use our usual 55dma to distinguish between flattening and steepening episodes, and then add a one-day lag. On the flattening side, Figure 12.8a shows that bear flattening is somewhat USD bullish, as would be expected, as it often goes hand in hand with a hawkish Fed. Bull flattening is USD bearish. But the information ratios are not overly high. Figure 12.8b shows the same analysis for the steepening side. The bull steepening is most USD bearish and is in line with a dovish Fed. A bear steepening is most USD bullish. Here, the information ratios are much stronger. Some of the flattening and steepening regimes achieve higher information ratios than the simple rate differentials do, though this could partly be because the rule on the slope are just lagged by one day, rather than by two, which is possible because there is no European leg to worry about. However, these models are less frequently switched on, and we therefore keep our focus on rate differentials in our composite model.



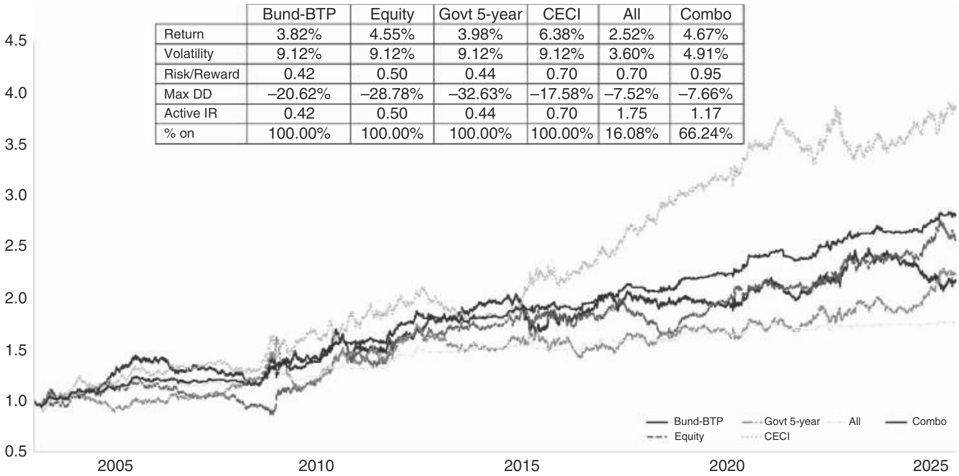
	Bull Flatten	Bear Flatten	Twist Flatten	Buy & Hold
Return	–0.47%	0.15%	0.38%	0.66%
Volatility	4.11%	3.60%	2.30%	8.30%
Risk/Reward	–0.11	0.04	0.16	0.08
Max DD	–35.35%	–29.70%	–14.48%	–56.70%
Active IR	–0.24	0.10	0.55	0.08
% on	23.46%	19.52%	9.03%	100.00%

	Bull Steepen	Bear Steepen	Twist Steepen	Buy & Hold
Return	–0.83%	1.36%	–0.04%	0.66%
Volatility	3.99%	3.60%	2.16%	8.30%
Risk/Reward	–0.21	0.38	–0.02	0.08
Max DD	–43.04%	–14.36%	–14.06%	–56.70%
Active IR	–0.47	0.86	–0.07	0.08
% on	19.15%	19.26%	6.37%	100.00%

FIGURE 12.8 (a) USD likes bear flattening. (b) and likes bear steepening even more.
Source: Bloomberg and authors' calculations.

12.3 A COMPOSITE MODEL

Having discussed both sides of the smirk, i.e., risk aversion, but also business cycles (both through data releases and rate differentials), a good model would likely need to cover both aspects of the smirk. We therefore put all four of our trading rules together: BTP-Bund spreads, relative equity performance, data changes, and the five-year yield differentials. Each model is either +1 or –1, and our aggregate model that only trades when all four models point in the same direction, i.e., having a +4 or –4 read, has an active information ratio of 1.75, compared to the unconditional Sharpe, which is just long EUR (carry adjusted), which is flat at just 0.02. It is switched on around 16% of the time. Trading all four models and aggregating them results in an information ratio of 1.17. This model is switched on more frequently: around 66% of the time. This is shown in Figure 12.9. For a similar version, see Miani et al. (2025), which includes



	Bund-BTP	Equity	Govt 5-year	CECI	All	Combo
Return	3.82%	4.55%	3.98%	6.38%	2.52%	4.67%
Volatility	9.12%	9.12%	9.12%	9.12%	3.60%	4.91%
Risk/Reward	0.42	0.50	0.44	0.70	0.70	0.95
Max DD	–20.62%	–28.78%	–32.63%	–17.58%	–7.52%	–7.66%
Active IR	0.42	0.50	0.44	0.70	1.75	1.17
% on	100.00%	100.00%	100.00%	100.00%	16.08%	66.24%

FIGURE 12.9 Aggregating all four models works best.
Source: Bloomberg, Citi and authors' calculations.

terms of trade and finds strong performance across a range of G10 currencies. Japan is the exception, which we discuss as special in a section later.

12.4 THE USD IN BUBBLY TIMES

We have written in the equity section that a simple technique to identify bubbles can be helpful, not to get bearish equities, but to trade equities with better risk control. Identifying equity bubbles is also useful for FX trading. It is probably not surprising that periods of building US equity bubbles are typically USD bullish, before sometimes becoming USD bearish late in the bubble, as shown in Figure 12.10. The USD bullish move is partly because US rates also tend to move higher. Interestingly, even without rates moving higher, like in the 1997–1998 episode, the USD nonetheless moved strongly higher, in that instance because risk aversion. But more generally speaking, it is the building of the bubble that sucks capital into the United States, both in the form of portfolio investment and also in the form of FDI. US equity markets typically outperform during these bubbles. This is often the case, even if other equity markets around the world also do quite well. One reason may be that some of these bubbles are driven by technological innovation, and the United States has been at the center of the last episodes of that, as was especially the case during the internet revolution in the 2000s, but also during the current AI-driven investment cycle. However, we note that pullbacks in the DXY during those periods can be quite pronounced. The "bubble regime" can therefore not be used as the main framework to trade the USD, even though it may well determine the USD direction in the bigger picture.

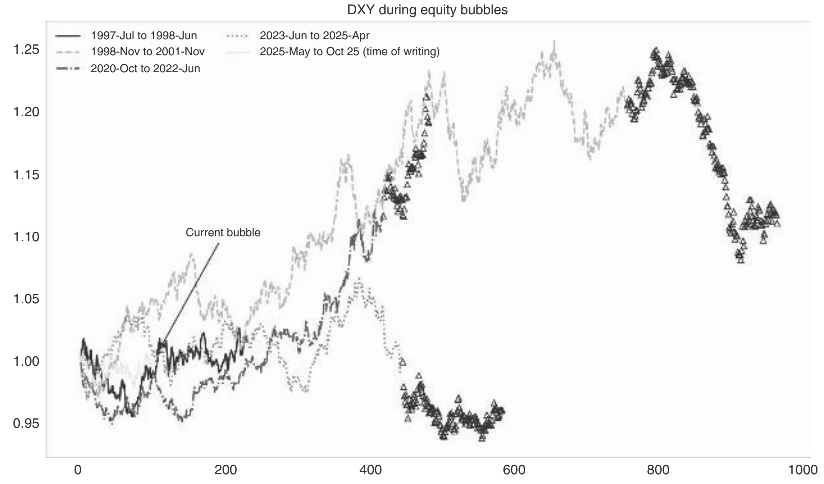


FIGURE 12.10 USD rising in equity bubbles.
Source: Bloomberg and authors' calculations.

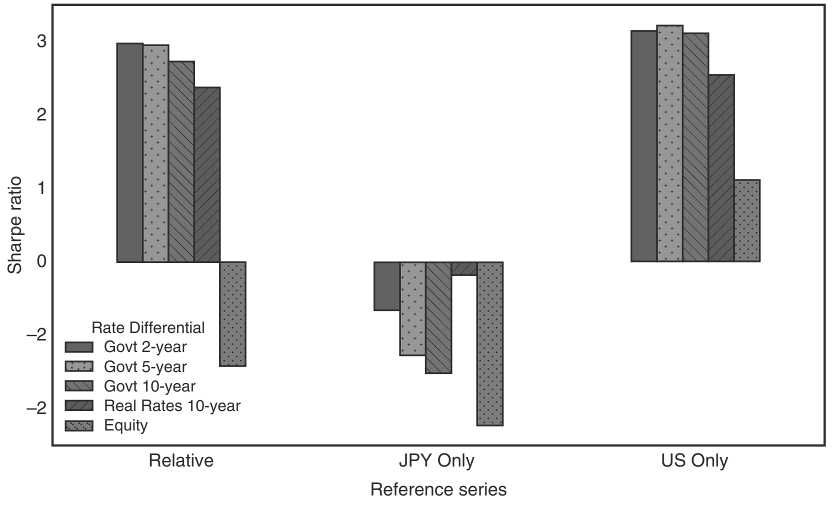
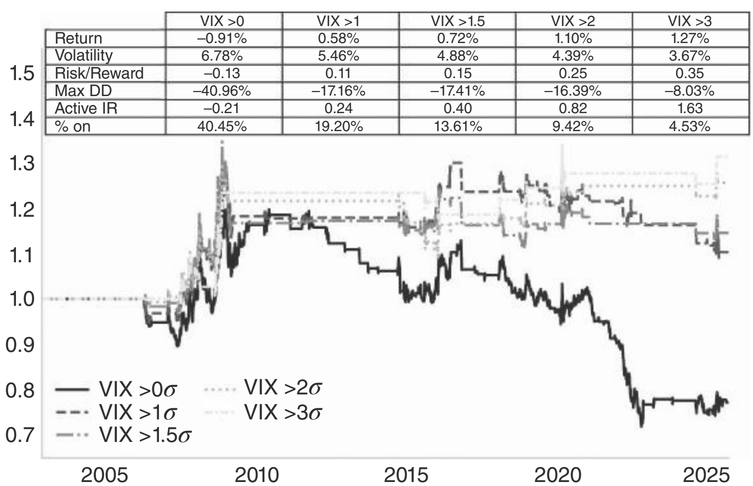


FIGURE 12.11 US rates drive the yen.
Source: Bloomberg and authors' calculations.

12.5 JPY IS SPECIAL

We have shown at the very outset of the chapter that the JPY is somewhat special, as it appreciates with risk aversion, which the majority of the G10 currencies do not do. However, when it comes to the rates factor, the JPY shows similar patterns to other currencies, and the main difference is that the USD bullish impact of higher US rates is much more consistent than for the trade-weighted USD. Figure 12.11 repeats our perfect foresight analysis that we have shown for the euro and TWI. Again, this is just for illustrative purposes and can not be traded. The Sharpe ratio of positioning with the direction of rate moves is higher for the JPY than for most other FX, coming in at >3 for most sectors of the curve. There is little difference between two year, five year, and ten year. Interestingly, the Sharpe ratio is slightly higher when just using US rates rather than focusing on rate differentials with Japan. Just using the JGB direction is negative, given that the correlation of JGBs with US rates is positive, but with the volatility of JGBs being just a fraction of the US. Relative equity returns are much less relevant, partly because Japanese equities benefit strongly from a weaker JPY.

One reason why the rates strategy has a higher Sharpe ratio than for other currencies is that JPY typically strengthens with rising risk aversion. Given that US rates can fall quite sharply during periods of rising risk aversion, USDJPY retains its directionality with US rates during such periods, which is not necessarily the case for other FX. One reason for this trading behavior of USDJPY is that Japan has a very strong net international investment position, i.e., a large surplus of external financial assets over liabilities. This measure is aggregating both public and private sector assets and is essentially the result of an accumulation of current account surpluses over time. As a result, during a time of crisis, Japanese investors tend to repatriate foreign assets, pushing USDJPY lower. Maybe as importantly, given that the JPY is typically a funder of carry trades, during times of crisis, investors tend to unwind these types of trades, given that carry trades thrive on low volatility, which is quickly upended by a crisis. To unwind carry trades, investors have to buy back JPY, also contributing to upside pressure on the JPY. The pure risk aversion channel for USDJPY has become weaker, though. In the 2020 COVID crisis, USDJPY sold off hard, but only for around two weeks, troughing before the peak in VIX. This trough did coincide with the trough in US rates on March 6, though. As such, the US rates correlation has remained strongly in place. Figure 12.12 shows the result of buying yen after the VIX is x standard deviations above its three-year average, or higher. We lag buying until the close of the

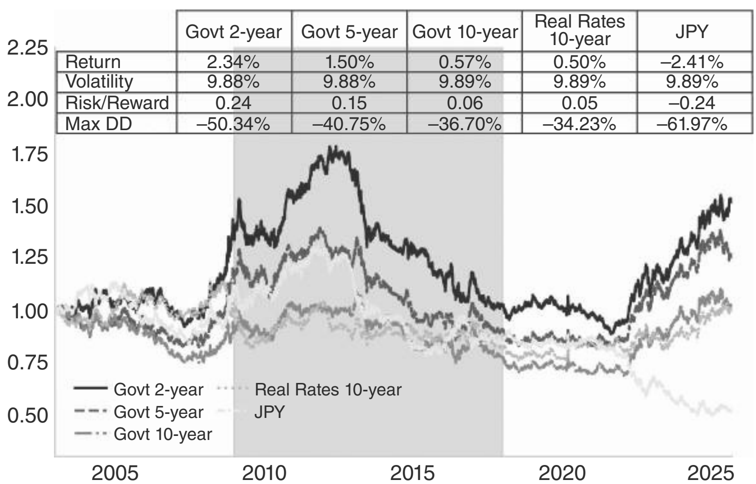


	VIX >0	VIX >1	VIX >1.5	VIX >2	VIX >3
Return	–0.91%	0.58%	0.72%	1.10%	1.27%
Volatility	6.78%	5.46%	4.88%	4.39%	3.67%
Risk/Reward	–0.13	0.11	0.15	0.25	0.35
Max DD	–40.96%	–17.16%	–17.41%	–16.39%	–8.03%
Active IR	–0.21	0.24	0.40	0.82	1.63
% on	40.45%	19.20%	13.61%	9.42%	4.53%

FIGURE 12.12 Only extreme VIX levels still push USDJPY lower with follow-through.
Source: Bloomberg and authors' calculations.

following day and hold the position for a week (five trading days). As can be seen in Figure 12.12, the most profitable period for this strategy was during the GFC. Since then, extreme moves, with VIX above 3 standard deviations, are still tradable, with 2 standard deviation moves less so.

However, if we operationalize the US rates strategy using our normal framework, the results are not overly impressive. As can be seen in Figure 12.13, trading USDJPY just in line with US rates is profitable, but barely so. Only being long USDJPY when US rates are above their 55dma, and vice versa, with a one-day lag, leads to an information ratio of 0.24, when using two-year US rates, which works best. That type of information ratio is obviously not tradable. Analyzing Figure 12.13 more carefully suggests two distinct regimes. Prior to mid-2012 and then again from 2021 onward, US rates worked very well as a driver for USDJPY. However, for almost 10 years in between, it did not. This interim period coincided with the rule of Japanese Prime Minister Shinzo Abe, who was in office from December 2012 to September 2020. His reforms put local Japanese factors into play rather than US rates. Furthermore, the initial part of Abe's reign coincided with US rates being glued to zero in the aftermath of the GFC. Without any Fed action on the one hand and an aggressive policy of weakening the JPY by Abe on the other, it is no surprise that the simple rates-based trading rule broke down. But, in general, this was an opportunity for macro funds, not a problem. Periods of strong, trending moves in a currency, especially when backed up by a fundamental story, are often when returns of macro hedge funds excel, even when a regime change means that some of the more quantitative models stop working.



	Govt 2-year	Govt 5-year	Govt 10-year	Real Rates 10-year	JPY
Return	2.34%	1.50%	0.57%	0.50%	–2.41%
Volatility	9.88%	9.88%	9.89%	9.89%	9.89%
Risk/Reward	0.24	0.15	0.06	0.05	–0.24
Max DD	–50.34%	–40.75%	–36.70%	–34.23%	–61.97%

FIGURE 12.13 Rate differentials weak but Even for USDJPY, rate differentials do not always work.
Source: Bloomberg and authors' calculations.

The Reign of Abe

PM Shinzo Abe launched Abenomics in 2012, as a response to Japan's decades-long regime of low growth and deflation. His economic agenda consisted of "three arrows": aggressive monetary easing, flexible fiscal policy, and structural reforms. For FX, it was mostly the first arrow, monetary policy, which was explicitly aimed at ending deflation and revitalizing growth, that was relevant. In particular, the BoJ shifted to an ultra-loose monetary policy under Governor Kuroda. In early 2013, the BoJ launched a massive quantitative and qualitative easing (QQE) program, buying long-term government bonds as well as equities (in ETF form) on a very significant scale. The monetary base expanded from around 125rn JPY in late 2012 to 360trn in early 2016 and kept expanding quickly thereafter. The BoJ's overnight call rate was cut into negative territory in early 2016. Ten-year JGB yields fell from around 75bp in late 2012 to around –25bp in early 2016. A weaker JPY was part of the desired outcome of these policies, as it was seen as growth-supportive, with a focus on Japanese exporters, but also was considered important in the fight against deflation, both through higher import prices and generally higher inflation expectations. USDJPY obliged, rallying from around 80 in late 2012 to 125 in 2015, as is illustrated in Figure 12.14. In this process, JPY became a very popular funding currency for the carry trade, further weakening the JPY. So far, so logical, and broadly still in line with our overall macro framework, that interest rates are the key driver, even if it was mostly on the Japanese side for a change. However, in late 2015, the JPY abruptly changed direction and strengthened. The reason was that the risk aversion angle took over again at that stage. Concerns about China's slowing economy, falling oil prices, and fears of a global recession caused investors to pull back from riskier assets, giving a bid to the JPY. USDJPY started to be more closely correlated with the Topix than with rates. On the Japanese side, when the BoJ surprised the market by moving to negative interest rates on excess reserves in January of 2016, it did not lead to renewed JPY weakness, as investors were doubting how much room for additional policy support the BoJ would still have, and whether monetary policy would be sufficiently supportive for growth and inflation. The market started to doubt Abenomics, leading to JPY buying as well. This tumultuous period in Japanese politics clearly overwhelmed the simple one-factor JPY trading model.

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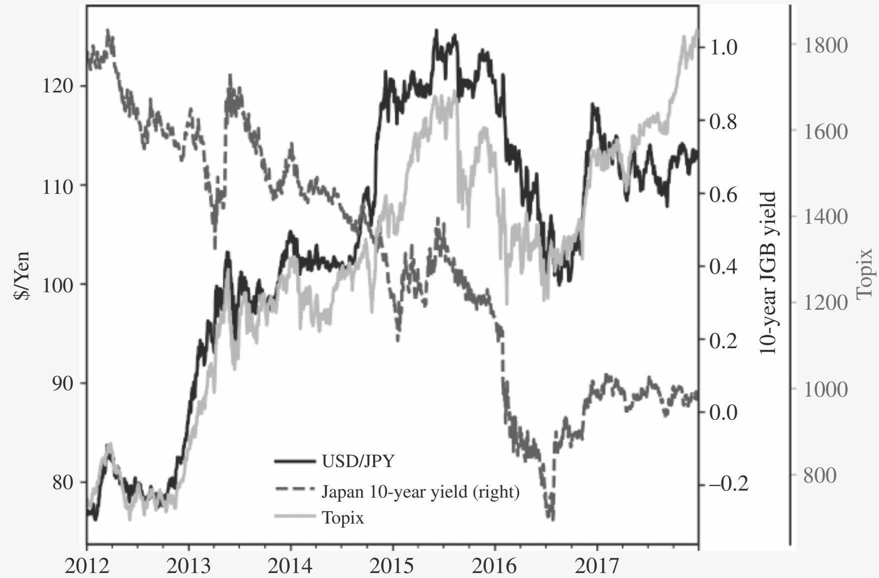


FIGURE 12.14 The reign of Abe.
Source: Bloomberg.

12.6 FACTORS FOR FX

Next, we go into a short discussion of the more important factors in FX space. Table 12.1 shows the information ratios of the various strategies. Clearly, carry has the best historical performance, followed by time series momentum (labeled TS Mom). Cross-sectional momentum, economic surprises, and value (PPP) also generate significant positive information ratios. The factors lean on work done at Citi, as, for example, reported in Saunders et al. (2025a).

TABLE 12.1 Carry the Most Attractive FX Factor

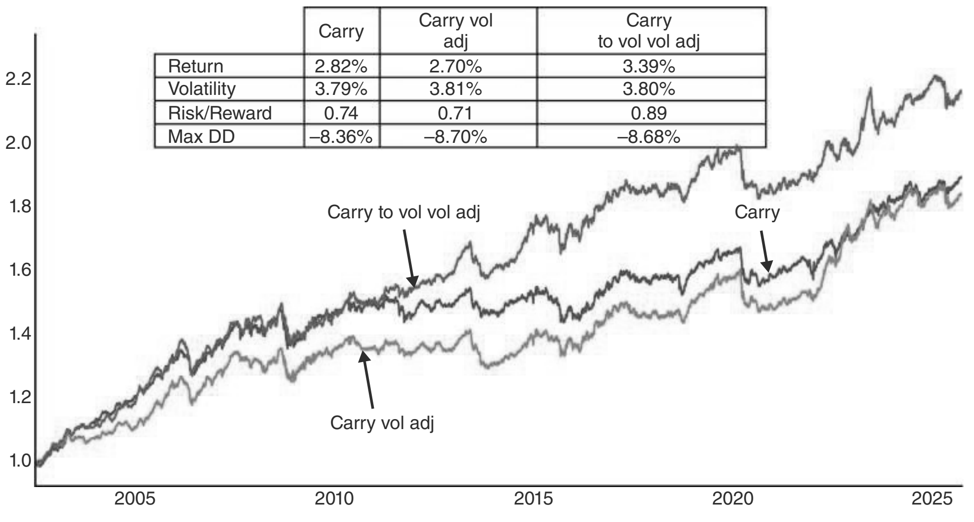
	Carry to vol	Eco surprises	Cross-sectional Mom	TS Mom	PPP	Model combination
Return	2.0%	0.8%	1.3%	2.5%	1.7%	1.7%
Volatility	2.2%	1.5%	2.1%	3.4%	3.1%	1.5%
Risk/Reward	0.89	0.56	0.64	0.71	0.55	1.20
Max drawdown	–5.1%	–4.8%	–8.3%	–5.9%	–7.7%	–2.6%

Source: Bloomberg, Citi, and authors' calculations

12.7 LIVING OFF CARRY

Given the inherent difficulty in translating macro fundamentals into successful FX trades, there has been a significant focus on trading factors, most famously the carry factor, but other factors are also making a showing in the literature. For us, the FX trade with the highest success rate is the carry trade. We have already written about the EM part in the emerging markets trading book. However, since that book was published, a lot has happened to the carry trade, and we have some important additions to our original framework.

Our favorite expression of the carry trade is to rank currencies by implied yields over implied volatility and then create a basket of the top five against the bottom five, where we set the size for each currency in line with the volatility over the previous three months. The sizing is extremely important. First, if we rank carry in a volatility-adjusted way, we also have to use volatility for the sizing, as we otherwise do not "earn" enough of the high carry in the less volatile pairs. Second, and more importantly, the volatility weighing typically results in a long USD position in notional terms, as the high-yielders are usually more volatile than the low-yielders. A net long USD position works well in terms of protecting returns when risk aversion leads to a pull-back in the appetite for carry trades. As the low-yielders typically yield less than the USD, increasing the position size of the low-yielders tends to add to the positive carry of the basket and therefore adds to the appeal of the carry trade. As illustrated in Figure 12.15, for a global basket that includes both EM and DM FX, the simple carry (top five against bottom five) leads to an information ratio of 0.74. Adjusting the positioning sizing by volatility results in a slightly falling information ratio of 0.71. If we rank the currencies based on carry to vol, the information ratio declines further to



	Carry	Carry vol adj	Carry to vol vol adj
Return	2.82%	2.70%	3.39%
Volatility	3.79%	3.81%	3.80%
Risk/Reward	0.74	0.71	0.89
Max DD	–8.36%	–8.70%	–8.68%

FIGURE 12.15 Volatility adjustments are the key to carry.
Source: Bloomberg and authors' calculations.

	G10 Carry to vol vol. adj.	EM Carry to vol vol. adj.	COMB Carry to vol vol. adj.
Return	1.04%	1.69%	1.95%
Volatility	3.94%	1.94%	2.19%
Risk/Reward	0.26	0.87	0.89
Max DD	–14.83%	–6.17%	–5.08%

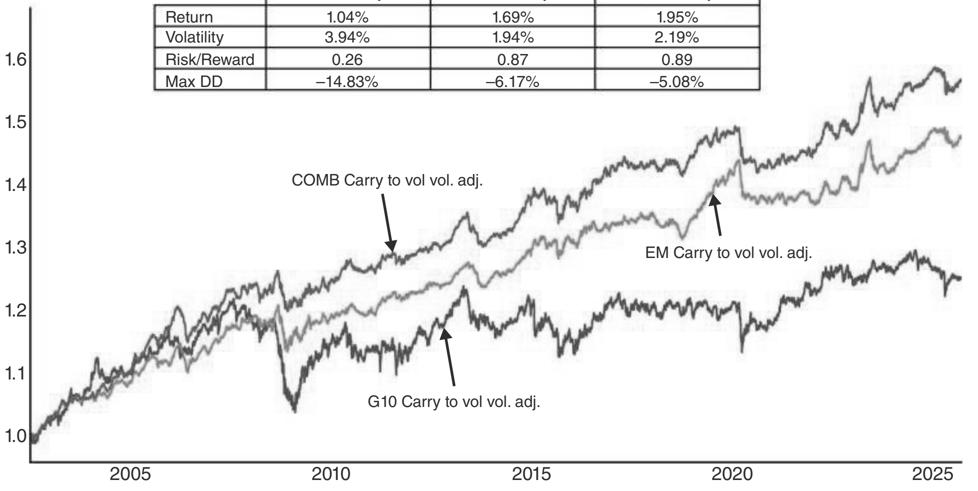


FIGURE 12.16 Carry works better for EM.
Source: Bloomberg and authors' calculations.

0.66 (not shown) as low-yielding but low-volatility currencies reduce the carry of the overall portfolio. However, adjusting also the position size of the carry to vol ranked basket, raises the information ratio to 0.89.

The carry trade tends to work better in EM than in G10, as per Figure 12.16. One reason is that low-yielders in EM space typically sell off during rising risk aversion, while low-yielders in G10 space tend to rally, adding insult to injury, when the long part of the carry trade is already bleeding. This is especially true for JPY, but also for CHF. The other reason is obviously that carry in EM is just higher.

Having said that, we would note that even if JPY is not a direct component of our EM carry trade, the direction of USDJPY matters more broadly and more so than the directionality of the SPX, as is illustrated in Figure 12.17a, b. Only owning our carry basket when USDJPY is below its 55dma results in an information ratio of 4.9 (no lag, no transaction costs). Our basket returns an information ratio of –3.5 when the opposite is true, creating a more decisive wedge than in the case of the SPX, where the SPX trading below its 55dma only leads to a negative information ratio of –0.81 (and above comes in at 2.15). Furthermore, the sensitivity to JPY remains clearly in place, while using the SPX for timing carry has essentially stopped working post-COVID.

What determines how well the carry trade does? Interestingly, our version of the carry trade does quite well in most environments. Long-lasting drawdowns are rare, as the short leg works quite well when risk aversion spikes. Still, we found some conditions when the carry trade does even better than normal. The first one is, somewhat unsurprisingly, linked to the extent of carry that is on offer. Figure 12.18 shows the carry inherent in our EMFX carry basket on January 2 of each year in relation to the performance of the carry trade over the next 12 months. Clearly, high carry is supportive of higher returns for this factor. If carry drops, initially returns are not

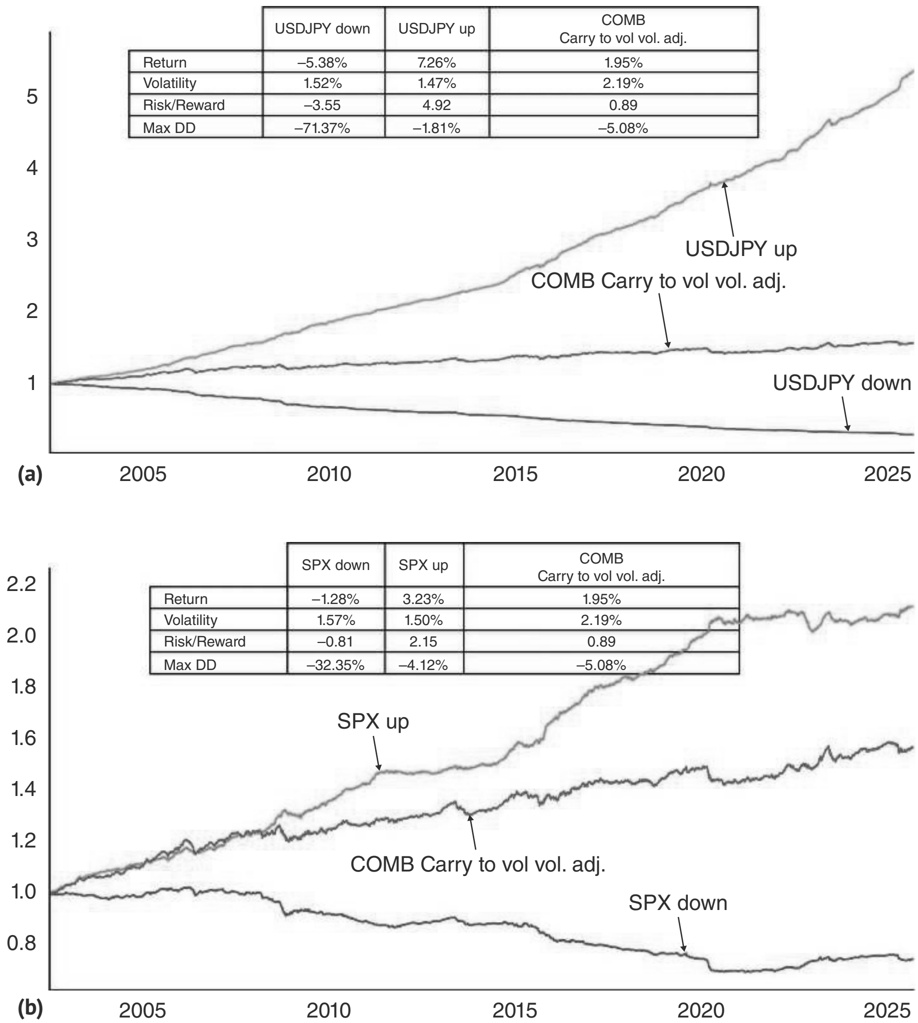


FIGURE 12.17 (a) Carry likes a weak USDJPY (b) ... and strong equities, but less so.
Source: Bloomberg and authors' calculations.

necessarily falling, but once carry has fallen below 400bp, returns are much less attractive. Still, even at a carry level around 400bp it takes a crisis like 2008 or 2020 for the factor to lose meaningful money.

Low volatility is also very helpful for the carry trade. Volatility is very difficult to forecast. While it is broadly mean-reverting, a simple mean-reversion strategy does not do overly well, given that volatility can remain depressed for a very long time before moving higher again. Having said that, one regularity is that fixed income volatility comes off late in the hiking cycle as well as during the time when the Fed is on hold. With FX volatility and US rates volatility highly correlated, this makes the late hiking cycle and the Fed pauses an attractive environment for carry trades.

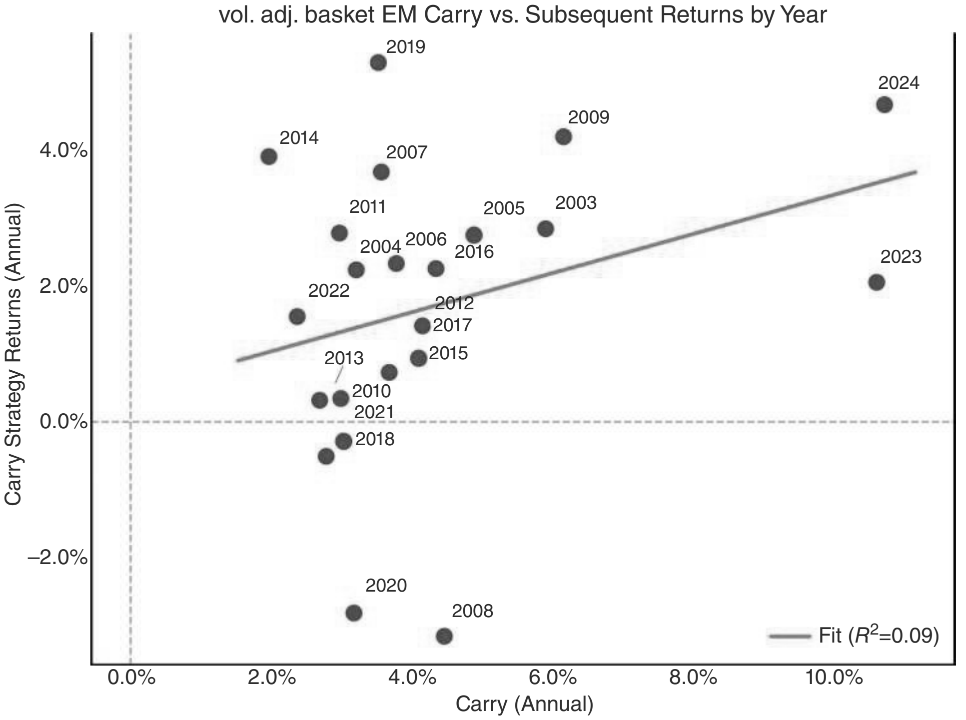
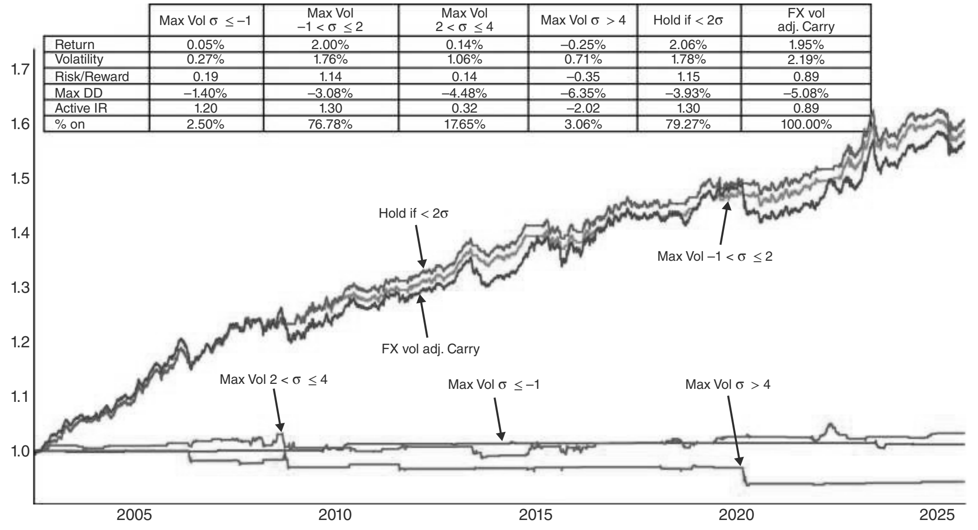


FIGURE 12.18 Size of carry matters.
Source: Bloomberg and authors' calculations.

While we already use volatility for the construction of our carry basket, we also like to use volatility for further adjustments to the carry trades. This can be done by more frequent re-weighting, where rising volatility leads to smaller positions. This is one reason why the daily reweighted FX carry baskets display better performance than the monthly reweighted portfolios, when abstracting from transaction costs. An alternative approach to take volatility into account would be to cut the carry trade when volatility rises by a certain extent. Periods of high volatility do not mean-revert particularly fast. Therefore, spikes in volatility are usually good times to reduce risk, even after the initial spike has already occurred. As already mentioned in the credit chapter, our favorite volatility measure is to take the maximum of the one-year rolling z-score of the implied volatilities of fixed income, FX equity, credit spreads, as well as oil. Taking the maximum makes sure that most crises are captured, even if they take place in a less central asset class. Using this measure to cut out of the carry trade when it spikes above two significantly raises the active information ratio of the carry strategy, from 0.89 to 1.30. The 2 to 4 standard deviation range still has a positive information ratio, but at 0.32, it is significantly lower. For outright negative information ratios, a spike to four standard deviations is needed, which occurs around 3% of the time in our sample (which is clearly not normally distributed, as illustrated in Figure 12.19. Of course, with the information ratio of the strategy generally staying above zero, investors may



	Max Vol σ ≤ −1	Max Vol −1 < σ ≤ 2	Max Vol 2 < σ ≤ 4	Max Vol σ > 4	Hold if < 2σ	FX vol adj. Carry
Return	0.05%	2.00%	0.14%	−0.25%	2.06%	1.95%
Volatility	0.27%	1.76%	1.06%	0.71%	1.78%	2.19%
Risk/Reward	0.19	1.14	0.14	−0.35	1.15	0.89
Max DD	−1.40%	−3.08%	−4.48%	−6.35%	−3.93%	−5.08%
Active IR	1.20	1.30	0.32	−2.02	1.30	0.89
% on	2.50%	76.78%	17.65%	3.06%	79.27%	100.00%

FIGURE 12.19 Carry likes low volatility.
Source: Bloomberg and authors' calculations.

save themselves the transaction costs involved in trading our volatility signal. But at the very least, a four sigma signal deserves to be taken seriously.

Positioning is also a factor that drives forward returns of the carry trade. While positioning is not easy to measure in FX, we are tracking Citi's proprietary flows into and out of the EMFX carry basket. We use the methodology laid out by Saunders et al. (2025b). We take into account both inflows into the long side of the basket and outflows from the short side of the basket. As we prefer the volatility-weighted carry basket, we also weigh inflows accordingly. Analyzing these net flows, we find that forward Sharpe ratios do have a correlation with the percentile of inflows into the carry trade for the previous few months. When inflows by leveraged accounts are above the 60th percentile in the previous month, forward Sharpe ratios fall significantly (though stay positive). This is illustrated in Figure 12.20, and works especially well for one-month forward returns, though in a weaker form, the pattern also holds for longer holding periods. Again, information ratios stay positive, which suggests that investors can also sit through some positioning signals. However, we also found that the worst outcomes tend to occur when both these factors strike at the same time, i.e., when volatility spikes at a time when positioning is heavy. Those events should be taken seriously, even by investors who prefer not to react to every trading signal.

In summary, the carry trade is a true and tested strategy that has stood the test of time. We would implement the trade in EM, with positioning size driven by risk instead of using equal notionals. We would size the trade the largest when carry is high, the Fed is in the low volatility part of its cycle, and inflows have been relatively more muted. But the default case should be to have the trade on. It takes a combination of crowded positioning *and* a volatility spike to undermine the carry trade more significantly.

Carry (vol adj)

Leveraged Flows Percentile	1m Flow percentile	3m Flow percentile	6m Flow percentile
0–20	1.34	0.66	1.26
20–40	1.82	2.71	0.05
40–60	1.67	1.11	1.47
60–80	0.14	0.49	1.38
80–100	0.55	0.57	0.79
<80	1.17	1.17	1.09
Uncond	1.04	1.03	1.02

FIGURE 12.20 Carry works best if uncrowded.
Source: Bloomberg, Citi, and authors' calculations.

12.8 VALUE WORKS TOO

Value is also an attractive factor in FX. Even a simple version of PPP works relatively well. Here, we rank the deviation of FX from its PPP level and then take a z-score. The factor goes long in proportion to the rank of each strategy, in line with the academic literature; see for example, Asness et al. (2013). Restricting the holdings to the top and bottom five currencies gives similar results. The factor is somewhat correlated with carry unless we make Balassa-Samuelson adjustments (Balassa, 1964 and Samuelson, 1964) for higher productivity (via residualizing differences by GDP per capita), but we find that strategy much less effective. Research by the Citi QIS FX team (Pickering and Ismail, 2022) shows that different valuation measures work at different times, but in general, we think it does not add too much value to make the simple PPP framework much more complicated. The additional alpha is quite limited, and PPP works reasonably well for both the EM and the G10 universes. Figure 12.21 shows the results for a broad universe of 22 countries using World Bank data. For FX, an information ratio of 0.48 is relatively strong, especially for a very simple model without much opportunity to overfit.

Just like for carry, for the PPP strategy, crowding is also a headwind. As per Figure 12.22, the forward one-month Sharpe ratio is above 1 when inflows from leveraged accounts were in the lower 60th percentile in the month prior but is negative if inflows were stronger than that. The pattern is also in place for longer holding horizons, but not as pronounced. This is one of the more important use cases for Citi's flow data, as is laid out, for example, by Saunders et al. (2025b).

Valuation is also useful for assessing times when a currency's positioning might be stretched or when intervention is (a) more likely or (b) can be successful—we discuss this in the context of the yen and Swiss franc in a separate section later.

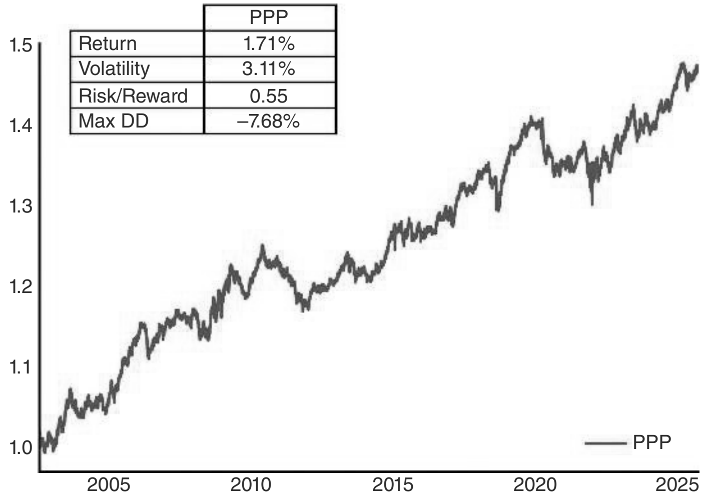


FIGURE 12.21 Valuation (PPP) can add value.
Source: Bloomberg, Citi, and authors' calculations.

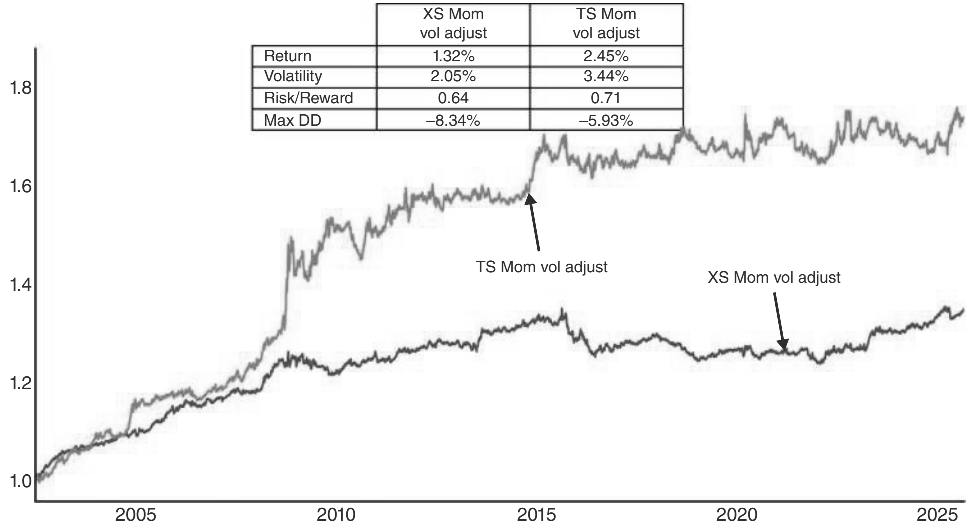
Value (PPP)

Leveraged Flows Percentile	1m Flow percentile	3m Flow percentile	6m Flow percentile
0–20	0.97	0.96	0.52
20–40	1.03	0.24	0.68
40–60	1.19	–0.05	–0.12
60–80	–0.37	0.49	0.35
80–100	–0.60	–0.02	0.21
<80	0.58	0.39	0.33
Unconditional	0.32	0.31	0.30

FIGURE 12.22 PPP also better if uncrowded.
Source: Bloomberg, Citi, and authors' calculations.

12.9 RIDING THE TREND

Given that macro-trading often has a trend following component, and given the popularity of FX in macro-trading, our prior expectation was that trend following should work quite well in FX. However, the truth is more nuanced. Before the financial crisis,



	XS Mom vol adjust	TS Mom vol adjust
Return	1.32%	2.45%
Volatility	2.05%	3.44%
Risk/Reward	0.64	0.71
Max DD	–8.34%	–5.93%

FIGURE 12.23 Momentum trades weaker than during glory days of early 2000s.
Source: Bloomberg and authors' calculations.

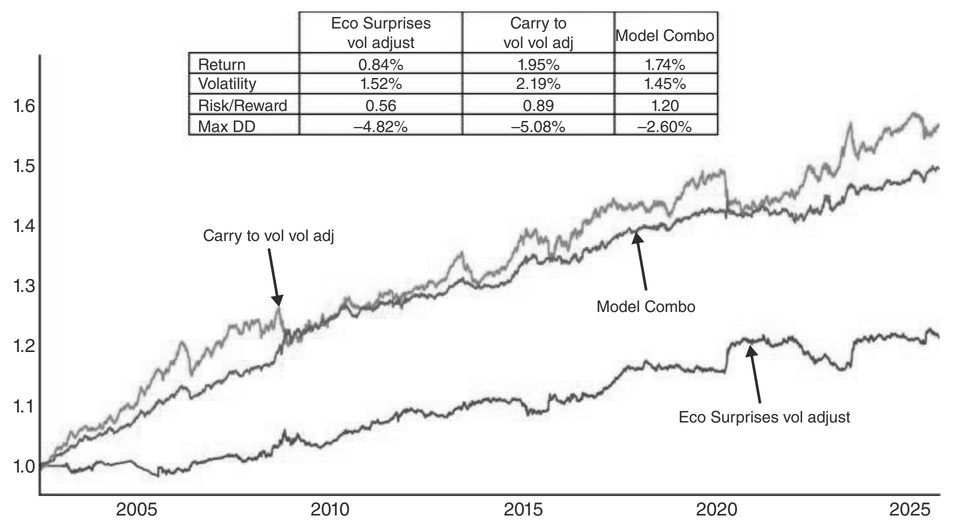
some relatively simple strategies worked well in FX, perhaps contributing to the popularity of FX trading by macro and systematic funds at the time. Post the financial crisis, some of the systematic strategies did not work as well, and the meager returns forced a retracement in stand-alone FX strategies and, in particular, in systematic ones.

In Figure 12.23, we show the classic cross-sectional momentum from the academic literature (see Jegadeesh and Titman, 1993 for an equity-focused version or Asness et al., 2013 for macro). We take a simple strategy where we normalize the last three month total returns for each currency. We then cross-sectionally rank and further volatility-adjust the positions. We implement the strategy weekly, given the high potential turnover. Blending faster and slower, longer periods can also be helpful to reduce transaction costs. This worked well up until 2009, but returns have been noisy ever since. Part of the reason was likely the low dispersion in monetary policy, which in turn was due to the slow recovery from the GFC, leading to fewer opportunities in jumping on the bandwagon of a country or group of countries with a differential macro backdrop. With higher interest rates came higher interest rate dispersion, and therefore, the strategy could make a comeback. Recent performance has been respectable, with momentum clocking in a 0.96 Sharpe since 2022, when the Fed started a hiking cycle in March of that year (with some other central banks ahead of the curve, including in EM). Still, the increase in the availability of data, the easy implementation of the strategy, and the ability for retail investors to take FX positions via brokerage and other accounts mean that we are unlikely to see a comeback of the returns of momentum strategies back to the glory days seen in the mid-2000s.

One area where returns have been more consistent in recent years is in trend following i.e., dollar-directional strategies, which take a position based on the sign of the most recent returns and typically use the dollar as a numeraire. We take the recent three-month performance versus the dollar and go long if it is positive, short if it is negative—again, positions are sized based on volatility. The signal in Figure 12.23 is implemented weekly, which gives very similar performance to a daily implementation. Given that the correlation between exchange rates is typically high, the time series momentum strategies tend to have quite a large long or short USD position. Correlation with the cross-sectional version is quite high, and returns have also been much weaker post the GFC, though overall the returns have been better than for the cross-sectional version.

12.10 HIDE IN SURPRISE

Lastly, we show that Citi's economic surprise index can be a useful diversifier to other FX strategies. The methodology is exactly the same as for the other factors. This time we sort currencies by the level of the Citi's economic surprise indicator. Its stand-alone information ratio comes in at 0.56. Empirically, it has done well in drawdown periods for the carry strategy, as shown in Figure 12.24. This may be just luck, but it could also be a more durable pattern if, in times of risk aversion, investors take solace in countries that are performing relatively well. Combining all factors, despite volatility adjusted carry's superior stand-alone performance, is the route to more consistent returns. In Figure 12.24, we show the performance of the combination of carry, value, trend, and economic surprises, all constructed in a similar way. This combination has delivered an information ratio of 1.20 (before transaction costs), an improvement over carry only.



	Eco Surprises vol adjust	Carry to vol vol adj	Model Combo
Return	0.84%	1.95%	1.74%
Volatility	1.52%	2.19%	1.45%
Risk/Reward	0.56	0.89	1.20
Max DD	–4.82%	–5.08%	–2.60%

FIGURE 12.24 Eco surprises a diversifier.
Source: Bloomberg and authors' calculations.

12.11 SUMMARY

We start laying out the USD smirk theory. Focusing first on EURUSD, we find that relative equity performance is superior to other measures of risk aversion to capture the left-hand side of the USD smirk. Our favorite way to implement the right-hand side is to use Citi's data change indexes in relative terms. Using rate differentials, we find that five-year nominal spreads add the most value. Analyzing curves, bear steepening and bear flattening are most USD positive. Combining the relative equity performance and BTP-Bunds spreads, with the data change indexes and rates, results in a high information ratio for trading EUR. JPY is special in that it relatively consistently appreciates with risk aversion. This means that rate differentials are the key for USDJPY, while relative equity performance is not as helpful. For the riskier currencies, our main framework is the carry factor. Carry usually works, especially when both ranking and position sizing are volatility adjusted. Warning signs are a strong JPY, rising volatility, and heavy positioning. Outside of carry, valuation also works well as an FX factor. Using Citi's economic surprise indexes also adds alpha, while the performance of simple trend following strategies has deteriorated.

CHAPTER 13

How to Trade FX: The Tactics

Having established some of the structural frameworks that we use for FX, we now move on to investigate a few more tactical frameworks. In particular, we go through some of the major events like the start and end of a Fed cycle, or FX interventions, and show some results on backtesting flows and technicals.

13.1 THE FED AND THE USD

Given that the USD is a relative price, it is not surprising that the correlation of the USD with the US business cycle is quite low. Narrowing the lens somewhat and just studying price action around the beginning and end of Fed cycles is more revealing. Figure 13.1 shows the performance around the first cut, starting in 1986. It turns out that the USD goes mostly sideways after the first cut. However, it relatively reliably weakens going into the first cut. To be precise, it falls on average by around 4% during the three months ahead of the first cut. The usual disclaimer is, of course, that it is not trivial to figure out the first cut three months in advance, though at least the final part of this rally is easier to catch.

If we just focus on the start of easing cycles that led to soft landings, we also find initial USD weakness, as can be seen in Figure 13.2. But in the soft landing case, this is followed by a much stronger bounce back of the USD than is the case across all easing cycles. The combination of the US avoiding a recession and, therefore, avoiding more aggressive Fed cuts, typically is USD positive. At the same time, in both 1995 and 1998, there was an ongoing emerging market crisis, which also underpinned the USD. Although some of these events are unlikely to repeat in a similar sequence, weakness in other geographies may be a feature of soft landings, rather than a bug. The basic idea is that the United States is more likely to achieve a soft landing if someone else in the world has a hard landing, and this US outperformance generates USD strength. In the 2024 US soft landing episode, China was relatively weak, and presumably, quickly falling inflation, which made a soft landing possible, would have been much slower if China had not been growing below trend. This may suggest that a strong USD is a typical feature of a soft landing. Given how difficult it is to distinguish soft landings from hard landings early on, it is reassuring that the pattern of USD weakness into the first cut also holds for soft landings. The only issue is that investors have to be more nimble with their USD shorts after the first cut if a soft landing appears likely.

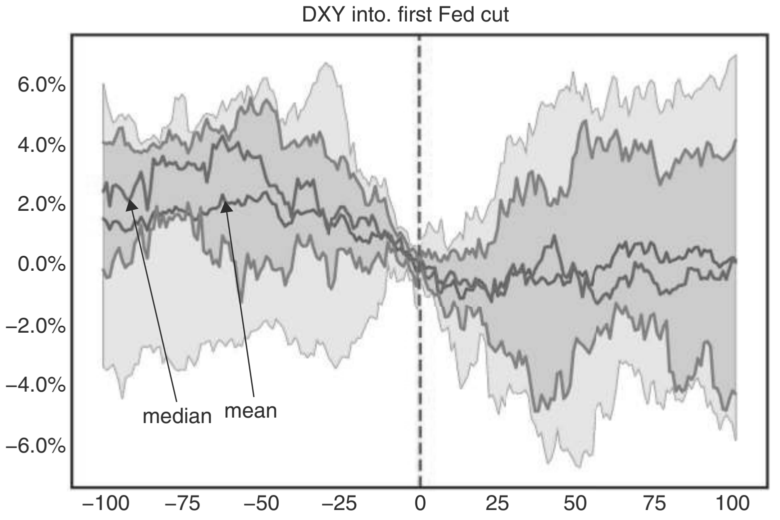


FIGURE 13.1 USD sells off three months into the first cut.
Source: Bloomberg and authors' calculations.

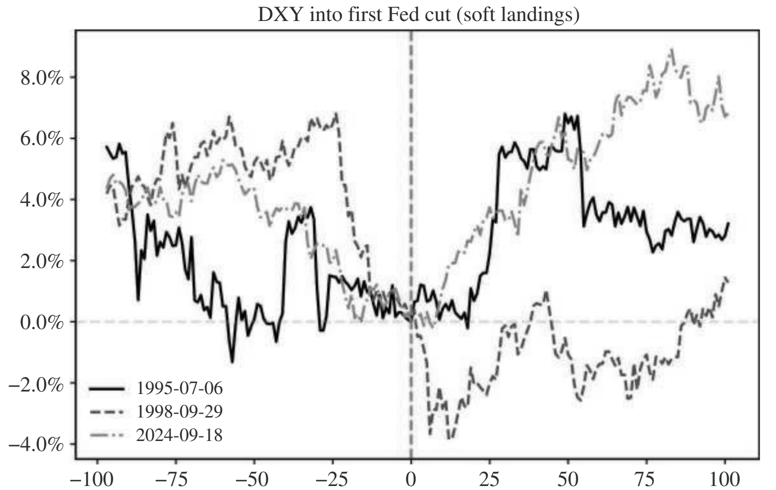


FIGURE 13.2 USD weak into first cut, stronger out of it, in soft landings.
Source: Bloomberg and authors' calculations.

Focusing now on the last cut, the USD trades mostly weaker into it, but then stabilizes and slightly rallies after the last cut, as can be seen in Figure 13.3. The reason may be that, just like in the case of equities and bonds, the uncertainty over additional cuts only gets resolved in the following meeting, when the Fed does not act, and the USD stops weakening in response.

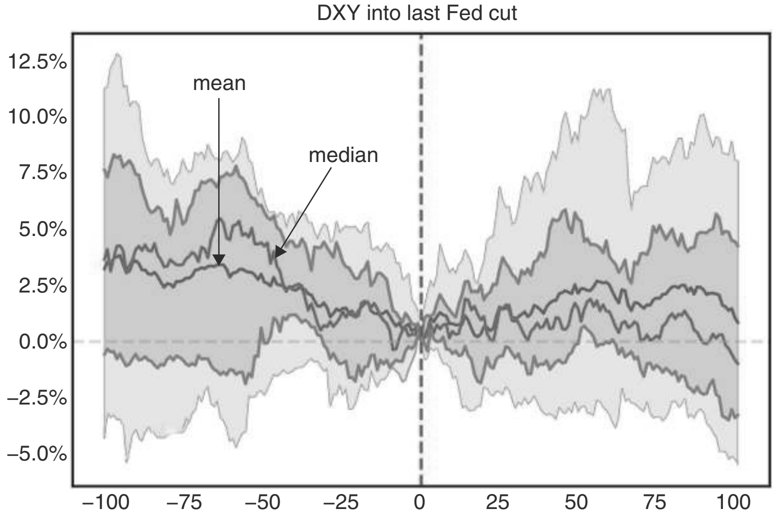


FIGURE 13.3 USD rallies after last cut and into first pause.
Source: Bloomberg and authors' calculations.

13.2 INTERVENTION—BURNING RESERVES, BUYING TIME

Intervention is a big topic in FX. We have already discussed FX intervention at length in the emerging markets book. Our main finding was that there is some short-term success, in the sense of short-term outperformance of the currencies in question following USD selling interventions and vice versa. The impact is stronger when buying USD than when selling USD, and the size of the intervention matters. In G10 FX intervention is less frequent than in EM. The reason is that an overly weak FX carries higher costs for EM than for G10. This is the result of a more significant inflationary impact of a weak FX in EM than in DM, partly because the share in the CPI of various commodities, priced in USD, is much higher. The other reason is that developed markets typically have not issued any meaningful amount of debt denominated in foreign currencies, making it much less likely that a weak domestic currency impacts debt sustainability. On the strong side, many DMs have as much to lose as EMs in terms of deteriorating export competitiveness, and interventions to weaken the FX are more common in G10 FX space too.

FX intervention comes in various forms. In developed markets, the main distinction is between bilateral and multilateral intervention, with the presumption being that the likelihood of success is much higher with the multilateral kind. The other distinction is whether the intervention is sterilized or not. Economic theory suggests that sterilized intervention is extremely unlikely to succeed, while nonsterilized intervention has a higher likelihood of working. The other question is whether there is a difference in terms of intervening against excessive weakness or excessive strength. The motivations are, of course, quite different. Unlike in the emerging market space, where countries more often intervene against excessively weak currencies, usually during a crisis, developed markets intervene more often against an excessively strong currency, typically in a bid to help export competitiveness or to thwart deflationary pressures. More rarely, developed markets intervene against excessively weak exchange rates. But moves have to be quite extensive and often stretch over several years before an eventual impact on inflation spurs policymakers into action. To investigate whether FX intervention "works" the first question is what constitutes "success." Just as in our investigation of EMFX intervention, we mostly care about whether there are profitable trading opportunities upon the announcement of the intervention. We also have a relatively short horizon for our investigation, as even short-term follow-through can be tradable.

In G10, two countries stand out as more intervention-affine. Japan and Switzerland. Next, we investigate Japan, where the BoJ acts as the agent for the Ministry of Finance (MOF) in FX markets. The BoJ has a long history of relatively frequent interventions against an overly strong JPY. As discussed by Takashima (2025), the run-up to an intervention is highly choreographed in Japan. First, MOF states that excessive and disorderly movements in FX are undesirable. This is followed by statements that suggest close monitoring of the FX, and a warning that officials will take appropriate action if necessary. Then the BoJ typically checks rates. The last warning mentions that determined action against speculative moves will be taken. Only after the intervention occurred will officials eventually publicize that they have actually intervened.

Post-COVID, starting in 2022, the BoJ has intervened against "excessive" JPY weakness, given that inflation and the weak JPY had undermined the popularity of the Japanese government. This is only the second time in the more recent market history that the BoJ is buying JPY, with the first time taking place in the aftermath of the Asian crisis. To be precise, the JPY purchasing episodes occurred from November 1997 to June 17, 1998, from October 21 to 24, 2022, and then again in April and July 2024. In Figure 13.4 we show the price action for 1997–1998 next to the SPX, which was one of the key drivers 20 for USDJPY, and also for the 2022 and 2024 episodes, but next

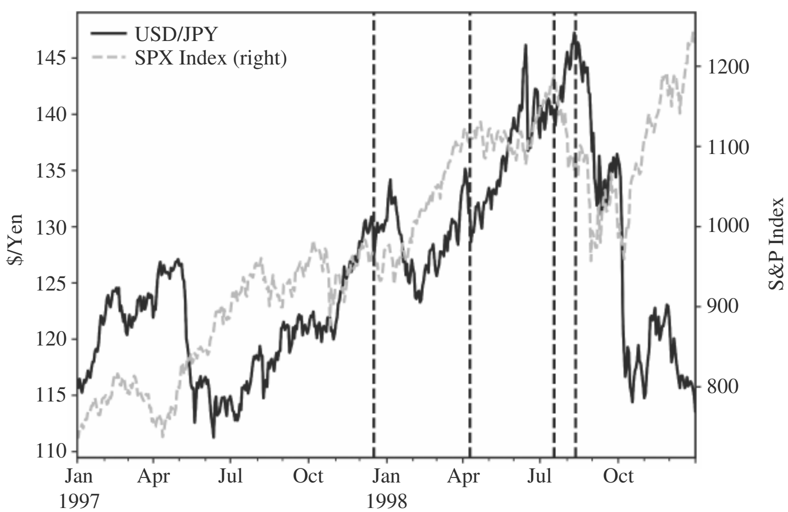


FIGURE 13.4 In 1998, intervention only worked, when fundamentals (here, the SPX) turned.
Source: Bloomberg and authors' calculations.

to US rates, which have more recently been the more fundamental drivers of USDJPY (as JGBs have a much lower volatility than USTs).

During the Asian crisis, USDJPY was under strong upside pressure, especially as concerns about the Japanese banking system became more prominent. In November and December, the BoJ intervened, selling USD, but with little success. December 17, 1997, was one of the larger one-day moves in USDJPY, but it was largely a one-day wonder. The intervention (combined with a fiscal package) on April 9, 1998, had a two-day price impact, mostly because authorities intervened again the following day in large size. But by June, it was clear that USDJPY would not easily turn down in reaction to BoJ intervention. On June 17, 1998, the United States intervened jointly with Japan to strengthen the JPY, which went hand in hand with negotiations between the United States and Japan on reforms that were hoped to end the Japanese crisis. As a result of the intervention, USDJPY fell by more than 5% in one day. But again, it was a one-day move. As USDJPY returned to pre-intervention levels, Japanese authorities (verbally) intervened on August 12, 1998. While this day marked the peak in USDJPY, the main reason for the stronger JPY was a change in fundamentals. The fallout of the Russian crisis had started to impact US markets, sending the SPX sharply lower, especially in the aftermath of the Russian default on August 17. The resulting flight to safety finally strengthened the JPY. As former Japanese Vice Minister of Finance Eisuke Sakakibara noted, what saved Japan, and maybe the rest of Asia, was the Russian crisis (see Sakakibara, 2000). The strengthening of the JPY also led to major stop-outs of large carry-trade positions by hedge funds, exacerbating the move in early October, relative to the SPX moves. Ex post, it appears that joint intervention in June 1998 more or less set the high in USDJPY, even though traders who would have followed the intervention would have lost money chasing it, and may have stopped out, as USDJPY went above pre-intervention levels, though only barely and for only one day. The lesson is that intervention only works in the very short term, and that to be successful, fundamentals have to change too, here measured by the behavior of the SPX.

A similar story unfolded in 2022 and 2024. This time, the intervention was unilateral, as there was no major negative global impact from the weakening JPY. During those years, USDJPY was much more driven by US rates rather than by risk aversion/US stocks. Again, the first intervention in September 2022 was not tradable. The one in October was tradable, largely because fundamentals in the form of US rates turned lower as well, as per Figure 13.5. Again, the move lower in USDJPY was outsized relative to US rates, making USDJPY the better trade, if and when fundamentals are turning, maybe partly because intervention is an additional tailwind and partly because by the time authorities feel the need to intervene to strengthen the currency, it is likely already very cheap, oversold, and investors are positioned the wrong way. This is also discussed by Tobon et al. (2024).

Lastly, the 2024 episode provided similar evidence. The "success" of the intervention of late April 2024 was also a short-lived affair, and USDJPY made new highs shortly afterward. July 11, 2024 saw the next likely intervention, even though it was not fully confirmed at the time. This did mark the local peak in USDJPY, as can be seen in Figure 13.6. But again, that only happened because fundamentals also turned. US rates started to move significantly lower, in part due to the very weak July NFP print. And just like in 1998, once more there was a major stop loss festival going through in

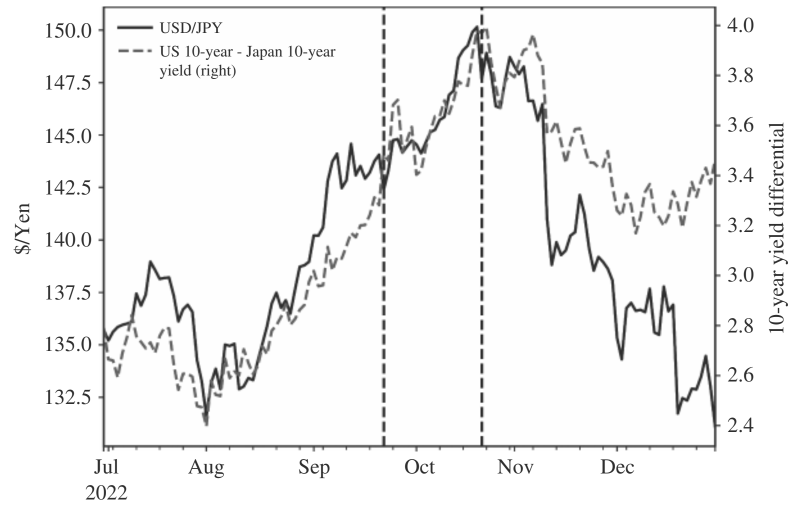


FIGURE 13.5 In 2022, intervention only worked when fundamentals, here US rates, turned.
Source: Bloomberg and authors' calculations.

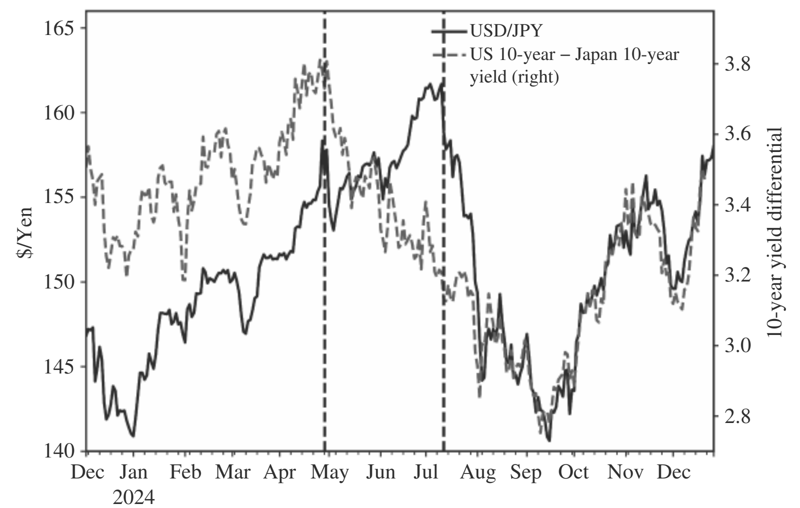


FIGURE 13.6 Same story in 2024.
Source: Bloomberg and authors' calculations.

carry trades, where JPY had been a popular funding currency. However, this time the move was smaller. USDJPY sold off by 13%, over a bit more than two months, while in 1998 it was around 23% in the first leg down. Also, interestingly, USDJPY changed direction again as soon as US rates moved up again. The key is to only go with the intervention if fundamentals also point in that direction. For policymakers, it means that intervention can be useful to build a bridge to the time when fundamentals do change. Otherwise, it will be a bridge to nowhere, and the market will fade very quickly.

13.3 GO WITH THE FLOW OR AGAINST? DO BOTH!

Flows are a cottage industry in FX land, and we have used the proprietary flows from Citi to find use cases. The main question is whether flows are a trend or a countertrend indicator, i.e., should investors go with the flow or fade the flow. The answer is a bit of both, but more of the latter, as can be seen in Saunders et al. (2023). There, we present the evidence to go with the flow. For this, we investigated whether buying currencies in line with the Citi flows is profitable. To be precise, we tested various inflow windows (from a look-back of 5 days to 10 days to 22 days). We did this for the various types of flows that Citi breaks out: flow from leveraged investors, real-money investors, corporates, and banks. We find that the best trend following behavior is for the 10-day bank flow, followed by the 22-day bank flow. This is interesting, as the bank flows are typically somewhat negatively correlated with the coincident price action, which gives following bank flows more of a mean-reversion flavor. Leveraged investors' five-day flows, which are positively correlated with coincident price action, also episodically work, but much less consistently. For all specifications, the pattern became much weaker after the COVID crisis, as per Figure 13.7, for somewhat unclear reasons. We would therefore be cautious in using flow data for trend following.

Although the evidence favoring flow is by no means very compelling, it at least sends a warning signal that investors should not fade the short-term flows. Having said that, the math changes somewhat when we lengthen the time frame of the flows we examine. As per Figure 13.8, fading the three-month flow of leveraged investors has a positive information ratio of 0.26. While this is too low for a tradable strategy, the fortunes of a "fading the flows" strategy drastically improve when volatility rises. Here, we use a combination of equity and FX volatility as the trigger. We test two strategies: moving up above average or above the 70th percentile for a signal. The holding period post signal is a minimum of five days or until volatility drops below the threshold. Using the 50th percentile, the active information ratio improves to 0.7,

	Bank 10d flow	Bank 22d flow	Lev 5d flow
Return	1.39%	0.84%	0.46%
Volatility	1.65%	1.68%	1.68%
Risk/Reward	0.85	0.50	0.27
Max DD	–2.59%	–4.09%	–6.09%

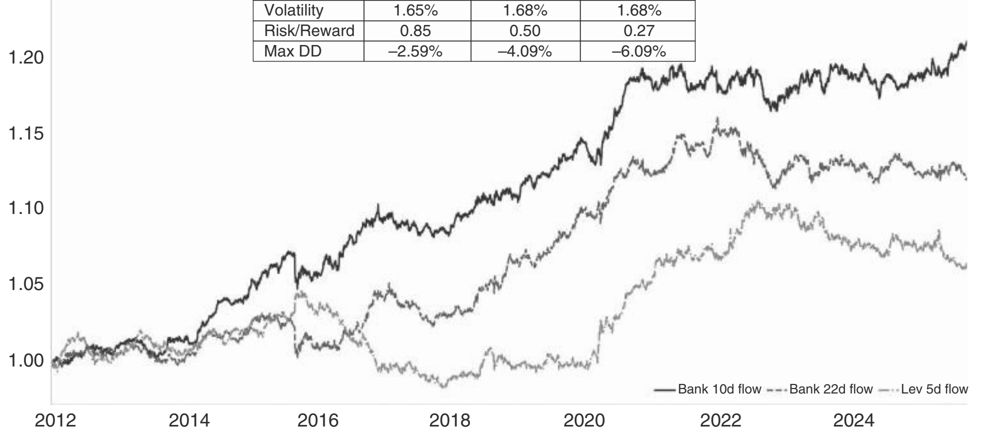


FIGURE 13.7 Following the flow now less profitable.
Source: Bloomberg, Citi, and authors' calculations.

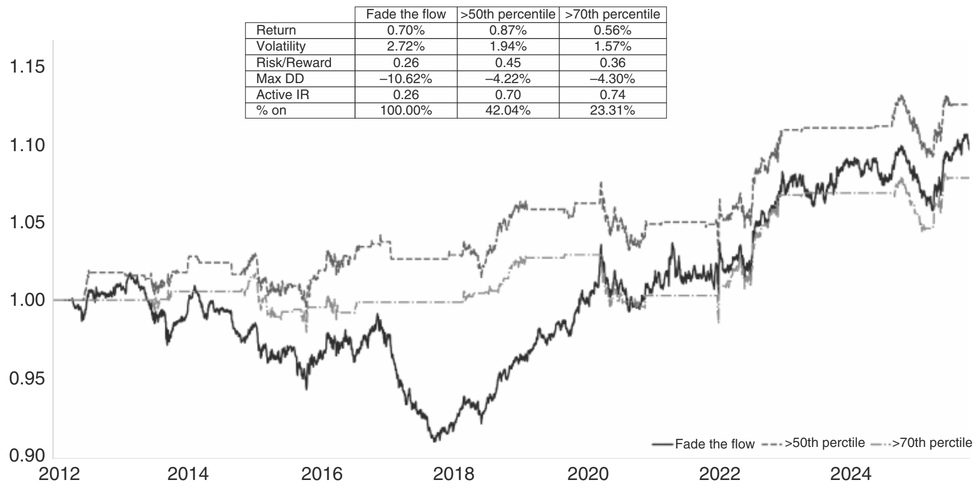
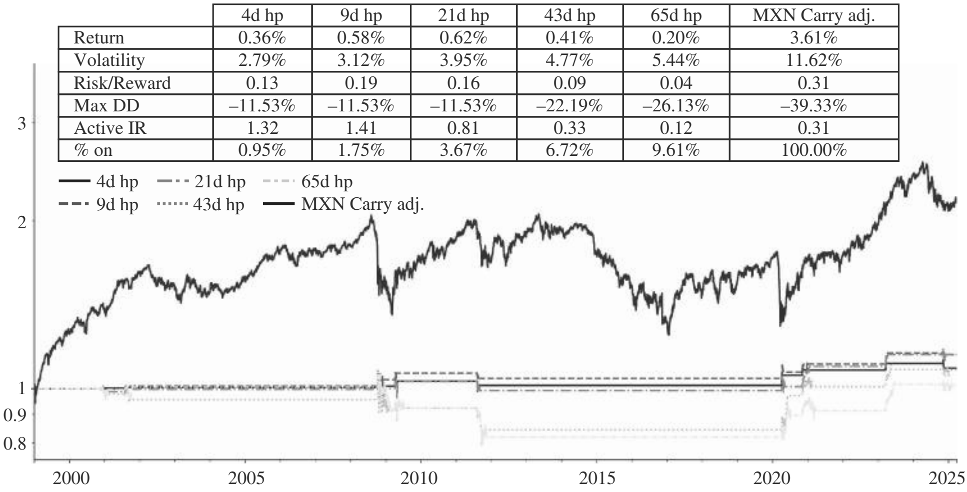


FIGURE 13.8 Fading flows is profitable, if used with a volatility trigger.
Source: Bloomberg, Citi, and authors' calculations.

and for the 70th percentile, it moves to 0.74. For this strategy, the performance has not been flattening out, but is still going strong. The trigger in 2024 was somewhat technical, going into the US election, when implied volatility rose ahead of the event, but without much volatility in spot. The April 2025 event around the "Liberation day" reciprocal tariff announcements was more in line with the underlying philosophy of the strategy (see also Chapter 4)

13.4 TECHNICALS: INTRADAY REVERSALS

There are a myriad of technicals that have been proposed over the years. An exhaustive study backtesting all of them would be fun, but likely also somewhat depressing. Often, technicals seem to be more art than science. We leave that for another book. One technical that seems to work relatively well is big intraday reversals. Although this is not FX specific or currency specific, we have tested it for USDMXN, driven by the observation that USDMXN sold off aggressively the day after the 2024 Trump election, only to end the day stronger. This was seen as a positive sign for MXN, as suggested by Willer et al. (2024a). Fundamentally, the question arose that if USDMXN can not move up on this election result, what can cause a sell-off? The precise study we ran was to analyze intraday sell-offs of 2% or more, where more than 100% of the sell-off is reversed by the end of the day, and USDMXN ends the day lower. Our study found that there is a high information ratio selling USDMXN at the close of the event day and holding it for up to nine days. The information ratio declines, but remains tradable on a one-month horizon. After two months, the information ratio is back to the unconditional one. This is illustrated in Figure 13.9. Interestingly, we also note that in this particular event, USDMXN only had follow-through on the downside for



	4d hp	9d hp	21d hp	43d hp	65d hp	MXN Carry adj.
Return	0.36%	0.58%	0.62%	0.41%	0.20%	3.61%
Volatility	2.79%	3.12%	3.95%	4.77%	5.44%	11.62%
Risk/Reward	0.13	0.19	0.16	0.09	0.04	0.31
Max DD	–11.53%	–11.53%	–11.53%	–22.19%	–26.13%	–39.33%
Active IR	1.32	1.41	0.81	0.33	0.12	0.31
% on	0.95%	1.75%	3.67%	6.72%	9.61%	100.00%

FIGURE 13.9 Intraday recoveries have follow-through.
Source: Bloomberg and authors' calculations.

one day. Part of the reason was a very strong USD environment. Selling EURMXN would have worked on a 10-day horizon. Furthermore, a Trump tweet about 25% tariffs on Mexico 14 days later, resulted in similar price action, in that a one-day spike higher in USDMXN was quickly reversed. The intraday high in USDMXN on election day was not conclusively taken out in this cycle. Clearly, the market had priced enough Trump risk for MXN already.

13.5 NO BASIS FOR THE BASIS

One interesting question is how funding markets play into FX. The FX OIS basis is a measure of the cost of obtaining US dollar funding relative to other currencies in the foreign exchange swap market. It represents the spread between the implied USD interest rate from FX swaps and the actual USD OIS rate. This basis typically reflects the relative demand for US dollar funding and is a barometer of global liquidity conditions and cross-currency funding stress. In theory, it could be expected that a widening basis would be USD positive because it signals that USDs are scarce. And to the extent that it coincides with rising risk aversion, which may cause investors to close positions funded in USD, a wider basis should also be correlated with a stronger USD. However, on average, there is no strong correlation, as per Figure 13.10. Maybe that is because at times the basis widens due to regulatory factors, or because of certain hedging flows or structured product flows that are going through the market. At the risk of ending this chapter on a downer, we conclude that, for directional trading, focusing on the FX basis is not overly useful. Instead, hedge funds typically trade the basis directly. For more detail, see, for example, Williams et al. (2025).

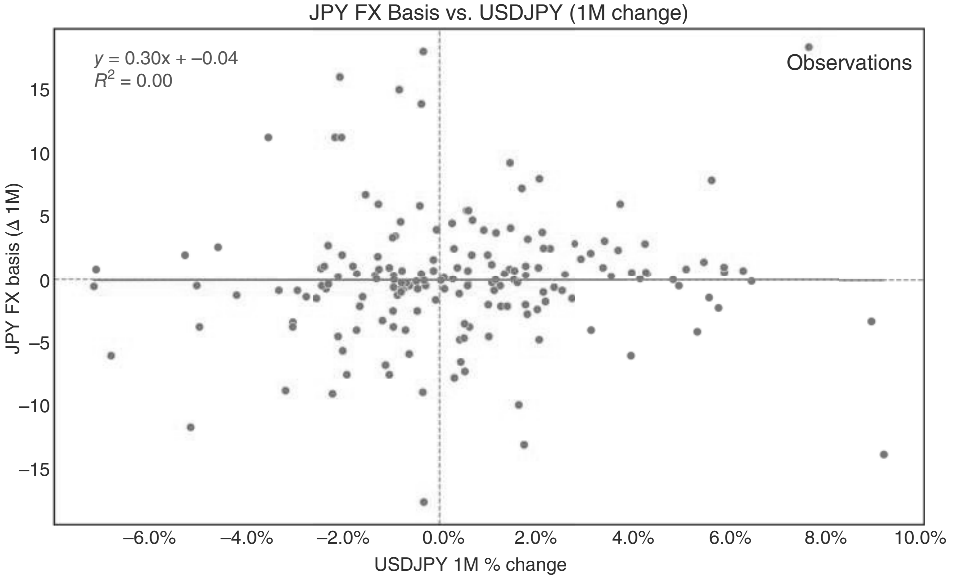


FIGURE 13.10 Basis barely matters for spot.
Source: Bloomberg, Citi research, and authors' calculations.

13.6 SUMMARY

Tactically speaking, we find that the USD does not have a strong relationship to the Fed cycles but does usually weaken into the first cut. The cuts that result in soft landings are USD positive. The USD also tends to strengthen after the last cut. In terms of FX intervention, we focus on Japan, as one of the few countries in G10 space that intervenes more frequently. Broadly speaking, intervention only buys time. It can be a bridge to fundamental change, but it will not change the direction by itself. Investigating FX flows, we find that there are short-term momentum effects, but the better strategy is to fade recent flows whenever a volatility shock hits. Some other technicals are also promising in FX, with large intraday reversals performing well in USDMXN, for example. The FX basis typically has no major impact on FX in a directional sense.

CHAPTER 14

How to Trade Commodities

Commodities are a very important asset class for global macro-trading. This is true for several reasons. First, as already discussed, there is a very direct impact from commodities on inflation, which is key for the trading of inflation itself and for the trading of government bonds more generally. Commodities also impact currencies through their effect on the terms of trade. The reason why commodities are more important than other traded goods for inflation or for trade flows is largely that commodity prices tend to be more volatile than prices of noncommodities, which means that their impact on inflation or on the terms of trade is outsized relative to their weight in the relevant indexes. Second, commodities can help macro investors determine where we are in the business cycle, and commodities are therefore important to generate trading signals even for asset classes that are not as directly impacted by commodities, like bonds and currencies. Lastly, commodities are also an important asset class to invest in, largely because they have important diversification characteristics. Especially during regimes of stagflation, oil can be an important part of a macro portfolio, as it can protect portfolios during such regimes. Gold can play a similar role. However, as we discuss later, the role of gold as an inflation hedge is somewhat nuanced.

14.1 COMMODITIES: THEY HAVE THEIR MOMENTS

For our full data set starting in 1975, we find that most commodity indexes have negative Sharpe ratios, while both the SPX and US bonds have significantly positive Sharpe ratios. This is illustrated in Table 14.1, which shows the return of futures indexes for commodities and excess returns over T-Bills for the other asset classes. For some commodities, we only have index data since the late 1980s. For copper and energy, the Sharpe ratio from then onward is actually positive, but barely so, especially for energy, and much below what both the SPX and government bonds had to offer.

The precise performance of commodities over the long term depends crucially on the starting point of the analysis. Starting in January 1970, instead of 1975, the performance looks better, but irrespective of the precise starting point, Figure 14.1 shows that at any point from the 1980s onward, the commodity performance has been poor, even more so relative to the SPX. Clearly, the inflationary 1970s did wonders for

TABLE 14.1 Long-term Performance Not Great

Asset class	Annualized return	Volatility	Best year	Worst year	Max drawdown	Sharpe ratio	Period
S&P 500	7.41%	17.11%	32.27%	–38.29%	–60.75%	0.49	1975–2024
US 7–10Y bond	1.87%	7.17%	18.32%	–15.95%	–37.58%	0.29	1975–2024
BCOM Agriculture	–2.03%	16.38%	38.30%	–31.86%	–85.87%	–0.04	1975–2024
BCOM Gold	0.60%	19.09%	109.97%	–42.46%	–92.79%	0.13	1975–2024
BCOM Energy	–0.99%	29.74%	107.64%	–48.67%	–97.01%	0.12	1988–2024
Copper Future	0.67%	25.29%	128.70%	–54.48%	–70.69%	0.15	1988–2024

Source: Bloomberg and authors' calculations.

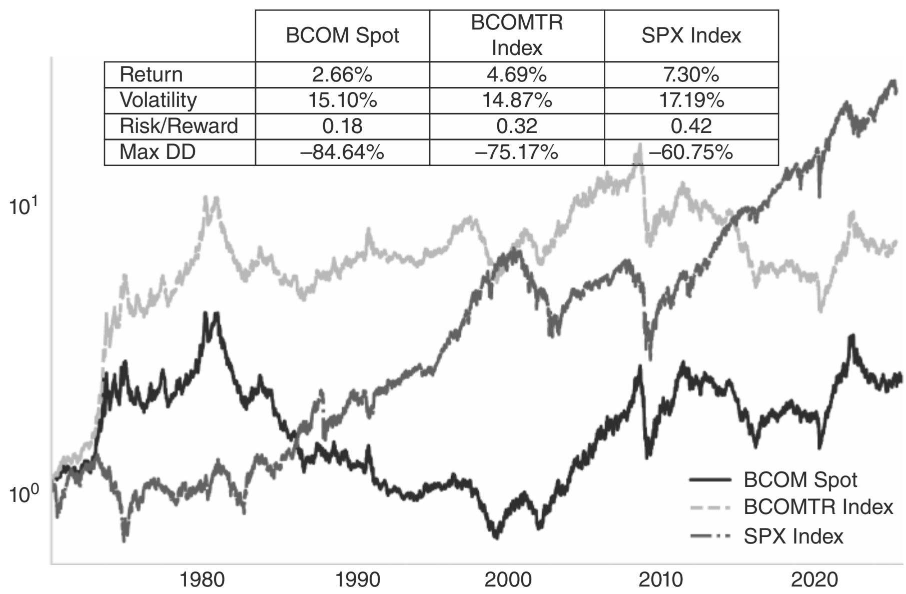


FIGURE 14.1 Commodities have lagged S&P since the 1980s.
Source: Bloomberg and authors' calculations.

commodities, while at the same time undermining equities. However, the all-time high of the Bloomberg index was set in mid-2008, as the so-called commodity super cycle was going strong, before smacking into the GFC in mid-2008. Even at the time of writing, and in spite of the inflation scare of 2022, the Bloomberg commodities total return index is still 50% below the peak set in 2008, while the S&P is up more than 150% over the same period. Commodities are therefore unlikely to be a good buy-and-hold investment, which makes them different from bonds and equities. Intuitively, this makes sense. Both commodities and equities benefit to some extent from inflation, especially when it accumulates slowly over time, pushing up most prices. High inflation benefits commodities more because very high inflation leads to a multiple contraction for equities. But equities also benefit from productivity improvements, whereas commodities generally get hurt by those productivity improvements that lower marginal costs and increase production of commodities. It should therefore not be surprising that in the long term, equities outperform commodities.

One reason for the poor performance of the Bloomberg commodity total return index is the negative roll yield, which became a major factor after 2004. In contrast to the total return index, the Bloomberg spot index is well above its 2008 highs. What changed? The financialization of the commodity market took place in 2004–2005. As investors noticed the appetite for commodities by China, pension funds, endowments, and other institutional investors began systematically allocating to commodities, and the GSCI and BCOM saw major inflows. As most investors wanted exposure via rolling indexes, they typically concentrated on the front end, keeping curves in contango. The negative roll yield is especially a problem with the energy complex, where both oil and natural gas often trade in contango, suggesting that the future position rolls down the curve, leading to a negative yield. This can be seen in Table 14.2, where the total return for energy is negative from 1992 to 2025, while spot is up 4.5% p.a. Although spot commodities are difficult to hold for investors, the large difference between spot and total return would suggest that, at least for some commodities, owning physical commodities in the portfolio could be preferable. Since 2003, investors have had access to an ETF that holds physical gold. However, the main oil ETFs hold futures, given the high storage cost of oil. Copper has both. There is a physical copper ETF, though management fees are high at more than 2%, given high storage costs. We will therefore mostly focus on futures.

TABLE 14.2 The Roll Is Part of the Problem

Asset class	Annualized return	Volatility	Best year	Worst year	Max drawdown	Sharpe ratio	Period
BCOM spot	6.34%	14.72%	68.16%	–28.92%	–53.18%	0.48	1970–2025
BCOM total return	8.22%	14.61%	128.47%	–35.65%	–73.25%	0.60	1970–2025

(*Continued*)

TABLE 14.2 The Roll Is Part of the Problem (*Continued*)

Asset class	Annualized return	Volatility	Best year	Worst year	Max drawdown	Sharpe ratio	Period
BCOM precious spot	7.27%	18.09%	43.77%	–30.30%	–51.63%	0.47	1992–2025
BCOM precious total return	6.76%	17.98%	42.66%	–30.80%	–53.04%	0.44	1992–2025
BCOM energy spot	4.48%	29.34%	79.86%	–40.96%	–82.45%	0.29	1992–2025
BCOM energy total return	–0.80%	29.10%	120.36%	–47.33%	–96.41%	0.12	1992–2025
BCOM industrial metals spot	3.77%	20.73%	89.35%	–46.94%	–64.78%	0.28	1992–2025
BCOM industrial metals total return	4.27%	20.27%	79.98%	–48.27%	–66.84%	0.30	1992–2025

Source: Bloomberg and authors' calculations.

Given the unimpressive long-term performance of commodities, the question is when commodities are needed in a portfolio context. Two episodes come to mind: the supercycle of the early 2000s, as well as the inflationary 1970s.

Moment 1: When Commodities Were Super

The super commodity cycle lasted from 2000 to 2008 and was marked by a broad-based increase in the prices of various commodities, including crude oil, base metals, precious metals, as well as agricultural commodities. The key driver was the rapid expansion of demand from emerging markets in general and China in particular. China had embarked on the process of a rapid and broad industrialization and urbanization, which massively increased demand for raw materials, both for housing as well as for infrastructure like roads, bridges, and railways. The rapidly expanding manufacturing sector also added to the demand for raw materials, although some of it was just substituting for what would otherwise have been used in the incumbent manufacturing centers in Europe and the United States. In the early 2000s, China grew more than 10% per annum in real terms, and its consumption of base metals grew from 3% to 40% of total global demand. Other large emerging markets added to the demand, and it was the time when the term "BRICS" was first coined (Brazil, Russia, India, China, and South Africa).

On the supply side, the situation was made worse by years of underinvestment. This was a consequence of the Asian crisis in 1997–1998, which had depressed demand for commodities, leading to low CapEx. Furthermore, the tech bubble late in the 1990s coincided with the Asia crisis to drive investments to a much larger degree into the new digital economy and away from the old, commodity-heavy economy. As usual, during bull markets, supply-side disruptions further added to the bull moves. This was the case in oil, where instability in the Middle East and Nigeria added a risk premium.

In the latter part of the resulting bull market, demand increased further due to the broad financialization of commodities. On the back of strong returns and relatively lower correlations with equities, commodities were increasingly advertised as a distinct asset class, attracting significant investment flows from institutional investors, adding fuel to the bull market.

The last contributing factor was the weakness of the USD. Between 2002 and 2008, the DXY lost more than 40% of its value. Although the relationship between the USD and commodities is complex, with causation not being overly clear, it is quite likely that the weak USD was at the very least a factor reinforcing commodity upside.

Soft commodities were also swept up in the madness, even though the short cycle times of softs, in theory, allow for a speedier supply response. But the higher oil price led to increased demand for biofuels, pushing up prices for soft commodities. This cost inflation, driven to a large degree by energy prices, was a general feature of this super-cycle, as it also impacted the marginal costs for base metals. The inflows into commodity indexes in the latter part of the bull market also helped all commodities, increasing the correlation.

The super commodity cycle of the early 2000s eventually came to an end in 2008, when the GFC severely depressed growth and demand for commodities. However, it took some time. The peak for commodities was only set in July 2008, well after the United States had entered a recession in late 2007 and after the demise of Bear Stearns in March 2008.

Moment 2: When Commodities Were Groovy

Moving to the 1970s, the Bloomberg total return index recorded impressive annualized returns of approximately 24%. As the 1970s were also characterized by high inflation, the standard narrative is that commodities are good hedges against inflation. However, the direction of the causality between commodities and inflation is not overly clear, given how important the oil shock was as a source of inflation. We therefore check a simple rule that goes long commodities in response to inflation. The condition we use is that inflation has to be high (above 3%) and has to go up (here implemented as the CPI being above its 5mma). Those rules clearly capture the 1970s bull market but do overstay their welcome, as they stay broadly long until the 1990s. The rule captures a part of the 2022 bull market too, and here the exit was much better. Overall, the unconditional information ratio moves from 0.32 to 0.66 and is only switched on 22% of the time, as per Figure 14.2. Not superb, but it is better than many other assets in environments of high inflation.

	CPI > 5mma	CPI > 3%	Both	Unconditional
Return	1.51%	3.24%	2.53%	4.69%
Volatility	10.97%	10.37%	8.05%	14.87%
Risk/Reward	0.14	0.31	0.31	0.32
Max DD	–80.28%	–58.65%	–54.66%	–75.17%
Active IR	0.20	0.45	0.66	0.32
% on	45.07%	49.04%	22.41%	100.00%

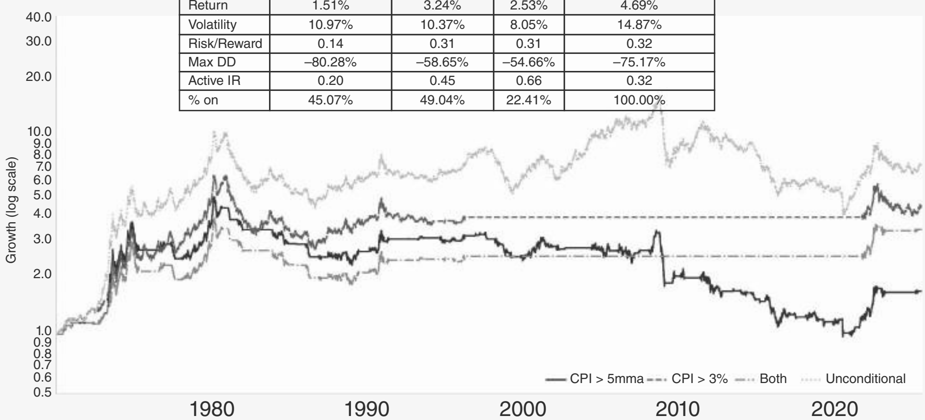


FIGURE 14.2 Commodities perform well in inflationary times.
Source: Bloomberg and authors' calculations.

The other point to make is that the 1970s were special because they were marked by the break of two important pegs. The first peg to break was in oil. In October 1973, OPEC imposed an embargo on oil exports to Western countries that supported Israel during the Yom Kippur War. By January 1974, this embargo, combined with production cuts, led to an increase in oil prices of almost 400%. Before this crisis, oil had been mostly pegged for decades, with only minor price fluctuations. A depegging event usually creates much more volatility than a continuous adjustment, as assets can become deeply under- or overvalued during the time of the peg, and buyers and sellers start to take the peg for granted, accumulating large positions where the financial logic is based on the peg persisting. Depegging events are therefore highly destabilizing as large positions, accumulated over decades, are suddenly off-site. The 1970s saw another oil crisis, when the Iranian Revolution in 1979 led to a disruption in oil supplies from Iran. Oil prices rose by another 400% by 1980. While this was not a depeg per se, going into the Iranian revolution, the spot market was very small, and oil was largely sold using long-term contracts, keeping prices relatively stable. The impact of the Iranian revolution, therefore, rivals the Yom Kippur War.

Second, the 1970s also saw the depeg of gold, or rather the depeg of the USD from gold. In the Bretton Woods system that had been created after the Second World War, the key currencies were pegged to the USD, which in turn was pegged to gold. However, given high spending pressures, by the late 1960s, the US gold reserves were insufficient to back the USD fully. The high inflation and unemployment of the 1970s created speculation that the USD peg to gold would break. In August 1971, President Nixon announced the suspension of the dollar's convertibility to gold. This meant that foreign governments could no longer exchange their US dollars for gold. Initially, there were still attempts to manage the exchange rates at fixed but adjustable levels, but by 1973 speculative pressures and widening balance of payment imbalances made these attempts unsustainable. The Bundesbank and several other European central banks abandoned fixed exchange rates against the USD in early 1973. Overall, the USD fell by 37% against the DEM between 1971 and 1973—no wonder that this was a heady time for commodities in USD terms. Given that Nixon's action implicitly devalued all fiat currencies, the impact on gold was very substantial. It appreciated by more than 300% between 1970 and the end of 1974. Even though Nixon implemented a 90-day freeze on wages and prices jointly with the end of convertibility to combat inflation, inflation rose to 12% in late 1974, given rising prices for imported goods and rising inflation expectations. Inflation was not helped by a 10% import tariff to incentivize consumers to buy locally to get the trade deficit under control.

It therefore stands to reason that both the inflation and the performance of commodities in the 1970s were a result of two important commodity pegs breaking. Given the more flexible pricing mechanisms in place for commodities since then, a repeat seems very unlikely, at least in this magnitude. It is therefore no surprise that the inflation shock in 2022–2023 was smaller and more transitory than in the 1970s. Gold in particular stopped working as an inflation hedge in mid-2023, when the Fed turned hawkish, before breaking out again in early 2024, when the Fed turned dovish. And oil moved up strongly with the Russian invasion of Ukraine, but reversed these gains relatively quickly too. The performance of commodities in the 1970s may therefore be somewhat misleading when estimating the optimal weight for commodities in portfolios.

14.2 COMMODITIES AND THE BUSINESS CYCLE

Outside of the two big moments for commodities, the business cycle and commodities interact in a less volatile way. For example, and unsurprisingly, commodities outside of gold tend to trade well during Fed hiking cycles. This is obvious for the case of base metals in general and copper in particular, given that growth must be sufficiently strong for the Fed to hike, leading to strong demand for base metals and energy. This is not the case for precious metals, where higher rates are typically a drag. However, overall, commodities that are pro-cyclical, like base metals and energy, more than offset the few commodities that could be considered more countercyclical, like gold. This is shown in Table 14.3. While the table confirms a clear prior that hiking cycles are good for the commodities outside of precious metals, there were two outliers. First, in 1997, the Fed did hike, but it was a solitary hike that then led to cuts in the aftermath of the Asian and Russian crises. Clearly, commodities were not able to go up in that negative global growth environment. And then there is the 2022 episode, where hikes were not driven by a positive outlook on the business cycle, but by inflationary fears. The surprise was not that commodities performed poorly, but that gold did not perform better. We will have more to say about that later.

Recessions are often a major headwind for commodities. And just like for equities, commodities bottom *after* the recession has started, and not before, at least outside of the 1970s, already discussed earlier, or the 1990 recession that included the Iraq war supporting energy. The other recessions of 1980, 1981, 2001, 2008, and 2020 were all commodity negative. This is illustrated in Table 14.4, which shows the lows between three months before a recession and the end of the recession. On a headline level,

TABLE 14.3 Commodities Do Well During Hiking Cycles

Tight start	BBG commodities	Crude (WTI) future	Copper future	Gold future
1986-Dec-16	14.1%	25.3%		9.9%
1988-Mar-29	10.2%	29.2%	–0.9%	–25.2%
1994-Feb-04	6.7%	23.7%	54.6%	–9.7%
1997-Mar-25	–15.3%	–41.3%	–28.3%	–22.7%
1999-Jun-30	28.5%	113.7%	2.3%	0.5%
2004-Jun-30	11.7%	49.9%	35.2%	6.6%
2015-Dec-16	12.2%	2.6%	25.4%	9.3%
2022-Mar-16	–10.6%	–10.7%	–12.3%	–9.1%
Median	11.0%	24.5%	2.3%	–4.3%
Average	7.2%	24.1%	10.8%	–5.0%

Source: Bloomberg and authors' calculations.

TABLE 14.4 Commodities Bottom Well After the Start of a Recession

	SPX draw-down	SPX trough days	BCOM draw-down	BCOM trough days	Gold trough days	Energy trough days	Base trough days
Nov-1973	–46.0%	336	–29.0%	–2			
Jan-1980	–17.4%	86	–21.0%	–61			
Jul-1981	–34.4%	407	–39.6%	460			
Jul-1990	–20.8%	102	–20.6%	–16	236	4	207
Mar-2001	–32.1%	204	–23.2%	245	32	245	245
Dec-2007	–56.6%	464	–57.3%	457	–100	445	450
Feb-2020	–34.0%	51	–31.1%	46	–81	86	51
Average	–34.5%	236	–31.7%	161	112	195	238
Median	–34.0%	204	–29.0%	46	–24	166	226

Source: Bloomberg and authors' calculations.

using BCOM, the sizes of the sell-offs are comparable to the SPX sell-offs. On average, commodities bottom after the start of the recession, just like equities, but earlier than the SPX, with the median being only 46 days after the start of the recession. However, excluding 1973 and 1990, the lag is much closer to the equity lag, at around 245 days. This is interesting, as the physical aspect of commodities would suggest that they are finding it more difficult to "look through the valley," which would be in line with much longer lags than is the case for equities. One of the reasons is that gold has worked in some recessions as a hedge, bottoming before the recession began. While we only have the history for total return indexes of energy and base metals since 1990, using that data shows that gold bottoms in the median case 24 days before the start of the recession, while base metals bottom 226 days afterward, slightly after equities. Energy is closer to metals than to gold, but somewhere in between. This is in line with the view that base metals are most directly impacted by a recession. Although there is a negative demand impact for energy too, supply-driven energy spikes have contributed to some recessions, making the correlations less straightforward.

Next, we investigate the interplay between gold and copper in more detail. Gold relative to copper does quite well around recessions, both into them and out of them, as per Figure 14.3a. When the recession ends, this performance quickly unwinds, though, with gold underperforming, as per Figure 14.3b.

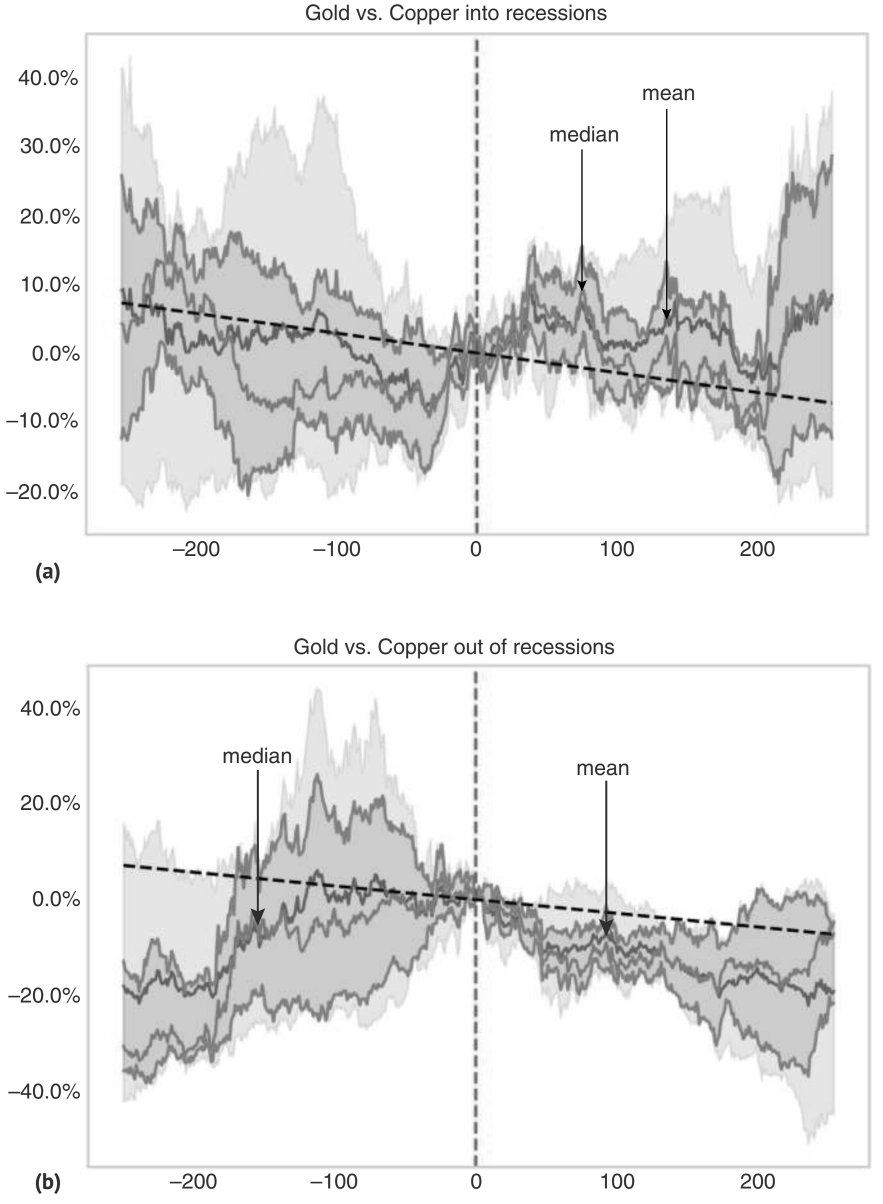


FIGURE 14.3 (a) Gold outperforms copper into recessions. (b) Gold underperforms copper out of recessions.
Source: Bloomberg and authors' calculations.

	Trend 22d MA	Trend 66d MA	Trend 200d MA	Trend 252d MA	Combo
Return	6.95%	6.71%	6.01%	7.44%	6.78%
Volatility	10.68%	10.98%	10.91%	10.87%	9.11%
Risk/Reward	0.65	0.61	0.55	0.68	0.74
Max DD	–21.27%	–26.94%	–31.47%	–30.01%	–23.13%

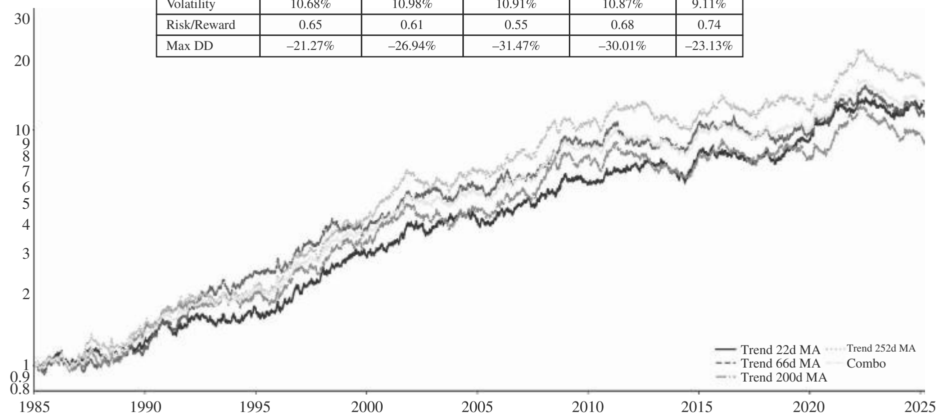


FIGURE 14.4 Trend-following is profitable.
Source: Bloomberg and authors' calculations.

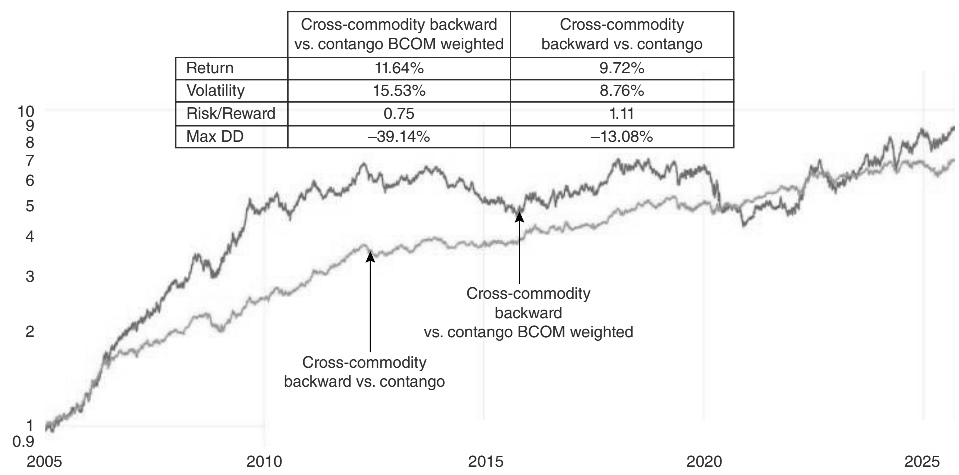
14.3 RIDING COMMODITY TRENDS

Trend following works relatively well in commodities, though, like in other asset classes, performance has somewhat deteriorated since the early 2000s. The strategy we investigate is quite basic: we take long or short positions across 18 commodity contracts based on whether they are above or below the x-day moving average. Positions are volatility-adjusted, and we target an ex ante volatility of 10%. As we use futures positions, returns are in excess of cash. We break out the strategy performance across short to long-term moving averages and also show the combination portfolio. This is shown in Figure 14.4, where most moving averages work quite well, and a combination of the various moving averages leads to a Sharpe ratio of 0.74.

14.4 CARRY IN THE GROUND

We now dig deeper into the roll yield. As commodities don't pay a coupon or a dividend, the implicit carry comes only from the shape of the curve. If the curve is upward sloping (in contango), a three-month contract after one month becomes a two-month contract. Assuming the curve remains unchanged over that month, the three-month position loses money rolling down the curve. Although in theory, the curve should change to offset these losses, in practice, this is not necessarily the case, just as in FX space, the FX depreciation typically does not fully offset the higher interest rate of the higher-yielding FX. In theory, the commodity curve should be determined by storage costs and interest rates such that buying the commodity and selling it in the future pays investors exactly the risk-free rate after storage costs, but in practice, short-term supply and demand factors can impact the short end of the curve, while hedging programs of energy producers and energy consumers can determine the backend. The cross-commodity carry strategy makes use of the fact that forwards are often a biased predictor of prices in the future. To be precise, the cross-commodity carry goes long the most backwardated markets and short the markets the most in contango across 7–10 futures markets. As can be seen in Figure 14.5, the information ratio is a respectable 1.11, where positions are volatility adjusted (but the carry is not volatility adjusted for the ranking of the carry across commodities). If positions are weighted in line with weights in the BCOM, the information ratios are lower, but still respectable.

Another carry strategy focuses on curve carry. The typical curve carry strategy captures "roll-down" in contango curves and roll-up in backwardation. This is done by taking short positions in the front of the curves and long positions in the back end of the curves. For concave curves in contango, the front end roll down from short positions compensates for the move in the shallower back-end contract. In backwardated markets, the long position further out the curve generates a larger P&L than the short position closer to delivery. The standard curve carry across commodities is exposed to more volatile and seasonal spreads. Curve carry is a USD-neutral strategy that is short front-end versus long back-end curves in equal amounts. The risk-adjusted curve carry sizes the positions for each commodity curve by its risk/reward (i.e., carry / volatility), such that all curves pack the same punch and sizes the front- and back-end weights so they are beta neutral. Figure 14.6 shows both variations, which deliver information ratios of 1.04 and 2.04, respectively. When we get more granular by commodity, we will show how the contango versus backwardation rule works in a time series context for single commodities.



	Cross-commodity backward vs. contango BCOM weighted	Cross-commodity backward vs. contango
Return	11.64%	9.72%
Volatility	15.53%	8.76%
Risk/Reward	0.75	1.11
Max DD	–39.14%	–13.08%

FIGURE 14.5 Backwardation is profitable.
Source: Bloomberg, Citi, and authors' calculations.

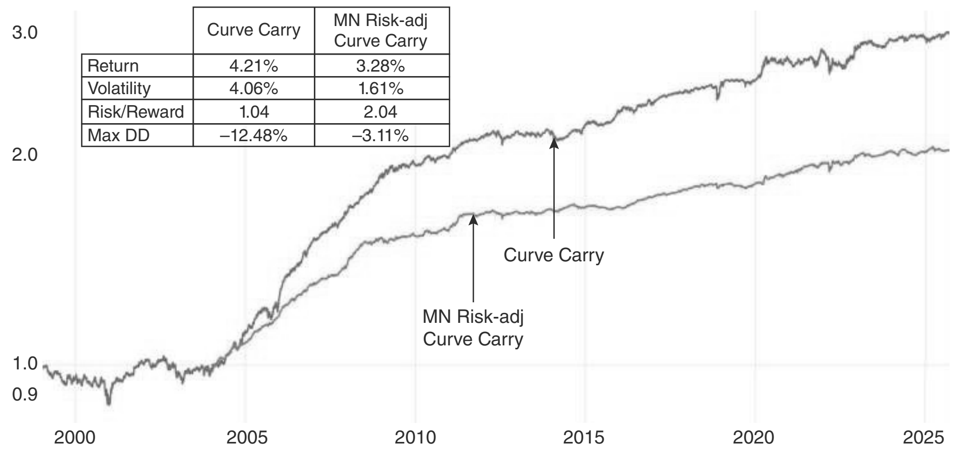


FIGURE 14.6 Carry is profitable—somewhat.
Source: Bloomberg, Citi, and authors' calculations.

14.5 BOTTOM'S UP

In terms of analyzing commodities from a more bottom-up perspective, it is important to realize that commodities are somewhat different from other assets, as supply and demand matter more. While supply and demand are also important for financial assets, for physical assets, such as commodities, both demand and supply can be much more inelastic than is the case for financial assets, where smaller changes in prices can bring markets back into equilibrium. Furthermore, supply is generally more volatile than it is for financial assets, as physical production problems, strikes, geopolitics, or weather can make a big difference, while the net supply of equities (IPOs, buy-backs) or bonds (issuance, buy-backs) is relatively smooth. The fact that many commodity producers are located in more politically and economically unstable parts of the world adds to the high likelihood of supply shocks. Demand shocks are typically slower-moving, linked to global growth in general or to the growth of one particular industry more specifically, which tends to play out over a longer period. Although demand shocks can sometimes be very significant, and volatility-inducing too, maybe most extremely so in the COVID crisis of 2020, demand side shocks are more comparable to what can be observed in financial assets.

Given this focus on demand and supply, it is no wonder that the key to getting the fundamentals right is to model the likely demand and supply numbers. This then results in a forecast surplus or deficit. To what extent a commodity price then reacts to a surplus or deficit situation depends on how plentiful inventories are at the time. Inventories consist of those held at the exchanges (globally, including China), which are the most easily observable, but also of trade inventories, which are held along the production chain by producers, wholesalers, or consumers, and strategic inventories, often held by governments or sovereign wealth funds. Inventories off-exchange are often not easily observable and have to be estimated. A related concept to the level of inventories is the extent of spare capacity, which is essentially inventories held below ground. This is especially important for the oil market, where a large part of the suppliers are organized in a cartel, which typically leads to production below maximum capacity. Supply shocks at a time of plentiful spare capacity lead to less price upside than supply shocks when producers are producing at maximum capacity.

Although modeling supply and demand is obviously highly commodity-specific, which is beyond the scope of this book, there are a few important considerations to keep in mind. First, the price impact of a given supply shock will be quite specific to the circumstances at hand, but as a general rule, the more concentrated the supply is across suppliers, the more likely it is that the commodity in question ends up at the receiving end of a significant supply shock. The second important characteristic of the supply side is the speed with which supply can expand. Many soft commodities can be grown relatively quickly, but starting a new copper mine takes anywhere from 10 to 16 years. On the demand side, the key consideration is how inelastic demand is for a given commodity. Often, a reversal of a price spike in a commodity is caused by the price spike itself, as a large enough price move leads to demand destruction. This is often the case in oil, for example, where large spikes in oil tend to contribute to recessions, reducing oil demand significantly. Oil is a product with relatively high demand elasticity, as gasoline is a meaningful part of the budget of consumers, and higher prices can quickly lead to a demand response. The same is true for many industrial processes, where energy is a significant part of the cost structure. For some other commodities, the cost of the commodity in question is only a very small part of the disposable income of the consumer and/or a minor input cost to an industrial process. Then the elasticity of demand is quite low, leading to a much larger price increase being required to bring demand and supply back into equilibrium. The structure of the futures market can also be relevant. The price impact of a supply shortage can be much more extreme if commodities have to be physically delivered than when futures are cash settled. Lastly, when visualizing how far prices could rise in response to a shock, it is important to think of prices in real terms. For example, WTI prices peaking at 120 USD after the Russian invasion of Ukraine compares to prices of 150 USD in early 1980 in the aftermath of the Iranian revolution, when adjusted for inflation, rather than the unadjusted spot price at the time of around 30 USD. Framing commodity prices in real terms opens the mind to how far a price spike can run. On the negative side, one important consideration is how easily the commodity can be stored if it is in surplus. Gold and copper are relatively easily stored, while energy products are difficult and expensive to store, which means that in periods of oversupply, prices can be under more downward pressure.

On the downside, one question is what the marginal costs of production are. This is, at least in theory, the price level at which producers would start to shut in capacity. In practice, the decision to shut in is also dependent on how high the costs to shut in and reopen are and what the price outlook is. However, typically, the market outlook is bearish when prices are falling close to marginal costs, which means that some shut-ins should be expected. The difficulty in using the concept of marginal costs to call for a bottom in prices is that it can be difficult to know where we are on the cost curve and who the marginal producer really is. More importantly, marginal costs are not constant. For many commodities, they strongly rise during a boom, as the inputs (specialized labor, required capital goods, etc.) tend to get quite scarce in a bull market. Citi's head of commodity strategy, Max Layton, has quipped that the bulk of the super cycle in the 2000s can be explained by a large rise in marginal costs of production, driven by energy prices. Marginal costs then reverse those gains in a bear market as the market dynamics for the inputs loosen up, together with the relevant commodity market. When prices have fallen to the boom-time marginal production costs, such costs have almost certainly fallen since then. As this concept is quite commodity-specific, we will discuss it in more detail, commodity by commodity.

A few of these general concepts are nicely illustrated by the biggest commodity trade of 2024, the time when Coco went loco.

Coco goes Loco

Over many years, cocoa futures have been trading in a range between 2,000 and 3,400. Then in 2023, cocoa surged from around 2,600 in early 2023 to above 10,000 in April 2024, an unheard-of rally. As is typical in these spikes, it was not that demand for chocolate suddenly exploded, but it was a supply problem. The first ingredient for the major price spike was that cocoa production is highly concentrated. Fifty percent is produced in Ghana and Cote d'Ivoire, both in Western Africa, making any supply issue in this area of the world highly significant for the cocoa supply. Then in 2023, not only one, but two separate supply shocks struck. First, an intensification of the swollen shoot virus disease significantly impacted cocoa supply by causing widespread damage to cocoa trees. The disease, which affects the vascular system of the cocoa plant, leads to reduced pod production, with yields falling by around 25% in the first year, dropping a further 50% in the second year, and potential tree death in year three if not controlled. Efforts to control the virus have included cutting down and destroying infected trees, further reducing the number of productive cocoa trees and exacerbating the supply shortage. The combined effect of reduced yields and the necessity of removing infected trees severely strained the cocoa supply chain. As it takes around five years for a cocoa tree to reach full productive maturity, the supply response is always slow. Secondly, poor weather exacerbated the impact. Irregular rainfall patterns, including both prolonged dry spells and excessive rains, stressed cocoa trees and disrupted their growth cycles. These weather conditions reduce the trees' ability to produce healthy pods, leading to lower yields. Dry spells can cause drought stress, leading to poor flowering and pod development, while excessive rains can lead to flooding and root diseases, further compromising tree health and productivity. While the weather had been more detrimental than usual due to the El Niño phenomenon of 2023–2024, there may also be a climate change-related component to the more volatile weather, suggesting that weather-related supply shocks may become more frequent. All in all, it is estimated that there was a fall in production of 700,000 tons in Western Africa, leading to a deficit of almost 500,000 tons for 2024, the largest in 60 years, compared to stocks entering the year of around 1.7 million tons, or three to four months of world consumption. At the same time, price elasticity on the demand side is low, as US consumers

(*Continued*)

(*Continued*)
only spend around 22bn USD per year on chocolate, i.e., around 67 USD per citizen per year, around 4% of what is spent on gasoline, for example. This would likely make a demand response quite small. Clearly, something had to give, and that something was the price, which quickly rose by 350%, as is illustrated in Figure 14.7. An exponential move up is often followed by an exponential move down, as prices tend to overshoot strongly, making commodities an interesting macro asset class.

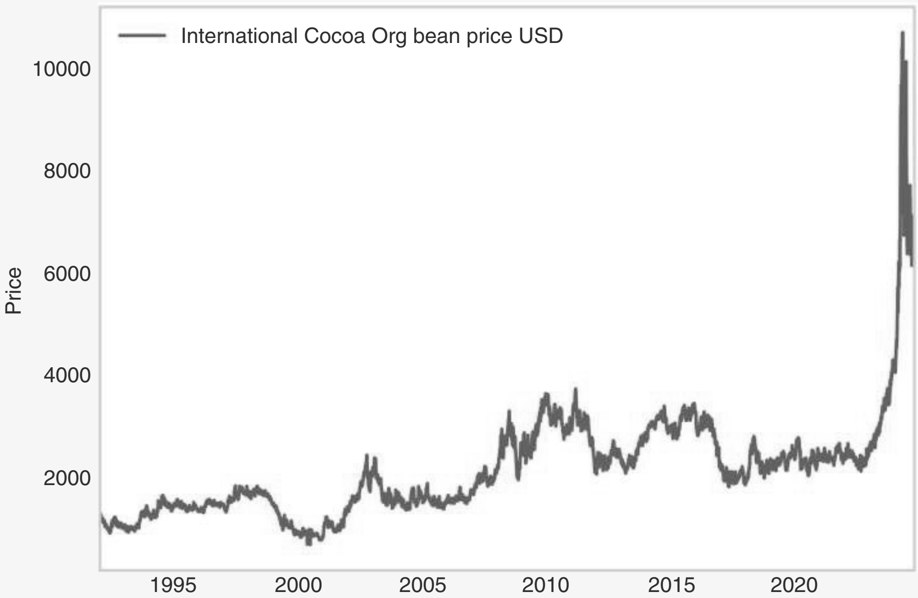


FIGURE 14.7 Coco going loco.
Source: Bloomberg.

14.6 ACE OF BASE

We now move to discussing our framework for base metals. Given our focus on the more liquid assets, our discussion here will mostly focus on copper, even though the basic framework can be applied to the less important base metals too. Unlike precious metals, base metal prices depend primarily on the demand of industrial end users rather than investment demand. Therefore, our basic framework for trading copper uses the global business cycle as the key input. Once more, we use Citi's global data change index to implement this framework. Investors should be long copper when economic data is improving. We implement this by buying the front-month copper future the day after our global data change index moves above its 55dma. This simple strategy results in an active information ratio of 1.06, compared to the unconditional active IR of 0.41, as per Figure 14.8. Segmenting this further by a strong versus weak USD (as usual parameterized by whether DXY is above or below its 55dma), only leads to a minor improvement, as the information ratio moves to 1.10 in the case of USD weakness in the strong data regime. Copper is mostly about the business cycle, and much less about FX.

Data on the global business cycle is also useful when trading the copper/gold ratio. The same logic holds. A strong business cycle supports copper, while gold becomes a more attractive hedge during soft spots, when rates usually fall. Here, the USD is more important, as a strong USD is usually associated with less demand for gold. The best case for the copper/gold ratio is when the data change index is positive and the DXY is above its 55dma. In that case, and lagging conditions by one day, the active information ratio is 1.05, compared to the unconditional one at 0.1. And this time around, both conditions lead to a significant improvement in the information ratios, as per Figure 14.9.

While we will discuss gold later, for now it is important to realize that the copper/gold ratio is not only a tradable macro asset but also an indicator of where we are in the business cycle, as it tends to confirm turns in the ISM on a higher frequency basis, though it is interestingly not always leading, and in some cases it lags by a substantial time. This is illustrated in Figure 14.10. While at the time of writing, there are questions of a structural change in the gold market, we think that eventually the correlations discussed here will come back as the key driver.

More copper-specific, it is well known that China has played an outsized role, being responsible for more than 100% of the growth in copper demand between 2000 and 2023. This suggests that Chinese data should have an outsized weight in any copper business cycle indicator. However, against such a refinement stands the fact that the economic data out of China is generally of poor quality and, maybe more importantly, that the percentage of the demand growth for copper, which China has been

	Data MoM > 55d	Data MoM > 55d | DXY < 55d	Data MoM > 55d | DXY > 55d	Buy & hold
Return	12.79%	6.13%	6.66%	11.11%
Volatility	17.65%	11.76%	13.17%	27.17%
Risk/Reward	0.72	0.52	0.51	0.41
Max DD	–32.75%	–22.14%	–35.49%	–68.53%
Active IR	1.06	1.10	1.02	0.41
% on	47.27%	22.50%	24.78%	100.00%

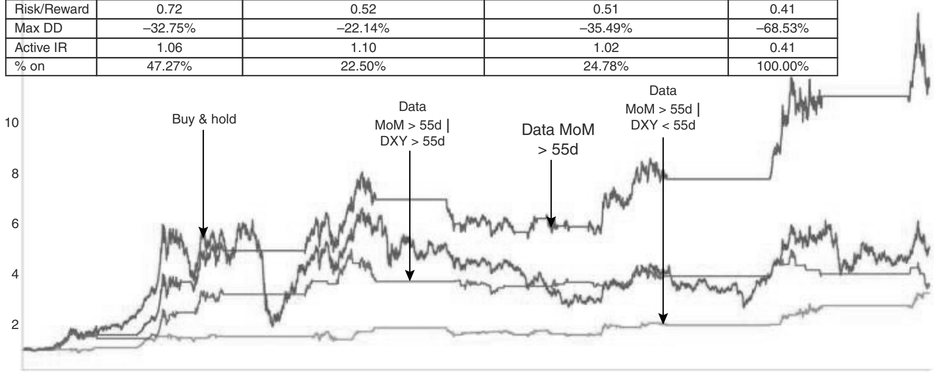
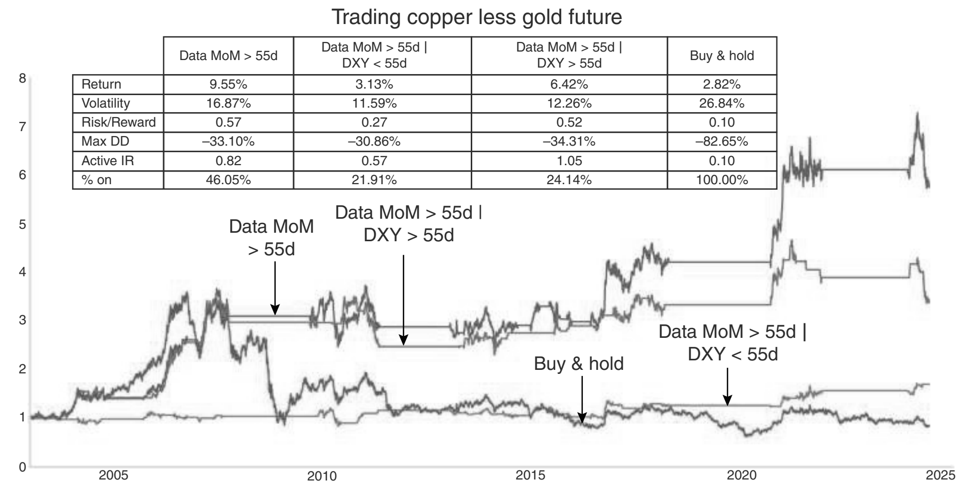


FIGURE 14.8 Positive economic data is key for copper.
Source: Bloomberg, Citi, and authors' calculations.



	Data MoM > 55d	Data MoM > 55d | DXY < 55d	Data MoM > 55d | DXY > 55d	Buy & hold
Return	9.55%	3.13%	6.42%	2.82%
Volatility	16.87%	11.59%	12.26%	26.84%
Risk/Reward	0.57	0.27	0.52	0.10
Max DD	–33.10%	–30.86%	–34.31%	–82.65%
Active IR	0.82	0.57	1.05	0.10
% on	46.05%	21.91%	24.14%	100.00%

FIGURE 14.9 Copper/gold ratio needs strong growth, strong USD.
Source: Bloomberg, Citi, and authors' calculations.

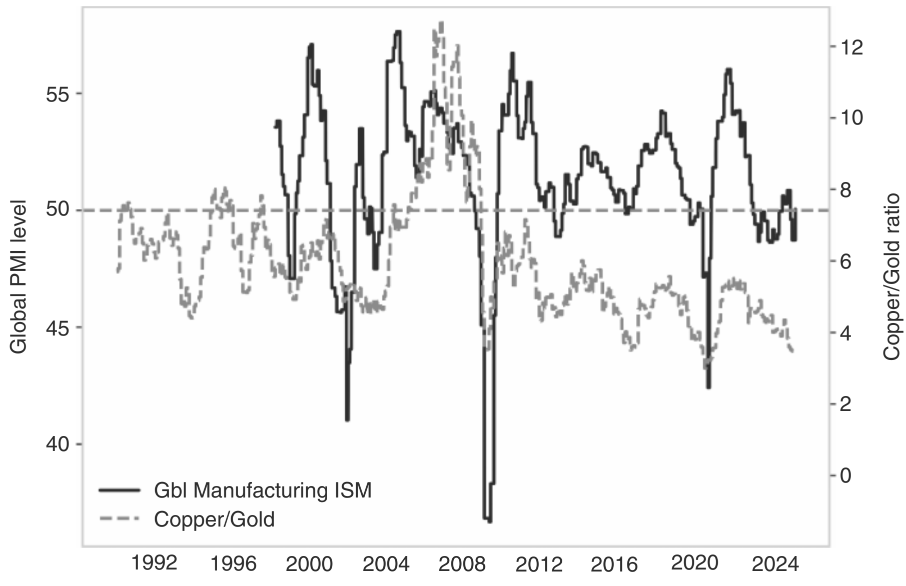
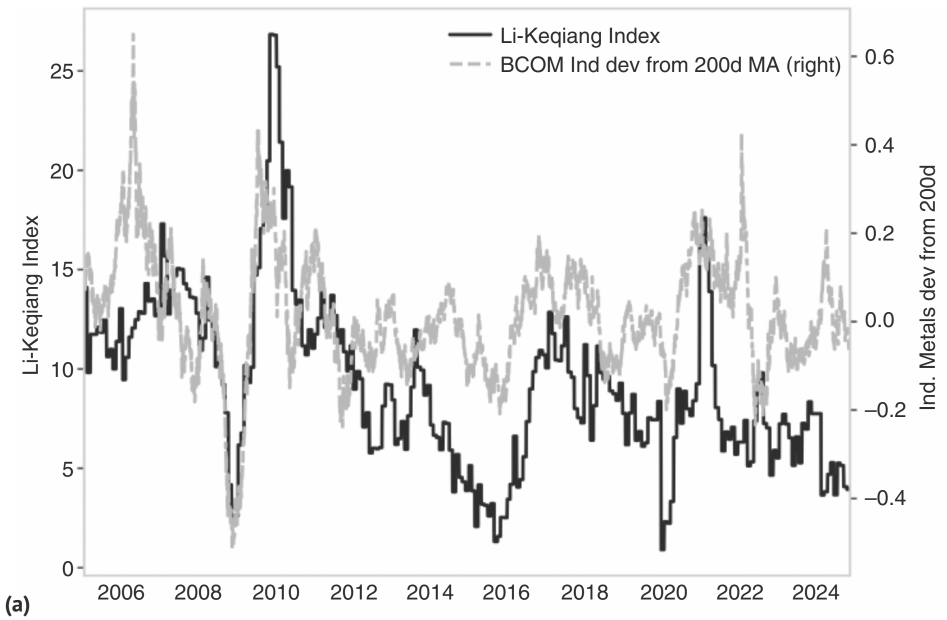
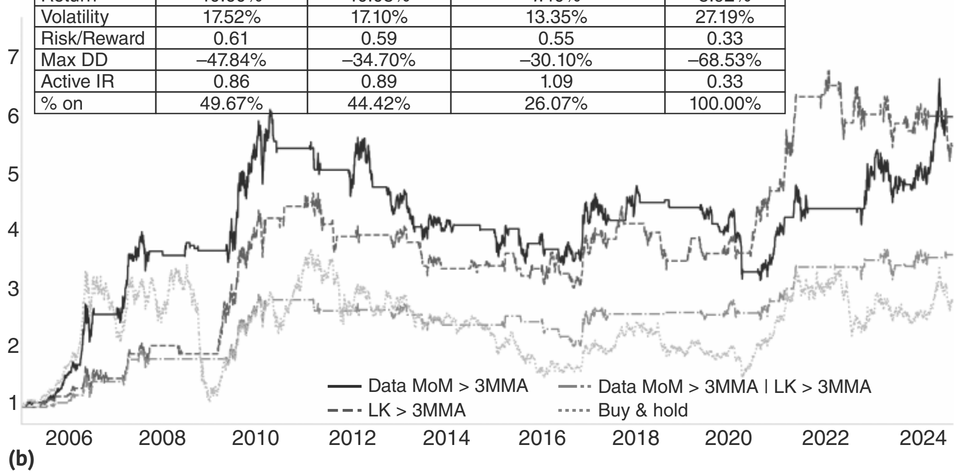


FIGURE 14.10 Copper gold ratio as a business cycle indicator.
Source: Bloomberg and authors' calculations.

responsible for, has come down over time, leading at first glance to a weaker correlation, as per Figure 14.11a. In the end, it is an empirical question. We therefore test whether the Li-Keqiang index, introduced in the chapter on Chinese equities, is helpful to trade copper. In particular, we test whether the Li-Keqiang index being above its 3mma is positive for copper. Two of the three components of the Li-Keqiang index are available in the following month, but the publication lag on the rail traffic has varied.

We lag the data by 15 business days in our backtests. In Figure 14.11b, we find that as a stand-alone strategy, it performs in line with the earlier data momentum rule (which now is trued up with the Li-Keqiang index by starting later and using a 3mma). However, using the Li-Keqiang indicator with a rule that switches on when both the data momentum and the Li-Keqiang index are above the 3mma improves the information





	Data MoM > 3MMA	LK > 3MMA	Data MoM > 3MMA | LK > 3MMA	Buy & hold
Return	10.60%	10.08%	7.40%	8.92%
Volatility	17.52%	17.10%	13.35%	27.19%
Risk/Reward	0.61	0.59	0.55	0.33
Max DD	–47.84%	–34.70%	–30.10%	–68.53%
Active IR	0.86	0.89	1.09	0.33
% on	49.67%	44.42%	26.07%	100.00%

FIGURE 14.11 (a) China business cycle as a driver for copper. (b) But it is not the most important driver.
Source: Bloomberg and authors' calculations.

ratio to 1.09. Reassuringly, the results are not overly sensitive to the precise publication lag that we use for the Li-Keqiang index.

Going forward, we would note that the energy transition could be the new China. By 2030, it is believed that copper demand from the energy transition will account for 45–50% of global copper demand, driven by the adoption of EVs, the expansion of electric grids, and the build-out of renewable energy systems like wind turbines and solar panels. This should result in a long bias for copper going forward. Downturns will be shallower, just as upturns will be stronger than in a normal business cycle, where there is no structural bid present. While this should not stop investors from trading the cycle, positioning should be more aggressive in upcycles than in downcycles. For an in-depth discussion, see Layton et al. (2023).

As already discussed previously, bottoms usually occur at prices that are not too far from the marginal costs of production. This is illustrated in Figure 14.12, which shows that it is rare for copper prices to fall significantly below marginal costs on a persistent basis, which we define here as USD/ton of production costs for the mine at the 75th percentile of the cost curve with respect to C1 cash cost (operating costs), plus sustaining capital costs. However, the chart also illustrates the difficulties in betting on this concept of a bottom. As can be seen, marginal costs change over time and are typically a lagged function of spot prices. This can make it highly uncertain whether prices have fallen by enough to lead to mine closures or whether miners will be successful in lowering operating costs, keeping capacity online. New stories of mine closures can be helpful to ascertain a bottom. As an aside, Figure 14.12 also illustrates our earlier point: it is interesting to observe just how important the rise in marginal costs (mostly energy) was for the copper bull market in the 2000s.

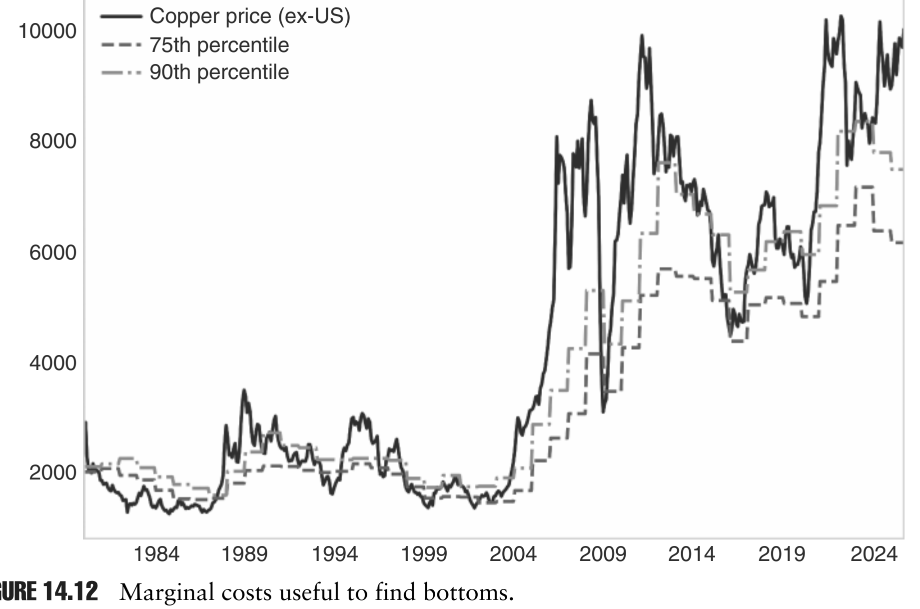


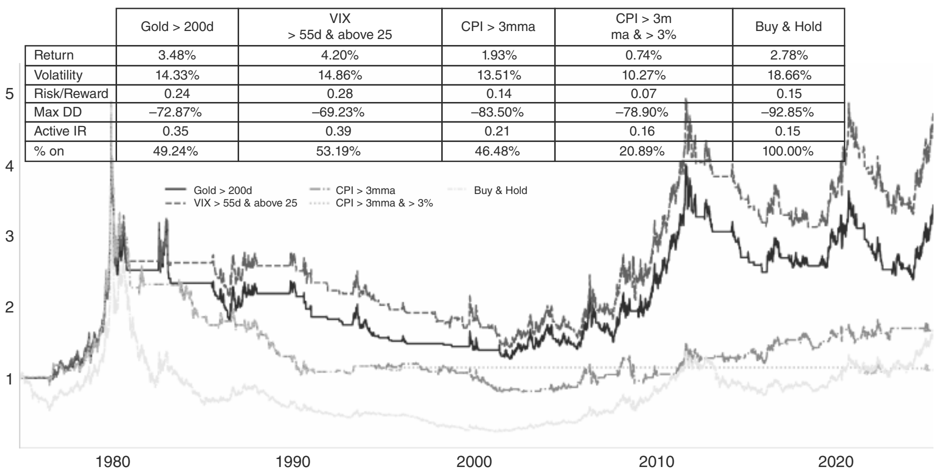
FIGURE 14.12 Marginal costs useful to find bottoms.
Source: Bloomberg, Wood Mackenzie, and authors' calculations.

14.7 GOLD: A 3,000-YEAR-OLD BUBBLE THAT REFUSES TO BURST?

We next move to precious metals. Just like for base metals, we once again focus on the most liquid metal in the precious metal area as well, i.e., gold. Although gold has its industrial uses, including jewelry making, it has long depended much more on investment demand than on industrial demand (for a very thorough discussion, see, for example, Layton et al., 2024). Given that gold has, as a first approximation, no yield, it is mostly seen as an asset for times of rising risk aversion, which makes competing risky assets with yield less attractive, or for times of high inflation (which makes competing safe assets, like bonds, unattractive). However, trading gold mostly in line with risk aversion or inflation has not worked overly well. Owning gold just when the VIX is above its 55dma, does result in a higher active information ratio of 0.37, above the unconditional level of 0.15, but just using the 200dma generates a comparable IR (with lower returns, though). Buying gold when VIX is above its 55dma only when VIX is above 25, does not fare much better, as is illustrated in Figure 14.13.

Similar to the case study for BCOM for the 1970s, we also investigate the impact of inflation on the trading of gold. We find that owning gold only when the US headline CPI is above its 3mma (i.e., moving higher) barely improves the active information ratio compared to the unconditional. We also check for threshold effects. If we are only long gold when inflation is above 3% and moving up (i.e., again above its 3mma), the information ratio also does not improve much. While gold reacted to the inflation spike in the late 1970s, it did not capture the post-COVID inflation episode. Gold does not seem to be a high Sharpe ratio inflation hedge.

The inflationary episode post-COVID is instructive in this regard. Gold traded very well during COVID, but then pulled back after COVID ended. With inflation taking off in early 2021, gold started to trade better again and traded up around 10% from the lows by early 2022. However, when the Fed started to hike rates in March 2022,

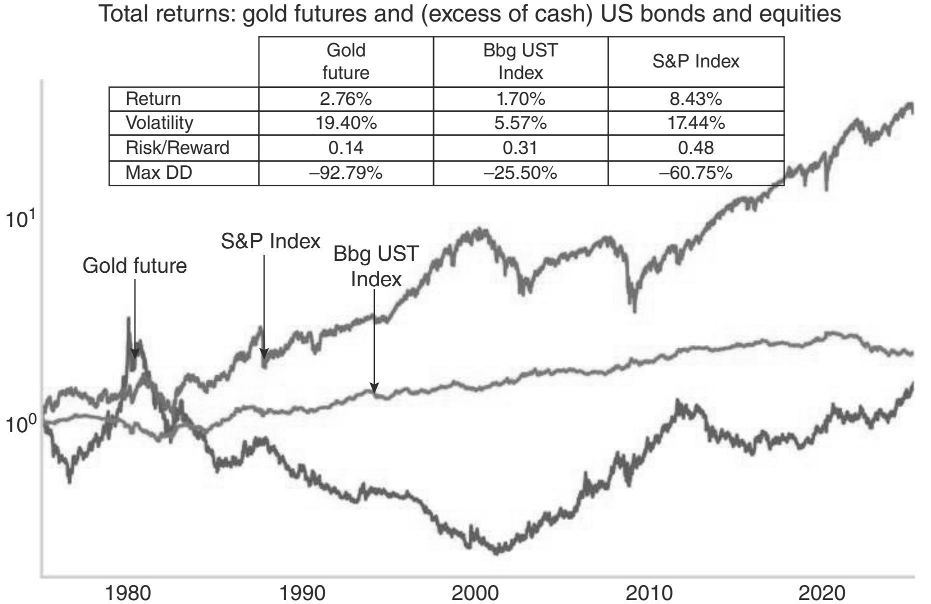


	Gold > 200d	VIX > 55d & above 25	CPI > 3mma	CPI > 3m ma & > 3%	Buy & Hold
Return	3.48%	4.20%	1.93%	0.74%	2.78%
Volatility	14.33%	14.86%	13.51%	10.27%	18.66%
Risk/Reward	0.24	0.28	0.14	0.07	0.15
Max DD	–72.87%	–69.23%	–83.50%	–78.90%	–92.85%
Active IR	0.35	0.39	0.21	0.16	0.15
% on	49.24%	53.19%	46.48%	20.89%	100.00%

FIGURE 14.13 Gold is not a great inflation hedge.
Source: Bloomberg and authors' calculations.

gold sold off and was making new local lows by late 2022, also driven by a strong USD, which was benefiting from higher US rates. It took a peak in the USD and the end of the Fed hiking cycle for gold to aggressively rally again. This big rally happened after the peak in inflation was long in. The lesson is that it is not so much inflation that supports gold, but inflation that is not addressed by the Fed, that is bullish for gold. As soon as the Fed started to raise (real) yields, gold fell even in the context of high and rising inflation. And when the Fed went on hold and eventually considered cuts, gold traded very well, even with inflation falling quickly. Gold benefits from excess liquidity, rather than from inflation per se.

One difference from the 1970s is that back then, the USD was very weak, while the post-COVID episode saw mostly a range-trading USD. Gold does seem to be a hedge for USD debasement fears much more than for inflation fears. While the two may be linked, they are not the same. The USD link is typical for many commodities, but it is the most straightforward in gold, as gold is, in some sense, just another currency, though one where supply cannot be manipulated by governments. The fact that supply is much more controlled than it is for fiat currencies implies a trend appreciation over time against fiat FX, though not necessarily against higher-yielding assets denominated in those fiat currencies. Gold, therefore, holds its own against the USD over time, but not against a total return index for US equities or even bonds, as per Figure 14.14. This suggests that using gold as a hedge has to be done for very specific times rather than as a long-term portfolio holding. At the time of writing, gold is rallying hard even with a Fed that is only moderately dovish and a USD that is only moderately weak. However, a big reason for this is the fear that the Fed may become unreasonably dovish down the road, undermining the USD. The basic concept of too loose a monetary policy as a driver for gold is therefore very much alive and kicking.



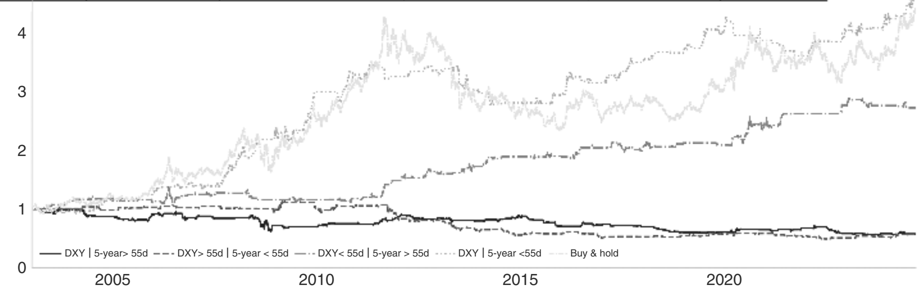
	Gold future	Bbg UST Index	S&P Index
Return	2.76%	1.70%	8.43%
Volatility	19.40%	5.57%	17.44%
Risk/Reward	0.14	0.31	0.48
Max DD	–92.79%	–25.50%	–60.75%

FIGURE 14.14 Gold hedges USD debasement, but worse than equities or bonds.
Source: Bloomberg and authors' calculations.

More cyclically speaking, we focus on both real rates and USD. Using our usual framework, we go long gold when US five-year real rates are below its 55dma, and when the DXY is below its 55dma too, one day lagged. Usually, we always lag our strategies by one or two days to make them, at least in theory, tradable. But here we first show the contemporaneous strategy, before we show the lagged one. In contemporaneous terms, this simple strategy works well, raising the active information ratio to 1.60, compared to an unconditional one of 0.47, as can be seen in Figure 14.15. Most of the work is done by the DXY rather than the TIPS leg. However, somewhat shockingly, the results completely break down when we use our usual one-day lag. This suggests that forward-looking views by investors on DXY, and to a lesser extent real rates, can be implemented in gold, even though mechanically reacting to how real rates

Contemp returns

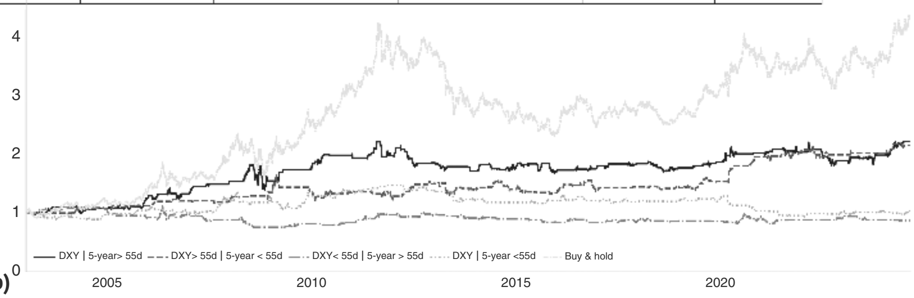
	DXY | 5-year> 55d	DXY> 55d | 5-year < 55d	DXY< 55d | 5-year > 55d	DXY | 5-year <55d	Buy & Hold
Return	–2.04%	–2.17%	4.93%	7.52%	8.34%
Volatility	9.54%	8.57%	7.80%	8.98%	17.56%
Risk/Reward	–0.21	–0.25	0.63	0.84	0.47
Max DD	–46.38%	–58.04%	–23.16%	–20.57%	–45.56%
Active IR	–0.40	–0.54	1.37	1.60	0.47
% on	28.24%	22.05%	21.32%	27.72%	100.00%



(a)

Returns with 1-day implementation lag

	DXY | 5-year> 55d	DXY> 55d | 5-year < 55d	DXY< 55d | 5-year > 55d	DXY | 5-year <55d	Buy & Hold
Return	4.16%	3.89%	–0.32%	0.54%	8.29%
Volatility	9.77%	8.31%	7.81%	8.99%	17.56%
Risk/Reward	0.43	0.47	–0.04	0.06	0.47
Max DD	–24.77%	–26.75%	–33.74%	–36.20%	–45.56%
Active IR	0.80	1.00	–0.09	0.11	0.47
% on	28.25%	22.05%	21.33%	27.70%	100.00%



(b)

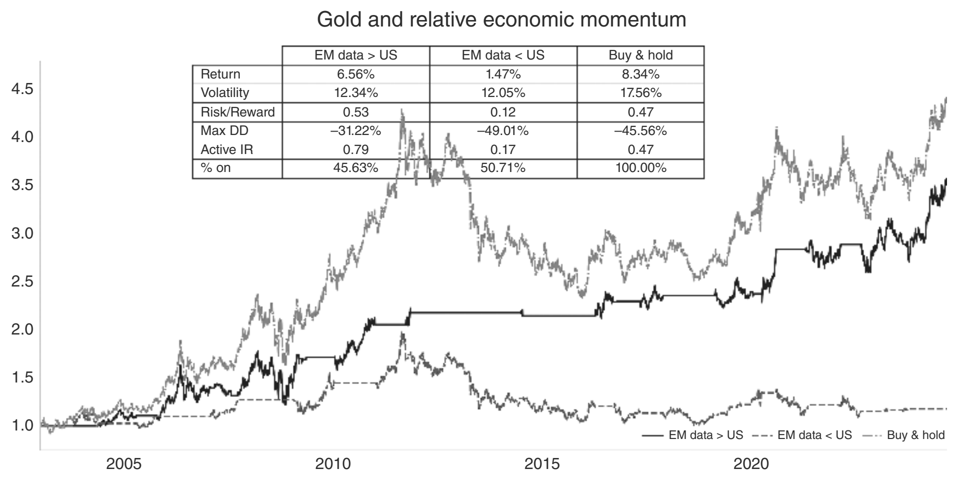
FIGURE 14.15 (a) Gold likes lower real yields and USD, (b) but the strategy breaks down with a one-day lag.
Source: Bloomberg and authors' calculations.

and gold have been trading is unlikely to work. However, we would also note that the sensitivity of the simulation to a one-day shift undermines the confidence in the model overall to some extent, in spite of the fact that economic logic is strongly in line with its basic tenets.

A similar rule, which works with a one-day delay for gold, is that if the relative data momentum favors EM over the United States, it is bullish for gold. The strategy raises the information ratio, but comes at a lower return. The underlying reason why that strategy works is similar to our earlier study. Times when EM data outperforms US data are likely periods with a weak USD, and at times also a period of lower US rates. It is therefore not overly surprising that gold does well in such periods. One reason why the rule is superior to our previous attempt is likely that the data change indexes are moving more slowly than DXY and real rates. Slowing down the trading signals works well in this case, as is illustrated in Figure 14.16.

A more tactical tool for trading gold is that the market tends to get quite overbought when the gold bugs take over in earnest. These strong gold market run-ups tend to be unsustainable. We measure this by taking times when gold moves 20% above its 200dma for the first time in three months to eliminate overlapping periods. As Figure 14.17 shows, there tends to be a reversal in short order. This is a useful indicator for taking profits and trading nimbly even within structural bull markets, but is highly tactical.

On a longer-term view, just like for copper, we find that there is a very bullish underlying structural demand driver, here in the form of gold demand by various central banks. In order to diversify FX reserves away from the USD, central bank purchases have ramped up. While a desire for diversification is universal, central banks of countries in the crosshairs of US foreign policy are especially keen and likely in a rush too. The reason is that the United States has militarized the USD for some time, ratcheting up the pressure further in the aftermath of the Russian invasion of Ukraine. With Russia losing access to the settlement systems for USD as well as to its considerable



	EM data > US	EM data < US	Buy & hold
Return	6.56%	1.47%	8.34%
Volatility	12.34%	12.05%	17.56%
Risk/Reward	0.53	0.12	0.47
Max DD	–31.22%	–49.01%	–45.56%
Active IR	0.79	0.17	0.47
% on	45.63%	50.71%	100.00%

FIGURE 14.16 Gold likes strong EM, weaker United States.
Source: Bloomberg, Citi, and authors' calculations.

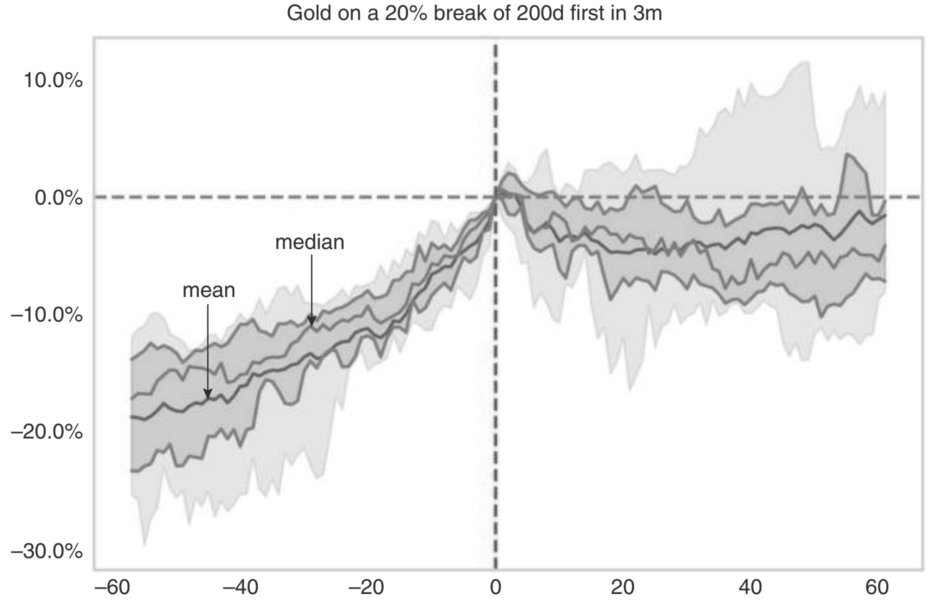


FIGURE 14.17 Gold with tactical short term tops when it goes 20% above the 200d.
Source: Bloomberg and authors' calculations.

USD reserves, quasi overnight, it is understandable that jurisdictions at odds with the Western worldview do think that holding USD reserves creates undue leverage for the White House. The problem is that there are not many other options, especially when excluding EUR, JPY, and a few other currencies of countries that are part of the Western bloc and may well join a US-led sanction regime. Currencies of large economies that are part of the global South generally do not offer a solution either, as those currencies are usually traded in the form of NDFs and are not freely exchangeable. This is no coincidence, as countries with more authoritarian regimes tend to prize control over economic efficiency, making capital controls a frequent part of the FX regime. This roadblock is therefore unlikely to change. Consequently, gold is one of the most obvious ways to diversify central bank reserves away from the USD. This leaves us with a similar conclusion to copper. Given structural tailwinds, we do not stop trading the cycle. But bullish episodes will be more bullish than our model would say, while bearish episodes are less bearish.

Brown's Bottom, Xi's Top?

The importance of central banks for the gold market should not be underestimated. Prior to the most recent decision by many central banks to increase gold reserves, some central banks were reducing them, as the end of the gold standard

(*Continued*)

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removed the key requirement to keep reserves in gold. Some central banks, therefore, wished to diversify their reserve holdings away from gold and into other reserve assets like UST.

Most famously, between 1999 and 2002, the BoE sold around 400 tons, or approximately 50% of its gold reserves, through a series of public auctions. After a long period of underperformance of gold through the 1980s and 1990s, Chancellor of the Exchequer Gordon Brown argued that the management of the reserves should be "modernized" by investing in higher-yielding assets, such as government bonds. Such a move would also increase liquidity, in case reserves would be needed on short notice for FX intervention. The gold auctions were announced on May 7, 1999, which led to a 15% sell-off in gold over the subsequent three months, a large move for a relatively historically low volatility asset like gold. Concerns about central bank sales became so widespread that in September 1999, fifteen European central banks signed the Washington agreement on gold, which limited the amount of gold central banks were able to sell each year, in order to avoid large, uncoordinated gold sales. Leasing, where central banks lend gold to other market participants for a lease rate, was also restricted, making shorting gold more difficult and expensive. Gold subsequently spiked by 27% over the next two to three weeks, partly on short covering. However, it then mostly unwound that spike, coming close to retesting the lows in early 2001. The low in gold prices is referred to as Brown's bottom, given how important the BoE selling was for the last leg lower in the gold bear market. From there, gold started a multi-year bull market, as illustrated in Figure 14.18. At that time, the BoE had concluded around 65% of the announced plans for its gold sales. It had already reduced auction sizes from 25 tons to 20 tons. The last gold auction by the BoE was held on March 5, 2002. The BoE concluded its gold sales slightly ahead of its full goal, selling 395 tons rather than the 415 tons announced, partly in response to prices having started to rise, being up around 16% from the early 2001 bottom. Gold has never looked back since. The episode illustrates not only the importance of central banks in the gold market. A broader lesson is that big supply overhangs have an impact, but the market does not wait for 100% of the supply overhang to have been digested. As a rule of thumb, removing 50% of a large supply overhang is enough for the relevant asset price to bottom. Lastly, if prices fall far enough to get the most patient holders out of the market, a turn in price performance is likely. Since the announcement of the gold sales, gold is up more than 800%, while the US Treasury market, as the main beneficiary of flows from the diversification exercise of the BoE, is only up around 150%. Going forward, we think just as the BoE selling set the bottom, in the bigger picture, the PBoC will set the top. Whenever China is done diversifying into gold, an important local peak may be in.

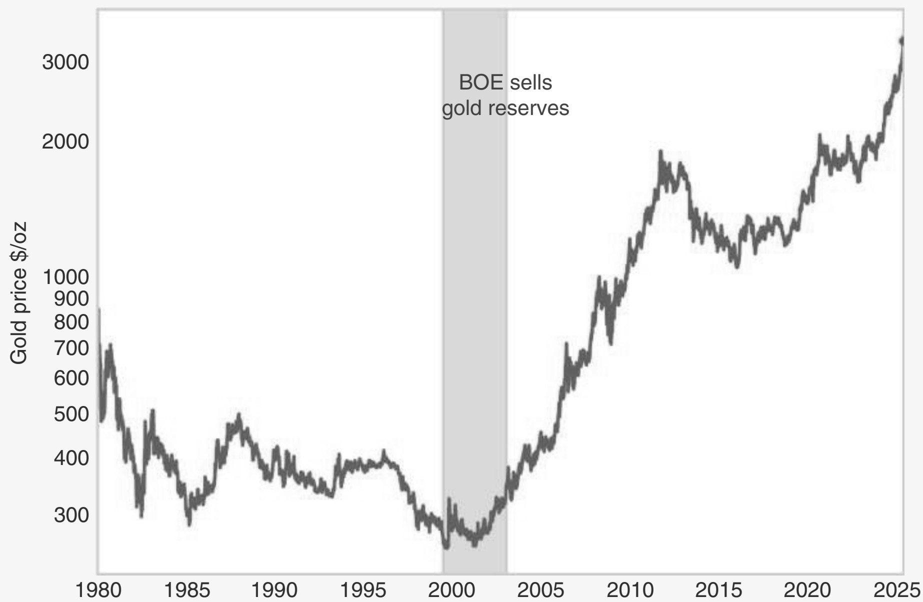


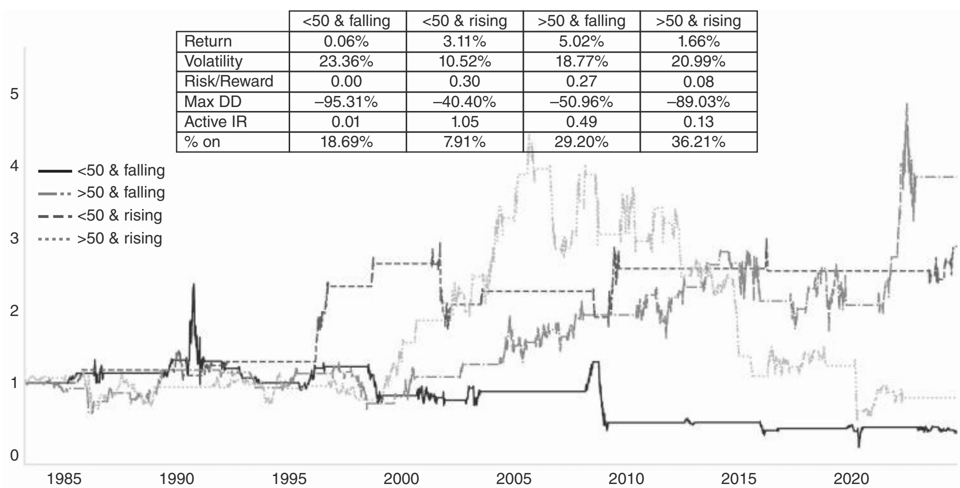
FIGURE 14.18 Buy when central banks are done selling.
Source: Bloomberg.

14.8 HOW TO TRADE WITH ENERGY

Once more, we focus on the most liquid and most global commodity in this sub asset class, i.e., on oil. Although commodity specialists trade the differential between WTI, Brent, and other grades all day long, from a global macro perspective, we care less about the basis between various grades and use Brent as a proxy for the global oil market, even though many of our studies are based on BCOM Energy. Oil is, in our view, the most important commodity for macro-trading. The key reason is that changes in oil prices are a key driver for inflation, much more so than any other commodity. The reason is not just the higher weight in the CPI basket and its importance as an input cost, but also its much higher volatility compared to other commodities. The impact of oil on inflation is not only on the headline level but, with some delay, also on a core level. Furthermore, oil prices also impact activity and the BoP for both exporters as well as importers. As such, a view on oil is important, even if for investors not trading oil directly. Sadly, our approach of finding rules of thumb to provide a framework does not lend itself as well to trading oil as it does for other commodities discussed earlier. The reason is that supply shocks explain more of the variability in oil prices than is the case for base metals or gold. This suggests a large role for geopolitics, but also for weather patterns, which is harder to catch in our business cycle-centric framework. Furthermore, around 40% of global supply is controlled by OPEC, a global cartel outside of the reach of the Western authorities in charge of curtailing collusive behavior, and more than 50% of production is controlled by OPEC+, i.e., adding countries coordinating with OPEC. We already discussed geopolitical events in our chapter on geopolitical events in an earlier chapter. We study OPEC actions later in this chapter.

Still, like for all commodities, the global business cycle also matters for oil. As usual, we use the manufacturing ISM and show how the energy index performs when it is above its 3mma, We find that the information ratio for long oil is highest when the PMI is rising (i.e., above its 3mma), but still below 50, i.e., when the economy is coming out of a soft growth patch or a recession. A PMI above 50 is also generally positive, with the main negative information ratio occurring when the PMI is below 50 and falling, as is illustrated in Figure 14.19. Overall, information ratios are not great, though. Trading WTI with the business cycle is not enough.

Another important tool is the oil curve. In the very big picture, curves that are in contango are bearish, while markets in backwardation are bullish. If oil is plentiful enough to push the market into contango, it can eventually pull the whole curve lower. The negative roll yield will add to negative returns. So far, so intuitive. To backtest this rule, we need to know which part of the curve should be inverted. To find out, we backtest various specifications. We show the relevant backtest in Figure 14.20. The winner is 12 months versus 1 month, which meaningfully improves the active information ratio relative to the unconditional, even though it is still relatively low at 0.49. Furthermore, the rule has not worked as well as in the past, ever since COVID. We still think that whenever 1 month/12 months moves into contango, investors need to be careful and potentially reduce longs.



	<50 & falling	<50 & rising	>50 & falling	>50 & rising
Return	0.06%	3.11%	5.02%	1.66%
Volatility	23.36%	10.52%	18.77%	20.99%
Risk/Reward	0.00	0.30	0.27	0.08
Max DD	–95.31%	–40.40%	–50.96%	–89.03%
Active IR	0.01	1.05	0.49	0.13
% on	18.69%	7.91%	29.20%	36.21%

FIGURE 14.19 Business cycle helpful to trade oil, but information ratios low.
Source: Bloomberg and authors' calculations.

	(3, 1)	(6, 1)	(12, 1)	(6, 3)	(12, 3)	WTI Futures (BCOM)
Return	14.33%	11.55%	17.53%	16.15%	12.48%	7.03%
Volatility	36.38%	36.40%	36.36%	36.38%	36.42%	36.43%
Risk/Reward	0.39	0.32	0.48	0.44	0.34	0.19
Max DD	–69.57%	–76.58%	–63.60%	–67.55%	–73.99%	–97.78%
Active IR	0.40	0.32	0.49	0.45	0.35	0.20
% on	93.88%	94.24%	94.26%	94.13%	94.22%	94.37%

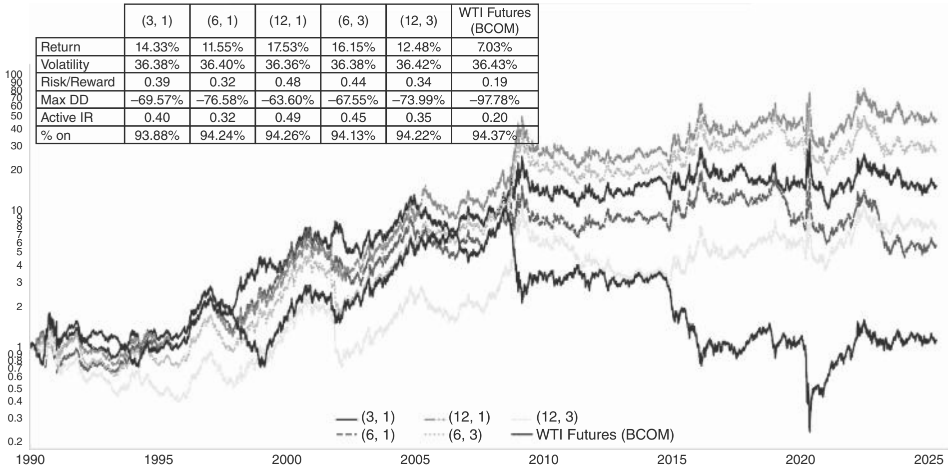


FIGURE 14.20 Backwardation is bullish.
Source: Bloomberg and authors' calculations.

Oil Goes Negative

Just like the cocoa example is an extreme case of how a supply shortage can play out, the behavior of oil during the COVID shock illustrates the case of severe demand destruction. In early 2020, when the COVID shock hit, oil demand unsurprisingly dropped extremely sharply, as, overnight, airlines were grounded, other forms of transport were significantly reduced, and industrial activity slowed sharply. Given that well closures are costly (and with the reopening of wells being even more expensive), oil producers were slow to shut in production. Spot WTI quickly fell from just above 60 USD in early 2020 to around 20 USD by the end of March. When the S&P 500 bottomed on March 23, most other risky assets bottomed as well. But not so oil. As oil demand did not spring back to life as quickly as demand for US equities, storage filled up, and by April, there was almost no unfilled storage capacity left. As a result, spot prices briefly dropped into negative territory, reaching negative 38 USD on April 20, 2020, before moving back into positive territory to around 10 USD the next day. Spot WTI never looked back and ended the year at just above 48 USD. Partly because the NYMEX WTI contract is deliverable, and partly because the front contract did not have much life left in it (back then, the front contract was May 2020, with the last trading day of April 21, and first delivery May 1), it also traded deeply negatively at around –38 USD, as illustrated in Figure 14.21.

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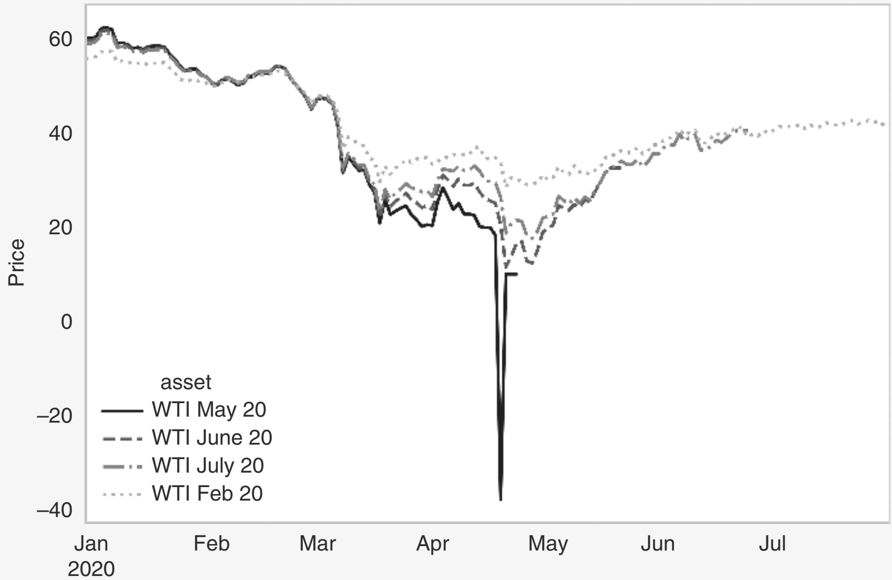
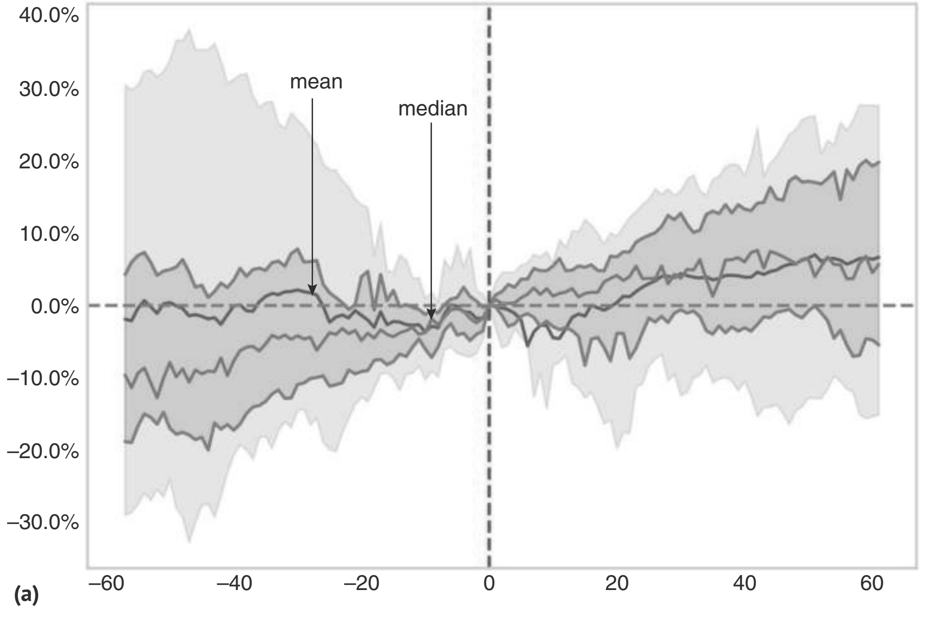


FIGURE 14.21 Negative oil prices—which contract to buy mattered a lot.
Source: Bloomberg.

Although it seems like an absolute no-brainer trade to buy oil on that day, especially given that the S&P had already clearly bottomed, the curve had moved into very extreme contango. This made the trade to buy oil futures, when the prices went negative, not as obvious as it seems. On the day when the May contract hit –38, the June contract traded at 20.40. But the lesson is that when things go completely nuts, the curve will just not price enough steepness. By the time of its expiry on May 19, the June contract settled at 32.50, an almost 60% gain in about a month. Taking less risk with respect to how long the storage deficit could keep prices extremely pressured, and buying the February 2021 contract to have exposure into year-end would have resulted in a 45% return to year-end, compared to 31% for the S&P 500. Only holders of USO, the oil ETF, had to live with much smaller gains, as USO management had shifted in late April from a focus on front-end contracts to a more equal distribution of contracts, missing out on a part of the aggressive recovery at the front end.

14.9 HOW TO TRADE OPEC MEETINGS

OPEC meetings are the FOMC days of the oil market. Important information gets released in a lumpy fashion, which can set price trends for more than just a day or two. As usual, we are not just interested in what fundamentally happens on OPEC day (hike or cut), but we care more about the price action on OPEC day. This is partly the case, as OPEC decisions often get leaked ahead of time, which means that decisions are often already fully priced. Still, we find the price action on OPEC day interesting. As per Figure 14.22a, OPEC cuts that display positive price action on the day usually have strong follow-through. If oil prices sell off on the day of the OPEC cut, there is



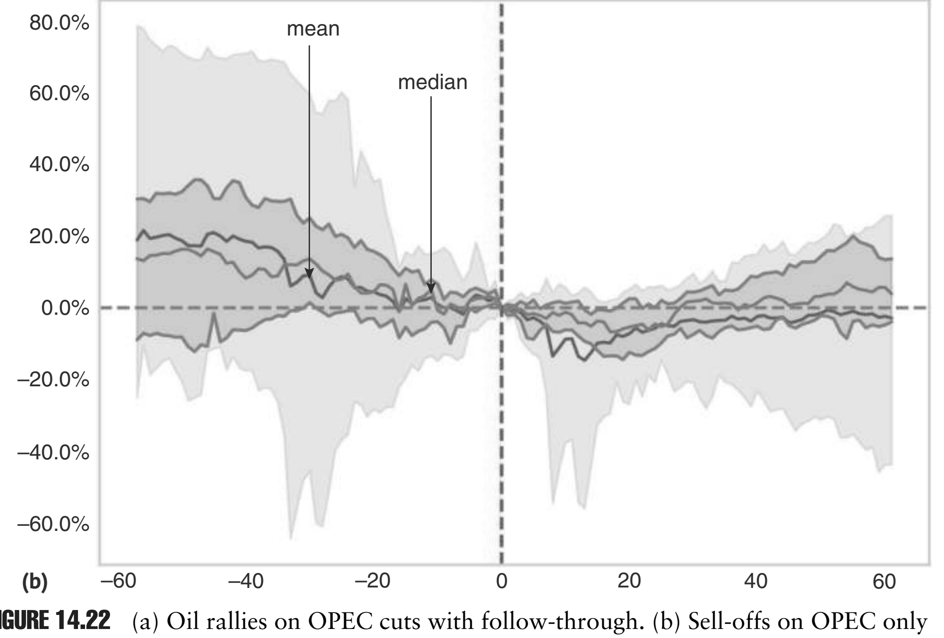


FIGURE 14.22 (a) Oil rallies on OPEC cuts with follow-through. (b) Sell-offs on OPEC only have short-term follow-through.
Source: Bloomberg and authors' calculations.

more limited follow-through on the downside, and the market turns up around a few weeks later, as per Figure 14.22b. It is also interesting that OPEC meetings do not seem to change the trend. In the first chart, prices already rally into the date of the OPEC cut, while in the second chart, they sell off into it.

On the other side of the equation, we look at output increases from 2003 to now. The charts look very similar to the ones when output gets cut, which is interesting. This is illustrated in Figure 14.23. Again, we get some follow-through with the price action of the day in either direction. Furthermore, again, the short-term trend does not

FIGURE 14.23 (a) Oil rallies on OPEC boosts also with follow-through. (b) Sell-offs on OPEC boosts only with short-term follow-through.
Source: Bloomberg and authors' calculations.

change. The market rallies into the bullish OPEC days and sells off into the bearish ones, starting around one month prior. Interestingly, the follow-through on the downside is short and shallow. This could be because production increases are fully expected at times. Alternatively, this could be similar to our finding in credit, where high supply also does not seem to matter much, as supply is the consequence of strong demand. Usually, OPEC only increases supply when the cartel understands that the market can bear it. Mistakes can happen, and famously did in November of 1997, for example, when OPEC hiked supply in the middle of the Asia crisis, but the mistakes appear to be the exception rather than the rule.

14.10 POSITIONED FOR FAILURE

Moving to a more short-term perspective, we investigate how well positioning in oil works for trading purposes. Overall, we conclude that it worked relatively well in the last decades. This analysis leans on the work published by Willer et al. (2024c). Figure 14.24a shows the combined positioning by speculators' positions in Brent and WTI as long positions over short positions. Whenever this number falls below 2, the subsequent upward move in Brent has often been quite meaningful, as per Figure 14.24b. To be precise, in the median case, prices have been 20% higher after 100 trading days, though we also note that the skew is initially negative. Positioning is particularly powerful when it comes together with a trigger that may change the market's cherished consensus view. This was, for example, the case in late September and early October of 2024, when Israel killed the Hezbollah leader Nasrallah on September 27th, which Iran answered with a large strike of ballistic missiles on Israeli territory. Although the market typically fades violence in the Middle East quickly, it is much harder to do with maximum short positions, and oil was unsurprisingly higher by 9% in the five trading days after the initial event.

14.11 SUMMARY

Commodities are, in general, not great buy-and-hold investments but can episodically generate significant alpha. In particular, most commodities do well during Fed hiking cycles and poorly into and during US recessions, with gold being an exception. Gold outperforms copper into recessions and during the early part of recessions, only to reverse after the end of the recession. Trend following and carry both work, though not as well as in the early 2000s. Commodity-specific factors revolve around how concentrated the supply is, how quickly supply can be increased, and how high the elasticity of demand is. For copper, we find that high-quality long positions require our global data change index to be rising. An upswing in the Chinese economic cycle also helps. Similarly, investors should go long the copper/gold ratio when the data change index is positive and the USD is in a strong regime. By itself, gold traditionally trades well in low rates, weak USD regimes, but this rule is hard to implement. Gold also does well when EM data momentum is larger than that of the United States, partly because this is often correlated with USD weakness and lower US rates. For oil, PMIs

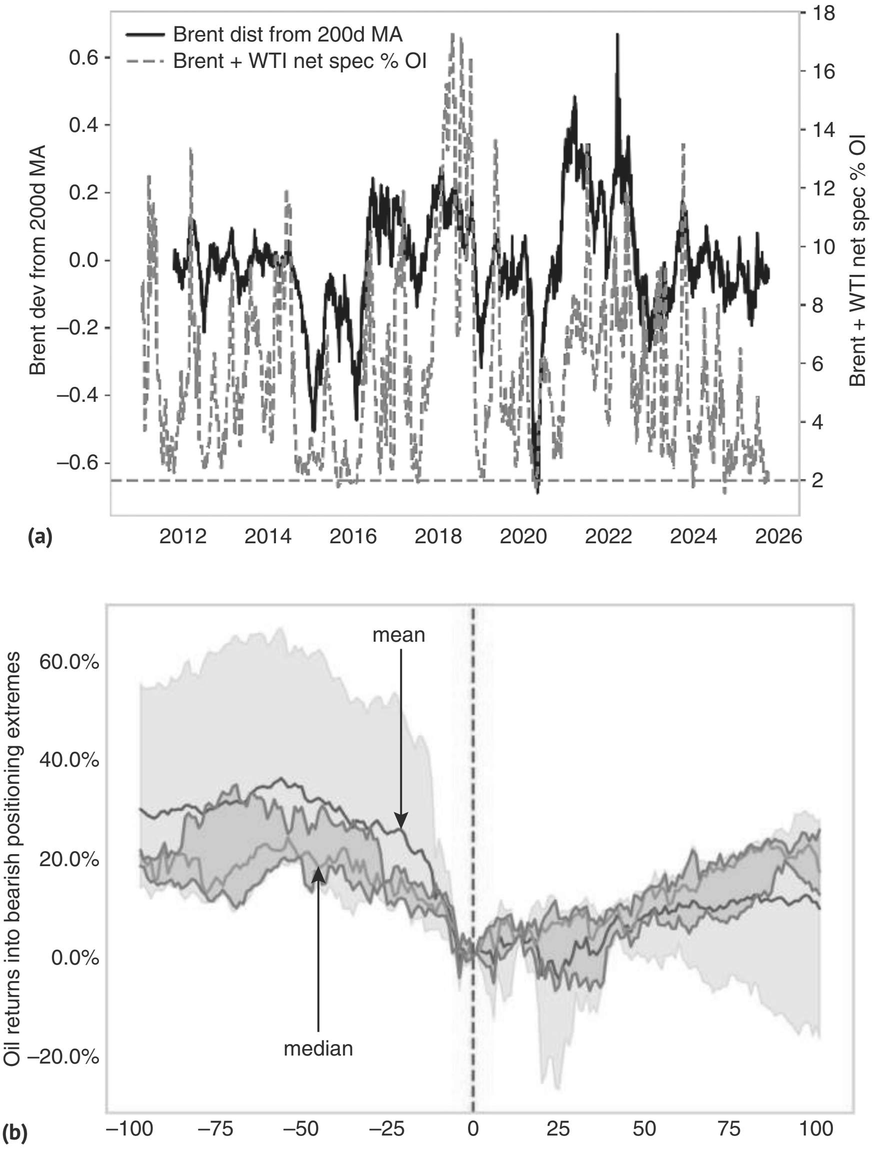


FIGURE 14.24 (a) Combined Brent WTI Long/short ratio below 2 (b) … is a bullish signal.
Source: Bloomberg, Citi research, and authors' calculations.

are also helpful, but less important than for base metals. There is some alpha from trading it in line with the curve shape: backwardated curves are bullish, while curves in contango are more bearish, though this effect has become weaker. More tactically, we find that price action on OPEC meetings has follow-through and that positioning is a reasonably consistent counter-indicator for oil.

CHAPTER 15

How to Allocate

Having studied the most relevant public asset markets individually, we now want to share an overarching framework for asset allocation across macro-regimes, followed by views on how to put it all together in a portfolio.

15.1 A REGIME INVESTING APPROACH

Regime investing has gained popularity and increased prominence in recent years. The prevailing wisdom over the great moderation period from the early 1980s to the late 1990s was to implement a 60/40 portfolio of equities and bonds, as declining macroeconomic volatility and inflation helped both equities and bonds to deliver positive returns, which were negatively correlated when it mattered most, i.e., when stocks sold off into recessions. However, the perception that policymakers had acquired the necessary tools to dampen the impact of such economic shocks was mostly a pipe dream. The financial crisis of 2008, subsequent European sovereign debt concerns, and the global pandemic provided clear evidence that macro volatility had not been tamed. On the contrary, investors quickly discovered the important insights of the economist Hyman Minsky, who suggested that in a financial market (and financial crisis) context, stability breeds instability. Not only had macro volatility not been tamed, but maybe it was structurally not possible to tame it, as investors react to a perception of reduced risk by increasing leverage. One innovation that improved on the 60/40 model was risk parity, which allocates risk based on volatility rather than with fixed weights. Those models beat 60/40 as the use of leverage allowed for larger bond positions, which protected investors better during the GFC. However, both 60/40 and risk-parity portfolios had ignored the lessons of the stagflationary 1970s and were therefore not suitable for the inflation episodes post-COVID. This was the latest reminder that it is quite crucial to identify the relevant macro-regime.

Early approaches to regime investing tended to fall into two camps. The first emphasizes simple discrete classification, usually using the level of growth and inflation to arrive at four distinct states of the world. This has the benefit of clear economic intuition, but ignores the uncertainty and potential for transition to a different regime. This can be seen in Figure 15.1, which shows monthly observations of this standard approach split into quadrants. When growth or inflation is close to the long-term trend level, the precise categorization of a period in a specific regime at times is somewhat

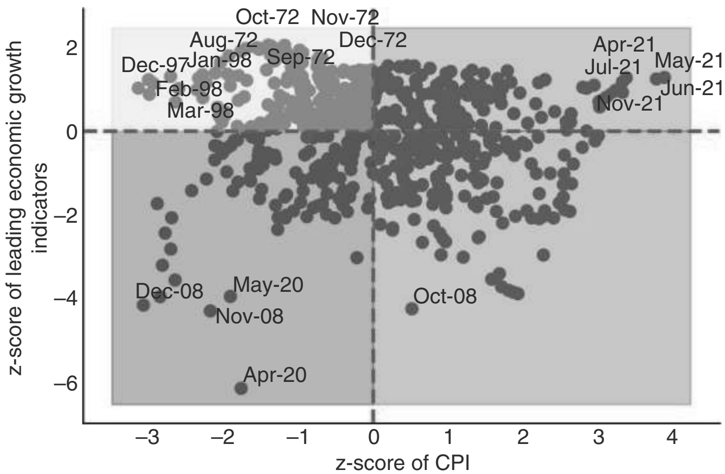


FIGURE 15.1 Regimes using only inflation and growth can result in arbitrary regime clustering.
Source: Bloomberg, Citi research, and authors' calculations.

random and can just depend on a rounding error. In addition, in this framework, it is very difficult to generalize beyond just growth and inflation, which misses other important dimensions of the macroeconomic environment. The second approach is very statistical and data-intensive, using hidden Markov models to detect regimes and transition probabilities. These models can lose intuition on which regimes are being detected and can struggle to estimate out-of-sample or across training windows.

The limitation of the quadrant approach also becomes evident in recession periods. Contrary to economic intuition, both equities and bonds (and credit) have similar above-average Sharpe ratios in times when both inflation and growth are below average. The reason for this counterintuitive result is that the levels of growth and inflation do not fully characterize the macro environment. Recessions come in two distinct phases. The first is "early recession" or the contraction phase, when growth is below average and weakening, resulting in a classic risk-off, when bonds outperform equities. Secondly, there is also a late recession or recovery phase, when risk assets have already sold off and central banks are stimulating. Here, markets are forward-looking and can see the light at the end of the tunnel. These tend to be a very strong environment for equity and credit investments. The simple rule to capture these effects is to split recessions into early and late, by checking if growth indicators are above or below their 3mma. The main findings are illustrated in Figure 15.2 across asset classes, where all data start in the early 1970s except the EM indices, which start in 1988 for equities and 1993 for sovereign bonds (shown for illustrative purposes). The returns are not FX hedged due to data availability, even though we usually prefer to separate FX and asset performance.

Late recessions are the best macro regime for equities, which is also unsurprisingly true for credit. More risky markets like EM do better than their developed counterparts. Overall, the worst macro regime for risky assets is early recession, although for US equities, stagflation is even worse. Early recession is where the 60/40 allocation

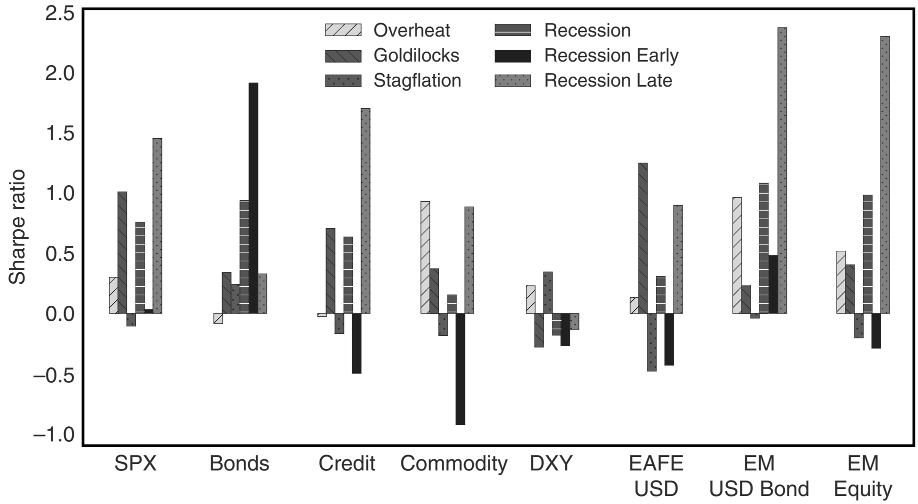


FIGURE 15.2 Equities do well in recessions but only after they have posted significant declines.
Source: Bloomberg, Citi research, and authors' calculations.

comes in, because bonds have their strongest returns in that regime before it peters out in the late recession regime. As mentioned earlier, stagflation (weaker than average growth, higher than average inflation) is the other regime equities struggle with, as do other asset classes. We will discuss this case in more detail later. During the overheating regime (higher growth, higher inflation than average), commodities do the best, and EM markets also tend to outperform. During Goldilocks (higher than normal growth, with lower than normal inflation), equities perform the best once more, and international equities outperform the United States as the dollar tends to weaken. Regimes clearly matter for macro returns.

15.1.1 A Playbook for the Future

Given the limitations listed earlier, our preferred approach is to characterize the macroeconomic environment using a wider range of variables drawn from asset pricing theory, following the early work of Rosemary Macedo and published in Saunders and Willer (2021). We use statistical tools to take a broader approach, effectively asking the question, which periods in the past are most relevant to determining the likely forward returns for asset classes (or factors, sectors, or strategies). Effectively, this jointly estimates the regimes, their characteristics, and transition probabilities, which are embedded in forward returns from the most comparable starting points in history. We use the nearest-neighbor algorithm to answer this question and gain insight into the forward-looking implications of current macroeconomic readings. Thus, we can incorporate additional relevant macro drivers without losing the economic intuition from regime investing.

Figure 15.3 illustrates the approach. The top panel shows four of the seven variables we think are important for characterizing the macro-investing environment.

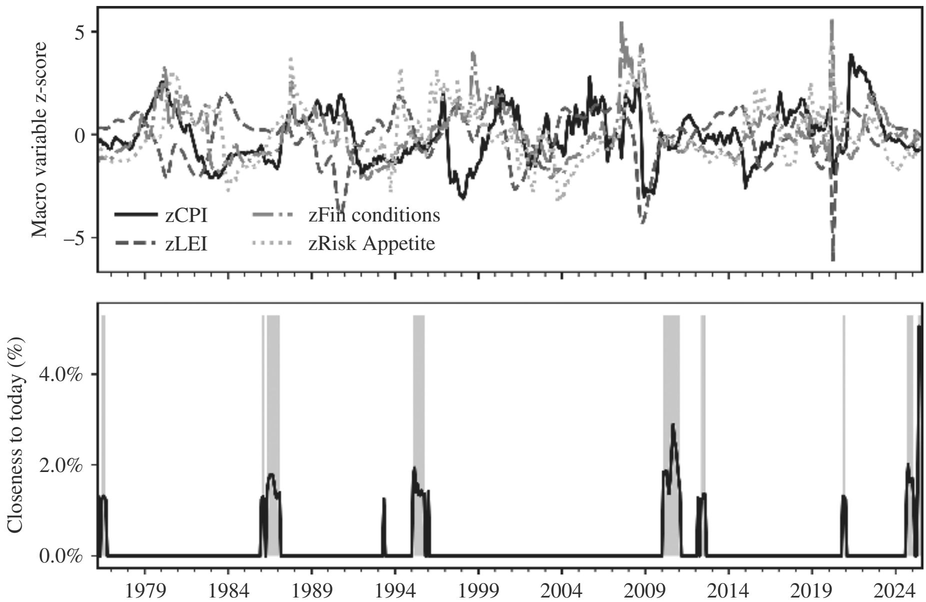


FIGURE 15.3 Playbook for the future. Which periods look like today?
Source: Bloomberg, Citi research, and authors' calculations.

Growth and inflation levels, financial conditions, and risk appetites (proxied by implied or realized volatility). We also incorporate changes in growth, inflation, and financial conditions as variables likely to affect either expected cash flows or investor discount rates. The bottom panel shows the weights on the most similar periods in September 2025, where we have picked Five years (60 months) of data. The macro-environment as of September 2025 had inflation below 10-year averages after a surge, growth close to trend, financial conditions loose, and volatility subdued. This rhymes with 1976 to 1977 (easy monetary policies, eventually leading to higher inflation), the mid-1980s, and the 2010 to 2012 jobless recovery. Generally, those were positive environments for equities, and that has played out at the time of writing.

15.1.2 If You Like It, Then You Better Put a Label on It

The nearest-neighbor approach is effective but is not always easy to interpret in terms of standard economic theory. There is also a need for scenario analysis and to generate asset class expectations based on economists' or portfolio managers' forecasts of the future regime rather than just the data at hand. In response, we developed a clustering approach in Saunders et al. (2024a), based on the cross-correlations of our macroeconomic variables. This enables us to classify the recession quadrant effectively, and we find that the approach gives a richer mapping than the discrete quadrant classifications. We use a k-Means clustering algorithm and find 10 macro environment clusters that follow a natural 80/20 rule. Eighty percent of the data fall into common clusters, and these map well to intuitive definitions. The other 20% are risk or unusual

Macro Regime Clusters

	zG7 CPI	zG7 LEI	zG7 CPI diff	zG7 LEI diff	zRisk Appetite	zFin conditions	zFCI diff
Normal (n=124)	−0.08	0.38	−0.19	0.03	−1.02	−0.59	−0.04
Recovery (n=84)	−1.08	−0.23	−0.18	0.16	−1.46	−1.37	−0.02
Goldilocks (n=94)	−0.96	0.82	0.21	0.19	0.18	−0.1	0.06
Growth shock (n=53)	0.58	−1.78	−0.05	−0.29	0.1	−0.19	−0.34
Tightening Fin Cond (n=46)	1.31	0.04	0.0	−0.4	0.31	1.54	0.56
Stagflation (loose) (n=23)	1.92	−0.98	0.19	−0.55	2.42	2.19	−0.26
Stagflation (n=10)	−1.13	−2.87	−1.67	−1.61	2.75	4.09	2.76
Inflationary boom (n=63)	1.57	0.97	0.64	0.02	−0.47	−0.05	−0.02
Recession (n=48)	−1.75	−1.3	−0.16	0.4	1.05	0.76	−0.77
Fin Mkt shock/crisis (n=48)	−0.91	0.6	−0.04	0.12	1.96	0.69	0.24

FIGURE 15.4 Grouping periods into the top 10 clusters.
Source: Bloomberg, Citi research, and authors' calculations.

clusters. Figure 15.4 illustrates the median z-scores across the top 10 clusters with our stylized names.

We show how assets perform in each environments in Figure 15.5, which delivers a few key insights. First, there is a normal environment, when growth and inflation do not deviate wildly from their long-term trends. All assets tend to deliver positive returns. The Goldilocks regime maps similarly to the traditional quadrant approach, growth shock maps to early recession, and recovery to late recession. There is an additional macroeconomic cluster where inflation is above average and growth is close to trend, which is less benign, as the tightening of financial conditions (either from monetary policy or from credit spread widening) means that stocks, bonds, and credit can all sell-off. This regime can lead to stagflationary conditions like in 2022–2023. Of the less common environments, which we use more for stress-testing of portfolios, we see two flavors of stagflation. There is one where central banks and markets expect "transitory" inflation e.g., 2022, and one where financial conditions have to tighten meaningfully and inflation hedges do not work (see discussion and Table 15.1). Recessionary times are actually one of the best times to invest; economic data is at its nadir, and asset valuations have cheapened. Markets anticipate a recovery in the data and tend to rally, especially risk-sensitive equity and credit positions. Equities see a change in leadership and momentum strategies, especially in equities, which tend to see violent reversals as the laggards become leaders. One example is financials in March 2009 or cruise ship/travel companies in April 2020.

With stagflation and tightening financial conditions, which can lead to stagflation, showing the most negative Sharpe ratios, we investigate next how asset allocators can protect against stagflationary outcomes.

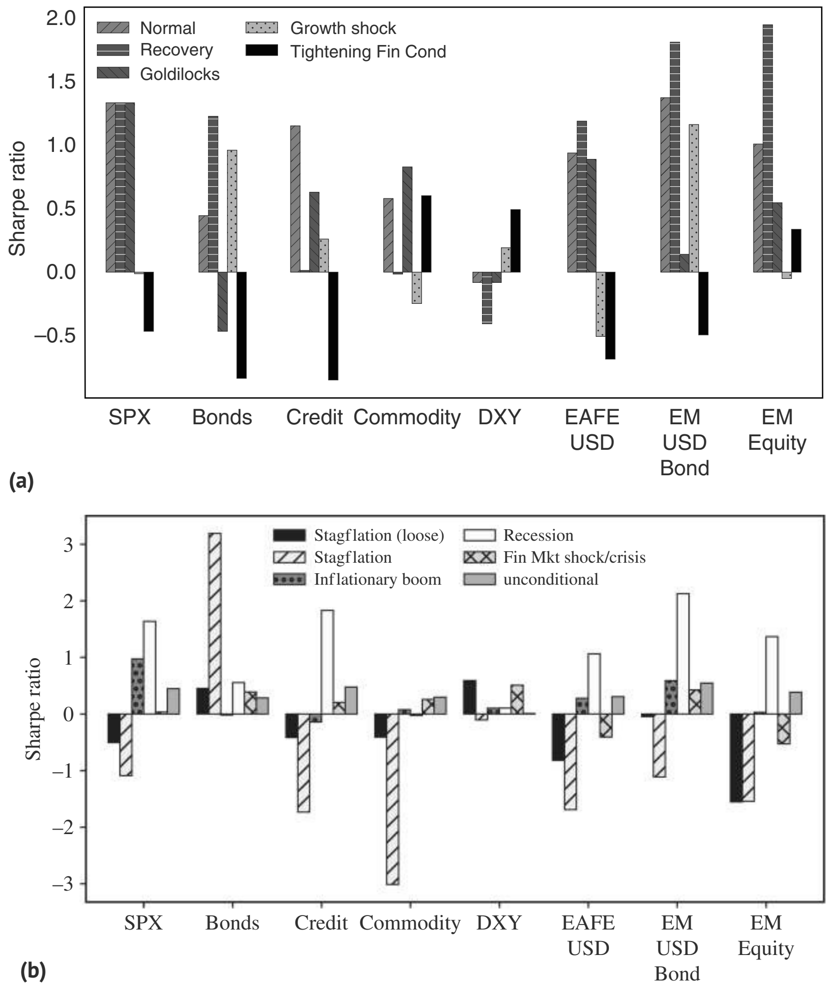


FIGURE 15.5 (a) Tightening financial conditions as the most difficult cluster. (b) Stagflation also a painful cluster.
Source: Bloomberg, OECD and authors' calculations.

15.2 HOW TO PROTECT AGAINST STAGFLATION

As mentioned previously, the workhorse for asset allocation for many decades has been the 60/40 portfolio, where 60% was invested in equities, and 40% in fixed income, mostly government bonds. The key feature of that allocation was the negative correlation between bonds and equities during early recession periods. In the late 1990s, Bridgewater popularized the concept of risk parity, which is also largely based on the negative correlation between returns from government bonds and equities. The innovation, based on the portfolio optimization work of the 1950s and 1960s, was that the use of leverage allowed for allocating the same amount of risk to bonds, commodities,

TABLE 15.1 During Fed Hikes, Not Many Inflation Hedges Work

	Levels		Returns since Jan 2001					
	CPI start	FF start	Gold	Bitcoin	TIPS	DXY	CTA	SPX
Jan-21	1.3	0.25						
Mar-22	7.9	0.50	3.7%	36.4%	2.7%	8.0%	6.9%	25.9%
Jul-22	8.4	2.50	–6.0%	–31.0%	0.1%	16.9%	13.3%	13.4%

Source: Bloomberg, and authors' calculations.

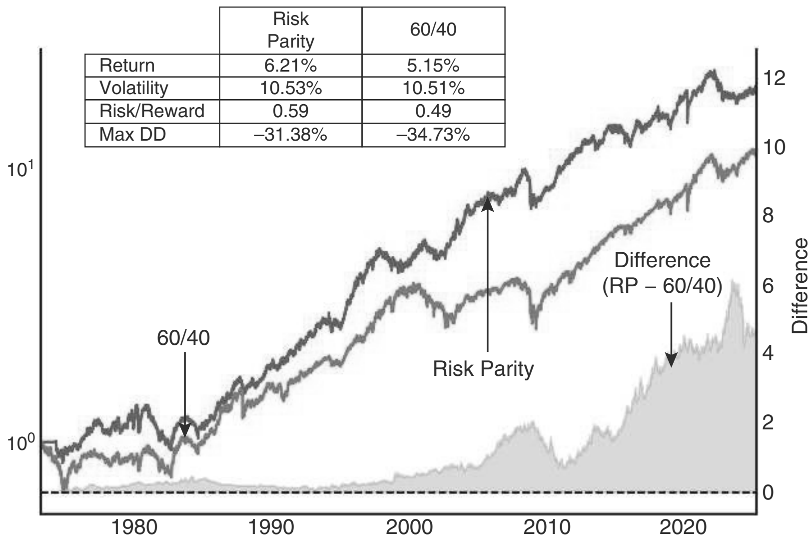


FIGURE 15.6 Risk parity outperformed post-GFC, but not during inflation post-COVID.
Source: Bloomberg and authors' calculations.

and to equities, irrespective of notional weights. This technique improves diversification, as it allows investors to allocate the same amount of risk across different asset classes, moving away from focusing too heavily on equities. Figure 15.6 shows that for long periods of time, risk parity portfolios have had superior returns to the classic 60/40 benchmark.

However, in the post-COVID period, risk parity has underperformed. The reason is that the best of all worlds for risk parity is when the correlation between bonds and stocks is negative and when the cost of leverage is low. This was the case for most of the aftermath of the 2008 crisis, where the world remained mainly concerned about further deflationary shocks. However, in the aftermath of COVID, that changed, as the policy response to COVID resulted in an extremely inflationary shock. The post-COVID inflation led to a change in the correlation between bond and equity returns, which became positive, as can be seen in Figure 15.7a. This was especially the case in 2022, when inflation was high, and the Fed reacted aggressively to higher inflation,

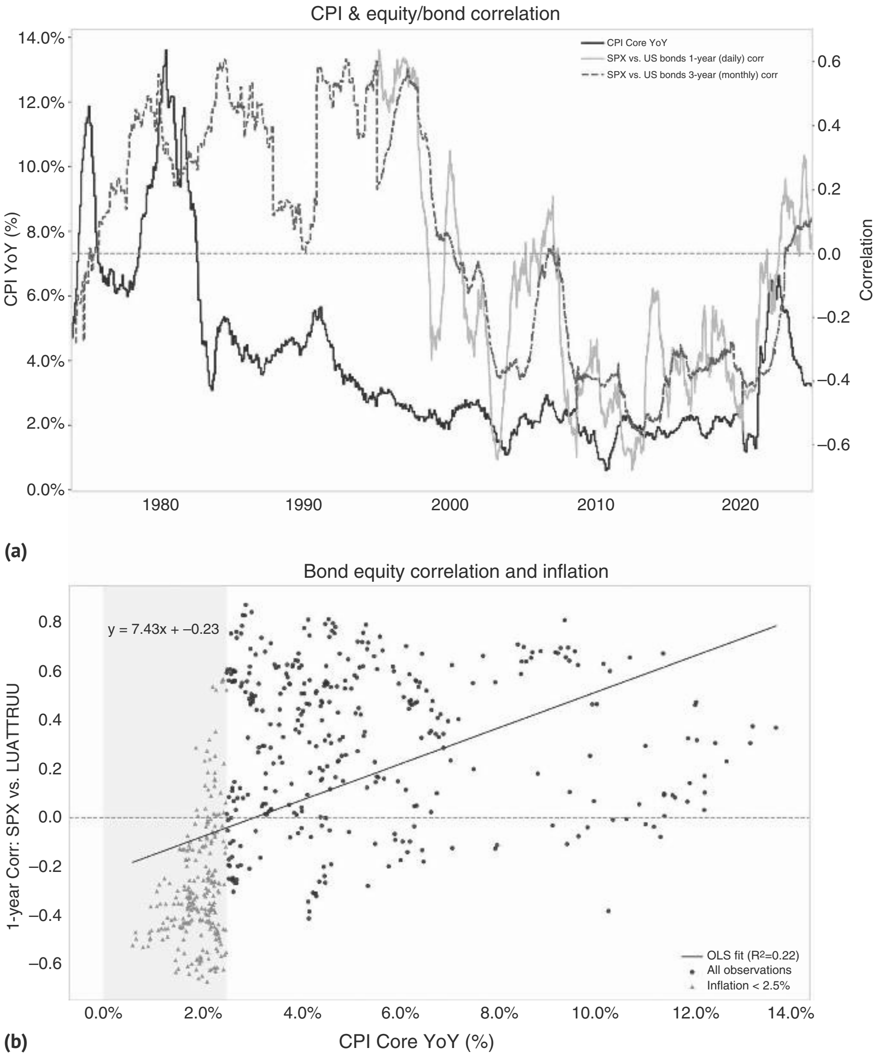


FIGURE 15.7 (a) CPI important for stock-bond correlation. (b) For CPIs below 2.5%. stock–bond correlation mostly negative.
Source: Bloomberg, FRED, and authors' calculations.

after initially dragging its feet, wrongly assuming that inflation would be transitory. Eventually, the inflation spike did turn out to be transitory, allowing the Fed to cut interest rates in 2024, and leading to the correlation at least stabilizing, with a downward bias. While the correlation between the US core CPI and the equity/bond correlation is not as stable as we would like, Figure 15.7b shows that for inflation levels below 2.5%, the equity/bond correlation is predominantly negative, while above 2.5%, it is mostly positive.

In the future, the important question to ask is whether inflationary or deflationary shocks are more likely. Unfortunately, there are some important contenders for inflationary shocks. The most important challenges that could generate inflationary shocks are the costs of the energy transition, which may generate "greenflation," the costs of structurally higher defense spending, as we exit pax Americana, and the general idea that governments have to a large extent socialized risks like COVID, creating an expectation among voters that the next shock would again be absorbed by the government rather than the private sector, at a time when government debt is already extremely high. There are clearly also deflationary shocks in our future, which include most plain-vanilla recessions, but also the unstoppable march forward of AI, which means that government bonds will likely remain a part of well-diversified portfolios. However, if inflationary shocks are more likely than they were in the past, a significant part of the bond allocation should instead be invested in inflation hedges rather than just in deflation hedges. This is before we even consider the problem of the bond glut, where debt issuance may be too high for back ends to participate as much in central bank cutting cycles as they did in the past.

The problem with inflation hedges is that they are much less plentiful than deflation hedges. This is especially true for episodes where the Fed is actually addressing inflation with rate hikes. An interesting case study in this regard is 2022. When inflation was rising, while the Fed was still dragging its feet, there were several classic inflation hedges that worked well, including gold, bitcoin, TIPS, and even equities. This was the case from January 2021 to March 2022. However, when the Fed started to hike rates in March 2022, even with inflation still going up until peaking in June 2022 (published in July), those hedges stopped working. As illustrated in Table 15.1, which shows the cumulative returns for some of the traditional inflation hedges, gold, Bitcoin, and SPX all had meaningfully negative returns between the start of the Fed hiking cycle in March 2022, and peak headline inflation in June 2022. Even TIPS had negative returns, as the negative impact of the Fed hiking cycle more than offset the higher carry from higher inflation. Equities did well between the start of Fed hikes and peak inflation, but that happened to be a bear market rally in a fairly pronounced bear market. Within public markets, the USD was the only asset that consistently performed well both when inflation was rising and was not being addressed by the Fed, and when inflation was rising, with the Fed hiking to belatedly address inflation. Given that the USD also performs well during most deflationary shocks, as risk aversion tends to drive up the USD, it may be an interesting hedge, especially when it is also positive carry. We would also note, though, that given that FX is a relative price, the strong USD performance in 2022 was helped by the fact that the inflationary shock was somewhat US-centric, given the strong US fiscal response. If this is not the case, the USD may not work as well as in 2022. As always in FX, many other considerations will also play a role, as we have discussed in the FX chapter.

We therefore extend our search for inflation hedges to factors and macro strategies. We find that trend-following and momentum strategies (in FX, and also in equities) work well. The same is true for the carry factor, also a favorite macro hedge fund strategy, though in FX carry, the high stagflation case is not profitable, as is illustrated in Figure 15.8. Value works in fixed income and FX, but not so much in equities. Overall, CTAs and macro hedge funds are therefore likely the best inflation and stagflation

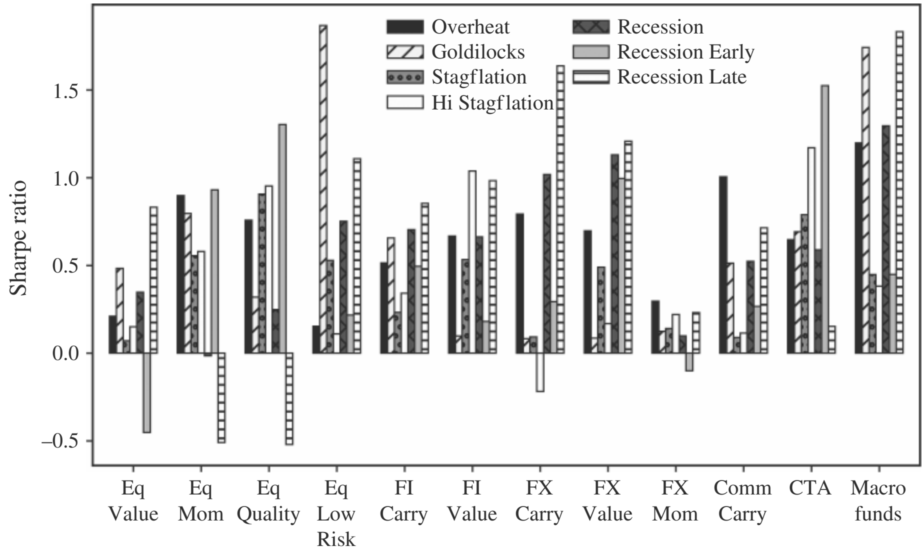
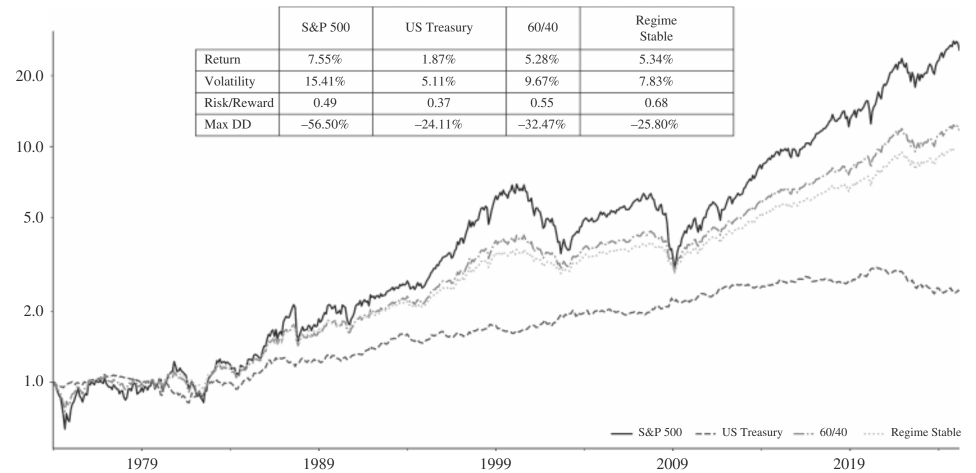


FIGURE 15.8 Several macro factors work well during stagflation.
Source: Citi Research, AQR, Barclay, Bloomberg.

hedges for investors. This is likely to remain the case going forward as bear markets driven by inflation are typically relatively slow-moving and longer-lasting. This suits CTAs much better than environments like 2020, when the bear market was very sharp and short-lived. Within the equity market, the quality factor performs very well. Carry also tends to work well, especially in fixed income.

15.3 STRATEGIC ASSET ALLOCATION FOR REGIME AGNOSTICS

Strategic asset allocation is its own field and beyond the scope of this book, as most of our tools and models are much more tactical in nature. However, we would be amiss to not point out that the regime approach is also extremely useful for strategic asset allocation. We address this topic in Saunders et al. (2022). The main argument is that, in the long run, portfolios should be regime-agnostic. Compared to the 60/40 portfolio, this means partly adding assets that can do well in the stagflation regime, which is the blind spot of the 60/40 portfolio. As discussed earlier in this chapter, this means adding real assets, trend-following, quality equities, and carry strategies to reduce regime dependence. Of course, the optimal strategic asset allocation depends on how much investors are willing to trade-off in terms of returns for regime stability. But for most preference structures, the prescription is to move an allocation away from equities and/or bonds and to those alternative asset classes. For example, a portfolio that is 50% equities (property or other real assets can also be included here), 30% bonds, 10% CTAs, and 5% quality and carry (each) results in positive performance across all regimes. Of course, stagflation still performs worse than other regimes, but returns are now (barely) positive. As per Figure 15.9, the simulated information ratio



	S&P 500	US Treasury	60/40	Regime Stable
Return	7.55%	1.87%	5.28%	5.34%
Volatility	15.41%	5.11%	9.67%	7.83%
Risk/Reward	0.49	0.37	0.55	0.68
Max DD	–56.50%	–24.11%	–32.47%	–25.80%

FIGURE 15.9 Regime-agnostic portfolio outperforms 60/40.
Source: Bloomberg, AQR, Barclay, authors' calculations *Source:* Bloomberg and authors' calculations.

is now 0.68, up from (also fixes 0.61 to 0.68). 0.55 for the 60/40 portfolio. Returns come in at 5.3%, slightly beating 60/40, but not equities.

The previous discussion was set in a long-only context. However, it is also possible, of course, to use shorts to hedge portfolios. An interesting aspect is whether credit can hedge equities or vice versa. We investigate that next.

15.4 THE CHOICE OF EQUITY VERSUS CREDIT RISK

One way to answer the question of how to choose between equity and credit is to calculate the ex post optimal Sharpe ratio portfolios across bonds, equities, and credit. We do this using rolling 10 years of data on returns, correlations, and volatilities. Figure 15.10 shows that in the 1970s, corporate bonds made up a significant proportion of the optimal Sharpe portfolio, at times running at 100%, as equities and bonds struggled with supply-induced inflationary shocks. However, this changed in the late 1990s as both fixed income and equities took larger weights in the portfolio. As an aside, we would note that this is also when higher-quality duration-matched data on credit excess returns became available, which may exaggerate the change in the optimal weight for credit. Since the GFC of 2008, credit's role in the portfolio has been greatly diminished, averaging just 12%, and currently is not sitting far from zero. Still, even since 2008, having corporate bonds as part of the optimal portfolio has increased the Sharpe ratio in our hypothetical exercise. However, for trading purposes, this analysis is too big picture. We go into more detail next.

In order to understand the predictors of credit versus equity performance on a vol-adjusted basis, we run a regression on the likely candidates: volatility (equity and bond), credit spread levels and recent changes, equity momentum and fundamental factors, the shape of the yield-curve, recent (i.e., US 2yr note) changes, and inflation.

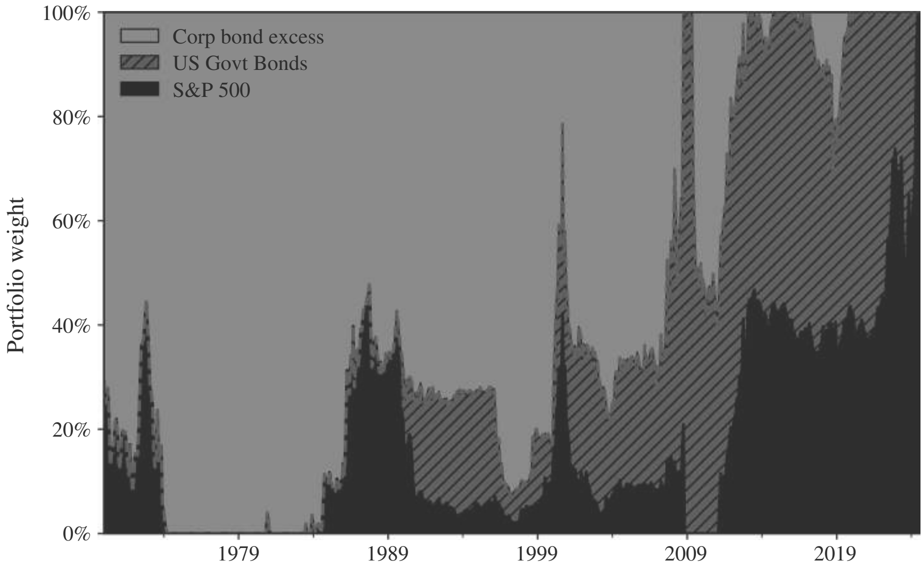


FIGURE 15.10 The ex post optimal share of credit has fallen sharply.
Source: Bloomberg and authors' calculations.

The results are in line with intuition; significant factors for choosing equities are a high VIX and recent equity outperformance. High spreads are more positive for credit. Credit spread changes are subsumed by the equity moves, however. High inflation favors credit, other things being equal, but it is not statistically significant. Interestingly, these factors can be captured in a real-time strategy, where we use the candidate variables in a ridge regression with an expanding window and then take positions on volatility-adjusted credit versus equity in proportion to the percentile of the forecasts. We adjust the volatility multiplier monthly. Typically, markets are range trading in relative terms, except during credit events, when credit will underperform, as can be seen in Figure 15.11. The backtest picks up those periods, as a rising VIX in the face of still relatively low credit pushes the model forecasts to meaningfully favor equities. Given the subpar liquidity profile of the investment-grade index, we also show results for CDX versus equity and find lower returns and Sharpe ratios. The PNLs have a similar profile of outperformance around credit events, but the Sharpe using the CDX is likely too low to trade it.

15.5 THE CHOICE OF GOVERNMENT BONDS VERSUS EQUITIES

While the choice of credit versus equities is extremely important in a hedging context, the main question in asset allocation is much more whether to favor government bonds over equities or vice versa. For longer-term investors, valuation is the obvious starting point for this investigation. As illustrated in the equity chapter, using valuation to time equity markets at the extremes usually leads to shorting a bubble, which is painful. Charts of long-term returns and starting valuations for equities do show

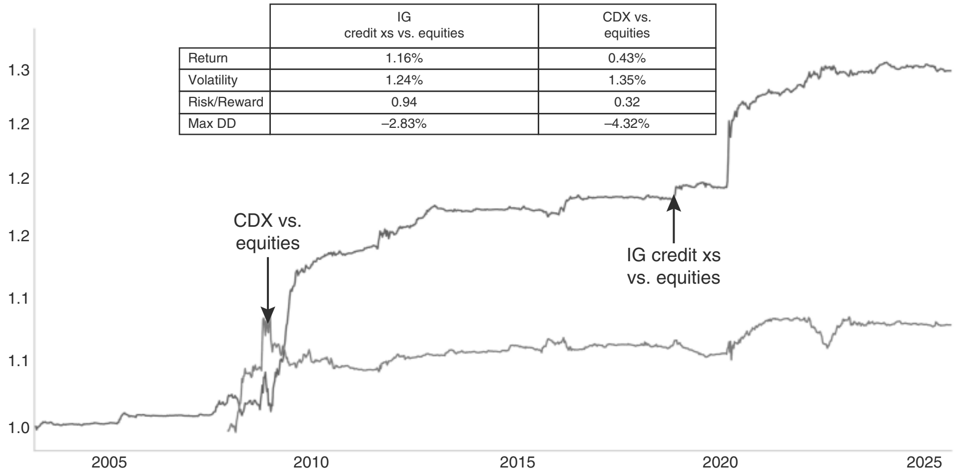


FIGURE 15.11 Equities are favored during rising VIX and with low credit spreads.
Source: Bloomberg and authors' calculations.

mean-reversion, but overlapping data makes it difficult to draw clean statistical inferences. Still, there is a question of whether focusing on cross-asset class valuation is more promising. To investigate this, we focus on the earnings yield gap, which can be expressed as earnings yields versus government bond yields. Operationalizing such a framework and accounting for structural changes (ever richer bond yields from the 1980s) is challenging. We therefore focus on a rolling three-year z-score of this measure. For long-term investors, we study the question by looking at extremes in the valuation of equities versus bonds and by imposing a long one-year holding period. We use earnings data from Bloomberg and lag it by three months to account for delays in earnings releases. We couple this with the level of the S&P to get a real-time measure of E/P, and we subtract 10-year government yields. The resulting z-score is shown in Figure 15.12. EYG is nicely mean-reverting. So far so good.

However, the backtest is less promising. Figure 15.13 shows the PNL of taking opposite positions when equities are particularly expensive relative to bonds, i.e., the z-score is –2 or lower. This strategy has added value at the tail end of the dot.com era and ahead of the 1987 crash. Recent signals would have been shorting an equity bull market, though. Overall, the information ratio is similar to buy and hold. On the other side, buying equities relative to bonds when the ratio was cheap has not been helpful. After the 2000 bubble burst, investors would have entered equities much too early during a bear market, where falling earnings kept equities cheap. The same happened in 2007, as declining bond yields pushed equities into the cheap zone before the financial crisis. The results are similar for longer-term z-score calculations up to 10 years. Broadly speaking, this analysis strengthens our view that valuation is a poor timing tool not only for equities outright, but also for the allocation between bonds and equities. While extremes in relative pricing versus bonds can help to highlight bubbles or unusually frothy conditions, there are many false alarms, and bullish equity signals tend to arrive in the midst of earnings deterioration or recessions, leaving investors exposed

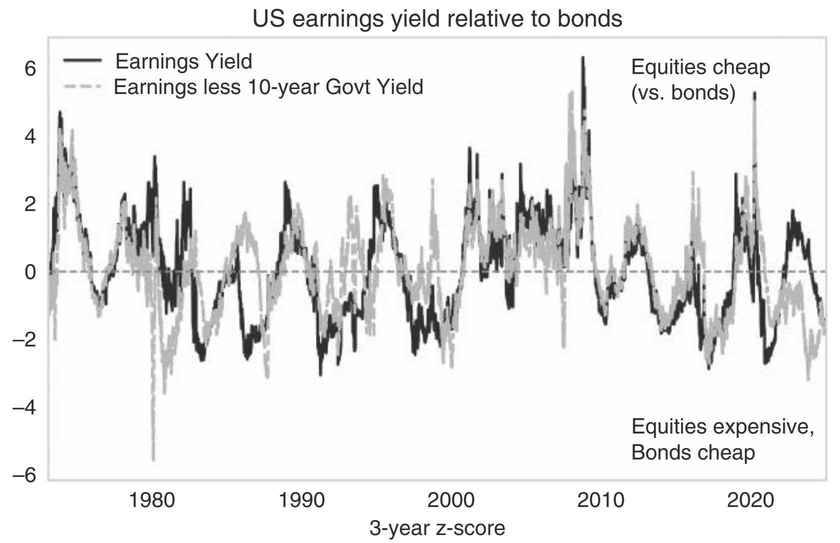
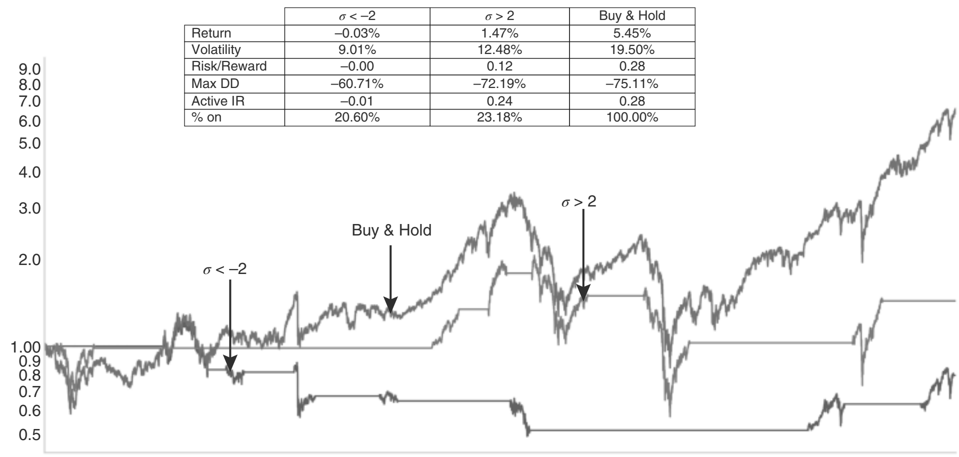


FIGURE 15.12 Earnings yield gap is mean-reverting.
Source: Bloomberg and authors' calculations.



	$\sigma < -2$	$\sigma > 2$	Buy & Hold
Return	–0.03%	1.47%	5.45%
Volatility	9.01%	12.48%	19.50%
Risk/Reward	–0.00	0.12	0.28
Max DD	–60.71%	–72.19%	–75.11%
Active IR	–0.01	0.24	0.28
% on	20.60%	23.18%	100.00%

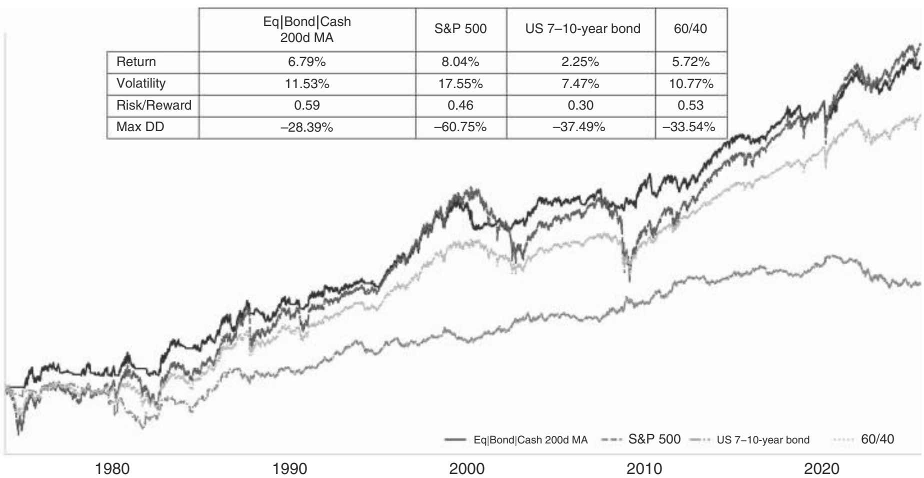
FIGURE 15.13 Using earnings yield premium not a great timing tool.
Source: Bloomberg and authors' calculations.

to drawdowns. Valuation, therefore, works better as a risk-management overlay—a warning flag at expensive extremes—than as a systematic buy/sell signal. For long-horizon investors, it may guide expectations of future returns, but it is a risk gauge, not a trading signal.

So what are investors supposed to do if valuation is unreliable? As usual, the answer is momentum. Historically, the simplest rule to allocate between bonds and equities is just to use the 200dma. If equities are above their 200dma, be long equities. If not, check if bonds are above their 200dma. If they are, buy bonds, risk-adjusted to have the same risk as was the case when fully invested in equities. If neither asset is above its 200dma, buy T-Bills. This simple rule results in an information ratio of 0.59, above the SPX of 0.46 and 60/40 of 0.53, as highlighted in Figure 15.14. Of course, the hope would be that our rules discussed in the equity chapter on how to buy bottoms would allow investors to not wait until the moving average breaks above the 200dma, but to move back into risk earlier.

In addition to the valuation or technical-driven frameworks, there are also structural developments to discuss, which are broadly a headwind for government bonds. First, the market capitalization of government debt has been rising very sharply, as governments have reacted to various crises (GFC, COVID), as well as plain vanilla recessions by protecting the private sector and socializing losses. It appears likely that this behavior will continue, suggesting that the market capitalization of public debt jumps significantly higher during a crisis, followed by stabilization, only to jump again when the next crisis hits. Presumably, this process will continue until markets become a binding constraint. The equity market, on the other hand, has been shrinking with negative net issuance due to a combination of strong buyback activity and a sluggish IPO market. Although market capitalization has increased, this is due to price appreciation, not issuance, as can be seen in Figure 15.15.

Second, the way in which debt indexes are constructed adds to the problem of too much debt supply. Most indexes use market capitalization to determine the weights. This means that the bonds of countries that are prolific issuers are getting a higher and higher weight. Given that credit risk is correlated with the size of outstanding debt, this means that indexes force investors to have exaggerated weights in countries that have the worst credit profile. Credit risk in government bond markets of developed markets has so far only been tested for brief periods, and most famously during the Liz Truss moment in the United Kingdom, but over the next decade, credit risk inherent in government bonds could be a more important factor. Although equities will not be



	Eq|Bond|Cash 200d MA	S&P 500	US 7–10-year bond	60/40
Return	6.79%	8.04%	2.25%	5.72%
Volatility	11.53%	17.55%	7.47%	10.77%
Risk/Reward	0.59	0.46	0.30	0.53
Max DD	–28.39%	–60.75%	–37.49%	–33.54%

FIGURE 15.14 A simple 200dma system works much better.
Source: Bloomberg and authors' calculations.

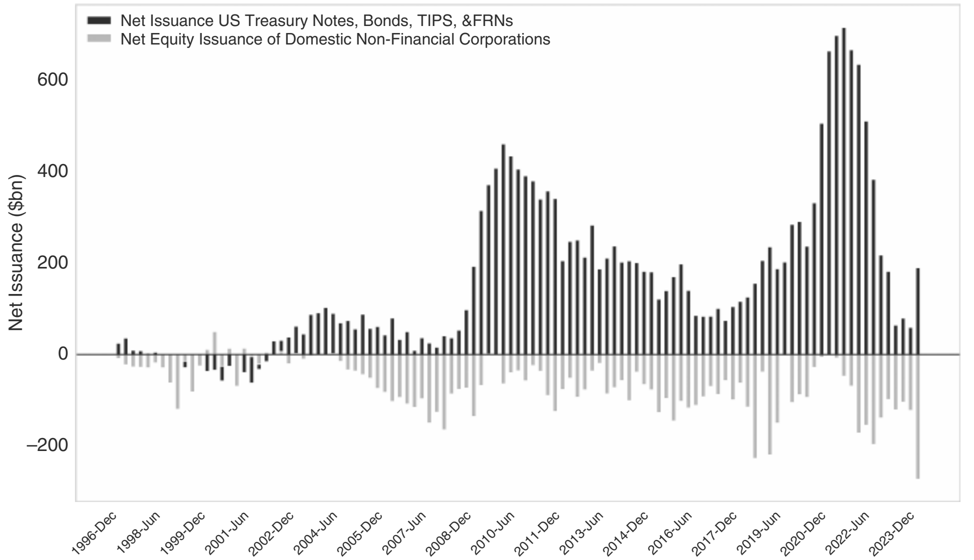


FIGURE 15.15 Net issuance favors equities over bonds. *Source:* Citi Research
Source: Bloomberg and authors' calculations.

immune if credit fears in the government bond market were to rise, it is noteworthy that equity market indexes do the opposite to debt indexes in that successful companies enter the indexes and increase their weighting, while unsuccessful ones do the opposite. The index weightings are therefore broadly favorable for investors in equity land and unfavorable in debt land.

15.6 DON'T RISK YOUR BUDGET, BUDGET YOUR RISK

Volatility is an important tool for asset allocation. Given the uncertainty surrounding future returns, it is notable that volatility tends to be persistent. In the FX chapter, we use this persistence to size positions across currencies with different risk characteristics. However, volatility does not just live in the cross-section (i.e., across assets) but also changes through time and environments (the time-series). This is something we explored before for the FX carry strategy with our volatility warning indicator. Here, we broaden the discussion. The main academic work on how time-varying volatility affects assets was Moreira and Muir (2017) but, as is often the case, was well known by practitioners for many years before.

To study time-varying volatility, we group each month by realized volatility using the last 22 trading sessions and group those into quintiles. We then calculate volatility, returns, and information ratios for the subsequent month and repeat the process. Our findings are illustrated by Figure 15.16, where the first panel shows volatility in the subsequent month in relation to the z-score of the volatility of the preceding month. Volatility is quite persistent. If it was high in the previous month, it tends to realize high again, while low readings beget low readings. Importantly, returns do not seem to follow such a strong pattern; the second panel shows that they remain around the same

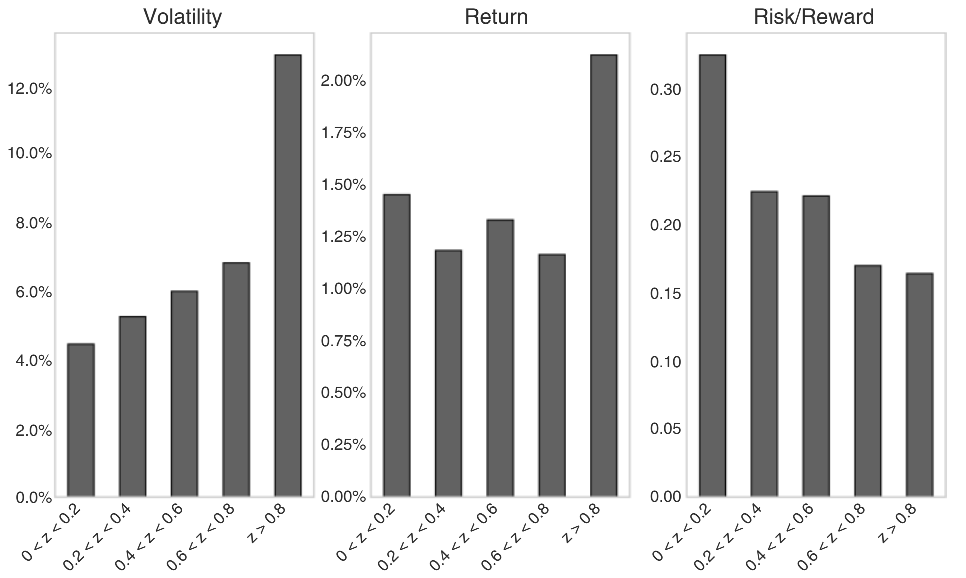


FIGURE 15.16 Higher risk, higher reward.
Source: Bloomberg and authors' calculations.

level, except for an uptick in the higher volatility quintile. This uptick is interesting and suggests that when realized volatility remains high for a second month, it can be in the context of upside volatility. Still, the increased returns do not compensate for the jump in volatility, resulting in lower information ratios even in the high volatility quintile. When maximizing information ratio, it pays to reduce positions when volatility strikes. We would note, though, that this is slightly different with implied volatility. On a tactical basis, a higher VIX delivers a slightly higher information ratio than unconditional, although lower implied volatility periods are even stronger, as discussed in Chapter 9, and shown in Figure 9.3b, where the difference between realized and implied is significant.

Applying the same framework to rates, inference for fixed-income markets can be fraught, given that daily data only starts in 1988, when bonds were already in a structural bull market. Earlier studies using yield data to proxy returns show a structural change in bond volatility in the mid-1980s (see Harvey et al., 2018). With these caveats out of the way, our study is illustrated by Figure 15.17, which shows the returns in each quintile of volatility over time. Interestingly, bonds do not have high returns following a period of high volatility. On the contrary, here returns are well sub par for the highest quintile, resulting in a very low (but still positive) information ratio. However, the pattern is not as nicely monotonically increasing as it is for equities, in that the second most volatile quintile has the highest information ratio. It takes extreme volatility to hurt bonds. This finding is mostly driven by the recent period, as the post-COVID rates bear-market features prominently. Even though investors should not put too much weight on one single episode, to us the 2022 episode is important, partly because the 1970s downturns are not always captured in historical analysis and because changes

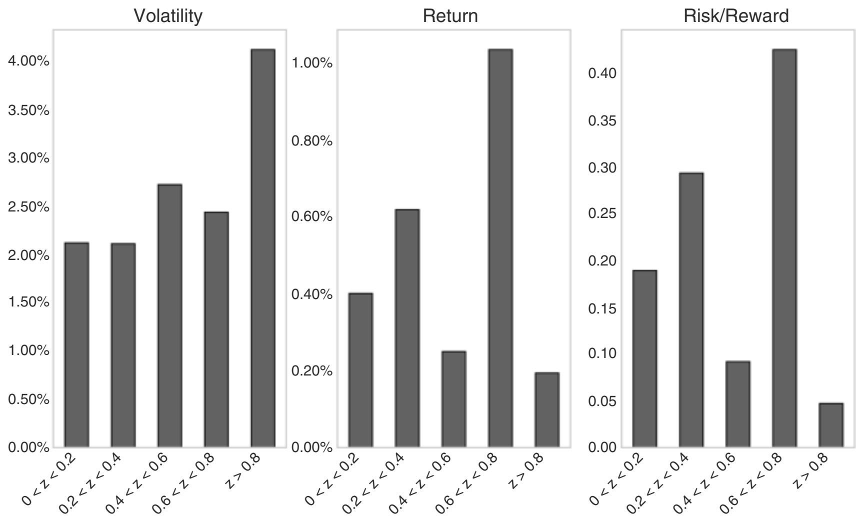


FIGURE 15.17 Higher risk without the reward.
Source: Bloomberg and authors' calculations.

in market structure since then make the 1970s less relevant today. We therefore prefer to reduce sizing in fixed income markets when volatility is moving up.

Finally, we broaden our analysis across other asset classes to understand in which assets volatility targeting is helpful. In particular, we analyze popular asset classes and scale positions to target the long-term volatility of each asset class. Each month, we examine the last month's realized volatility and then lever our hypothetical portfolio up if it has come in lower than the long-term average or down if it is higher in order to target a constant volatility through time. In equity and fixed-income markets, we see a benefit to scaling up risk in tranquil times and de-levering when realized volatility is higher than average, as shown by the volatility-targeted Sharpe ratios being higher than unlevered, as can be seen in Figure 15.18. The advantage in equities is highest for the United states, but still present in Germany (though not in Japan). We do not find a benefit for credit spread returns or commodity futures. Here, volatility targeting detracts value.

Overall, volatility targeting is a popular strategy, which tends to improve information ratios, at least for the more important asset classes, even though it does not necessarily improve total returns. As also discussed in the equity chapter, this strategy can lead to some momentum in markets—for example, when equity volatility falls from high levels, there tends to be a persistent bid as investors who reduced sizing have to ramp up their positions again.

15.7 TACTICAL, TACTICAL, ASSET ALLOCATION

So far, most of this chapter is firmly in the realm of tactical asset allocation, rather than strategic asset allocation. To get even more tactical, we analyze the role of quantitative funds in markets and whether there are any shorter-term signals for the various

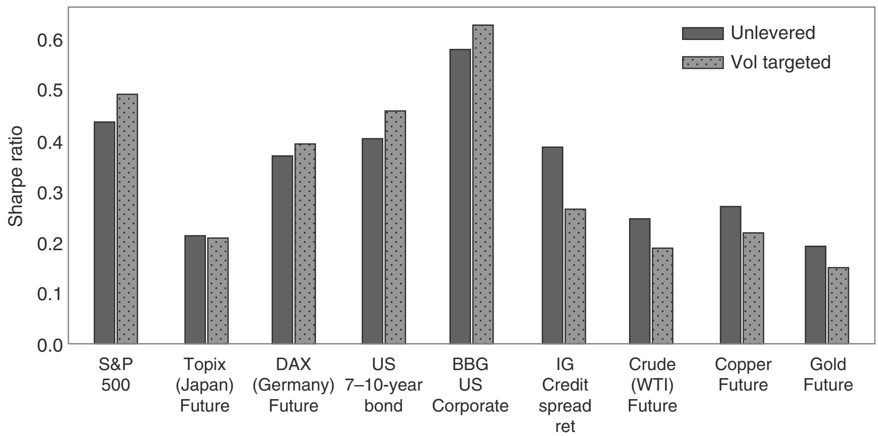


FIGURE 15.18 Targeting volatility works in some, but not all, markets.
Source: Bloomberg and authors' calculations.

asset classes. We focus on commodity trading advisors (CTAs), which have become an important player. Barclay Hedge estimates that as of mid-2024, CTAs had around 360 billion under management. With relatively high turnover, this suggests that CTAs are an important force in many markets. CTAs typically trade futures across asset classes and geographies and mostly employ trend following systems, based on moving averages. Next, we simulate a simple version of such a system by employing the 22-, 66-, 200-, and 252-day moving averages. We assume that for each moving average crossed, the CTA model adds one unit of risk, up to a max range of –4 (max short) to +4 (max long). This simple model does quite well to approximate publicly available PNL streams of CTA indexes. As a cross-check, we also use regressions of a CTA index on 12 of the most liquid futures contracts to infer current positioning. Given high turnover, we use the previous 22 trading days of return data (one month) to estimate exposures. When we use both simulated positions to forecast CTA returns, we find that the simple model using four moving averages does better than the one that infers positions from futures contracts. We then plot monthly forward returns for the moving average measure, depending on the inferred positioning (see Figure 15.19, left panel). We find that when using the moving average-based model, trend following basically works. This is especially true for commodities, where returns for the max short bucket are very negative, while the ones for the max long bucket are positive. FX and bonds also have the lowest returns in the max short bucket, which are roughly flat, and have decent returns in the max long bucket. For equities, there is no clear pattern, though, as the max short bucket has small positive returns. It is the second-lowest bucket, which actually has negative returns. While far from perfect, our moving average model is likely money-making. This would suggest that the value of CTA positioning as a counter-indicator is really extremely limited. Investors are better off going with the trend.

When hunting for counter-indicators, CFTC data does a better job than moving average CTA models, at least for equities. The right panel of Figure 15.19 shows monthly forward returns by z-score of the CFTC non-commercial positions, mapped

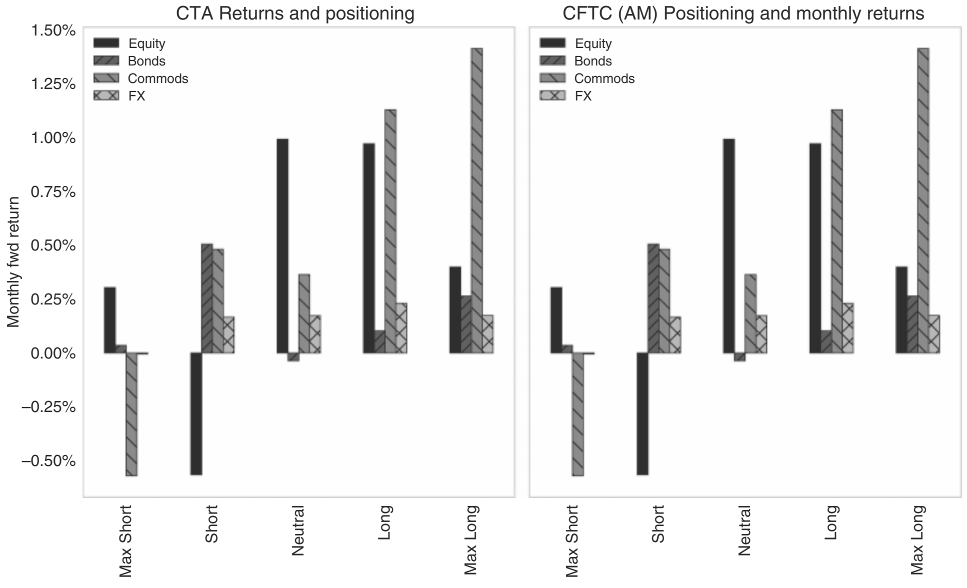


FIGURE 15.19 The trend is your friend, mostly.
Source: Bloomberg and authors' calculations.

to the equivalent positioning of our trend-following rules. A z-score of 2 or above is max long, 0.5 to 2 long, less than –0.5 is short, and less than –2 is max short. We look at month-end numbers for asset manager/managed money categories. Fading extremes works for equities (see also 9 using a daily signal). However, bonds and commodities show follow-through, and investors should go with the trend. FX is neither here nor there. For commodities, we would note, though, that the reality is highly contract-specific (see also Willer et al., 2023). Investors should therefore take extremes in CFTC data seriously when trading equities and certain commodities, but for other asset classes, it is not a good reversal indicator.

15.8 SUMMARY

In this chapter, we describe the regime investing approach in some detail and lay out how to improve upon the traditional four-quadrant approach. We focus on stagflation as a regime in which most public assets do not work well. Instead, trend-following and momentum strategies are attractive, as are carry strategies. These patterns are useful for regime-agnostic allocation. When deciding whether to buy equity or bonds, valuation measures are not overly compelling. Simple moving average rules tend to work better. Volatility targeting adds alpha in the most important asset classes, but not everywhere. More tactically, our data shows that trend following works and that CTA positioning should not be faded. CFTC positioning is also more often a trend signal than a mean reversion signal, with the exception of equities.

CHAPTER 16

Non Traditional Assets: Private Assets and Crypto

The book has so far focused on the traditional public markets, partly because our approach is mostly quantitative and therefore relies on sufficient data for our backtests and models. However, for asset allocators, non traditional assets have become more important. This is both the case for crypto and for private assets. We briefly discuss how both crypto and private assets fit into portfolios. We start with crypto.

16.1 MACRO AND CRYPTO—A MATCH MADE IN HEAVEN

Macro investors have been drawn to crypto like flies to the fire. The reason is partly that crypto has FX-like characteristics, which macro investors are very familiar with. The other reason is that macro investors are oftentimes quite momentum-minded, preferring trend following to mean reversion. With crypto being quite trending, it has been a match made in heaven. Lastly, crypto burst on the scene when the global environment was still quite difficult for macro investors, increasing the focus on this new shiny object. Of course, since the early days, crypto has come a long way. In particular, given that crypto has already survived two crypto winters, it is likely here to stay, and the risk of it going to zero has likely disappeared. With the endorsement by US regulators, this is now even more likely. In the following, we provide a framework for thinking about how to include crypto in the investment process, focusing on Bitcoin.

Bitcoin entered mainstream portfolios when spot ETFs were approved in February 2024. This led to a burst of enthusiasm and flows into this widely used investment vehicle. The ETF has broadened the investor base to include those who do not want to deal with custody and storage issues around physical holdings or with the basis risk from futures-based products. These flows have at times been a powerful tailwind for crypto assets as shown in Figure 16.1: every \$1bn of weekly flows has been associated with a 3.5% increase in price. Given the early stage of adoption with financial advisors and other institutional investors studying its portfolio allocation effects (see next section), these flows may continue, as more and more products are offered across different investor bases.

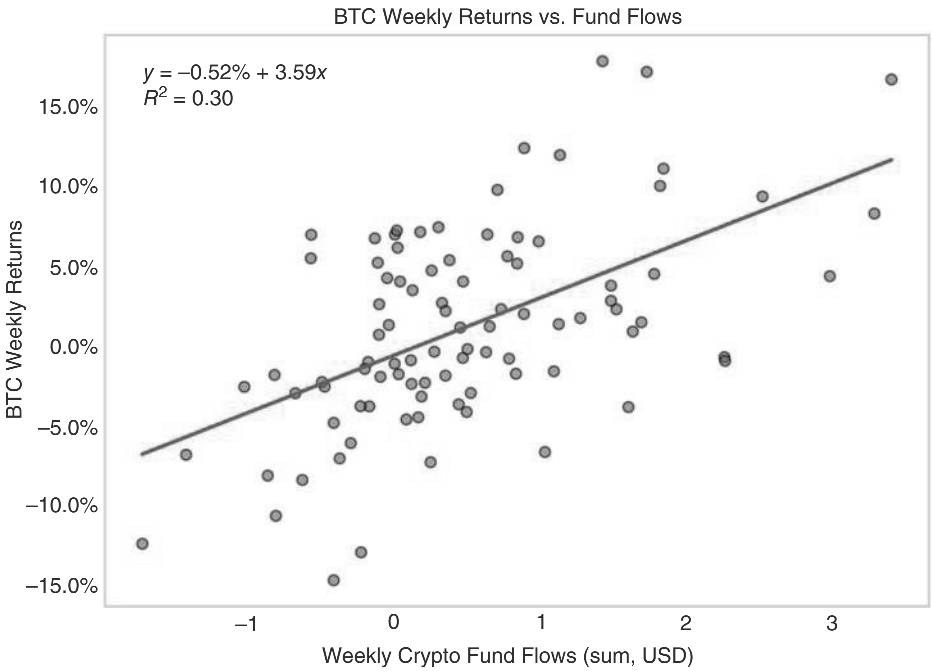


FIGURE 16.1 Bitcoin ETF flows show a strong relationship with price moves.
Source: Bloomberg and authors' calculations.

The first issue to analyze is the nature of Bitcoin in a portfolio context. Is it a safe asset, an inflation hedge, or more of a procyclical tech stock? We find that the long-term correlations are much more consistently high and positive with (technology) stocks than with gold, despite the digital gold moniker. This is especially the case in the post-COVID world, as is demonstrated in Figure 16.2. The correlation with gold is the second most consistent one, though. The negative correlation to DXY had been somewhat consistent, but has recently flipped to (barely) positive. For rates, the pattern is least stable. Furthermore, as per Table 16.1, Bitcoin is very consistently a negative performer when equities experience drawdowns, which is not the case for gold. The interesting point to make is that we are observing the opposite of what many observers expected. The basic idea was that over time, Bitcoin would graduate from a technological project to a safe asset. But post-COVID, the correlation with equities increased, rather than decreased, while the correlation with gold fell. While we are open to the idea that the correlation can go up with gold and go down with Nasdaq as Bitcoin matures, for now, there is little evidence of this happening. Bitcoin is mostly a beneficiary of plentiful liquidity and, therefore, is likely to remain correlated with other speculative investments, like tech stocks.

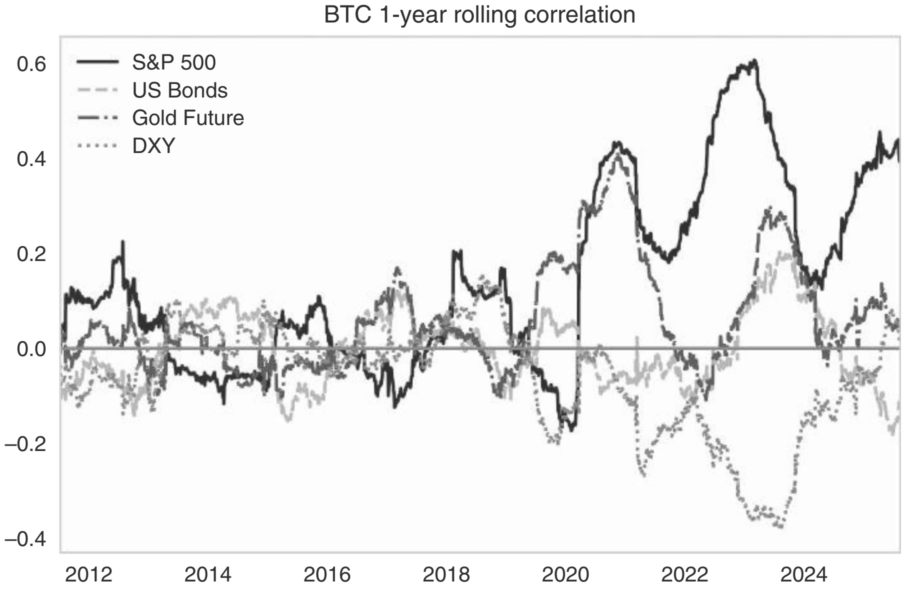


FIGURE 16.2 Bitcoin mostly a tech stock.
Source: Bloomberg and authors' calculations.

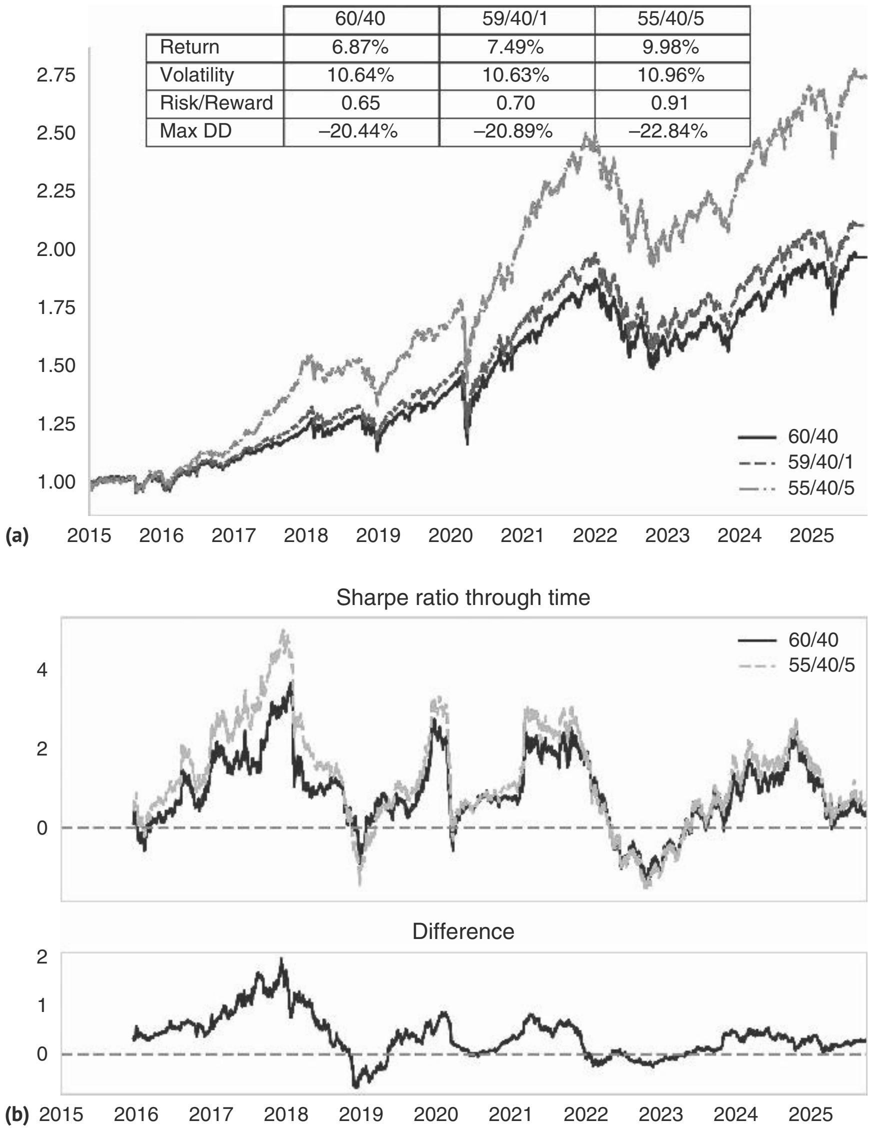
TABLE 16.1 Bitcoin Drawdowns with Equities, Unlike Gold

Start date	S&P 500	US bonds	Gold future	Bitcoin
Mar-2020	–12.6%	2.6%	1.4%	–25.5%
Sep-2022	–9.4%	–3.7%	–3.2%	–4.1%
Dec-2018	–9.2%	2.0%	4.5%	–6.7%
Apr-2022	–8.7%	–3.1%	–2.2%	–15.8%
Feb-2020	–8.4%	2.5%	–1.3%	–7.4%
Jun-2022	–8.3%	–0.9%	–2.2%	–41.1%
Sep-2011	–7.0%	1.7%	–11.4%	–37.3%
Oct-2018	–7.0%	–0.7%	1.6%	–5.3%
May-2019	–6.6%	2.1%	1.5%	62.0%
Dec-2022	–6.1%	–0.9%	3.8%	–3.4%
Aug-2015	–6.0%	0.0%	3.4%	–19.1%
Mean	–8.1%	0.2%	–0.4%	–9.4%
Median	–8.3%	0.0%	1.4%	–7.4%

Source: Bloomberg and authors' calculations

16.2 HOW MUCH TO ALLOCATE TO BITCOIN?

Given that Bitcoin is unlikely to compete with safe assets in a portfolio, we analyze how much of the allocation to risky assets should instead be allocated to Bitcoin. Most of the analysis starts with historic performance. Allocating to Bitcoin would have enhanced the portfolio returns of a traditional 60% equity/40% bond investor. We begin the analysis in 2014 when the first "TradeFI" investment trusts were available. Figure 16.3 shows the higher returns (and volatility) from adding Bitcoin at the expense



	60/40	59/40/1	55/40/5
Return	6.87%	7.49%	9.98%
Volatility	10.64%	10.63%	10.96%
Risk/Reward	0.65	0.70	0.91
Max DD	–20.44%	–20.89%	–22.84%

FIGURE 16.3 (a) Historical analysis shows Bitcoin *would* have improved performance. (b) Improvements to traditional portfolios were front-loaded.
Source: Bloomberg and authors' calculations.

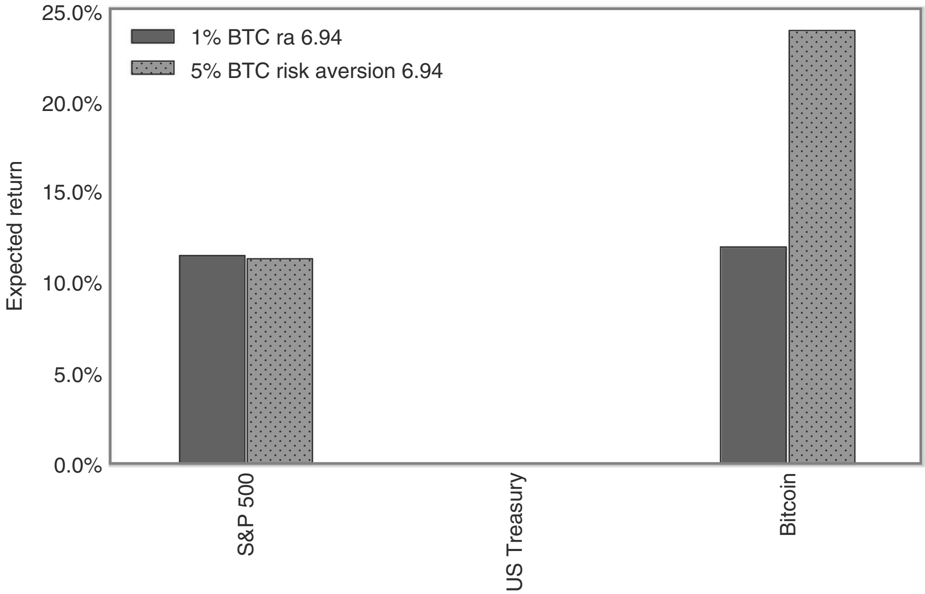


FIGURE 16.4 Black-Litterman suggests double-digit returns are required for 1% Bitcoin.
Source: Bloomberg and authors' calculations.

of the equity allocation. A 5% allocation would have increased the information ratio by 0.26, as can be seen in Figure 16.3a. The analysis here is for the United States, but the effects are even more pronounced using global indices. However, we also show the rolling one-year Sharpe ratio, which suggests that the benefits have been much larger for those who invested early in the adoption cycle, as the rolling Sharpe ratio peaked in 2018 and has been less pronounced since (and has also spent more time in negative territory), as per Figure 16.3b.

However, such analysis is riddled with survivorship and hindsight bias. Bitcoin caught the attention of financial markets precisely because of its exceptional performance in the early days. Almost any allocation to the nascent asset class would have added value, just as buying Amazon back in 1995 or a pre-IPO Google allocation would have improved your portfolio. A better way to think about an allocation today is to reverse the forecasting process. Given Bitcoin's volatility and correlation pattern today, what return would be needed to justify, for example, a 1% allocation. We perform such a reverse optimization in the spirit of Black-Litterman (see Black and Litterman, 1990), calibrating investor risk aversion using the performance of a 59/40/1 portfolio. As per Figure 16.4, to justify a 1% allocation to Bitcoin, the expected annual return would have to be around 12%. For a 5% allocation, a return just below 25% would be required. Bond correlations over the full period are still negative with equities (and crypto), so required expected returns for bonds are small.

16.3 FOLLOWING THE CRYPTO TREND

Given the high uncertainty on *how much* to allocate to Bitcoin, we now investigate the question of *how* to allocate to Bitcoin. If the macro funds are partly drawn to Bitcoin because it trends, then presumably trend-following strategies must work

	Trend 22d MA	Trend 55d MA	Trend 66d MA	Trend 200d MA	Trend 252d MA	Combo	Bitcoin
Return	60.35%	63.83%	62.58%	56.77%	65.42%	61.79%	63.22%
Volatility	47.93%	49.15%	49.64%	54.71%	56.54%	47.76%	66.88%
Risk/Reward	1.26	1.30	1.26	1.04	1.16	1.29	0.95
Max DD	–68.62%	–55.13%	–59.16%	–65.67%	–64.79%	–54.18%	–83.40%

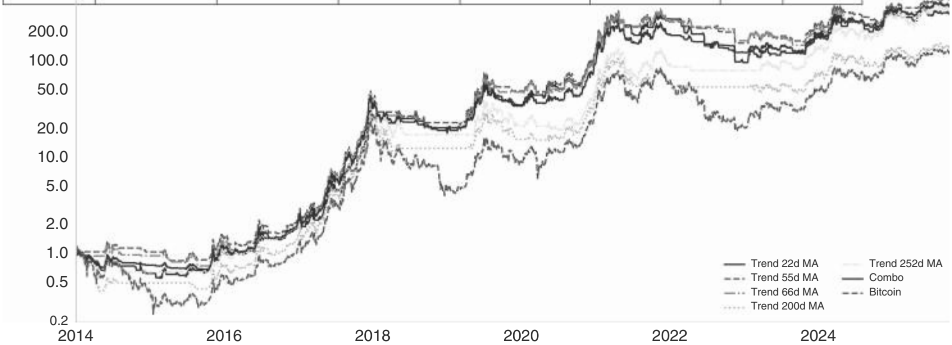


FIGURE 16.5 Trend following and Bitcoin—match made in heaven.
Source: Bloomberg and authors' calculations.

quite well. We demonstrate this in Figure 16.5, where we show the information ratios of various moving averages. Using a 3mma improves the Bitcoin information ratio from 0.95 to 1.26. Typically, trend-following strategies work by avoiding unpleasant tails and come at the price of slightly lower total returns. Interestingly, for Bitcoin, actual returns are almost on par, making a trend following strategy a good framework for Bitcoin investing.

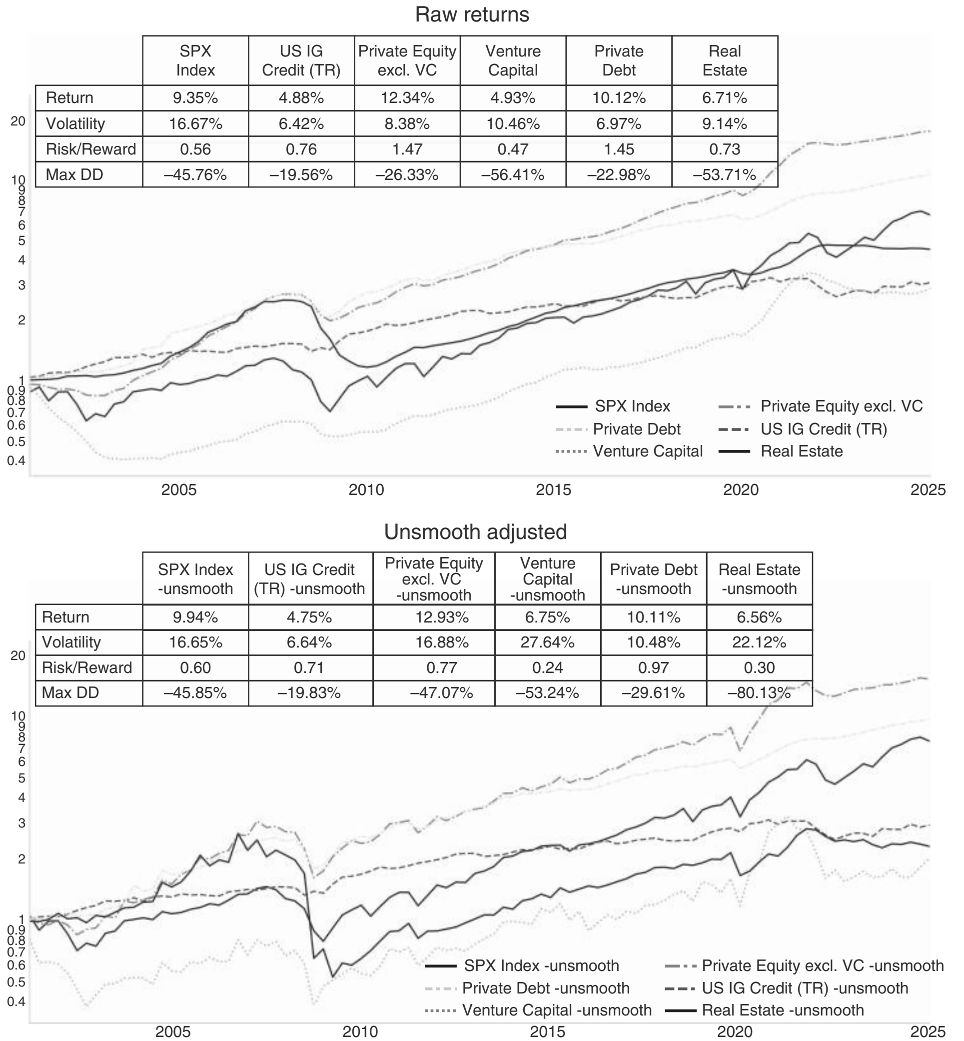
16.4 PUBLICIZING PRIVATE ASSETS

Private assets have been growing aggressively in the previous decades, gaining market share from public markets. They are now around 10% of the total global market cap. Growth has initially occurred in private equity, but more recently, private credit has also taken off. The reason is partly regulatory. An increased regulatory burden after the popping of the 2000 bubble and the corporate governance failures of 2002 had made IPOs less attractive. Similarly, more aggressive regulation of risk-taking and balance sheet usage of banks in the aftermath of the GFC has turbo-charged the development of private credit. The phenomenon feeds on itself. Increased market relevance leads to an increasing pool of capital available to fund private assets, reducing the need to tap deeper public markets. This is especially the case, as it is debated whether the illiquidity premium of private assets should be positive (what if funds are needed on short notice?) or negative (illiquidity can save investors from panic selling at the lows). Even though we are of the view that the liquidity premium should be positive, growth is likely to continue given all these tailwinds. Although data issues are limiting the analysis of private assets, we think they have become so important to asset allocators that we want to address how to think about the allocation decision in this regard.

16.5 A SMOOTH RIDE?

The first issue to address is return smoothing, which occurs due to infrequent reporting and general uncertainty with respect to where valuations truly are. This can reduce the measured volatility. Evidence for the smoothing of returns comes from their high autocorrelation, which is significantly in excess of what is observed in public markets. We use this phenomenon to unsmooth returns to make private assets more comparable to public assets. The unsmoothing process adjusts returns by using the first-order autocorrelation by blending current returns with those observed in the previous period.

We show the information ratios for public and private assets, smoothed and unsmoothed, in Figure 16.6. The advantage for private equity returns over public



	SPX Index	US IG Credit (TR)	Private Equity excl. VC	Venture Capital	Private Debt	Real Estate
Return	9.35%	4.88%	12.34%	4.93%	10.12%	6.71%
Volatility	16.67%	6.42%	8.38%	10.46%	6.97%	9.14%
Risk/Reward	0.56	0.76	1.47	0.47	1.45	0.73
Max DD	–45.76%	–19.56%	–26.33%	–56.41%	–22.98%	–53.71%

	SPX Index -unsmooth	US IG Credit (TR) -unsmooth	Private Equity excl. VC -unsmooth	Venture Capital -unsmooth	Private Debt -unsmooth	Real Estate -unsmooth
Return	9.94%	4.75%	12.93%	6.75%	10.11%	6.56%
Volatility	16.65%	6.64%	16.88%	27.64%	10.48%	22.12%
Risk/Reward	0.60	0.71	0.77	0.24	0.97	0.30
Max DD	–45.85%	–19.83%	–47.07%	–53.24%	–29.61%	–80.13%

FIGURE 16.6 Performance advantage falls with unsmoothed returns.
Source: Bloomberg and authors' calculations.

equity returns remains, but it is smaller after smoothing, dropping from 0.9 points to 0.17 points. The information ratio remains attractive at 0.8. Venture capital, on the other hand, has a worse information ratio than public markets post-unsmoothing, though we would also note that dispersion among venture capital funds is famously high, and the successful funds likely have a higher Sharpe ratio even after unsmoothing. For private credit, the advantage falls from 0.7 points to 0.26, with the information ratio still a high 0.97, though we would note that the asset class has not gone through a proper down cycle yet, given very high structural growth. The unsmoothed information ratio for real estate is about as poor as for venture.

The correlations of unsmoothed returns are high with public equities and among the different private assets themselves, as can be seen in Table 16.2. This is especially the case for private equity (PE), venture capital (VC), and private credit. For real estate and infrastructure, correlations are much lower, though data for infrastructure (and private credit) only start in 2008. The high correlation for PE, VC, and private credit is also observable when focusing only on the worst quarters for public equities returns. This suggests limited diversification benefits.

In terms of factor exposures, PE, VC, private debt, and real estate all have significant factor exposure to equities and to the size factor (small caps). This is illustrated in Figure 16.7. Both the value and growth factor also play a role in explaining returns. Interestingly, the positive beta to the size factor has diminished for PE but not venture capital. At this stage, PE is much more involved in large caps, such that the sensitivity to the size factor is moving close to zero.

TABLE 16.2 Positive Correlation with Equities Limit Diversification Benefits

	S&P 500	US Treasury	BBG commodities	IG excess	Private equity excl. VC	Venture capital	Private debt	Real estate
S&P 500	1.00	–0.38	0.38	0.74	0.66	0.49	0.55	0.25
US Treasury	–0.38	1.00	–0.37	–0.45	–0.36	–0.29	–0.32	–0.22
BBG Commodities	0.38	–0.37	1.00	0.49	0.50	0.29	0.50	0.29
IG excess	0.74	–0.45	0.49	1.00	0.46	0.25	0.52	–0.02
Private Equity excl. VC	0.66	–0.36	0.50	0.46	1.00	0.77	0.74	0.66
Venture Capital	0.49	–0.29	0.29	0.25	0.77	1.00	0.43	0.43
Private Debt	0.55	–0.32	0.50	0.52	0.74	0.43	1.00	0.45
Real Estate	0.25	–0.22	0.29	–0.02	0.66	0.43	0.45	1.00

Source: Bloomberg and authors' calculations

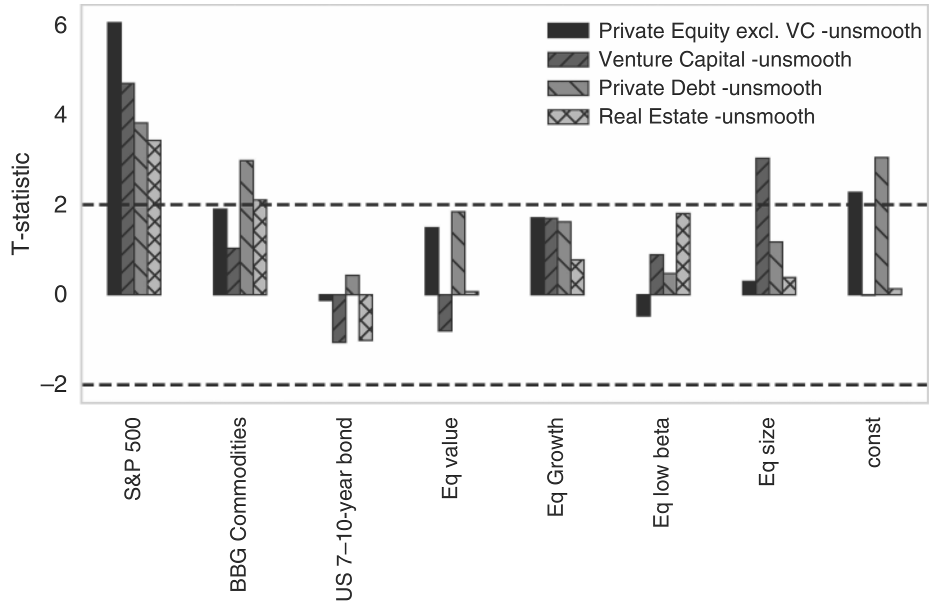


FIGURE 16.7 Private assets with strong US equity and growth factor exposure.
Source: Bloomberg, Prequin, and authors' calculations.

16.6 INFRASTRUCTURE (AND REAL ESTATE) FOR STAGFLATION

The next question is in what regimes the private assets are sensible positions to carry. In Figure 16.8, we show the forward Sharpe ratios for private assets by regime. We find that private equity does best in overheating and late recession, followed by Goldilocks. It does relatively poorly early in recessionary environments. In stagflation, information ratios are positive, but subpar. Overall, this is unsurprisingly a very similar profile to what public equities offer and not too different from other private assets. In particular, the distribution of returns across regimes is very similar to that of private debt. Infrastructure and, to a lesser extent, real estate offer decent Sharpe ratios during inflation (overheating and stagflation), although the evidence for infrastructure is tempered by the shorter history (the index starts in 2007, not 2000). Still, given that infrastructure investments tend to include projects that often have inflation plus pricing, we would give it the benefit of the doubt. The better returns in stagflation periods are notable because it is a difficult regime for most public markets.

Overall, we note that traditional asset allocation models would propose large positions in many private assets on the back of the observed historical returns and volatility, even when using unsmoothed returns. For real estate and infrastructure, the correlation structure adds to the appeal. Of course, liquidity needs may be an obstacle for many investors, and valuations need to be taken into account at the time of the investment decision.

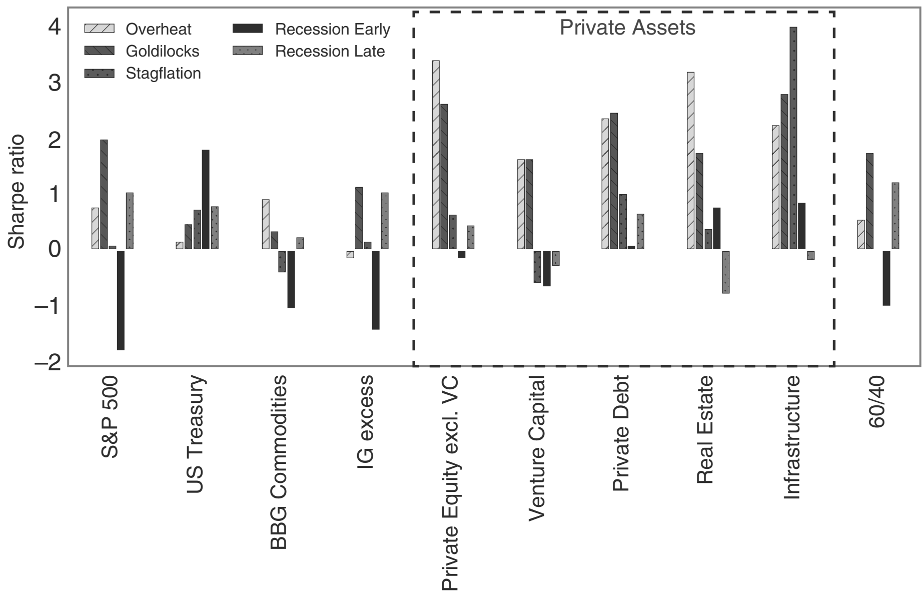


FIGURE 16.8 Infrastructure and real estate help during stagflation.
Source: Bloomberg and authors' calculations.

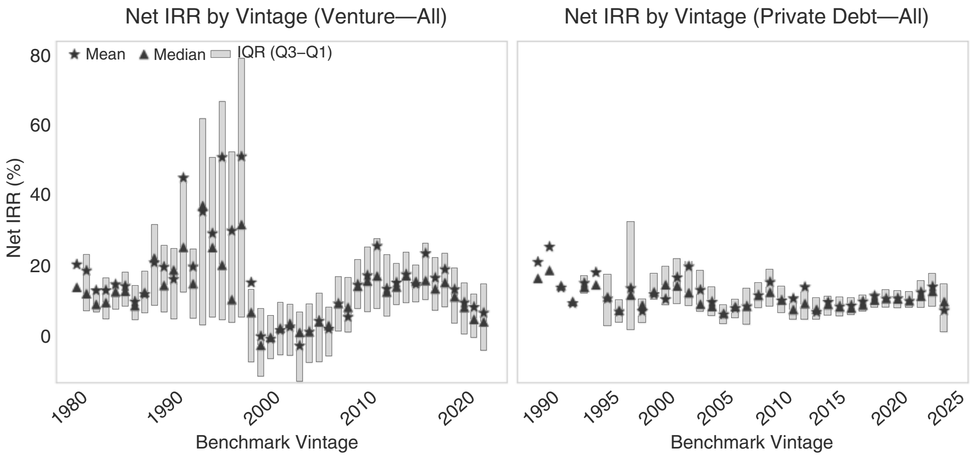


FIGURE 16.9 Manager dispersion is wide in VC.
Source: Prequin and authors' calculations.

16.7 ALL MANAGERS ARE NOT CREATED EQUAL

Manager selection is also key, as there can be wide dispersion in returns across managers. In Figure 16.9, we show the inter-quartile ranges for managers' net IRRs by benchmark vintage from Prequin. The range is especially wide for VCs, on the order of 20%. Private equity and especially venture capital investors should devote more resources to finding best-in-class managers relative to other asset classes. This is much less the case for private debt.

On the valuation front, we note that private equity IRRs seem to be correlated with starting valuations in public markets, as per Figure 16.10a. On this front, the going

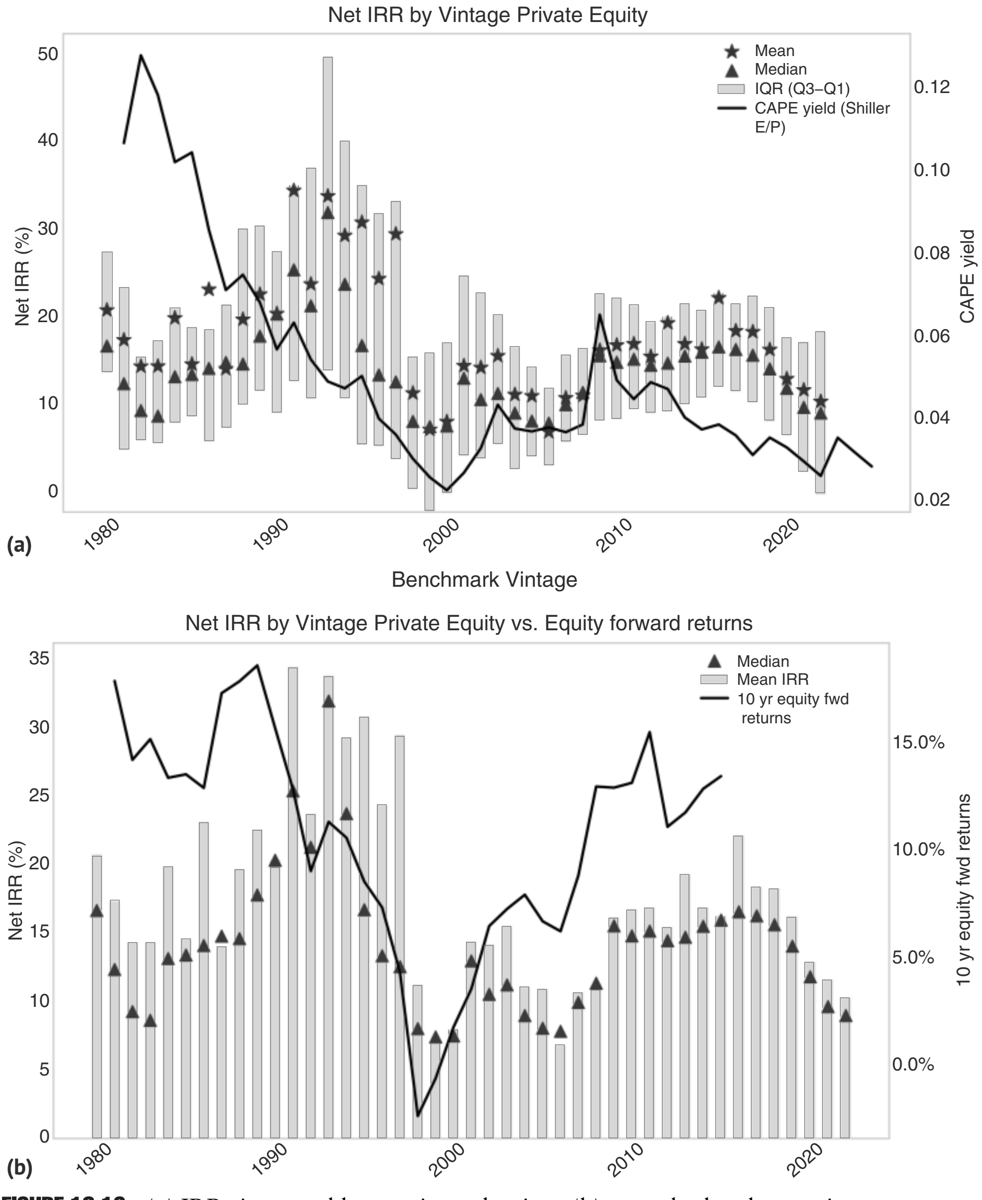


FIGURE 16.10 (a) IRRs impacted by starting valuations (b) … and related to equity forward returns.
Source: Bloomberg and authors' calculations.

may be somewhat tough for a while. On a similar note, IRRs by vintage are correlated with forward 10-year US public equity returns, as can be seen in Figure 16.10b. We had demonstrated in the equity chapter that the Cape Shiller PE is not overly useful for short-term trading, but it is important for 10-year forward returns. This would also suggest a more difficult environment. Having said all that, we are somewhat valuation agnostic in our approach, though this is easier to justify in public markets with a clear exit strategy. It could be that the unprecedented speed of technological innovation is a more important driver than higher starting valuations, though.

16.8 SUMMARY

Crypto has become an important asset class for macro funds. Having survived two crypto winters, we expect that crypto is here to stay. We focus here on Bitcoin. Although adding Bitcoin to the typical 60/40 portfolio would have been very positive for the Sharpe ratio in the past, this does not say much about future returns. At this stage, low double-digit returns are required to justify a one percent allocation in a 60/40 portfolio. We prefer a trend following approach going forward and find that Bitcoin lends itself well to that. We also discuss private assets: we find that even after unsmoothing returns, private equity and private debt still have higher Sharpe ratios than their public cousins. The same cannot be said for venture capital. During stagflation, infrastructure and, to a lesser extent, real estate generate attractive returns, while most other private asset classes have a high equity sensitivity and therefore do not prosper in this regime. Both crypto and private assets deserve a place in a macro portfolio. We expect to have many years of research ahead of us in both areas, as well as in our bread-and-butter macro trading work. Thank you for reading this book!

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